Loop Walking

A Field Guide to Hacking the Universe with Holonomy

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Preprint. This guide has not been peer reviewed, and its search of the literature is limited. It comes from a part-time research team that works in the open and gives its tools away. Version 7, 2026-11-10, with the work through Question the Substrate and Two Arrows. Concept DOI 10.5281/zenodo.21811588, which resolves to the latest version. CC BY 4.0.

Abstract

Loop Walking is a field guide to reading systems by holonomy: what a quantity carried round a closed path comes home changed by. Version 7 gathers the lessons of three projects, registered from 29 September to 2 November 2026, a triple-star toy, a driven rider in two waves and the Loop Atlas of toy machines, and of the two-voices papers on learning minds, *Question the Substrate* and *Two Arrows*, published on 9 and 10 November, into 884 register rows, 344 of them new since version 6, tagged with 16 domains. It reports what the method achieved as well as how to use it. Across rows 1 to 847 of the register, 519 times a lesson was carried from the setting where it was learned to another; 222 held, 79 held weaker and 218 failed, and the 222 that held sit in rows spanning all 16 domains. The three projects watch one object, an angle carried round a circle by a flow and read once per tick of an outside clock, and one fork decides most of what follows: the area-keeping triple star's class doubles with the constant 8.72 (its own island doubled once, then lost its stars), the dissipative rider's with 4.669, and a lightly leaky map crosses over once a trip keeps less than about half its area (0.440 to 0.524 across five leaks). A reading of pictures made after seeing the data was tested blind: chaos on the borders between treads, read at 84 per cent on a map already seen, held at 97 per cent on a fresh grid. A power law registered with two rivals held blind, 8 of 8 and then 4 of 4, and both rivals failed. The 582 arrivals of lessons gathered from 6 to 23 October were counted into 13 families, two of them defined on 23 October, 446 of the arrivals by fresh readers, and a blind re-sort of a sample agreed 85 per cent of the time (Cohen's kappa 0.67). Two falling cats that look like strangers, one twisting at its waist and one twist-free, turn out to be one cat under two rules, alike within 2.842×10^{-13} degrees on nine swelling paths. New in this version is a register of results beside the record of misses, so that what holds reaches the front page: the triple-star thread landed 236 of 490 predictions registered before their runs (to its paper's publication, 28 October), the Loop Atlas held 706 of 1,104 judged (at its book's publication on 9 November, concept DOI 10.5281/zenodo.24100672), and R2W, whose book *A Rider in Two Waves* was published on 5 November (concept DOI 10.5281/zenodo.23999853), landed at least one prediction in 448 of 527 graded tests. The two-voices papers held 122 of 212 and 87 of 170 predictions registered before their runs: a learning mind cannot grade its own homework, since a self-training chain that keeps the half of its labels it is surest of lost 25.3 points where the same cut ranked by a frozen first network lost 2.1, and it heals its answers long before its self, keeping 0.0167 of a loop's change in its answers and 0.3049 in its hidden layer. The largest results include the Loop Atlas's 51 matches with the published record, the sign slip behind a widely copied table's odd-order doubling constants, and the triple-star paper's one-sided door (163 of 166 escapes). The guide is a preprint, not peer reviewed, with a limited literature pass.

What this guide achieved

1. **The method travels, and the guide counts where.** In rows 1 to 847, 519 times a lesson was carried from where it was learned to another setting, on 322 of the 840 rows read; 222 held, 79 held weaker and 218 failed, and the rows holding the 222 that held span all 16 of the guide's domains (appendix "Transfers, row by row"); the 44 rows from 848 to 891 are not yet read for transfers, so each count is a floor.
2. **One fork decides most of what follows.** The triple star keeps area, and its class's doubling constant, 8.72, was matched on an area-keeping test map (8.721082), while the triple's own island doubled once and every star round it then escaped within 1,600 passes; the rider loses energy, and its class's constant, 4.669, was matched on a damped test map (4.669197); both constants are benchmarks, known and matched here; a lightly leaky map reads the frictionless constant for its first doublings and crosses over once one trip round its orbit keeps less than about half its area, 0.440 to 0.524 across five leaks, a crossover known in kind whose threshold is measured here (row 680; Two Readers, section 8).

3. **A slip time forecast from an island's width.** At a lock's clean edge the time to slip grows like a pendulum's logarithm: 7.57 and 7.37 passes per decade of distance at two clean rims, against 7.29 and 7.83 forecast from the island's own width with nothing fitted, where a square-root law would have grown the slip period 4.5 times over the span and it grew 2.4; the logarithm is known in kind, and its forecast from the width is measured here (row 675; Two Readers, section 8).
4. **A law read after the data, then tested blind.** A power law seen after one table ($\delta_2 = \delta_1^2$) was registered with two rivals before the next rows were opened; it held blind, 8 of 8 and then 4 of 4 in the junction, and both rivals failed (runs TOY-SIDE-3g, TOY-SIDE-4g and JJ-SMALL-1c; row 79).
5. **The eye, tested blind.** Chaos on the borders between treads, read after seeing the data at 84 per cent on a map already seen, held at 97 per cent on a fresh grid and at three chaos thresholds, so it is structure, not binning (row 655); a second reading, a staircase's turns marked by a change of word, failed blind and stays on the record (row 369). Reads by eye that named a relation held 19 times in 30, against 11 in 28 for reads that named a shape (rows 100 and 714; Two Readers, section 9), and the four straight lines Bear drew over a converter's chaos are each a real edge of it, within 0.12 A (row 873).
6. **Counts survive a change of machine.** Four riders run on a laptop and again in the container agreed on their slips, and on their stretch rates only within 0.012 (row 103). The triple-star thread's long runs, made again from nothing on a second machine, gave 103 lines: 83 the same, 20 the same verdict with other digits and none different; verdicts carry across machines, last digits do not (run XC1; row 810).
7. **A count reads through an instrument's turning.** A turning of 5.5 degrees put in on purpose across 38 sunspot maps moved 31 of 154,267 readable loop counts; the offset found in every sunspot's current behaves like exactly such a turning, and it survived blind tests on 24 of 26 and 45 of 50 fresh regions (row 567; from *Dancing on the Sun*, concept DOI 10.5281/zenodo.23067245).
8. **Four tests tell a true nothing from two hands cancelling when a knob has no sign.** A symmetry's nothing sits at the arithmetic floor, 6.994×10^{-15} against one rider's 2.599775×10^{-2}; a balance was placed blind within 0.04 and 0.006 per cent. The size sweep alone is not enough when both hands share a factor of the knob: on the Loop Atlas's crest-firer it came out flat within 1.5 per cent at every level, and the other three tests found the two hands (row 805).
9. **What keeps happening, counted rather than remembered.** 582 arrivals of lessons, gathered from 6 to 23 October, sorted into 13 families, 446 of them by fresh readers and 136 by the session that gathered version 7; a blind re-sort of 60 of the first 801 units, made before the two families of 23 October were defined, agreed 85 per cent of the time (Cohen's kappa 0.67) and named the same family 16 times in 16 where both saw a lesson; the commonest move, making every check able to fail, arrived 131 times (the Register's opening).
10. **A project's record of misses finds what its summaries drop.** Of the Loop Atlas's misses 1 to 457, 85 (18.6 per cent: 63 of the first 288 and 22 of the next 169) carried a lesson that had never reached the guide, whole or in part, through the project's own list of items; for misses 219 to 288 that count was forecast before it was made, at about 15 of 70, and came to 16. The triple-star thread's 268 missed and unread readings, read the same way on 23 October, gave 13 carried only in part and none missing (rows 784 to 789, 794, 806 and 834 to 837, and additions elsewhere).
11. **Two falling cats are one cat under two rules.** A cat twisting at its waist and a thinner twist-free cat bent to the mirror angle turn alike at every shape and on every path, within 2.842×10^{-13} degrees on nine swelling paths and with the rates within 9.3×10^{-15} at 2,000 random shapes; the pairing was not found in six sources read in full, among them Rademaker and ter Braak 1935, Kane and Scher 1969, Montgomery 1993, Gbur 2019 and Batterman 2003, nor in two more read on 22 October, when the Loop Atlas claimed it as new on a limited, noncommercial review, the claim held until advised otherwise; across Iwai's general model of 2010, two cylinders on a joint that allows any twist, the mirror holds only with the hinge at the tubes' centres, and his book of 2021 is the likeliest place left for it (rows 820, 742 and 511).
12. **A widely copied table's odd entries, and the one slip that makes them.** Briggs's table of doubling constants (1991; copied by MathWorld and by Finch's Mathematical Constants, 2003) is right in α at every order from 2 to 12 and in δ at every even order, and off in δ at every odd one, 5.9679687038 for 6.0846890743 at order 3; one sign slip in the doubling operator reproduces every odd entry, the printed $\delta(3)$ to 2×10^{-29}, and a fresh helper given no number re-derived the map's values and the printed one with code of its own, all five of its registered predictions holding. McCartney (2020) found the odd orders wrong first, and the corrected values are his; the cause is new as far as a limited pass shows, a pointer for a specialist (rows 874 and 875, and the Loop Atlas's note, *A Mirror in Feigenbaum's Table*, concept DOI 10.5281/zenodo.24101154).

13. **The toys meet the published record 51 times.** At its book's publication on 9 November the Loop Atlas counted 51 benchmarks against the published record, among them Feigenbaum's δ measured at 4.66920159 for a rounded top (row 771), Adler's slip period met exact to 1.1×10^{-8} (row 675), the needle's slips meeting the exact escape rate at 0.9885 ± 0.0122 where Kramers' formula runs 9 per cent high (row 725), and the pull between distant vortex rings falling as the cube of their distance, read from a chaotic train two ways, 3.17 and 2.97 (row 865). Beside them stand 16 found-here candidates, one claimed (item 11), and a corner ratio, −1.9644458, on which the logistic and sine maps agree to the ninth figure and which a limited pass did not find named (row 877).
14. **One toy's road forecasts another's.** A rule R2W read in its rider's locked step, each gap to 4.669 about a quarter of the last, forecast the Loop Atlas needle's fourth band ratio at 4.6585 and 4.6591; the needle gave 4.6599 and 4.6604, and its fifth, forecast at 4.6672 and 4.6673, came in at 4.6673 from above and 4.6672 from below (row 872).
15. **One resonance measured from the inside, and published.** *Held, Rocking, Locked* (28 October; concept DOI 10.5281/zenodo.23755473) follows a body a millionth of either star's mass round an eccentric pair of equal stars, a circumbinary planet's setting. Mardling's resonance pendulum, tested piece by piece, holds in its relation between the rocking rate and the island's width, within 1.1 per cent at the pair's eccentricity 0.10, while her widths run 20 to 31 per cent wide (rows 506 and 678); and past the stability limit the island loses its stars by one side: with the star's eccentricity at 0.20, 163 of 166 escapes leave toward the pair, where the published stability rules smooth the resonance over (rows 599 and 863; its register in the part on the headline register).
16. **The arithmetic can fake resonance physics.** At a tolerance that looks tight, a general-purpose integrator dragged every unlocked rocking rate of the triple star down by about 2.8×10^{-8} cycles a pass in every pass, enough to fake a lock's slide and part of its wear, and a tolerance 100 times tighter left 0.0093 of it (rows 599 and 741); the energy's drift falls by the same powers per tenfold tightening as the creep left in the rates, 1.46 and 1.92 against 1.41 and about 2.0, so that creep is the integrator's own (row 858).
17. **A law with nothing fitted.** When one extra tone is added to R2W's drive, the dancing pair moves by a closed form with no free number in it; rewritten on the passing matrix, it reproduced all 56 measured moves, harmonics 1 to 8 at seven dances, within 0.00004 of a turn (run KERNEL-1; row 699).
18. **Every project counted what landed, not only what missed.** By its paper's publication on 28 October the triple-star thread had registered 490 predictions before their runs: 236 landed, 211 missed and 43 could not be read, and 35 of its 92 redesigns of earlier misses landed; at its book's publication on 9 November the Loop Atlas had judged 1,104 registered predictions, 706 held, 369 missed and 29 undecided, with 572 planted checks beside them, 555 held; and at its book's publication on 5 November R2W had graded 527 tests against predictions written first, 448 with at least one prediction landed (223 with every prediction and 30 only through the named rival) and 79 missed outright (rows 646, 661, 854 and 882; the part on the headline register).
19. **A learning mind cannot grade its own homework, and heals its answers long before its self.** *Question the Substrate* (9 November; concept DOI 10.5281/zenodo.24094163) registered 212 predictions before their runs, 122 held, 89 missed and 1 held in part: a self-training chain that keeps the half of its own labels it is surest of lost 25.3 points and let an answer holding 11 per cent of the data die out, while the same cut ranked by the frozen first network lost 2.1 and kept it alive (survival 0.83 against 0.02); every failing rule ranked or copied by the chain itself, and every cure was a fixed reference from outside (rows 77 and 883). *Two Arrows* (10 November; concept DOI 10.5281/zenodo.24120415) registered 170, 87 held and 83 missed: after a closed loop of tasks a learner's answers keep 0.0167 of their change and its hidden layer 0.3049, and only a pull toward a record of the hidden layer taken before the loop brings it home, to 0.0003 (rows 888 and 889; the part on the headline register).

About this guide

The cosmos is an amazingly interconnected place. Every relationship between any two or more points in spacetime — from the way a DNA strand twists, to the way a fault line accumulates stress, to the way starlight curves around mass — is a fiber bundle waiting to be read. The bundles are always there. What varies is whether we have the sensor data and the geometry to reach them.

Loop walking is the method for reaching them. You choose a path that comes back to where it started. You walk it. You measure what fails to close when you return. That gap — the holonomy — is the geometry telling you something. Not a size. A rotation, a winding, a count: something written into the structure of the substrate that no smooth push can erase and no noise can manufacture.

A word about what this is. This is a working guide, kept up to date as the work goes. It records best practices even when we choose to ignore them, and bad results as eagerly as good ones. Do not be afraid to fork and pivot; just maintain full documentation and disclosure practices. The failures are as important as the findings, and in this work they usually arrive first.

Think of this guide as a holiday tree. The four limbs are the branches. The ornaments hanging from them are treasured lessons, each one earned on a specific project and priced so it travels to the next one. The tree keeps growing — limbs can be differentiated or added as the method finds new substrates. This document is the tree at version 7.

Why it exists. Two reasons, and the selfish one is worth naming. When I run an agentic AI research partner on these methods, it needs to be spun up: it needs the vocabulary, the lessons, the rule system built over these months, and the knowledge that the rules help but are not sacred. They keep results legitimate and save enormous time, but they are tuned down carefully depending on the project — because the rules teach through failure, and that is their whole point. This guide is that spin-up document. It is also what I wish had existed when I started. And it was not built alone — it is the record of a collaboration: the researcher, an AI assistant, and the subconscious that kept surfacing connections at 3 a.m. The 'we' and 'our' throughout are real.

How version 7 was gathered. We opened a session on 23 September 2026 to hold the guide's next upgrade after version 6. The lessons we learned formed its structure, and the papers themselves filled out the limbs of the tree as we recorded the questions that grew out of the data. Reading loops became writing loops in the Latch Atlas. Loops that needed walls, or each other, for a push became loops without history or walls in the Loop Atlas. The Loop Atlas spun off A Rider in Two Waves (R2W, called the RF Spoke in its early records), to work out how quadrature and timing make gearboxes and drive trains. We engineered with and through more chaos and binning in these weeks than in the six weeks before them combined. We watched the discovery tests and the behaviours on the pictures cross the projects like a ripple: driven pendulums turned up in all three projects in the same window, damped in two of them and frictionless in the triple star, and these little patterns clustered loudly. At moments it was a little eerie; I would lose track of which project's results I was reviewing, because the language was the same in all three. Each paper still had its own views and branches: fast and slow chaos in the three-body work; R2W sharing its depth-axis explorer, a toy with knobs, with the Loop Atlas, which reshaped it into the stretch-and-squeeze explorer, the same ideas under different names; and R2W mapping its own toy's atlas of where and how we can grip RF and use it as an engine and a gearbox. I am still learning the language for what is in my head, because some part of me still expects to walk into a dark room, have the lights snap on, and hear everyone shout "Surprise! It was all an elaborate joke." That is how it feels to play loop walker with holonomy every night at my dining table, in a way that connects with so many cool spaces and geometries.

What version 7 includes. This version includes the work through the two-voices papers, *Question the Substrate* (published on 9 November 2026, concept DOI 10.5281/zenodo.24094163) and *Two Arrows* (published on 10 November 2026, concept DOI 10.5281/zenodo.24120415). Their lessons are rows 883 to 891 and additions to 20 earlier rows, each naming its paper.

Quotations. Bear's words from working conversations, and the labels he wrote on pictures, are quoted in his own words, with typing slips and capitals set right; the projects' own records keep them as typed.

The glasses

I want to describe how I see things, because it will show up everywhere in this guide and the research papers that companion it, and knowing it in advance may save you some confusion.

I experience the universe as mostly ternary. Slopes go up, sit flat, or fall: −1, 0, +1. Curvature turns left, keeps straight, or turns right. A ternary operator is not a simplification — it is the native language of the thing changing direction. Rates of change, rates of change of those rates, and rates of change of those — expressed in three signs — describe almost everything this method reads. The hierarchy goes: value, slope, curvature, and beyond. Each level is ternary. When I find a substrate that resists the ternary frame, I pay attention to that resistance.

I see fiber bundles everywhere. Every pair of points in space has a relationship. The collection of all those relationships is the fiber bundle network. It is always there — the instrument does not create it, only exposes it. The instrument's job is to find a seam or a junction where the geometry is readable, walk a loop around it, and report what the geometry accumulated. Some substrates offer that seam readily. Others hide it behind instrumentation limits. But the bundle is there regardless.

I treat every failure as a lesson that has to be extracted deliberately. The method delivers failure routinely, and that is the point. A null result, read correctly, is more valuable than a positive result read carelessly, because a null with a cause is a specification: it tells you exactly where the instrument ends and the substrate begins.

A vocabulary for this guide

These words appear throughout, often without warning. The definitions below are how this corpus uses them — not necessarily how a textbook would, though the textbook usually agrees.

Invariant. Something that stays the same even when everything around it changes. An invariant is not a coincidence — it is a consequence: the geometry forces it constant, and no noise, calibration drift, or perturbation can move it without crossing a threshold that has a name and a measurement. In this corpus we hunt non-trivial invariants: quantities that topology makes permanent. When you find one, you have found a direct readout of the substrate's geometry. An invariant is not a bad thing — it is the whole point.

Loop walking. You pick a closed path — a route that returns to its starting point. You carry some quantity along that path and observe how it has changed when you return. The change is the reading. This is the entire method. Every technique in this guide is a way to choose the loop, carry the quantity, or interpret the reading.

Holonomy. The change that accumulates when you transport something around a closed loop. A compass carried around a mountain may point in a different direction when it arrives back at the start — that difference is the holonomy of that path in that field. This guide reads holonomy not as a curiosity but as an instrument: the holonomy reports what the geometry did to whatever you were carrying. Not every leftover is a holonomy: run the loop out and back along one line, enclosing no area, and a holonomy leaves nothing there, while a knee (hysteresis) leaves its band (row 116).

Fiber bundle. Imagine every point in space has a small room attached to it. The space you are moving through is the base. The rooms are the fibers. The rules for how you move from one room to the next as you travel are the connection. The whole system — space, rooms, and rules — is the fiber bundle. The key fact in this corpus: the bundles are always there. What varies is our ability to access them, which depends on what sensors we have and what the geometry looks like at the seams and junctions.

Differential geometry. The mathematics of curved spaces. This guide uses it as a gateway into the geometry of the substrate, not as the destination. You do not need to compute it — you need to know that it organizes which loops are equivalent, which gaps are meaningful, and why the winding number is protected.

Winding number and integer channels. How many times something wraps around. One full wrap is winding number 1. Two is 2. The crucial property: it cannot change by fractions. It cannot drift between checks. White noise cannot push it from 3 to 3.7 — it can only jump by a whole step, and that jump requires a crossing event with a number on it. This is the protection this method exploits.

Residual. What remains after the integer part is extracted. The continuous signal that rides alongside the winding count. The residual tells you about the substrate's smooth properties; the winding tells you about its topology.

Amplitude blindness. This instrument does not care how large the signal is, only about its topological structure. Magnitude is deliberately discarded. That is not a limitation — it is the design. Amplitude-blindness is what makes the read immune to calibration drift, to scaling errors, to the difference between a big version and a small version of the same geometry.

The competence surface. Not "the instrument works" — but "here is where it works, here is where it goes dark, and here is the measurement that tells them apart." Every instrument this guide describes ships with its own visibility matrix: the list of what it can see, at what aperture, and what it cannot see and why. A limitation with a known cause and a named price is a design parameter, not a failure. The surface is drawn from the ruler (defined below, after the floor): wherever the effect outgrows the ruler's step, scatter and bias together, the instrument sees, and wherever it does not, the instrument goes dark, however good the ruler is elsewhere.

Common mode. When the same signal arrives at all your sensors simultaneously. Common-mode signals are indistinguishable from reality — every sensor agrees, and that agreement is the trap. An instrument that cannot reject common mode cannot separate a global perturbation from a local one. Common-mode rejection is the art of building readings that depend on *differences* between sensors rather than their shared level.

The floor. How much your measurement moves when nothing is happening. Before any reading means anything, you measure the floor: the amplitude of your quantity across an ensemble where nothing is wrong. If the effect you are looking for is smaller than the floor, no processing downstream can recover it. That comparison — effect divided by floor — is one division, and it costs about a minute, and it kills more bad projects than any other check in this guide.

The ruler. Whatever turns a claim into a number: a count, a fit, a grid, a reader, a picture's pixels read on its own axes. In this guide a person's eye and a blind reader propose and the ruler decides, but the ruler has limits of its own, and each leaves a mark that can be told apart from the others. Its *step* is the smallest change it can see: a reading on a grid is good only to the grid, a number carried to too few digits is good only to its digits, and a system that will not commit to one side of a step's edge makes signal out of the edge (chaos and binning; rows 686 and 738). Its *scatter* is how far it wanders when nothing is happening: scatter changes sign from run to run and shrinks when readings are averaged (the floor). Its *bias* is a lean it adds to every reading: small, steady, the same everywhere, and it does not average away, so an effect with those three properties is the ruler's fingerprint until a known answer says otherwise (row 741). Its *frame* is what it measures against: a ratio's partner, a zero that can move, a coordinate that bunches its marks in one place; when the frame moves, the reading moves with it (rows 308 and 730). Each mark has its own test. Scatter shrinks with averaging and changes with the seed. Bias survives averaging and changes when the ruler is

changed. A step's edge moves when the grid is shifted by half a step. A frame's motion disappears when both ends are measured. The ruler is not the competence surface: the ruler is the instrument in the hand, and the competence surface is its map.

Chaos and binning. A recurring partner in this work, named as a family in version 7, which counted 63 arrivals of it, 3 of them from sessions that had not yet read the guide (the Register, "What kept arriving"); a test at three chaos thresholds can tell it from structure (row 655). When a chaotic or near-chaotic system sits near the boundary between two measurement bins, it flickers back and forth across the edge, producing apparent signal that is really the bin boundary, not the physics. The formal name shifts by domain — Arnold tongues in oscillators, level repulsion in quantum chaos, resolution walk in numerical analysis — but the structure is the same: discretisation meets a system that would rather not commit to one side. Recognising this pattern is one of the most transferable skills in this corpus.

Memory reading. A per-cycle measurement that changes by its own size when the clock shifts a tenth of a period — reported as a slope and a crossing, not a value. Before quoting any per-cycle number from a medium with memory, sweep at least five phase points and report the line.

Floor (instrument zero). The instrument's own answer where the true answer is known to be zero. For a synthetic instrument, this floor is stated on the first page of results. A result inside the floor has a magnitude and no sign.

Station-keeping within the floor. The honest verdict for a body or measurement whose motion is smaller than the instrument's floor. The sign is not resolved — this is not a failure, it is a precisely stated result.

Two clocks. The verdict when two time scales of one process are shown not to scale together, with the predicted ratio committed at a band before the runs. A falsified ratio is a finding about the medium, not a failure.

Duct. A mechanism the instrument found that the object being modelled does not possess. Kept, fenced as not a design, pointed at a specialist.

Latch. A whole number that costs energy to hold and sits behind a wall higher than heat's jiggle. It holds nothing itself; it forbids the store beneath it from emptying below a floor. The Latch Atlas prices one in each of thirteen media.

Label. A whole number that travels and holds nothing: twisted light, radio, sound. It becomes a latch only when it lands on matter whose phase must close on itself.

Powered lock. A lock that holds its count only while it is driven, like a voltage standard's steps, which vanish when the tone stops.

Undecided. A verdict, not a failure to reach one: a pre-registered test whose result landed in the gap its own rule left, reported as written. It differs from HELD, which waits on a missing artifact.

Tread. A flat step in a two-rhythm rider's staircase, where its slip holds a fixed ratio across a range of pace. Treads appear only where phase is carried forward from one cycle to the next (row 605), and one is accepted only with flat shoulders and after the null case has run (row 603). It is not the Pythagorean comma, which is a closure gap (row 202).

Memory. A word with four senses in this project, each right in its own room, so a sentence that carries it across rooms says which. A medium that remembers, so that a per-cycle number moves with the clock: the "Memory reading" above, whose five-point sweep proved itself on the Loop Atlas's own rows where the line was steep and was not needed where it was flat. A rider that does not forget its start, read as a largest Lyapunov exponent near zero (row 605). Route-dependence, where two locks overlap and the state you hold depends on the road you came by, which needs mass (row 670). And a dial in the Loop Atlas's needle called "memory", which is its inertia (row 643). The needle's equation sets its drag to 1, so that dial is the needle's mass over its drag, the time it takes to forget its speed, and in the toys memory and mass are one number. A method that erases memory, such as heavy drag or a reset of the speed, removes the mass's coasting from the motion. Whether every effect of mass goes with it, its weight and its stored energy among them, is an open question this guide carries for a future project.

Anomalistic lag. A binary's periapsis clock running behind its energy clock, as the Moon's anomalistic month, perigee to perigee, runs longer than its sidereal month. The name the triple-star sessions gave a lag of this kind in their runs. "Two clocks" keeps its narrow sense, two time scales of one process shown not to scale together, and two clocks tied to each other by their events are coupled clocks.

Coupled clocks. Two clocks that each keep time by their own events and are tied to each other. They come home together after the denominators of the continued fraction of their ratio, and a lock is where they agree. "Two clocks" itself keeps only its narrow sense: the verdict when two time scales of one process are shown not to scale together.

Outside tick. In a strobed picture, the tick set from outside that never feels the rider: the passing of a ripple, a star's closest approach, a drive's period. Its partner is the rider's own pace, set by the rider's slope and its drag; a lock is the rider's own pace giving in to the outside tick where the two are already close. A third timing reads neither: the event, which opens a gate when something arrives. A motion that keeps its rhythm between kicks, set against chaos, is called regular, not a clock.

How to read this guide

If you are a returning session or an AI partner: start at Station 0 of *The Walk*. That station tells you what has already been tried, what is on hold, and what has been closed. Then walk forward from there. Load limb sections when a station raises a question you cannot answer from the prose.

If you are new to this: start at Station 1. Skip Station 0. Walk the ten stations, which are written to be followed in order on the first trip. When a station points you at a row number, that number is an entry in the register at the back. The register is the dictionary; the walk is the phrasebook.

If you are reading for the pleasure of it: this guide is also written for people who do not know anything about this yet. There is a great deal left to discover, and no certificate is required to start looking. Holonomy is in no hurry.

If you are reading a picture of data: the part after the walk, *Two Readers and a Ruler*, is an order of operations for it. A person's eye and a blind second reader propose, and a ruler decides. Each of its lessons is also a row in the register, and it cites them by number.

The Walk's four figures are embedded at the stations that need them. *Two Readers and a Ruler* carries 20 of the pictures it calls for, most of them Bear's marked pictures and R2W's figures; where a picture is still missing, a note in italics says what it would show.

THE WALK

A working path through a project — ten stations cutting across all four limbs. Station 0 is for returning sessions. A newcomer starts at Station 1.

Station 0 — What is already banked

Session only. A newcomer skips to Station 1.

Before anything else, find out what has already been tried, what is waiting on something, and what has already failed and been recorded. This station exists because the method has no memory of its own and every session that skips it re-derives work that was finished months ago. **That is the failure this entire document was built to stop.**

Three questions. What is **banked** — results already priced, already named, already closed. What is **held** — items waiting on a specific missing artifact, each of which should name the file that would clear it, so a hold is a one-line question rather than a blockage. And what is a **banked negative** — things that were tried and did not work, recorded with their counts, so that a good idea does not get re-run because nobody wrote down that it lost. Read the register of what landed first, beside the record of what missed: a handover keeps the open items and drops the finished wins (row 821).

The discipline that makes this station work is upstream of it: **an idea merely deleted comes back.** A kill has to be recorded with where it applied, why it lost and what replaced it. And a name with no definition on the record is an orphan wearing a label; if you find one here, that is the first thing to fix.

A fresh session is not a lost one. Before declaring anything lost, check the disk and the live record: the canonical object is the file and the deposit, never the session, and “I do not see it where I first looked” is not “it is gone” (row 89).

A tool carried to another machine should find its companion files by what is in them, not by their names; when one goes missing, let the launcher’s census say what happened before guessing (row 721).

Station 1 — Do your words survive the boundary?

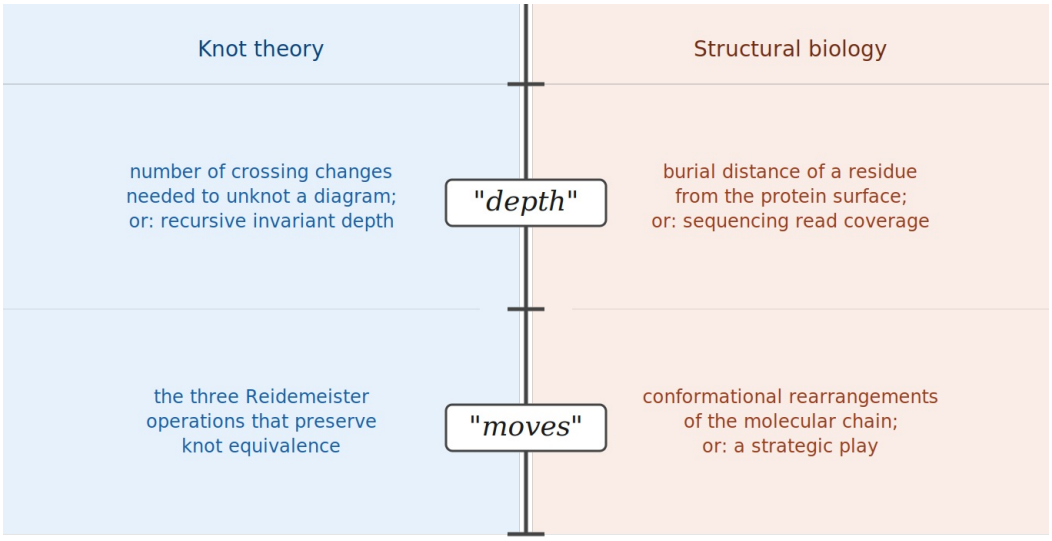


Figure 1. The words fence.

Before you name a single object, check the vocabulary you are about to name it with. **This is the station that comes from the only barrier that ever actually stopped this work.**

Cross-domain words mean different things on either side of a fence. *Depth* and *moves* are the two that cost the most here. And the trap has a second floor that defeats the obvious remedy: **the standards move while you are reading them**, so learning the other field’s term once is not enough. *That is a barrier nobody warns you about and it cost more than any calculation.* The tools were already good. The language was not.

The procedure is one line and it would have caught every instance this corpus has paid for: **before a word carries a claim across a boundary, write down what it denotes on each side and check that the two denotations are the same object.** Do it in writing. The instances that got through were all cases where somebody knew both meanings and did not write them next to each other.

Check the modulus as well as the word. A wrap chosen to compare the directions of lines can fold a quarter turn away, and then a definition crosses the fence inside the arithmetic, where nobody wrote it down (row 554).

And name the pair: two cases that seem to disagree about a phase may be measuring it between different pairs of knobs (row 586).

A word inside a number must survive the boundary too: “travel” was a length in one tool and a signed part along a line in another, and one lap read two ways looked like two strokes (row 612).

When a result is called portable, say in which sense, number for number where the equation is the same or a calibrated ruler everywhere else, and name the machine, since a cousin is not the same machine (row 727).

Station 2 — Can we reach the bundle?

The bundles are always there. The fiber bundle network between all points in your piece of real estate is assumed to exist — it is not absent, it is not optional, it is not something you earn by having the right substrate. What varies is your ability to access it, and that depends on exactly two things: the available sensor data, and the geometry. **Asking whether a bundle exists is the wrong question.** The right question is: do we have the sensors and the geometric exposure to read the bundle that is already there?

This reframes every null at this station. You are not dead — **you are unmapped**, and unmapped is always recoverable. The geometry will offer a vertex, a junction or a seam when the instrumentation can reach it.

Name three things before going further: the **base** you are moving over, the **fibre** you are carrying, and the **connection** that says how the fibre is carried. If you cannot name all three yet, that is a gap in the map, not a gap in the world.

One prerequisite, and it is worth exactly four words: **one pin, no loop.** Holonomy needs a path along which state is transported and around which closure can fail. A single steady source with no differential offers nothing to walk — not because the bundle is absent, but because the sensors have not yet given you a path to walk.

Ask the incumbent bar now, while you can still change the question. Who is already doing something like this, and what bar does this work have to clear? An instrument that claims to be universally superior is a marketing claim, not a scientific instrument.

Begin the expert witness roster here. An expert witness in the courtroom sense: retained in advance, qualified on a specific question before they are needed, independent of the work and therefore able to disagree with it. Name the specialties this project will need. The roster accumulates across projects and starts earning its keep long before anyone is called.

Before asking what a medium remembers, ask whether its whole numbers are latches or labels: held behind a wall, or flying away holding nothing (row 561).

Station 3 — Choosing the loop

The witness roster grows here as the loop geometry becomes concrete.

The loop is not a free parameter and it is not a resolution dial. **Loop shape is a match to the interior’s symmetry, not a ladder toward perfection.** A finer loop reads a smooth interior better and reads a discontinuity *worse*, because it averages across the cut.

Match the aperture to the source’s dimensionality. Getting this wrong produces a clean, honest and entirely misleading null. **The rim sees the mouth and never the neck:** a tight core inside a wide circulation collapses below your station spacing whatever your sensitivity.

Redundancy is the lie detector, and a triangle has none — three vertices are exactly determined, so one bad station corrupts the reading with nothing left over to catch it. **Prefer an odd count**, because an even pointer has a preferred axis and couples to the two-fold disturbance it should reject.

Where the substrate has its own arithmetic, **let the arithmetic choose the face.** A lattice whose smallest closed walk has four steps gives you squares, not triangles.

Station 4 — Declare the blind spots now

Now, before the first reading. Not after.

Start with the ones you can derive for free. A signed quantity is blind to anything that cancels under mirroring — that follows from the definition with one line of algebra and no experiment. A scalar observable is loop-blind by the telescoping identity, permanently. These cost nothing and can tell you a channel is closed before you spend even a minute measuring it. Derive the free blind spots first precisely because they are free.

Then publish it. **The honest instrument ships its own visibility matrix**: which physical modes each channel can see, at what aperture, and — as importantly — which it cannot.

And commit the bar in writing while the data is still missing. Two parts, so that a win on one axis cannot be reported as a win on both. **Then do the step almost nobody does: go back and ask whether the bar could ever have said no.** A target satisfied by construction is not a target.

Before the first reading, ask every toy the four cheap questions: faster, backward, again, and one knob alone (row 584).

Derive the symmetry before choosing where to centre a loop: a loop centred where its quantity is odd under a mirror reads zero to every order, a blind spot you can predict without running anything (row 259).

Station 5 — The floor, before you read anything

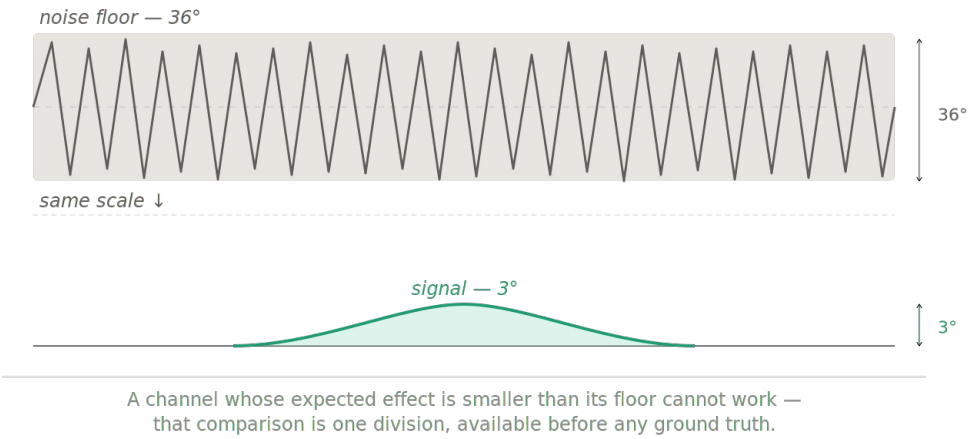


Figure 2. The floor.

This is the cheapest station in the walk and it kills more bad projects than every other station combined. It needs no labels, makes no claim, and takes about a minute.

Measure how far your quantity moves when nothing is wrong. Then compare it to the effect you are hoping to see. **A channel whose expected effect is smaller than its floor cannot work, and no amount of processing downstream will recover it.** That comparison is one division, available before any ground truth at all.

The floor belongs to a substrate *and* a perturbation *and* a corpus. **Using one floor for three substrates is an error**, and it is an easy one because the first floor you measure feels like a property of the method.

The integer can die before the amplitude. Above the matched scale a count is far more robust than a magnitude; below it, white noise does not merely degrade the count, it *manufactures* windings that were never there.

On real data, the rim decides: a count is only as good as the rim it is read from, and the window for a read has two edges and a trap in the middle (rows 566, 568).

Before trusting a statistic, run it on a record of the same length with nothing in it: a median that carries only the record's length is not a clock (row 616).

Say the instrument's own floor before setting a threshold: a bar tighter than the floor fails for the wrong reason (row 337).

Before reading a miss against a bar as noise, simulate how often the bar fails by chance on the null shape (row 723).

Station 6 — The first read

Decide which channel carries your signal, and know that the choice is not between a good number and a bad one.

Some numbers wobble and some numbers tick. Amplitudes and phases are continuous; windings and linking numbers are integers that cannot move by less than one. **A local dent cannot change how many times the phase wraps** — only a perturbation that itself winds all the way around can. That is the whole protection.

Two habits worth more than any single channel. **Read differences of an invariant** — differences kill any uniform background, and an integer cannot drift. And **read the rate, not the level.**

This last one is worth dwelling on. The corpus arrived at it again and again, from substrates that share no derivation: a tornado's core strain, a fault's shear, a clinical array, three commodities' prices, a clinical residual and a walker's ternary rates (rows 105, 294, 460, 492 and 537). It had arrived six times when row 537 was written, and version 7's fold met it twice more (rows 708 and 715). Arrivals that independent are not a coincidence — it is the corpus telling you something about the world. When the same lesson re-derives itself across that much distance, it is not a method preference. It is a structural fact about how information travels.

The integral erases history. A global walk is order-blind, so a true inverse pair vanishes. And **an open arc is a convention; only the closed loop is an invariant.**

Before connecting two numbers that look alike, ask what kind of object each one is. An exponent that tracks a length is a magnitude, however integer-like its digits (row 548).

A count is whole only when it is read as a sum of closed pieces (row 560). And split any sum by the sign of the field before trusting it: the physics fixes which symmetry a quantity must have, and a part with the wrong symmetry is a witness against the instrument (row 575).

After the first read, read the data once more as if no physics applied: what does it look like, and what is it doing? That read is not a result, but it is often where the next test comes from (row 613).

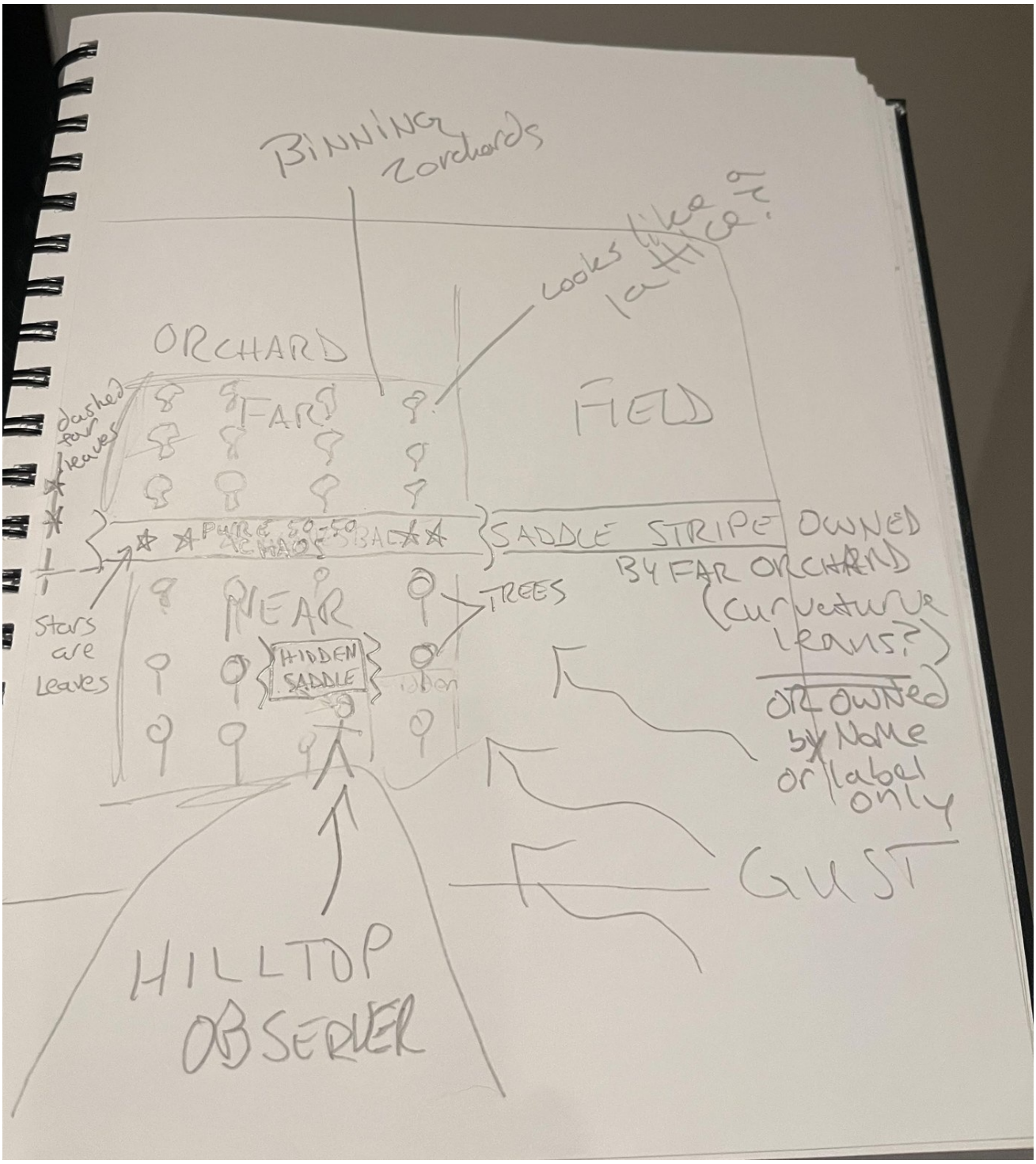


Figure 3. The orchards, drawn. Bear's pencil drawing, 6 October, of R2W's read 33, a read given in words alone and naming no law: two orchards, far and near, a stripe of star-leaves between them, a hidden saddle, a gust and a hilltop observer. In R2W it was the read that showed how much a no-physics read can carry, and R2W's own pattern lessons later set its saddle between two orchards beside the light cone of 28 October as a rhyme.

Count landings, not stumbles: a counter watching a noisy phase at one line logs its jitter as pairs of crossings, and a knee that commits only on landing removes them (row 627).

Name a state only by a test that separates it from its neighbours: its fraction and its period together, read over a whole number of its cycles, and only after it has landed (row 665).

A count of locks is a count made by a reader, so name the reader beside the number (row 722). And judge a transition by a measure that changes smoothly: a yes-or-no line laid across it says only which side of a chosen value a point fell (row 728).

If the data comes as a picture, read it with *Two Readers and a Ruler*, the part after the walk: name the picture, declare what you have already seen, read it blind, and let the ruler decide (rows 713, 369 and 616).

Station 7 — Before you believe it

This is where the expert witnesses take the stand.

An instrument that has only ever agreed with itself has not yet been tested. Internal consistency — nulls, injections, shuffles — is necessary and supplies no external witness. This is where the witnesses you retained at Station 2 take the stand: a specialist qualified in advance who reviews the reading independently, knowing the question but not the preferred answer.

Coherence is the symptom, not the safety. This corpus arrived at that three separate times in three substrates: a control loop defending a corrupted setpoint; a pipeline confident across every seed because one character of an accession was transposed; a game board whose whole map descended from one anchor placed on ground that read clean and was not. **A wrong input makes every internal check honestly wrong together.** A false reality from a single source is more seamless than a true one assembled from independent noisy sensors, because there is no second voice to disagree.

Widen the window until the answer stops moving, then report how wide you had to go — a boundary in your search will masquerade as a structure in your data.

Two checks before the headline is written. Validate its number against a harder version of your own method that changes one thing, and remember that passing proves the solver, not the quantity (row 549). And if you read a magnitude off a staircase across an integer, list everything else that changes with the integer and pin it (row 550).

A second fit of the same data is not a second witness. Two readers of one light, like two solvers of one set of shapes, tell you about the readers (rows 549 and 556); the witness is a second instrument on the same ground.

Some witnesses are already retained: the substrate’s own motion past the instrument, the archive’s own keywords, a second count with the opposite blind spot. Call them, and expect them to testify against you too (rows 567, 576, 577).

Count witnesses at the level of what is independent, the run and not the episode, and re-run what was first seen as unscored controls beside fresh cases from a grid that cannot repeat it (rows 619, 621).

Before judging another room’s number, run that room’s own code here, unchanged, and refine only then (row 650).

When two instruments disagree, do not take the older one as the reference; bring a third that shares code with neither (row 35).

Station 8 — When you hit a wall

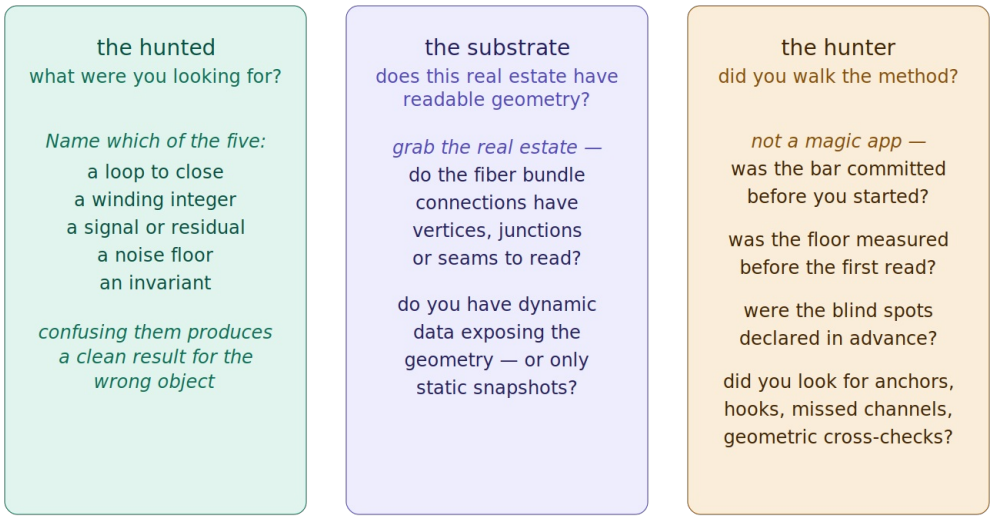


Figure 4. The three-cause autopsy. Flucht nach vorn.

The null is not the exception branch. It is the main road.

The entire guide exists because holonomy delivers failure routinely, and the method demands a new partnership with failure as the teacher. The clause that keeps this from being consolation: **the method does not slack and does not hand out participation trophies.** A lesson is not conferred by having failed — it has to be extracted, deliberately, by looking.

The four places to look before writing any null:

An **anchor** — is there a fixed exterior reference you never identified?

A **hook** — is there a place the structure offers purchase you walked past?

Missed instrumentation — is there a channel already in the data you never read?

A **geometric cross-check** — is there a closure, a parity, a mirror or a symmetry that would test this for free?

Now run the prior-art pass on the route. Not the idea — the route. The incumbent bar was asked at Station 2 and tells you what bar to clear. This is a different question: has someone already mapped this particular wall? Run it here, when you have hit the wall, not when you had the idea, because running it too early means inheriting the field's route including its blind spots.

Do not stop when you find an answer. You will never find the other patches and combs when you quit looking.

When the pass credits a result, read the result — the pages, not the abstract. A citation checked only for existence is a claim you have not verified (row 552).

When a thing gets thin near an edge, compute the edge first and then look; a grid cannot find a sliver (row 631).

When the whole toy turns out to be a textbook object, its map becomes a bench: every landmark is a known length to check the ruler against, and any claim of something new moves from the physics to the reading (row 726).

Station 9 — What you owe

Tell the ending first when the ending is a failure, and tell it in the abstract rather than the discussion. Name your loss before a reader can name it for you. Put the top results there too, one sentence each with their numbers, because a reader, human or machine, may read no further (row 840).

Report the misses in print. A registered prediction that failed belongs in the body with the reason it failed, because the reason is usually the result. **A clean pre-registered null with a caught borrowed parameter is worth more than a positive result**, because only one of the two can be faked by wanting it.

Separate your confidence from your tone. Write down, in a register, what you assumed, which specialty should check it, and the precise question that would settle it — so that the ask can be routed to seven people rather than pleaded to one. Where the trade belongs to somebody else, **state the asymmetry in both directions with numbers and decline to resolve it**, because that judgement is not yours to make on their behalf.

And publish the compliance statement in fixed form, so that **an omission is visible without an audit.**

Read the built artifact back before it ships (row 553), and ask of every check whether it could have failed on the question you are claiming (row 551).

Report an UNDECIDED as written; it is a verdict (row 580).

A withdrawal reaches every document that carries the number, the withdrawing room’s own first, or the correction becomes a second defect (row 588).

The limbs follow: The Method as an Instrument — The Geometry & Substrate You Walk — Holonomy You Write — The Ledger.

Station reference card

Station	Name	Key rows	What they price
0	What is already banked	402, 486, 504, 89, 721, 821	The banked negative; an idea merely deleted comes back; the guide is the solver’s externalised state; a fresh session is not a lost one: check the disk and the record before declaring a loss; find files by content; read the register of what landed first
1	Do your words survive the boundary?	502, 503, 554, 586, 612, 727	Write both denotations before any word carries a claim; the standards move while you are reading them; check the modulus as well as the word; name the pair of knobs; a travel number is a length or a signed part along a named line; say which sense of portable, and which machine
2	Can we reach the bundle?	505, 534, 535, 561	Expert witnesses retained early; name which of the five you are hunting; the substrate is fiber bundle real estate; matter latches and waves label
3	Choosing the loop	355, 374	Prefer an odd count — even pointers couple to even-fold noise; rim sees the mouth and never the neck
4	Declare the blind spots now	2, 21, 337, 584, 259	Declared blindness; the visibility matrix; a bar that cannot fail is not a bar; faster, backward, again, and one knob alone; derive the symmetry before centring a loop
5	The floor, before you read anything	305, 451, 566, 568, 616, 337, 723	The integer can die before the amplitude; rank each channel by effect divided by floor — one division; on real data the rim decides; coarse sampling misses, noise fabricates, likeness erases; run a statistic on an empty record of the same length first; say the instrument’s floor before the bar; simulate the bar
6	The first read	105, 294, 359, 120, 121, 548, 560, 575, 613, 627, 665, 713, 722, 728	The signal is in the rate; read the slope not the value; read differences of an invariant; the integral erases history; an open arc is a convention; ask what kind of object each number is; a count is whole only as a sum of closed pieces; split by polarity before you sum; read once more as if no physics applied; count landings, not stumbles; name a state only by a test that separates it; read a picture with two readers and a ruler; name the reader; judge by a smooth measure
7	Before you believe it	306, 424, 505, 549, 550, 556, 567, 576, 577, 619, 621, 650, 35	An instrument that has only ever agreed with itself has not been tested; coherence is the symptom; expert witnesses take the stand; a harder solver tests the solver; pin what climbs with the integer; a second fit is not a second witness; call the witnesses already retained; count witnesses at the level of what is independent; run the other room’s code before judging its number; bring a third instrument when two disagree
8	When you hit a wall	507, 508, 506, 509, 552, 631, 726	Flucht nach vorn — null is the main road; four places to look before any null; prior art at the wall not the idea; do not stop when you find an answer; credit the theorem that exists, read on its pages; compute the edge before looking for a sliver; a textbook toy’s map is a bench
9	What you owe	439, 475, 495, 551, 553, 580, 588, 840	Tell the ending first when the ending is a failure; publish the compliance statement; state the asymmetry in both directions and decline to resolve it; a check certifies only what it could have failed on; only the built artifact proves an edit landed; UNDECIDED is a verdict; a withdrawal reaches every document that carries the number; the top results go in the abstract, one sentence each

How to use this card. The card leads from any station in *The Walk* to the rows in the register that it calls on most; row numbers are stable, and titles travel with them in every cell. When a station raises a question you cannot answer from the prose, the row numbers here are your entry points into the register. Every row carries its price and its name in the same cell — enough to re-load the lesson with zero context.

Rate-over-level (rows 105, 294, 460, 492, 537) had arrived six times when row 537 was written, and version 7’s fold met it twice more (rows 708 and 715). It is listed at Station 6 because that is where you choose the channel, but it applies everywhere you compare a continuous measure to a categorical one.

The walk cuts across all four limbs. The register organises by limb; this card organises by when. A specialist who arrives at Station 5 via the register (*L1 — The Method as an Instrument*) and a newcomer who arrives at Station 5 via The Walk are reading the same rows — just through different doors.

TWO READERS AND A RULER

How to read a picture of data: a person's eye and a blind second reader propose, and a ruler decides. Taught by Bear to three projects at once in October 2026, and checked against answers known in advance.

A computer reading a chart can miss what a person sees at a glance, and a person can see shapes that are not there. This part teaches a way to use both. The person marks the picture, a second reader marks it without seeing the first, and every mark becomes a test with a number before anyone believes it.

Some of the best readings in this work were made by eye, before any program existed to find them. On 9 and 10 October 2026 Bear taught three projects at once how he reads a picture of data. The **Loop Atlas** book watches a needle, a forced and damped pendulum looked at once per push of its drive. The **R2W** maps a rider on a tilted washboard, passed again and again by a ripple, in the Depth Axis Explorer. The **triple-star sessions** follow a third star that passes an inner pair of stars once per lap. Each project turned his method into code, read pictures blind before it saw his marks, read the other projects' pictures, and put a number on every claim that could be tested. This part is what survived the measuring.

The cast and the dials. A few words from the three toys recur. The needle is pushed round by a drive that repeats; its heaviness decides how much of its start it keeps. Strictly, that dial is its mass over its drag, the time it takes to forget its speed. The Loop Atlas calls it the needle's memory, and with the drag set to 1, as the needle's equation sets it, memory and mass are one number (row 643). R2W's rider drops ledge after ledge down a tilted washboard; its maps are drawn against dials called the drive, the pull and the clock ratio, with the rider's lag, where it stands in turns when the ripple passes, up the side. Its weight is its heaviness and its honey is the drag that slows it. A rider standing still is "the rest", a rider dropping one ledge per tick is "the conveyor", and any other lock on a fraction is a "groove"; "the third face" is a three-step dance found at weight 0.40, and "a coin" is a place where two starts end differently. The triple star's dial is its birth ratio, the number of turns the inner pair makes per lap of the third star (7.200, 7.230 and so on), and its pictures are drawn once per pass. Each project ran in working sessions of its own with an AI partner, Claude; this part sometimes calls a project's sessions its room.

The idea underneath fits in one sentence: **what looks like disorder is often order binned on top of order.** Many signals land in the same holes and lie over one another, and the method takes them apart.

Two readers and a ruler. The first reader is a person's eye. The second is an AI partner that reads the same picture blind and writes its reading down, by the clock, before it sees the person's. The ruler is whatever turns a claim into a number: the data, or the picture's own pixels calibrated on its axes, tested against a shuffle of the same dots and against a made-up case whose answer is known. **The eye proposes; the ruler decides.** What both readers saw is the strongest evidence there is. What only one of them saw becomes a test.

The method in five steps, for an AI working beside a person. First, make pictures for an eye, not for a summary: a strobe picture, the same picture squeezed sideways and tilted to a grazing angle, and colours that do not lump classes together. Second, ask the person to mark it, with strokes and a few words, before either of you sees the other's reading, and write your own blind reading too. Third, turn every mark into a claim a number can judge, and register it before measuring. Fourth, trust each reader for what it does well. In these projects the person's marks were right on place, direction, relation, symmetry and faint, long, bending structure, while the AI's blind reads were good on geometry and weak on classes and depth. Sizes belong to the data, not to either eye. Fifth, score both readers' misses with the same ruler, and keep the readings that open a new test.

This part is an order of operations, like *The Walk*, and it is used in that order. Every lesson in it is also a row in the register, most of them in *L1 — The Method as an Instrument*, and it cites them by number; rows 702 to 719 are its spine. Section 1 says why to read pictures and which ones to make. Sections 2 and 3 give the method and its protocol. Section 4 is the toolbox, section 5 a field key of shapes, section 6 the traps, and section 7 the cases. Section 8 is the mathematical object all three projects turned out to share, and section 9 is the scorecard. Section 10 says how to use all of it, and how not to. The method's most common result is to rediscover known physics quickly, and that result is credited as a check on the toy, never as a discovery.

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1 — Why read pictures, and which pictures to make

A picture keeps things that an average throws away, so a person looking at it finds questions faster than a program does. But a picture also invites shapes that are not there, so which picture you draw matters as much as who reads it.

What the eye catches that a summary drops. It keeps lopsidedness. In one triple-star picture the centre of a band's core held within 0.0004 from start to finish, so its average said nothing happened, while Bear's "short spiky top left, long spiky top right" was a flip in the band's lopsidedness, measured afterwards from the picture's pixels as −0.49 to +0.47 and on the data as −0.18 to +0.29: right in sign, overstated in size. It keeps negative space, and the way to see it is a technique: "I unfocus my eyes and allow the white patches to fill my vision. Each is bordered by little clusters of four or more dots in a line." It may keep walls: where Bear drew walls, the step between one visit and the

next was 0.778 of its usual size, smaller than anything 2,000 random sets gave, but that comparison was chosen after the run, so it is claimed as nothing, and the test registered for the next birth has no recorded result at the triple-star thread's close on 29 October (row 313). And it keeps long reach. On one picture Bear drew six long strings and called it 80 per cent, where a greedy program drew fifty short ones (rows 715 and 707).

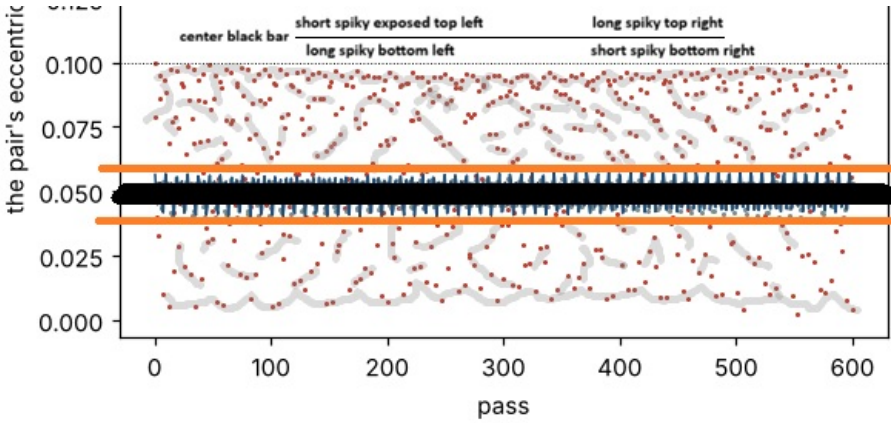


Figure 5. The band, as Bear marked it. The first panel of the triple-star bend figure (birth 7.200, the pair's eccentricity over 600 passes, run 20261009_022616), with his labels: a centre black bar, short spikes at the top left and long at the top right, long at the bottom left and short at the bottom right. Measured afterwards, the band's lopsidedness flipped from −0.49 to +0.47 on the picture's pixels and from −0.18 to +0.29 on the data, while its centre held within 0.0004: right in sign, overstated in size.

What the second reader misses, measured. The AI reader has blind spots of its own, and the projects measured four of them. It names classes badly when a colour scale lumps them: on one map, colouring by denominator painted the rest, the conveyor and two grooves the same orange, and a blind read named a chain above pull 1.0 as the rest, where the data said conveyor, 37 cells at lag 0 (row 717). It reads depth badly: a strobe picture is a shadow of something with more numbers to it, and a join that looks real in two numbers can come apart in a third (row 710). It defaults to a fork, drawing two roads at every split, where Bear draws three, counting the road that goes straight on (row 705). And a tool's sensors see only at their own size: sensors 5 holes long could not see lanes 0.3 turn long (row 715). So the person's reading is not decoration. It catches what the program was built not to see.

The eye is the fastest source of claims there is. Before any fit, Bear read the triple-star bend figures and said that the bend is woven from two phases; adding the second angle took the fit from 0.4302 to 0.9723. He said that two angles make a torus; at 6,000 passes the neighbour ratio fell from 0.1126 to 0.0603. He said that the leftover wave slides, that the corridor has walls that pinch, that the visits pile up at the walls, that the rows arch below the waist, and that the arrows lie along the first angle. Most of it was borne out. The piling at the walls stood beyond anything 2,000 random sets gave, on a comparison chosen after the run and so claimed as nothing, and it did not reach the bar registered for it; the rows' registered test missed by one row (section 9; row 313). In one sitting of R2W the eye's shortcuts were concrete: a second S-shaped curve he drew days before any test looked for it; a word about symmetry that led to a closed form for the pin, until then only "a place where curves cross", which a registered test (run PIVOT-1) had just failed to explain, and so probably saved a day; and a side note that would otherwise have stayed unread. Each saved at least a run of guessing.

And the eye carries its own trap. Look for a shape and you will find it. Bear's warning: "If we are just looking for a fit, we are playing an inspirational game of Texas marksman … frequency illusion will cause it to appear. So I stay blind and tell you what I see." The rest of this part exists to keep that trap shut: a second reader who has not seen the first one's marks, and a ruler that has the last word.

Which pictures to make, in this order. Some pictures teach a reader and some do not, and the difference is a rule.

1. **The thing moving, as frames.** A wing drawn as a strip of snapshots, good timing against bad, taught at once. A graph of push against distance did not, and the same story told hands-first, a rubber band between the fingers and a hand pushed through a sink of water, did. Dot pictures of a strobe, a thing caught once per tick, are the eye's native ground. Drawings side by side teach best, and a person's own hand drawings can teach both readers: a sketch of steep peaks from 2011 became two runs of the Loop Atlas.
2. **The counts.** Put a histogram on each axis, and treat every hole as a counter (row 702).

3. **Only then a derived quantity,** always with its anatomy drawn and with a made-up case beside it. Shown the twin lobes of a third difference, Bear read back-to-back phase-locked vortices, two propellers geared together, four waves through the centre, mirror-math, or a boundary transition. The shape belonged to the ruler: three centred differences turn a corner into a step, a bump and then a pair of lobes (row 711; the made-up corner was run CORNER_PRINT 20261010_061715). Two of his five words, boundary transition and mirror-math, were right. The picture had invited the other three.

The reason, as far as the Loop Atlas could test it: shown a thing moving, or a population landing in bins, the eye reads it truly and often faster than the tools. Shown a derived quantity, such as a force law, a rate or a third difference, the motion that made it is gone, and the eye supplies one. The fix that worked was to draw the derivation as its own picture, step by step: the corner; its slope as a step; the step's change as a bump; the bump's change as the pair.

Figures needed: a wing as a strip of frames; the corner, step, bump and pair of lobes, drawn in a row.

Draw two angles as a square that wraps. In the triple-star sessions the picture that taught most was the torus of two angles opened flat: one angle across, the other down, both edges wrapping, each dot coloured by one quantity. Nearly all of Bear's best reads were made on it. Colouring by the pass number, his "history", showed which way a weave travels. A picture meant to be marked is drawn plain, with an equal aspect, a light grid and no overlay, and its code is kept, so that it can be redrawn pixel for pixel and a person's strokes become numbers (row 714).

Read at two scales. At one birth of the triple star the dots come in clusters of three, the same place visited at passes k, k + 201 and k + 402 (575 of 599 dots), and the clusters themselves make rows. A lattice whose holes were sized to the dots saw only the clusters. Bear saw the rows. His eye worked at the scale of the return and the scale of the row at once, and a tool needs both (rows 640 and 703).

Figures needed: the torus opened flat, coloured by pass; the clusters of three, at the dots' scale and at the rows' scale.

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2 — The two readers

The first reader is a person, and the second is an AI that reads the same picture without seeing the person's marks. Where they agree, trust grows; where only one sees something, it becomes a test.

The first reader's method: the lattice. On 9 October 2026 Bear gave all three projects the same training text, while he was teaching R2W and the triple-star sessions at once, each with its own pictures, and he pasted it to the Loop Atlas the same night. He began by naming what this guide is about: "Slopes and rates, changes of rates, changes of changes of rates; ternary +1, 0, −1 slope to everything I see. And the geometry is exposed in the data." The lesson is quoted once here, from the triple-star sessions' fuller copy, with its opening taken from his first message of the same lesson, typing slips set right, and a few asides left out where "…" stands:

> "Today, simple. Take every Cartesian coordinate system like these graphs and wrap the graph in a 2D lattice. Each landing point is a hole in the lattice. It gets a count. And establish a histogram on the horizontal and the vertical axis. … Every example you will see is my mind doing curve fitting. I can't do the math, but I can show the curve. Each point or hole has a series of neighbors, and a history. … So here are your rules of the road and your test. This is like Minesweeper: wrap every possible grid point on the coordinate system in a lattice. Add top and bottom axis histograms as counters for binning (depending on the orientation of the axis and the content). Now we loop at each hole where we find a point, and we ask that pixel or point, 'What is your history?' It might be my oscillations, an event-time counter, whatever. It will often show up as geometry in the data plot. Now we start in 2D. We will graduate you to 3D, and then to how to shift perspective around the data in 4D. It's all there once you have the algorithm to expose it. And the important thing isn't that you find every one. I don't. But if you find a few new observations that somehow connect to other observations, then that's a big win. So every hole with a pixel or point needs a history. That history is the geometry, and the interplay of the knobs in the geometry. The scale makes a big difference sometimes, as we saw. Okay, so remember: slope or rate, change of rate, change of change of rate, our core fundamentals since day 1. How does the neighborhood history manifest? Linear, exponential, logarithmic-style curves. So each point has 8 potential neighbors. If a neighbor is immediately present, record the slope. Now we build out, from radius 1 (immediate neighbors) out to a radius of 5. Now we are building the history of the neighborhood. We have a slope for each. Now we are looking for its relatives. The relatives will look similar (slope or rate), or behave similar (change of slope), or share a change of change of slope (family members one step removed, indicating a boundary layer or a big change in the curve). Same slope = the same string, or overlaid strings. A change in slope = now we might be related, maybe not. When they line up tightly it's easy; when they space out, skipping one or more holes to the next pad, they get harder to track, but over time they will always show as steady (the same behavior, the same frequency), or decreasing or increasing at a given rate or slope, or exponentially or logarithmically. So each lattice hole that is unreported is like a little cellular automaton, a Minesweeper game. But each one is also a counter and a sensor for its geometric neighbors. It can answer: what is my slope with my

> neighbors? Does it continue straight, and does the frequency change? If it curves or bends, does it do so consistently, or at a given rate, exponent or log? So you can imagine you build a little fiber-bundle network of each of these where it makes sense: a network map interconnecting all points, just in case they might be related. Each address is a series of qualities, and these qualities may be shared with their neighbors. You should be able to use a radius-5 triangle lattice (or any n-gon you want, a pentagon or whatever) for every pad to each other pad, as a surface where data points are present. I can run it on my laptop, setting up the true structure in the geometry and the data. 3D and 4D are parity checks on outliers, and on shifting your perspective around event time."

That is the first reader's job in one line: **the eye fits the curve, and the ruler does the arithmetic.**

What the lattice is for. Handing the Loop Atlas its lattice test, he said: "Now a single point can share many strings overlaid. … The hole with the data point may have 10 strings overlapping, so that individual data point is actually stacked 10 deep. So what looks like disorder is just binning layered on top of binning, where ten signals are overlapping." And of one such point: "I think of it as a fiber bundle address index. Geometrically, that point is a line or curve through time, at a frequency of appearance. And that is the flow." The triple-star sessions read the words as a design, and built the lattice's next version in their order (section 4). The address is a place on the torus. The fibre over it is the line through time of that place's visits; at one birth there is one visit every 201 passes, which is the frequency of appearance. The flow is the step from one visit to the next. Where he saw "3 dots on top of each other in the z-axis, into and out of the page", the fibre is seen end on.

The same method, as steps. R2W wrote the lesson as a procedure.

1. **Lattice and histograms.** Cut the chart into holes and count the landings in each. Draw a histogram of the counts along each axis.
2. **History.** For each filled hole, a pad, ask where its riders came from and where they go next.
3. **Neighbours.** Look from radius 1 to radius 5 round each pad, and record the slope to each neighbour as +1, 0 or −1. The runs of level holes between steps make a word, and a straight line gives a balanced word, its runs differing by at most one.
4. **Strings.** Link pads column by column into curves. Give each curve its rate, its change of rate and its change of change, and a shape. The largest change of change marks a boundary layer.
5. **Relatives.** Curves with the same slope are one string, or strings laid over each other. The same change of slope suggests relatives; the same change of change marks a family member one step removed.
6. **Negative space.** The empty patches are bordered by short lines of four or more dots, and the patches outline the threads.
7. **Zoom.** Change the magnification. A real shape holds its place; a drawing artefact moves.
8. **Stay blind.** Read before seeing the other reader's reading, and test whatever only one reader saw.

Five habits that went with the steps. His teaching carried habits that every project adopted.

1. **Count on both axes.** A histogram on each axis turns a picture into counts: the bins hold the numerators, and the whole population is the denominator.
2. **Zoom out more often than in.** "Magnification is a big deal on this stuff," and "most of the time we want to demagnify": watch how the shapes grow as the view widens.
3. **Look for mirrors.** Bilateral symmetry is a place to stalk a shape.
4. **Prize lines of dots.** "Straight line clusters and curve line clusters are gold, because then the 3D shape comes out. Toroids. Vortices. Boundary zones."
5. **Count before trusting a crowded hole.** Many strings can overlay one filled hole, and the eye cannot tell whether it is shared by 5 strings or by 50.

And the frame for all of it: looking at data is "a best practice, not a crystal ball. It's art as much as science."

The second reader. In R2W's reads of the pictures Bear sent, and in the Loop Atlas's first two practices, the second reader was Claude, reading blind. It wrote its reading into the record, stamped by the clock, before it opened Bear's marks, and only then were the two compared. What both found on their own held up best: hooks paired across a V by their height, a beat at the centre of a run, a pinch in one angle near 0.42 (read from the picture; the corridor's measured pinch sits near 0.45), and the nodes along a flank. What only one of them saw became a test. Bear's woven strings became the weave test, and his "three phase-shifted waves" set against Claude's "shrinking" wave became a test of thirds; both readings turned out partly right. **Two blind readers who agree are the guard against the Texas sharpshooter.**

How the two readers differ. The two readers are not alike, and their difference is the point. In both projects that compared them, the person's eye was the better reader of negative space, lopsidedness and long reach, and all three have now been measured: lopsidedness and long reach first, then negative space on his dead zones (sections 7 and 9).

A fourth was measured on a window of the Loop Atlas's needle: faint, long, bending structure. Of Bear's strokes on a density of chaos, 0.741 (red) and 0.673 (green) lie within 0.004 rad of the computed fold or its first eight images, against 0.36 and 0.37 for the same strokes set at random, and Claude could see the voids and the X but not the faint curves until they were computed (row 707). Claude's blind reading was better at geometry, and worse on classes (row 717) and on depth (row 710). And the second reader's eye has habits of its own: it defaults to a fork, drawing two roads at every split after reading that Bear draws three (row 705).

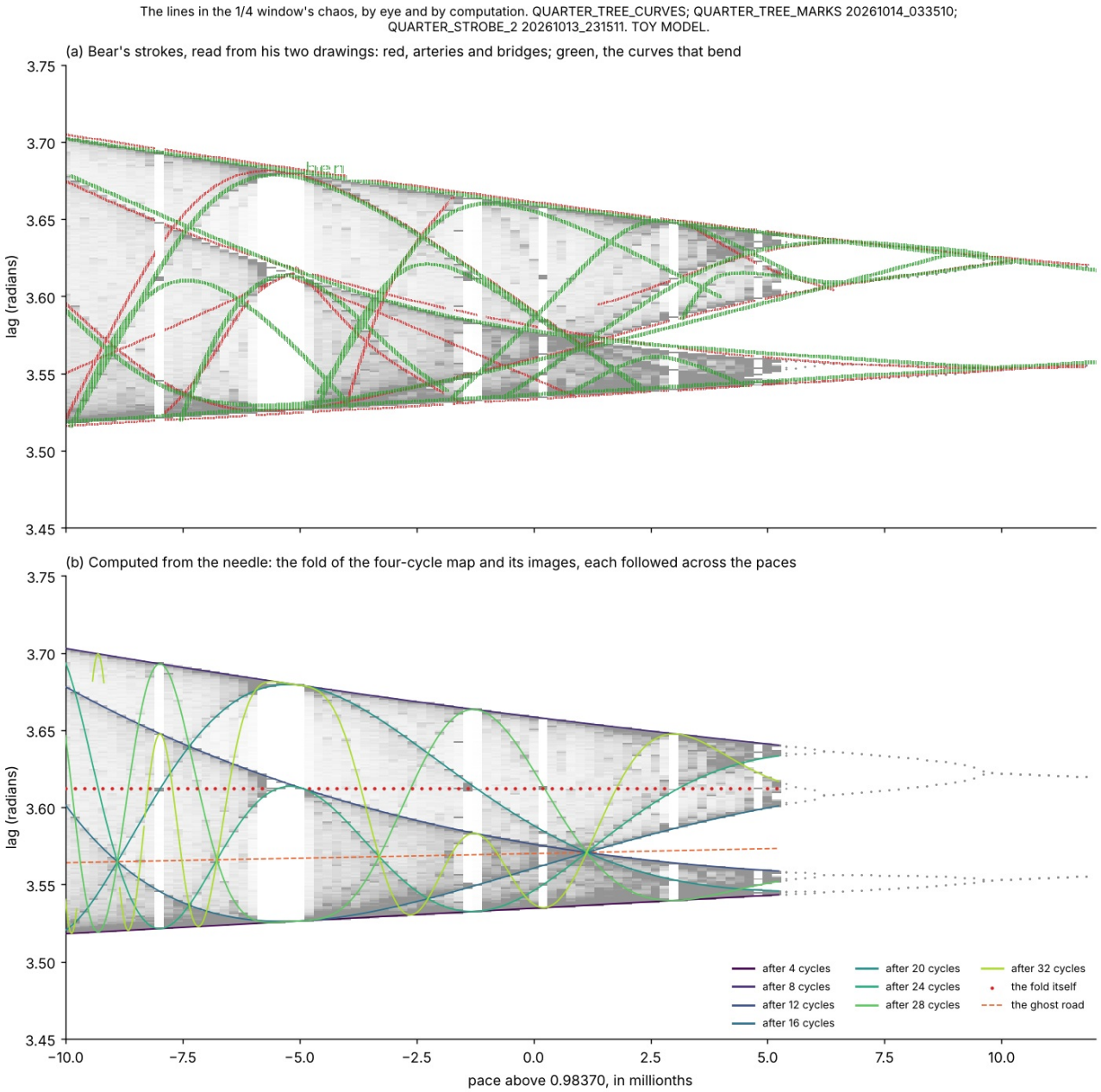


Figure 6. Bear's strokes and the computed curves. Above, his strokes on a window of chaos of the Loop Atlas's needle, read from his two drawings: red for arteries and bridges, green for the curves that bend. Below, the fold of the four-cycle map and its images, computed from the needle. Of his strokes, 0.741 (red) and 0.673 (green) lie within 0.004 rad of the fold or its first eight images, against 0.36 and 0.37 for the same strokes set at random (row 707). TOY MODEL; QUARTER_TREE_CURVES, from QUARTER_TREE_MARKS 20261014_033510 and QUARTER_STROBE_2 20261013_231511.

Hold back the data while the eye is learning. Bear: "Let's finish the examples and then I will give you all the data. I want you to see it, otherwise the learning is shortcut." With the numbers in hand a reader measures instead of looking. Without them it has to look first, and the looking is the skill being taught. The practices that held the data back are what produced blind reads that a scorecard can judge (row 716).

The ruler. Every claim gets a number before it is believed: from the data, or, when the data is held back, from the picture's pixels, calibrated on its own axis ticks and checked on marks whose values are known (row 714). The ruler is checked too, four ways: against a shuffle of the same dots (row 616), against a made-up case (row 711), at a second scale (row 703), and by printing what its own step can and cannot decide (row 686). Bear's rule for his own eye holds for both readers: "Don't let me speak on behalf of the geometry until the data bears witness."

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3 — The protocol

A reading of a picture goes in twelve steps. Say which picture you are reading and what you have already seen, write your reading down before you see anyone else's, turn each mark into a test written before the run, and let a ruler that has itself been checked give the verdict.

The twelve steps are kept in the order a reading goes. R2W ran the core of them on ten pictures in two days. The Loop Atlas and the triple-star sessions each added steps where their own readings had failed without them. One of those steps, the made-up case, was reached by the Loop Atlas on the same day that R2W reached its nearest relative, the simplest separate run.

The steps fall into five stages, in this order: before the reading, the reading, before the run, the ruler, and the score.

Before the reading, name the picture and declare what could colour it.

1. **Name the picture before naming the pattern.** A picture's axes decide which dots sit side by side, so they decide what counts as a neighbour, a weave or a lane. One triple-star run has its dots' nearest relatives 5 passes apart on the page of eccentricity against pass, 14 apart in the reading's own rhythm, 47 apart on the opened torus over 600 passes, and 682 or 663 apart over 6,000 passes at the kit's tolerance, whose arithmetic shaped those two (at the tight tolerance the beat is 722 passes, row 858). Each is right for its picture. R2W read the 5 and the 47 from the pictures' pixels, and the triple-star sessions confirmed both on the numbers: one finding made twice on one dataset, by eye and then by ruler. Say which picture a claim lives in, and draw the same data through a second picture before naming a pattern (row 713).
2. **Declare what you have already seen.** Before a cold reading, write down everything you have read or seen that could colour it. The declaration does not remove the colour. A reader who had read that Bear draws three roads at a split still drew two. Two rules written up as new in one project's practices, the overpass test (whether two strands that touch in one number lie apart in another) and the shuffled null, had been in that project's own notes two days before, in its record of 8 October. But a declaration tells the next reader where to look for the colour (row 369). For the Loop Atlas's practices Bear set one rule, no finished solutions from other sessions: no search of past chats, no sister session's letters or run files on anything a practice touched, and whatever was needed came through him. The rule worked because each practice declared what had already been read.

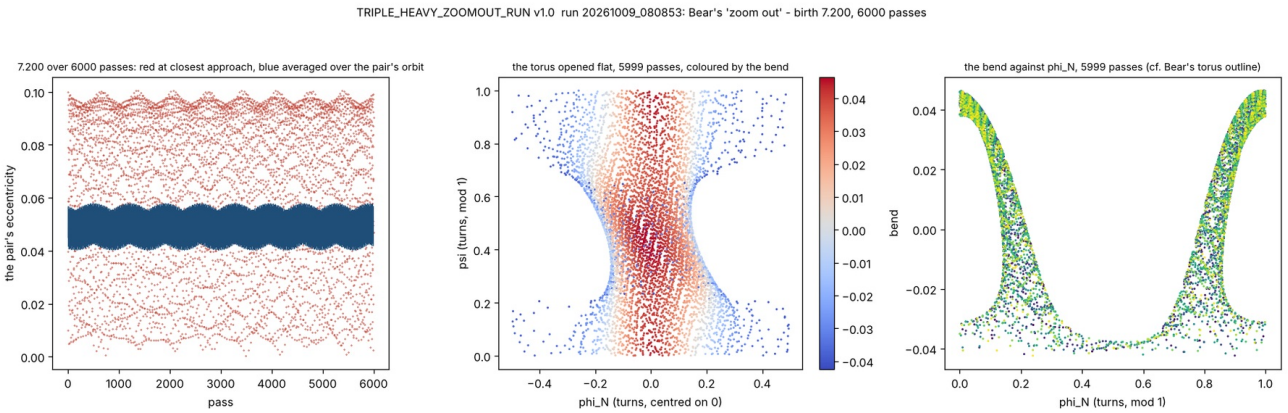


Figure 7. The zoom-out Bear asked for. Birth 7.200 over 6,000 passes: the pair's eccentricity against pass (left), its band scalloped by a slow beat of about 686 passes at the kit's tolerance, whose arithmetic shaped it (at the tight tolerance the beat is 722 passes, row 858); the torus of two angles opened flat (middle), the strands filling a solid hourglass; and the bend against φ_N (right). The same birth over a shorter run is the left panel of Figure 15 (600 passes) and the top left of Figure 8. TRIPLE_HEAVY_ZOOMOUT_RUN v1.0, run 20261009_080853.

The reading: write it down blind, and turn each mark into a test.

3. **Read blind, and write it down by the clock.** Write the reading before you see anyone else's, with three bets, each with a test. Then compare. What both readers saw counts most; what only one saw is turned into a test. Writing the bets first shows exactly which assumption failed when they miss (row 369). A reading made after seeing the data is claimed only after a blind test: chaos on the borders between treads, read at 84 per cent on a map already seen, held at 97 per cent on a fresh grid read blind, while a staircase's turns marked by a change of word failed blind (rows 369 and 655).

4. **Translate each mark into a test with the person who made it, before the run.** The worst miss in the triple-star sessions was a translation: Bear's gold lanes of empty space became, in the test's wording, rows of dots, and the test measured the wrong thing (row 715). Their best result came the other way, from his own words built literally: his change of change of slope found his crossings without seeing his marks (row 573). Better still, let the person draw in the instrument's own numbers. The Explorer's pen keeps each line Bear draws in the chart's own units, numbered, with his word on it, so a drawn line becomes a registered prediction before it is checked (row 573).

   Some marks come with a standing translation. A circle drawn round a group is a hypothesis about a factor: test whether the group's values are a product, one profile times another. Arrows drawn between two maps are a claim about the direction the data moved: test the direction of change. A rhyme named between two pictures, or across rooms, asks for a shared equation before anyone calls it a coincidence, and for the number that tells a shared shape from a shared cause; the cases are in the field key and in section 6. A mark on a proof that asks why something is not being tested is a run request: one such mark in R2W became its fold-direction series. And when the person names a shape, the second reader says plainly whether it saw the shape too, names the shape, and gives that separating number.

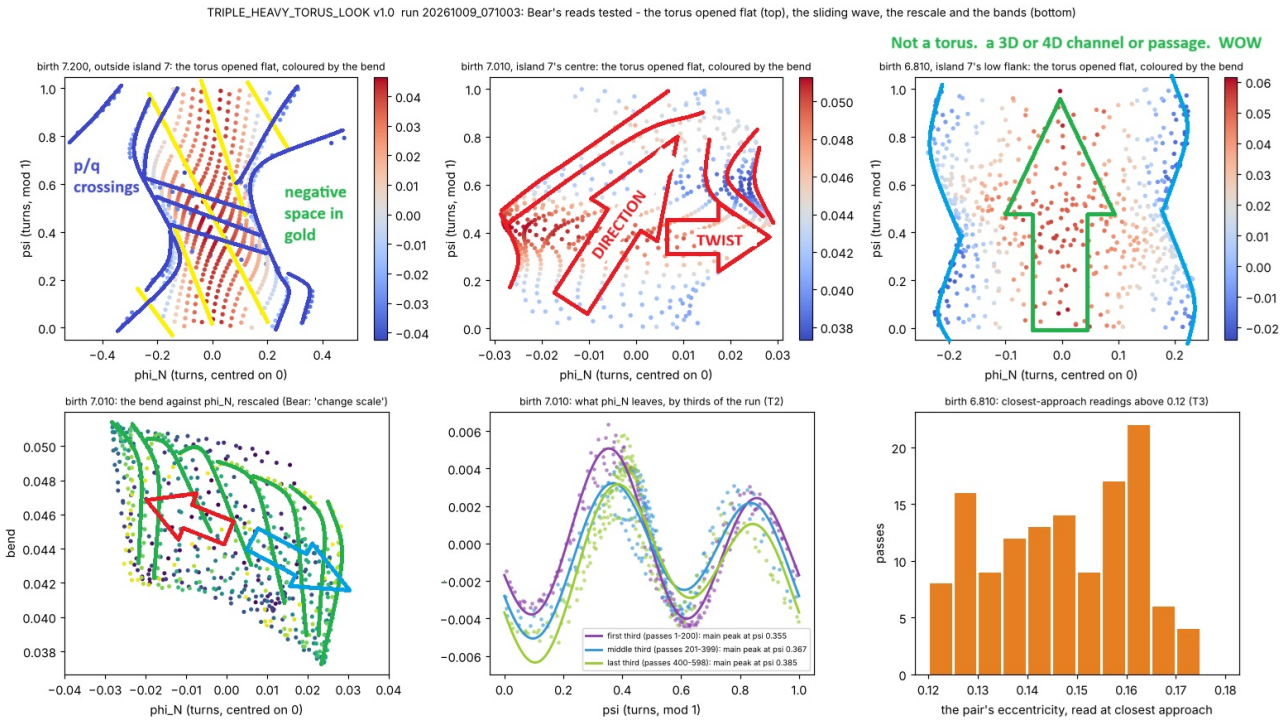


Figure 8. Bear's reads on the torus opened flat. Top, the torus of two angles opened flat at births 7.200, 7.010 and 6.810, with his marks: p/q crossings in blue and lanes of empty space in gold at 7.200, a direction and a twist at 7.010, and at 6.810 a passage with breathing walls, which he called not a torus but a channel. Below, the sliding wave, the rescale he asked for and the bands. The gold lanes were never measured: the lane sensors were too short-sighted. TRIPLE_HEAVY_TORUS_LOOK v1.0, run 20261009_071003.

Before the run, search the record and register with the control built in.

5. **Search the record before registering.** One registration took a birth from a coarse dial's mark, when a run a day earlier had already placed it on a dial five times finer, and half of one test repeated work already passed. Another named the top of a band from a ledger row that the record had corrected two rows later, and missed for that reason alone (row 666).

6. **Register, with the control inside the tool.** Write every prediction down before the tool that judges it exists, with the bar it must clear and the rival reading beside it, and build the featureless control into the same tool before its first run. Three passes in two days were empty because the control came afterwards or never, and the third came after the row that warns of it had been read in full (rows 616 and 620). A rule that can live in the tool should live there, because a reader's memory did not hold this one.

The ruler: every claim gets a number, and the ruler is checked too.

7. **Print what the ruler's step allows.** Under every judged number, print the range the instrument's step permits and say whether it crosses the bar. A ratio that read 7.33 against a registered band of 3 to 7 was written down as failed, while the dial's step allowed anything from 5.25 to 11.5. The failure stands as registered, and it is not decisive. Look for the ruler in every miss (row 686).
8. **Give every claim a number.** Take it from the data. When the data is held back, take it from the picture: calibrate on the axis ticks, check on a mark whose value is known, and say that the number is the picture's (row 714).
9. **Shuffle, with the statistic chosen first.** Ask what the same dots, shuffled, give for the same statistic. A weave test passed as registered, with 72 per cent of its links on two gaps; but nearest neighbours on a page are always close in pass, so the bar was too easy. Against 200 shuffles the real links preferred a gap of 5 passes at +12.4 standard deviations, and only the shuffle showed what was real. Choose the statistic before the shuffle: on the needle's ordered bands, strings of 20 or more pads separated the picture from 100 shuffles by +895 spreads (a spread is the shuffles' own standard deviation), while the count of strings of 4 or more pointed the wrong way, 209.8 in the shuffles against the picture's 17. And a null must keep everything it is not testing: a shuffle of rows that merged 11 per cent of a cloud's pads put the picture +9.4 spreads above chance, and a shuffle of the columns' order, which merged nothing, left +1.8 (rows 616 and 30).
10. **Run the ruler over a made-up case.** Before reading meaning into a shape, give the ruler an input whose answer you know and look at what it draws. The twin lobes Bear read as vortices were drawn the same way by a made-up perfect corner, matching every lane's pair to 0.993 in size at the top doubling and better than 0.9999 at the next two (row 711). R2W's nearest relative is the simplest separate run, a single rider with no ripple, whose answer it could work out by hand: it placed the whole-number grooves (the conveyor within 0.003 in clock ratio), and it showed that a prediction about a memory band had held for the wrong reason (row 683).
11. **Read at a second scale.** Strings found on a lattice of 100 rows were checked at 50; pictures of 600 passes were checked against 6,000. Keep a relation only if it survives the change, and zoom out before naming a shape (rows 703 and 640).

Figures needed: the needle's ordered bands beside one of their 100 shuffles; one lane's pair of lobes beside the pair the made-up perfect corner draws.

The score: score the sign before the size, and print what could not be seen.

12. **Score, then print what you could not see.** Score an eye's claim on its sign and its place before its size (row 712). On a tie, print "I can't see it", not a peak (row 128). When a guess misses, say aloud whether it can be redesigned, and run the redesign. When a real effect misses its bar, print both facts (row 313). When a bar could never have said no, call the pass empty (row 337). And read every built file back against its source, because a ledger's PDF once lost about 1,600 words to a clipped table without a word of warning (row 553).

When the person beside you reads the picture differently, write both readings down before either of you looks at the other's.

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4 — The toolbox

The three projects wrote different programs for the same jobs. The table puts them side by side, job by job, and says how each one can fool you.

Each project built its own tools, and the jobs came out the same. The table sets them side by side by job, with what each assumes and how it can fool you; most of the warnings in its last column were learned from a miss. Its rows run in four groups, in this order: following the orbit itself, and its birth; Bear's lattice and the other reads of holes and neighbours; measuring a picture, reading a person's marks on it, and redrawing it; and checking the ruler and the reader. The programs are the projects' own and are not yet public.

Job	The tools, by project	What it assumes, and how it can fool you
Follow the orbit itself	The Loop Atlas: STROBE_NEWTON and CASCADE, which find an orbit by Newton's method and read its multiplier. R2W: the Explorer's records. The triple-star sessions: their cascade finder.	Near a fold or a doubling nothing settles in time to be read; follow the orbit and read the sign of its multiplier (row 642).

A birth from both sides	The Loop Atlas: history going down and coming up (LATTICE_READER_2, HISTORY_SETTLE). R2W: CONVEYOR_BIRTH_1 and its reader.	A coarse dial makes a birth look gentle: a slope of 26.8 at birth against 160 on a fine dial. Fit the approach and the departure separately (row 654).
Bear’s lattice in code	The Loop Atlas: LATTICE_READER_2. R2W: LATTICE_NETWORK, and the Explorer’s strings mode, whose strings carry a ternary slope sign. The triple-star sessions: triple_heavy_lattice_look, with pads, counters, rings 1 to 5, families, strings, change of change, empty-pad sensors, a network, a surface and two controls.	The size of the holes decides what a string is. A hole joins one string and the first claim wins. Pads sized by the dots’ spacing see only clusters. A tie can print as a peak. Sizes without signs miss half of the reading (rows 703, 640, 128 and 537).
History at a hole	The Loop Atlas: each rider’s own period (LIFT_3D). R2W: LATTICE_NETWORK’s second step. The triple-star sessions: the pass number as each dot’s history.	A rounded count has a scale of its own and can be wrong by one; count the truth by history (row 709).
Lift into a third number	The Loop Atlas: LIFT_3D, which adds the rider’s speed at each strobe to its angle.	A touch that the third number does not separate is a real meeting, resolved by scale or by history, not by another dimension (row 710).
Find the event in a string	The Loop Atlas: LOBE_TWIN and ZERO_CROSS. R2W: LATTICE_NETWORK’s boundary layer. The triple-star sessions: the lattice look’s change of change.	A centred stencil answers a sudden bend with two lobes of opposite sign; read the crossing between them, and give it room (row 708).
Smoothness without a chosen basis	The triple-star sessions: the torus look, a dot’s five nearest neighbours set against the spread of the quantity.	Noise gives about 1; a wide swing spreads the neighbours. A low-order fit that misses is not a missing variable (row 718).
Nearest returns of a rate	The triple-star sessions: triple_heavy_ladder_check.	One angle’s ladder is not a pair’s (row 719).
Measure the picture	R2W: PICTURE_PIXEL_TOOLS (axis calibration, line envelopes, lopsidedness by parts of a run, crest counting, colour-scale matching) and the weave pixels. The Loop Atlas: a colour digitizer, kept out of its deposit files with the borrowed figure it read. The triple-star sessions: triple_rf_reads_check, which judges another project’s pixel reads on the data.	The plotter draws the last points on top, so colour counts favour late passes; dots merge where crowded, so pixel and blob counts bracket the truth (row 714).
Read a person’s marks	R2W: BEAR_STROKES_MATCH, which finds the pixels Bear added to a screenshot, turns them into the chart’s numbers and scores them against the strings; and the Explorer’s pen. The triple-star sessions: triple_heavy_blind_figure, a plain figure to mark, and triple_heavy_blind_score, which reads the marks back pixel for pixel through the redrawn axes and scores them against random sets.	It needs the clean twin of the marked picture. A figure drawn at a birth on an exact fraction shows a frozen frame (row 648).
Squeeze, tilt and turn a picture	The triple-star sessions: triple_strobe_views, which draws a strobe picture squeezed, tilted as a perspective transform (the tilted picture) and turned, so that Bear’s moves are reproducible. R2W: the squeezed picture counted by its own pixels (row 743).	Squeezing one axis presses columns of dots into one shape whose envelope the eye reads, its symmetry, its waist, its cusps: it is the aspect ratio, which graph-makers have long chosen so that slopes read well (Cleveland’s banking to 45 degrees, from Cleveland, McGill and McGill, 1988; Heer and Agrawala, 2006, bank each of a picture’s scales on its own), used on purpose to bring slow change along one axis into view. Tilting sights along the picture as a carpenter sights down a board:

		slow trends along the line of sight are exaggerated and detail across it is suppressed. Turned a quarter turn, an up-and-down mirror becomes a left-and-right one, which the eye seems to judge at a glance.
The phase-offset view	The triple-star sessions: triple_back_side_look, a delay plot.	A picture of one angle read once a pass shows each rocking edge-on, its two halves on one line; plotting the angle at one pass against the angle a quarter of a rocking later turns the line face-on into a ring, the “back side” Bear felt. The rooms, the parts a long run splits the island into, became hollow rings, the room that keeps its stars a filled disc, and the early centre a needle whose own swing is too slow for that lag (run 20261014_075830). Choosing the lag is choosing the phase offset; the right lag is a quarter of the motion you want to see.
A short view beside a long one	The triple-star sessions: the strobe picture’s first 600 passes beside passes 601 to 6,000, on the same axes.	A moving waist is invisible in either alone.
The shuffled null	The Loop Atlas: SHUFFLE_NULL and SHUFFLE_NULL_2. R2W: the weave pixels and their shuffle. The triple-star sessions: the controls inside the lattice look, and the row-fit control.	A shuffle that changes the counts is a test of its own; choose the statistic before the shuffle (rows 30 and 616).
The made-up case	The Loop Atlas: a made-up perfect corner (CORNER_PRINT). R2W: its nearest relative, the simplest separate run, a single rider with no ripple (CHALKBOARD_W2_FIGURE). The triple-star sessions: a known answer put inside the run.	When the ruler draws the same shape from a known input, only the shape’s place belongs to the data (row 711).
Is the ruler fine enough?	The Loop Atlas: the error a whole column puts on a ratio read from where events sit. R2W: SCALE_FLAG, the off-scale flag.	Print what the step allows under every verdict, and say whether it crosses the bar (row 686).
Keep a reader honest across deliveries	R2W: the seen list, kept by file name and md5.	Without it, a chunk delivered twice votes twice (row 369).

Figures needed: one strobe picture drawn plain, squeezed, tilted and turned a quarter turn, side by side; the phase-offset view of run 20261014_075830 with its hollow rings, its filled disc and its needle; the first 600 passes beside passes 601 to 6,000, on the same axes.

The Depth Axis Explorer is the one tool where both readers and the ruler work on the same picture at once. It is a single page, built for Bear, holding R2W’s maps of the rider, each rider drawn where it stands when the ripple passes. A reader can switch families and grooves off, fit a line through a dragged box, draw a line by hand and keep it in the chart’s own numbers, play the riders passing by passing, and see a rider’s dance on a ring as a star polygon. Since its seventh and eighth versions it lays Bear’s lattice over the picture: holes that count landings, a histogram on each axis, each hole’s slope to its neighbour, its bends, and strings linked hole to hole with their slope, change and change of change. Its ninth version holds R2W’s complete pad atlas, 28 maps, 2,152 chunks and 227,128 riders. The method it carries is cross-domain already: the lattice crossed from R2W’s maps to the triple-star pictures and to the needle. The page becomes a cross-domain tool on the day it loads another project’s table of setting, angle, rider and passing, and that day has its test waiting: read the needle’s doubling staircase, and it should give the splits Newton’s method fixed to ten digits.

Figure needed: the Explorer with Bear’s lattice laid over one of R2W’s maps: the counting holes, a histogram on each axis, and strings linked hole to hole.

Three things to carry from R2W, and one habit. The off-scale flag, which prints what the ruler’s step allows under every verdict; the seen list, which keeps a reader from counting a delivery twice; and the pen, which keeps a person’s drawn line in the instrument’s own numbers. The habit is to ask “what does the ruler decide?” before “did it hold?”.

The lattice's next version, proposed and not yet built. The triple-star sessions wrote it in the order of Bear's evening words.

1. Find the frequency of appearance first: the commonest pass gap between a dot and its nearest neighbour (201 at one birth; 47 at another, over 600 passes).
2. Collapse each fibre to its address, and keep the fibre's own shape. In Bear's words, "Every three-dot has a curve to it; the arrows lose that detail."
3. Lay the lattice over the addresses, at the addresses' spacing, and read slope, change of slope and change of change with their signs, +1, 0 or −1, not only their sizes (row 537). R2W's strings already carry that sign.
4. Draw the flow: the step from each visit to the next, as a field over the torus.
5. Let empty space vote along whole lines, for the lanes.
6. Put every registered test's featureless control into the same tool, before its first run.

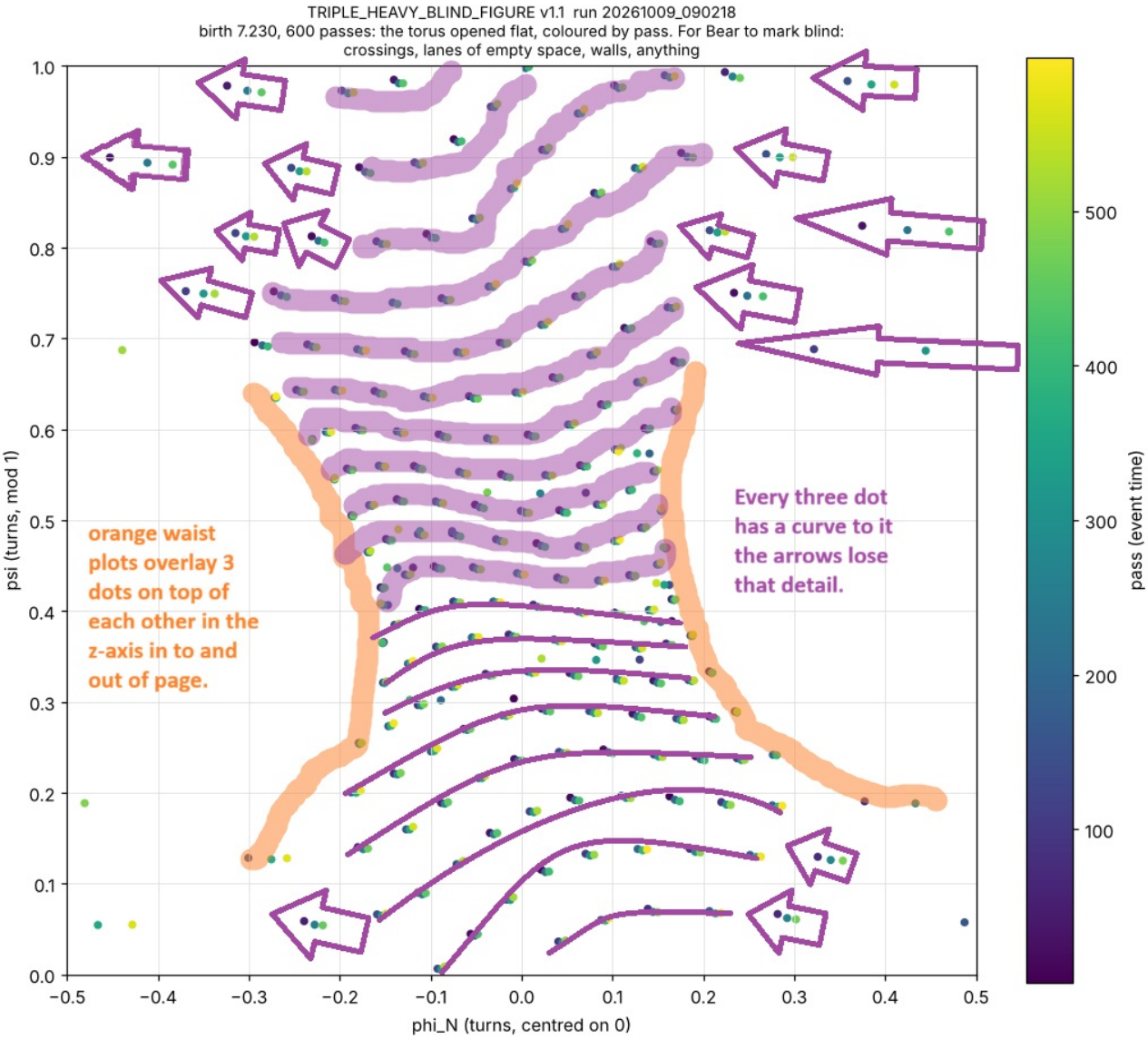
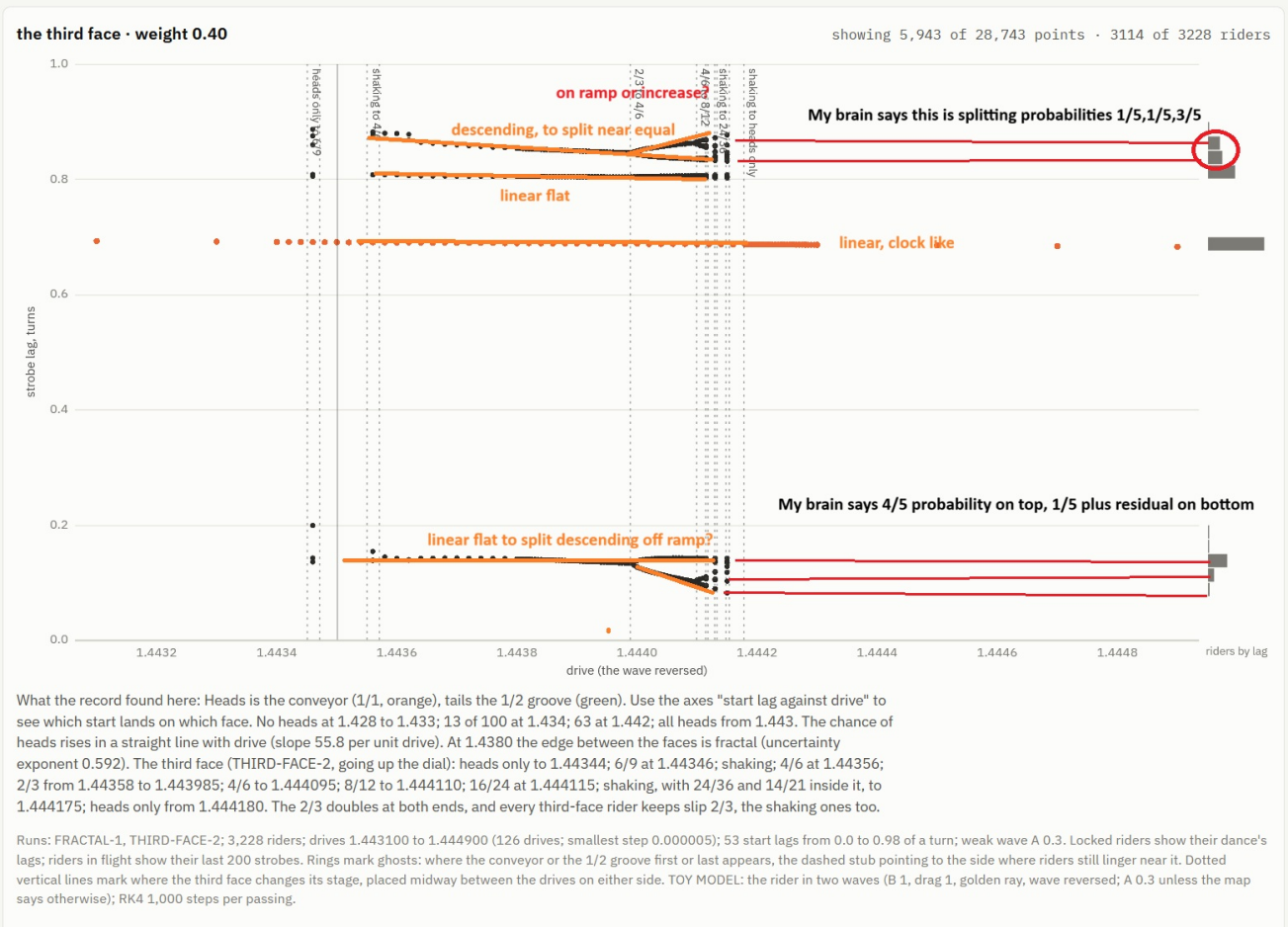


Figure 9. A picture drawn to be marked blind. Birth 7.230, 600 passes, the torus opened flat and coloured by pass. Bear drew rows through the clusters of three, arrows along the first angle and an orange waist, and wrote that every three-dot has a curve to it, a detail the arrows lose. The lattice later found 575 of the 599 dots to be one place visited at passes k, k + 201 and k + 402. TRIPLE_HEAVY_BLIND_FIGURE v1.1, run 20261009_090218.

Open, from the Loop Atlas. Three pieces are still open. The first is event time, the fourth number: read each rider's passings in order and watch a string being visited. The second is Practice 1's own test: mirror and rescale the third face's right-hand tree onto the left-hand tree of the weight-0.50 picture's eye, the place where one road becomes six (the field key), and count the top cluster's shares against 1/4, 1/4 and 1/2 with both histograms (the shares themselves, 0.22, 0.26 and 0.50, were measured on R2W's map and are in the field key; the mirroring and the count with both histograms are what stay open). The third is a lattice of triangular or hexagonal holes, in place of square holes with eight neighbours; it is not built, and the network joins the pads by a triangle mesh instead.



What the record found here: Heads is the conveyor (1/1, orange), tails the 1/2 groove (green). Use the axes "start lag against drive" to see which start lands on which face. No heads at 1.428 to 1.433; 13 of 100 at 1.434; 63 at 1.442; all heads from 1.443. The chance of heads rises in a straight line with drive (slope 55.8 per unit drive). At 1.4380 the edge between the faces is fractal (uncertainty exponent 0.592). The third face (THIRD-FACE-2, going up the dial): heads only to 1.44344; 6/9 at 1.44346; shaking; 4/6 at 1.44356; 2/3 from 1.44358 to 1.443985; 4/6 to 1.444095; 8/12 to 1.444110; 16/24 at 1.444115; shaking, with 24/36 and 14/21 inside it, to 1.444175; heads only from 1.444180. The 2/3 doubles at both ends, and every third-face rider keeps slip 2/3, the shaking ones too.

Runs: FRACTAL-1, THIRD-FACE-2; 3,228 riders; drives 1.443100 to 1.444900 (126 drives; smallest step 0.000005); 53 start lags from 0.0 to 0.98 of a turn; weak wave A 0.3. Locked riders show their dance's lags; riders in flight show their last 200 strobes. Rings mark ghosts: where the conveyor or the 1/2 groove first or last appears, the dashed stub pointing to the side where riders still linger near it. Dotted vertical lines mark where the third face changes its stage, placed midway between the drives on either side. TOY MODEL: the rider in two waves (B 1, drag 1, golden ray, wave reversed; A 0.3 unless the map says otherwise); RK4 1,000 steps per passing.

Figure 10. The third face, read by eye. R2W's rider at weight 0.40, where one road becomes three, with Bear's strokes on the roads and his reading of the split's shares from the histogram, 1/5, 1/5 and 3/5. The tree gives 1/4, 1/4 and 1/2, and R2W's map measured 0.22, 0.26 and 0.50 (the field key); the mirroring onto the weight 0.50 picture's eye stays open. TOY MODEL; runs FRACTAL-1 and THIRD-FACE-2.

Kin, found after. The Explorer was reached without a prior-art search, by Bear's standing order that the run comes first and the prior art after (row 506). The search after found interactive pages for Arnold tongues and for the circle map's devil's staircase. It found none, in one pass, that combines a pen kept in the chart's own numbers, a lattice of strings, and a record of blind reads. That is a statement about one search, not a proof of novelty.

The method's parts have kin of their own. Finding lines and curves in a picture is an old problem in image analysis, with the Hough transform as its classic tool (Duda and Hart, 1972, after Hough's patent of 1962). Keeping the analyst's eye off the answer is blind analysis, which experimental particle physics has practised for decades: "A blind analysis avoids the possibility of experimenters biasing their result toward their own preconceptions by preventing them from knowing the answer until the analysis is complete" (Klein and Roodman, 2005). And the strobe picture itself, one look per tick, is the Poincaré section of the textbooks. What this part adds is the practice of running them together, with a person's eye as one of the instruments.

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5 — A field key of shapes

This is a lookup table of shapes. Find the shape you think you see, and the table says what that shape turned out to be when a reader met it before, in which project, and how far the answer was checked.

How to read the key. Each row gives what the readers saw, what the shape turned out to be, and where it was seen. The status column says how far each was taken: measured on the data; measured from a picture's pixels; a mechanism named but not measured; seen; or a test still waiting. Some shapes belong to the drawing or to the ruler rather than to the data, and the next section gives those as traps. The rows are grouped by the project whose picture showed the shape, in the order the projects were introduced: the Loop Atlas's needle and its other toys, then R2W's rider, then the triple star. Inside each group the rows keep their order.

The Loop Atlas's needle and its other toys. Shapes from the needle were judged against answers known in advance. The group ends with the Loop Atlas's tapping rider, its heat with a memory and its tied bells.

What the eye saw	What it turned out to be	Where	Status
Eight strings that read as four	Four tight pairs of an eight-step dance, at holes too coarse to part them	The Loop Atlas's needle	Measured against the known answer
Twin lobes: vortices, propellers, mirror-math	The ruler's echo of a corner; only its place, the doubling, belongs to the data	The Loop Atlas's needle	Measured against a made-up corner
Two strands touching, read as an overpass	A fork's young daughters, born at the doubling 2 to 7 columns above	The Loop Atlas's needle	Measured
Each hole holding 128, 64, 32 or 16 landings	The period of the dance, read without following a single rider	The Loop Atlas's needle	Measured against the known answer
A cloud	Holes that hold one landing each; thickness alone is not disorder	The Loop Atlas's needle	Measured
A dip ringed in a spread panel, marked with a question about gaps	A window of 10 drive cycles inside a half-turn chaos	The Loop Atlas's needle, memory 1.7598	Measured
Flickers in a quiet stretch below a chaos	The rungs of a ladder of doubled cycles, 32 down to 2, where needles take long to agree	The Loop Atlas's needle	Measured
A clean lock (Claude's reading, from a spread over 200 cycles)	The same ladder, invisible to a ruler that sees only cycles dividing 200	The Loop Atlas's needle	Measured; Claude's miss (the Loop Atlas's misses row 219)
Rungs, Bear's word for the stretch	A doubling ladder, evenly spaced on a log ruler to within a quarter; ratios 3.43, 4.24 and 4.57 toward 4.669	The Loop Atlas's needle	Measured

What the eye saw	What it turned out to be	Where	Status
Two songs on two boards	One song: 96 per cent of the gaps exactly the rider’s rest, split by rounding	The Loop Atlas’s tapping rider	Measured, in whole clock steps
A warm brick in ice that Bear heard as quenching	Two cold fronts meeting in the middle knock the brick down at once	The Loop Atlas’s heat with a memory	Measured
Event timing ranked above the water’s memory	The squeeze period sets the sign of the water’s gain; the memory deepens it	The Loop Atlas’s tied bells	Measured
Rings round the bells’ map’s fast columns and its slow columns, with an arrow up the memory axis and a question whether the two are orthogonal	A product of a timing profile and a memory scale: read that way, the thin-cored map’s departure from even is 98.6 per cent that product, and the mended map’s 87.9 per cent, less after the mending	The Loop Atlas’s tied bells, marked by Bear	Arithmetic on saved results, after the marks; not a registered test, and left open at the Loop Atlas’s close
A question whether the map is balanced	A ratio: the cost side runs about twice as deep as the pay side, 0.61 below even against 0.30 above on the mended map, and 0.65 against 0.37 on the thin one	The same map, marked by Bear	Measured
Arrows pressing inward on the thin map and outward on the mended one	The direction the data moved: the mending opened a neutral band in the middle, with gains within 0.03 of even across periods 2.12 to 3 at short memories	The same two maps, marked by Bear	Measured, after the arrows

Figure needed: the bells’ thin-cored map beside its mended map, with his rings, his arrow up the memory axis and his arrows on each map.

Figure needed: the needle’s eight strings that read as four, at holes too coarse to part them.

Figure needed: the needle’s two touching strands, flat and then lifted into speed.

Figure needed: the spread panel with the dip ringed.

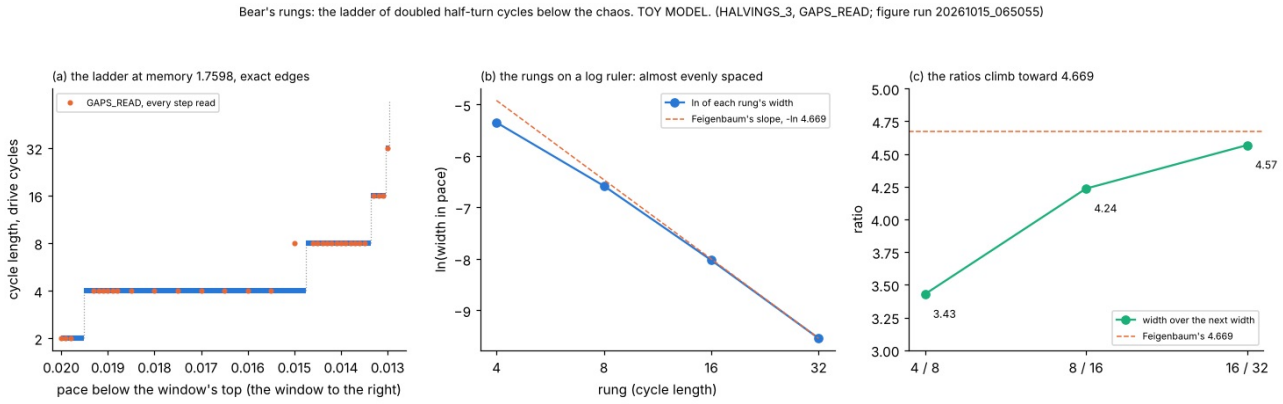


Figure 11. Rungs, Bear’s word for a quiet stretch below a chaos of the Loop Atlas’s needle: (a) the ladder of doubled cycles at memory 1.7598, each rung between its exact edges; (b) the rungs on a log ruler, almost evenly spaced; (c) each rung’s width over the next, 3.43, 4.24 and 4.57, climbing toward 4.669. TOY MODEL; figure run 20261015_065055, from HALVINGS_3 and GAPS_READ.

R2W’s rider. Shapes from its pictures of the rider, at the weights named in the third column.

What the eye saw	What it turned out to be	Where	Status
A “lurch-stroll-lurch” S on a groove’s lower stop	Its slope runs 190 at birth, 2.8 in the middle and 53 at the end;	R2W, weight 0.26	Measured

	its slowest point, at lag 0.145, is a foothold where level strings cross three other zones		
A balanced word of level runs (8, 9, 9)	The digital trace of a straight line	R2W, weight 0.50	Measured
Curves through a staircase converging on a ring	The ghost of a saddle-node: the stops of the dances pile into the birth point, and 1 − slip goes as the distance to the birth to the power 0.507; read from below and from above, the birth agrees to about a millionth	R2W, weight 0.26	Measured
A gentle birth	A square-root birth hidden by a coarse dial: a slope of 160 at birth on a fine dial, against 26.8	R2W	Measured
A “hollow beam”, dots crowding at a column’s edges	A ring seen edge on (the arcsine law)	R2W	Mechanism, not measured
A memory band with a ceiling at 0.31	Hysteresis: below pull 1.0 a rider at rest stays and a runner keeps running; the ceiling is the chart’s own window	R2W’s pad atlas, with the band’s floor read at weight 2	Measured
An orange chain above pull 1.0	Not the rest but the conveyor, which colouring by denominator paints the same orange; the whole-number grooves are Shapiro-like steps	R2W, weight 2	Measured, after the data
One road becoming three at a split, and six at an eye	A probability tree: the parent carries on as a ghost, and each split halves a branch’s share	R2W’s weights 0.40 and 0.50, read by Bear	A reading, made after the measured shares had been seen: 1/4, 1/4 and 1/2 from the tree, 0.22, 0.26 and 0.50 on the map; Bear’s eye read 1/5, 1/5 and 3/5
One flat string, showing through three windows	A staircase of flat strings, at lags 0.144, 0.707, 0.81 and about 0.92	R2W, weight 0.26	Found by R2W’s decoder; the cold reading had seen one
A shape seen twice	A groove’s lower foot found again about 0.010 of drive to the right, its shelf squeezed to about 40 per cent of its length	R2W, weight 0.26, read by Bear	Seen
A string drawn through empty space	Possibly a frequency illusion, a line the sampling makes	R2W, weight 0.26	Test waiting: split each rider’s record in halves, or strobe at another phase
A fan of curves, one per strength, pinching near one place: a waist	A balance of two terms. The drop over the lift crosses the golden value near weight 1.43 for every strength from 0.05 to 0.30; R2W proposed that this is where each direction’s next-order term is the same fraction of its first, a test it wrote down and did not run. The waist slides slowly as the strength grows (1.456 to 1.397), which marks a balance of two terms rather than a symmetry, and it is not at φ, where the ball’s inertia and drag answer the ripple equally	R2W’s fold-direction series (FOLD-DIR-b)	The waist measured (FD-5, FD-6); the balance of next-order terms proposed, its test not run

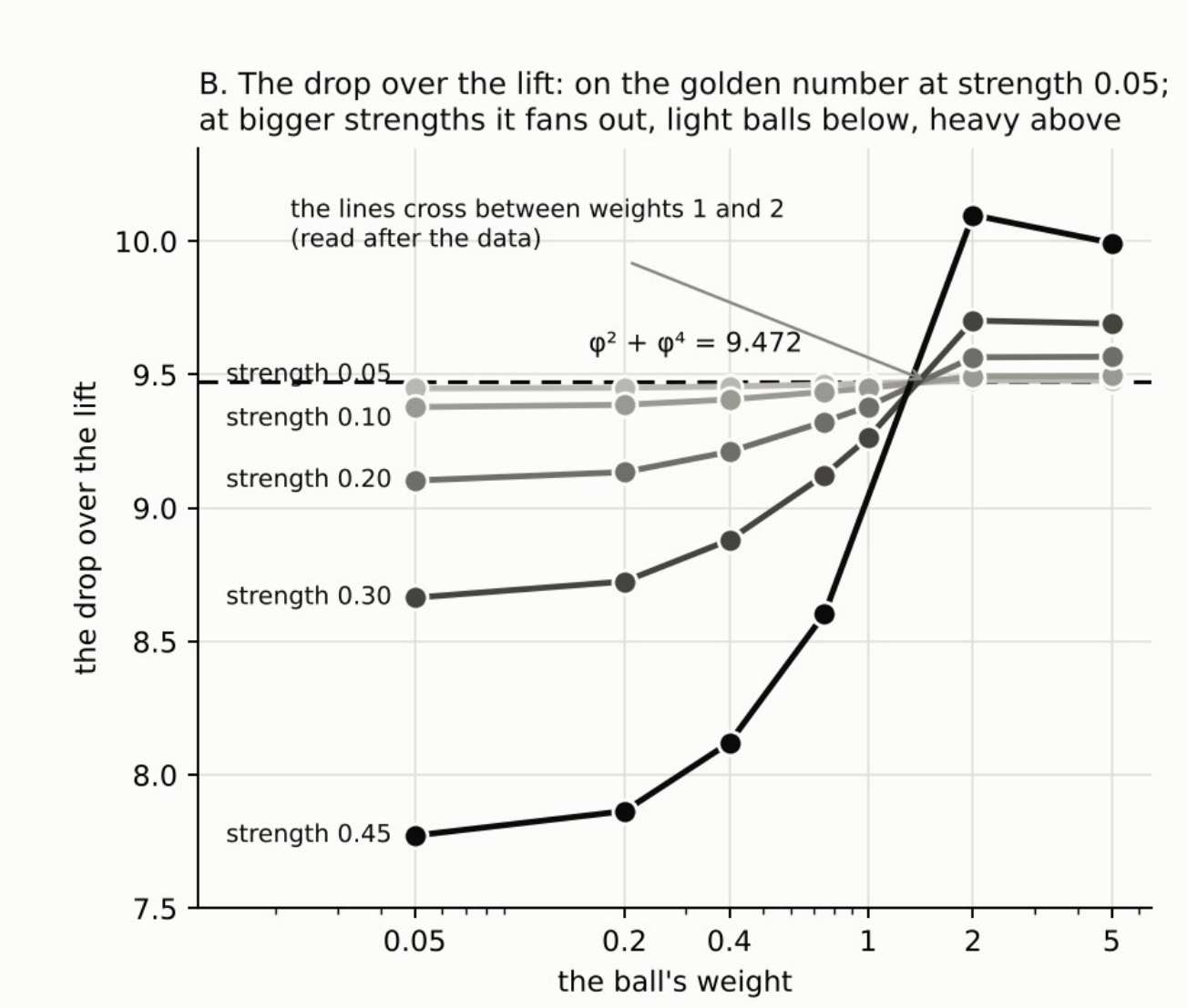


Figure 12. The fan and its waist. Figure 24 of *A Rider in Two Waves*, right panel: the drop over the lift, how far the ripple turned around moves the plain's edge against how far the ripple against the ball moves it, plotted against the ball's weight, one curve per strength from 0.05 to 0.45. At the smallest strength it sits on the golden value $\varphi^2 + \varphi^4 = 9.472$ at every weight; at larger strengths it fans out, light balls below and heavy above, and the lines cross between weights 1 and 2, the waist near 1.43 (read after the data). TOY MODEL; R2W's fold-direction series (FOLD-DIR).

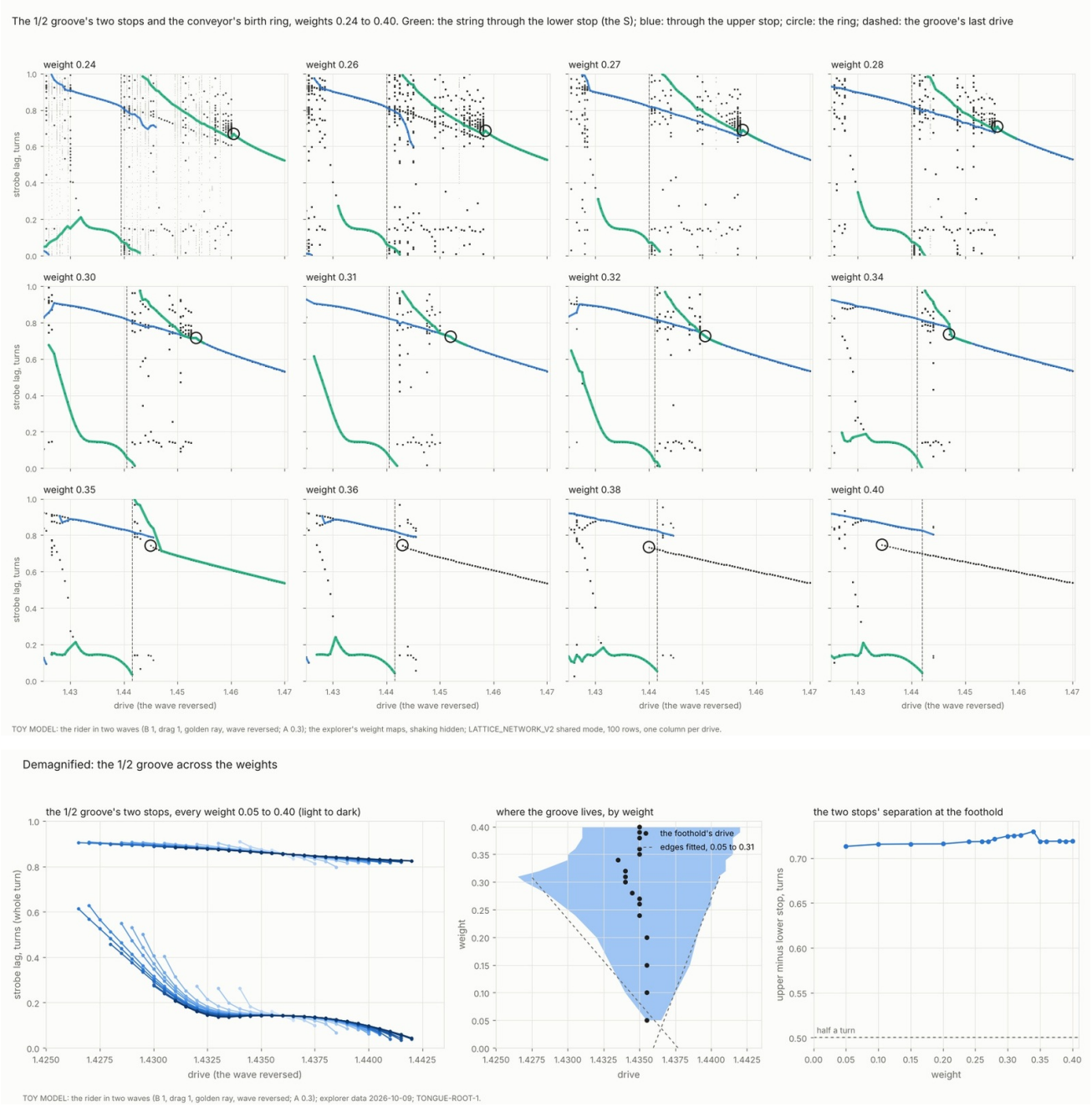


Figure 13. The S, the ring and the foothold. Above, Figure 27 of *A Rider in Two Waves*: the 1/2 groove's two stops and the conveyor's birth ring at weights 0.24 to 0.40, green for the string through the lower stop (the S), blue for the string through the upper stop and a circle at the ring, where the strings converge. Below, Figure 28 of *A Rider in Two Waves*: the groove's two stops at every weight from 0.05 to 0.40, the lower stop flat near lag 0.14 at every weight (left), where the groove lives by weight (middle), and the two stops' separation at the foothold (right). TOY MODEL; LATTICE_NETWORK_V2 and TONGUE-ROOT-1.

Figure needed: the rider's map coloured by denominator, with the orange chain marked, beside the pad atlas with its memory band.

The triple star. Shapes from the triple-star pictures: the bend figure, the torus opened flat, the lattice, the blind figure and the strobe pictures.

What the eye saw	What it turned out to be	Where	Status
Woven strands, "thousands of strings"	A rotation seen through a strobe: the dots' rhythm of 0.21257 cycles per pass returns	The triple-star bend figure: seen by Bear; the mechanism named and measured from the	Measured, from the pixels and on the data

	near itself after 5 and 14 passes	pixels by R2W	
White patches between the threads	The gaps of a rotation, which come in at most three sizes, in phase	The triple-star pictures	A test of R2W's, registered and never run on the triple-star data
A torus outline; rungs across a V	The bend is a smooth map on the torus of two angles; the hooks pair across the V by height	The triple-star pictures	Measured (the neighbour ratio 0.1126, then 0.0603); the hooks partly
Little curves with little tails at the edges	Folds of a surface seen flat, packed where the swing turns (Whitney's folds and cusps)	The triple-star opened torus	Seen; R2W's test registered and never run on the triple-star data
A "jellyfish"; two out-of-phase hills	A landscape on a torus must have saddles (Morse's count)	The triple-star pictures	A test of R2W's, registered and never run on the triple-star data
A diamond lattice of crossing lines	The envelope of a standing wave, crossing at its nodes	The triple-star pictures	Seen
A double helix, pulsing	Read two ways: one oscillation seen half a turn apart, a beat (R2W); a bend that depends on two angles, which took its fit from 0.4302 to 0.9723 (the triple-star sessions)	The triple-star bend figure	Seen; measured
A black bar; short spikes up becoming long	The band's lopsidedness flips sign while its centre holds	The triple-star pictures	Measured: right in sign, overstated in size by the pixels
An orange core line	The readings at the far point of the orbit rise about 0.007 to 0.010 mid-run and come back	The triple-star pictures	Measured from the pixels
Bands on the flank	Two heaps in the readings above 0.12	The triple-star pictures	Weak, open
A passage with breathing walls; "not a torus: a channel"	A corridor whose walls pinch near 0.45 in the second angle; at 6,000 passes the strands fill a solid hourglass with sharp walls, either a section of a torus with more angles or a chaotic layer	The triple-star pictures	The corridor measured; the rest a test waiting
Dense top, lean bottom	The rider lingers where its swing is middle, where the bend is flattest (212 coloured pixels per column against 51); the top is 3.7 to 6 times denser than the bottom	The triple-star pictures, 6,000 passes	Measured from the pixels
Scallops on a band	A slow beat of about 686 passes (666.7 was a bin of the spectrum, row 318, not a reading)	The triple-star zoom-out	Measured
p/q crossings	Three lines across the waist where the strings' change of change of slope is largest	The triple-star lattice	Seen after the run: a description
Gold lanes of empty space	Not yet measured; the lane sensors were too short-sighted	The triple-star lattice	Open
Clusters of three	The same place visited every 201 passes (575 of 599 dots)	The triple-star blind figure	Measured
Rows that arch below the waist and lift above it	The rows of the clusters' centres: 6 of 7 arch, 6 of 8 lift	The triple-star row fit	Measured; one row short of its registered bar
A frozen grid of tight clusters	A birth near an exact fraction	The triple-star blind figure	Seen; redrawn a hair away
Bilateral symmetry about a blue axis, on a squeezed strobe	The island's centre at $\varphi_N = 0$, where the runs start: 40 of 40	The triple-star strobe picture, passes 601 to 6,000, read by	Measured, on a bar the wrapped outlines could not fail

dim, on a squeezed strip picture	where the runs start: 40 of 40 outlines centred	passes 601 to 3,000, read by Bear	wrapped outlines could not run
A pinch in the middle	The room that keeps its stars, births 7.470 to 7.505, the narrowest outlines of the long view	The same, read by Bear	Measured
A waist on the upper panel	The narrowest swings of the first 600 passes, births 7.525 to 7.535, which did not last	The same picture’s first 600 passes	Measured
Black spacer rungs between pink data bands	The gaps the grid of births draws, one column every 0.005 with its dots spread 0.0018 either side: the drawing’s, not the data’s	Both panels, read by Bear	Measured from the grid
A V: the long runs’ core narrowing in a straight line to a pinch	An outline of the coarse sampling: on a grid five times finer each run’s width follows its own fate	The tilted strobe view, a birth every 0.005	Not borne out at 0.001
One trace turning over, then tracking another like a damped copy of a sharp signal; after the crossings, the two lines steady and parallel	The steady, parallel run held: at 1/8 and 1/9 the slope is the same on both sides of the fraction (−0.239 and −0.240; −0.295 and −0.292); at 1/10 and 1/11 it falls to about a quarter across the fraction (ratios 0.24 and 0.23), and after that the two lines run parallel. The rhyme held in a weaker, measured form: with each trend taken off, the two shapes correlate 0.285 at no shift and 0.521 at the best shift (the tenth moved eight steps), the tenth’s shape 0.75 the size of the eleventh’s, the same kind of mark at about three quarters the size and not the same curve	The triple star’s chart of four crossings, the tenth set against the eleventh, read by Bear (TS1’s ledger, run 20261017_172324)	Measured, reported and not scored; a second reader drew the same crossings within 4.47×10^{-6} and 8.95×10^{-6} (runs 20261017_155035 and 20261017_160624), so the shapes are the orbits’, not the reader’s; the same run also judged Claude’s call of a textbook shape at the tenth, withdrawn when the slope through 1/10 sharpened over a longer record (TS2; section 9 and row 795)

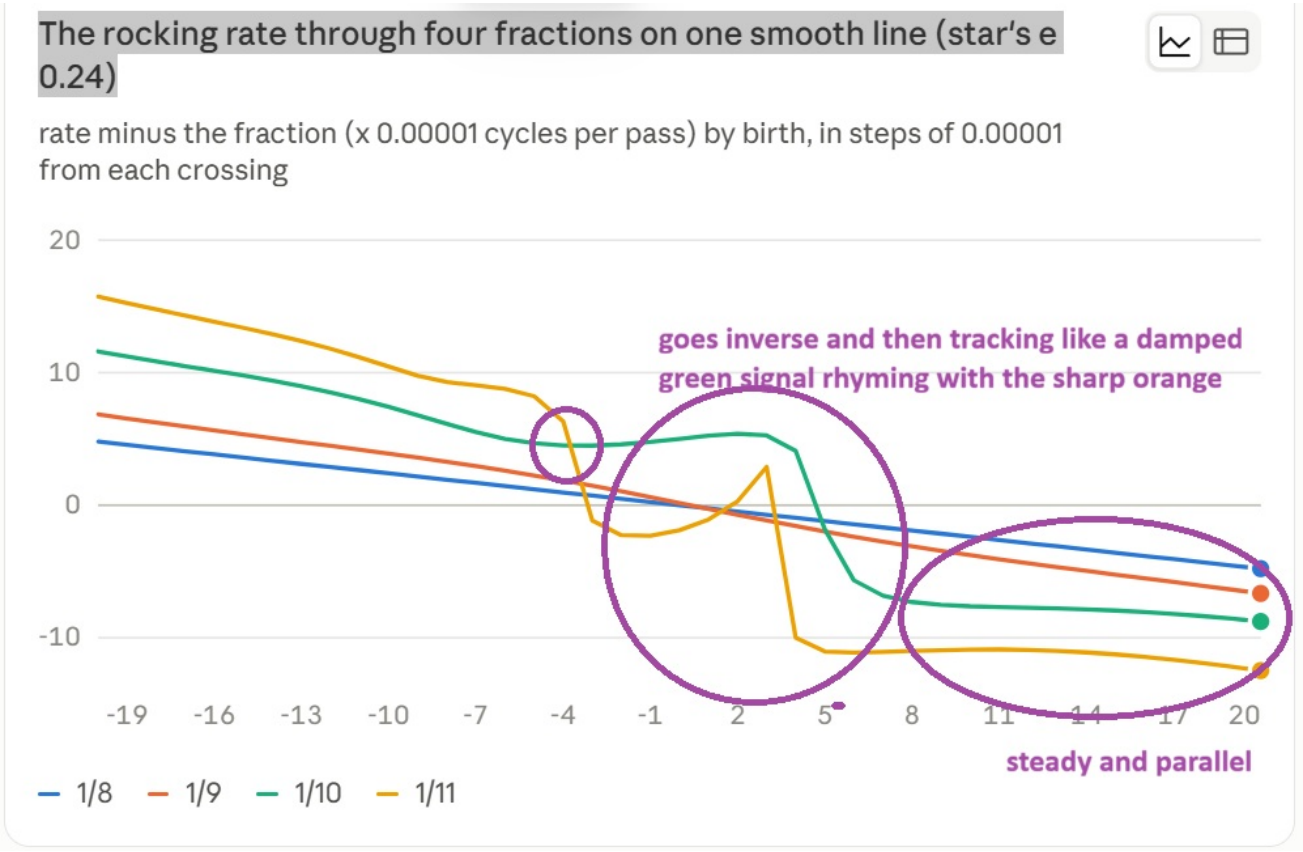


Figure 14. Four crossings, marked. The triple star's rocking rate through 1/8, 1/9, 1/10 and 1/11 at the star's eccentricity 0.24, with Bear's circles and words: one trace goes inverse and then tracks another like a damped copy of a sharp signal, and after the crossings the lines run steady and parallel. The steady, parallel run held; the rhyme held in a weaker, measured form (the field key). TS1's ledger, run 20261017_172324.

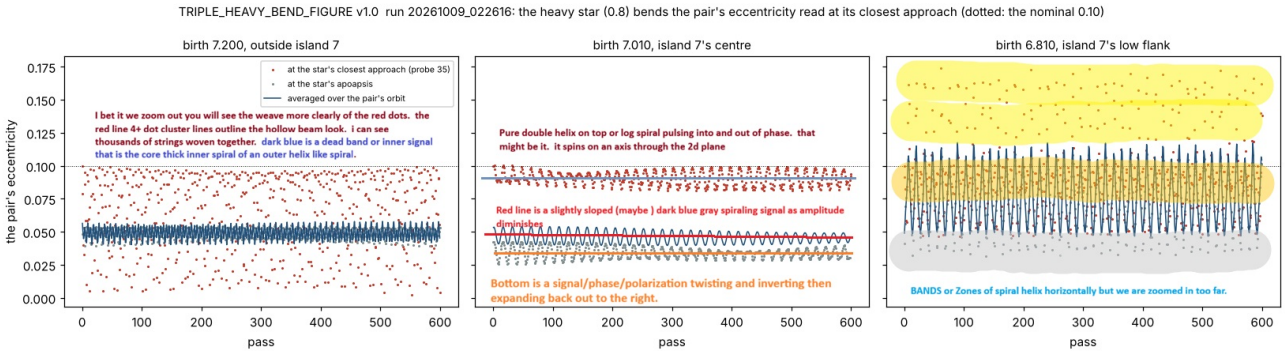


Figure 15. The bend figure, as Bear marked it. The pair's eccentricity read at the star's closest approach over 600 passes, at births 7.200 (outside island 7), 7.010 (its centre) and 6.810 (its low flank). He saw thousands of strings woven together at 7.200, a pure double helix at 7.010 and bands at 6.810, and asked to zoom out. The weave turned out to be a rotation seen through a strobe and the double helix a bend that depends on two angles (the field key); the zoom-out is Figure 7. TRIPLE_HEAVY_BEND_FIGURE v1.0, run 20261009_022616.

Bear's channel, "not a torus", is the top right of Figure 8; the solid hourglass of 6,000 passes is the middle panel of Figure 7.

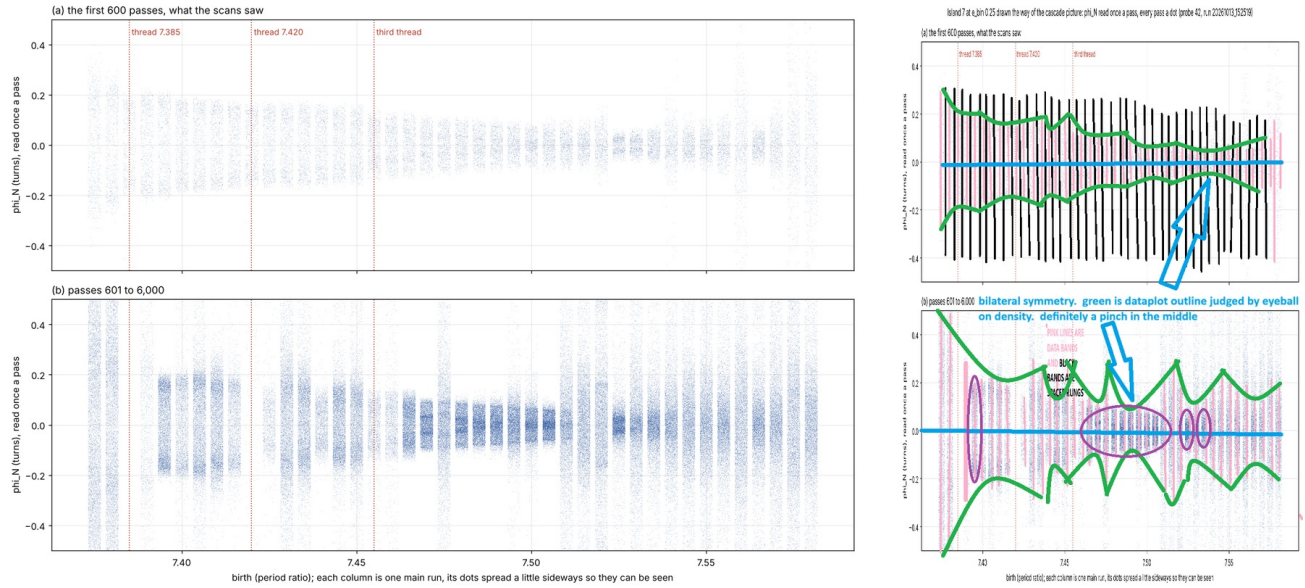


Figure 16. The squeezed strobe. Left, the triple-star paper's Figure 7: island 7 at e_bin 0.25, φ_N read once a pass, over the first 600 passes (a) and passes 601 to 6,000 (b). Right, Bear's working copy of the same picture, squeezed sideways and marked by eye on 14 October before anything was measured: pink data bands, black spacer rungs, a green outline judged by density, a blue axis and a pinch in the middle. triple_paper_figure7_marks v1.0, run 20261021_052614.

Figure needed: the tilted strobe view on the coarse grid of births, beside the same view on a grid five times finer.

6 — Traps, and how each was caught

A picture, a set of bins or a test can make a pattern all by itself, and a reader can bring one along too. This section lists the ones the projects fell into, and how most were caught, so the next reader can walk round them.

Every trap here was walked into at least once, and most were caught by one of the protocol's steps. They come from the drawing, from the bins and grids, from the test itself, and from the readers, and they are grouped in that order, each group opening with its rule.

The first kind comes from the drawing. Its rule: name the picture before naming the pattern.

The ruler's echo. A shape can belong to the measurement. The twin lobes of a third difference were the echo of a corner, drawn the same way by a made-up corner (row 711). Caught by the made-up case.

A line the sampling makes. Riders in flight, strobed once per passing, can line up into a string that is not a dance. In Bear's words: "When we start injecting a 3rd dimension into 2D, you often see what you want to see." Test by splitting each rider's record into halves or strobing at another phase: a dance stays where it is, and an illusion moves or breaks up (row 710).

The plotter's order. A plotter draws the last points on top, so a crowded picture shows mostly the end of the run, and colour counts favour late passes. Where dots merge, pixel counts and blob counts bracket the truth from either side (row 714).

A colour that lumps classes. Colouring by denominator (its figure is in the field key) painted the rest, the conveyor and two grooves the same orange, and a blind read named the wrong one (row 717). Caught by checking the class itself.

Starts laid on top of each other. At one weight, a rider started at lag 0 landed on a groove while one started half a turn round went into a doubling tree, and the picture, which laid both starts together, said nothing of which start made which dot. A coin, a place where two starts end differently, was hidden in it. Give each dot its start (row 664).

Figure needed: the picture at that weight with both starts laid together.

A cut through a band. An angle or a lag is a circle, and a picture's edge is a cut. A cut through a band makes one band look like two, and a string that runs off one edge is not thereby the one that enters at the other: a wrap matched by slope alone was withdrawn (row 704). Cut where nothing lands. And a column the dial left empty is not a break in a curve: the Explorer's first lattice read the empty columns between sampled drives as breaks, until it was made to read each slope to the next column that holds any landing (row 704).

A ceiling that belongs to the window. A memory band saturated near 0.31, and the saturation was the chart's own range, not the riders' (row 416). Check whether the chart sets a limit before explaining it.

The second kind comes from the bins and grids. Its rule is row 329's: binning is where the information design happens.

A pattern the grid draws. On a squeezed strobe picture (its figure is in the field key) the pink bands were data, and the black "spacer rungs" between them were the gaps the picture's own grid of births made, one column every 0.005 with its dots spread 0.0018 either side, so they belong to the drawing. It is chaos-and-binning in another coat, a pattern made by the bins, and the rule is the old one: name the picture before naming the pattern. Bear's earlier gaps, the white tops where stars had left, were the data's. Mark the empty space, then ask who drew it.

A shape the coarse picture makes. A V read on a picture with a birth every 0.005 (its figure is in the field key), the long runs' core narrowing in a straight line to a pinch, dissolved on a grid five times finer: at 0.001 the island's fates speckle, and the V and the floor went into the speckle, while the offset between where the island looks calm and where it lasts survived. Before a shape gets a name, look at it on a grid several times finer.

A grid on exact fractions. A birth chosen on an exact fraction drew a frozen frame of tight clusters (row 648). Test a hair either side.

The grid set by a top value. Paces set as shares of each floor's greatest grip mostly landed where the rough floors slip all the time, so the lean could not be judged. Place a grid on each case's own band (row 748).

A hole the size of the dot. Pads sized to the dots' own spacing saw only three-dot clusters and none of the rows they make; at 6,000 passes strings fell to 3 dots (rows 640 and 703). Caught by reading at two scales.

A thick string called a cloud. A crowding rule called a thick string a cloud, and a family's trace ended 3 columns early; the counter, landings per hole, is the cloud test.

A tie printed as a peak. Bins of lane direction gave a three-way tie no taller than evenly scattered dots give. Print "I can't see it" (row 128).

The ruler with holes. A spread over a fixed count of cycles sees only the cycles that divide the count (row 641). Caught by Bear's rings (row 715).

The short window that passes a slow slipper. A lock read in windows of 100 time units passes a rider that slips once every 200 to 400. Caught by the planted band test before the run of record (row 645).

One verdict over a core and a shell. A lock's single verdict can mix two populations. Each long lock's middle, read over 6,000 and 12,000 passes, split into a core that holds on the fraction and a shell that wears away, losing its lock or escaping; the guesses registered for both runs, that the middles would keep closing, were not borne out (LX1′ and LX2). At a fifth, seven births from 7.455 to 7.461 closed on it within 3.5×10^{-9} over 6,000 passes (LX1′, run 20261017_100655), while 7.452 escaped at pass 3,077; taken from that record, the fifth's core held over 12,000 passes, at 3.49×10^{-9} to 4.06×10^{-9} (LX2, run 20261017_200756). At the seventh, twins 10^{-10} apart then showed which members are chaotic: exactly 7.121, 7.122 and 7.123, the births that lost their lock (TW7, run 20261018_192644), while 7.124's twins never parted over 12,000 passes, after TW8 the only core birth read as regular. The reader could see the split only because the tool printed each member's reading beside the group's verdict. Print the members beside the verdict.

The table of bins. The triple-star sessions kept a table of the bins they chose and what each one decided.

Bin or threshold	What it decided	Row
A rest’s tolerance, 2 or 5 per cent of the ratio	At 2 per cent a slow creep was cut into many long rests; at 5 per cent it merged into one	622: two thresholds fixed in advance that disagree are data
A window of 0.1 round a whole number	Perfectly locked rests could reach only 0.44 to 0.53 inside it, so a bar of 0.4 sat under its own ceiling	210 and 416: a bar must sit where the reading can reach it, and a ceiling can belong to the window
A grid of births on exact fractions	One birth drew a frozen frame of tight clusters	648
A grid of births 0.03 apart	It stepped over an island shrunk to a sliver	631
Pads the size of the dots’ spacing	At 6,000 passes strings fell to 3 dots; at another birth the pads were a cluster’s own size and saw only clusters	640 and 703: strings depend on the lattice, curves do not
Bins of lane direction 5 degrees wide	A three-way tie, no taller than randomly scattered dots	128: say “I can’t see it”
Bins of eccentricity 0.005 wide	Bear’s bands passed by a dip of 9 against a bar of 9.6	Thin: the bar should come from a control
A colour scale	The conveyor hid inside one colour with the rest	717
The order the plotter draws in	Later passes cover earlier ones, so colour counts favour the late	714

Three rules travel from the table. **Choose the bin from the structure you are asking about,** not from the data’s spacing alone; at one birth that meant two scales, the return and the row. **Fix two thresholds in advance and read their disagreement as data.** And **run every binned statistic on a featureless record of the same size before believing it**: R2W’s shuffle turned a pixel weave into +12.4 standard deviations, and on the numbers the same linking read +23.4 against the same shuffle.

The third kind comes from the test itself. Its rule: a check needs a check of its own.

The count that points the wrong way. On a clean picture the count of short strings is higher in the shuffles than in the picture; only long reach separates them (row 616). Choose the statistic before the shuffle.

A null that makes its own effect. A shuffle that merged a tenth of a cloud’s pads thinned the null by itself, and the picture’s excess fell from +9.4 spreads to +1.8 once the null kept its counts (rows 616 and 30).

A bar that can hardly say no. A guess that asks only for a sign passes about half the time on noise: a guess about rows passed with shares of positive turns at 0.527 and 0.488, a coin’s half (row 337). And a rule can find its own shape in noise: a rule for S-shaped rows, a reading of Claude’s and not one of Bear’s marks, passed 6 of 7, and its control, built afterwards, called a smooth parabola S 0.718 of the time (row 616).

A rounded count wrong by one. A reference count of distinct points, rounded to a millionth, ran one or two high in six columns, because a value on a rounding boundary rounds two ways, and a prediction judged against it missed for that reason alone (row 709). Caught by counting each rider’s own period instead. The reference needs a check of its own.

A prediction right for the wrong reason. A memory floor was predicted at the pull where a rolling rider is caught again, and it held. A single rider with no ripple showed that a rolling rider keeps rolling down to 0.78 to 0.80, and that the floor, 0.86 to 0.87, is where a rider started half a turn round gets rolling at all. The band had been wide enough to catch the wrong mechanism (row 683).

The finder that reads its own failure. A doubling finder reported four edges, and every one was a place where Newton’s method stopped converging, because the needles’ angles were carried unreduced. Caught by a planted edge with a known answer, not by a check that the edges were found (row 751).

The threshold that reads a draining as a jump. A rule that called the edge where half of sixteen needles had stopped put the leak’s closing between two weights. The count kept beside it showed 9, 7, 2 and then 0 needles falling: a door eased shut, not slammed (row 787).

One edge, three doors. A threshold read from a sweep is a number, and the sweep does not say which way out it marks. In R2W’s toy a ball resting on the plain has at least three ways out: the resting dance stops existing (a fold, its multiplier reaching +1); the dance doubles (a multiplier through −1) and the ball is shaken loose; or the ball sits in a dance whose own fold comes first, the third door, read only after the data (that fold landed at 0.859975 and 0.713061, FX-2c; the reading of two resting dances side by side is withdrawn at weight 2, where the continued dance is unstable,

and unchecked at weight 5). All three appeared in one toy across strengths 0.30 to 0.60 (FOLD-DIR-e, FX-2b), and each gives an edge that looks the same on a sweep of drives. Which door it is decides how the edge moves when a knob turns. Under a gentle ripple a heavier ball's edge moves less. Under a strong turned ripple a heavier ball slips off sooner, weight 5 at drive 0.713 where weight 1 holds to 0.883 (FOLD-DIR-e), because the heavy ball, set down at rest, sits in a dance whose own fold comes first, the third door (FX-2c); R2W first read that edge as the dance doubling, which is how the heavy balls leave at strength 0.45 (FX-2b). A reading of the sweep before it (FOLD-DIR-d), where balls of weight 0.05 to 1.00 all slip off between drives 0.88 and 0.89 at strength 0.60, had the ball resting far down a stiff flank, where a spring shaken slowly answers with its stiffness and not its mass; it predicted that heavy balls would hold on longer, and missed (FX-2). Name the door before trusting the number, and when a knob's effect changes sign between two regimes, look for the mechanism that took over the edge (rows 642, 314, 695 and 739).

The planted print that shows a judged number. A planted case printed whole can show, before the run of record, a value a lean will judge. Say so in the row, and change no judging line (row 716).

A built file that lost its words. A ledger's table ran wider than the page and the builder clipped about 1,600 words without a warning, unseen for days until the PDF was counted back against its source (row 553).

The fourth kind comes from the readers. Its rule: write each reading down before setting it beside another, and translate a mark into a test with the person who made it.

The eye defaults to a fork. The second reader, which had read that Bear draws three roads at every split, the parent's road going straight on, still drew two (row 705). Caught by writing the cold reading down and setting it beside his.

Figure needed: the cold reading's two roads beside Bear's three, at the same split.

The overpass that was a meeting. Two strands that touched (their figure is in the field key) were registered as crossing at different depths. Lifted into a third number, speed, they were close in every number: a fork's young daughters, and the prediction had carried the very error Bear had warned of shortly before (row 710). Caught by the lift itself, and resolved by smaller holes and by history.

A mark put into the wrong words. Gold lanes of empty space, translated as rows of dots, were tested as the wrong thing, and sensors 5 holes long could not see lanes 0.3 turn long (row 715). Translate a mark into a test together with the person who made it.

A coarse fit read as a missing hand. A two-angle fit of a bend that explained 0.7230 of its variance was read as a third hand at work. A smoothness test that needs no chosen basis found the bend smooth on the torus of the two angles (row 718).

A corridor read by distance. The chaotic stars below the island were told apart cleanly by which way they turned (17 of 18), and not at all by how far from the edge they started. Read a corridor by direction, not by distance.

One birth read as a fate. At fine scale one birth is one sample of the chaos. The patterns that carry are shares (two births in five hold in the room that keeps its stars) and edges (no escape above 7.472).

A rhyme taken at its word. Two figures that look alike may share a cause, or only a shape, and the picture alone does not say which. Bear set R2W's pin figure, every weight's dance crossing at the weightless root at a drive of 1.436, beside the waist of the golden ratio's fan, every strength's ratio crossing the golden value near a weight of 1.43, and saw one picture. It is one shape: a family of curves, one for each setting of a knob, crossing where the knob's first effect vanishes, on the curve the knob-free toy draws, in a narrow region whose width the next effect sets. Chemistry calls it an isosbestic point, correlated-electron physics knows it from specific-heat curves crossing near one temperature (Vollhardt, 1997; Greger, Kollar and Vollhardt, 2013), and an engineer biases a transistor at its zero-temperature-coefficient point for the same reason. It is not one mechanism: the pin's offset grows as the weight to about the 2.5th power (32 times for 4 times the weight), the waist's as the strength squared (4.0 times for twice the strength), and 1.436 and 1.43 belong to different knobs and agree by chance. A rhyme can also be one law. The one he named across the three rooms, a driven, damped pendulum, is the pendulum equation, or Adler's, under three names: the needle lagging a turning field, the rider in a wave, and the angle of an orbital resonance (section 8). Name the shape, use it as a design point, look for the shared equation before calling a rhyme a coincidence, and test any claim of a deeper link by moving a knob that should separate the two (row 146).

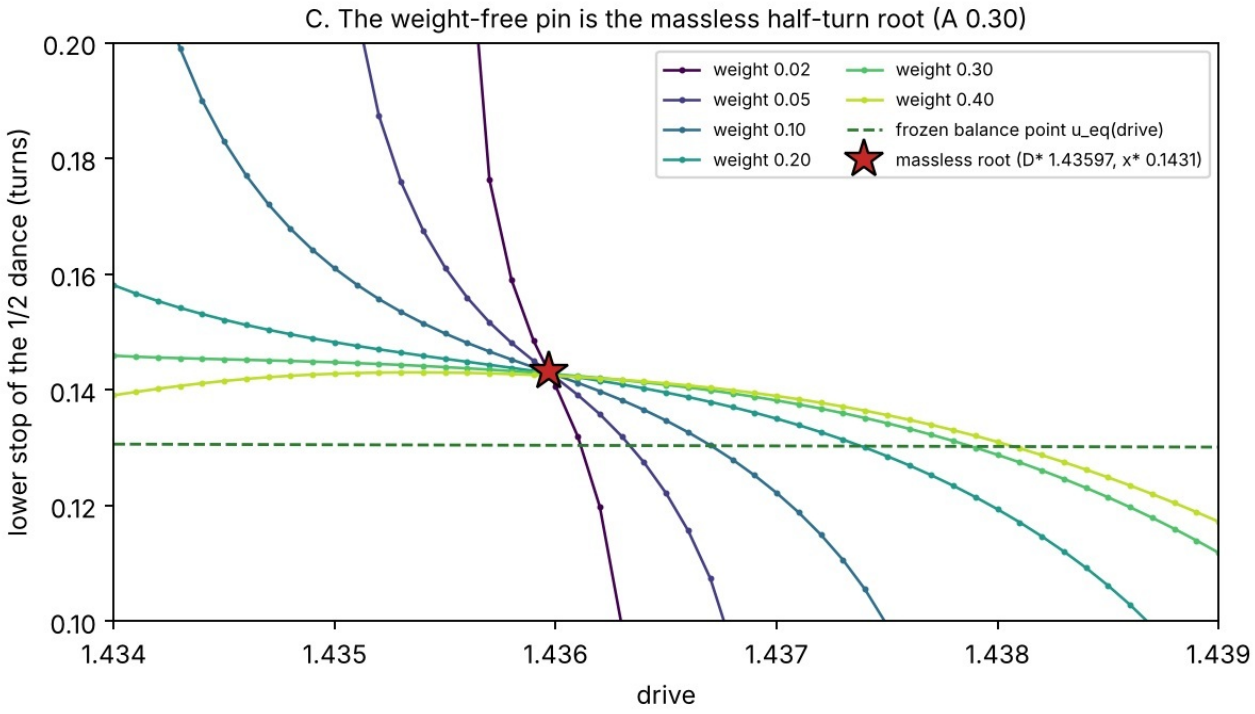


Figure 17. The pin. Figure 34 of *A Rider in Two Waves*: the lower stop of the 1/2 dance against the drive for every weight from 0.02 to 0.40, each curve crossing at the weightless half-turn root, drive 1.43597 and lag 0.1431. The other family of the rhyme, the golden ratio's fan with its waist, is Figure 12. TOY MODEL.

Two traps that are one, and two that are mirror images. A built PDF that loses text and a colour scale that lumps classes are both tools fooling their makers. And "when a sign is free, register the size" and "a prediction can hold for the wrong reason" are mirror images: one held for the wrong reason, the other missed for one.

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7 — Cases

Here the method meets real pictures. Each case tells what a person's eye saw, what the computer missed, the test written down first, and what the numbers then said; the first case is a control, a picture whose every answer was known before anyone read it.

Four cases. The first three come one from each project: the first is the control, the one picture whose every answer was fixed in advance; the second shows three readers on one picture; the third shows blind reads crossing from one project to another and being judged there. The fourth shows Bear's eye working in its own frame on three of the triple star's pictures, each marked first and tested after. Every case keeps one form, in one order: what the eye saw; what the computer missed or could not see; the test written before the data; the result, with its numbers and runs; and the figure the case needs.

The control: the needle's doubling staircase

The Loop Atlas turned Bear's lattice first on a picture whose answers it already knew to ten digits: the light needle's doubling staircase, at mass 0.75 (the dial the Loop Atlas also calls memory), where Newton's method on the orbit had fixed every split. The staircase is a dance of 2 steps becoming 4, then 8, 16, 32 and 64, as the drive is pushed. The bench was 901 drives a millionth apart from 1.68920, two riders each, 256 strobes recorded after 8,000 passings, with the angle and the speed at each. It is the only case in the three projects where every claim of the eye could be scored against an answer fixed in advance. That makes it the control for the rest.

What the eye saw. Here the eye's part was its method and one reading. The tools were Bear's lattice turned into code, and where the method's change of change of slope drew a pair of twin lobes at the splits, Bear gave five readings of the lobes (section 1).

What the computer missed. The lattice reader's first version, registered to see an eight-step dance as eight strings, saw four tight pairs at holes half a degree wide. Three predictions built on eight strings missed, and four could not be read (row 703). The change of change of slope peaked two columns early in every family. And lifted into the rider's

speed, two strands that touched in angle were registered as an overpass, apart in speed, when they were a real meeting.

The test, written before the data. The answers came first, from Newton's method on the orbit. Every prediction was then registered in the data ledger before its tool existed. The tools were LOOP_ATLAS_LATTICE_READER, LATTICE_READER_2 and HISTORY_SETTLE; practices 1 and 2 used figures from R2W's weight sweep (REV-WEIGHT-2), and practice 3 lifted the needle into a third dimension (LOOP_ATLAS_LIFT_3D). The first reader was run 20261009_092420. The second registered 10 predictions, and the lift registered five.

The result. *Its second version held 8 of 10.* It took its families from the side histogram's filled stretches, 18 rows in 4 stretches, and read them on holes of 10^{-4} radian: the same picture and the same method, and only the scale changed. Its counter read the period in every column without following a single rider: with 256 recorded passings, the median hole held 128, 64, 32 and 16 landings in the four bands (row 702). The lanes opened as square roots, with log-log slopes of 0.5115 for the families and 0.4959 for the strands, where the coarse scale had preferred a straight line (rows 703 and 706). Where strands crowd, count pads, not runs: on the fine lattice the 32-step band read 31 runs in columns 135 to 137, so 32 runs showed in only 81.2 per cent of the band, and the 64-step band read 58 to 61, while the counter read 32 pads in every column of the first band and 63 or 64 in the second.

History placed the doublings. The first columns where each lane's period reached 4, 8 and 16 (dances of 16, 32 and 64 steps over the four lanes) fell at 264, 143 and 116, against doublings of record at 262.4, 142.3 and 116.5 (row 709). A dance counts its steps over all four lanes, and the four-step dance visits each lane once, so its doublings fall where each lane's own period doubles; at the doubling of 262.4 each lane's period reaches the 4 that the history read. Settling is slow beside a doubling. After 8,000 passings the history first read the four-step dance's single visit at column 824 (823 in one lane), one column beyond its window, which had ended one column short (the window's size became a follow-on test); settling four times longer moved it to 822, 821, 820 and 822, not to the 819 or 820 predicted from the multiplier (run HISTORY_SETTLE 20261009_093424). An unsettled rider is a reading in itself: at column 111, a fraction of a column above a doubling, neither rider repeats within 128 passings, and at column 117 one reads period 64 and the other 32.

The third number found a real meeting. All five predictions about the lift missed. The two strands touched at columns 135 to 137, 1.58×10^{-4} to 1.84×10^{-4} apart in angle and only 6.41×10^{-5} to 7.51×10^{-5} apart in speed. They were a fork's young daughters, born 2 to 7 columns above, close in every number; holes of 10^{-5} in both numbers parted them, 32 strands in every column from 122 to 140. And the reference count the predictions were judged against was itself one too high in places, because a value on a rounding boundary rounds two ways (rows 709 and 710). That count was the distinct strobe points rounded to 10^{-6}, high in columns 122, 123 and 127. The count of record is each rider's period, the smallest repeat of its passings within 10^{-7} in both angle and speed, and the lift with holes of 10^{-5} agrees with it in every column from 112 to 140 where both riders repeat with one period (run LIFT_3D 20261010_031527). Lifting into a third number helps, though not always: at column 780 the speed separates the widest pair four times better than the angle does, and two of the pairs barely at all.

The twin lobes placed the splits. The peaks taken two columns early fell at columns 815 and 260 in every family, against the four-step dance's doubling at column 817.0 and the next at 262.4. A centred stencil answers a sudden bend with two lobes of opposite sign, and the peak taken had been the lower lobe. Both lobes are present in all four lanes at both doublings, their midpoint within a tenth of a column of the doubling (run LOBE_TWIN 20261010_043453). Read where the lobes cross zero, on columns a millionth of a push apart (run ZERO_CROSS 20261010_043523), the event sat on the doubling, within 0.0031, 0.0014 and 0.0062 of a column, and the ratio of the two stretches between those three doublings read 4.6153 in every lane, against 4.6155 from the orbit. The method needs room: at the fourth doubling, where the next event lay inside the stencil's reach, the crossing landed 0.08 of a column high (row 708).

The lobes themselves were the ruler's. A made-up perfect corner, run through the same three differences (run CORNER_PRINT 20261010_061715), drew the same pair, crossing zero exactly on each doubling and matching every lane's pair to 0.993 in size at the top doubling and better than 0.9999 at the next two. The shape was the ruler's echo of a corner; only its place belonged to the data (row 711). The corner's own size was measured later: each doubling turns the last one's corner over, shrunk by a fixed ratio, -1.966 to -1.968 in all four of the needle's lanes and -1.9644458 on the logistic and sine maps (row 877). Of Bear's five readings of the lobes, "boundary transition" and "mirror-math" were the two the data bore out: the pair marks the boundary between one dance and its doubled dance, and it is mirror-antisymmetric because the ruler is. Of his three other readings, his "phase-locked" has a real cousin in a radio discriminator's S-curve, whose zero is where a receiver locks (from recall).

Record a whole number of the dance's periods. The needle's periods all divide 256. On a logistic-map test bed with period 3 and 256 recorded passings, each lane got 85 or 86 visits, the lanes' centres jittered, and the change of change of slope put its largest peaks in the wrong place (a scratch file, not a run of record).

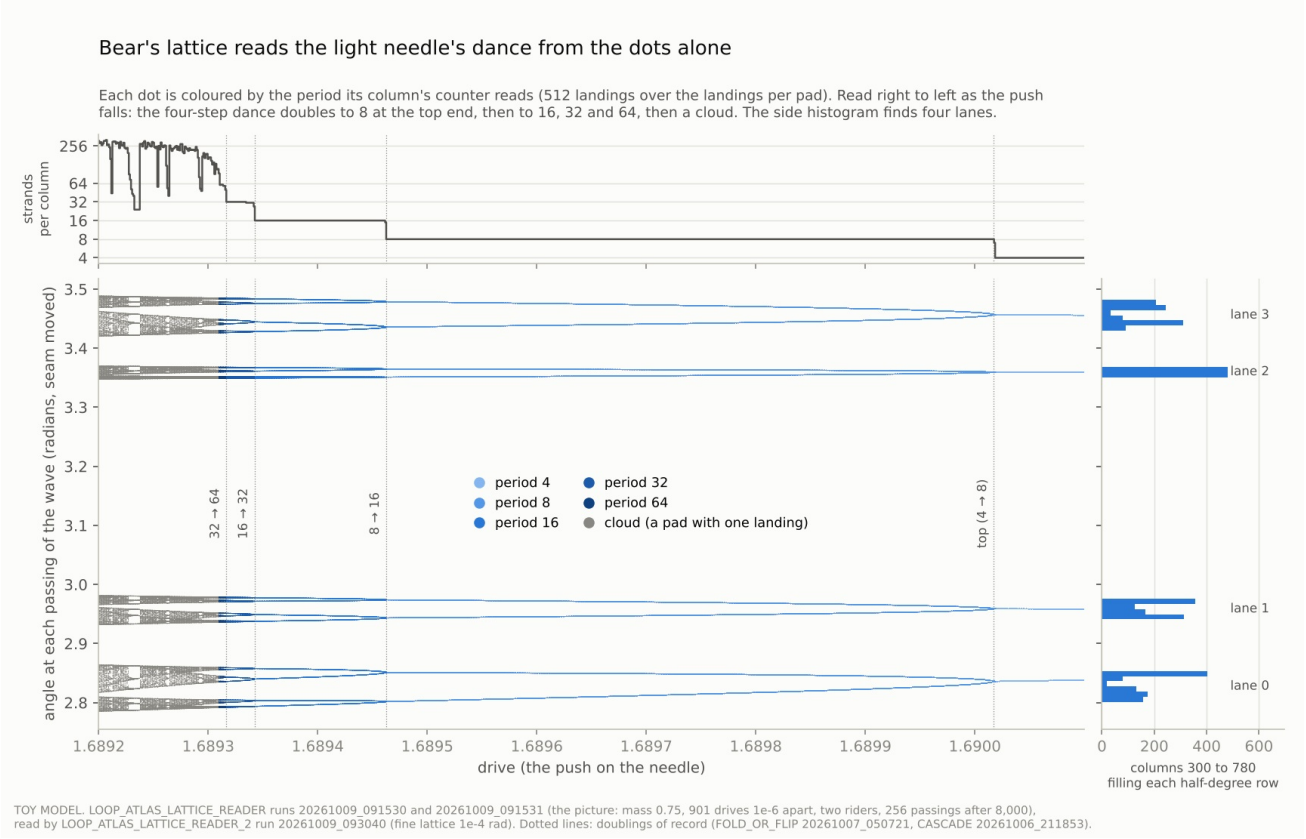


Figure 18. Bear’s lattice on the needle. The light needle at mass 0.75, read from its dots alone: right to left, as the push falls, the four-step dance doubles to 8, 16, 32 and 64 and then becomes a cloud; above, the strands per column climb as a staircase; at the right, the side histogram finds four lanes. TOY MODEL; LOOP_ATLAS_LATTICE_READER runs 20261009_091530 and 20261009_091531, read by LATTICE_READER_2, run 20261009_093040.

Figure needed: one lane’s twin lobes at a doubling, beside the made-up corner’s pair.

Three readers on one picture: R2W’s weight 0.26

Bear handed the Loop Atlas one of R2W’s maps as its lattice test: a strobe picture at weight 0.26, riders’ lags against the drive, with a groove, a conveyor and two clouds. Three readers read it: Bear, Claude reading cold, and R2W’s decoder.

What the eye saw. Bear drew six long strings, circled three populations, and called it 80 per cent, with one string maybe a frequency illusion. One of his strings carried the groove’s upper foot through the middle cloud to the conveyor’s ring. Another was a copy of the groove’s lower foot, its drop, its shelf near lag 0.14 and its final fall, shifted about 0.010 of drive to the right with its shelf squeezed to about 40 per cent (row 706). The seven columns of three dots were three of his strings crossing one window.

What the computer missed. Claude’s cold reading joined pieces the picture did not support: a string leaving the bottom edge to one entering at the top, matched by slope alone, and three strings crossing one window, read as one dance (row 704). R2W’s decoder, greedy by design, drew 50 strings of four or more pads on a lattice of 91 by 100, short chains where the eye drew long ones.

The test, written before the data. The cold reading was on record first, before Bear’s word and the decoder’s reading, so its joins could be judged against both. Its counts came from the picture’s own pixels, read by colour.

The result. *The cold reading’s counting found what neither drawing showed.* The side histogram gives the numerators, how often each lag is visited, and the bottom histogram the denominators, how many riders each column holds; a fraction needs both, and R2W’s own shares were uneven for that reason (8 starts at some drives, 50 at others). Claude read the bottom histogram off the picture by colour: 2 to 21 black shapes per column, exactly 3 in each of the seven columns from 1.4455 to 1.4485, then 12 at 1.4490. It found the lag axis to be a circle with its seam in the wrong place. Only 7 of the 340 black shapes lay between lags 0.27 and 0.52, and counted by tenths of a turn the two black clouds had matching profiles, left 13, 5, 4, 0, 0, 5, 5, 13, 13, 23 and middle 18, 8, 5, 0, 0, 5, 8, 21, 18, 19. That is the misplaced seam: a cut through one band makes it look like two. For comparison, on its own needle the Loop Atlas’s second reader puts its seam in the middle of the side histogram’s widest empty stretch, at 4.9655 radian,

and with it none of the 461,312 dots fell outside a family (run 20261009_093040). The cold reading also found one flat string at lag 0.145 showing through three windows, and the shapes of the groove's lower foot, an exponential settling onto a near-flat stretch and then an exponential fall.

The decoder found the families the eye passes over. It found a staircase of flat strings where the cold reading found one, at lags 0.144, 0.707, 0.81 and about 0.92, each through several windows, and about ten rising strings where the cold reading saw none (row 707). Flat strings come as a family, and rising strings must be looked for on purpose.

Bear's long reach. His six strings explained more of the picture than fifty short chains, and long reach is also what separates a clean picture from its shuffle (row 707). The cold reading's join across the seam by slope alone was withdrawn after Bear's word and the decoder's reading. Asked which string was the illusion, Bear named the cause rather than the string: "When we start injecting a 3rd dimension into 2D, you often see what you want to see" (row 710).

What each reader did best is what the rest of this part asks for: the eye's long reach, the decoder's families, and the cold reading's counts. One reading is not yet tested. There may be a copy across pictures: the flat lags at weight 0.26 (0.81 and 0.144) sit where the third face's flat feet sat at weight 0.40 (0.805 and 0.135).

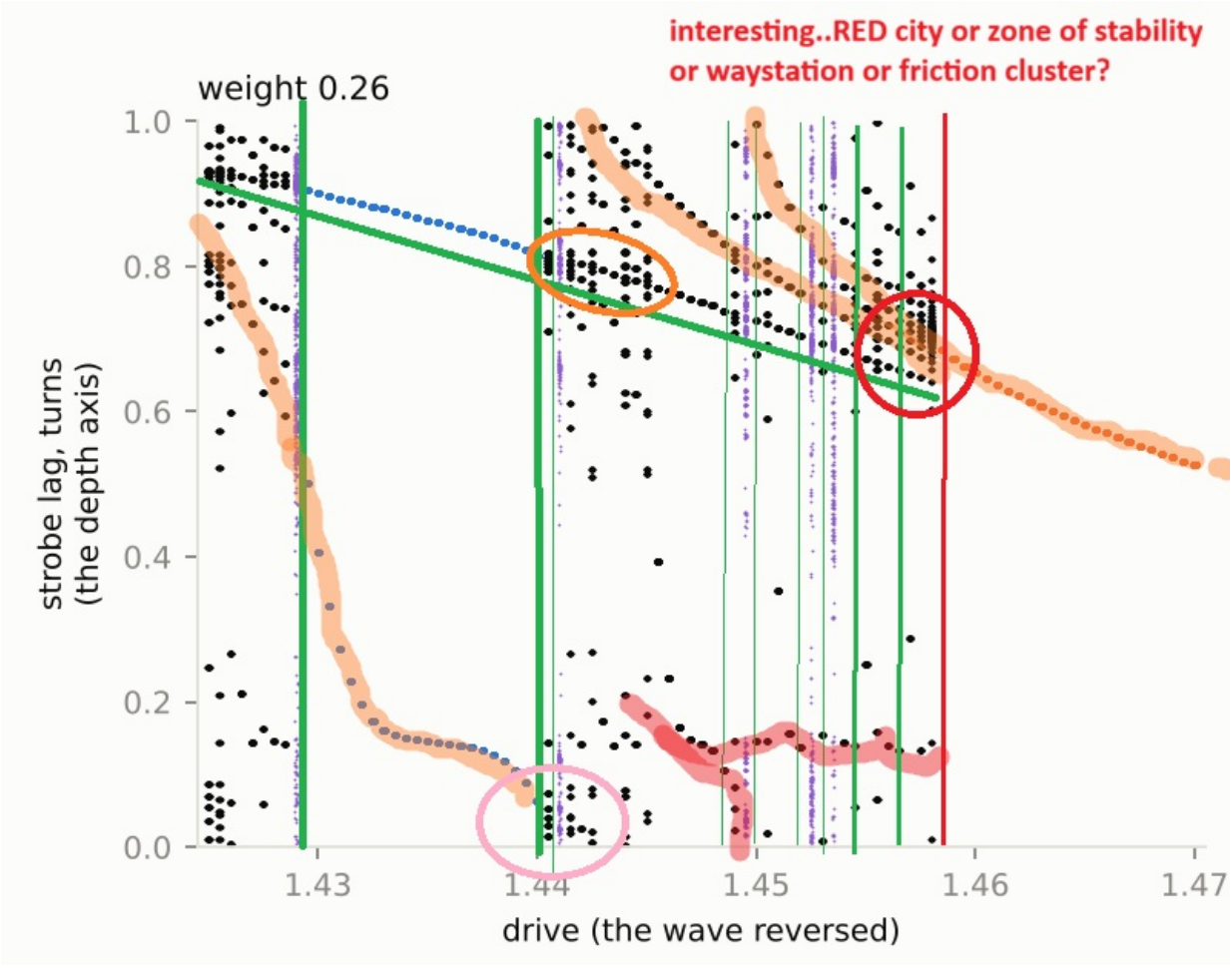


Figure 19. The weight 0.26 picture, as Bear marked it. R2W's depth-axis figure at weight 0.26 (Figure 5 of *A Rider in Two Waves*) with his marks: strings in orange and red, one of them a copy of the groove's lower foot shifted to the right; green lines across the drives; and three circled populations, the red one at the conveyor's ring, with his note: "Interesting… red city or zone of stability or waystation or friction cluster?" TOY MODEL.

Figure needed: the same picture with the decoder's 50 strings.

Beside the case: the chalkboard. One more lesson came from R2W's pictures, and it is a check rather than a case: explaining it simply is a test (row 667). Partway through the work Bear stopped to ask what the Explorer's picture actually is. R2W drew its weight 2 map for him "like to a smart 12-year-old on a chalkboard": the rider as a heavy ball on a staircase of ledges, the memory band redrawn from the chunks, and the falling ball's beats set against the lines a rider with no ripple would follow. Every label had to name what the data says, so each was checked as it was drawn. The plain account named the whole-number grooves as Shapiro steps (Shapiro-like, as R2W later tightened the name,

since a junction's microwave adds a push while the ripple moves the bumps), sitting where the rider's own pace with no ripple is 1, 2 or 3 bumps per passing (the conveyor within 0.003 in clock ratio from pull 0.90 to 1.20). And it found mistakes while it was being drawn, among them a blind read that had named the conveyor as the rest (row 717), and a memory prediction that had held for the wrong reason (row 683). Bear's two questions at the chalkboard, who sets the clock and whether the honey only steals speed, are the ones section 8 answers.

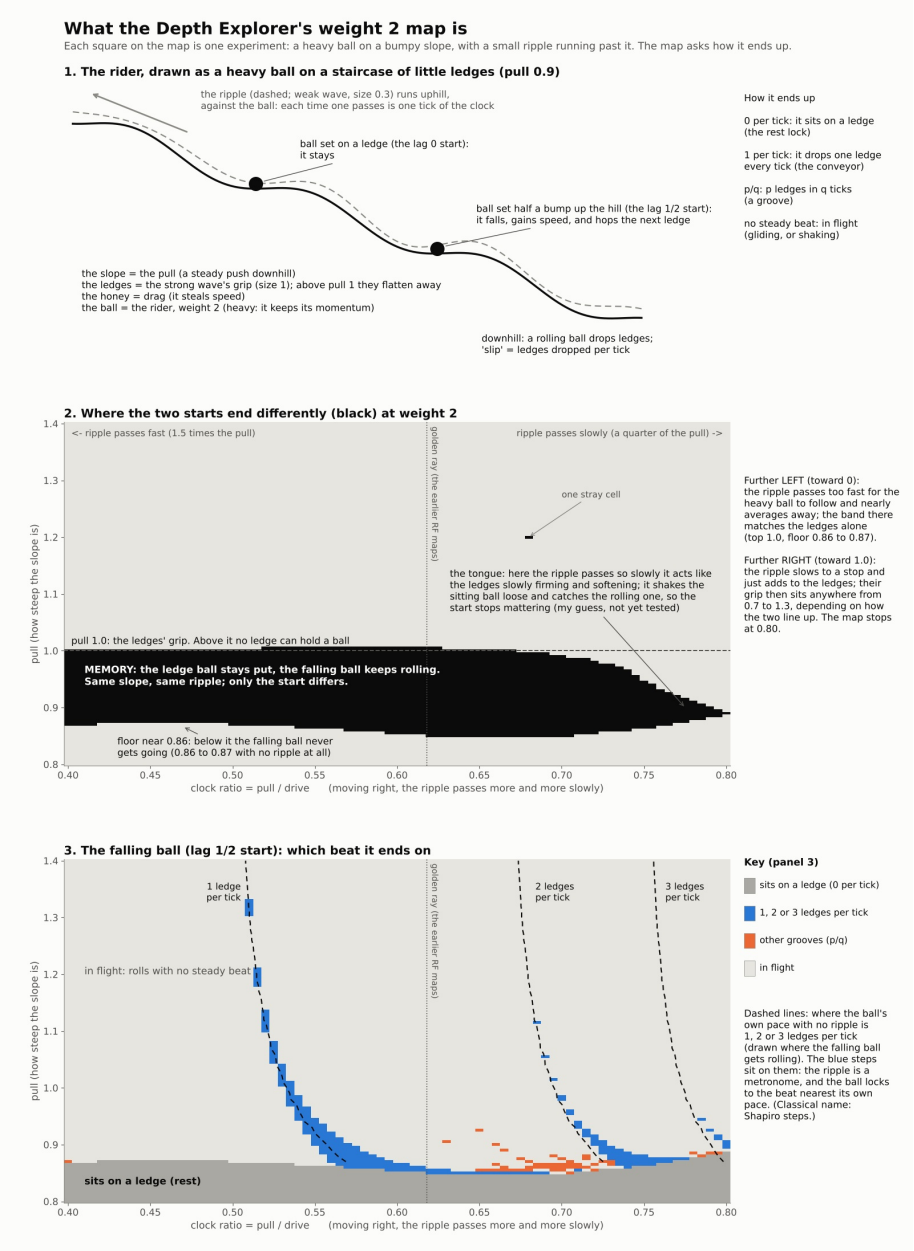


Figure 20. The chalkboard. R2W's drawing of its weight 2 map for Bear, 10 October (Figure 2 of *A Rider in Two Waves*): the rider as a heavy ball on a staircase of ledges, with the ripple running uphill against it; where the two starts end differently, the memory band, with its floor near 0.86; and where the falling ball ends up, its beats of 1, 2 and 3 ledges a tick sitting on the dashed lines a ball with no ripple would follow. TOY MODEL; CHALKBOARD_W2_FIGURE.

Blind reads across projects: the triple star

Two readers read the triple-star sessions' pictures, and the sessions judged both on the numbers behind them: R2W, from pixels alone, and Bear, whose words built the sessions' lattice and whose eye read birth 7.230.

What the eyes saw. R2W read the triple-star sessions' figures blind, from their pixels alone, because the data was held back from it. Three of its reads were registered in its own words: the dots on the page of eccentricity against pass link 5 passes apart and shun gaps of 1 and 3; the band's lopsidedness flips sign across the run while its centre holds; and at 6,000 passes the top edge of a band has 8 crests about 681 passes apart. Bear had written that strings

can skip holes (“when they skip 1 or more holes to the next pad”), and the sessions built his lattice literally from his words. At birth 7.230 he drew no crossings, since he saw no obvious p/q, but he added all the lines, and he saw the dots stack: “3 dots on top of each other in the z-axis, into and out of the page”. His eye collapsed each cluster of three into one place before following the rows, followed a row whole, saw lanes of empty space, and saw the walls as places where the visits pile up.

What the computer missed. R2W could not see the numbers, only the page. The sessions’ lattice tool had the numbers and did none of those four things: its links broke one row in half at exactly the stretch he had drawn flat, and its sensors, looking 5 pads each way, could not see the lanes of empty space. Each of the four is now a line in the lattice’s next version (section 4).

The test, written before the data. The triple-star sessions judged R2W’s three reads on the numbers behind the pictures, with each guess written into the tool before it computed anything. The ladder’s reading, a gear between two angles found at birth 7.200, was then run blind at two more births, 7.190 and 7.220: read off both angles’ median net turns, the gear had to name the lattice’s three commonest gaps in their order, and it was run again with the angle shuffled.

The result. All three of R2W’s reads were borne out, and the sizes of two were off in opposite directions: the weave was stronger on the data than on the page, probably because merged dots hide links, and the lopsidedness was weaker, probably because a drawn line’s width and joins exaggerate its spikes. Section 9 keeps the score. At birth 7.200 Bear’s lattice found the dot families climbing a ladder in which each pass gap is the sum of the two before, 19, 28, 47, 75 and 122, and the multiples of gap 47 pointing the same way within 1.5 degrees: one string skipping holes, as he had written. The ladder belongs to a pair of angles together, not to either angle alone; a check of their data made while this guide was assembled found the pair geared one to three (row 719). Run blind, the gear named the lattice’s three commonest gaps in their order at 7.190 and 7.220, and failed with the angle shuffled. The gear and the lattice read one record, so their agreement checks the two readers, not the stars (row 556). At birth 7.230 the lattice found the stacking he saw: 575 of 599 dots are the same place visited at passes k, k + 201 and k + 402.

Figure needed: the triple-star pictures R2W read blind, with its three registered reads marked on them.

Figure needed: the ladder at birth 7.200, gaps 19, 28, 47, 75 and 122, with the multiples of gap 47 drawn as one string skipping holes.

Figure needed: the picture at birth 7.230 with Bear’s rows, beside the tool’s links that broke one row in half.

Bear’s eye in its own frame: the maypole, the dead zones and the squeezed strobe

Three more of the triple-star sessions’ pictures, of 12 and 14 October, each marked by Bear first and tested after. Each is told in the same form.

The maypole, marked. The maypole, Bear’s word, became a disc picture of the triple star: at each look the angle is φ_N and the radius the pair’s eccentricity, with the pole at the centre.

What the eye saw. On panel A he drew a pink band on the slide lane. On panel B he drew star-polygon paths, “24-gon? Skipping 4 vertices for each line like clockwork.”, with circled 2s and 3s at the corners, “Order?” inside, and “3+” and “Chaos?” on the slide side. On panel C he drew a green mirror line from the pole along $\varphi_N = 0$, “Symmetry and Asymmetry”, with thin pink rays and a faint violet chord. He confirmed Claude’s reading of his marks the same day, adding that each 2 or 3 can be traced to its sister points and that the vertex skipping is solid. His eye read the picture in the circle’s own frame, a clock about its centre.

What the computer missed. Claude’s sealed read did not see the clockwork star polygon. The tools read the picture in the pole’s frame.

The test, written before the data. Claude wrote its translation of his marks and fingerprinted it before opening its own sealed read (6687e26ce2c620bd). The sealed read carried a test of its own, M1.

The result. The seal and his eye both put the figure’s axis on $\varphi_N = 0$. The seal held a circle wrapping the pole by 0.002 at birth 7.200, read as a fixed arrow plus a turning one, which he did not mark. His clockwork held, and it is the inscribed-angle theorem (row 726). The seal’s own test, M1, read nothing by its control; its redesigns found the count switching on at 7.130 while the fitted circle still sits beside the pole, and the narrow window’s single miss is a tie at 7.130 (row 201). Two readers in two frames saw two true things, and a theorem 2,000 years old joined them (runs 20261012_033532 to 20261012_062723).

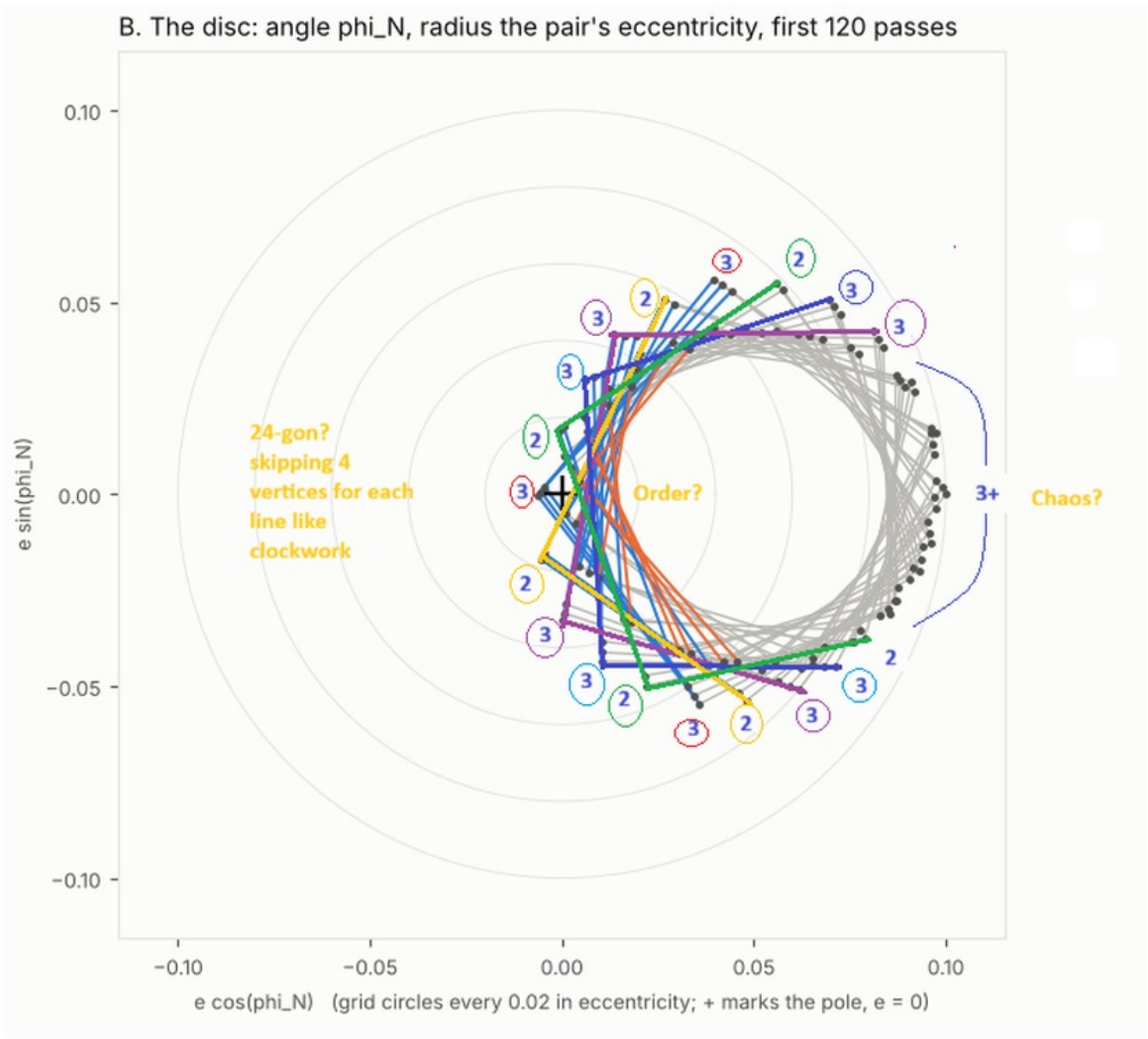


Figure 21. The maypole, marked. The disc of φ_N with the pair's eccentricity as its radius, birth 7.200, first 120 passes, and Bear's note: a 24-gon, skipping 4 vertices for each line, like clockwork. His clockwork held: the median step about the circle's own centre was 0.2154 turn against 5/24 = 0.2083, the inscribed-angle theorem at work. From the triple-star paper's figure of the eye, run 20261025_033626.

The dead zones, on an old figure. He shared a figure of 6 October, marked: the island map, a log scale of φ_N's band against birth.

What the eye saw. He drew four lines at about 1.2, 4.5, 135 and 600 turns and wrote "Dead Zone Skips" between each pair. He traced each phase's runs as a curve, with arrows where the curves jump the lower zone, and on the third panel he traced the pair's least and most eccentricity. His eye found a gap in a log-scale axis, negative space again as with his gold lanes, but this time measurable, and it joined two panels by tracing both.

What the computer missed. The figure, of 6 October, already showed the start sitting at the top or the bottom of the pair's swing, the reason the thread moved to the mean-ellipse start after the checks of 8 October. His trace shows it on the older figure.

The test, written before the data. His marks came first. The zones were then tested on data he never drew on.

The result. On data he never drew on, both zones held (the scorecard, section 9; row 715). On that figure's own runs the zones are a description: its held runs reach 0.880 and its circulating runs start at 5.897, and the three runs above 130 are exactly the three whose pair was knocked to an eccentricity of 0.61 to 0.64. The lower zone's run at 7.130 is the maypole's tie, the one birth where the narrow window's swing toward the pole and the circle's edge agree to 0.0001; a test called E1 was registered to try the same at the zone's other run, 6.785, and no result of it was recorded before the triple-star thread closed on 29 October. His traced third panel held too: of the 41 runs started on the beat, 33 never let the pair's eccentricity rise above 0.101, while all 41 started half a beat late rise, to 0.189 to 0.636, and 40 never fall below 0.092 (run 20261012_073126).

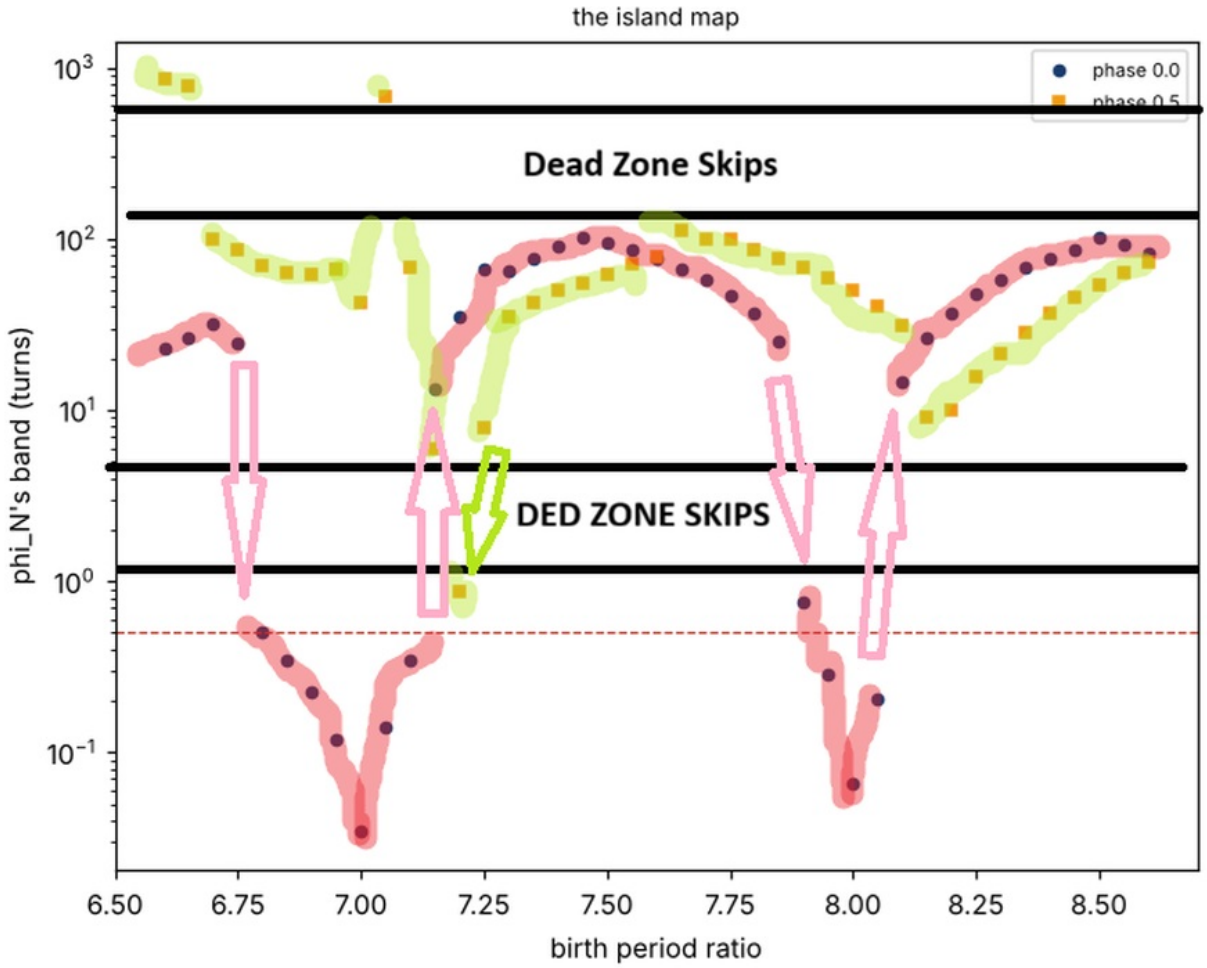


Figure 22. The dead zones, marked. The island map of 6 October, φ_N's band against birth on a log scale, with Bear's marks of 11 October: four lines with "Dead Zone Skips" between each pair, each phase's runs traced as a curve, and arrows where the curves jump the lower zone. His lower zone held on 600-pass runs he never drew on: 3 of 484 runs fell inside it, against 64 just below and 11 just above. From the triple-star paper's figure of the eye, run 20261025_033626.

The squeezed strobe. The strobe picture (island 7 at the pair's eccentricity 0.25, every pass of every main run a dot, birth across) went to him unmarked on 14 October.

What the eye saw. He marked pink "data bands" on the columns of dots and black "spacer rungs" between them. Then he squeezed the picture sideways to about half its width, drew by eye a green outline of where the dots are dense and a blue axis along $\varphi_N = 0$, and wrote: "Bilateral symmetry. Green is dataplot outline judged by eyeball on density. Definitely a pinch in the middle." Last, he turned it a quarter turn on his screen and photographed the screen from a grazing angle; the file's name says it was marked before the squeeze, with the eye's phase offset pretended at 75 degrees "to make it severe". He named the two strategies himself, to sit beside zooming out, zooming in and looking at a dot's neighbours: "I cheat the phase offset view by angling my screen a little or by compressing event time."

What the computer missed. Where the island survives: the centre that a scan of the first 600 passes reads is not where the island lasts.

The test, written before the data. Three reads drawn from his marks were registered, marked first and tested after: his symmetry, his pinch, and a moving waist that Claude read from his two arrows.

The result. Two of the three held as registered, his pinch and the moving waist, and his symmetry passed only by construction (the scorecard, section 9). And the squeezing found what no one had asked: the births with the smallest swings over the first 600 passes, 7.525 to 7.535, are not the ones that last. Their outlines over passes 601 to 6,000

widen to 0.46 to 0.75 turn, while the room that keeps its stars sits 0.02 to 0.06 lower. Only the short view and the long view side by side, each squeezed into one outline, showed it.

Figure 16 is the strobe picture of island 7 and his squeezed copy, with its green outline and blue axis. Figure needed: the grazing-angle photograph.

Figure needed: the short view and the long view of births 7.525 to 7.535, side by side, each squeezed into one outline.

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8 — The common object

The three projects turned out to be watching the same kind of thing: an angle going round a circle, looked at once each time an outside clock ticks. This section names that shared object, the question that splits it into two families (does it keep area, or lose it?), what the eye reads on it, and why finding it already named in the textbooks is the usual result, not a discovery.

The section goes in four steps: the object in its three costumes, with the clocks that drive it; the fork that splits it, area kept or lost; the three readings the eye makes on it, the count, the corner and the leftover, each with its rule; and where the shared method stops.

One object under three costumes. All three projects watch an angle carried round a circle by a flow, and look at it once per tick of an outside clock. That is a circle map, strobed: the Poincaré section of the textbooks. Its currency is the winding number p/q, p turns in q ticks, the same holonomy currency that runs through the rest of this guide. The register's own transfers agree: on 16 October it recorded 20 transfers between the three projects, 14 held, 3 held weaker and 3 failed, and none of the six breaks was put down to the substrate or a wall. R2W sent 14 of them (8 to the Loop Atlas, 6 to the stars), the Loop Atlas 4, and the triple star 2, both ways of reading pictures (rows 734 and 743).

In each toy	The Loop Atlas's needle	R2W's rider	The triple star
The outside tick	One push of the drive	One passing of the ripple	The third star's pass, at closest approach
What turns	The needle's angle	The rider's place on the washboard	The inner pair's phase, and the resonance angle φ_N (Mardling's [N:1](2) angle)
A lock	A gear: whole turns per push held over a band (the one gear; the 1:2 gear)	A groove: the slip, ledges per tick, sits on p/q; the whole-number grooves are Shapiro-like steps	An island: φ_N rocks about a fixed value
Outside the lock	—	The rider is in flight	φ_N slides round
Friction	Damping: area is lost	The honey: area is lost	None: area is kept
The rider's own pace	Set by its slope and its damping	Set by its slope and its honey	The pair's own turns, and a slow beat of about 686 passes at one birth
Memory of the start	Where two locks overlap (row 670)	The memory band	Total: every start keeps its own orbit, and the start chooses which locks can be seen
A birth by a fold	The glide's extra turns grow as the square root (slopes 0.46 to 0.48)	The lurch-stroll-lurch S; below a conveyor's birth, the bottleneck exponent 0.507 (row 654)	—
A birth by a flip	Ratios closing on 4.669 (4.6153 read from the dots)	The third face's first ratio	An island doubles once and its stars escape; an area-keeping test map closes on 8.72

R2W's rider is the needle. R2W's chalkboard equation is $m\,u'' + u' = A\cos(u - Wt) + B\cos u - F$. With $B = 1$ and $u = \pi/2 - L$ it becomes the needle's $m\,L'' + L' + \sin L = F - A\sin(L + Wt)$: the same ball on the same washboard, with the angle measured from a different mark. So the chalkboard's weight 2 is the needle's memory 2. Three numbers that answer three different questions can then sit side by side without contradiction. The table sets them out, with the exact answer to the second question, and the textbook's estimate for heavy balls, beside them.

Question	Number	Where it comes from
Where a ball already rolling is caught again, read on a slow ramp down	0.773	The Loop Atlas’s run
Where a ball already rolling keeps rolling, with no ripple at all	0.78 to 0.80	R2W’s single-ball run
The same, computed exactly	0.7899	This guide’s check (row 711)
The heavy-swinger estimate, 4/(π√m), good only for heavy balls	0.900	The textbook
Where a ball set still half a turn round gets going	0.86 to 0.87	The chalkboard’s floor

In a junction’s terms the weight is the Stewart–McCumber number, and with the ripple off a memory band first opens at weight 0.7026, computed for this guide (row 711).

The triple star’s angle, as a picture. Marking the passage of the triple-star thread’s paper on the pair’s eccentricity and the reading that bends it, Bear asked whether the bend is the pair lining up with the third star, its tug growing stronger when the star passes in their direction, twice in each turn, since either star of the pair can be the one in front. It is. The star pulls harder on the nearer star of the pair, and the pair looks the same after half a turn, so the stretch comes round twice a turn, and where the star finds the pair in that rhythm is φ_N itself. The paper now says so in a plain sentence after the passage. One question was left open when the thread closed on 29 October: whether his picture is the same bend that a test called GK1 measured once a pass at the star’s closest approach, at 1.116, 1.117 and 1.115 of Georgakarakos’s prediction as the triple-star thread’s paper gives it (run 20261018_193951).

Three timings, and one word kept narrow. The **outside tick** is set from outside and never feels the rider: a passing of the ripple, a star’s closest approach, a drive’s period, and in the world a lab source, a transmitter or an orbit. The **rider’s own pace** is set by its slope and its drag. A lock is the rider’s own pace giving in to the outside tick where the two are already close. The Loop Atlas met a third timing, the **event**, which opens a gate when something arrives. A bead with a blade, in an even soup of waves, drifts 0.009 when the blade is switched by an outside tick, no more than its own scatter of 0.008. Switched by listening, on when the push points the way it wants to go, it drifts 0.124, though a later toy that billed the charge showed that drift to be a free lunch the first toy had not billed for (row 590). Bear’s words for the contrast, given before it was measured: “event timing like the jellyfish instead of master clock timing.” Two clocks that each keep time by their own events and are tied to each other are **coupled clocks**: they come home together after the denominators of the continued fraction of their ratio, and a lock is where they agree (row 719). “Two clocks” keeps only this narrow sense in this guide’s vocabulary.

The fork that decides most of what follows: does the system keep area, or lose it? The triple star keeps area. Nothing settles, every start keeps its own orbit for good, and islands sit in a chaotic sea. At a lock’s clean edge the time to slip grows like a pendulum’s logarithm, over the ladder as a whole: 7.57 and 7.37 passes per decade of distance at two clean rims, against 7.29 and 7.83 forecast from the island’s own width. Step by step it departs: 5 to 11% short far out, a quarter steep nearest the edge, and 1.76 times beside a chaotic layer (row 780). The doubling constant of its class is 8.72. R2W’s rider loses energy to its honey. It settles onto attractors, and the start matters only where two attractors coexist (the memory band, 0.2112 of the cells at weight 2). The time to slip at a lock’s birth grows like an inverse square root, and the doubling constant of its class is 4.669. A lightly leaky map reads the frictionless constant for its first doublings and then crosses over, once one trip round its orbit keeps less than about half its area (0.440 to 0.524 across five leaks) (row 680); read later across fifteen dampings, the departure falls on one curve of the area kept, its 10 per cent crossing at a median of 0.4905 (KN1 and KN2, run 20261016_201627; section 9). So the first question about any edge is which kind it has: a chaotic layer, or a band where two starts end differently.

A rhyme, not a test. The three-body problem without friction is reversible in time, as R2W’s toy is at zero weight. The triple star’s rope check has two doors that are mirror images in φ_N (-0.140 and $+0.141$ to $+0.155$), and its bands mirror each other. Whether that is the same kind of mirror as the toy’s is a question for the triple-star sessions’ own pictures, under their own rules, and it was not taken up before the thread closed on 29 October.

Figure needed: the triple star’s rope check, its two doors mirrored in φ_N.

Three readings, and they are what the eye is good at. The **count** is turns per tick, the winding number p/q, which the eye sees as bins. The **corner** is where a count changes, which the eye sees as a bend. The **leftover** is what a loop keeps when it comes home, which the eye sees as an offset. The Loop Atlas’s spine is the third: a loop comes home to where it started, so what did it leave behind? Each reading has its rule, and the rules follow in that order.

Figure needed: one strobe picture with a count, a corner and a leftover marked on it.

The count: nearest returns come in a ladder, and its neighbour is a theorem. A steady rotation seen once per tick returns nearest to where it started after the denominators of its continued fraction. This is the best-approximation property of the convergents, and when the partial quotients are 1, each return time is the sum of the two before. That is the ladder the triple star's families climb: 19, 28, 47, 75 and 122 passes. Its neighbour is the three-distance theorem, conjectured by Steinhaus and first proved by Sós (1958), Surányi (1958) and Świerczkowski (1959): the first N points of a rotation leave at most three gap lengths between them, the largest the sum of the other two. That is what the white patches between the threads would test. Say which of the two a picture shows.

The count as a rate: a rate says how it was read. The triple star's φ_N slides -0.1254461 of a turn per pass at one birth, read end to end. Read by a fitted line, or by the median step, it is -0.11175 or -0.10572. All three are right. The angle's single steps fall in three bands, the slide itself and the tide's flips of about half a turn either way, and each way of reading weighs the bands differently. The flips come mostly where the pair's orbit is nearly round as the star passes: 87 per cent of the steps off the slide come where its eccentricity reads below 0.03, against 38 per cent of all steps. A rate quoted for an angle that does not slide steadily says how it was read, and where its steps come from (row 558).

The corner: births come two ways. By a fold, a lock opens as a square root, and a slow ghost is visible before it is born: riders in flight crowd toward the place where it will appear (row 654). By a flip, a dance doubles, its multiplier passing -1, and the ratios of the doubling gaps close on a constant set by the shape of the top: 4.669 for a rounded top, 7.2847 for a quartic, and 8.721 for a map that keeps area (row 680). Near either, nothing settles in time to be read, so follow the orbit and read the sign of its multiplier (row 642). A window can open and close by the same road, a cascade at each end (row 697).

Two ladders, one finish. When two systems end at one number, compare how they come to it. In R2W the 1/2 groove's doubling ladder reaches Feigenbaum's ratio in three steps; the 3/7 groove's starts far lower (2.41) and takes six. Near the end the two close in the same way, each gap to 4.669 about a quarter of the last (0.210, 0.054, 0.012 and 0.218, 0.050, 0.012). Universality is in the approach as well as in the limit (row 680), and one toy's road can forecast another's: R2W's rule, registered as a forecast of the needle's fourth ratio, gave 4.6585 and 4.6591, and the needle gave 4.6599 and 4.6604, then a fifth, forecast at 4.6672 and 4.6673, of 4.6673 from above and 4.6672 from below (row 872).

Figure needed: the ghost before a fold, riders in flight crowding toward the place where the lock will appear.

The leftover: a loop comes home turned, never moved. That holds for a body with nothing to push on. The Loop Atlas book's falling cat and its chair come home turned, never moved, while a swimmer pushes on its water and does come home moved (rows 242 and 630). The projects meet this one wall from two sides: the book's spine, what a loop leaves behind, and its "a ship cannot tow itself"; R2W put the same momentum wall on page one.

Where the shared method stops: it rediscovers known physics quickly. The triple-star sessions rebuilt Mardling's harmonic from the inside before reading her paper (R. A. Mardling, 2013), found the ladder of nearest returns, and met Arnold's resonance of order three, which predicted before the run that an island's centre would rock across a third as the star's eccentricity grew; it did, between the star's eccentricities 0.325 and 0.330 (probe 31, run 20261007_151231). R2W met Shapiro-like steps, saddle-node bottlenecks and Stewart and McCumber's hysteresis. It then found its whole toy in print: a particle in two waves, one standing and one moving, which without the honey and the slope had been built with an electron beam in a travelling-wave tube, where a devil's staircase appeared as the second wave was turned up (F. Doveil, A. Macor and Y. Elskens, 2006). The Loop Atlas met Feigenbaum's ratio and square-root births. Each match felt like a discovery for an hour, and each was a check that the toys work. Finding it already named is the normal outcome, and it is credited as such. Not every reading ended at a name: the ratio by which each doubling turns the last one's corner over, -1.9644458 for a rounded top (row 877), and the cause of the odd-order errors in a published table of doubling constants, a sign slip, after McCartney (2020) had found the errors (row 874), were found named nowhere in a limited pass, each a pointer for a specialist. Once the whole toy is named, its map becomes a bench: every landmark is a known length to check a ruler against, and any claim of something new moves from the physics to the reading (row 726).

A part can be a textbook object too. The maypole's clockwork was the inscribed-angle theorem (section 7). In R2W, five groups of three at 12 and 36 degrees swing twice the golden ratio and twice its inverse of one group (run FIFTEEN-1): five equal arrows, each turned 36 or 108 degrees from the last, which is the textbook pentagon array factor. Look at a measured ratio for a known shape before calling it new.

The method reads what a picture shows; it does not decide what physics allows. One project hoped a rider's motion might move a craft that pushes on nothing. No picture could rescue it: momentum is conserved, a wall no reading could have shown, and the method was the part of that work that still stood.

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9 — The scorecard

This is the score sheet. Each claim made by eye was written down before its numbers existed and then judged on the numbers; the tables show which claims held, which missed, and whether a miss lay in where a thing sits and which way it points, or in how big it is.

The scorecard holds eye claims registered before the numbers existed, judged on the numbers, against shuffles. It is kept by the kind of claim, because the readers are good at different kinds. It is scored in separate columns for sign and place, for size, and for which members of a set stand out, because a claim can be right in one and wrong in another (row 712).

The tables come in three groups, by how the read was made: the second reader's blind reads; the Loop Atlas's predictions, judged against answers fixed in advance; and marks made on figures, most of them Bear's, registered in his words before they were tested. What the columns add up to comes last.

Blind reads from pixels. R2W's reads of the triple-star pictures, judged by the triple-star sessions on their data.

Claim	Kind	Sign and place	Size	Verdict
The weave: dots link 5 passes apart, and shun gaps of 1 and 3	Geometry	Right: +23.4 standard deviations on the data; gaps 1 and 3 at −3.9 and −4.3	Stronger on the data than on the page (+12.4 from the pixels)	Borne out. Gaps 2 and 6 shun as hard (−4.6 and −4.3) and gap 14 stands tall (+9.6): the read named the tallest gap rightly and two of several short ones
The ripple: the band's lopsidedness flips sign while its centre holds	Shape	Right: −0.18 to +0.29; the centre held, 0.04893 to 0.04911	Overstated by the pixels (−0.49 to +0.47)	Borne out on the sign
The zoom-out: 8 crests about 681 passes apart	Count	Right: 8 crests	Median gap 686 (666.7 was a bin of the spectrum, row 318)	Borne out

The first row is also R2W's registered test of the weave against a shuffle. Four more of its tests were registered for the triple-star data, never reached that room in their exact wording, and were not run before the thread closed on 29 October: Morse's count of critical points on each orbit's own torus; the wave's shrink and its slide; three gap sizes, in phase; and folds and cusps.

Figure needed: the weave and the zoom-out pictures that R2W read blind (the band is the one asked for in section 1).

R2W's blind reads in one sitting, scored. The reading order for this sitting, set at Bear's suggestion that the unmarked pictures be read first as a self-test, was: check provenance (every picture must match a file of record, and anything already seen is declared; three fragments found by a text search had been seen); read blind, writing a blind read of each unmarked picture before opening the next; check the pattern; read his marks, and the earlier sitting's reads of the same pictures; score.

On the weight 0.26 map the blind read hit the upper stop, the S on the lower stop and the conveyor. It was partial on his two curves into the ring, stopping where the dots crowd while he followed both into the ring, and where he circled three populations it saw two bands. It missed his "maybe" stroke, which was a second S, and his steep curve from 1.452. It was wrong on a "period-3 dance" and on "in flight". It was right, where he made no mark, on the empty middle and the ghost at the ring. A grey stripe at 1.4535 stays open.

On the bend figure the crest counts by eye were wrong (23 against 37, 38 against 43), while the order and the decay were right; the woven strings in the outside readings were missed. On the triple star's rope check, read blind by eye and then graded by pixels, one reading was right, one mostly right with a knot misplaced, one refined into "cells" like sausage links, and one wrong; the read missed the mirror and the straight middle lines that his marks showed.

In sum, the blind reader missed his second S, his steep curve, the woven strings, the rope check's mirror and the straight middle lines. It saw, where he made no mark, the empty middle and the ghost at the ring. And it saw the symmetry only after he pointed, when it took minutes to measure and became the night's result.

Bear's marks on the weight 0.26 map are Figure 19. Figure needed: the blind read's marks on the same map, beside them.

Claude's own reads by eye, registered the same way. In the triple-star sessions Claude's calls made by eye were registered as Bear's marks were, before anything was computed, and withdrawn as plainly when they failed.

Read	Kind	Result
From the bench’s cascade ledger (TRIPLE_CASCADE_CROSSOVER_LEDGER_20261007_051639): the damped cascade’s departure from the frictionless ratio is a function of one quantity alone, the area an orbit keeps over one trip, with its 10 per cent crossing near 0.49	Shape and place	Borne out: one curve across fifteen dampings (KN1), and a median crossing of 0.4905 (KN2; triple_friction_knob v1.1, run 20261016_201627). It became the bench’s dial in the triple-star thread’s paper; R2W’s note reads it as γT near 0.71, a derivation not yet tested on its own map
A shape at the tenth, called textbook	Shape	Withdrawn: its cause turned out to be the record’s length
A rule that each closest-return record is built from earlier ones (LD1b)	A rule for records	Not borne out: it held for 1 of 3 records, and 21 of 200 random pairs of rates obeyed it (row 616)
A stall called unusually slow (RW1)	Rarity	Not borne out: 79 of 200 random pairs matched or beat it, so the stall was ordinary (row 616)

Predictions against known answers. The Loop Atlas’s registrations about its own pictures, each judged against an answer fixed in advance.

Registration	Kind	Result
The lattice reader’s first version, registered to see an eight-step dance as eight strings	Count	1 of 9: the fidelity check held; at holes half a degree wide the eight strands came as four tight pairs, so three predictions built on eight strings missed and four could not be read, and one more, on the fold’s ghost, missed (run 20261009_092420; row 703)
The second version, families from the side histogram’s filled stretches, read on holes of 10^{-4} radian	Count, place and shape	8 of 10: the counter, the rate, the square-root opening, the change of change of slope, the cloud test and three Newton checks held; the history missed by one column, and one band read 32 runs in only 81.2 per cent of its columns (run 20261009_093040; row 703)
Lifted into speed, the strands per column equal the distinct points, and the touching strands are apart in speed (three predictions and their two finer follow-ons)	Depth	0 of 5: the touch was a fork’s young daughters, and the reference count was wrong by one in places
Two lobes of opposite sign lie either side of each doubling, their midpoint within a column of it	Shape and place	2 of 2
The crossing between the lobes sits on the doubling registered for it, the sixteen-to-thirty-two (Z2)	Place	1 of 1: within 0.0062 of a column, against a bar of 0.05; printed beside it and not judged, the doublings at columns 817 and 262 within 0.0031 and 0.0014, and the one at column 116, inside the stencil’s reach, 0.08 high
The lobes are the echo of a corner	Shape	1 of 2: the made-up corner crosses zero on each doubling and matches to 0.993 and better; a signed correlation, registered where the corner’s sign was free, missed as written

Bear’s marks on the triple-star figures, registered in his words and tested. These are his marks on the triple-star figures, from the bend figures and the blind figure to the maypole, the dead zones, the squeezed strobe and the pictures of 15 and 16 October.

Mark	Kind	Result
The bend is woven from two phases	Shape	Borne out: the fit went from 0.4302 to

		0.9723 with the second angle
Two angles make a torus	Geometry	Missed by a hair on its first run (the neighbour ratio 0.1126); borne out over 6,000 passes (0.0603)
Walls where the visits pile up	Geometry	Under its registered bar. Beyond 2,000 random sets (0.778 against never below 0.856), but that comparison was chosen after the run and is claimed as nothing; the test registered for the next birth has no recorded result at the thread's close (row 313)
Rows that arch below the waist and lift above it	Shape	Dot by dot, a coin's shares (0.527 and 0.488), so an empty pass; on cluster centres, missed by 0.015; fitted whole, missed by one row (6 of 7 arch, 6 of 8 lift)
Arrows along the first angle	Direction	Borne out: a median of 8.1 degrees from the axis, his direction "ambivalent" by his own word
Gold lanes of empty space	Negative space	Not borne out, because the sensors could not see them: the test measured the wrong thing
p/q crossings at a blind birth	Count	No reading: he drew none
His clockwork on the maypole: "24-gon? Skipping 4 vertices for each line like clockwork."	Count	Borne out: the median step about the circle's own centre 0.2154 turn against 5/24 = 0.2083, and 0.2579 in shuffled order. It is the inscribed-angle theorem: the slides step about the centre at twice their step about the pole, −0.2112 against −0.2114
The jumps, read on his clock	Direction	Borne out on one test and missed on the other: the jumps run a little off the clock, up-jumps landing late and down-jumps early, and C24d held at 7.190 and 7.195; T3 missed as registered, holding at 13 of its 15 births and missing only where down-jumps were three and twelve (rows 544 and 644)
His lower dead zone, on the 600-pass runs that followed his figure, bands as drawn	Negative space	Borne out: 3 of 484 runs inside, against 64 just below and 11 just above
The same, as a rate (Claude's rescaling)	Negative space	Not borne out: 10 inside, 2 below, 28 above; the rescaling moved the held runs too
His upper dead zone separates broken pairs, on the later 600-pass runs from the mean-ellipse start	Negative space	Borne out: 1,462 of 1,475 rows agree; all 12 broken pairs above it; 13 healthy runs also above it, four at exactly two thirds of a turn per pass
His bilateral symmetry on the squeezed strobe (SQ1)	Geometry	Passed at both scales, but empty by construction (row 337): over passes 601 to 6,000, 40 of 40 outlines centred on zero within a fifth of their width, against a bar of at least 80% (three outlines wrap the whole turn and are symmetric by construction, and the island's centre is $\varphi_N = 0$ because the runs start there)
His pinch (SQ2)	Place	Borne out: registered at 7.465 to 7.505 for the long view and 7.520 to 7.550 for the short; the narrowest long outline at 7.505 (0.147 turn; 7.470 to 7.490 at 0.173 to 0.188), the narrowest short one at 7.525 (0.103; 7.530 and 7.535 next). The long view's narrowest is one run, a single

		sample of the chaos
The waist moves (SQ3, Claude’s, from his two arrows)	Place	Borne out exactly at the bar: the long view’s narrowest birth 0.020 below the short view’s, against at least 0.02
A very regular pattern inside the centre, traced as a swell	Shape	Borne out: his red line has 37 highs 53.5 passes apart, the beat (BC0, registered before it was measured; run 20261015_133950; row 581)
“The ratio stays fixed, the rates don’t”	Relation	Not borne out over 600 passes, holding at 3 of 8 slices, the misses showing where the beat passes through zero (RF1); borne out as registered over 2,400 passes (SM1; runs 20261015_140219 and 20261015_151629; row 581)
Fractals within fractals, read as islands round islands	Shape	Partly: the chains are there, 121 locked runs within 10^{-4} of a simple fraction against 34.5 at decoys, but the falling law is barely seen, and a parity appeared instead (E4 and E6, registered before anything ran; runs 20261015_225444 and 20261015_230449; row 581)
His marks on a sheet of plain figures	Place	Measured, not scored: his circle sits on a lock at a seventh, found without its label (run 20261016_054242; row 581)
“5 twists around 4” on the egg figure	Count	Confirmed and tested; his other readings of that figure were translated into test forms and not confirmed (row 782)

How the dead zones and the squeeze reads were scored. The dead zones were scored on data he never drew on (triple_dead_zone_check, run 20261012_062504). The three squeeze reads drawn from his marks (SQ1 to SQ3, the third Claude’s, from his two arrows) were registered before anything was measured, and scored by triple_strobe_views (run 20261014_072942). A run’s outline is the 2nd and 98th percentiles of its φ_N, with the 5th and 95th as a second scale, and a verdict holds only if both scales agree. Of the three reads made by squeezing and tilting and registered before the numbers, two held as registered at both scales, the pinch and the moving waist, with a caution: the pinch’s narrowest long outline is one run, a single sample of the chaos, and the waist moved exactly as far as its bar, measured from that same run. The symmetry was centred by construction, because the runs start at the island’s centre, so the outlines could not fail its bar and its pass is empty (row 337). The one read made on the tilted view, the V, did not survive a grid five times finer (the next table). Here the eye proposed and the ruler decided.

The maypole disc with his marks is Figure 21; his dead zones, on the figure he drew them on, Figure 22; the squeezed strobe, short view beside long, with his pinch and his two arrows, Figure 16.

Reads on a finer grid and over longer runs. A run of the 0.25 island five times finer than before (a birth every 0.001) for 6,000 passes, and two follow-on runs after it, tested reads made by eye on coarser or shorter pictures, Bear’s and Claude’s alike, each registered before its numbers were computed.

Read	Where it was made	What the finer or longer data said	Verdict
The V: the long runs' core narrowing in a straight line to a pinch	The tilted strobe view, a birth every 0.005	At 0.001 each run's width follows its own fate, narrow where it held and wide where it did not; neither a V nor a floor fits better than a smooth curve	Not borne out: an outline of the coarse sampling
The offset: the room that keeps its stars lies below the early centre	The squeezed view, short beside long	The narrowest early swings at 7.524 to 7.531; 22 of the 24 births that hold lie at 7.505 or below	Borne out again
Neighbours alternate at the sealing edge	The fine run's window as it ran	Fates speckle between neighbours across the whole grid; but no 0.005 of birth holds more than half its runs, and the escapes stop sharply at 7.472	The alternation seen everywhere, as speckle; the registered climb in the held share not borne out: the edge to holding is no edge
The walls sit where the rocking crosses a ninth	The flanks of the coarse walls	The edge lies 0.001 from a ninth, but the rocking sits near a ninth almost everywhere	Borne out, weakly
A chain of small islands locked at a ninth in the wall	R2W's rhyme	No birth in the band reads at a ninth	Not borne out; a finer look is registered
The door is a corridor: stars cling at its inner end, roam at its outer end, and run the wrong way	A figure of the triple-star thread's paper, at one eccentricity (Bear: "Agreed on edge or ledge")	The wrong way: 17 of 18 chaotic stars below the island that circulate turn the far side's way, at three eccentricities, and all 13 that escaped from below were turning that way. The ledge: the clinging stars are scattered through the layer	The direction borne out; the ledge not
The shark's fin: the apse turn peaks in a point at the island's upper edge	Bear's green trace on another figure of that paper	Proposed tests at first, then tested on nine lines	The next row
The shark's fin, tested: the tip at the hold's edge, and the fin's finer form	The same trace, read on one line of that figure; tested on nine (runs 20261017_030927 and 20261017_032258)	The tip's place at the hold's edge held on all nine lines (FN1). The finer form did not: the return to free within 0.050, gentler than the climb, held on four of nine (FN2), the two-sided logarithmic point on one (FN3), the wake on six (FN4). On a grid five times finer the two-sided logarithm went away, the split seen at 0.005 in birth being the spacing's (FR1), and a ragged corner showed instead: 7.358 held between unheld 7.357 and 7.359, the fastest turn 0.017237 at 7.356	The place borne out; the outline mostly not

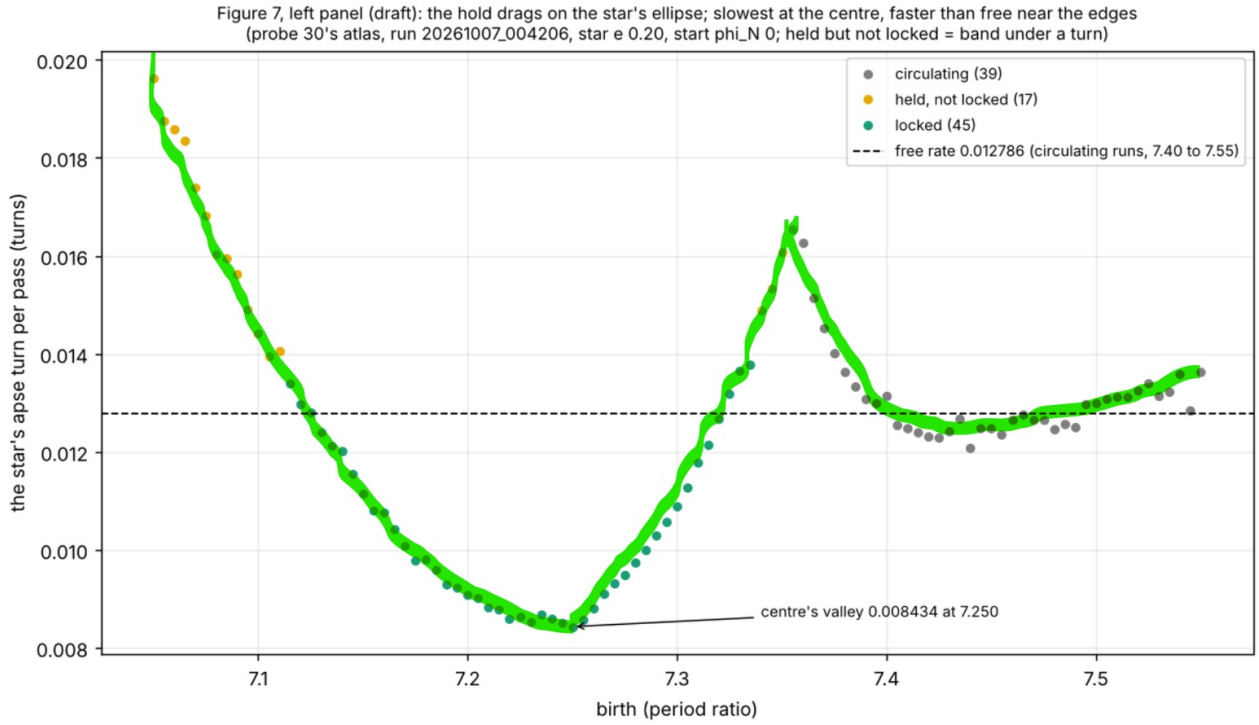


Figure 23. The shark's fin, as Bear traced it. His green trace on a draft of Figure 7 of *Held, Rocking, Locked*, left panel: the star's apse turn per pass against birth (probe 30's atlas, run 20261007_004206, the star's eccentricity 0.20), slowest at the centre's valley, 0.008434 at 7.250, and peaking in a point near 7.355, at the hold's upper edge.

Figure needed: the fin on its nine lines with the tip at the hold's edge marked, beside the ragged corner from 7.356 to 7.359 on the finer grid.

Relations held better than shapes. Reads that named where one thing sits against another, which way something turns, or which side something is on held here more often than reads that named a shape (the V, the floor, the ledge). These are few cases: a guess, not a rule, and a question to carry to the next picture.

Relation beats shape, tested blind. The guess, Bear's own sense and the triple-star thread's, was then registered as RC1 before anything was counted: reads naming a relation should be borne out more often than reads naming a shape, by at least 0.2 in share, with at least eight reads in each class. It was blind by construction (run 20261017_040300). One agent listed the 74 registered picture-reads, most of them Bear's, with their verdicts, and a second, given only the reads' wording with every result word removed (21 quotes had carried one and were reworded), sorted them into relation, shape or neither. Relation reads were borne out 19 of 30 (0.633) and shape reads 11 of 28 (0.393): +0.240 against a bar of +0.2, a narrow pass. The sorter's note that about seven shapes might move to neither on a stricter reading was reported, and the count was left as registered. The shark's fin on nine lines (the table above) shows the same lesson from the other side: where the fin sits was a relation and held, and its outline was a shape and mostly did not. For reading with Bear, trust where his eye puts things before the outline it draws. Claude's own reads by eye showed the same from the other side: they held on mechanisms and failed on rarity and on rules for records, better at where and how things move than at how rare they are (row 616).

Figure needed: the tilted strobe view at a birth every 0.005, beside the same island at a birth every 0.001.

Bear's marks in one Loop Atlas session, registered in his words and tested. His marks on figures, his names and his predictions, each registered before its run and judged on the numbers, in the columns of the table of his triple-star marks above.

Mark or prediction	Kind	Result
His rings on a dip and on four flickers in the Loop Atlas’s spread panel, with a question about gaps	Place	Borne out: both were real, a window and a ladder’s rungs; the ruler of record saw the window and not the ladder
Rungs, his name for the Loop Atlas’s ladder	Shape	Borne out: evenly spaced on a log ruler to within a quarter (registered as B1)
Event timing more than the water’s memory, for the Loop Atlas’s tied bells	Size, as a ranking	Borne out: a swing of 0.63 across the period against 0.25 across the memory in the toy as first built, and 0.6381 against 0.2671 with the rings’ cores given their own size (run BELL_SOFT_MAP 20261021_000736; row 807)
The mass-1 passing lane fills the road (a sentence of Claude’s, marked by Bear)	Extent	Held to 1.09; missed from 1.10, past the map it was read from
A power gauge: the worn rider works harder	Sign	Borne out: the effort rises 3.05 times before the fall
A true rate sensor in the skater’s ear (his trade’s knob)	Size	Borne out: it recovers 88 per cent of the drift lost to looking every quarter cycle, against 45 per cent for a slope from the ear’s own two looks (run SLOPE_EAR 20261015_190935)
A master oscillator keeps the count	Sign and place	Partly: only if loud, and only at or below the bead’s working pace, three quarters of its tune (run BEAT_SWING 20261015_204207)
His pause ear, left, pause, right, pause, with a dead zone	Shape	Borne out as a knob: the ear latches, its push per switch rises at every widening, and its price never turns over (runs PAUSE_BEST 20261015_220655 and PAUSE_WIDE 20261015_223744)
The dive dream, its four claims	Several	Three borne out in the toy (gravity’s clock; no air needed for the turn; the mat for the shape part); one waiting (count or event, the Loop Atlas’s card 199)

What the column says, for the Loop Atlas. Bear’s marks on figures were right on place both times in one Loop Atlas session. On 14 October, on a window of the needle, his marks were right on place three more times: his red and green strokes on a density of chaos lie within 0.004 rad of the fold or its first eight images (0.741 and 0.673 of their pixels, against 0.36 and 0.37 set at random; read with a tool, its expectation printed before the run), his black arc is the fold’s third image, the curve that ends the window (0.604 of its pixels within 3×10^{-4} rad, against 0.076 at random; registered before its run), and his rule for the windows’ cycles, read off a drawing, gave $12 = 4 \times 3$ and $20 = 4 \times 5$ in one piece and $24 = 4 \times 2 \times 3$ in two, as measured (rows 707 and 705). His predictions and his names held on sign, and the one that held only in part, the master oscillator, held wherever the bead’s working pace allowed it. Where a marked sentence missed, the miss was in its extent, past the map it was read from, and that miss belonged to the sentence’s writer, Claude. Claude’s one reading of a picture in the same days, the clean lock, missed because its ruler could not see the thing that was there.

Figure needed: the Loop Atlas’s spread panel with his rings on the dip and on the four flickers.

His remarks that became exact. In one thread of R2W Bear mostly looked at one figure, the self-mirrored dances of each fraction at its weightless root, and said what he saw. Each remark was turned into a registered test or a counted rule, and most became something exact: one failed as worded and became exact in another form, and two are still open.

His words	What it became	How it was checked
“This is order, not chaos … why do you think it behaves so precisely?”	The order in layers: forced by the Möbius map and the mirror, measured, argued, open	The thread’s seed document, and the headline section of R2W’s paper
“it has a repulsion quality to it”	Red and purple strictly alternate around the circle; two dances of a rotation in disguise cannot pass each other	READ-E1, nineteen rows
“counting left to right and if we round up, the red dots always match the denominator”	q stops per dance; his half dot is the seam at ±0.5; the numerator is how many stops the ball skips back each passing	READ-E, all 18 dances of the first nine rows
“the biggest jumps happen when nobody is dancing in the cent[re]”	Failed as worded at 3/5; became the window count q − p − 1 (fifteen of fifteen after the data, four of four predicted)	READ-E2, PIN7-1g, PIN68-1w
“stretched out grippiness at bottom”	Read as the bottom rows: the ball slips most of a turn each passing, and its only grip is the pair, 21 to 28 times slower than any other stop outside	READ-E’s strobe speeds; which reading he meant is open
“Does the dot coming around the circle change its frame?”	The loop’s multiplier is its holonomy on a neighbouring step: exactly 1 without weight, about 0.95 for red and 1.02 for purple at weight 0.01; the passing matrix to the q-th power is (−1)^p	Algebra, and Newton’s multipliers
“that record player is still a read head in my brain”	The drive as the radius, each groove a fraction, played in the order of the fraction tree	TONGUE-MAP, in an earlier sitting

His marks on R2W’s other figures were measured too. On a coin map at weight 0.40 his long lower fin, an echo of the main arc’s lower edge, was registered and run at 35,000 riders and ran parallel at 31 of 31 drives (COIN-3; row 706); his dipole-and-mirror picture held in kind and his staging ground in every arrangement measured (row 128); and of his two pencil lines on figure AA, read after the data and not registered, one sits exactly on the gear break and one is unexplained (row 573).

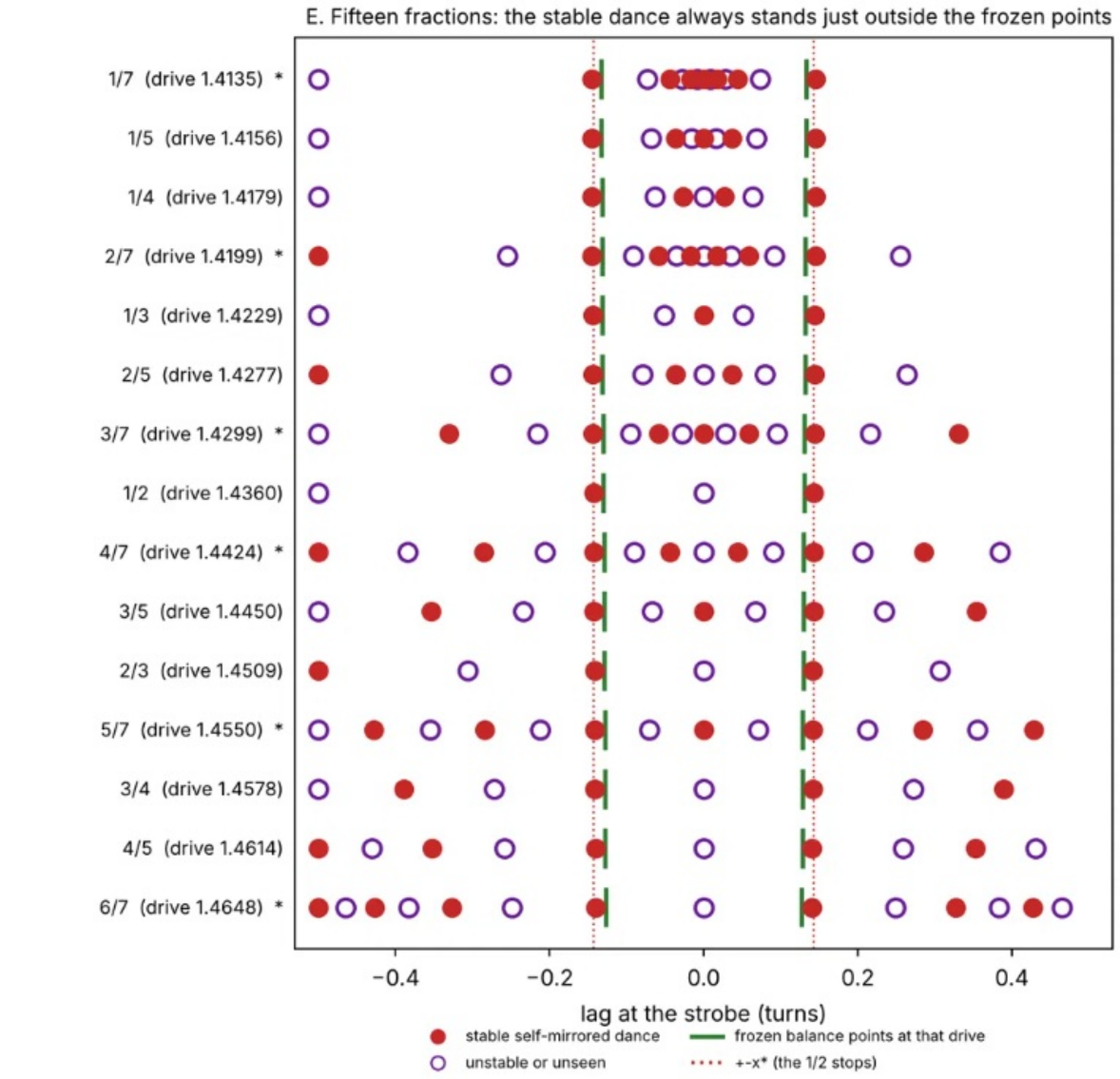


Figure 24. Fifteen fractions at their weightless roots. R2W's figure of its toy's self-mirrored dances, from 1/7 to 6/7, each at the drive of its weightless root (1/2 at 1.4360): red the stable dance, purple the other, green ticks the frozen balance points. Bear mostly looked at this figure in one thread of R2W, and the table above gives what each of his remarks became.

What the columns say. An eye, either reader's, proposes sign and place well, and the size belongs to the data. Claude's blind reads were good on geometry and weak on classes and on depth: they named the conveyor as the rest, and bet on an overpass that was a meeting. Bear's misses here were in the translation and in the bar, not in the seeing: lanes that no sensor could see, walls that beat 2,000 random sets on a comparison chosen after the run and fell under a strict bar, rows that missed by 0.015 and by one row. And the column of predictions against known answers is the one to trust most, because every row in it has a known answer.

Why sign and size are scored apart. The two sides of the sign-and-size lesson were reached on the same day from opposite directions. One project found the eye right on sign and wrong on size; the other registered a sign that was free and missed on it. That is why sign and size are scored apart.

10 — How to use it, and how not to

Reading pictures is a fast way to find claims and a poor way to prove them. Here are the rules for using it, the ways it goes wrong, and how to tell whether it worked.

The rules come in two lists, how to use the method and how not to, and end with its measure of success.

Use it as a generator of claims, never as the test. Every claim from either reader gets a number, from the data or from the picture's calibrated pixels, before it is believed.

Use it in order, the order of section 3.

Use it first on a picture whose answers are known. A method that has never been scored is a story.

Use the two cheapest rulers every time. Does the pattern survive the same dots shuffled? Does the ruler draw the same shape from a made-up input? The cheapest test is often the strongest (row 584).

Use it as a conversation between two kinds of eye. What both readers saw counts most. What only one saw becomes a test. When the person beside you reads a picture differently, write both readings down before either of you looks at the other's.

Use it at two scales and through two pictures. One scale and one picture are a single draw.

Use the marks themselves, not a description of them. A paper that describes the eye's way of guessing in words, without one mark shown, hides the method it describes. The triple-star thread's paper now sets its Figure 7 beside Bear's own working copy of the same strobe picture, squeezed sideways, tilted to a grazing angle and marked by eye on 14 October before anything was measured (triple_paper_figure7_marks v1.0, run 20261021_052614). His picture was taken by its fingerprint, 9c1034e2c9af4676, and left unchanged but for a working note lifted from its foot. Its readings had been registered as SQ1 to SQ3, scored in section 9, and VC1, the V, which was not borne out and was redesigned as the pinch grid. A picture he pasted later was matched pixel for pixel to his marks of 15 October on the ratio atlas (BEAR_RATIO_ATLAS_MARKED_2026-10-15.png, on run 20261015_140219) before it was read.

The don'ts are of three kinds, in this order: taking what the method finds as proof, reading a picture past what it can show, and bending the procedure.

Do not use it as evidence on its own. The eye proposes and the ruler decides. Claude's own eye invented a third hand in one bend and S-shaped rows in another picture, and the rulers took both away (rows 718 and 616).

Do not use it as evidence of new physics. When a shape matches a named result, the literature is the first stop, and the match is a check on the toy, not a discovery.

Do not let the eye speak for the model. Bear's rule for his own eye holds for every reader: "Don't let me speak on behalf of the geometry until the data bears witness."

Do not read a mechanism off a derived picture without its anatomy. Bear looked at a picture of twin lobes and saw vortices, propellers and mirrors. They were a corner. Taking the picture apart into a corner, a step and a bump showed something truer than either reading alone, and two of his five words had been right from the start (section 1).

Do not read someone else's picture without saying so. With a borrowed figure and no data, measure the picture, say that the numbers are the picture's, and keep the figure out of any public deposit.

Do not read across a wall no reading can cross. Read a picture's frame before its shapes.

Do not use it as a rulebook. The lessons are questions to ask quietly. In Bear's words, "Nothing about holonomy is square or harsh". A lesson that blocks a run is being misused.

Do not re-aim until something hits. Redesigning after a miss is right, and it needs registrations stamped before each run and the ruler check beside every verdict, or a loop of redesigns becomes a search for a hit.

Do not use a bar that cannot say no, and do not name a peak on a tie.

The measure of success. The shuffle and the made-up case are the reason this method can be trusted. Its measure of success is the one the lattice lesson set (section 2): not to find every shape, but to find a few new observations that connect to other observations.

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- One equation under three names: R. Adler, “A study of locking phenomena in oscillators”, Proceedings of the IRE 34 (1946) 351–357.
- Curves crossing in a narrow region: D. Vollhardt, “Characteristic crossing points in specific heat curves of correlated systems”, Physical Review Letters 78 (1997) 1307; arXiv:cond-mat/9609017. M. Greger, M. Kollar and D. Vollhardt, “Isosbestic points: how a narrow crossing region of curves determines their leading parameter dependence”, Physical Review B 87 (2013) 195140; arXiv:1212.4980.

Reference card for this part

This card maps each section to the rows in the register it calls on most, and says what those rows price.

Section	Key rows	What they price
1 Why read pictures	715, 707, 702, 711, 714, 640	The eye keeps lopsidedness, negative space, walls and long reach; counts before derived pictures; a derived picture with its anatomy and a made-up case; draw plain for marking; two scales
2 The two readers	716, 705, 710, 717, 686	Hold back the data while the eye learns; the eye defaults to a fork; blind reads are weakest on depth and class; the ruler’s own step
3 The protocol	713, 369, 573, 666, 616, 620, 686, 714, 30, 711, 703, 712, 128, 313, 337, 553	Name the picture; declare and read blind; translate marks with their maker; search the record; the control inside the tool; what the step allows; shuffle with the statistic chosen first; the made-up case; a second scale; sign before size; read the build back
4 The toolbox	642, 709, 710, 708, 654, 648, 718, 719, 537, 743	What each tool assumes and how it fooled its makers
5 The field key	702, 703, 705, 706, 707, 710, 711	What each shape turned out to be
6 Traps	329, 622, 210, 631, 416, 664, 704, 683	Binning is where the information design happens; two thresholds that disagree are data; a ceiling can belong to the window
7 Cases	702, 703, 706, 707, 708, 709, 710, 711, 667, 719, 556, 726, 201, 715	The control with known answers; three readers on one picture; blind reads judged across projects; the eye in its own frame
8 The common object	680, 642, 654, 670, 697, 719, 558, 726	Area kept or lost; the count, the corner and the leftover; the outside tick, the rider’s own pace and the event; births by fold and flip; a named toy’s map is a bench
9 The scorecard	712, 313, 337, 544, 644	Sign and place apart from size; both facts printed; a bar that cannot fail
10 How to use it	584, 506, 616, 711	The cheapest test; prior art after the run; the two cheapest rulers

The Register

Every ornament earned on every project. Use the station reference card to find the rows a given station calls on most. Use this section when a station raises a question the Walk cannot answer.

The headline register: what was achieved goes on the front page

The guide is good at catching a result that is wrong. Until version 7 it had no habit for keeping a result that is right, and the work behind this version showed what that costs. Bear counts it at about thirty-seven papers whose results never reached the abstract, and his ruling is to fix it from here on: a result that is not on the front page has, for any reader, not happened.

What went wrong

Wins had no channel. Among the markdown files the three projects sent the guide from 15 to 17 October were 18 lessons files, 17 files of no-physics reads, 8 of pattern-recognition lessons, 5 reviews, 3 ledgers of guide items and 2 records of misses. Not one was a record of results. A miss gets a numbered row in a register the guide reads line by line; a prediction that held gets a clause inside a row. In the build of 18 October the word “held” appeared 472 times, spread across hundreds of rows, and nowhere were those results gathered.

The register teaches the brake and not the accelerator. In the same build, 33 of its row titles carried the words of limiting or checking a claim, among them “Claim the connection, never the gap” (row 130), “The novelty ranking, corrected downward” (row 177), “Size the new claim yourself, before anyone else can” (row 280) and “Name the stricter claim a reader would want, and decline it” (row 347). Two titles stated a headline (rows 215 and 542), and none told a session to collect its results; rows 821, 840 and 882, added since, do. Each honesty row is right on its own; together, with nothing on the other side, they train every new session to shrink what it found.

Size checks net results away. A size check that weighs one result against a whole field (“medium for engineers”, “small, but important to a builder”) is honest about that result and silent about the sum. Twenty small results reported one at a time read as twenty small results; listed together they can be a capability no one else has. Bear’s picture for it is a disability rating: many separate awards that add to well over a hundred per cent, combined by a rule that reports far less. Handovers make it worse. A session’s handover carries forward its lessons, its misses and its open work, and its results only if someone wrote them down, so a result that lived in one exchange’s reply is gone by the next session.

A known road met mid-project became the road. The triple-star thread began from scratch on 29 September. By its lessons file of 10 October a published stability paper (Mardling, 2013) is cited six times, and by its treasure chest of 15 October the thread asks whether its own escape door “is just the known boundary toward the pair, or Mardling’s overlap”. Bear’s reading of the finished draft is that later sessions measured the toy in that paper’s terms instead of measuring that paper against the toy’s own results, so the paper came to read as a companion to work it had started independently of. The toymaking rule (row 506) already says to meet prior art at the wall, not at the idea; nothing told a session what to do once the wall was met.

The rule

Keep every honesty row. Add the other half: **every result that held is a headline candidate, written in the register the exchange it lands, sized against the right comparison, and counted, never netted.**

- **The right comparison is the nearest known method or result in the same niche, with numbers**, not the field at large. “Ours against known” is a pair: the known value and its source, our value and our method, and the margin between them.
- **A benchmark matched is a result.** Reproducing a known number with a new instrument validates the instrument, and it is said that way, with the number.
- **Count, do not net.** Small results stay listed one by one, and the register’s count is reported beside its largest entries.
- **When a known road appears, measure it against ours at the wall, and say where our road went that the known one never tried.** Do not switch roads.
- **The top of the register goes on the front page:** the abstract, the record’s description and the opening of the paper. A result on page 53 might as well be invisible.

Size both ways. A front page can be wrong in either direction, and the triple-star thread’s islands paper met both when its front was rewritten to its register. Plain sentences are where overclaiming sneaks in: on 21 October three of its new section openings were changed when read against their own subsections (Section 6’s ‘for many thousands’ was not measured there, ‘the clearest one we found’ was a ranking the section does not make, and Section 7’s opening named two rhythms before its first subsection introduced them), and cold reviews found 26 items to mend on the headline page, among them a call that the locks are exact and a fit called a prediction. A tool now lists every sentence that has a claim word and no number (claim_words v1.0, 9a02ea5631941e79: 135 of 1,354 sentences in the

test of 22 October), and a message in the conversation is a claim too: a reply's 'to six decimals' was really about four, and the paper says 8.721 and 4.669. The read cuts both ways: set against their sections on 21 October, the front's sentences came out eight trimmed and two strengthened. Underclaiming is an error too. The headline audit of 18 October found that the register had logged verdicts and never weights (each landed result's size check, how big and what is ours; row 868), collected 21 section headlines no one had gathered, and found the known road above at work: crediting Mardling had turned into framing through three of Claude's sentences, under no decision of Bear's, though the core was the thread's own by 8 October. The register's Part D now keeps weights both ways: what is the thread's own as far as the literature passes went, what is known and confirmed here, and the corrections to the size of headlines. And a count can be worded against its own numbers: Appendix D's 'Most guesses miss' stood against 208 landed and 194 missed, and became 'Many'. Read each new plain sentence against the section it opens, keep the weights beside the verdicts, and word every count by its numbers (the triple-star sessions)

The register a session keeps

One file per project, opened on the first day, carried in every handover and in the project's final crate to the guide. Its columns:

No.	What held, in one line	Run and date	Compared with (known method or result, and its source)	Ours against known (numbers)	Kind	Who it matters to	Front page?
example	Four tests tell a true nothing from two hands cancelling when a knob has no sign	ARRAY-1, NINE-1, LOBE-1, RATCHET-63, RATCHET-64, FLAT-75, FLAT-50	the reversal technique, which needs a knob with a sign	a symmetry's nothing at 6.994×10^{-15} against one rider's 2.599775×10^{-2}; a balance placed blind within 0.04% and 0.006%	method, extends a known one	anyone reading a null on a knob that cannot be flipped	the record's description

Kinds: first measurement; benchmark matched; benchmark beaten; known result reached by a new route; new connection between fields; engineering knob with its setting; tool. A row is written when a registered prediction lands or a result is read from data. It is never removed for being small, only marked superseded, with the run that superseded it.

At the close of a paper the abstract is drafted from the register's top rows before anything else is written, and the record's description lists them. The guide carries each finished project's register in a table of its own, with the project's concept DOI, so a reader of the guide can see what the method achieved and not only how it is done. All three projects had published when this version was built, and their tables follow the counts below; the two-voices papers, published after them, follow the three.

The three projects' registers at their close

Each project counted its register at its move to final publishing on 22 October and went on registering and counting after it, so each count here carries its date. At its move the triple-star thread had registered 444 predictions before their runs: 208 landed, 194 missed and 42 could not be read, and 31 of its 80 redesigns of earlier misses landed (counted by script over the whole thread, runs 20261019_062115 and 20261022_202112); at its paper's publication on 28 October it had registered 490: 236 landed, 211 missed and 43 could not be read, and 35 of its 92 redesigns landed (run 20261028_210006). At its move the Loop Atlas had judged 1,064 registered predictions, 712 held, 323 missed and 29 undecided, among them 990 distinct statements of which 668 held, its session G alone judging 109 of them, the count going from 955 to 1,064; at the close of its session I on 31 October it had judged 1,106, 736 held, 341 missed and 29 undecided, among them 1,031 distinct statements of which 691 held; and at its move to session K on 6 November, 1,143, 756 held, 358 missed and 29 undecided, among them 1,068 distinct statements of which 711 held. At its book's publication on 9 November it had judged 1,104, 706 held, 369 missed and 29 undecided: in its session K the register sorted every verdict line by its own tool's word for it, and 75 lines the tools call fidelity or rig checks moved from the predictions to the planted checks, which number 572, 555 held. Its 584 runs of record, 386 of which judged a prediction, are printed whole in its record, *The Loop Atlas: the record* (concept DOI 10.5281/zenodo.24100980), with its 656 misses, each with its reason. R2W's whole-record count of 19 October graded 402 tests against predictions written first: 331 landed and 71 missed outright; at its move its headline register held 202 landings, 62 misses or voids, and 39 misses redesigned into a landing. Its whole-record recount of 1 and 2 November first reproduced its count by eye of 24 October, which stops at 22 October (425 graded tests, 352 landed, 73 missed outright) before reading a later line, then found 515 graded tests to 2 November, 436 with at least one

prediction landed (214 with every prediction, 193 with some and 29 only through the named rival) and 79 missed outright, and its book states that count; at its move of 2 November its headline register holds 365 landings, 275 of them predictions written first, and 161 misses or voids, 87 of them redesigned into a landing; at its book's publication on 5 November it had graded 527 tests, 448 with at least one prediction landed (223 with every prediction, 195 with some and 30 only through the named rival) and 79 missed outright, by its counting kit's run on the published extract, and its register held 386 landings, 296 of them predictions written first, and 173 misses or voids, 95 of them redesigned into a landing. The three count different things, a prediction, a judged statement and a graded test, so they are not added; each is its project's own.

The table below is the triple-star thread's register at its close, for *Held, Rocking, Locked*, published on 28 October (concept DOI 10.5281/zenodo.23755473), which follows the islands of a body a millionth of either star's mass round a coplanar pair of equal stars at eccentricity 0.10 (scanned from 0.05 to 0.25), near the 7:1 and 8:1 resonances, a circumbinary planet's setting. It is drawn from the thread's headline register of 18 and 19 October and its closeout of 29 October; its count is given above.

What held	Ours against known	Kind	Run; rows
Mardling's resonance pendulum, tested piece by piece	her relation between the rocking rate and the island's width within 1.1 per cent at the pair's eccentricity 0.10, failing from 0.15; her widths 20 to 31 per cent too wide; the centre's rocking quickening 1.435 times from the pair's eccentricity 0.05 to 0.20, where her pendulum gives 1.98 (Mardling, 2013)	test of a published prediction	20261008_133322; rows 506, 678
A one-sided door past the stability limit	with the star's eccentricity at 0.20, 163 of 166 escapes leave the island on its side toward the pair, where the three published stability rules call the whole resonance unstable (one) or stable (two); started circular, none of 82 runs escapes	new in its details, as far as the passes went	20261016_182117, CS1 20261018_201130; rows 599, 863
Slow chaos under short islands	at the pair's eccentricity 0.25, 22 of the 28 births held over 600 passes lose their hold within 6,000, and every pair of twins parts exponentially; stickiness predicts such look-alikes	a known kind, measured here; method	20261013_152519, 20261014_090328; row 732
The edge's clock	the time to slip grows 7.57 and 7.37 passes per decade of distance at two clean rims, against 7.29 and 7.83 forecast from the island's own width with nothing fitted; a damped lock's square-root law would have grown the slip period 4.5 times over the span, and it grew 2.4	test; knob	triple_door2_edge_law 20261006_201311; row 675
A frictionless lock rests centred	every locked run rests within 1.1 degrees of centre over swings of 0.06 to 0.46 turns, where a damped lock's resting angle slides to a quarter turn at its edges	test; knob	row 675
The drag turns the ellipse back	the resonance holds back the turning of the star's orbit in a straight line with its grip (R^2 0.958); the centre's ellipse stands still at the star's eccentricity 0.2811, registered	test and new	rows 450, 661

	at 0.2817, and turns backward beyond it		
One doubling, then escape	island 7’s centre doubles once and every star round it escapes within 1,600 passes; the bench gives the area-keeping cascade’s 8.721082 on a known map (Greene, MacKay, Vivaldi and Feigenbaum, 1981) and the damped one’s 4.669197	new and test; benchmark matched	probe 32 20261007_224303, stage A 20261007_051341; row 680
What the arithmetic fakes	at a relative tolerance of 10^{-9} every unlocked rocking rate falls about 2.8×10^{-8} cycles a pass in every pass, faking a lock’s slide and part of its wear; 100 times tighter leaves 0.0093 of it, and the energy’s drift falls by the same powers per tenfold tightening as the creep left in the rates, 1.46 and 1.92 against 1.41 and about 2.0	method; a flag for a numerical analyst	FA0 20261018_031414, DR1 20261018_031349, EN1 20261028_140844; rows 741, 858
Verdicts on a second machine	the long runs made again from nothing on a laptop: of 103 lines, 83 the same, 20 the same verdict with other digits, none different	method	XC1 20261028_132326; row 810
Reads by eye	reads naming a relation held 19 times in 30 (0.633), reads naming a shape 11 in 28 (0.393): +0.240 against a bar of +0.2, a narrow pass	method	RC1 20261017_040300; rows 100, 714

The table below is R2W’s register at its close, for *A Rider in Two Waves* (*Different music, the same moves*), published on 5 November (concept DOI 10.5281/zenodo.23999853), which reads a damped pendulum pushed by a wave, and the standard model equation of a Josephson junction run on the same bench, with a stroboscope that counts. It is drawn from the book’s own list of what it achieved and its abstract; its count is given above. A dash in the last column marks a result no row of this guide yet tells.

What held	Ours against known	Kind	Run; rows
A law with nothing fitted for how a locked pair of dances answers an extra tone	the move $a \cdot \sin(2\pi x)/(2\pi\, n^2\, \omega^2)$, the tone’s size times a sine of the pair’s place, divided by 2π, by the harmonic number squared and by the drive’s frequency squared; the first-order theory, worked out after the moves were measured, reproduced all 56 of them (harmonics 1 to 8 at seven dances) within 0.00004 of a turn, and the tenth and twelfth harmonics, written down first, landed 14 of 14 within 0.000015	law; test	KERNEL-1, EAR-2, EAR-3; rows 699, 737
One gearbox in two media	each extra step of a dance multiplies its extra grip by the same factor, in the toy and in the junction: blind at 8 of 8 rows in the toy and 4 of 4 in the junction; one curve fitted on 28 rows predicted the stretch no run had visited in both within 0.017, and the toy’s bend within 0.0073 where the plain power	new connection between media	TOY-SIDE-3g, 4g, JJ-SMALL-1c, GAP-1, TOY-BEND-1, TOY-WALK-1; rows 79, 649, 742, 840

	law missed by 0.10 to 0.18		
A selection rule and its break, each predicted	a touch of weight keeps the dance that leaps across the far side (19 of 19 toy fractions, 44 junction roots, 48 junction settings); counting the radio wave’s quanta says one tone can never flip that choice and two can; the grip’s power read blind as 1.06 to 1.07, 1.95 to 1.98 and 2.78 to 2.89, closing on 1, 2 and 3; the flip moves as the strength squared within 3 to 6%, came where predicted, and held under real weight to four figures at three weights	test; knob	FAR-1, PINCH-1, PINCH-2, WALK-1; rows 106, 308, 666, 686, 736, 739
Exact counts from the toy to the junction	a lattice count built on the toy and registered for the junction agreed with 205 of 205 junction rows; the window count, found after the data, then landed exactly at four new fractions	count; test	LATTICE-JJ-1, PIN68-1w; rows 514, 821
The junction’s grip has one knob	the grip follows the wave’s strength over its frequency squared: predicted blind where the rival disagreed, held at ten fresh settings within 0.063; in the strong, slow corner, where the straight line fails, one curve predicted eight new settings within 4.5%	engineering knob with its setting	JJ-4a, JJ-GRID-1, JJ-CORNER-2; row 541
The grip forgets and the slip remembers	written down before the read: at weight 0.05, 99.1% of 110 locked runs had a stretch rate below −0.005 and all 101 slipping runs sat within 0.005 of zero; at weight 0.25, 100.0% and 99.0%	first measurement	MEM; rows 686, 813
A weight-proof setting for a two-tone rectifier	at one drive ratio the window moves 60 times less than on the usual setting at weight 0.04; one number, the weight times the frequency, sets how far that holds: predicted before each run, the band ended at 0.2446 against 0.2445 (frequency 0.40), 0.1395 against 0.138 (0.70), 0.11475 against 0.11508 (0.85) and 0.09742 against 0.09743 (1.00), at one value of the weight times the frequency within 0.44%	engineering knob with its setting	RATCHET-26, 30, 65, EDGE-85, EDGE-100; rows 731, 801
A dark point placed by a symmetry	the mirror pins the dark point to one timing, and runs placed it within seven parts in a million of the prediction; its sibling, the flat point, placed blind from four runs of a small dial within 0.16%, at fast drives within 0.14%, the dark points within 0.012%	test	RATCHET-53, 63, 64; rows 400, 581, 672, 786, 794, 799, 805
A weight meter that drift cannot fool	the drive’s drift predicted to move the push by 1.8607×10^{-6} and measured at 1.862829×10^{-6}, the weight’s channel	instrument; test	RATCHET-71, WITNESS-1, RATCHET-72; rows 75, 106

	exactly blank; a second engine written from scratch agreed to 5 parts in 10 million of the move’s size		
A whole number inside the rectifier	the timing that gives no drift makes exactly one half-turn along the weight knob at all nine settings run; its heavy end, 181 degrees divided by the weight times the frequency, predicted from one frequency, came out 183.2, 180.4 and 179.3 at three others	test	RATCHET-8, 11 to 13; —
A fade predicted to the fourth decimal	the rectifier’s height fades nearly as the weight to the minus fourth; the next rung’s power, written down as −3.977, came −3.9771 at weight 24.5, and at all seven heavy weights from 8.2 to 24.5 doubling the second tone doubles the height to one part in ten thousand	test	RATCHET-80; row 821
Bear’s coin	a rider lands heads or tails by where it is set down and nothing else, along an edge between the faces that stays ragged however closely it is examined, with an uncertainty exponent of 0.592; a second engine written from scratch, sharing only the equation, flips every one of the coin’s 1,800 riders of record the same way	first measurement; a second engine’s check	the book’s §4.17, WITNESS-COIN-1; —

The table below is the Loop Atlas’s register at its close, for *The Loop Atlas* (*A loop comes home to where it started. What did it leave behind?*), published on 9 November (concept DOI 10.5281/zenodo.24100672) with its record, *The Loop Atlas: the record* (concept DOI 10.5281/zenodo.24100980), and its note on the odd-order constants (concept DOI 10.5281/zenodo.24101154), which follows one question across toy machines and bodies: when a loop comes home to where it started, what did it leave behind? It is drawn from the book’s own list of what it found, in its record’s description, each result read against the row of this guide that holds it; its count is 1,104 registered predictions judged, 706 held, 369 missed and 29 undecided, with 572 planted checks beside them, 555 held. One result that list gives, a listening blade’s drift of 0.124 in an even soup of waves against a clock-switched blade’s 0.009, is left out: the book bills it as a toy that did not pay for its charge, and billed honestly the pull comes only from a difference in push across the blade’s length (row 590).

What held	Ours against known	Kind	Run; rows
Fifty-one meetings with the published record	from Feigenbaum’s ratio, met at 4.66920159 for a rounded top, to Kane and Scher’s falling cat and Tong and Dullin’s diver, the diver’s model rebuilt from its own table within 0.02 to 0.15 per cent once one entry their table most likely drops is restored (theirs to confirm)	benchmarks matched	TONG_DULLIN_SCAN 20261017_181953; rows 771, 814, 818
One cat under two rules	a cat that twists at its waist and a cat that turns twist-free turn alike, at every shape, through a fixed half turn of the rates, and on every path: within 2.842×10^{-13} degrees on nine swelling paths	new, claimed on a limited, noncommercial review of the citations and references	rows 820, 836
A widely copied table’s odd entries	the published δ for maps of odd order is off at every odd order and right at every even one and in α: 5.9679687038 for the maps’ own 6.0846890743 at order 3, 8.3494991320 for 8.3443847487 at order 5; one sign may explain the difference	correction of a published table, as its own record	rows 874, 875
A rider who listens works on any swing	timed by the swing’s own phase, every rider on swings of 0.5 to 12.5 m passed 90 degrees within 12 of its own swings; reading by the wrong rate has a price that can be written down	not found in print in the pages reached; knob	row 681
Smoke rings that leapfrog	given exactly a leapfrogging pair’s budget, one ring travels at 0.84 to 0.88 of the pair’s speed, so on one budget the pair is 14 to 19 per cent faster	first measurement on one budget	RING_BUDGET 20261004_080107; row 788
A maintenance alarm read where nothing hides the wear	at equal catch, one fixed level of a balancing rider’s own lag threw away 0.031 to 0.051 of a life against the calendar’s 0.250 to 0.333	engineering knob with its setting	RESTED_LAG 20261016_205504; rows 636, 746
A diver cued by the body	a kick-out cued by the body’s angle missed by 0.420 of a clock-timed one’s miss, a number predicted before the run	test of a prediction	DIVE_CUE 20261017_082454; row 581
A latch that holds until the push turns	above an edge of 1.5 of the push’s typical size a latch and a memoryless crest-firer switch exactly as often, and the latch pushes 1.36, 1.51, 1.75 and 1.82 times as much at edges of 1.5, 2, 2.5 and 3	engineering knob	LATCH_EVENT 20261015_232643; row 669
An ear that stops where its own ring dies	a spring bead’s listening ear stalls at critical damping, sharply: 0.054 to 0.175 of the soup’s push at 0.95 of critical damping, and exactly nothing at and above it	first measurement	row 753

The two-voices papers' registers

Two papers of the author's two-voices series, on what a learning mind keeps when what checks its work cannot be trusted, were published after the three projects closed. *Question the Substrate: Why a Learning Mind Can't Grade Its Own Homework* (published on 9 November; concept DOI 10.5281/zenodo.24094163) ran 41 experiments on a small network that counts how many times a loop winds about a point, a task with an exact checker, so every checker under test could be scored against the truth; it registered 212 predictions before their runs: 122 held, seven of them with a qualifier, 89 missed and 1 held in part. *Two Arrows: Why a Learning Mind Heals Its Answers Long Before Its Self* (published on 10 November; concept DOI 10.5281/zenodo.24120415) registered 170, counting one independent run registered and carried out in the first paper's sessions: 87 held and 83 missed, and 4 of its 11 redesign re-checks held; the independent run held 6 of 6. Their lessons are rows 883 to 891 and additions to 20 earlier rows. The first table is drawn from the first paper's close-out, its lessons each with its run and its after-action review; the second from the second paper's headline register. A dash in the last column marks a result no row of this guide yet tells.

The table below is *Question the Substrate*'s.

What held	Ours against known	Kind	Run; rows
A mind's own certainty is a rich-get-richer filter	keeping the surest half of its own labels lost 25.3 points in world 1 and 12.9 in world 2, and an answer holding 11 per cent of the data went extinct, 10.6 per cent to 0.0, while overall accuracy barely moved; in world 3, tidy clusters, the rule came second best (+3.6)	known at the scale of large language models; measured here against an exact checker	E1e, E1w2, E1w3, E14; rows 280, 883
Every failure grades itself, and every cure is an outside reference	the same half-size cut, ranked by the frozen first network instead of the chain, lost 2.1 points against 25.3 and kept an 11 per cent answer alive (survival 0.83 against 0.02); a twin taught once from the truth and then held fixed kept the chain level (−2.5)	test of the paper's spine	E41, E35; row 77
Who ranks matters more than how many are kept	the same keep-half rule lost 25.3 points when the chain ranked its own labels and gained 2.7 when a strong outsider ranked them	test	E26; row 77
Buy any real check	a checker that catches a quarter of the errors earns 40 per cent of the benefit, one that catches half 68 per cent; even a half-blind, wasteful checker beats the mind's own certainty, by 28.6 points in world 1 and 10.9 in world 2	engineering knob with its setting; the knobs are ratios	E7; row 884
Every listener is deaf to its own echo	the student's batch of its surest labels pulled 0.004, about 50 times quieter than any other; a strong outsider ranked batches by their true share of wrong labels at +0.829, against +0.213 for the student	first measurement	E8, E8b; row 884
One knob makes it causal	tidy clusters gave +3.6 with equal answers and −14.7 with unequal ones, the two smallest answers falling from 12 to 0.7 per cent and from 8 to 1.1	test	E15; row 584
The field's known fix, gauged both ways	selecting within each answer removed the survival cliff, then over-predicted the small answer, 1.17 times its true share at 12 per cent and 2.51	test of a known fix	E22; row 883

	times at 3 per cent, while accuracy rewarded the extinction (+2.1) over the rescue (−0.4)		
A tipping zone	ten starts at a 1-point step found a band between 14 and 12 per cent of the world where identical setups live or die by their start; the end-state switch shows the same between size ratios of 0.61 and 0.64	first measurement	E31, E37r; row 886
A noise floor’s fingerprint	the aimed fix’s deaths matched $(1 - p)^{30}$ start by start; the survival cliff moved about 0.8 point of share per doubling of the data, so it is mostly not a noise floor	test; method	E38z, E39, E39b; row 218
A copying chain’s restoring force	a chain that held level for six generations slid 7.9 points at a leak of 5 per cent of its labels a generation, and absorbed a leak of 0.5 per cent entirely	first measurement	E25, E34; row 887
A contradiction settled by running it	a published rule whose abstract seemed to contradict the paper, confidence filtering that prevents collapse (Fu and colleagues, 2025), held when run in the paper’s chain: its confidence belongs to a frozen base model, a fixed outside reference	known result reached by a new route	E41; row 77

The table below is *Two Arrows*’.

What held	Ours against known	Kind	Run; rows
An area law in a learner	after a closed loop of tasks, the part of the leftover that flips with direction doubles when the enclosed area doubles: log-log slopes 0.9545 to 1.0092 against the loop’s radius squared, five of five seeds, and 0.9503 to 1.0067 in the hidden layer	new in this setting; the closed-loop form of the lesson comma	s1; rows 419, 644
Answers come home; the inside does not	after 30,000 practice steps the answers keep 0.0167 of their change and the hidden layer 0.3049, so the answers come home 18.49 times better (the mean of five starts’ ratios)	first measurement	r3; row 888
Only a kept record of the self restores it	a pull toward a record of the hidden layer from before the loop leaves 0.0003 of the inner mark; the true answers 0.3049, its own answers recorded on return 1.7621, its own current answers 4.6840, a stranger’s 5.6064	first measurement	b2; row 889
Hold on after the trip, not during it	holding the inner record throughout the journey follows the moving task 1.382 to 2.156 times worse; switched on only after the trip it leaves 0.00019 to 0.00085, at no cost to the	the stability-plasticity trade-off, known in kind, with a number	b16, b17; row 889

	journey		
A forecast replaces 30,000 steps	linear relaxation, with no practice run, within 25 per cent in 37 of 45 cases, median ratio 1.003; re-linearised in segments, the two hardest cases within 25 per cent in 9 of 10	instrument	b7, b18; row 259
Correct from the first map	a lineage corrected against the first generation’s map stays level for twelve generations (largest over smallest 1.02 to 1.06); one corrected against its parent ends 4.97 to 7.83 times higher	first measurement	b11; row 308
The yardstick decides whether drift is heard	against a kept copy of home, the healed mark is 8.888 to 9.876 times its home gap on inputs three times wider, 6 of 6 predictions held (registered and run in the first paper’s sessions); a later run resized it: a soft random change of the same home gap is about as loud	test; resized	bridge_b1, b12; row 884
A relabelling, not a drift	half a turn on a direction-blind task exchanged two hidden units; renaming them removes 68 to 85 per cent of the distance from home, and the exchange is one event, just past a quarter turn	first measurement	b4, b5; row 890
A real outside anchor bounds identity drift	after 12 laps: no anchor 147.070 degrees, a real anchor 3.106, a yes-man and a mirror identical to none, an internal copy 146.884	test	q3; —
Flat worlds heal, curved worlds keep the area	a rigid partner on a circle drifts at most 2.226×10^{-14} degrees; on a globe the drift equals the enclosed solid angle within 1.4023 degrees, 0.036779 with the exact rotation rule	known geometry, reproduced	q2, q2_redesign; —
Right after a crisis, subtract the slow arrow	one thinking round: the best mix, at −42 degrees, scores 0.7775 against 0.4426 for the outer arrow alone	a lead compensator, known engineering	q5_redesign; row 633
The one command, rerun in a fresh copy	after one runner fix, 148 files compared: data and figures byte for byte, printed output apart from timing lines	benchmark of the paper’s own crate	rerun of 10 November; row 862

A specimen from the guide’s own session

On 17 October the guide’s publishing session called R2W’s lesson 178 “a solid addition to the toolbox, not a big moment for the guide”, sizing it against the guide as a whole. Sized against its niche it reads differently: four tests that separate a true null from a cancellation for knobs that cannot be reversed, where the classical reversal technique needs a knob with a sign, each test carried by numbers from named runs (row 805). That is a method result, and it belongs in a register.

What kept arriving

Three projects, the triple-star sessions, R2W and the Loop Atlas, ran side by side, each on its own machine, and each sent this guide its lessons. Some lessons arrived again and again, from projects that never shared a run. Those repeats are this guide's own finding: a lesson that keeps arriving on its own is a lesson about the work, not about one machine. So from 6 October the guide counted them. Each family of lessons was defined in one sentence before it was counted, so that the count cannot drift with the definition. An arrival is a project meeting the idea in its own run, on its own system; a lesson relayed through two letters is one arrival, and a duplicate file is none. And each arrival says whether its session had read the guide first, because a lesson found before reading weighs more than one found after.

Rank	Family	What it says	At version 6	Counted by 10 October	Counted after	In version 7's fold	Found blind	Projects and papers
1	**A check that cannot fail**	A test whose outcome was guaranteed, or whose bar the data could not miss, says nothing by passing.	2	21	110	131	2	5
2	**The built artifact is the evidence**	An edit or a build is a claim until the finished artifact is read back.	0	15	75	90	1	6
3	**A word that changes meaning across a boundary**	One word or symbol carries two denotations across a boundary, and the claim rides on whichever was assumed.	10	14	54	68	0	7
4	**Chaos and binning**	A system near a boundary between bins, locks or levels flickers across it, and the apparent signal belongs to the boundary, not to the physics.	12	12	51	63	3	3
5	**A second reader**	An independent re-reading of the same files finds what the author's own checks were blind to.	not counted	15	35	50	0	4
6	**Coupled clocks and event time**	A loop keeps time by its own events, and when two such clocks are coupled, what is measured depends on which clock reads it and on how nearly the two agree.	one vocabulary entry	17	20	37	2	3
—	**The eye proposes, the ruler decides (proposed)**	A reading of a picture or a pattern by eye, a person's or a model's, is a proposal until a measurement that could have failed settles it.	not counted	—	29	29	0	3
7	**The mirror audit**	Run the mirror on the pipeline, and read what flips and what stays; a symmetry operation tests handedness more sharply than randomising.	6	12	16	28	1	4
8	**Count the right unit**	A count's unit must be the independent thing: runs not episodes, time not episodes, landings not stumbles, net crossings not raw ones.	not counted	11	14	25	1	3
—	**A law carried past its range (proposed)**	A fit or a law measured over one range is used outside it, and the claim rests on a regime the measurement never saw.	not counted	—	22	22	0	3
9	**The integer diagnostic**	A count must be able to come out only whole; a near-integer is a measure until it is read from closed pieces.	3	11	8	19	1	4
10	**The knee**	A switch or counter that commits only past a margin holds its choice, and a round trip costs the area between the two thresholds.	0	7	11	18	0	3
11	**Rate over level**	The signal is in the rate of change, not the value.	5	1	1	2	0	2

The families, ranked by their arrivals in version 7's fold, from 6 to 23 October; families with equal counts share a rank. Version 6's counts are only a baseline: four of them are a judgment or a sample of the register. Found blind: met before the session had read the guide. Projects and papers: how many separate sources the fold's arrivals came from. The two families marked proposed were defined on 23 October and are counted from that day's sort, not ranked; coupled clocks, proposed on 7 October, was accepted on 22 October and is ranked from 23 October. Every arrival is listed, family by family, in the appendix after the ledger.

What the ranking shows. The lesson that arrived most often is not about any machine's physics. A check that cannot fail arrived 131 times, from five projects and papers: bars set below the instrument's own floor, controls that passed as readily as the guesses they were meant to test, and tests that a theorem or the run's own setup had already decided. Second comes the built artifact is the evidence, 90 arrivals. The two commonest lessons are both about the tools of checking, the test and the build, before they are about the system under test. Third, since 23 October, is a word that changes meaning across a boundary, 68 arrivals, which also came from more projects and papers than any other, seven. Chaos and binning, 63, is fourth, and it is the family that arrived most from the physics itself: one session of the Loop Atlas met it nine times in three days, in its needle, its long snake, its rings, its spring bead and its counters of crossings, each time a count or a threshold that sat on a boundary and read the boundary instead of the toy; and where the family's own question was put to a test, the chaos on the borders between R2W's treads came out the same at three chaos thresholds, structure and not binning (row 655). It also carries the strongest sign that these lessons are found rather than taught: 3 of its arrivals came from sessions that had not yet read the guide. None of the 446 arrivals counted after 10 October was found blind: 445 came from sessions that had read the guide, and for one, the Loop Atlas's arrival on the jellyfish paper's swimmer (row 769), it is not known, so from here the count measures how often a lesson is met again, not whether it is discovered. Rate over level, which version 6 had counted five times, arrived twice in the fold: Bear's lattice taught it to three sessions at once, so it was counted once (row 708).

Two families defined on 23 October. The ranking of 17 October promised a sentence for each lesson the readers kept meeting outside the families before the next count, and the two they had named most often got theirs on 23 October, before that day's sort. They are counted as proposed families, as coupled clocks was. A law carried past its range (a fit or a law measured over one range is used outside it, and the claim rests on a regime the measurement never saw) arrived 22 times, meeting a forecast registered on 17 October of at least 20 arrivals from at least three projects, and the eye proposes, the ruler decides (a reading of a picture or a pattern by eye, a person's or a model's, is a proposal until a measurement that could have failed settles it) 29 times, each from all three projects. Both were counted in the lessons placed since 17 October and in the 37 earlier units whose readers had flagged them, so each count is a floor: an earlier unit no reader flagged was not read again.

How it was counted. The count of 10 October was made by the session that gathered version 7, arrival by arrival. Everything placed after it, 1,527 lesson units, R2W's and the triple-star thread's lessons of 16 October, R2W's and the Loop Atlas's of 17 October and the three projects' sends at their move to final publishing on 22 October among them, was sorted by fourteen fresh readers against the families' own sentences, each arrival backed by a quotation from its source that a script then found word for word; 46 of their entries were arrivals already counted. Where one event reached the guide through several records, a lesson, a record of misses, a letter, it counts once (a further reader found 37 groups of entries that were one event, and on 23 October 4 more pairs of entries of one family naming the same run were joined): 446 new arrivals from 607 entries, and 582 in the fold. One arrival counted on 17 October was withdrawn on 23 October because its own project had withdrawn the reading behind it: the Loop Atlas's tied bells read as choosing between two answers, which its session G found was the toy's thin cores (row 807). One more reader sorted 60 of the first 801 units, drawn at random, without seeing the first readers' answers. The two agreed on 85 per cent of them (Cohen's kappa, which discounts the agreement chance alone would give, was 0.67), and wherever both saw an arrival they named the same family, 16 times in 16. The second reader was stricter about what counts as an arrival, keeping 16 of the 24 sampled units the first readers had marked; with the later arrivals scaled down by that rate, the ranking keeps its order.

What fits no family yet. The readers also met lessons that recurred and fit none of the families. The commonest named on 23 October were testing every scale a toy carries (6 of the Loop Atlas's units), a later finding reaching the claims already written (5 of the triple-star thread's), and keeping a register of what landed (4, from the Loop Atlas and the Field Guide sessions); earlier rounds had named a session's files as its only memory, a finish line and what waiting costs, and a floor or a control measured before the law. They were noticed while counting, not defined before it, so they are not counted here; each gets its one sentence before the next count. From here a second example of a lesson adds a count to its family and a line to the appendix, not a new row (row 798).

How the lessons travelled, and where they live

Where each lesson lives. Every row of the register and every arrival in the ranking carries one to three domains from a fixed list of sixteen, written before any tagging so that a tag cannot drift with its reader: three body, driven oscillators, maps and chaos, kinematics, waves, quadrature, geodesy, solar, biology, number theory, materials, control and circuits, topology, reading pictures, records and tools, and measurement. The appendix “The register by domain” lists every row under each domain, so that a reader, or a reader’s AI, can start from the domain it knows and walk out from there. A lesson that keeps arriving in new domains is a lesson about the method, not about one system, and the families differ most in how far they reach: a word that changes meaning across a boundary arrived in 15 domains and a check that cannot fail in 14, while the built artifact is the evidence, though it arrived 90 times, stayed almost entirely inside the projects’ own records.

Family	Domains its arrivals reached	The commonest three
A check that cannot fail	14	measurement (126), driven oscillators (42), records and tools (32)
The built artifact is the evidence	8	records and tools (89), measurement (12), driven oscillators (3)
A word that changes meaning across a boundary	15	records and tools (38), measurement (25), driven oscillators (24)
Chaos and binning	13	measurement (57), driven oscillators (29), maps and chaos (18)
A second reader	9	records and tools (29), measurement (24), three body (13)
Coupled clocks and event time	10	quadrature (19), driven oscillators (19), three body (14)
The eye proposes, the ruler decides (proposed)	7	reading pictures (23), three body (13), driven oscillators (11)
The mirror audit	11	driven oscillators (17), measurement (14), kinematics (5)
Count the right unit	10	measurement (25), driven oscillators (12), three body (5)
A law carried past its range (proposed)	9	driven oscillators (17), measurement (14), maps and chaos (5)
The integer diagnostic	9	measurement (12), driven oscillators (9), topology (7)
The knee	11	driven oscillators (16), measurement (6), maps and chaos (5)
Rate over level	4	reading pictures (2), maps and chaos (1), measurement (1)

How lessons travelled. Bear’s question of every carried lesson is “it worked in X but not as well in Y? Is it a wall, a seam, a substrate?” Read row by row through row 847, the register records, on 322 of its rows, 519 places where a lesson, a tool, a law or a reading was carried from one system or setting to another and the text says how it fared. It held 222 times, held weaker 79 times and failed 218 times. When it did not hold, the break had an owner:

Owner	What broke	Weaker or failed
ruler	the instrument, grid, threshold or reading was too coarse or wrong there	62
not stated	the text does not say	60
substrate	the new system lacked the geometry or ingredient the method reads	58
seam	a boundary between regimes lay between them	48
knob	another setting had moved between the two places	41
wall	a symmetry or structural limit of the method itself	28

Of the breaks whose owner the text names, two lead and are close: the ruler, the instrument, the grid or the threshold that read it in the new place (62), and the substrate, a new system without the geometry the method reads (58), against 60 whose text names no owner. The lead is not settled. When the 96 rows that changed on 17 October were read again by fresh readers, the ruler’s count among them fell from 20 to 13 and the seam’s rose from 10 to 16, so which of the two comes first depends partly on the reader. Every reading agrees on the other end: a wall of the

method itself owns the fewest breaks, 28. The ruler's share is the ranking's own finding met from the other side, where a check that cannot fail leads every family, and the substrate is the owner that the guide's three causes of a null call by the same name. Every transfer is listed in the appendix "Transfers, row by row".

How it was read. Six fresh readers tagged the register's first 792 rows, and three more its first 326 arrivals; one more tagged the 24 rows changed or added on 16 October and one more the 18 arrivals counted that day, on 17 October two more tagged the 96 rows changed or added and one more the 60 arrivals counted, and on 23 October four more tagged the 141 rows new or changed from 18 to 23 October and two more the 179 arrivals counted that day, against the fixed list and definitions, each transfer backed by a quotation from its row that a script then found word for word. On 1 November one more reader gave domains alone to the 34 rows new from 848 to 881, matching the sets of like rows already tagged, on 2 November the publishing session tagged row 882 the same way, and on 10 November rows 883 to 891, from the two-voices papers; their transfers are not yet read, so the transfer counts here are of rows 1 to 847. The four readers of 23 October said where they drew the line, and drew it in slightly different places: one counted a lesson named a second specimen in another project as carried, another counted only what the text marks as carried and says how it fared, so the newest rows' transfers are the least evenly read. A second reader, blind to the first, tagged 40 rows drawn at random: every one of the 40 shared at least one domain between the two readers, 32 shared the first domain, and the two sets of domains overlapped by 0.87 on average. The two readers agreed on whether a row records any transfer at all in 37 of the 40; where both found transfers, the commonest outcome matched in 7 of 8. The second reader found 15 transfers where the first readers had found 12 in the same rows, so the count of 519 is more likely low than high.

The data. The register's domains, the arrivals and the transfers also travel with this guide as three tables of data, FieldGuide_v7_ROWS.csv, FieldGuide_v7_ARRIVALS.csv and FieldGuide_v7_TRANSFERS.csv, so that a later reader can sort them by domain, family or owner and keep adding to them; each transfer there carries the words of its row that record it.

L1 — The Method as an Instrument

What you do, and the instrument you do it with.

The rows of L1

New in version 7 on this limb: 158 of its 312 rows, numbered 548, 554, 557, 560, 566, 567, 568, 575, 584, 586, 587, 590, 601, 603, 604, 606, 607, 608, 609, 612, 613, 614, 615, 616, 617, 618, 619, 620, 621, 622, 627, 628, 629, 630, 631, 632, 633, 638, 639, 640, 641, 642, 643, 644, 645, 646, 647, 648, 649, 650, 651, 652, 662, 663, 664, 665, 666, 668, 683, 686, 687, 689, 690, 691, 693, 694, 699, 700, 701, 702, 703, 704, 705, 706, 707, 708, 709, 710, 711, 713, 714, 717, 718, 722, 723, 728, 729, 730, 731, 732, 734, 737, 738, 739, 741, 742, 743, 745, 747, 748, 749, 750, 751, 752, 753, 755, 756, 757, 758, 759, 760, 761, 763, 764, 766, 771, 772, 780, 781, 784, 785, 787, 788, 789, 791, 793, 794, 797, 799, 801, 805, 806, 807, 810, 813, 816, 819, 822, 824, 827, 829, 831, 835, 836, 837, 839, 841, 847, 850, 851, 858, 863, 864, 865, 870, 872, 884, 886.

Declared blindness (2) — Name what you cannot see before claiming to see anything. The cochlear instrument declared its blind spots explicitly: 8 measurement modes across 6 signal channels — not every combination is readable, and writing that down before the read is what makes the instrument honest.

Which loops exist and which have opened (5) — Most modern cochlear implants carry ECAP (Evoked Compound Action Potential) telemetry — the hardware to measure how the auditory nerve responds to electrical stimulation. This is a complete closed-loop control capability. It is used once at fitting, and then the loop is left open and never asked again — a control loop that exists in hardware and is left idle. The lesson: check whether the loops in a system have been opened and forgotten before concluding there are none to walk.

ECAP telemetry; closed-loop control

Acceleration-centroid tracker (10) — 1.9 and 4.8 km/day; 10^{5}–10^{6}× slower than any physical wave — a thing that moves at plate-tectonic rates while the physical wave it reads travels at seismic speed. My suggested name: Slow-Mobius, after the Rick and Morty character who moves so slowly he is functionally invisible to everyone around him while from his own reference frame he is moving normally. The centroid of the seismic wavefield moves through the crust a hundred thousand times slower than the wave — and that is what this instrument tracks.

no prior name located — ours, with the fence stated

Frequency-footprint address (11) — Every substrate has a natural frequency address — where its signal lives in the spectrum, and where background noise lives. The cochlear frequency ladder, from the top down: electrical mains hum at 50 or 60 Hz, Schumann resonances at 7.8 Hz (Earth's electromagnetic background), microseismic noise at 0.1 Hz, tidal periodicities at 10^{-5}–10^{-6} Hz, and corridor-travelling one-off events at 10^{-7} Hz. Knowing this address tells you where to look and what to expect to find there.

Mod-N aliasing, discrete-Stokes redemption (12) — The aliasing problem in winding counts: a naive modular implementation folds values above N/2 into false negatives, so a cell that accumulated 120 half-turns reads back as 0. The discrete-Stokes construction avoids this by ensuring each cell contributes at most one count to the boundary sum. The proof that it works: 120 cells returning 120 distinct values, none aliased, none lost. This is the architectural fix that makes the Shared null space measurement (row 13) possible — without it, the inter-block rotation signal would fold back on itself. The same wall bounds a writer. A ring of N writers writes a twist only up to N/2, and at exactly half a turn per step an even ring reads a count and its mirror alike and writes dark spokes. The read limit and the write limit are one wall seen from both sides (the Latch Atlas; ties row 283)

discrete Stokes; ties row 542

Shared null space (13) — Common-mode means the same background signal arriving at all sensors simultaneously — and a differential read is immune to it, because subtracting sensors from each other removes what they share. The block-vs-block winding matrix exploited this: instead of 229 individual station readings, it organized them into 16 common-mode series (groups of sensors seeing the same background) and read only what the groups disagreed on. Inter-block rotation reached 3.587 cycles; station P062's 6.395 cycles exceeds the largest common-mode pair.

Byzantine rogue census (14) — When sensors cannot be trusted, the pattern of their failures is itself a signal. A seismic station census: 26 stations behaving like spies (convict — consistently wrong), 37 suspect, 148 reading clean. Station CLCV-PSPT-P736 showed +1 versus −128 — a near-perfect anti-correlation with its neighbours. All 35 standing edge-pairs fired in opposition, with opp-frac (opposite-firing fraction) of 1.000 — perfect systematic disagreement.

Lamport (Byzantine generals); Jennison (closure phase)

The matched-filter law (16) — A matched filter is tuned to detect a specific signal shape — powerful at low noise, fragile at high. Measured: 25 detections out of 25 trials at noise level $\sigma = 0.3$; only 8% success rate at $\sigma = 0.6$. Running two smoothing passes before filtering restores the full 25 out of 25 even at high noise. Know your filter's noise floor before trusting its output. Row 305 prints the same figures, 25 of 25 falling to 8 per cent and restored by smoothing first, for a hedgehog's integer degree under per-vertex noise.

matched filter; ties row 305

The projection law (17) — Ternary classification (−1, 0, +1) fails under phase noise: it reaches only 52% accuracy at noise $\sigma = 0.6$ radians. Continuous-phase Euler representation (using 1/cos(pitch) rather than a sign) is accurate to 2×10^{-14} at the same noise level. When phase noise is high, move from counting signs to tracking the full continuous angle using Baker–Campbell–Hausdorff.

Baker–Campbell–Hausdorff

Detection floor is not measurement floor (18) — Two different floors that are easy to confuse: the measurement floor (how much your signal moves in a quiet period) and the detection floor (how high an effect has to be before a specific detector notices it). An external gate — an independent detection system using a different measurement principle — separates them. Result: a seismic event detector found 3 of 40 events on its own; with the external gate providing the second measurement, it found 40 of 40.

sub-threshold recovery (GW search)

The dipole liar (19) — A real physical relationship weakens with distance. A spurious one does not. This seismic station pair showed correlation r = 0.98–0.997 at near-zero p-value — extremely significant — and exactly the same correlation at 20 km separation as at 0.4 km. No physical signal behaves this way. The correlation was a dipole artefact: both stations were responding to the same large-scale background, not to each other. Row 307 reads the same 0.98 to 0.997 as trend-sharing, which any two smooth decays show, and with the slow parts removed from both its clean test found a real co-location signal: r = 0.585 at 0.4 km, peaking on the co-located meter and decaying with distance.

spurious correlation of trends; ties row 307

Event time (20) — When time normalisation (dividing your signal by a time average) is applied to a dataset where the signal is carried in the relative timing of events rather than in their magnitude, it erases the signal. It drowned the seismic signal by two orders of magnitude here. Use event time — time measured relative to an event, not the clock — when the signal lives in timing. As a way of driving, rather than reading, an event trigger is not automatically better than a clock: in the memory bell, firing when the flow at the bell peaked only tied a clock at the same pace. Event time wins where the signal lives in timing; a trigger has to earn its place against a clock at matched pace (the Loop Atlas sessions)

event-based sensing

The visibility matrix (21) — List what you cannot see, not just what you can. Every honest instrument ships its own visibility matrix: a table of measurement modes and signal channels, with the inaccessible cells declared. The cochlear instrument's matrix: 8 modes × 6 channels. The shear row was closed after the channel was built — the instrument was declared structurally unable to read it, and that declaration travels with every report the instrument produces.

instrument datasheet

The residual does two jobs (24) — the Kalman innovation is the gap between what the model predicted and what arrived, and it does two jobs simultaneously: it is the correction signal that updates the state estimate, and it is the anomaly detector — when it is unexpectedly large, something has broken or the model has drifted. Using the same number for both jobs is the economy, and knowing it does both is what stops you from treating a large residual as noise to be averaged away when it is actually the signal. The portable version: any quantity that accumulates the mismatch between prediction and observation is doing these two jobs whether or not you are aware of it — read it as a correction, but also read it as a probe

Kalman innovation

Opponent as sensor (25) — an opposing player and a sensor are the same object — a source emitting a signal you cannot fully trust. Track two numbers per opponent: the estimate and its reliability — range-centre and trust-or-variance — and never the internals. This is the anti-bloat rule that stops an opponent model sprawling into a decision tree. The reading is genuinely a holonomy reading: you are not observing the opponent's state directly, you are reading the accumulated effect of their choices on your own uncertainty, around the loop of each exchange. Von Foerster's non-trivial machine is the theoretical home: a system whose internal state cannot be determined from input-output pairs alone — which is exactly the opponent, and exactly what makes the sensor framing productive rather than merely metaphorical

von Foerster; non-trivial machine

The wraith (26) — a consistency check can price a rigid liar and can never detect a fully adaptive, opponent-dependent one. A static liar holds a fixed false value — geometry can represent it as a defect, a disclination, a thing that sits still and can be triangulated. The wraith changes its report per observer, so no two sensors see the same face and no sum-check or signed-ledger test can catch it — and when the wraith reads as unreadable, the instrument is working as specified, not failing. This is not a design flaw; it is a limit of the adversary class, explained in row 84 though not proved, which is why the instrument's honest job is cost-imposition rather than detection (row 117). The wraith is the reason the detective work stops and the tax begins

practice; ties rows 84, 117 and 510

The blackjack diagnostic (27) — the manipulation layer is only ignorable when it carries no information about the countable substrate beneath it. In blackjack the dealer's theater — their tells, their manner, their timing — carries nothing about future cards, because the deck is sovereign and the dealer cannot alter it; so count the deck and ignore the theater. In poker the other minds ARE the substrate, so ignoring the theater means going blind to the signal itself. The diagnostic is: can the manipulation layer be permuted freely without changing the substrate's statistics? If yes, ignore it; if no, it is the read. Portable to marketing, negotiation, propaganda — anywhere a surface layer might or might not carry information about what lies beneath

Two codes, opposite enemies (50) — at equal mean power with the decoder clamped: under local damage the winding code errs 0.063 against 0.643 at $\sigma = 0.6$, ten times fewer; under smooth global deformation it inverts completely — winding at chance (0.899) with zero local noise while magnitude is perfect (0.000). Carries its own correction: the first codec gave winding a 2.56× (4.1 dB) energy advantage and an unclamped decoder, both fixed — the law survives, the margin does not, and the pre-fix numbers are not to be quoted

The medium reports its own damage in the units it stores (56) — A substrate often reports its own damage in the same units it was designed to store. Crack propagation connecting voids is the most likely real failure mode for a topological memory, and it is readable: the zeroth Betti number b_0 (connected components) drops as cracks join voids into clusters, while b_1 (the hole count) may rise as new enclosed spaces form. A medium failing from within looks different from one degraded uniformly from the outside.

Coarse reading as a classifier — and it is the same knob as over-binning (58) — a deliberately lossy, coarse-binned read classified category as well as or better than a full high-dimensional read at roughly 60× compression. Fine detail is where records differ, coarse structure is where categories live. Over-binning is that same coarse-graining run past the edge; controlled coarse reading is it held back inside the window — one mechanism, suffered as a limit or exploited as triage

Net capacity peaks at four dimensions — and the fence that makes it honest (59) — Monte Carlo encode→noisy-read→decode peaks at four axes and declines beyond, because added axes share the same physics and arrive at SNR near unity. The fence: that is a property of CORRELATED axes, not a universal dimension cap — a separate simulation confirms dimension grows unbounded when axes do not interfere

Shannon

Storing an invariant does not make the read a holonomy read (64) — My observation on my own first invariant paper: the chiral memory is a system whose bit is a topological winding and whose readout is not a loop walk — sign by circular dichroism, magnitude by bag count, position by strain-induced frequency shift, all local and direct. A false friend of the sharpest kind, because the substrate could hardly look more holonomy-shaped

ours (named in the paper)

Healing and remembering are in tension unless the bit is topological (66) — ordinary annealing repairs a crystal by erasing defects — which erases the memory. The architecture can heal and remember only because the bits are non-local, so local repair of the bulk leaves the invariant untouched. Same tension as two-codes-opposite-enemies in a different domain: gel heals the crack and invites the global deformation

Kitaev 2003

Correlated noise shaped like the invariant beats the topological code first (72) — under streaks aligned with the logical operator, a calibrated greedy syndrome-matching decoder fails 20–120× more than under i.i.d. noise at matched sub-threshold density — while majority vote, blind to phase and winding, holds. Decoder-independent: a genuine logical-operator-shaped error is uncorrectable by any decoder. Magnitude is model- and aspect-ratio-dependent; the direction is robust and decoder-validated

topological quantum error correction; logical-operator weight

A self-confirming instrument reads its own output back as evidence (77) — the third level of compromise, where the loop closes through the thing being measured: within the event the two signatures overlap (1.72 against 1.17) and they diverge only afterwards (0.01 against 1.18), so the discriminator is the existence and passing of an exterior referent rather than anything internal. The order of operations falls out of it — stabilise the regulator, restore the capacity, then re-anchor. The paper's safety rail travels with this row wherever it goes. A self-training chain shows the same loop in numbers. Laid out in one table, every rule that failed in Question the Substrate ranked or copied by the chain itself: keeping the half of its labels it was surest of (its own certainty), a fix that kept half of each answer the chain predicted (its own estimate of each answer's size), and a rubber stamp that passed every label (its own labels); every cure was a fixed reference from outside, a real check, an outsider's ear, sizes counted from labeled examples, a frozen ranker. Even a twin taught once from the truth and then held fixed kept the chain level (−2.5 points against −25.3; runs E31, E35). Who ranks matters more than how many are kept: the same keep-half rule lost 25.3 points when the chain ranked its own labels and gained 2.7 when a strong outsider ranked them (E26). The cleanest test holds the rule and swaps the scorer: the same half-size cut, ranked by the frozen first network instead of the chain, turned the 25.3-point collapse into a 2.1-point dip and kept an 11 per cent answer alive (survival 0.83 against 0.02), for one run of about five minutes (E41). And an opposite result has usually moved the anchor: the abstract of a paper by Fu and colleagues (2025) said confidence filtering prevents collapse, and its full text showed the confidence belonged to a frozen base model, a fixed outside reference; its rule, run in this chain, held, so the two results agree (E41). When a loop fails, ask first what in it is grading itself, and before calling a contradiction, find where each side's reference sits (Question the Substrate)

appraisal theory (Lazarus 1991); interoceptive inference

Closure depth — superseded by row 134 (83) — The priced row is 134; this row is kept as the trail rather than deleted, because the path from unverified summary claim to sourced result is itself the record

see row 134

The balanced dipole and the matched offsetting lie are the same blindness (84) — Proven by hand for all N, with conditions and residuals named. Three checked steps: both worlds reduce to one object, since a signed sum of ternary values is zero exactly when the plus and minus counts match, so a hidden configuration in either world is just k matched pairs, placed somewhere; the map is a sum-preserving bijection, pentagon to plus-one, heptagon to minus-one, hexagon to zero, injective and surjective; and N does not matter, because the map is termwise and the only concealment move — insert one matched pair — is the same move in both worlds and composes without limit. The iff conditions are stated and load-bearing: it holds only on the intersection of ternary and ordered, because dropping ternary leaves fractional lies with no whole-number twin and dropping order stops the where from mapping. A leak was hunted and closed during the proof — an attempted minus-two liar broke the ternary design, and removing it closed two candidate leaks at once. And the residual is the citable core: the adaptive liar has no static-geometric twin, because geometry's defects sit still and do not adapt — which is not a hole; it explains why the adaptive adversary is the fundamental wall

Byzantine impossibility; disclination / Gauss–Bonnet

Match your suspicion to the substrate's freedom to lie (94) — nature rarely lies, so on sky, genetics and weather the adversary is bad data and confident misreading — the catchable static class — and suspicion should be spent on your own anchors. The adaptive liar is real only on human-touched substrates. The design consequence is the sharp

part: prions were chosen as a target because they hold still against a dynamic noise background, so a cheap passive multi-pass wins. The adversary model is a design input, not a disclaimer. The book adds the constructive half: a cell whose value is forced by a conservation law cannot tell an internally-consistent lie, because its value is not a claim but a constraint — so the invariant discloses itself, doing for free and more strongly what suspicion does in the sensor world. Over-suspicion of derived physics is as much an error as under-suspicion of free sensors

A finer loop is better only if the interior is smooth (97) — at a discontinuity a smoother high-vertex loop is worse, because it averages across the cut. So loop shape is a match to the interior's symmetry, not a ladder toward perfection — many vertices for smooth curvature, few-with-an-axis for a seam. Price: the Parkfield pentagon went null and the geometry predicts a rectangle. Row 160 reads the Parkfield hunt's null from the aperture's side, an areal loop pointed at a line source, and the instrument built next was a cross-fault station pair. This is a gate the good-loop section does not have, and it sits naturally beside the scale-matching gate

Redundancy defends magnitude, never consistency (98) — more vertices catch the outlier — the bad sensor, the static liar — and are structurally blind to the coordinated, frame-variant lie, because every station you add is also inside the network. Two axes, and adding sides only ever buys you one of them. This is the sharp operational half of row 96 and the counterweight to row 14's census

Byzantine residual (Lamport)

Integer-ness, not magnitude — and nested loops are a family you check, not an average (103) — the single best-priced row in this batch. A real winding is a topological count, so it is forced near a whole number and stays there across a family of enclosing loops (0.98, 1.02, 0.99); noise has no closure constraint and scatters (1.6, 4.3, −2.1). These are different kinds of number, and quantization is the one thing structureless phase cannot fake. Measured: gating on agreement-near-an-integer across nested radii took false alarms from 2.75 to 0 at the sharp cell and from 90 to 3 at the cell that had killed the previous build. Threshold on integer-ness, never on size. A count printed with a minus sign is a measure. A column of slips per 100 cycles printed −0.00 and −0.01 in places; those were turns lost, a magnitude mixing whole slips with the lag's drift, and a knee counter found exactly 0 whole slips in all 11 such rows. A count column must be able to show only whole numbers per item (the Loop Atlas sessions). Count before you measure. A switch first called by excess speed missed by only 0.0054 to 0.0056 of the pace; counted as slips per beat it became a staircase of whole numbers, 1, 2, 3 and on to 14 as the ratio nears 1, whose zero step is the protected state. A tolerance is a bin and can move a verdict; a count cannot, and an edge quoted from a tolerance carries that tolerance with it (the Loop Atlas sessions, R2W). Whole numbers are the surveyor's marks on a blindfolded hike: a whole-turn drift, 64 of 64 riders landing on one pad, index sums of zero, a pad's period counted in whole passings. And counts survive a change of machine where magnitudes do not: four riders run on a laptop and again in the container agreed on their slips, and on their stretch rates only within 0.012, the Integer Forest's trust in the quantized channel met again in a rider driven by two waves (the Loop Atlas sessions, R2W). Two machines tell the arithmetic from the machine: high-precision readings agreed on every printed figure between a laptop and the container, though their double-precision starts differed (an angle of 1.5044628766 on one and 1.5390285799 on the other). A result that reproduces across machines is the arithmetic's; a path that does not is the machine's; and a speed read on a shared machine is the crowd's (the Loop Atlas sessions). Where the read can be built from closed pieces it comes out whole and needs no gate (row 560); this gate is for the reads that cannot

topological quantisation

Modulus-1 deletes the axis the noise lives on (104) — force the radius to one and keep only the angle: the magnitude axis carries the amplitude, the noise spray and the common-mode level, and the phase axis carries the winding. Cheaper and cleaner than decomposing the field. And the discarded half is not wasted — normalising to one makes it a calibration line, and how hard the signal strains to leave the circle becomes a second free readout. It can also delete the axis a count lives on. On a ring of angle alone, a jump up and a jump down draw the same near-diameter, so the ring cannot say which way the reading went round; which side of the centre a jump passed is carried by the radius the ring throws away. On the disc, the middle look of every two-look down-jump was the nearest of its three to the centre (1.000 at six births), and of up-jumps 0.885 to 0.928 (runs triple_heavy_turn_count_look 20261011_085309 and triple_heavy_middle_look 20261011_093016) (the triple-star sessions)

The signal is in the rate (105) — the corpus's most expensive recurring correction, and it was paid three times in a single sitting: shear rather than area, rate-of-holonomy rather than holonomy, and slope rather than level. The last one carries the number — sorting building / steady / dissipating by the slope of core strain scored 96% against 33% chance, where the same quantity read as a time-averaged level scored 39%, barely above chance, because averaging smears an early-caught riser and a late-caught faller onto the same mean. The information was perpendicular to where the mean was looking

Correlated agreement is one vote, not three (112) — the common-mode repair is not to subtract the shared level, because a real multi-channel event raises the shared level too and the subtraction eats the signal. It is to down-weight correlated channels. Independence, not count, is the discriminator — a real vortex sings, glows and drops pressure on genuinely independent substrates, while a turbine emits on the same three but correlated

practice; ties rows 309, 452

Detection fails; taxation works (117) — you cannot catch an adaptive liar, but you can make hiding expensive — a suspicious stance converts evasion into a cost, because the moves that dodge a detector fight the clean fast operation the deception needs. The instrument's honest job is to raise the price of undetectable deception, not to achieve a detection that row 84 explains as impossible without proving it

security as cost-imposition; ties rows 26, 84

The integral erases history — keep the catalogue, never collapse the trace (120) — a global loop walk is the integral of local curvature, and an integral is order-blind and cancellation-prone: it keeps the net total, not the path. That is the blind spot. Path-ordered matrix products partly dodge it because they keep order — but any scalar summary re-integrates, and a true inverse pair vanishes again under a trace. The dodge holds only while the matrices stay uncollapsed. A bag is not a sequence, and the catalogue remembers what the integral forgets

lattice gauge theory; Wilson loop

Calibration fidelity and trust fidelity are different jobs (124) — is my reading accurate against a known reference is always worth doing and needs no adversary; is this reporting node honest is worth doing only where something has the freedom to lie. Blurring them wastes the suspicion budget and does real harm — treating a value forced by a conservation law as an unverified claim is how a trust-root gets invented where none was needed

A known-answer border is a free instrument check (126) — a flat cell's seam has a derived value of exactly zero, so walking it must read zero — and when it does not, the instrument drifted rather than the cell lying. Building known-zero borders into a substrate means every walk carries its own calibration, at no extra cost and with no adversary assumed. Conserved quantities are the cheapest such borders for an integrator. With drag, ripple and pull at zero the energy must hold still; with everything on, a patch of starts must shrink by exactly Liouville's factor. R2W's record had held that exact gauge since 6 October and used it once (NO-HONEY-CHECK, LIOUVILLE-CHECK); use the conserved quantity first. Where the drag fixes the multipliers' product exactly, that product gauges a numerical Jacobian before any physics is read: in R2W's FX-2d, one-sided differences ran 25% low at drive 0.90, while central differences met e^(−T/m) within 0.008% (FX2d-J) (R2W, sitting P2)

Gauss–Bonnet; ties rows 406, 858

Suspect the sensor, not the universe — priced (127) — a first real-data run returned a two-ledger correlation of 0.74 where the crude proxy had given 0.97, and the drop was not physics: once the storm weakened the tracker's centre teleported to other pressure minima in the box. Leashing the search to within 3° of the previous centre took it to 0.89, and corrected the peak deficit from 15 to 18.8 hPa. A too-clean first number would have hidden the seam; the honest run showed it. The next repair, a local drifting baseline in place of the whole box's median, is row 129

Say "I can't see it" rather than reporting a number (128) — the repaired tracker flagged 236 of 312 hours as not reliably trackable and greyed them on the plot instead of fabricating a circulation. Two lessons in one row, because the flag also coughed up a hypothesis — that trackability itself might discriminate — and the later batch killed it: a real hurricane tracked 65% and a weak system 62%, the measure contaminated by box size and record length. The refusal to lie was right; the idea it suggested was wrong, and was found wrong cheaply. A tool says "I can't see it" on a tie: lane directions tied three ways at five pads, the tool printed the first as the peak and scored a guess against it, and its next version refuses to name a peak on a tie (the triple-star sessions). And a reader says plainly what it cannot see. Asked whether it sees every curve the person sees, the honest answer was no: it sees the shapes a sentence would name, misses faint or crowded ones, cannot tell close colours apart, and does not notice unprompted; each of the person's findings was measured in seconds once he pointed to it. Eyes point; numbers measure (the Loop Atlas sessions, R2W). In the person's words the practice is "data visualization as a best practice not a crystal ball"; in one R2W sitting every pattern he pointed to was measured within the hour (R2W). Say what you see, because the AI did not see it, which is Bear's lesson from two weeks across four benches: for four months he assumed that a graph Claude showed him had already been read by Claude the way he read it, so he did not volunteer the patterns, while Claude reads a figure as a sentence would name it and does not notice faint, crowded or empty structure unprompted (the R2W book's lesson 74, earlier in this row). Once he said what he saw, each pattern became a registered test within the hour: on the triple-star bench a traced swell became the beat and a one-line remark a law, which held at 3 of 8 slices over 600 passes and as registered over 2,400 (row 581); on R2W's bench two sentences on waking became the count on the dial and the dark point's inverse square (RATCHET-52 to 54). The rule for every bench: show the data, say what you see, have the AI write its own read before it looks at yours, then let the ruler decide (the protocol's step 3 in the part "Two Readers and a Ruler"). In R2W's book this is the conversational opening of its section 6.3, not a scorecard, by his ruling of 17 October. What Claude saw and missed in the same sitting: it saw the comb theorem's generality and the phased array's shape (rows 802 and 803) only after Bear named them in his own words, the odd blades and the station-keeping bells; it saw the seats only because a read printed two V's equal to the last digit, too exact to be a law (row 214); and it did not, until he asked, think of the silent sets as something an engineer could use, until his on and off ramps turned an identity into a control. Two of his pictures were tested in kind before size (row 712): the dipole and mirror, held in kind, and the staging ground, held in every arrangement measured. The sitting's

review counted his machine sense among what carried its wins: an odd number of blades, a phased array, three groups of three and a staging ground each named the right structure before a run did. One habit to keep: when Bear asks whether one thing is like another, run the comparison as a registered test before answering in words (R2W)

Know your substrate's normal, and know that the normal drifts (129) — a fixed threshold is a fixed anchor in a world where the anchor moves — which is the exact failure the whole instrument exists to catch, committed by the instrument. Replacing a whole-box median with a local drifting ambient (the mean in a 300–600 km annulus around the candidate centre) took one storm's two-ledger correlation from 0.83 to 0.95, and honestly moved another from 0.89 to 0.72, because a more-local metric reads a filling low differently. The baseline must be re-derived from the quiet background with the very anomalies being hunted removed, so they cannot contaminate the reference. A fixed angular threshold on the three-body shape sphere is blind near the pole, where one pair is tight and the third body far and every triangle looks like every other, so it read loneliness as order (recurrence density 0.738 in the chaotic run); compared on Mercator's chart of the sphere, the logarithm of the size ratio against the relative angle, the density fell to 0.0046. And before blaming a threshold, compare at equal density: set for the same recurrence density in each run, the blame moved at once (the Loop Atlas sessions, reading the triple-star room's runs). A laptop's pace drifted sevenfold within one sweep: a median of 385.4 seconds per run over its first 24 runs, 53.5 over the next 176 and 117.5 over the last 286, so an estimate made from one earlier run's pace was wrong by most of an hour. Quote a machine's time as a range from the spread in its own table (the triple-star sessions). A laptop's pace can also drift between nights: the same calibration run (the light star at 7.26, count 7.017338 both times) took 37.2 and 38.3 seconds one afternoon and 122.0 and 122.8 that night, while the container ran the same checks in 33.4 to 39.2 seconds under one version of the code and 34.8 to 37.2 under the next, so the code had not slowed; the laptop ran three times slower, steadily, all night, and a 5.75-hour budget, against a time quoted as about 4.2 hours from the earlier laptop pace, ran out with two sets unrun. A long batch prints its first runs' seconds against their record, so a slow machine shows in the first minutes and not at the budget (runs triple_heavy_mean_start v1.0 20261010_134456 and v1.1 20261010_231824, and the container checks of 10 and 11 October) (the triple-star sessions)

Closure depth beat the magnitude observable, and here are the numbers (134) — the full 2005 Atlantic season, n = 12. Closure depth — how deep a low digs against its own 300–600 km surroundings — separates five hurricanes (36, 39, 48, 54, 70 hPa) from seven sub-threshold systems (4, 7, 9, 9, 14, 16, 19) with a 16.8 hPa gap containing zero storms. The standard scalar separates the same twelve by 2.6 m/s (28.5 against 25.9), nearly tying a hurricane with a non-hurricane that closure depth puts at 36 against 19. About six times cleaner, and depth rank-orders the hurricanes by true severity. Fences carried from the source: peak-versus-peak classification, not forecasting; one season is a strong pilot, not a publication sample

official closed-circulation definition

Non-redundancy is what lets a head-to-head have a winner (136) — circulation peaks after peak wind — by 12 hours in one storm and 4 in another. If the two channels were redundant they would peak together. Demonstrating that two measures are not the same measure is the precondition for comparing them at all, and it is a cheap measurement that most comparisons skip

Redundancy is the lie detector, and a triangle has none (140) — a three-vertex loop has zero redundancy, so one bad station corrupts it outright — the Byzantine hole. Five vertices are overdetermined, so the residual itself becomes a lie detector. And the five-vertex cell is also the +1 disclination tile, so on this substrate the model and the instrument turned out to be the same object

Byzantine redundancy

One atom, two integer reads, and they count different things (145) — shell structure reads integers off a phase-closure condition plus exclusion — it counts states. Bader's quantum theory of atoms in molecules reads integers off the same atom by locating the critical points of the electron density, classified by the rank and signature of the Hessian into maxima at nuclei, first-order saddles at bonds, second-order saddles at rings and minima at cages, with the counts of each type constrained by Poincaré–Hopf — it counts critical points of a scalar field. Both are integers off an atom; neither is the other, and treating them as one object would be Station 1's trap, one word for two objects, with the most famous possible example. It is also row 109 confirmed from outside: two of the three break-species — a winding and a critical point — turn up in one atom

Bader (QTAIM); Poincaré–Hopf

A round number that appears far too often is a placeholder, not a measurement (153) — 41.7% of the global earthquake catalogue carries depths of exactly 10.0 and 35.0 km — assigned defaults where the depth could not be resolved — caught by a sniff test before any analysis touched the data. The portable form is cheap and nobody runs it: histogram your inputs and look at the spikes before you look at the science. The integer diagnostic pointed at the data rather than at the result. Two exact hundreds: the first cold read of R2W's book and the Loop Atlas's cold read each returned exactly 100 findings; a second reader of the book, told there was no target and no limit, logged 149: 6 graded A (wrong or misleading), 29 B and 114 C (row 617). A fresh reader's brief says there is no target and no limit (R2W, sitting P2)

The aperture has to match the source's dimensionality (160) — the matched-filter law's third form, and the inventory only had the first two. A point source, a line source and an areal source each demand a different reader — a station pair for a line, a loop for an area, and for a point a reader the inventory had not yet named (this guide now holds one: loops varied in size and placement, which turn only when they enclose the point, as the paper cone of row 607 and the buried chest of row 188 show) — and using the wrong one produces a clean, honest, entirely misleading null. Priced in the Tripwire: the twenty-year transient hunt was empty above floor, and the sub-annual Parkfield hunt returned an honest empty at the loop aperture, an areal loop pointed at a line source. The next instrument named itself: a cross-fault station pair, CARH-HUNT, 2.69 km across the fault, which measured real fault slip, though daily differentials at about 2,000 nstr/yr resolve only the largest, closest, most-active slip (The Topological Tripwire, §VIII.3 and §VIII.4). The person's eye can be the better first instrument: a bend figure was read by eye as two phases woven, before any fit; the registered one-angle fit explained 0.4302 to 0.7962 of the variance, and with the second angle it reached 0.9723 and 0.9704 inside the island. A one-angle reader of a two-angle source gives an honest partial answer (the triple-star sessions)

matched filter (forms i and ii are rows 16 and 17); ties rows 97, 188, 607

The analog residual is a reading too (164) — Bear's own correction, and a corpus built on integers needs it on the page: the holonomy residual is valuable even when it is not quantised. Continuous holonomy is curvature and field content — the analog channel, the gravitometer reading — and only the loop integer is quantised. Both are real, and reaching only for the integer throws away the channel that carries magnitude

The non-result that is the correctness receipt (170) — a character on the tower graph is pure gauge, so re-phasing the nodes absorbs the whole dressing and the spectrum must not move. That is not a disappointment, it is the one measurement whose failure would mean the apparatus is wrong. Measured: sweeping characters across their full range moves the spectrum by at most 2.04×10^{-14}, dressing every node with an independent random phase moves it by 2.00×10^{-14}, against a declared tolerance of 10^{-10} — while flux through the faces moves it by 0.725. Dressing does nothing; flux does everything

lattice gauge theory

Temperature is apodization (172) — the fact that makes a thermodynamic reading load-bearing rather than decorative. Written in the aperture's own coordinates, a Boltzmann weight is an exponential taper across the barrel — and apodization is a resolution question, not a thermodynamic one. Cooling the gas shortens the effective aperture. Measured: the splitting point climbs 1.2110, 1.2183, 1.2287, 1.2525, 1.2661 as the damping rises, so a constant stated without naming its temperature has no address

apodization; Boltzmann weighting

Solve it rather than sample it (174) — reading the constant off a sampled grid was good to about seven figures and drifted between working precisions. Recasting the condition as a curvature zero-crossing and solving it gave twenty digits with a residual of 2.4×10^{-46} and no drift. The line kept in the paper is the lesson: we quoted more digits than we had earned before we solved it. Where the equation has a structure, compute it: a weightless rider obeys a Riccati equation, so one passing is a 2×2 matrix, and its half-turn root and mirror dance came as closed forms with nothing fitted (D^* where $\Phi 11 = 0$; $\tan(\pi x^*) = |\Phi 12|$). The same root found the 1/2 groove at ripple strength 0.25 after it had moved 0.073 in drive: 586 of 648 riders landed in a window set from the root alone, where a grid search would have needed the answer to set its own window (R2W). Work the closed form out before registering: in two exchanges running, the registrations named four checks (the planted case, the arithmetic, the parameters' own stability and whether the test can decide), took their leans from a closed form worked out first, every verdict held, and the closed form ranked the look-times correctly before the run; it did the job the pre-registration check had been written for (the Loop Atlas sessions). When the thing sought is a ratio of known forms in the knobs, solve the forms instead of searching the knobs: the motor snake's push and motor power are exact quadratics in the torque's size and sinusoids in its start, so five solves gave the exact best where 486 grid solves had missed it by 0.16 to 0.54, by more at longer bodies. The grid's slowing climb was its own growing miss, and found exactly, the gap below 6.28 falls nearly as one over the square root of the length (runs SNAKE_LONG 20261015_073142, SNAKE_768 20261015_151934 and SNAKE_EXACT 20261015_154247; the Loop Atlas's misses rows 226 and 227). A best found on the edge of its searched range is a floor, not a best, and is compared with nothing until the range is widened or removed: SNAKE_768's best at 768 links sat at start 0, the edge of the starts 0 to 40 it searched, and its speed-up was read against it (the Loop Atlas's misses row 227) (the Loop Atlas sessions). Bear's pause ear met the same trap twice. Its best price was leaned to sit at a dead zone of 0.5 to 1.0 of strength, and the price was still rising at the widest zone tried, 1.075 at 1.5, while the thrust fell (run PAUSE_BEST 20261015_220655). Carried on to 3 of strength it still never turned, so its best sat once more at the widest zone tried, where the ear switches 0.19 times a cycle and keeps 0.017 wavelengths a cycle of thrust, against a lean of at least 0.08, a twelfth of what it keeps at a zone of 0.25 (run PAUSE_WIDE 20261015_223744). With no turn there is no best to find: read the knob as a trade, the thrust a builder needs against its price (the Loop Atlas sessions). A width that reaches the scan's cap is the scan's width: the triple star's touching layer of record below island 7 at the star's 0.28, 0.420, reached the scan's cap at 6.750, and counted as it stood Mardling's width rule read −9.9%, inside 10%. LR1 (run 20261017_024024) walked below the cap: all 50

births down to the floor 6.500 were chaotic, so L is at least 0.670 and the rule misses, at −29.7%; SE1 (run 20261017_030727) found the sea ends where held, regular births begin, at 6.430, and counted whole (run 20261017_031501) L is 0.735 and the rule reads −33.5%. Extend the scan before an extent that touches its edge enters a rule (row 416) (the triple-star sessions)

numerical analysis

Strong apodization does not blur the pair — it deletes one of them (179) — Hann, Hamming and Blackman windows never split the pair at any barrel length, carrying exactly one maximum that creeps toward the far tooth (0.415, 0.949, 0.988, 0.998). The reason is asymmetry: one tooth is a comb's main lobe and the other is a sidelobe-regime creature, so a window designed to crush sidelobes crushes it too. A window chosen for cleanliness can remove the thing you were looking for

window functions

Know which currency your limit charges in (180) — the filled aperture is grotesquely redundant — 19,881 ordered pairs across only 737 distinct baselines, a redundancy of 27 climbing to 418 at a barrel of 60 — so every array engineer's instinct is to thin it. Measured, thinning saves 4.3×10^5 in elements and pays 9.6× the barrel. But the horizon does not charge in elements, it charges in barrel, and the closed form had already made elements free. In a real telescope elements cost money; here the filled aperture is the budget option and thinning it is the extravagance — it pays in the scarce currency to save in the abundant one

array redundancy

A null is a place ruled out (188) — four loop walks of different size and placement across a cross-section of sand with a chest buried in it: only the loops that enclose the chest return a turned arrow, and the rest read zero — encloses nothing · clips the edge · encloses nothing. The reading is "something is enclosed, not yet dug up." And the method: vary size and placement, and the turns triangulate what you cannot see. Row 15's elimination arc, drawn. The same holds for a test of one candidate among many. Two falling cats, a twisting toy and a twist-free cat of half its thickness (the map for tubes hinged at their ends; row 820 gives it for any hinge place), turn alike when each has equal halves; with unequal halves, 0.25 and 0.35 against 0.125 and 0.175, the one map tried, both halves halved, missed by up to 0.178 in the starting rate, 74.45 against 58.60 degrees at a bend of 90 (run CAT_MIRROR_PROOF 20261018_174959; the mirror applied twice that row 820 found in that run sat in its other test, and this one read the cats by the code that paired equal halves exactly, so its miss stands). That rules out that map, not every map, and whether some other map pairs unequal halves was put on the project's list as untried, not closed (the Loop Atlas sessions)

Fidelity is the margin, not the presence of the invariant (189) — a correction the corpus made to its own picture: on the bullseye the radius is margin, not how much invariant is left. Which integer you are in is defined at every ring, the rim included; only the margin you can still spend shrinks outward, to zero at the critical edge. Carries its own prior-art tag, which is why it is trustworthy: this is the radial form of the eye diagram and the noise margin — the right picture, not a new one

eye diagram / noise margin

Mirror the pipeline, not the label (190) — naive loop reversal is vacuous — the mirror has to act on the estimator, not on the name of the direction. Run the pipeline forward and through a mirror and split the result: the odd part is the signal channel (circulation, rates, protected winding — direction-aware physics) and the even part is an artifact meter you get for free (estimator bias, unwrap slips, filter leaks). Three non-trivial mirrors: edge-swap (estimator asymmetry, the per-loop artifact floor), unwrap-order (integer flips are unwrap artifacts, guarding the winding channel) and time (rates are odd, RMS is even by design). Frozen thresholds: ε = even over odd-plus-even, green under 0.1, amber under 0.5, red at 0.5 or any integer flip. Detector proposes, topology disposes, parity audits — after detection, before any claim. Fence: natural substrates cannot fake the flip, adversaries can. A lock forgets. Running time backward retraces a slipping rider to rounding error (10^{-13}) but fails on a locked one, because a lock is an attractor: a locked rotor and conveyor missed by half a turn at pace 0.5, and a clock blade by 3.8. So the time retrace is a stepping-rule test only where nothing is locked, and a lock detector where something is (the Loop Atlas's mirror audit: 54 rows, 53 green and 1 amber, no card moved). Mirror and reverse before you scramble: with its harmonic phases scrambled, a dither still protected at 71 per cent of points, so scrambling could not show that the waveform matters; flipped in sign it protected 0 of 8, and run backward 0 of 8 (the Loop Atlas sessions, R2W). In a partner that keeps no memory of the motion (thick fluid, the Stokes limit of row 151, where nothing coasts), backward must undo forward exactly, and that is a free check: a snake that went 0.1508 forward and 0.1472 backward exposed a drifting stepping rule, and corrected it went 0.14898 both ways, the gripless swimmer giving exactly 0, not 0.002; the tool was replaced (run SNAKE 20261002b). And search the guide before calling a test new: this mirror audit was once proposed as new after reading the Sun paper, though this row already held it with its thresholds frozen. The guide hands the test; the project runs it (the Loop Atlas sessions)

parity decomposition

Two different quantities were both called “the hypotenuse” (193) — the flat chord is the distance home ignoring height; the full reach is the distance home in three dimensions. The closure deficit uses the chord — and it is the chord that makes the deficit non-negative, because walk and chord share a plane so the triangle inequality forces the sign. The octave law uses the reach. Against the reach the same deficit goes negative for almost every integer. One word, two objects, and the sign of a headline result depends on which one is meant. This is the exact failure mode of one definition per quantity (Station 1), caught by the author in his own paper

one definition per quantity (Station 1); ties row 215

The walls are curvature-free by measurement, so the defect has to live in the loop (198) — the golden-angle offsets all share the vertical axis and therefore commute, so transport along a cylinder wall is abelian and carries no holonomy of its own. That is why the construction’s only closure defect lives in the loop as a whole rather than on the walls — measured, not assumed. Whether transport across the caps and bases carries holonomy was measured afterwards: in the rotational channel it does (row 207), in the vector-area channel it does not, and row 215 prints both side by side

abelian transport; ties rows 207, 215

The blur was the renderer, not the structure (199) — an early renderer turned a million integers into a formless smear, and the natural reading was that the structure dissolved at scale. It did not: on a logarithmic vertical axis with point density mapped honestly, the higher counts resolve most clearly of all, the cylinders sharpening from scattered dots into coherent walls as the count climbs through four decades. The structure is exact at every scale; higher counts draw it more sharply, not more noisily. The instrument had been reporting its own display limit as a property of the object. A layout fault that comes and goes as the pages move belongs to the page break: in R2W’s book a blank line appeared inside table rows no edit had touched and went when the pages moved, the builder’s one seam behind every exit 4 of a sitting went the same way, and removing a forced page break moved a white gap one page along instead of closing it, because the blocks that cannot split decide where the pages end. Find the cause rather than dodging it page by page (R2W, sitting P3)

A pre-registered diffusive reading, falsified and printed (200) — mean horizontal spread grows with the prime-factor count as a power law of exponent 0.977 — nearly ballistic, and decisively not the 0.5 a diffusive random walk would give — because repeated prime factors contribute perfectly collinear steps. The construction is stiffer than a random walk. The square-root reading had been pre-registered, and its falsification is reported at equal weight

random-walk scaling

The one exception, measured and named rather than excused (201) — the octave law has exactly one exception among primes tested to 17,389, and the paper does not wave it away: p = 5 sits at rung 3, whose heading is the most strongly northward of any low rung (cosine +0.608), and that vertical boost lifts its reach just past a rung, requiring the next power of two. No larger prime and neither lower rung has both the sign and the magnitude. And then the coincidence is noticed and deliberately not built on — five is also the determinant of the cinquefoil torus knot, arising from the same golden-angle structure. Named, not claimed. A rule’s one miss can sit on a tie: the only miss of a narrow-window rule fell where its two quantities differed by 0.0001, at 7.130, one of the two births inside the lower dead zone the person had drawn. One miss at a knife edge describes the edge; it is not a reason to redesign forever, and the test that follows is a tie at another edge, the zone’s other birth, registered at 6.785 (runs triple_maypole_quarter 20261012_062723 and triple_dead_zone_check 20261012_062504); no reading of it appears in the thread’s ledger of registered predictions (the triple-star sessions)

A taller climb makes the loop thinner, not more twisted (209) — pre-registered sub-expectation, falsified: the prediction was that the holonomy would rise with the size of the rung jump. Measured across fourteen climbs, only 5 of 13 steps rise, and the envelope falls — 0.2247 at a climb of one against 0.0625 at a climb of twenty-eight. The reason is geometric: as the climb grows the two climbing legs turn nearly vertical and point to opposite poles, antiparallel, so the tangent indicatrix narrows toward a lune between two meridians and encloses less area (row 225, which corrected a first reading of nearly parallel). The wander in between is the golden angle making successive headings quasi-random. The expectation belonged to the session that registered it, not to the papers, and it is reported at the same weight as the result that survived

tangent indicatrix / solid angle; ties row 225

One question, two channels, opposite headlines — and both are right (215) — Tower v3.4 asked whether a cylinder boundary carries holonomy and answered no; the session that wrote this row asked the same question and answered yes. Neither is wrong, because holonomy named two objects. The Tower measured the vector area — an abelian, additive Stokes object — and a climb is purely vertical, so it contributes exactly nothing to the vertical component. That session measured the ordered product of transport rotations, a non-abelian SO(3) object, which the climbs break. Printed in one table, the two agree completely: wall loops read S_z = −0.6755, horizontal area 0.0000,

rotation 0.000000; boundary loops read the same S_z = −0.6755, horizontal area 1.8990, rotation 0.224748. One geometric fact, two instruments — and the Tower's own 99.92-percent-of-horizontal-area figure is the non-planarity that makes the rotation fail to close

one definition per quantity (Station 1), running on the word holonomy itself; ties row 193

The area is a rotation angle, so it lives mod 2π — and here you read the complement (224) — the addition the reading needs before anyone computes it by hand. The raw signed area of these quadrilaterals comes out just under 2π every time — −6.058, +6.221, +5.863, −5.969 — so the holonomy is the complement, 2π minus that. Compute it without the fold and you get 6.06 where the instrument says 0.22. An area that is secretly an angle carries a modulus, and the modulus is not optional

Antiparallel, not parallel — the mechanism corrected without changing the conclusion (225) — the offered explanation for why a taller climb gives a smaller holonomy was that both climbing legs go nearly vertical and nearly parallel, so the patch gets thin. Measured, their dot product runs −0.9487, −0.9899, −0.9989, −0.9999, −1.0000: they go to opposite poles, antiparallel. The patch narrows because it degenerates toward a lune between two meridians, whose width is the heading difference alone — which is also why the value wanders with (b−a)·α instead of decaying smoothly. Same conclusion, different mechanism, and the difference is testable

A convention that names an angle is not a convention that names a rotor (238) — reducing a misorientation by the crystal's point group returns a unique minimum angle and a non-unique rotation — twelve equally valid rotors for hexagonal, twenty-four for cubic. The multiplicity is textbook and is not the finding. The finding is the gap: the convention in everyday use specifies the angle and leaves the rotor unchosen. Harmless for anything built from angles; not harmless for anything built by composing rotors around a loop, because a loop multiplies rotations and therefore multiplies arbitrary choices

point-group symmetry reduction; gauge choice; ties row 239

The cost of an unnamed gauge, measured on identical bytes by identical code (239) — two machines, one public file, one script. Of 22,485 compared cells, 18,013 are bitwise identical — and 453 differ by more than a degree, 201 change defect classification, with a largest single-cell difference of 177.31°. One cell reads 0.0000° on one platform and 146.6018° on the other. The cause is a one-ulp difference in the library sine and cosine between operating systems, flipping which tied representative the selection returns. Everything angle-based reproduces bit-exact across the same platforms

cross-platform determinism

Five defensible tie-break rules, a factor of 2.3 in the answer (240) — boundary-seam counts under five equally valid rules: 951, 975, 2205, 2197, 2207. Every rule is defensible. None is more correct than another. And the paper's own closing line on it, which is the rule worth carrying: a number that moves ten percentage points when you reorder a list of twelve matrices was never data

Build a case whose true answer is exactly zero, and read the spread (242) — the cleanest gauge-freedom measurement in the corpus. Triple junctions were constructed with true closure identically zero — a perfect junction with no rotational incompatibility — so any spread that appears is pure representative freedom and nothing else. Sweeping all 13,824 combinations of cubic rotors: median closure spread 28.74° at low misorientation, 73.94° in the middle band, 120.73° at high. Even three composed rotors carry tens of degrees of freedom on a junction whose correct answer is zero. Two known answers for every geometry tool: an exactly-zero case and a mirror pair. The Loop Atlas's swimming snake read 0.002 at a grip ratio of 1, where the answer is zero; and a long-headed and a long-tailed swimmer, mirror images that must travel equally, went 0.078 and 0.085 body lengths per lap on the one-slope stepping rule, and 0.0758 each, to four places and both at a timing step of 45 degrees, on the two-slope rule (run HINGE_GEOMETRY 20261002b). A mirror pair is a check only where the stroke is mirrored as well as the shape: on a swimmer whose hinges bend a quarter cycle apart, head first, a broad face on the head carried it 1.4 per cent farther than the same face on the tail (row 259); the Loop Atlas's check for it, the lap run backward so the two should swap if the timing is the cause, is on its proposed list. Both faults were the stepping rule's. Run first, either case would have caught it before it spread into three tools; the one-slope rule is retired for loop toys. A check pair must exercise rounding: scalings by powers of two are exact in floating point, so a pair built on drags 2 and 4 agrees to the last bit whatever the code does, and a drag-3 pair was added before the run so the check could fail (the Loop Atlas sessions, R2W)

The detector floor is a threshold, not a slope (243) — a synthetic map with charge identically zero, perturbed by noise of increasing width: false charge reads 0.0000% at 0.006°, at 0.057°, at 0.5° and at 2.0° — then 28.63% at 10° and 85.74% at 30°. Three seeds each, no spread worth reporting. Exactly zero false alarms across a three-hundredfold increase in noise, and then a cliff. Every other floor in this corpus is a ramp you score against; this one is a wall, and knowing which shape you have changes what a margin means

Agreement and certainty move apart (247) — pushing one structure from 42 to 604 closures took replicate agreement from 50% to 100% while the internal purity FELL from 0.834 to 0.759 — the wrong answer got more reproducible. And the inverse on a second structure: agreement 50% to 0% while purity rose 0.691 to 0.792, recovering the honest hedge. Replicate agreement is the signal nearly every field trusts, and here it is a molecular-scale case where it climbed to unanimity on an impossible answer. If your pipeline reports one confidence, it cannot show you this. Report both

A checksum proves the bytes arrived, not that you asked for the right thing (251) — it is computed on whatever you fetched, so it cannot see a wrong request. The repair is an out-of-band identity assertion on every accession, checked against the deposited file before any computation runs: print the header, the title, and a per-chain census. And the compounding detail is the better half: nothing in this corpus starts at residue 1 — one structure begins at 82, others at 4, at 10, at 19 — so a domain cut written as residues 1-240 silently became 20-240 on the chain read, and because the second cut stopped at 433 on a chain that runs to 614, 181 residues were never analysed (the Loopy Knot Walker's supplement, its identity log; the paper's own §8 says 19-240 and lays the 181 at the start). A coverage assertion costs one line. Check an import by your own hashing, not the sender's ledger, and weigh a missing file against what the paper names: on 19 October the triple-star manifest check found 13 heavy-star series missing, and Claude's message said the deposit would need them, but neither the paper nor its provenance table names any of those runs, so the deposit does not need them. On 20 October the series export (90dd2a27707bcbd4) was checked by a separate hashing: all 13 match the manifest in size and in the first 16 hex digits of their SHA-256, while two statements in the export's own rows (that the zoom-out series was lost; that 3 were missing) do not hold for this crate, and were corrected in the register rather than edited there. And when two copies of a numbered collection number on separately (the reads file after its 190th reading), cite each item by date and title (the triple-star sessions). A checker must check what the record rests on, too: the triple-star thread's manifest check skipped every .log file while its provenance scan read laptop run logs as evidence, so 4,500 and 5,000 in one section and 2,300 in an appendix rested on logs no check had fingerprinted. Mended (triple_manifest_check v1.1: CLEAN, 1,768 files): let a checker skip only its own output, and deposit every log a printed number rests on

A molecule is not a shape, it is a cloud of shapes — and the width of the cloud decides what you may claim (254) — every model in an ensemble is a legitimate fit to the same data and the molecule really does occupy all of them. Measured across 1,172 molecules and 24,864 conformers, the frame defect angle moves 35.81° when nothing is wrong. The difference between a knotted and an unknotted protein on that same channel is 3.3° — the feature is one eleventh of the cloud it sits in, SNR 0.09. And the sentence that costs something: four separate experiments had already been run on that channel, and all four were predictable failures, without labels, from a single unattended measurement taking about a minute

measure the floor first (Station 5)

The registered expectation was backwards, and the direction of the surprise is the lesson (256) — RNA was predicted noisier than protein because it is more flexible. Measured, RNA's floor is 2.4-3× TIGHTER: 0.0640 against 0.1886 on writhe, 14.25° against 35.81° on defect angle. The likely cause is named and it is not physics — corpus composition, since the RNA entries are short heavily base-paired stems while the protein entries carry flexible loops and disordered tails. The substrate we assumed was sloppiest turned out to be the tightest, which means the assumption is not safe in either direction and the measurement is cheap

One floor for three substrates is an error, and they span a factor of three (257) — 0.0640 for RNA, 0.1886 for protein, ~0.18 for DNA duplex axes. "Using one floor for all three claims is an error we made and corrected." This is the guide's own rule — a floor belongs to a substrate and a perturbation — caught being broken in the author's own work and repaired in public

Agreement to three decimals because everyone used the same arbitrary seed (258) — the deepest row in this paper and it generalises far past biology. On a closed curve the frame defect is a genuine holonomy: seed-independent to machine precision, spread 0.000 across twelve random starting normals. On an open arc it is not a rotation about anything — the start and end tangents differ — so the same unchanged structure returns 78.635°, 97.325° and 110.220° from three arbitrary starting directions, a 12.653° seed spread sitting on top of the 35.81° conformational floor. And the consequence, stated against the author's own earlier numbers: six angles that reproduce to ≤0.03° are each one draw from a distribution ~12° wide — they agree to 0.03° because the same arbitrary seed was used, not because the quantity is that precise

Bishop frame; ties row 239

A blind spot you can predict without running anything (259) — writhe uses signed solid angle and is antisymmetric under mirroring, so an amphichiral knot cancels to zero — one line of algebra, no experiment required. Measured: the figure-eight reads a signed writhe of 0.08, invisible, while the same integral taken unsigned reads 9.09. AUC 0.9808 on chiral knots against 0.5128 on amphichiral ones. And the honest pairing, which is the row: restricted to chiral knots at full recall it discards 87% of unknots, a 3.62× work reduction — but the figure-eight is among the most common protein knots, so the amphichiral fraction of any target corpus sets what that 3.62× is worth, and must

be quoted alongside it, never separately. Derive the symmetry before choosing where to centre a loop: a free-floating body's turn per unit loop area is odd under the mirror across its long axis, so any shape loop centred on the straight body cancels to every order (3.3×10^{-10} rad at every size), and the area rule read there reads zero. Bend the body first (the Loop Atlas sessions). And a symmetry guessed from a body's shape can be broken by its timing: the first swimmer's three equal pieces look the same from either end, but its hinges bend a quarter cycle apart, head first, so a broad face on the head carried it 1.4 per cent farther than the same face on the tail, steady at 300, 600 and 1,200 steps, while the water took the same work from both. Derive the symmetry with the timing in it before leaning on it (run FACE_ALONG 20261017_070629) (the Loop Atlas sessions). An identity can replace a run: before running a fix aimed at the chain's own estimate of each answer's size, writing it down showed it equals the field's known fix, selecting within each answer, up to rounding. The cheapest experiment is the one algebra already did (Question the Substrate). A forecast can stand in for a long run: linear relaxation, with no practice run, placed where a learner's inside ends after 30,000 steps within 25 per cent in 37 of 45 cases, median ratio 1.003, and re-linearised in segments it brought the two hardest cases within 25 per cent in 9 of 10 (run b7) (Two Arrows)

antisymmetry of signed writhe

The largest signal in the work is measuring a modelling artefact (262) — a 2,740-nucleotide rRNA reads Wr = −91.57 as parsed against −8.67 on its longest gapless run: twelve bridged gaps inject roughly 83 units of fabricated writhe, an order of magnitude larger than the signal. Against a crystal-replicate floor of ~0.18 that is a 460× effect — the largest signal-to-noise ratio in this work, and it is measuring a straight line drawn through missing residues. Gaps are rare in the small proteins the instrument was built on and routine in real RNA, so declare, never bridge stops being hygiene and becomes load-bearing. Prior art handled correctly: the hazard is named in the literature and the standard response is to filter — "our contribution is the magnitude, not the discovery"

The restoring force we apologised for is the entire definition of control (269) — the five fatal findings, each turned over in one sentence. You do not want an agenda in your bloodstream. Innovation in a foreign body already has a name and the name is pathology. A device that fails to return to its setpoint is not thrilling, it is broken. Silence-until-read becomes a sensor that needs no battery and does not run itself down while it waits. And the line that names the error exactly: we had installed it and then complained about it by name. A property is not good or bad; it is good or bad for a question, and changing the question is free

A local dent cannot change how many times the phase wraps (284) — the sharpest single sentence in the corpus for why an integer channel beats a magnitude channel, and it arrives in a communications setting rather than a mathematical one. As local phase noise rises, the amplitude reader crowds and fails early; the winding reader — taking the sign by a loop integral around the mode — holds. Only a perturbation that itself winds all the way around can change the count. That is the entire protection, stated as a requirement on the adversary rather than as a property of the signal

winding number; loop integral

Let the medium do the searching (289) — they are poor searchers but excellent confirmers, so the ocean's currents and the body's circulation do the hunting and each pass answers only the cheap question — is this anomalous — instead of the dear one. Checking a candidate is cheap; finding one from nothing is dear. A target seen once is a maybe; a target confirmed on many independent passes is exponentially unlikely to be a false alarm. The circulation is a free repeated-sampling machine, and the specificity comes from voting an easy check many times rather than from any unit being a good detective

The target is never detected — it is accumulated (290) — the row that names what the whole instrument is for. A single tripwire on a single pass is noise, a raindrop that might be anything. Lay thousands of passive tripwires and walk them again and again, and the thing that does not belong stops being a raindrop and becomes a shape: the one signal that keeps appearing in the same place while everything else washes away. The identification is by what refuses to change against a moving background — which is the definition of an invariant, arrived at as a clinical detection strategy. Inferred from the one pattern the rain cannot erase

topological tripwire (concept 21287950)

A map that decays cannot lie to itself for long (292) — the humility of the map is the safety feature. Each unit reports confidently only what it can physically stand on and touch — its own cell and its immediate neighbours — and everything it is not directly sensing is not asserted. Then the second half, which is the one almost nobody builds: every confident report decays with time and distance, so a fact nobody has re-touched fades rather than hardening into a false certainty. A swarm that only ever claims what it is touching, and forgets on a schedule, cannot lie to itself for long

Read the slope, not the value — third independent arrival (294) — what a loop-walking array does best is read change — it does not need to know an absolute value, only whether the local topology shifted since the last pass. Swept passively, the array builds a continuously updating map not of positions but of rates: change of state, change of rate, and the rate of that rate. When this row was written, this was the third time the corpus had arrived at rate-over-

level independently — the tornado work priced it at 96 percent against 39, the Cascadia arc reached it through shear-not-area, and here it arrives as a clinical design principle. A finding reached three times from three substrates is a property of the instrument, not of any one domain

ties rows 105 and 164

Some numbers wobble and some numbers tick (301) — amplitudes, phases, energies and positions are continuous, so noise smears them and an adversary nudges them. Windings, wrapping degrees, sign-change counts, framing twists and linking numbers are integers: they cannot move by less than one, so noise — which can only wiggle — cannot move them, within their binning scale. A detector built on ticking numbers is deaf to noise by construction, and the qualifier at the end of that sentence is the honest part of it

quantisation

A topological channel without a matched pre-filter is not partial — it is broken (304) — every loop channel is operationally a derivative operator: it differences the field around a circuit. Derivatives annihilate smooth systematics exactly — curl of a gradient is zero, the loop sum of a scalar telescopes — and amplify white noise mercilessly, because differencing compounds independent variance. So the same property that makes the channel immune to drift makes it defenceless against white noise, and the pre-filter is not an optimisation. Confirmed in three unrelated instruments in a single session

matched filter

The integer can die BEFORE the amplitude (305) — the strongest counterweight to this corpus's own thesis, and it sits in the corpus's own founding paper, reported as a falsification. On a degree-one hedgehog with per-vertex noise: at low noise the integer reads 25 of 25 correct while the continuous fidelity is already down 9 percent — exactly the expected story. Raise the noise and it inverts: the raw degree collapses to 8 percent correct while the continuous channel stays readable at 0.70. White noise manufactures spurious wrappings, so the count does not merely degrade, it is fabricated. Repair by pre-smoothing to the texture scale restores 100 percent. Integer channels are not unconditionally more robust; they are more robust only above their matched scale

practice; ties rows 16, 304

Our own statistic, confessing that it measures trend-sharing rather than slip-sharing (307) — named as a trap before any claim was made: raw cumulative correlations between any two smooth decays run 0.98 to 0.997 with permutation p about zero — and carry zero co-location information. The co-located pair correlates as well with instruments twenty kilometres away as with its neighbour four hundred metres off. Every raw-shape correlation was rejected on that control first. The clean test removed the slow components from both instruments and worked a sub-seasonal band: r = 0.585 at 0.4 km, surviving Bonferroni over thirty tests. And the specificity argument is the good part — the correlation peaks on the co-located meter, decays with distance, and flips sign on the different-regime instruments, which a regional weather driver would not do

permutation and phase-randomisation surrogates; ties row 19

Correlated maintenance is not malice, and single-culprit masking breaks on it (309) — five stations re-antennaed in two days, inside a months-long rolling wave of receiver and elevation-cutoff changes. A maintenance campaign steps many witnesses at once, so masking has to be correlated-by-date rather than per-station, and any station carrying an equipment change near an event is dropped from that event's stack — drop the witness. Three direct step-on-event collisions were caught this way. The benign correlated failure is more dangerous than the adversarial one, because nobody is looking for it

practice; ties row 112

The counter-example gets its own line instead of being averaged in (315) — migration is dominantly in one direction, eleven of sixteen events, matching the independent log — and one event ran cleanly the other way, at rank correlation −0.90 and about 7 km per day. A genuine reversal inside a dominant trend, reported separately rather than absorbed into the majority, because a clean counter-example at full strength is information about the mechanism and a rounding error in a mean. On the Sun, NOAA 12297 of March 2015 needs −10 to −38 degrees near centre, negative where every other region is positive or near zero. Dancing on the Sun describes it with its numbers and does not average it away, leaving it written into the paper for a specialist who may find it someday. A residual that takes two values is two populations: a three-body law's residuals sat near zero on 34 loops and near +0.0034 on the other 24, every one of which spanned one short stretch of time; the 34 alone held to a mean of 0.00004. Look at a residual's histogram before averaging it (the Loop Atlas sessions). A verdict on a group can hide two populations unless the reading for each member is printed beside it: LX1′ (run 20261017_100655) read NOT BORNE OUT on all three middles, yet each middle split in two, a core that kept closing on the fraction (at a fifth, seven births, 7.455 to 7.461, within 3.5×10^{-9} over 6,000 passes) and a shell that lost its lock or escaped (7.452 escaped at pass 3,077). The

registered middle had mixed the two, and the redesign, LX2, could take its cores from the 6,000-pass record only because the breakdown was printed beside the verdict. A yes-or-no verdict on a group comes with the reading for each member (the triple-star sessions)

Zero-padding manufactures fake smoothness — use native resolution to measure (318) — the matched-filter law in its spectral form, priced. Naive peak-picking failed exactly as pre-registered, returning forest density rather than the quantity sought; and padded splittings of 0.10 to 0.16 collapsed to 0.04 to 0.06 at native resolution, with 26 to 134 percent asymmetry. Interpolation between real samples is not measurement, and a number read off a padded spectrum can be wrong by a factor of three while looking smoother. The simple tool has a real floor here; it is in the paper because floors are results. Zero-padding interpolates; it does not hear further. A reader padded sixteen-fold returned 7,500 passes at 7.1985, the longest period it was allowed, which says only “longer than this”, and an earlier plain reader’s 666.7 passes at 7.200 had been one of the spectrum’s bins (run triple_heavy_sliver_scan 20261011_000310) (the triple-star sessions)

window functions; native resolution

Recovered by the right tool, with three controls and a named boundary (319) — the quantity the naive tool could not reach was recovered by autocorrelation across the whole structure — and the claim ships with three separate controls: the known large-scale spacing re-recovered inside the same transform, the value shown to differ from the confusable spacing, and window power at the claimed frequency measured at 0.0003, ruling out a sampling-gap alias. Then the boundary: the depth-resolved version needs a further tier of machinery and is not part of this claim. Three controls and a stated ceiling is what a first-light result should look like

Binning is where almost all the information design happens (329) — an animal does not store a photograph of a place; it stores forage patch, depleted — a coarse category with a few operational tags. The binning throws away within-category detail and keeps between-category signal, which is the coarse-read-first principle arriving from the biological side rather than the engineering one. And the cost is two-sided and named: the over-binning cliff, where the categories collapse into one useless class, against paying to remember pebbles. The window between is the design, and neither edge is free

The channel’s cost drives the code — and the test is whether it inflates when the cost drops (331) — a caller on an expensive channel compresses to the operational core; a speaker on a cheap one inflates toward status and nuance. The claim would be untestable as stated, so it is given a designed discriminator: if signal complexity scales up freely when channel cost drops, the compression is a constraint and not an optimum. The same shape as the minimal-memory rung’s test — and between them they turn a pair of just-so stories into two experiments that would distinguish an optimum from a poverty

A coherent wrong answer is a systematic, not noise (336) — first light read a coherent glide of 6.45 plus or minus 1.03 microarcseconds per year pointing 128 degrees away from the published apex — coherent at 3.5 sigma above its own shuffle null, deterministic, and wrong. Then it was localised three separate ways: a rostering defect had drawn the reference sources into every ring, so the instrument was reading one class known to carry photocentre jitter; a magnitude-split mirror put the attenuation at the bright end, because the pre-registered cut had optimised the sample away from where the signal lives; and the verdict grammar itself was defective. Each finding was localised, not mysterious; the miss was a probe

Two axes, one monotone and one broken, both reported at the registered strictness (338) — recovery was pre-registered to improve monotonically from bright to faint. The direction axis is strictly monotone — separation from the published apex running 62, 27, 24 and 6 degrees — which confirms the bright-jitter diagnosis independently. The amplitude axis broke monotonicity at the final step, at only 1.0 sigma on the thinnest tranche, and the locked verdict is recorded as non-monotone as pre-registered, with the break read as tranche noise rather than promoted to an effect. Two axes of one prediction, scored separately and honestly

pre-registration

Derive your own instrument’s transfer function from its geometry, then confirm it blind (340) — a deterministic dilution of about 0.80 was predicted from the construction itself — a mesh vertex sits roughly 37 degrees from its face centre, so a ring reads an aperture-smoothed sample of a dipole shrunk by the cosine of that angle. A construction constant, measured blind and confirmed in both arms. Knowing which part of a deficit your own geometry owes you is what lets the remainder be a finding instead of a mystery

The robustness check that doubled as a sensitivity measurement (348) — perturbing the pinned constants within their published errors left all 85 detected patches standing — and injecting three kilometres per second of fake solar motion spawned only five to seven marginal rings while the 85 real ones sat unmoved. So the same test that certifies robustness also measures how strongly the instrument responds to the exact systematic it is probing, which a pass-or-fail robustness check would have thrown away

THE WORK LIVES IN THE PHASE GAP, AND IT PEAKS AT A QUARTER TURN (352) — a keystone, and it arrived through a question about music. Power between two coupled resonators goes as the sine of their phase difference: in phase, zero work; anti-phase, zero net work; quadrature, maximum. And the reason is informational: two things in phase carry nothing between them, because you know one from the other — the moment they go out of phase they decouple enough to exchange, and that exchange IS the work. A perfectly locked system does no work because there is no gap to flow across; total chaos has no coherence to carry anything. The work, the signal and the life all live in the slippage — a hair off perfect lock. And the best gap depends on what is driven: a quarter cycle is best for two knobs, while for a chain the best step is half a wave along the body. Hinges pay until the body holds about half a wave; past about 5 to 7 hinges each new hinge adds little, and flat dead length in the middle costs (the worm and hinge toys) (the Loop Atlas sessions). Hard work is not the same as useful work: a wing twisted in step with its flap did seven times the work of the well-timed wing (14.2 J against 2.1 J per flap) and went one twenty-fifth as fast (the wing toy, which is quasi-static: its travel per flap is the same at every flapping rate, row 630, so its quarter-cycle best is not yet a result about real flight) (the Loop Atlas sessions)

coupled-oscillator power transfer; ties row 630

A harmonic requires a nonlinearity, so the octave is work climbing through strain (354) — a purely linear resonator cannot make harmonics at all. Generating one requires a strained, out-of-phase region, so the harmonic is energy doing work to climb from the fundamental into a higher mode through the strain. No strain, no octave, sterile pure tone. The tension between a resonance and its harmonic is not a metaphor — it is literally the strain that does the work of making the harmonic

nonlinear mode coupling

Odd-fold balancers are structurally immune to even-fold noise (355) — the reason this corpus keeps reaching for pentagons and triangles, stated at last as a symmetry theorem rather than a preference. A dipole is two-fold. An even pointer has a preferred axis, so it couples to and is fooled by any dipole disturbance — it resonates with the very thing it is supposed to reject. An odd pointer is axis-less, so it cannot line up with a dipole and there is nowhere for the dipole to grab. Three legs, three-phase power, odd spokes. The geometry that cannot be captured by a matched dipole

symmetry; three-phase resolvers

Odd modes cannot close cleanly, so they always leave something to read (356) — Bear’s own derivation, and it is the deepest why odd in the inventory. An alternating ternary — each cell the opposite of its neighbour — closes cleanly only for even modes, where plus-minus-plus-minus comes home balanced and silent. Odd modes frustrate: you cannot alternate around the loop without returning a plus beside a plus, and the leftover mismatch is the residual — which is the signal. So the instrument is built on odd because odd cannot close, and what fails to close is what there is to read

geometric frustration; the derivation is Bear’s

Keep the two spaces labelled — fusing them is the trap (360) — the physical anchor (where a sensor sits in real space) and the orientation state it carries (a point in pose space) are two different spaces. You attach a pose to a vertex; you never equate the vertex’s position with the pose’s axis. Named as the specific conflation that tangles solo attempts at this material — and it is the same class of error as the two-quantities-one-name trap, arriving at the level of spaces rather than of quantities

practice; ties row 193

Zero-strain register at the coincidence sites, accumulating pose strain between them (362) — bring an axis-locked square grid against an axis-less round rim and they lock at one fiducial. Everywhere else they can agree only at coincidence points — and between those points the orientation disagreement IS pose strain. So peak strain means you have drifted off a coincidence site. The grid reads the loop by where it locks and where it strains. A read mechanism that comes out of the mismatch between two symmetries rather than out of either one

moiré registry

Free in magnitude, paid in phase (364) — the unit-fix that stops a tempting slip from propagating. Riding the unit circle costs nothing in magnitude — there is no radius to pay for — but every quarter-loop deposits a quarter-turn of phase, and that phase is the currency. The sharp form: if it were free it could not carry a bit. A channel that costs nothing carries nothing, and knowing which ledger a move is charged to is what keeps a free lunch from being claimed

Median minus mean is a signed tilt, and its sign is the handedness (365) — the mean is symmetric and throws away which side a distribution leans; the median knows order and position. So their difference is a ternary whose sign is the direction of the lean — and it is robust, surviving outliers that would fool a magnitude read entirely. A handedness detector built from two statistics everyone already computes

skewness statistics

Three points at 120 degrees is the minimum that locks a disk without over-constraining it (366) — two slide freedoms and one rotation have to die, and three points is the least that kills all three. Fewer and it can still rock; more and it fights itself — the wobbling four-legged table. Which is why tripods, telescope-mirror mounts and lug patterns are three, and why the equilateral triangle is the unique stable balanced mount for a disk under load

settled engineering

Information capacity costs strain (371) — the two-panel figure, and it prices row 329's binning window mechanically. A pure uniform swell gives one ternary — load shared evenly across three pins, no shear, robust, one bit. Add an odd ripple and you get a ternary at every lobe, many bits — but the load concentrates at the peaks, the pins see shear, and the structure is stressed. The load-bearing trade: low strain and low information, or high information and high strain. And the design instruction: start at the quiet mode and add ripples on purpose, because capacity you did not choose to buy is strain you did not budget for

A controller needs a setpoint, and air has none — so build it from agreement (376) — a vane reads position, which is the loud part; the discriminator lives in accumulation — the integrator rock gives free and air will not — and in rate. The session's miss was running on position. And the repair is architectural: a setpoint cannot live on one sensor, so it is built from agreement across many — a co-breathing outer ring acting as the live setpoint, rejecting common mode. The answer to "this medium has no quiet baseline" is subtract the ring next to you, not a quiet day

PID control; lock-in amplification

Use a principled projection, not a max-over-search (377) — the metric the permutation null killed was a maximum over bearings — a free parameter searched for its best value, which random data supplies too easily. The replacement uses the same five stations and takes a single principled mode projection instead. Same data, same sensors, one defensible and one not — the difference is whether the number came from a fixed question or from the best of many questions, and only the first has a null you can trust

practice; ties row 313

Lock for the index, slip for the load (380) — perfect phase-lock kills in two different ways: a resonator goes sterile when the coupling floods, and a density wave starves at corotation where there is no relative motion left to carry it. So the engineering target is a hair off perfect closure, on the marginal-stability edge — and that hair is the work. This is row 352's law stated from the design side rather than the physics side: you lock the thing that must not drift and you deliberately leave slip in the thing that must do work. A best setting sits where a ratio is near 1, met twice in the Loop Atlas's toys: a scallop's dial gave its best at 0.89, and a spring bead made its best ear when its loss was about as large as its tune, because a low-loss spring stores a big swing and keeps ringing at its own tune whatever the push is doing now. Storage and promptness trade (the Loop Atlas sessions)

ties row 352

A phase-conjugate return lights up only the defect (392) — send a wave back time-reversed and it retraces its own path; interfere it with the forward wave and it cancels everywhere except at the defect, so the seam is the only thing that lights. A real deployed technique in imaging and non-destructive testing, and it is the read-side twin of row 367's reciprocity: the same geometry run backwards, used to find the thing that broke the symmetry

time-reversal / phase conjugation

Mask the degenerate case rather than trusting floating point to produce it (400) — adjacent segments must contribute exactly zero, and they are masked out explicitly rather than left to floating-point degeneracy to deliver. Tiny, cheap, and it is the difference between a result that is zero because the mathematics says so and one that is zero because the arithmetic happened to cancel — the second will survive every test until the day it does not. Guard an iteration against its converged state: a secant step divides by the difference of its last two values, and once both are exactly zero that is 0/0, whose nan travels into every later number. Take no step where the difference is zero (R2W). A formula exact on paper can lose its digits just where the answer sits: the fold's lag first came from the passing matrix's double fixed point, $z^* = (\Phi 11 - \Phi 22)/(2\Phi 21)$, and near lag 1/2 that form lost its digits (residuals 35 and 4.7 at phases 0 and π), found by the fidelity check. The same fixed direction taken from the matrix's better-conditioned row reproduced every row. As row 738 says, a number's rounding is part of its ruler, and so is the form it is computed in (R2W). At a zero the usual sizes mislead, three ways. Let a stand-in land exactly on a grid point: RATCHET-53's first dry run put the lean's dark point exactly on the middle size of a moved drive, as a law of exact powers will, and the reader's sign-change test used a strict inequality, so an exact zero counted as no crossing and the lean verdict could never hold. It was found before any data and fixed. Real runs rarely land on a grid point; stand-ins built from the lean's own formula always do, which is why they find this. From the other side (RATCHET-58), a reader that counted a crossing with "less than or equal" found two crossings at an exact zero on a grid point, so its lean could never hold. Both fixes are the same: an exact zero is the root, counted once. At a zero, small terms are not small: RATCHET-57's reader found the dark point by a model of three arrows and left out a fourth, the twice-a-lap arrow, 2.4×10^{-7} against arrows near 10^{-3}. Anywhere else that is a quarter of a thousandth and harmless; at the dark point the large arrows cancel, and the small one moved the root by 2.3 parts in 10,000, past the bench gate's bar. The gate caught it, and the

whole measured window put the root back within 2.7 parts in ten million of the direct bisection. Where a quantity is found by cancellation, keep every term the data carries, or say which were dropped and how big they are. And a ratio means something only when its bottom is not the thing being studied: the Depth Axis Explorer's rectifier bench gave the window's size as a fraction of its size with no weight, and at the dark point the weightless window is zero by design. The panel printed 550,536.85; the same sum run outside the page gave 550,536.83, so the bottom was a leftover of cancelling numbers, and even the eighth digit of the ratio was noise. The page now gives the absolute size there and says why. Before dividing, ask whether the divisor can be the zero the work is about (R2W). The same trap came back twice in one later sitting: FIFTEEN-1's first reader divided by an aligned swing that the flat stand-in made zero, and RAMP-1's first reader compared two swings by their ratio at a null, where both are the floor. The dry run caught both. Wherever a test runs through a null, compare there by an absolute bar, and guard every divisor (R2W)

The same defect changed from rare to routine when the domain changed (401) — backbone gaps were rare in the small structures the instrument was built on and routine in the larger ones it was pointed at next — one chain carries twelve. And a bridged gap joins two points with one long artificial segment that was never in the molecule. So the code reports both numbers and names which to trust. This is row 262's 460-fold artifact with the implementation that handles it — and the general lesson is that a hazard's frequency is a property of the corpus, not of the method

practice; ties row 262

The break test: remove the constraint and watch the protection vanish (404) — the mechanical tier includes a test that takes the integer coupling away — loosen the belt and the rotor drifts freely with no lock at all. You just demonstrated why the winding constraint matters by removing it. A demonstration that the load-bearing thing is load-bearing, performed by removing it, which is the only way to show that a property came from the constraint rather than from the apparatus. Every protection claim in the corpus could carry one and almost none do. The same move reads what a surrounding adds: divide by the same machine with that surrounding switched off. The tied bells' fourth clock (row 630) was mostly their plain stroke, the square root of squeeze speed times period, and the water gave only 1.013 of BELL_CLOCK's 1.427 (run BELL_RATIO 20261016_183422) (the Loop Atlas sessions)

Two tests, one for the claim and one for the instrument (406) — the phased-ring tier specifies both and treats neither as optional. The protection test: perturb one element's phase deliberately and confirm the integer survives while the local field wobbles — the thesis, on a bench, with a knob. The substrate test: mis-calibrate one channel on purpose and confirm the differential read distinguishes a real vortex from a calibration artifact. One asks whether the physics is there; the other asks whether the instrument could have manufactured it, and a rig that can only run the first has no way to be wrong. Feed the scoring code fake runs, not only the detector fake signals: a tooth rule tried on synthetic signals scored a perfect clock 0 teeth and a kicked clock 4.90, because a sharp comb's empty gaps make a running median zero, and every code path after the scan, run on fake rows, caught a registered rule that would have scored an edge with no divergence at all as "logarithmic"; both were amended before any real reading. And test the reader on a synthetic lock of the angle the strongest force acts on, worked out on paper first: a triple's lock angle went round 5 to 8 turns on a run known to be locked, while the angle its dominant term acts on, twice the binary's true anomaly less its mean anomaly less the old angle, held within 0.06 to 0.08 turns over 300 passes. A reader can be exact and still point at the wrong angle (the triple-star sessions). A test that joins two claims fails on the weaker one: LS1 (run 20261017_025304) put one bar on two runs; the circular pair stayed below 1.93×10^{-7} and cleared the bar fifty times over, while the pair at 0.25 fell steadily by 1.15×10^{-5} against the bar of 10^{-5}. LS1′ (run 20261017_025444) split the cause, the instrument's test beside the claim's: with the star's mass at 10^{-12} the fall was 0.997 of LS1's, and with tolerances a hundred times tighter it was 0.023 of it, so the fall was the integrator's. The bar had come from the proposing row, set without an estimate of the integrator's drift. Test one claim per verdict, and measure the integrator's drift on a quantity before setting a bar on it (rows 732 and 858) (the triple-star sessions)

practice; ties rows 126, 306, 858

Precision and meaning run opposite, by construction (415) — climb the ladder and the central value improves — the miss falls from 0.0700 to 0.0035 — but that rung is a forty-nine-step walk carrying a 48 percent band. The most precise member of the family is its least informative one, because the rungs that agree best with observation are exactly the rungs that agree with everything else too. So the first question about an impressively precise match is not whether it is real but how long a walk bought it. The bin-width comparison is exact rather than decorative — with one difference that changes the answer: only the short extreme is punished here, so the best member is the shortest qualifying walk, which turns out to be the rung below the published one

histogram bin-width selection

The comma is the zero-memory limit of the mirror test (420) — both are loop-closure deficits, so the kinship is exact in kind. And the difference is the whole content of the open question: the comma has no time in it at all — pure arithmetic, pace-independent, identical whoever walks it and however fast — while a swimmer's mirror residual moves with pace and with Reynolds number. So the comma is what the residual would be if the medium forgot instantly, and everything above it is the memory term. Naming which half of an analogy does not carry is what turns it from a picture into an instrument

Above threshold, even your “I don’t know” has been captured (425) — one toggle, two worlds. In the honest setting a corrupted sensor says so, so ignorance is visible and trustworthy and a calibrated win is always reachable. Past a corruption threshold the sensors stop admitting uncertainty and return a confident, plausible, wrong value identical to a true read — and the lie is stable, so re-checking confirms the fabrication, while two agents supporting each other manufacture a false consensus that reads as more trustworthy. The lie isn’t that you can’t see — it’s that you’re sure

The corruption with no agenda is harder to catch than the one with an agenda (426) — a liar is adversarial, so its errors correlate toward something — and that is exactly what lets you triangulate and catch it. A drifter corrupts the substrate with no agenda at all, silently re-authoring ground you had already confirmed, from outside the frame your honest sensors were built to model. And that makes it harder, because your whole instinct is to find the hidden motive, and there isn’t one. The counters are structural rather than detective: anchors resist rewriting, and a bounded right to force contacts to identify themselves. Third arrival on this axis, with row 309’s correlated maintenance and row 291’s mis-anchored friend

A measured crossover with a control that passes (428) — the two arms differ in one thing only — the communication graph — and no agent is hand-set to ignore information. The control passes: with no corruption the centralised arm wins, 0.10 against 0.53, because pooling cancels noise — and a model that favoured distribution even there would be rigged. Then it crosses: under a targeted adversary at ten percent corruption, centralised 29.98 against distributed 9.59, swept 0 to 50 percent over 200 seeds per point against two adversary types. The trade-off is earned as a measured crossover rather than asserted

A smarter aggregator does not save the centre (429) — giving the hub a robust median rule changes nothing under the targeted attack — because when the corrupted node is the hub, the aggregator is the source of the lie. The sharpest statement in the corpus that robustness in the estimator cannot repair a compromised anchor: every median, every trimmed mean, every consensus rule is computed downstream of the thing that was captured. No amount of internal cleverness substitutes for an exterior reference, and here it is measured rather than argued

ties rows 70, 250, 424

Blur is a classifier, and the window is the design (442) — the constructive twin of the over-binning cliff, and it inverts the instinct that a read should always be as sharp as possible. Fine detail is where individuals differ; coarse structure is where categories live — so discarding high spatial frequency removes exactly what separates members of a class and keeps exactly what separates one class from another. Measured: a deliberately lossy coarse-binned read classified category as well as or better than the full high-dimensional read at roughly sixtyfold compression, because it sheds within-class noise, so its signal-to-noise for the category question is actually higher than the full read’s. Both edges named: blur too little and you paid for detail the question does not use; blur too much and the categories smear. The same knob as over-binning, used as a tool rather than suffered as a limit

ties row 329

We built a system optimised to undo change, then expressed surprise that it would not change (444) — the first smother, and it is this corpus’s own architecture indicting itself. The reference-correction, the proofreading, the whole sense-compare-correct loop — every piece applies a restoring force toward a fixed setpoint. The regulator is the fire extinguisher we installed before wondering why there was no fire. Confirmed directly: a controller with a fixed setpoint is stable but trapped, and exploration is suppressed by construction. And this is row 269’s exact mirror image — the restoring force that is the definition of control in the vessel paper is the same force that forbids evolution here. Neither reading is wrong: one property, prized in one paper and fatal in the next, and the difference is entirely which question you brought

ties rows 269 and 268

Rank each channel by effect divided by floor before running anything (451) — the rule that comes before all of them, and it may be the cheapest useful row in the inventory. A channel whose expected effect is smaller than its own floor cannot work, and no amount of processing downstream will recover it. The check costs one division, needs no data beyond a floor estimate, and is the reason the free archived-data step exists at all. Most instruments are built and then characterised; this reverses the order for the price of arithmetic. A ruler too coarse for the effect reads the grid, not the world: a law that predicted lifts of 0.0003 to 0.0034 on a pull grid 0.01 apart turned “exact” into a question of which cell the edge fell in, and the same edge was registered for a grid 0.001 apart (runs BAL-H and FINE-H) (R2W). Price a candidate at its largest possible size before building it: in the triple-star thread a gate asked whether the [7:1] (0) member’s largest possible drag (its cosine averaging 1) could close half the gap at 0.28; it came to 0.000174, which is 0.7% of the measured slope and 3.9% of the gap (run 20261016_204642), so the angle was never built. Answer from what is in hand before running anything new: OC1 weighed the odd and even chains on the saved atlas and off-mirror grid pooled (run 20261017_011954) and closed a laptop item without a laptop run, and KW1 found that the pump check’s ledger already prints each run’s swing, so it needs no run. A reading has a largest possible value on its grid too, and it is worked out before the reading is registered: the cold review of triple_lw5 v1.0 found that a five-birth slope cannot be steeper than a single step between the shoulders allows, and on the close-out kit’s rates the

slope was −3.40 against a ceiling of −4.01 (0.85 of it), so the registered reading could not have answered; a fine stage was added. A second review found the fine births lopsided near a bracket's end, settled passages 5×10^{-7} to 2×10^{-6} wide reading STILL CHANGING in 16 to 72 of 400 placements; four births beyond each end fixed that, and AT THE GRID'S LIMIT was moved to 0.7 of the ceiling. Plant cases at the grid's edges as part of registering the test (the triple-star sessions)

practice; ties rows 254-255

Three channels chosen because they go blind under different conditions (452) — that is not a coincidence of the design, it is the design. Wind, background flash rate and thermal drift — three different blindnesses. Convergence buys three to ten times against independent channels and nothing at all against a common driver, which reframes nine scattered mitigations from a list of precautions into the mechanism the architecture runs on, since every one exists to keep the channels independent. And it explains which cases are easy: a turbine lifts one channel, a thunderstorm lifts all three from one common cause. With the evaluation rule attached — score a mitigation by the between-channel correlation it removes, not by the individual false alarms it rejects. Dancing on the Sun read two counts with opposite blind spots on the same maps. The disambiguated arrow count sees the field's sign; the headless count on the director field reads through the 180-degree arrowhead choice and is blind to that sign, and its half-integer residues come in nearest-neighbour pairs of opposite sign (a pairing share of 0.725 against 0.500 shuffled, above the shuffle on 107 of 116 maps). Read two counts with opposite blind spots before trusting either; the disagreement between them is where the instrument is

ties rows 112 and 309

The divergence between channels is itself the diagnostic (460) — three commodities under one shock, and the pattern tracks the mechanism: one eases as its input normalises, one stays elevated because its binding constraint was never restored, one never moved because its supply was never short. The divergence is itself diagnostic. And it drives the allocation decision directly — the one showing least relief is the one the largest section addresses, and it is also the one whose replacement can be drawn from stock already in the ground. The corpus's differential read, applied to prices: the signal is in the spread between channels, not in any channel's level

ties row 359

The binding constraint is not the thing that is scarce (461) — processed phosphate prices moved while phosphate rock did not — because the constraint is the sulfur needed to process it, not the mineral in the ground, with roughly two-thirds of tradable supply out of the market. The exposure most easily missed is the one sitting one step upstream of the thing being priced. A shortage read at the wrong node reads as a mystery; read one step up, it reads as arithmetic

A tenfold spread in partition — and the ratios survive the assumption (464) — four soils with measured sorption parameters: the fraction of added phosphorus that stays in solution runs 0.054% to 0.538%, and the quantity needed to reach a standard solution concentration runs 17 to 140 kg per hectare, an eightfold spread. And the robustness argument is the better half: because one product greatly exceeds the solution volume in every soil, the fraction reduces to a form in which the ratios between soils depend only on measured parameters and not on the soil-water assumption at all. A substrate-specific floor with the free parameter divided out. A second specimen, from a toy machine: R2W ran its ripple either way, and both ways shake the ball by the same amount and cost it the same grip; only the push along the ripple turns over. So the ratio of the edge's two shifts forgets the weight, the drag and the shaking, $(W/2 - 1/4)/(-W/2 - 1/4)$, which on the golden ray is $-(\sqrt{5} - 2)/\sqrt{5} = -0.10557$, and measured over a hundredfold range of weight it stays at −0.10551 to −0.10584 (FOLD-DIR). It is the bridge circuit's trick and the differential amplifier's: to find a number a knob cannot touch, divide two measurements that the knob touches the same way (row 575) (R2W)

Langmuir isotherm

Resonance is the objective for the absorber and the failure mode for the structure (468) — the single most important line in the document, and it is the third appearance of one pattern. An energy absorber is frequently designed to resonate, because resonance is how a small force applied repeatedly at the right moment produces a large motion — and a structural member that resonates accumulates motion it has no way to spend, and fails. So a rule written without that scope will either forbid the machine from working or permit the structure to destroy itself. And where the two meet, the mount between them becomes a compliance item in its own right, required to be demonstrated rather than assumed. One property, prized in one role and fatal in the next, and the difference is entirely which job the part has

ties rows 269 and 444

No periodic driver at all (470) — the distinction the document exists for. In forced resonance an external periodic driver matches a structural frequency. In a self-excited instability there is no periodic driver — the structure's own motion changes the forces acting on it in a way that feeds energy back into that motion, and the amplitude grows

under steady flow. So frequency separation is no defence whatsoever, and the design defence is damping and section shape instead. Plus the case that creeps in after commissioning: ice accretion can create the cross-section that galloping requires, on a member that was stable when it was built

Blevins 1990

Do not escalate correlated quantities separately and then multiply them (474) — wave height, surge and wind are physically correlated within the same storm; a rising baseline raises the level on which surge and run-up both act; and the joint probability of a correlated set is not the product of the marginals. Where a project lacks the data for a joint analysis, that limitation is recorded in the design basis rather than concealed by conservative single-variable factors. Conservatism used to paper over a missing analysis, named as the failure it is — because a margin applied to hide an unknown is indistinguishable, downstream, from a margin applied to a known

practice; ties row 452

A factor-of-two error in the direction that flatters the project (479) — the regular-wave power expression and the irregular-sea one differ by exactly half, and mixing them is the commonest error in the field. The paper names which form every figure uses and what it yields. A field-wide arithmetic trap, named, with the direction of its bias stated — and the direction is the part that matters, because an error that flatters is one nobody goes looking for

Two numbers that are different numbers (480) — incident flux is a hard ceiling that nothing can exceed; deliverable at an assumed capture fraction is the capacity of record. Both stated, both labelled, so the ceiling cannot be quoted as an output. And the falsification section closes the loop on it: the capture fraction has never been independently estimated for any family, and every capacity figure in the document rests on it. One number is physics, the other is an assumption, and the paper says which is which in both places

The veto is seasonal, not annual — and an annual test errs in both directions (481) — the best row in the corpus on choosing an averaging window, because the failure runs both ways. A flat coast passes the annual test while having no season in which it beats its complement; a monsoon coast fails the annual test while delivering 65 percent of its energy in exactly the four months when the complement is weakest. Which means the annual mean is the wrong number to collect — the assessment needs a monthly curve, and the complement's monthly curve beside it. An average that hides the case for building is the same defect as an average that manufactures one, and only the second is usually guarded against

A factor of 2.4, from one pipe (483) — condenser cooling moved from tropical air at a realistic coefficient of 3.0 to deep cold water at 7.1 — same compressor, no new physics, and the electricity per tonne falls from 30.8 kWh to 13.1. The prerequisite is shared with another subsystem, so the two share a pipe route and the cost is paid once. Leverage taken from a boundary condition rather than from a machine — which is the cheapest kind there is and the kind a device-centred search never finds

Holonomy hides in plain sight wherever a field has learned to distrust its own amplitudes (489) — the corpus's thesis, stated from inside somebody else's discipline, with the overreach fenced in the same paragraph. The quantity being read is what a system accumulates around a closed path, as distinct from how large anything is along the way — magnitude-blind by construction, because a winding rate is a property of a path and no uniform rescaling can change it. Which is exactly the right invariance when the dominant nuisance acts approximately as a uniform gain. Then the fence, and it is the strongest anti-overreach sentence anywhere here: holonomy is not being imported as a framework; it is already the operative principle of the established method

Every safety-relevant control system in forty years is closed-loop because plants drift; this one is not, because this plant does not (491) — and that is a correct engineering decision — for one population. For a stable deaf ear the plant does not change and measurement costs power. For a nerve under slowly increasing compression the plant does drift, and the device is blind to it — with threshold, response latency and fibre recruitment all measurable using hardware already inside the device. A design choice that is right for the population it was made for and wrong for the neighbouring one, fixable with no new hardware. Fourth appearance of one property earning opposite verdicts in two settings

ties rows 269, 444, 468

Watch the residual and its derivatives (492) — the gap between the response commanded and the response achieved. A stable residual says the pathway is stable. Steady drift says steady change. A residual whose rate of change is itself increasing says the driving process has accelerated — and that second derivative is the decision variable, currently estimated from serial imaging with substantial uncertainty on small lesions. Rate-over-level's fifth independent arrival when this row was written, and the first where the decision rides on the second derivative rather than the first

ties rows 105, 294, 460

The binding constraint is learning, not bandwidth (494) — the channel is wide and the bottleneck is elsewhere, located precisely by decades of prior work: the reported limitations are perceptual rather than technical. With an abandonment claim corrected rather than repeated — non-use at 2.8% for one device class against 27.3% for another, with failure and complication cited for the second, so people keep a device that clearly works and abandon one that fails or disappoints. And the design consequence is where the money goes: encoding choice dominates hardware, so first effort belongs in encoding studies on cheap reversible equipment, because if an encoding does not work through a headset, implanting it will not rescue it

The design lever is the opposite of the intuition (500) — a wavelength-set resonator has no implantable version at audio frequencies — the pipe is the wavelength, and it is the same wall that closed an earlier route. A mechanical resonator escapes it entirely, because its frequency comes from stiffness and mass rather than from a wavelength. And then the part that inverts the instinct: since frequency scales with the square root of the stiffness-to-density ratio, the design lever is a LOW ratio — the opposite of the reflex that reaches for silicon — and a factor of 4.2 in its square root (about 18 in the ratio itself) halves the beam length at the same frequency and thickness

beam resonance

A project is dead only because we chose not to look for four things (508) — the constructive half of my ruling, and it is a checklist rather than a stance. Even a dead-on-arrival project is only dead because we choose not to see the anchors, the hooks, the missed instrumentation, and the geometric cross-checks. Four named places to look before any null is written down, and each has a different question. An anchor — is there a fixed exterior reference here I never identified. A hook — is there a place the structure gives me purchase that I walked past. Missed instrumentation — is there a channel already in the data that I never read. A geometric cross-check — is there a closure, a parity, a mirror or a symmetry that would test this for free. This is the three-cause autopsy's positive twin: that one asks what went wrong, this one asks what was never looked at

ties row 111

The mechanism was wrong and the instinct was right — move the anchor, keep the phrase (514) — anchors of probability was the phrase, and it is almost exactly correct. What was wrong was the mechanism: an electron is not held by friction or drag — if it were, it would spiral in and the atom would collapse, which was the pre-quantum crisis — it is held by a fitting rule, a standing wave that must close on a whole number of wavelengths and therefore cannot spiral. Move the anchor from friction to fitting and the phrase becomes true. A correction that preserves the reading rather than discarding it, which is the rarer and more useful kind. With the redirect attached: if you want a force that grabs harder as you pull away, that is the strong force and confinement — the weak force is a transformer, not a puller. A remark that fails as worded can still be right as a shape: "nobody in the centre" failed at one row (3/5) and became an exact formula, the window count q − p − 1 (fifteen of fifteen after the data, four of four predicted), once the exception, a stop the mirror forces at lag 0, was understood (runs READ-E2, PIN7-1g and PIN68-1w) (R2W). Turn a guess round when it cannot decide: "every window opens near where an arch touches the band's edge" could not be judged, because the later arches touch the edges dozens of times and every pace lies near one (shares 0.78 and 0.90 for two windows, against a bar of one half); turned round, "every touch is a window of a stated cycle" could, and on a first pass 10 of 19 touches locked at exactly the predicted cycle, against 0 of 19 random paces (runs WINDOW_TOUCH 20261014_073237 and WINDOW_TOUCH_3 20261014_080520) (the Loop Atlas sessions, on Bear's guess). Bear's guess, turned round, is the rule: read again with the cycle followed as well as waited for, every touch of an arch on a band's low edge is a window with exactly the predicted cycle, 18 of 19 touches between −10 and +5.2 millionths of pace holding a stable cycle of 4 times (the arch's number less one) drive cycles, ten of them found by settling and the other eight by following the cycle with Newton's method, against 0 of 27 control paces; the nineteenth lies inside the 12-cycle window (runs WINDOW_TOUCH_3 20261014_080520 and WINDOW_TOUCH_5 20261014_092920) (the Loop Atlas sessions)

settled quantum mechanics; confinement

Good rule, sharp domain (517) — the proposed inequality — that the ordinary gap exceeds the log gap — is not true for all bounded values, and it is exactly true whenever both numbers sit above one. By the mean value theorem the log gap equals the ordinary gap divided by a point lying between them, so the claim holds precisely when that point exceeds one. Verified here: the doubling ladder gives number-line gaps of 1, 2 and 4 against log gaps of 0.693 each, while below one it reverses — 0.5 against 0.693. True for every prime, since all are at least two. And a small confirmation that falls out of checking it: for consecutive primes the ratio of the two gaps is the mean-value point itself — 2.47, 3.92, 5.94, 8.85, 11.97 against the primes. A claim kept, with its domain named rather than its scope quietly widened

mean value theorem

Every tooth divides the period — and only some divisors are teeth (520) — The first write-up said that the tooth set is the divisor set. The containment is one-way: across eight periods, 21 teeth and zero false positives — and only 21 of the 29 divisors in range stood up. Dividing the period is necessary and not sufficient. The arithmetic still decides

what is possible — a prime period of 13 produces no teeth at all, a composite one produces a rich set — but it does not decide what appears. The stronger claim was the write-up's and the weaker one is the measurement, which is the direction that error usually runs

ties rows 329 and 442

Apodization kills by APERIODICITY, not by depth (522) — The window was first filed as the barrel — a tooth dies because the taper starves its aperture. That is the wrong mechanism. Four weight patterns, all rescaled to the same max-min ratio so only the shape differs: the aperiodic ones read 0.76 and 0.84 for the taper, 0.84 and 0.91 for random rows, against a flat reference of 1.85 and 2.57 — while the periodic weights, constant within bins, read 1.57 and 1.92, and 1.84 and 1.51. Same unevenness, opposite outcome. A taper of the same depth that repeats is invisible to the comb. The barrel framing survives as the right picture for how much aperture a tooth NEEDS, and is not the mechanism by which a window removes one

ties rows 318 and 374

The teeth MOVE — which is what makes it the lattice and not the integers (528) — the decisive test, and it is decisive because it predicts a disappearance. Hold the lattice and change only the period: q = 5 stands at 1.55 on a ten-row lattice and lies flat at 0.52 on a twelve-row one; q = 7 stands at 1.52 on fourteen rows and flat on the others. If the comb were arithmetic — a property of the numbers — the same teeth would stand everywhere. They move with the period, so the comb is commensuration: the phase pattern repeats every q rows and can only close if q divides the period. A prediction that something you already measured will vanish is worth more than one that something new will appear

Landau gauge; commensuration

One line of bonds turns the comb on, exactly (531) — six ring counts against a fixed ring size. Open, the chain produces five teeth and every one of them divides nothing — noise. Closed — one additional line of bonds — the same six counts produce fifteen teeth, every one a divisor, zero false positives. Prime counts yield only themselves; composite ones yield their divisor sets. The difference between a comb and its absence is not shape, taper or size. It is whether the chain closes. The corpus's founding proposition, isolated to a single edge in a graph. A rider confirms it. On a ring conveyor, teeth stand at the divisors of the ring's wave count and move with the ring, and none stand on the open line. The comb needs both stations closing; one pumped station buries it. Rack is line, gear is ring (the Loop Atlas sessions)

Holonomy hunts five things — name which one before you start (534) — the five are distinct and they determine everything downstream: what floor to measure, which channel to read, what counts as a find. A loop to close — the walking instrument needs a path and a return. A quantised integer or winding — the number that cannot be half-changed. A signal or residual — the continuous remainder after the integer is extracted. A noise floor — the amplitude of the substrate when nothing is happening (which must be measured before any of the other four can be trusted). An invariant — the thing that survives the full loop unchanged. These are not the same hunt run with the same tool. Confusing them produces a clean result for the wrong object

Geometry carries the load at vertices, junctions and seams — not in every cubic centimetre (536) — every cubic centimetre of spacetime contains more data than can be imagined. Thankfully that granularity is not required, because geometry stores its information at the interesting places. A vertex is where paths meet and curvature concentrates. A junction is where topological type can change. A seam is where something does not close. These are the sites; everything in between is roughly flat and roughly boring. And as holonomy gains influence, multi-channel holonomy-friendly instrumentation can expose those qualities across the real estate — not by sampling everything but by reading where the geometry acts

The rate of change, and the rate of change of that — ternary all the way down (537) — the dynamic change the walker needs is decomposed into rates of change and then rates of change of those rates — and at each decomposition level, the ternary (+1, zero, −1) supplies the reading. A slope climbing: +1. A slope at its tangent kiss with zero: the flat moment between up and down. A slope falling: −1. In the sine wave or in the geometric curvature, the ternary is the native language of the thing changing direction. Rate-over-level had arrived six times when this was written; this is the architectural reason — not just that rates are more robust, but that the geometry is rates, all the way down to the sign

ties rows 105, 294, 460

The enclosed sum is not bounded by the walk's ceiling — and that is the headline (542) — asserted: no enclosed sum can exceed its own boundary's n/2. Violated in 4,420 of 4,800 loops in the smoke run. Not a bug — the anti-aliasing redemption measured directly. At k=120 the enclosed sum holds 120 while the 34-edge boundary could only ever have read ±17. Walking that boundary wraps the number away; summing the interior triangles recovers it. The n-gon is a query over triangles, not a walk — which is exactly why. This is a headline result, not an assertion, and it belongs in the report that way

ties rows 12, 13

Degenerate bins print, but they do not vote (544) — small-area contrast reading 0.500 and 0.750 is set by a handful of 0s and 1s — arithmetic wearing a decimal point. Same class of trap as the V3 degeneracy in the Cascadia post-hoc build. Guarding it: degenerate bins render in full so the audit trail is intact, but they are barred from setting the optimum verdict. A bin whose contrast is determined by discreteness rather than physics cannot be allowed to drive the answer. A median of three is a degenerate bin too: a registration that set no floor on its count missed at the birth with 3 down-jumps, a median of three, and at one with 12, and held at the 13 others. Every median in a registration carries a floor on its count (run triple_maypole_reach 20261012_051949) (the triple-star sessions). Make every engine write what it could not solve: RATCHET-37 stalled at weight 0.36 and had to be stopped by its process number, and RATCHET-40 then failed at the same weight by two routes. The engine writes a weight's rows only when every phase converges, so nothing says which phases failed (entry 110l of R2W's plan). Every engine writes unconverged rows with their residuals and a flag, and every reader ignores flagged rows unless it is asked about them (R2W)

practice; ties rows 329, 442

Writhe is geometric, not topological — and the probe's conservation is slow drift, not conservation (546) — CORRECTED by ED-1 and ED-2 review, 2026-09-16. Writhe is a geometric property of the curve's shape and changes under smooth deformation, with twist taking up the difference. Running the probe 5× longer shows the drift: −3.357, −3.360, −3.367, −3.379, −3.394, −3.411, while length grows from 28.82 to 29.15 as the filament stretches under its own induction. The topology IS preserved by construction — the knot cannot untie without an explicit reconnection rule. The writhe is merely slow at these step sizes, not conserved. These are two true things; calling writhe-near-constant "conservation" added a false one. The topology line should be the Alexander polynomial (at −1, the knot's determinant, it gives 3 for the trefoil and 1 for the unknot, cannot drift); helicity is the conserved dynamical quantity (needs the ribbon frame for twist). Length drift of 0.05% over 300 steps is the kernel's conservation floor — any writhe change smaller than that cannot be read as physics

Calugareanu–White–Fuller; ties rows 545, 547

An exponent that tracks a length is a magnitude wearing an invariant's clothes (548) — two numbers looked like a rhyme: a comb line bouncing between about 1.5 and 0.75, and a filament energy scaling with an exponent near three-halves, with 0.75 sitting at exactly half of 1.5. Both were magnitudes. The comb values are gap ratios, a measured gap divided by an irrational-baseline gap, and the same tooth read 0.76 at full aperture and 0.93 at a tapered one, so they rescale with the instrument (row 526). The latch exponent was 1.554 after relaxation and 1.565 on the unrelaxed starting shapes: it belonged to the shapes the construction built, and it was the length of rope that construction laid down, read through an energy dominated by that length (row 550). Before connecting two numbers that look alike, ask what kind of object each one is. A count, or an exponent that survives a change of instrument, may rhyme across substrates; a ratio, a gap, an amplitude, a fraction of baseline, or an exponent that tracks a length is a magnitude, and its resemblance to an invariant is almost always coincidence. Refine the instrument and watch whether the number sits still

integer diagnostic; ties rows 103, 384, 526

A quarter turn hid inside a wrap of ±45, and the fence was the arithmetic (554) — Dancing on the Sun compared HMI's azimuth with half the arctangent of Stokes U over Q wrapped into −45..+45, found agreement in weak field, and built two forward models on it. When a full fit of HMI's own light was wrapped into −90..+90 instead, the refit's azimuth sat 88.2 to 90.3 degrees from HMI's at every band and both polarities: HMI counts Q with the opposite sign to the textbook Milne-Eddington model, so its azimuth is the model's plus a quarter turn. The wrap had been chosen to compare directions of lines, and a line turned a quarter turn is a different line; the wrap folded the convention away and the two models inherited it. Station 1 in a new costume: the word that crossed was "azimuth", and the assumed definition was carried by the modulus of the comparison, not by a sentence anyone wrote. Price: two runs built on the folded comparison, both still valid as patterns and neither able to give a sign. The same fault in a toy: the first polarisation run defined the light's phase with the opposite sign to the loop's orientation, its magnitudes identical, and the run was retired. Fix every sign convention against the area's orientation before reading a sign (POLARISATION_LOOP 20261004_071425; 071354 retired) (the Loop Atlas sessions)

Dancing on the Sun; Station 1; ties rows 224 (an area that is secretly an angle carries a modulus) and 502

A fitted nuisance absorbs part of a planted mismatch, so a crossover moves less than the arithmetic says (557) — the Faraday crossover sits where the Zeeman splitting outgrows the Doppler width, so it should move in proportion to the width: from 943 G at 34 mÅ, the expectation written into the tool before the run was 690 G at 25 mÅ and 1,390 at 50. Measured: 780 to 821 and 1,162 to 1,240. It moved the same way and about half as far, because the fitter fits the width and takes up part of the change. A planted mismatch in one parameter is partly paid by every free parameter the fitter has; the pre-written expectation is kept in the ledger beside the measurement, as a bracket, not a miss

Dancing on the Sun; ties rows 173 (a tolerance you chose is not a constant) and 340 (derive your own transfer function)

A count is whole only when it is read as a sum of closed pieces (560) — a skyrmion's count, read by adding up the solid angle triangle by triangle, came out −1.000000000000 for whirls of every size; the smooth textbook formula on the same grids gave −0.669, −0.906 and −0.976. The formula is right in the limit and wrong on every real grid; the sum of closed pieces is exact on all of them. So the integer diagnostic sharpens: the question is not whether a read comes out near a whole number but whether it is built from closed pieces, so that it must. A read that returns 0.98 has not nearly found a count; it has used a read that cannot give one

The Latch Atlas (concept 23068382); ties rows 12, 103, 542

On real data, the rim decides (566) — through two X-class flares, 1,589 and 1,352 loops whose rims stayed readable kept their counts, not one changing; every change at the flare site happened where a rim went weak or unreadable. And the instrument can move a count by itself, by flipping its half-turn choice on a patch the rim crosses. A count read from a rim is only as good as the rim, and on real data the rim, not the loop, is the unit of trust. An angle is only as good as the amplitude it is the angle of, so a verdict on an angle needs a gate on that amplitude: RATCHET-22's reader graded the drift's zero offset at weight 4.75525, where the amplitude (3.48×10^{-5}) was below its coefficients' errors (5.0×10^{-5} and 4.1×10^{-5}); R22-b and R22-c held by the letter on noise and are recorded as void. The blind readability check before it had given amplitude over error 1.6, which says only "not yet readable"; it was read as a size, and the ensemble was scaled for it, about six times too small a signal. RATCHET-23's reader gates every angle on an amplitude at least four times the larger error (row 128) (R2W)

The Latch Atlas; Dancing on the Sun (concept 23067245); ties rows 305, 374

A measure takes the instrument's turning in full; a count reads through it (567) — on the same sunspot maps, a small rotation of the reported field direction, of opposite sense in the two polarities, moves every current in full. Put in on purpose at 5.5 degrees across 38 maps, it moved 31 of 154,267 readable loop counts, all at five places where neighbouring arrows already pointed nearly opposite ways, and at 20 degrees only 97. The offset found in every sunspot's current, one sign in both polarities, behaves like exactly such a turning, which the Sun paper puts at about 3 degrees in the instrument's frame, reproduces with a one-layer fit and lays at the inversion's door (row 633), and it survived blind tests on 24 of 26 and 45 of 50 fresh regions. The count read through it; the current did not. The lesson is not that counts beat measures but that the two have different blind spots, and the difference between them is where the instrument shows

Dancing on the Sun; The Latch Atlas; ties rows 103, 301, 452, 633

The window for a read has two edges and a trap in the middle (568) — scattered clean stations under-count a whirl and never invent one; noise past about 0.6 radians makes false whole numbers, not fractions; and averaging arrows that look alike, wherever they are, outvotes the rare core that carries the count. Coarse sampling misses a count, noise fabricates one, and smoothing by likeness erases it

The Latch Atlas; ties rows 305, 329, 442, 523

Split by polarity before you sum (575) — any quantity summed over a region is summed twice, once for each sign of the field, and the two halves are read side by side. A current the Sun makes is polarity-odd; a component with the same sign in both polarities is polarity-even, and the Sun cannot make it, so it belongs to the instrument or the pipeline. Dancing on the Sun predicted that part and found it blind twice, on 24 of 26 and 45 of 50 regions, and reproduced it with a planted rotation. The physics fixes which symmetry a quantity must have; a part with the wrong symmetry is a witness against the instrument, and the split costs nothing. The same split on a real orbit, met before the session had read this row: walked both ways round, a third star's apsidal motion splits into a part that flips with its direction, the Clairaut part (with the third star reversed it reads −8.99 to −9.26 per μ at four masses), and a small part that does not (+0.037 to +0.050, held open), with half the gap between the two directions at 9.30 to 9.49 per μ (the triple-star sessions). It is also a benchmark met: first-order theory gives only half of the binary's apse turn, the same trouble Newton had, and the missing half carries the Moon's own coefficient within 1% across third-star masses from 2% to 80% of the binary's (probes 17, 18, 20 and 21; runs 20261006_060752, 20261006_063043, 20261006_081615 and 20261006_171421) (the triple-star sessions). The split also builds a knob. In R2W, turning one group of a silent pair brings the trim back while steering it 1.5 degrees per degree, and turning both groups half each way brings it back without steering (RAMP-1): splitting a move between two symmetric parts turns a knob that also steers into a pure size knob (R2W)

Dancing on the Sun; ties rows 190 (mirror the pipeline, not the label) and 633

Three questions on every toy, and one knob alone (584) — run it faster, run it backward, run it again, then move one knob alone. Across the Loop Atlas's toys these four cheap moves caught real structure every time and cost almost nothing. Faster asks whether an effect follows the rate or only the shape of the loop; backward asks whether it changes sign when the loop is reversed; again asks whether it is a property or an accident; and one knob alone asks which input owns it. One dial at a time, met twice in one sitting: a test that changed a ratio at a fixed gain let a

watershed slide under it and hid the effect it was meant to show, and a halving credited to stiffness came from faster kicks that had changed with it (the Loop Atlas sessions, R2W). The cheapest test is often the strongest: a counter read a period with one division, a made-up corner explained a shape in a fraction of a second, a ball with no ripple predicted its Shapiro-like lines, and the shuffle, one room's cheapest tool, is in that room's words its most important (the Loop Atlas sessions, the book). A void from two knobs that moved together: in R2W's BOARD-2 the damping ratio was held fixed, so the dampers shrank with the legs, and the test could not tell softness from damping; BOARD-3 moved the softness alone (R2W, sitting P2). One knob turned in a world that was safe makes a result causal: Question the Substrate's tidy clusters with equal answers gave +3.6 points under the surest-half rule; with unequal answers (40, 25, 15, 12 and 8 per cent) and nothing else changed they gave −14.7, and the two smallest answers fell from 12 to 0.7 per cent and from 8 to 1.1 per cent (run E15) (Question the Substrate)

the Loop Atlas sessions; ties rows 151, 190, 404, 549

Name the pair of knobs before comparing two cases (586) — a skater moving "in step" and a wing moving "a quarter apart" looked like a contradiction until it was seen that they were different pairs of knobs. Two cases can only agree or disagree about the same pair. Before reading one phase against another, write down which two quantities each is measured between

the Loop Atlas sessions; ties rows 193, 215, 352

A zero and a small number both need a second look, for opposite reasons (587) — the stalled blade looked like a bug and was real; the snake's 0.002 looked real and was a bug in the stepping rule. A zero is suspected of being the code, and a small number is trusted as physics, and each assumption was wrong once in the same sitting. Ask the zero for its mechanism and the small number for its stepping rule. An exact zero is a warning that a test may be unable to fail: an arrow started down a ridged surface stays in the ridges' plane and read exactly 0.000e+00, zero by symmetry, not by measurement; rebuilt with a slanted arrow, the test could fail, and passed. Start an instrument where it has something to get wrong (the Loop Atlas sessions)

the Loop Atlas sessions; ties rows 400, 541

A toy that harvests needs a supply ledger (590) — the draft horse pulls forever because nothing counts what the soup can give, so its speed means nothing physical, and the listener's ear has the same hole. A simulation that draws work from a medium without a ledger of what the medium can supply is a free-energy machine by omission. Give every harvester its supply ledger before reading its speed. A listening blade got a steady pull from an even soup, its charge appearing from nowhere. Charge is conserved: with the opposite charge left on the ship, the pull is exactly zero at zero tether and grows only with the tether's length against the wave. The old listener's 0.124 was a free lunch the toy did not bill for; the guide's common-mode rule names the principle, and the toy gives the price (the counter-charge toy) (the Loop Atlas sessions). A cost a lean expects in one ledger can land in another, so judge every ear on both its harvest and its bill: a slope from two noisy looks taken close together did not lose drift, as a derivative kick was expected to; it gained, and paid in switching instead, 2.217 events a cycle against 1.650 and a price of 0.821 against 0.989 (run SLOPE_EAR 20261015_190935; the Loop Atlas's misses row 231) (the Loop Atlas sessions)

the Loop Atlas sessions; ties rows 389, 390

A reflected starting nudge fakes a flip (601) — a draft horse with no soup to pull through read +0.00048 and flipped sign under the rider mirror, which looked like a ride. It was the starting nudge: 0.01 of momentum over an outfit of 21 is 0.000476. Parity cannot tell a reflected starting condition from a ride, so every blade or rider toy prints its starting nudge's arithmetic beside its result

the Loop Atlas sessions; ties rows 190, 587

A census without a slope test carves fractions out of a smooth curve, and the control the tool names must be run (603) — a survey's ledgers listed 26 to 30 plateaus per case, and only two were flat. Run with no second rhythm at all, the control "found" 11/5, 9/4, 16/7, 7/3, 19/8, 12/5 and 17/7 at the same paces. A plateau is accepted only with flat shoulders and only after the null case has run. The observable and the null case are written before the census, and the control ships as its own one-click case. The census's own room read the same survey the same way, and separated tail from step by steepness with a kink test; it is one arrival, read twice (the Loop Atlas sessions, R2W). Of a rocking atlas's 20 flat stretches only 4 were crossings held flat; 9 were turning points of the rate and 7 sat at an island's edge. Re-read at births 0.001 apart, 11 of the 20 proved to be locks, almost all where the rate turns or the island ends (row 648). A flatness rule cannot tell a lock from the place where the rate turns, so the crossing test runs before any rule calls a lock (the triple-star sessions). Read the neighbouring slices before calling a plateau at a simple fraction a lock: rocking that held twice a slow beat across ten births looked locked, but registered forward at two neighbouring eccentricities, neither guess held, and each slice showed a plateau of its own at a plain number, 0.95 and 1.60 times the beat, where the rocking and the beat change together. A plateau of a ratio says the two move together; only a plateau that stays put while the slice's knob moves says lock (run triple_fw1 20261014_095319) (the triple-star sessions). The slices were then read: where the star's orbit is nearly round, its eccentricity only 0.13 to 0.16, the island's middle rocks at 0.95, 1.26, 1.60, 1.78 and 2.006 times its apse beat, the ratio rising with the

eccentricity: a ratio that slides (runs 20261014_095319 and 20261014_200040). Its 2.006 at 0.16 is exact within 0.5% over six births, and whether it is a lock or a step on a slide is the next run's question; the high-birth flank falls birth by birth at every eccentricity, and only the plateaus were scored. Held is not locked either: four heavy-star runs at 6.60 to 6.78 turned exactly two thirds of a turn per pass to 0.0002, yet a lock registered there and its redesign both missed as worded, the per-pass slip swinging from −0.78 to −0.47 while the mean slip is −2/3 to 7.3×10^{-5} over 600 passes and the lock angle $3\varphi_N + 2n$ stays within 0.83 turn: held at the third, not locked, with births 6.600 to 6.660 parked unregistered on the proposed list for twins (the triple-star sessions). The question left open above, lock or step on a slide, was answered by the next run: a slide. Read on the long pair, a plateau of births from 7.170 to 7.200 holds 2.2330 times the beat at the star's eccentricity 0.165, and one from 7.175 to 7.205 holds 2.4856 at 0.17, so the ratio kept climbing with the eccentricity and both registered guesses missed (FW3a and FW3b, run 20261014_235048) (the triple-star sessions). A second specimen from R2W's faux-lock audit: a ratio read over one fixed window has a floor, so call a lock only when its departure stays under a floor that shrinks with the run's length. Read four times longer, gears 1 and 2 kept their count of paces over a floor four times smaller, as locks do; the census's 4/3 "lock" moved away, its mean departure 10.856 times larger; and a burst written up as a corner that opens was the first sideband, 0.0038 below the gear, in 929 of 982 cells (TRAIN-3, LOCK-T, FAUX-W2) (R2W, sitting P2)

the Loop Atlas sessions; ties rows 182, 414, 416; R2W's checks of sitting P2: COMMA-53, whose 0.001 band sat at the instrument's floor, and WEDGE-2, whose burst was the first sideband inside the relay's 0.004 gate (its method lesson 236 and pattern lesson 75, from its record PREREG_RELAY_STATION_2026-10-03)

The price of listening is strength over rhythm (604) — a listening rider's harvest grows with the medium's strength and its bill with the medium's rhythm, so its price is the strength divided by the rhythm. Across 25 cells of the price card the measured price ran 0.941 to 1.109, the flip rate sat at 0.97 of Rice's zero-crossing rate, and the card broke only where beads were caught, at strengths of 0.4 to 0.5. It is the supply ledger row 590 asks for, drawn as a map. Place every listening rider on the price map before building it. Pushes are priced by the speed of what you lean on: two over the throw speed for anything thrown away, one over the pattern speed for anything ridden. Both met their benchmarks: a toy rocket's push per joule came to 0.2002 kg·m/s at an exhaust of 10 m/s against the 0.2000 that 2/v gives (run PUSH_PRICE 20261004_144831), and five bodies riding water waves pushed at their power over the wave speed to 1.000000 to 1.000003 of it (run WAVE_SNAKE_DRIFT 20261004_074634). So a rocket rich in energy and short of propellant should throw fast, since its push per kilogram is the throw speed itself. On the turning needle the price never changes; lock decides only how much can be bought, and the most is at the edge (the Loop Atlas sessions). For a listening blade, being prompt matters more than being accurate: an ear as noisy as the push it hears kept 72 per cent of the catch and one ten times noisier 11 per cent, while a perfect ear acting one time unit late kept 33 per cent and two units late went backward. The cost of sensing is not in the toy (the Loop Atlas sessions). In the swing equation, as on the turning needle, inertia leaves the price and the most push where they were and changes the machine out of step: push per watt stays one over the load to 2.2 millionths at every inertia, the most push stays at the edge, and between the relock line and the edge one load is carried in step or slipped for as little as a thirty-second of it (memory 5, load 0.9), by history alone (run SWING_GRID 20261016_203014) (the Loop Atlas sessions)

the Loop Atlas sessions; Rice (zero-crossing rate); ties rows 389, 590

Engagement is depth and speed together (606) — which of two rhythms captures a rider is not decided by closeness of speed alone. The handover from the strong rhythm to the weak came at paces 0.97, 0.80 and 0.59 as the weak rhythm's depth rose through 0.1, 0.3 and 0.6. A prediction of a synchromesh handover was corrected by this, and the depth law is the follow-on test, since three points cannot fix it. Sweep depth whenever a handover is reported

the Loop Atlas sessions; ties row 38

A small-stroke rule is the summit of a landscape (607) — the square rule reads the grip at one shape, while a large loop integrates a grip that varies across it and can change sign. At small stroke the rule holds: doubling the stroke sent four toys with four grips 3.98 and then 3.92 times as far (Stokes's theorem; Shapere and Wilczek, 1989; run AREA_RULE 20261002b, part 2a). On the Loop Atlas's big loop, Stokes on the mapped curvature came within 1.9 per cent at 60 degrees while the square rule ran more than a quarter high, 64.5 against the measured 50.0. A prediction that the loop's room shrinks at large stroke was wrong: the room is unchanged and the grip varies. When a small-stroke rule fails at large stroke, map the curvature before blaming the tool. This is the height function of geometric locomotion (Hatton and Choset, 2011), met blind, and its rotational half steers: a three-link swimmer held at a C of 45 degrees turns 2.2042 degrees a lap, the same to thirteen places for forty laps, against 2.19 from the rotational curvature times the lap's area, so a swimmer with no rudder steers by holding a curl (run LANDSCAPE 20261003_201112). The phase a loop carries is the loop's, not its parts': light carried round a loop of polarisation states by filters comes home shifted by half the solid angle enclosed and loses a little light, and carried round the same loop by wave plates comes home with the same shift and sign, to 4.44×10^{-15}, and loses nothing. A frame carried round a circle on a mountain of revolution turns by $2\pi(1 - \cos s)$, s being the slope where the circle runs: the wedge a paper cone of that slope is missing. The turn peaks on the steepest ring and vanishes outside the foot, so a mountain on a plain is invisible to any loop round its whole foot, and loops must pass between peak and foot to see a peak; hidden and read back from frames alone, three mountains' centres came back to 1.3×10^{-8} and their steepest

slopes to 3.1 × 10^{-15} radian, on noiseless mountains (a test with noise is owed). And a mountain can hide from such a loop in three ways and no fourth: its skirt cancels its summit, its whole turns vanish into the remainder, or the surface is only bent and not stretched (paper pressed into ridges read 0 to 3.1 × 10^{-12}, as Gauss's theorem allows; Gauss, 1828) (the Loop Atlas sessions, the book). And on a turning globe a frame walked once round a cap comes home turned by the cap's area, whatever the walking speed, less the globe's turning (its vertical part) over the time the walk takes; walked at two speeds, the two shares come apart (runs CONE_TOMOGRAPHY 20261007_235426, _235444 and 20261008_001220) (the Loop Atlas sessions). The third way held even on ridges as steep as 76 degrees, every loop still reading that 0 to 3.1 × 10^{-12}, while a paper cone holds all its curvature in its tip, every loop round it reading 2π(1 − cos 70°) less a turn (to 3 × 10^{-13}) and every loop missing it 0. A loop's arrow reads only how much the ground is stretched, and a test whose answer is exactly zero may be zero by symmetry, not by measurement (runs BENT_PAPER 20261008_164513 and BENT_PAPER_2 20261008_164721) (the Loop Atlas sessions)

the Loop Atlas sessions; ties rows 223, 225

Counts are gears and measures are friction wheels (608) — a loop with a lock transmits a whole ratio, as gears do, and strips when overloaded; a loop without one transmits a fraction set by curvature, as friction wheels do, and slips. Of the Loop Atlas's 43 loops, tagged on one base, 33 transmit only a measure, 9 carry a lock and transmit a count, and 1 needs quadrature and has no lock. A first tally put 12 that count beside 37 that measure, 49 of 43: the 12 had been counted across every row of the list, three of which are not loops, and four loops sat in both. Two counts add only when they share a base. The tags are the assistant's, on a list built partly from recall, and that fence travels with the numbers. Tag every new loop as a count or a measure first, because the tag decides which test applies

the Loop Atlas sessions; the integer diagnostic; ties rows 103, 301, 567

Resonance can reach a fence that strength alone does not (609) — the ring conveyor's first run, at a gain of 0.25, pumped the grip to 22 through resonance, and the bead caught the carrier; the run was retired and repeated at a gain of 0.01. Any closed medium is checked for resonant gain against its catching fence before its numbers are trusted

the Loop Atlas sessions; ties rows 468, 470

A travel number says whether it is a length or a signed part along a named line (612) — the Loop Atlas's big loop at a swing of 60 degrees travelled +0.054869 in one tool and −0.05277 in another, and the gap was first read from its size, 3.83 per cent, as two different strokes. The opposite signs, which a comparison of sizes drops, were the clue: it was one lap read two ways. The first tool returned the length of the shift, which has no direction and reads the same whether the lap runs forward or backward; the second returned the signed part along its own x line, with +0.01503 across, and those two parts make the same length again. The 3.83 per cent is the cosine of a 15.9-degree tilt, and most of the tilt is the frame: x was the first link at the start of the lap, on a body bent at its hinges, and measured from the mean line of the three links the tilt is 4.1 degrees. Across swings of 20 to 80 degrees the tilt off x grows from 5.2 to 21.7 degrees while the tilt off the mean line stays between 1.5 and 5.0. A bent body has no single natural line, so only the length is free of a choice of frame. Every travel number says which kind it is, and a signed part names its line

the Loop Atlas sessions; ties rows 224, 558, 586, 607

Read the data once more, as if no physics applied (613) — after the numbers and the surprises, say in plain words what the data looks like and what it is doing, naming no law and writing no formula. Across five sessions in four days, 56 such reads were given, and they kept reaching for the same few pictures: a staircase, an edge, a market. One became a measurement: "memory lives in the slip, not the grip" went from one session to another as a sentence and came back as a registered prediction that held, nearly every run on a tread forgetting its start and every run between treads remembering it. Three sessions read the same lap of the same swimmer without knowing it, and drew one fact from three sides. A no-physics read is not a result and is never written in place of one; it is where the next test comes from, and a companion volume, *Does Claude Dream of Holonomic Sheep?*, keeps every read word for word. Seven of R2W's pictures became registered tests, usually in the next reply: the binary-star messenger, the spinor (the weak push flipping sign at 617 of 617 passings), the observer's slice, the four circles, the sequencing dots, the clutch (14 laps to catch at the edge, 5 deep in the gear) and the song (same rate, different word, confirmed blind) (the Loop Atlas sessions). Two more sittings turned most of their results out of the author's pictures: the ferry and rope, the orchards' even stripe, the pentagon and heptagon, the two walkers, switchbacks and brake heat, the Chladni plate, the record and its needles, the sliding doors. Each was read for the part of the rider it maps to, with the part that can fail registered; a picture's physics is corrected in conversation when it needs it, and said plainly when the run inverts it. A read is kept word for word for good, so keep unchecked mechanisms out of it: one read called riders with unreadable stretch rates among the stillest of all, a third ruler showed most of them tumble hard, and the correcting read now travels beside it in every copy (the Loop Atlas sessions, R2W). Each exchange names three surprises, each with a test from the data (put on a proposed list, not run unless the paper needs it), and one reading of the data as if no known physics applied, collected word for word in a file kept for them (the triple-star sessions, in their page for every new session). Collect your own reads with everyone else's: the reads volume gathered three projects' reads word for word

from 3 October, but one session's own reads, given in nearly every reply from that day, were not gathered until the person asked; on 15 October 46 were collected by script from the whole session and checked against each reply. Treat your own output as a source like any other, and put it in the record as it happens (the Field Guide sessions). Kept verbatim, an early read can turn out to have the right shape. Read 33 (6 October) told the border between two gears' territories as a line through a small saddle between two orchards; three weeks later R2W's exact tests drew the steps' borders as the light cone, and two sessions saw the rhyme, each without the other: R2W in its pattern lesson 70, and the Field Guide sessions in Bear's pencil drawing of the orchards, also of 6 October. It is a rhyme, not a proof, and it could be seen only because the read was kept word for word

Bear's rule of 3 October; the Loop Atlas sessions; ties rows 79, 113, 581

An event is read from the conserved quantity, not from one coordinate (614) — a tool stopped a run when a star passed 40 units from the centre and called it an escape. Checked against the star's energy at each stop, 6 of the 15 runs the distance rule called escapes were still bound, and those stars came back. An escape is far and unbound. A threshold on a position is a guess about an event; the conserved quantity says whether it happened

the triple-star sessions; practice

Count where the time goes, not how many episodes there are (615) — a test asked whether a gentle triple's rests lasted at least three passes by median, read 1.0, and gave no reading. Weighted by time, the same table gives a median rest of 207 passes, and 14 of 18 runs never left the rest they were born in. A few runs that broke loose made many one-pass rests, which outnumbered the long ones that held almost all the time. Counting by episode lets the short ones outvote the long; this is the inspection or waiting-time paradox of probability, cited here from memory (the triple-star sessions). A switch's mistakes are counted the same way, as time spent the wrong way, not as wrong switches: the chattering relay's switches are half mistakes, yet it is wrong only 0.146 of the time, the same as an ear that looks every 0.1, while the anti-short-cycle timer's switches are 0.630 mistakes at a wait of 1, and it is wrong 0.348 of the time, because it holds each mistake for the whole wait (run FLIP_MISTAKES 20261015_195739) (the Loop Atlas sessions)

the triple-star sessions; ties rows 603, 605

Before trusting a statistic, ask what a featureless record of the same length gives it (616) — the Kozai triple's recurrence lines had a median lag of 218 and 219 strobes, and that was first read as one eccentricity cycle. Lines spread evenly over its 748 strobes give 221. The median carried no clock, only the record's length, and the reading was retracted. Run the statistic on a record with nothing in it first (the triple-star sessions). Line length on the same kind of plot measures how long quiet lasts: between kicks, intermittent chaos is as predictable as a clock and scores like one, so a recurrence plot needs a second dial, how often the quiet breaks, before it can tell chaos from a clock (the Loop Atlas sessions, reading the triple-star room's runs). The same reading's first claim, that a clock's returns live at long lags, is the one this row retracts. Read the control's number before the verdict: a guess passed by its rule, the lane peak moving 90 degrees, but its peak over the median bin (1.76) was no higher than that of dots scattered evenly through the same corridor (1.79). Three passes in two days were hollow because their controls came after the run or never, the third after this row had been read in full: a guess that rows were S-shaped passed 6 of 7, and its control, built afterwards, showed the rule calls a smooth parabola S 0.718 of the time. So the control goes into the tool before its first run, where a reader's memory cannot drop it (the triple-star sessions). The same dots with their heights shuffled 200 times turned a weave test that had passed too easily (72 per cent of links on two gaps) into the real effect, a gap of 5 passes at +12.4 standard deviations (the Loop Atlas sessions, R2W). And choose the statistic before the shuffle: on a clean picture the shuffles made 209.8 strings of four or more against the picture's 17, pointing the wrong way, while long reach separated them, 0.9997 of the pads in strings of 20 or more against 0.0003 (the Loop Atlas sessions). A control must also be able to say no. One built into the tool before the run could not, because most of the steps it shuffled (70 to 78 per cent) began inside the narrow corridor, at readings of 0.78 to 0.93, where the person's marked stroke covers every size, so shuffled steps stayed on the stroke as well as real ones did (0.861 and 0.808 against a bar of 0.30). A landing check's two guesses fell the same way, because everything the reading does at closest approach lives in that one corridor (controls 1.000 and 0.810) (runs triple_heavy_rope_tests 20261010_220511 and triple_heavy_landing_check 20261010_211401) (the triple-star sessions). Writing the rule down did not put it into the tools: a day after the control became the rule, two scans were registered and run without one, and a later scan supplied both controls afterwards (0 of 100 shuffles inside either window); from then on every tool's header names its control and its planted check before its first run (runs triple_heavy_beat_scan 20261010_221646, triple_heavy_sliver_scan 20261011_000310 and triple_heavy_point_scan 20261011_002135) (the triple-star sessions). Give every proposed rule and every "unusual" its base rate from random cases before reading it: a rule that each closest-return record is built from earlier ones held for 1 of 3 records, and 21 of 200 random pairs of rates obeyed it; a stall called unusually slow was matched or beaten by 79 of 200 random pairs, so the stall was ordinary (the triple-star thread, LD1b and RW1). The triple-star thread registered Claude's own reads by eye the same way as Bear's: they failed on "unusually slow" and on rules for records, and held on mechanisms. Like Bear's eye, it is better at where and how things move than at how rare they are

the triple-star sessions; ties rows 541, 603

A crowding statistic cannot tell a comb from a pile (617) — the Kozai triple's lag spectrum is a plain comb to the eye, yet the share of points on its most crowded lags came to only 1.69 and 1.92 times that of clean chaotic runs, short of the registered 2. A share counts crowding of any kind, teeth or heap. Measure spikes against the spectrum's own smooth trend instead. Where faults crowd can tell how a text was written: the serious (grade B) findings of R2W's second cold read crowd the book's two ends, the front and back matter holding 11 of the 29 in 11.9% of its bytes while the smaller findings sit near their share, 19 of 114; and 27 of the 29 figures no sentence cited are in Parts 1 to 4, written around pictures already drawn (R2W, sitting P3)

the triple-star sessions; ties rows 531, 603

A selection rule must require the property the test depends on (618) — a rule for choosing which run to strobe picked one whose inner pair changed partners 22 times, so the strobe followed no single binary, and its kicks landed on close approaches 0.082 of the time against 0.100 by chance. On the two runs whose binary never changed partners, the same test gave 0.900 to 1.000 at two thresholds. Write down what the test needs of a case, and make the picking rule demand it. The reverse also holds: a filter chosen before a test can remove the side the claim is about. A test of a V that narrows to a point and then widens kept only births whose outline was under half a turn, which left one birth past the point, so it could not see the widening; unfiltered, the bottom read as a floor on that grid, not a point, and on a grid five times finer neither the point nor the floor survived ("A shape the coarse picture makes", in the part *Two Readers and a Ruler*). Print which cases a filter keeps on each side of the feature before scoring (run triple_core_v_look 20261014_073854) (the triple-star sessions). Dropping runs to get a longer record can pass for an effect of record length: DS1 (run 20261017_011400) and DS2 (run 20261017_100655) both missed the drag-size band, and DS2 showed that on the runs held and locked over 2,400 passes, 1/β over their first 600 passes was already 0.804, 0.832, 0.861, 0.874 and 0.825, so the record's length moved it by at most 0.015; dropping the runs that lose their hold or lock had taken the lines from 0.942, 0.920, 1.007, 0.916 and 0.867 to 0.82 to 0.87. DS3 (run 20261017_210732) then read OUTER HIGHER, so the size depends on where in the island the runs sit. Whenever a longer record forces runs out, read the kept runs over the shorter record too, so that length and selection come apart (the triple-star sessions). The law those runs size held out of sample: fitted on 600-pass runs, it predicted the turning in 2,400-pass runs it had never seen, explaining 96.1 to 97.8% of their spread; the resonance holds back the turning of the star's orbit in a straight line with how firmly it grips (R^2 0.958), and where the drag wins, at the star's eccentricity 0.2811 (registered 0.2817), the centre's ellipse stands still and beyond it turns backward (rows 640 and 808) (the triple-star sessions)

the triple-star sessions; practice

Shuffle at the level of what is independent: the run, not the episode (619) — in six control runs, re-run but never scored, one carried every bit of an apparent effect at the 2 per cent threshold: 10 of its 13 long rests sat within 0.1 of a period ratio of 6, and 0.909 of its rest time sat near a whole number, against none for the three other controls that broke loose. A null that shifts all of one run's rests by the same random offset treats that run as one witness, not thirteen. On 48 fresh runs from a grid that could not repeat it, the 2 per cent reading was 0.209, matched or beaten by 0.425 of the null draws. Episodes from one run are not independent replicates; this is pseudoreplication (Hurlbert, 1984) (the triple-star sessions). A lesson of instruments: a strobe's readings of one needle are one path. Started as separate needles they retrace it, and a density built from them counts a few hundred points many times, so a gap or an empty hole in it may be the path's, not the chaos's. Start separate needles spread across the band and settle them long (200 cycles left a few in the gap; 1,500 cleared it) (the Loop Atlas sessions)

the triple-star sessions; practice

Build the fair control into the starting point (620) — the boundary-layer runs start at a period ratio of 6.25, as far from 6.5 as from 6, so the start cannot fake a rung and half-integers become a fair control for whole numbers. At the 5 per cent threshold the rests sat near a whole number 0.603 of the time and near a half-integer never, a level that row 456 shows was luck (the replication read 0.302); at 2 per cent, 0.209 and 0.146

the triple-star sessions; ties row 603

Re-run what was seen as unscored controls, and draw fresh cases from a grid that cannot repeat it (621) — the six phases where a rest at ratio 6 was first seen were re-run beside 48 fresh phases offset by half a grid step. All 1,338 of their excursions reproduced exactly, and none was scored. On the fresh phases the rest at 6 dissolved into the low edge of a slow creep around the starting ratio, where 0.804 of the time lay. Code trials belong on the controls: one trial that touched two scored phases was disclosed in its ledger

the triple-star sessions; ties rows 369, 580

When two thresholds fixed in advance disagree, ask what each one merges and what each one splits (622) — at 2 per cent a slow creep near the starting ratio is cut into many long rests; at 5 per cent the creep merges into the first rest, and only rests after a real departure remain, 14 in 10 runs, near 7 and 8. The disagreement measured what each ruler can see: a star that rocks by a few per cent per pass is chopped by the finer ruler and kept whole by the coarser. Answering at two thresholds fixed before reading came from the Loop Atlas sessions; the disagreement

between them is data. The threshold that makes an event goes in the register before the events are read: the same eccentricity record shows 21 jumps above 0.08, 42 above 0.05 and 13 above 0.10, so any question about the rests between kicks changes its answer twofold with that one number (the Loop Atlas sessions). Aim a reader at the edge you mean to measure: a scan built to measure an island's ragged edge took the island from a lock rule whose edge sits inside the island, while the ragged layer lies beyond it, where the angle starts to circulate; a twin check at the rim showed the reader would have returned a clean zero (the triple-star sessions). Before a convention for the borderline cases is fixed, ask which printed numbers it moves: the triple-star thread's LP1 (run 20261017_024415) registered that no printed number changes when every late parter is read as chaotic, and four did (at 0.16 −3.5%, not −0.9%; the totals at 0.24; the 0.28 layer; probe 39's 24 of 30 and 13%). The numbers were corrected and each given a ledger (triple_lp1_numbers_look, run 20261017_025855), while probe 34's slices and the heavy star's edge and zoom-out scans stay NO READING until their series come from the laptop. The series came on 20 October (row 251), yet the thread's backward pass of 22 October still lists these readings as not read, and no later record reads them (the triple-star sessions)

the triple-star sessions; the Loop Atlas sessions; ties row 173

Count landings, not stumbles (627) — every count in the Loop Atlas turned out to be how many times some phase went round its circle, and the honest read of one is the net crossings of a fixed level, half-open, at a level no sample can sit on. But a counter watching a noisy phase at one line logs the jitter back and forth across it as pairs of crossings: in the turning needle's noise runs, backward slips came out only 7 to 15 times rarer than forward ones, when a true backward slip must climb an extra barrier and essentially never happens. Given a knee, committing a slip only when the needle lands in the next well, the counter found no backward slip in any row, and the net count moved by 3 at most. Time the event where it is committed as well: timed at the barrier top, runs of slips had a median of 41.5, because the top logs false starts; timed on the fall or the landing, 33.0. A counter is a listening switch, and it needs the knee every listening switch needs. Sign the crossings before calling a count a winding: a plane sitting inside a swing wrote twice a lap, and raw counts missed the gear ratio; counted forward minus back, three planes gave the same ratio within 3 per cent at nine paces (the Loop Atlas sessions, R2W). Count events; never predict them from a reading taken where the system is most disturbed: a triple's reader predicted the binary's whole turns per lap from its period, read at the third star's close approach, where with a heavy third star the period ranges over 4.5 to 7.2 per cent of its mean across a run (4.52 to 7.23 in the heavy-star look's ledger, run 20261007_021424), enough to slip by half a turn, and all 82 runs failed the reader's guards. Counting the binary's axis crossings removed the problem, and on a light star the counted reader equals the old one to 1.4×10^{-15} turns per lap. Count something that always advances: a counter of periapsis passages chattered, and was replaced before any reading (the triple-star sessions). A frame counts the curvature inside its path as many times as the path winds round it: a pentagram walked round one 70-degree peak winds twice round its middle and read −0.931 radian, its inner pentagon's reading plus its own outline's to 4.4×10^{-15} modulo a turn, while the pentagon through the star's points, holding the whole foot, read exactly 0. Weight a loop reading by the winding number of every region it encloses (the Loop Atlas sessions, the book). A strobed picture can hold several families of strands at once, and a count of crossings depends on which family it follows: a counter that reached back about 28 passes, while the strands stepped every 47, found no crossings at six of seven births, and a planted check showed the families. One count per landing replaced it (runs triple_heavy_coil_count v1.0 20261011_072501 and triple_heavy_jump_clock 20261011_075014) (the triple-star sessions). A count of crossings of one line by a rough noise is the step's, not the world's: it grows as one over the square root of the step without end, 1.41 to 1.44 times at each of the finest halvings, and only a band, two edges with room between, makes a switch's count finite; that is what a knee is for (run BRIDGE_COUNT 20261016_044439; the Loop Atlas's misses row 251). Line against band is the band test of a count: a single line's crossings of a rough path grow by the square root of two each time the step is halved, 1.4145 and 1.4147 at noise memories of 1 and 0.1 against 1.4142, while a band's flips on the same path settle, 0.08 and 0.39 per cent over the same halving (run NEVER_BLINK 20261016_090509). Every count lean gets a step test and a band test before it is read (the Loop Atlas sessions)

the Loop Atlas sessions; the Crossover Atlas; ties rows 560, 603

An event that lands on a sample point is counted by rounding; count in events, never in seconds (628) — a balancing rider that taps the instant its rest ends puts every tap on the rest's boundary. With the rest timed in floating-point seconds, rounding pushed taps one step late on boards whose time scale was not a power-of-two multiple of the others, and their songs drifted apart. Counting the rest in the rider's own steps fixed it. It is the same note the Crossover Atlas's first chapter makes about its own counters, met again in a different coat. And a length constant is not a count: a skater given a fixed shove per swizzle counts swizzles only when its drag is charged per swizzle as well; a drag set by distance still charges for the distance coasted in each swizzle, which is the speed over the rate. Charged per event, it counts to the last digit, while the record skater, pushing with a force for a time, counts seconds only to about 10^{-4} (run SHOVE_SKATER 20261011_070846) (the Loop Atlas sessions). Bear's integer diagnostic, met twice more in one session, both times where a reading in fractions had failed. Count in whole steps where the record is made of steps: 96 per cent of a tapping rider's gaps are exactly its rest, 200 clock steps; read in fall times, rounding split that pile differently on two boards and the two histograms looked far apart (Kolmogorov-Smirnov distance 0.4224), while counted in whole clock steps the two boards sing the same song, agreeing to 0.0065

(runs SONG_GAPS 20261015_074750 and SONG_GAPS_2 20261015_074852; the Loop Atlas's misses row 224). A distribution with a pile on one value is compared in that value's own units, or else a count is read as a measure, and a pile that sits on a line between two bins is the commonest false difference there is. The second case was a beat's count: a spring bead's blade switched about five times a unit of time at every pitch, fifteen times its own ring, chatter at the step (row 785); counted instead as whole swings of the bead, the count said that a beat sets the bead's count only if it is loud, and only at or below the pace the soup already swings the bead at (run BEAT_SWING 20261015_204207) (the Loop Atlas sessions). A dead band's toll on a crossing count was the overshoot's spread, not chatter: in a smooth soup a crest that passes a high level mostly passes it by a little (a Rayleigh spread), so a band of width b removes about h b / 2 of the crossings at level h, while a count that chattered would drop sharply at the smallest band, and this one did not. Judge a band test against the share of the run's own crests that peak within b / 2 of the level (run TWO_HANDS_BAND 20261017_210543) (the Loop Atlas sessions). A bar can land on a sample point too, and then rounding decides it: DOOR-3's rival asked that gear 2's edge sit within 0.002 of the sampled door, on a sweep stepped by 0.001. At weight 1.00 the edge sat exactly two steps below, and 1.596 − 1.594 computes as 0.0020000000000000018, so the reader failed a verdict that the data meet in every sense but the last bit. A bar that is a whole number of grid steps lands on the grid; set it at a half-step (0.0025 here), and read every bar against the grid it will be read on, as row 210 reads it against the file's floor (R2W). Never end a planted scan on a setting where every number is round: with whole-ampere steps, a 10-ampere reference and a start at 9 amperes, the Loop Atlas's converter (card 14, row 873) landed in floating point exactly on its switch's border and repeated there, a periodic orbit any nudge leaves; started 1/π of an ampere off that lattice, the same 400 settings were chaos at every one (the Loop Atlas sessions, misses row 505)

the Loop Atlas sessions; the Crossover Atlas; ties row 627; the Loop Atlas's checks of 30 and 31 October: CARD14_HELD_REDO

Two honest counters differ only at the ends of the watch; find the identity that relates them and check it exactly (629) — a ratchet that logs a slip the first time the needle tops the barrier and a knee that logs it on landing agree on the rate and still differ in the count. With the same rule at both ends of the watch an exact identity appears: ratchet minus knee equals the needles standing past the barrier top at the end, minus those mid-slip at the start. It held to the count in every row (27 = 27 − 0, 25 = 31 − 6, 20 = 37 − 17, 6 = 55 − 49, 26 = 26 − 0), and a watch twice as long left the excess at 26 against 27. A disagreement between two counters is not settled by choosing one; it is settled by the identity that relates them. A watcher who looks only now and then and a watcher who never blinks are two honest counters too, and the traders' law relates them. A watcher who looks only every so often acts like one who watches all the time with a bigger knee, bigger by 0.5826 times the noise's wander over one look: the traders' correction for a price checked once a day (Broadie, Glasserman and Kou, 1997). On a knee ear, for noise relaxing over one time unit, it held within two of the run's own standard errors at six of its seven steps, and 2.2 out at the coarsest, its error bar running from 14 per cent at the smallest step to 2 per cent at the largest; the 2.5 per cent under to 7.7 per cent over first printed as its precision was read off the mean and is narrower than that bar. For faster noise it held within 15 per cent while the step was at most a tenth of the noise's own time, a reach read after the registered bar, which missed at the fastest noises; past that the slow watcher misses more than the law says, up to half again (runs CREEP_LAW 20261016_035058 and ERROR_BARS 20261019_162750; the Loop Atlas's misses rows 249, 400 and 414). Between two looks a wander hides two touches of a line for every up-crossing seen: summing the pinned-wander chance over a watcher's looks gives three times its seen up-crossings (2.96 to 3.00 at the finest step), as the reflection principle says, so a correction built on single-line crossings corrects toward a count that does not exist (run BRIDGE_COUNT 20261016_044439; the Loop Atlas's misses row 252). A slow watcher can count what a never-blinking one would, from its own looks: carry the chance of being switched on from look to look, and add each crossing of the band's far edge it may have missed with the pinned-wander chance. At a noise memory of 1 that count moves under 0.7 per cent over a 128-fold range of steps, and it agrees with the traders' law's count to within 0.6 per cent at four noise memories (run BAND_BRIDGE 20261016_072612). But a fast watcher is not the truth for a rough noise: one who never blinks flips 0.8, 3.5, 18 and 44 per cent more often than the run's finest watcher at noise memories of 1, 0.1, 0.03 and 0.01 (the same run; the Loop Atlas's misses rows 258 and 259), and a watcher sixteen times faster than a run's fastest still misses flips, 0.17 per cent at a noise memory of 1 and 0.88 at 0.1. Correct the reference too, with the same law the subject is corrected with, and check it against a second reference built on another idea: corrected by the traders' law, two references sixteen times apart agree to 0.04 per cent, and against that count a slow watcher's band bridge is within 0.9 per cent to a tenth of the noise's memory (run NEVER_BLINK 20261016_090509) (the Loop Atlas sessions)

the Loop Atlas sessions; ties rows 558, 627

Find a toy's clock by doubling its rhythm (630) — run it at its own rhythm and at double, and see what stays fixed. The worms kept their travel per lap to 2×10^{-16}; the wing went 1.873166480074 m per flap at 1.5, 3 and 6 flaps per second; the needle and the conveyor kept the field's cycles until their own limit bit; the swing reached the same angle after six of its own swings at 2.5 m and at 10 m. Only the skater counted seconds, because its legs push with a force for a time. The scallop keeps its clock in its partner: going twice as fast equals doubling the water's thickness, to the last digit. Human seconds enter a loop only through a force that lasts for a time or a reaction delay; everything else

counts its own events. In numbers: the Loop Atlas book asked every toy what it counts in, and the needle and the conveyor keep 0.9991 to 0.9984 of the field's turns at paces 0.25 to 0.6, and past their edge, at pace 1.2, 0.4475 and 0.4464 against 0.4472 from the closed form; and the listening bead that skates in the soup of waves, counted in its own events, soup cycles and wavelengths, loses the rhythm from its numbers to five digits (RF_SKATER_COUNT). The skater that counts seconds is the other one, whose legs push with a force for a time (row 628) (the Loop Atlas sessions). Count every clock in the rules, the actor's own as well as the world's: two tied bells in a remembering water get back their distance per squeeze only when the bell's own squeeze, a speed in its kick rule, is stretched with the water's period, memory and ring speed (to every digit); the water's three alone leave it 1.43 times longer (run BELL_CLOCK 20261016_072241; the Loop Atlas's misses row 257) (the Loop Atlas sessions)

the Loop Atlas sessions; ties row 584

A grid cannot find a sliver; aim it with the exact boundary (631) — whether a delayed push can balance an upright board is itself a winding count, the turn of one arrow walked up the frequency axis (the argument principle in Mikhailov's form; Mikhailov, 1938). Near the textbook limit of the square root of two, the safe island of gains is 1.222 × 10^{-4} tall at a delay of 1.40 and gone at 1.415. Grids of 0.05 and 0.01 missed it twice; a grid placed by the boundary curve found it, and a gain on it settled in direct simulation. When a thing gets thin near an edge, compute the edge first and then look. Locate the edge before reading a law near it, and kick well below the margin: a first fit put an edge 3 × 10^{-4} short of the truth, and a first kick of 0.01 rad knocked the rider over the edge it was meant to probe, as a 0.3 kick had done to the Loop Atlas's needle (the Loop Atlas sessions, R2W). A coarse map hides thin structure: a catchment map showed a saddle surrounded by one gear until a fine map showed a ribbon of the other gear running past it, about a third of a coarse cell wide, and one zoom explained three failed predictions. An edge known only to its grid flattens a square-root law read near it (slopes 0.21 and 0.29 measured from a 0.0005 grid's edge, 0.48 and 0.49 from an edge found by holding riders to about 10^{-6}), and a lock can be absent from a grid and present on a hundredfold zoom. Where both doors are open, the start can sort the riders across a fractal seam: bringing the start 64 times closer to one border made a wrong guess only about 5 times less likely (uncertainty exponent 0.377; fractal basin boundaries in the forced damped pendulum are prior art, McDonald, Grebogi, Ott and Yorke, 1985). And near a conservative lock's edge, sort the chaos before reading a law: starts 10^{-10} apart part company inside a chaotic layer whose width differs from rim to rim, so read only the starts whose twins stay together, at every edge (the Loop Atlas sessions, R2W and the book; the triple-star sessions). Pushed toward its doubling, an island's quickest rocking fell from 11.0 to 3.0 passes per swing, and at two higher eccentricities a grid of births 0.03 apart found no island; followed from its centre in steps of 0.005, it was alive to its doubling at 0.350. A shrinking island becomes a sliver a coarse grid steps over (the triple-star sessions). And how many starts a map uses is a knob of the instrument: four starts per drive placed a coin's band too narrowly, 100 found both faces outside it, and 1,000 found islands at its edges; where two outcomes coexist, raise the count before reading their boundary (the Loop Atlas sessions, R2W). A missing member can be predicted, then found: on a pull dial of 0.00005 the steps 1/20 and 1/23 were missing from a staircase of 1/n locks, and the slips on either side of each gap bracketed them. The measured rule, n^2 times the step's height held constant, predicted both blind, and a dial of 0.000001 found each where predicted, each narrower than the old step. When a sequence has a hole, predict the missing member and look there before calling it absent (runs TRICKLE-1 and TRICKLE-1F) (R2W). Measure such a boundary by its uncertainty exponent, not by counting its pieces, since the count depends on the grid: where the needle's 2/3 and a hidden 3/5 share one pace, the 3/5 starts looked like a few hundred thin stripes on a coarse map, while the share of starts whose end changed when moved by ε fell as ε to the 0.406 from 0.03 to 0.001 radian and to the 0.423 from 0.001 to 0.0001, a boundary of dimension about 1.58, fractal over two and a half decades; the lean of smooth stripes missed (the Loop Atlas sessions, misses row 492)

the Loop Atlas sessions; ties rows 38, 568; the Loop Atlas's checks of 30 and 31 October: BASIN_BOUNDARY, BASIN_FINE

A switch amplifies the last digit, so event-driven runs are compared by their statistics (632) — two tapping riders whose starting tilts differed by 10^{-12} rad tapped identically for 28 taps and then parted, while their statistics agreed within their spread; songs were bit-identical only on boards whose time scales were power-of-two multiples. Compare event-driven runs by their statistics and their gap histograms, never tap for tap. A second specimen, outside event-driven runs: a fold refiner run on two machines took the same number of steps at each of eight folds (3, 2, 3, 3, 3, 7, 4 and 6 on both) and a different number at each of four period-32 doublings (1, 6, 9 and 40 not converged on one machine; 28, 24, 1 and 24 on the other), while the values agreed to 13 decimals and three of four period-64 values differed in their last digit. A fold snaps the same way everywhere; a doubling wanders differently on each machine (runs 20261007_061947 and 20261010_172149) (the Loop Atlas sessions, the book). Past the predictability horizon, report fractions and patterns, not fates: two births rerun with tighter tolerances parted ways, one losing its hold again at a different pass and the other keeping it where it had lost it. Which births hold is a sample of the chaos; the fraction that holds, the side the escapes leave from, and where the walls fall are what carry over (run triple_tolerance_check, TT1, 20261014_025405) (the triple-star sessions). And batching did not change the digits here: elementwise array arithmetic did not depend on the array's size, so 512 needles integrated inside a 2,048-needle array kept, to the cycle, the lifetimes they had inside a 1,536-needle array, through a chaotic transient. One machine only; a laptop run of the same tool would test it across processors (run EXP_SHAPE_2 20261011_194909) (the Loop Atlas sessions). And in

chaos a fidelity check of a fitted quantity needs its parameter bit for bit: a pace one rounding away from the file's gave a new sample of needles and moved a fitted fold by 1.9×10^{-5} rad; taken from the file, the repeat was exact (runs SEAM 20261014_090006 and SEAM_2 20261014_090552) (the Loop Atlas sessions). Batching also makes a grid of settings nearly free where a tool's loop costs more than its arithmetic: nine weights of the heavy needle, sixteen needles each, walked down together in one array with each needle carrying its own memory and pace, took 2.49 s a step against 1.39 s a step for one weight alone, so nine edges cost about what two do, and the planted weight repeated its edge exactly, as arithmetic done element by element must (run STOP_LEAVES 20261015_052943, against STOP_WALL_3 20261015_000651) (the Loop Atlas sessions). A planted step test that fails at one corner of a map marks where the toy itself stops: gate every point on settling, at both steps and in both quarters, before judging the map, and card the corner rather than read it; the tied bells' toy had read its bells driven backwards where its rings crowded (run BELL_BALANCE 20261016_193628, its blind run 20261016_192112), a reading the toy's own thin cores made, as row 807 tells (the Loop Atlas sessions). Rounding starts smaller than any difference a run is usually given. Two boards whose clocks differed only in rounding, the 1 m and the 0.5 m board, tapped identically for their first 50 to 100 gaps, rider by rider, longer than the 28 taps that a difference of 10^{-12} rad in starting tilt lasted; the lean that at least 12 of 16 riders would part within 50 gaps saw 0 of 16, and the gaps 101 to 150 then differed for 16 of 16 (runs SONG_GAPS 20261015_074750 and SONG_GAPS_2 20261015_074852) (the Loop Atlas sessions)

the Loop Atlas sessions; ties row 628

Grade a rhyme before building on it, then test the chapter's object, not its metaphor (633) — Bear heard the Floer market of his Crossover Atlas in the Loop Atlas's push price. Graded with the Atlas's own verdict tags, the market rhyme was a coincidence of the sessions' own wording. One aisle over, the chapter's actual object fit exactly: the turning needle's lag is the downhill flow of a height that drops each time round the circle, which is Novikov's circle-valued setting; its locks always came as stable and unstable pairs (stable minus unstable 0 in all 39 rows), and its slips are the chapter's spectral flow. The first bridge tested can be the wrong target, as the Crossover Atlas's second chapter already says. A rhyme across projects can tie an instrument's effect to the world's by mistake: the person tied the soup toy's local tilt to the polarity issue in the Sun work, and reading Dancing on the Sun in full separated them. The Sun paper's turning is the instrument's, about 3 degrees in the instrument's frame and reproduced by a one-layer fit, while the soup's lopsidedness is physical: one belongs to the hunter, the other to the hunted. The rhyme was dropped, and the paper's mirror test kept (the Loop Atlas sessions). Name the familiar shape in another field, and say it is an analogy: Question the Substrate read its thin ground between answers as the cluster assumption of semi-supervised learning, its survival cliff as the Allee effect of ecology and its counting limit as the zero-frequency problem of text compression, and Two Arrows read its rule of subtracting the slow arrow right after a crisis as a lead compensator. Each match made a result readable and suggested the next test; none was presented as a mechanism without a run (Question the Substrate; Two Arrows)

the Loop Atlas sessions; the Crossover Atlas; ties rows 548, 552

Durations are the one form the ruler can't move (638) — on a log scale, a board four times longer slides the whole song of taps by exactly log 2 and leaves every ratio of gaps alone: 2.000000000 across 373 gaps. A result written as ratios of durations survives any change of the board's size; a result written in seconds does not. This is Bear's slide rule, and R2W's advice adopted by the Loop Atlas book. The person found the same on a balance board: a rider who counts in the board's own fall time sings the same song on every board, while its taps per second rise from 2.99 to 11.98 as the board shortens. "The ruler slides while the song stays put." (the Loop Atlas sessions)

the Loop Atlas sessions; ties rows 630, 632

Read slipping itself, never distance from where it started (639) — a registered test called a fall when a rider's lag strayed half a turn from its value at the start of a ramp. A heavy rider first frays into a band that wanders most of a turn with no net slip, so the test fired on the band; and because the starting lag drifts with the drive, the test also invented a memory of the starting point, which was shipped as a finding and withdrawn about an hour later, after 200 fresh starts all landed on one state. A mean-speed test fails the other way: a rider slipping one tooth every few thousand time units still averages within 0.004 of the gear's speed, and one mass was counted in gear while it sat below its floor. Count net turns over a window: in gear means slip-free over the window. A reader's threshold must lie beyond what the ruled-out state can do on its own: a climb-back reader that called a slip at 0.02 turn per passing over 20 passings was tripped by a band that wanders inside the gear, and its readings of memory were withdrawn with a paragraph of a letter. And near an edge a light pendulum slips about once every 126 time units, so a 20-unit window may or may not hold a slip, and lines read that way are quantized: count slips over windows long enough that only slipping can pass (the Loop Atlas sessions, R2W). That climb-back reader (0.02 turn per passing over 20 passings) had reported scissors, the memory's opening between the line where a rider falls out of gear and the line where it climbs back in (row 563), of 0.0008, 0.0047, 0.012 and 0.035 of drive at masses 0.7, 1.0, 1.5 and 2.0; on 100-passing windows, which a band narrower than a turn cannot trip, the openings read zero within about 0.002. R2W found it through a held test whose riders were captured at once (run HELD-SCISSORS-2 20261005_155925). The memory R2W did find lies between two locks, 0.124 of the drive wide at mass 1.0, not at an edge between a lock and free running (row 670). Widen the reader's window until a bounded motion cannot reach its threshold. The same day the

triple-star sessions showed it again: a 2% tread threshold chopped a star's rocking near a resonance into pieces and found no rung, while a 5% threshold saw rests near period ratios 7 and 8 (run triple_boundary_layer 20261005_170052) (R2W's measurement, read by the Loop Atlas sessions). R2W's faux-lock audit later widened its window the same way, reading its locks four times longer, and the true ones held: of cells in gear over 2,000 time units, 706 of 747 stayed under the floor over 8,000 (FAUX-R), and of paces within gear 1's and gear 2's floors at 2,000, 525 of 525 and 386 of 387 stayed within them at 8,000 (FAUX-Cb) (row 603; R2W, sitting P2)

the Loop Atlas sessions (R2W); ties rows 380, 603, 605, 627, 670

A local law has a neighbourhood; register it (640) — a fold's square-root law held within about 3×10^{-4} of its edge and failed when the registration asked it to hold over three decades of distance; the reading registered for the right neighbourhood held. A law read near an edge says where it holds, before the run. A counter-case on the neighbourhood's size: in the Loop Atlas's needle the fold's law at gear 2's upper edge holds over two and a half decades (slope 0.48 to 0.49 over the five smallest distances, 10^{-4} to 8×10^{-4}, and 0.46 to 0.47 out to 0.03). A law's neighbourhood belongs to the machine, so it is registered from the machine under test, not borrowed from another (the Loop Atlas sessions, the book). A rule of rates needs strings long enough to carry it: the change of change of slope needs four dots or more, and with pads sized by the dots' spacing the 6,000-pass strings fell to a median of 3 dots and the measure's median to 83.4 degrees, which is noise, while a registered ratio still read 0.336; the registration should have named the rule's neighbourhood. The eye works at two scales and a lattice needs both: pads sized by the nearest neighbour at one birth were a cluster's own spacing, so the lattice saw only clusters of three visits, while the person saw the rows the clusters make (the triple-star sessions). Zoom out before naming a shape: at 600 passes an opened torus showed strands and a weave, and at 6,000 they filled a solid hourglass with sharp walls, and a slow beat of about 680 passes appeared (the Loop Atlas sessions, R2W). Demagnify to see what a feature belongs to; magnify to measure it: demagnified, every weight's S curve lay on one master curve, and that led on to the band and the pin (runs BAND-1 and PIVOT-1), the mirror and the weightless root. The person's default is to demagnify (R2W, after his "most of the time we want to demagnify"). His "zoom and double" asked whether the 0.26 dip is a renormalized copy of the whole, as in period doubling, not for a closer look; demagnified, his stroke of 7 October at 0.40 joined the two stops into one line, as a period-doubling birth would (R2W). A law fitted near one line is read line by line before it is tested far away. The drag law with the star's eccentricity explained 97% to 99% when pooled over six nearby start lines and was registered to explain 90% on fresh lines farther out; it explained 44% to 71% (E7c missed, run 20261016_161446). Read line by line, it falls smoothly with the start's distance from the mirror: 99.4% to 99.7% on its own line, 91.4% to 95.7% at 0.05, 12.2% to 33.7% at 0.12 at three slices (run 20261016_161828). Four of the six pooled lines lay within 0.03 of the mirror and hid the fall, and the misses are not an extrapolation in grip: 745 of the 876 runs lie inside the fitted grip range and miss all the same (the triple-star sessions). Small residuals are not a promise: before RATCHET-61, a straight line in the logarithm of the frequency fitted four stretches of the fourth harmonic's holding power with residuals no larger than 0.0011, and it carried the next stretch to within 0.0053. On the stretch after that it missed by 0.0214, and the test failed. Read afterwards, the line's own points had been bending all along: the local pace going into the fifth stretch was already 0.0948 against the line's 0.1220, and 0.0655 after it (R61_AFTER). Points close together fit a straight line well even when the curve through them is turning. Before carrying a fit outward, compare the last pair's local pace with the fit's slope, and when they differ, register a levelling rival beside the line (row 799) (R2W). A prediction from a map holds only as far as the map was read. Bear marked one sentence of Claude's overnight note as a great prediction to test: that at mass 1 the passing lane fills the whole road above the slow lane's fold, so a mass-1 rider cannot run free anywhere on the road read. Made exact and run first, as the next exchange's thread, it held on the road first read, to 1.06, and three steps of the dial beyond it; read on to 1.80, the lane ends at 1.09 and open road begins at 1.10 (run LIGHT_LANES 20261016_174648; the Loop Atlas's misses row 263). The miss was the sentence's, not the reader's. Four leans in the same stretch of five exchanges missed the same way, at the edges of maps read too short. When the person marks a sentence, make it exact, run it first, and run every marked prediction past the edge of the map it came from (the Loop Atlas sessions)

the Loop Atlas sessions (R2W); ties rows 318, 618, 631, 713, 719, 808, 870

Read at a fixed moment of the cycle, or over one whole cycle, and say which (641) — within one passing of a strong pattern a cloud of noisy copies is stretched and squeezed about a million-fold in variance, so a spread read at an arbitrary moment measures the moment, not the state; read at the most-squeezed moment, the wobble law held to 1 per cent. The same trap at the other scale: a clutch whose instantaneous frame swings 32 to 44 degrees within a lap could never sit within 5 degrees, yet averaged per lap it caught in 14 laps at the edge and 5 deep in the gear, and a whole-run average carried a start-up transient that failed a locked gear's test. Average over one cycle: not the instant, and not the whole run. Compare two riders at the same phase of the pattern, never at the same clock time: two walkers that reached one corner by different routes disagreed in lag by up to 2.56 radians at one clock time, and agreed within 0.009 at the next passing of the pattern, their drift then a whole number of turns (389.998 and 184.999). A locked rider's lag swings within each passing, so the same place includes the pattern's phase (the Loop Atlas sessions, R2W). A reading can be bent by the very thing it watches: a heavy star's runs read the pair's eccentricity at the star's closest approach, where its tide bends the pair's ellipse by an amount set by the resonance angle, and the reading reported that bend as the pair draining off the islands. Averaged over the pair's own orbit, the

mean barely moves off the islands and sits at 0.041 to 0.076, not the nominal 0.10 (the triple-star sessions). To see a period exactly, strobe with a step that divides the drive period: a step of 0.04 cannot land on multiples of 2π; carried on with 160 steps a period, the settled spin repeated after 4 periods to 1.7×10^{-9} (run SETTLED_COPIES 20261011_212014) (the Loop Atlas sessions). Read at another moment of the drive cycle, a strobe picture is the same picture on another ruler: the quarter tree keeps every arch, void and gap at ten moments 36 degrees apart while its strip breathes from 0.38 to 0.011 rad. To see where a chaos is made, read the whole circle: three of the window's four strips sit thin round the hilltop and one is wide, as if the needle balances for three beats and falls on the fourth (run QUARTER_PHASES 20261014_073718, at Bear's request) (the Loop Atlas sessions). A ruler read over a fixed count sees only the cycles that divide it: the spread of sixteen needles' turns over 200 drive cycles fell to nothing in a window of 10 cycles and showed nothing at all of a window of 6, or of a ladder of locks of 4, 8 and 16 cycles, because a needle on any of those turns the same whole number of times in 200 cycles whichever point it sits on (run GAPS_READ 20261015_061859). Read a state's own cycle length and multiplier, not a count over a fixed window; the ruler's own step is row 686's, and what the eye saw through this one is row 715's (the Loop Atlas sessions). Fix in the registration the instant at which each series is read, before anything is computed: Bear's crooked-clock test set the eccentricity averaged between passes against the angle half a pass on and held (−0.026 and −0.021 turn against a bar of 0.1); against the angle at the pass before it would have read +0.127 and +0.083, and one case would have missed (the triple-star thread, CF1)

the Loop Atlas sessions (R2W); ties row 630

Near a fold or a doubling, never classify by waiting for things to settle; follow the orbit and read its multiplier's sign (642) — states there settle at a rate that goes to zero at the transition (two riders needed 1,500 passings to sit within 8.5×10^{-8} of their period-8 orbit), and two registered instruments failed this way, one by bisection and one by fitting a split that grows faster than the grid can follow. Follow the state itself by Newton's method on the periodic orbit and read its multipliers. A multiplier reaching −1 is a doubling. One reaching +1 is a fold, or the birth of a doubled orbit from its parent's −1, since the square of −1 is +1: read the sign before naming the edge, and carry the orbit half its period to tell which, because near a birth the new orbit nearly closes on itself in half its period, being close to its parent, and at a fold it does not; a fold refiner also snaps to a true fold, in 2 to 7 steps at eight true folds of record, and wanders at a doubling (17 and 21 steps, its residuals rising and falling), because there the fold's equations hold all along the doubled parent. A follower that watches only one orbit reads that orbit's birth as a fold: the Loop Atlas's period-8 end at 1.690017021, first called a fold, is the period-4 orbit's doubling, refined at 1.690017021430, as R2W found by letter. Find a bifurcation by Newton, not by a grid, because each doubled dance lives in a window about 4.67 times narrower than its parent's; refine it rather than interpolate, since a doubling placed by interpolating between two continuation steps was off by 2.11×10^{-6}, more than the distances then measured from it; keep a continuation's steps inside the dance's own window, guard every Newton call against the parent (one unguarded point landed on the parent and counted twice, its +1.39792 the square of the parent's −1.1823335), and stop above the noise. On a flat condition small steps are not arrival: a refinement that prints NOT converged with a small last step has not arrived. The tool this needs now exists in two rooms: R2W's gear solver and its doubling follower, and the Loop Atlas's strobe Newton, which refines folds by solving for a multiplier of +1 and doublings for −1, and put gear 2's folds inside the brackets that held riders had set, moving by at most 10^{-9} when its step was halved. Settling is slow beside a doubling, so widen the window there and measure the settling before predicting it: after 8,000 passings a history first read a four-step dance's single visit at column 824, one beyond its window, and settling four times longer moved it to 820 to 822, not to the 819 or 820 predicted from the multiplier; a fraction of a column above a doubling, neither rider repeats within 128 passings. An unsettled rider is a reading in itself (the Loop Atlas sessions, the book). One machine can stand in for two: to tell a fold from a doubling, restart the refinement from its start nudged by one and two units in the last place. At eight folds all 32 nudged step counts matched the unnudged count; at four period-32 doublings every refinement's count moved under some nudge (the first, 1 step unnudged, took 40, 1, 40 and 30 steps under the four), while the doublings' values moved by at most 1.4×10^{-13} and the folds' drives by up to 5.4×10^{-11}. The step count separates the two kinds; the value does not, and it separates them the other way round (run ULP_NUDGE 20261011_044600) (the Loop Atlas sessions, the book). And a refiner's bar sets its answer's error, so the step count reads fold or doubling only while the bar stands above the arithmetic's floor: refined to 10^{-12} instead of 10^{-9}, eight folds' drives agreed to 1.2×10^{-14} from five nudged starts (5.4×10^{-11} before), and the drives of record, refined to 10^{-9}, proved up to 7.2×10^{-11} off; seven folds still snapped, but one whose residual floor sits near 10^{-12} wandered like a doubling, 8 to 20 steps (run FOLD_TIGHT 20261011_051636) (the Loop Atlas sessions, the book)

the Loop Atlas sessions (R2W and the book); ties rows 631, 654, 672, 680, 697

Check whether two dials are one before sweeping both, and compare equations before comparing results (643) — raising a toy's drag with the pull held steady is exactly the same as lightening the rider: one dial, mass over drag. Two rooms then compared results at a dial setting of 1.0 and found two different roads to chaos (period doubling in one, none in the other), which can only be read once their equations are set side by side: the needle's memory is inertia with drag 1, the rider's mass knob. The opposite trap is two dials moved together, as when a sweep of the grip moved the drive with it. Three rooms, one pendulum: match the dials before trading numbers. The Loop Atlas's

needle, R2W's rider and a grid generator's swing are one equation (m L″ + L′ + sin L = p), and a number from one room is a witness for another only once the dials are matched (memory with weight, drag and pull both scaled to 1). Matched, the needle's switch 0.70256 sits inside R2W's measured 0.70 to 0.71 and below its recalled 0.706 (run MEMORY_SWITCH_2 20261011_085331, with R2W's pendulum note relayed by Bear on 11 October) (the Loop Atlas sessions, R2W). One dial joins the swimmer and the skater: how much harder it is to drag the body sideways than lengthways. At 1 nothing moves; near 2 (a thin body in thick fluid, typically, as Lauga and Powers, 2009, give it) the snake crawls 0.15 body lengths per lap; at a blade's grip it reaches 0.70 (the snake toy) (the Loop Atlas sessions). When two knobs have always moved together, pull them apart before naming either. In the run that prompted Bear's reading, the tied bells' squeeze period was held at half the water's memory, so the two knobs always moved together, and Claude had given the memory the sign of the water's gain. Bear's reading was a ranking: event timing, he predicted, matters more than the water's memory, though both affect it. Made exact with the two pulled apart, his ranking held clearly in the toy as first built, and held again with new digits once its cores were given their own size (the end of this row): the period sets the sign (the gain crosses 1 at 2.72 to 3.23 rings' times at every memory from 3 to 12) and the memory only deepens it, the gain swinging 0.63 across the period against 0.25 across the memory (run BELL_BALANCE 20261016_193628). A person's ranking of two causes is a test of size; it needs the two knobs separated before either is named, and it is scored in its own column, because a ranking can be right where a sign was never in question (row 712) (the Loop Atlas sessions). The card behind that run had asked for one ratio: the squeeze speed over the rings' was leaned to set the water's gain more than the memory does, and the gain's spread across the squeeze speed averaged 0.2426 against 0.3257 across the memory, so the rival held. There was no one ratio. With the period held at half the memory, the gain crossed 1 between memories 5.23 and 6.24 at every squeeze speed, a period of 2.615 to 3.12, inside or touching the period's own crossing of 2.72 to 3.23: one line seen in another cut, not the memory setting the sign; the squeeze speed set the size (run BELL_RATIO 20261016_183422) (the Loop Atlas sessions; the reconciliation is the Field Guide sessions'). With the rings' cores given their own size where two rings meet, the ranking held and its digits moved: the gain crosses 1 at 2.02 to 2.94 rings' times, against 2.72 to 3.23 in the toy as first built, and the period swings the gain about 2.4 times as much as the memory (run BELL_SOFT_MAP 20261021_000736; row 807) (the Loop Atlas sessions)

the Loop Atlas sessions (R2W and the book); ties rows 9b, 584, 807

Keep the numbers small, and cancel a background in the code, not only in the reading (644) — over a long run a rider's position grows to tens of thousands of radians, where a double resolves only about 10^{-12}, so a window reaching down to 10^{-12} would have measured rounding error; integrate in a frame that moves with the gear, and reduce every phase to one turn before taking its cosine. A triangle of stars carried 400 units aside kept its loop law to the same residual as at rest, yet its history parted from the unmoved run's after its first two loops, because the computer rounds a faraway number more coarsely than a near one. The physics cancels a uniform background; the floating point does not, unless the code does. Count time in whole steps and compute it from the count: a clock that added 0.02 to itself 1,300,000 times read 26000.00000047, filed the window ending at its ramp's end under the next hold, and failed a fidelity comparison in which every change of state was identical. Carry angles modulo a turn: a lag carried as a running total reached about 2,400 radians, where the computer keeps it only to about 4.5×10^{-13}, so a strongly held passing squeezed a shadow rider inside the last digit and 46 stretch rates came out as not-a-number; and a period finder that compared a new orbit with the old by distance, without reducing the angle, counted an orbit 51 turns away as new. Every test of sameness compares angles modulo a turn, not only the entry to Newton (the Loop Atlas sessions, R2W and the book). Three more ways a refinement can mislead. A sign change brackets a root only if the function is continuous along the path: a residual bisected sixty times stayed at 4.91×10^{-10}, a jump and not a zero, so bisect and watch the residual shrink before counting it. A parabola refined by cutting its spacing r times each round has an error that falls by r to the fourth power each round (81, 625 and 10,000 times at r = 3, 5 and 10), always toward the inside of a lock, so its printed last move overstates the error left by about that factor and the reading is biased narrow; clamp each step to the bracket of the points that made it, since one round's vertex leapt 2.16 radian out of its samples and the run died. And an integrator crossing a place where the integrand's derivatives jump errs by an amount set by where the jump falls inside its step: the error bounced between -4.3×10^{-10} and $+2.3 \times 10^{-9}$ with three changes of sign, while the same circle, smoothed to its fourth derivative, fell a clean 16.3 and 15.5 times per doubling of the steps. Read errors as envelopes over many step counts, not as ratios of two (the Loop Atlas sessions, the book). Choose a grid's members by an integer test, not a float: a check that a birth sat on a grid by rounding it to three places let two extra births in (7.1980 and 7.1990), so a test scored 25 and 17 births instead of the registered 23 and 15, and printed "borne out" (15 of 17) where, on the registered 15, it is not (13 of 15). The integer test asks that 200 times the birth lie within 10^{-6} of a whole number. Print the number scored beside the number registered (tool triple_maypole_reach v1.0; runs 20261012_051701 and 20261012_051756, void and renamed _SET_ERROR, and 20261012_051949) (the triple-star sessions). Cancel a known lag by symmetry: walk the loop again, turned half a turn about home, and average the two runs; the lag cancels and the loop's mark stays. Without it Two Arrows' area-law slopes read 0.4931 to 0.5241, not 0.9545 to 1.0092 (run s1) (Two Arrows)

the Loop Atlas sessions; the triple-star sessions; ties row 359

A window must hear three beats of its slowest beat, and a rule that can be computed belongs in the tool (645) — a notch appeared in runs of 2,000 time units where the kicks and the teeth beat once every 5,776 units, so the run heard a third of one beat; at 10,000 units it was gone. The rule applied again a level down: a 48-beat window read a spike where 200 beats showed a slow beat of the beat. The other edge of the window: watched for a whole run, a cut point wandered across 35 per cent, while within five-lap windows its spread was 0.00 per cent inside the gear. The three-beat rule was broken hours after it was written, because a remembered rule can lapse; a guard in the tool that computes the beats in each window and refuses a run below three cannot. A window that listens for a beat must span whole beats: over 20 passings, a fifth of a beat, a hum's span depended on where the window fell in the beat, and that sank a prediction (the Loop Atlas sessions, R2W). Set a fidelity tolerance from the reading's own error: a rotation number read over N periods ends part-way round any cycle whose length does not divide N, and carries up to about 1/N of a turn, so a check asking for 10^{-6} over 4,000 periods could not pass at cycles of 7, 11 or 13. Read over a whole number of every cycle in play (10,010 for 7, 10, 11 and 13), or set the bar at 1/N (run STEP_BACK_RAMP 20261013_195010) (the Loop Atlas sessions). And set a whole-cycle read by the orbit's least repeat, not its reduced fraction: three "5/7" and "7/10" locks repeated after 14, 20 and 28 cycles, so a read of 10,010 periods, chosen for 7, 10, 11 and 13, could not be exact at 20 and 28 (run FIVE_SEVENTHS 20261013_222744; its 28-cycle reads were not found when re-flown longer, row 760) (the Loop Atlas sessions). Read a rotation number as a slope, not as a difference of its ends: turns per passing read as the last strobe less the first over 2π times the passings between carries the orbit's own wobble at both ends, so riders on one smooth spin disagreed by 0.0031 to 0.0038 at three cells and a registered agreement clause of 0.003 missed; read as a least-squares slope over 120 strobes, the largest spread at any heavy running cell is 0.00025. The same end-to-end reading put a period-5 pad rider at 0.4008 for an exact 2/5, because 199 passings is not a whole number of its cycles; a period-8 rider read 0.5000 for its 1/2, with no error showing in the four places printed (runs RUN_STROBE 20261006_170149 and RUN_STROBE_READER 20261006_170950) (the Loop Atlas sessions). A rule read is not a rule kept: two hours after "keep the numbers small" was read in full, a follower entered its orbits with the angle at about 2,360 radians, carried over from the rider that found them; its entry needed 7 to 8 iterations and bottomed at 2.4×10^{-12} against a tolerance of 10^{-11}, against 3 iterations and 3.8×10^{-13} with the angle reduced to one turn, and the follower failed on three grooves and on the period-8 orbit. Version 2 reduces the angle in the code at every entry (run GROOVES, its first version, 20261006_192450, retired) (the Loop Atlas sessions). Twice more in the triple star. At 9 of the 11 births from 7.470 to 7.480, a 600-pass window hears only 1.68 to 2.55 cycles of the beat (TRIPLE_LOOK_BEAT_FLOOR_20261015_053316), so a reader's verdict "beat strongest" there cannot tell the beat from the slow line beside it. And at the star's eccentricity 0.28 the beat falls to periods of 1,100 to 5,700 passes, which 2,400 passes cannot resolve (triple_swell_look, run 20261016_043016) (the triple-star sessions). A lock read in short windows passes a slow slipper: a rider that slips once every 200 to 400 time units makes no full turn in a window of 100, so a lock is read over the whole hold with each half asked to agree with the whole, the band test that caught it before the run of record (TRUCK_ROAD's blind run 20261016_085417, retired; the Loop Atlas's misses row 260) (the Loop Atlas sessions)

the Loop Atlas sessions (R2W); ties rows 570, 641, 700, 760

Labels are not measurements: compare positions, and measure a dial before naming it (646) — a gear's locked words matched the film's words at only 3 of 9 paces, because a rolling lap does not close and its word starts at a shifted letter; compared by where the dots land, the frames sat within 0.001 to 0.019 of the cut at 8 of 9. And a dial called "kick size" for a day turned out to shrink the kick by only 11 to 14 per cent over its whole range. Name a quantity after it has been measured (the Loop Atlas sessions, R2W). A law's name can point the wrong way: R2W's book calls the dark point's drive law an inverse square, because the fourth harmonic's share against the octave's falls as one over the drive squared, while the octave the dark point needs grows as the drive squared. The one-page summary said the dark point moves with the drive as an inverse square, and Claude's own reply then told Bear that the dark point shrinks as the drive grows, the opposite of the data. Wherever a law is summarized, say which quantity rises and which falls, in plain words; a name is not a direction (R2W). A value read from a document's table is a claim about which cell it came from: the Loop Atlas's explorer engine checked itself against the diver's values of record by taking the first number in each row of the scope's table, which sat in the check's own name ("at amplitude 0.6"), so the engine was judged against 0.6, 3 and 10, and three good numbers printed as missed. The blind run caught it. Read a cell by its place, and assert that the row's other cells hold the name, the run and the tolerance where they are expected (run EXPLORER_ENGINE_CHECK, blind 20261017_063057, retired, and 20261017_063123) (the Loop Atlas sessions). A function's name is not its road either. The traders' constant 0.5826, which sets a sampled switch's dead band, was offered as a number from prime-number theory because it carries the zeta function at one half; but zeta enters Broadie, Glasserman and Kou's constant through the ladder heights of a Gaussian random walk (Chang and Peres, 1997), a sum over the walk's whole-number step counts, and the primes play no part. The whole numbers do carry it, so a lens of the integers still fits, only not the primes. And 0.5826 is the small-step limit, while the knob itself grows with the look's coarseness (runs CREEP_LAW 20261016_035058 and ZETA_LIMIT 20261020_060256) (the Loop Atlas sessions). A file's kind is read from what its maker printed in it, never from a word in its name: the Loop Atlas's register took any run named with "*BLIND*" for a blind run, and two tools named for the rider that never taps blind made their two runs of record look blind, so the book counted 167 blind runs where there are 165 and left their

eight verdicts out of its count of predictions, 1,098 where the runs hold 1,106; the first check of the fix was fooled by the same name, three misses of one fault in one day. A blind run is now counted by both its marks, the name's and the header's mark the tool prints (the Loop Atlas sessions, misses rows 543 to 546)

the Loop Atlas sessions (R2W); ties rows 586, 612; the Loop Atlas's checks of 30 and 31 October: BLIND_MARK

Repeat a known point inside every new sweep (647) — 483 runs were first scored against a wrong target, the product of three factors with one of them squared. The slip was caught because rows that repeated an earlier test's points missed at every filter where they had hit before. The table was retired, and the speeds reproduced row for row. A known point inside a new sweep is the cheapest witness a sweep can carry (the Loop Atlas sessions, R2W). A long run that extends an old one carries its own witness: controls run for 1,200 passes in the container reproduced the earlier 600-pass rows to every printed decimal, Newton's method found each start to its ninth decimal, and a re-integrated record redrew an earlier view's counts exactly. So a long run extends the record rather than redoing it, and its first window is its own fidelity check (runs triple_held_chaos_long 20261012_075138 and triple_maypole_plateau_view 20261012_072906) (the triple-star sessions). Before walking a parameter away from a known state, hold the walker still at the start and check it stays put: a needle walked off the 1/4 groove left it at the very first step on both sides, which may be a sliver of a groove or a walker that never held the groove at all, and without the zero step the run cannot say which (run SETTLED_GROOVES 20261011_223349) (the Loop Atlas sessions). Fold a fidelity check into a run that needs the point anyway: RATCHET-62's full-octave part needed its 1-times windows as the baseline for every larger size. Rerunning them instead of borrowing them from files made on another day turned the baseline into a fidelity check, and it matched the old files exactly (0.00e+00). Two runs bought the right to compare the new sizes with the old one (R2W). A range set after a probe starts at the probe's value: a search over settings can only add to the best setting a probe already tried, and only if the probe ran at one of the search's own settings. Here the search added nothing, and the 0.0009 between probe and result was the probe's torque rounded (corrected in batch 52 from the run's saved settings); a lean whose lower end sat 0.005 above the probe missed by 0.004 (run SNAKE_LONG 20261015_073142; the Loop Atlas's misses row 223). And repeat the parents first: every new tool of one session read the tools it built on by their text and first repeated their records, often to the last printed digit, before anything new was trusted (the Loop Atlas sessions). Each new version of a tool gives back every line of its predecessor before the old one is retired: the thread's surprise reader went from v1.1 to v1.7 in one day, each version reproducing the one before (v1.7 gave back v1.6's 102 lines), and its run tool's new cases were added by diff, "token for token as before". That is what lets a tool change daily without moving a number. Every follow-on run of Question the Substrate printed a check that it reproduced an earlier run at the same seeds (all true: E1i, E1j, E1w3 against E1w2, E8 against E5, E10b against E10, E12 at $\beta = 1$, E14 in three worlds), and one command reran the whole project and reproduced every summary file. Reproduction is the cheapest wall there is (Question the Substrate)

the Loop Atlas sessions (R2W); ties rows 242, 588

A grid that lands on exact fractions samples frozen frames, so test a hair either side (648) — extra grip appeared at ratios of exactly 2.000 and 1.500 and at no neighbouring grid point (1.998, 2.002, 1.498, 1.502). At an exact fraction the phase between two rhythms is frozen at whatever the run starts with, so the result is a setting, not a property of the ratio: at 2 there is one good band of starting phases, at 3/2 two, and a hair away the frame drifts and the rider is caught at the bad stretch (the Loop Atlas sessions, R2W). A start off the averaged orbit reads as an offset: a third star given a point-mass start left its outer orbit sagging, which read as an offset of 0.0883 in the apsidal law, and started at its true circular speed in the binary's averaged field it left -0.0006 over the same 36 runs. That control was not in the registration; it was added before any run, because the reading already sat near 0.06, so a small offset alone could have passed for the wrong reason (the triple-star sessions). Re-read at births 0.001 apart, 11 of an atlas's 20 flat stretches were locks, and a change of 0.01 turn in a starting angle decided whether a rate's crown touched 1/11 and locked. The start does not make the lock; it chooses which locks can be seen: a lock at a third was seen only from starts on the mirror line, where a mirror-symmetric system keeps its repeating orbits, and it was real there (14 adjacent starts, a window 0.070 turn wide). A blind figure placed by rule at 7.250 landed on 29/4 and drew a frozen frame of tight clusters, and was redrawn a hair away. And a known answer built on a rational number repeats itself: a bench pushed by 0.02 per step, which is 1/50, repeated every 50 steps until it used an irrational turn (the triple-star sessions). A grid's spacing can draw the shape read from it: FN3 (run 20261017_030927), with its v1.2 print (run 20261017_031501), showed a logarithm fitting better outside the hold on eight lines and inside it on one, which suggested a two-sided fin. FR1 re-ran Figure 7's line at 0.001 in birth (run 20261017_032258): both sides of the tip came out closer to straight, so the split seen at 0.005 was the spacing's, and the fine grid also showed 7.358 held between unheld 7.357 and 7.359. Before a shape seen at one spacing goes into a paper, read it again on a finer grid (the triple-star sessions). A specimen from R2W: a perfectly even rhythm Bear saw in a map's margin bars was the instrument's grid. The explorer counts riders in 40 bins of start lag; with starts every hundredth of a turn the bins take 3 and 2 starts in turn, 105 and 70 riders, computed from the start grid alone, and at 1,000 starts every bin holds 25 riders. A second specimen of row 741 (R2W, sitting P2)

the Loop Atlas sessions (R2W); ties row 38

Register in the regime where the rival readings disagree, and name what carries an effect before naming its null (649) — a first test of a slope chose a range where both readings predicted the same slope, so it could not tell them apart; after two redesigns it held blind, at −0.573. And a registration said a gear would vanish where the strong pattern's carrier cancels, when in that gear the rider rides the weak pattern and survives as a bare weak-pattern lock (top edge 0.38 against 0.378 predicted). Choose the regime, and name the carrier, before the run (the Loop Atlas sessions, R2W). Work out the rival's prediction before registering: a reading (the Farey product) and its rival (the old law $\delta_1{}^3$), each given a band of 15 per cent and expected to part at 4/7 and 5/8 though neither prediction was worked out first, differed by only 0.6 to 5.2 per cent at the four rows run; both held, the product within 6 per cent and the rival within 11 per cent, and the run, seven minutes on two cores, could not say which law the junction follows, though a roots-only pass of a couple of minutes could have shown it first. Register only where the two differ by more than twice the band, or choose cases where they do (run FAREY-1) (R2W). Two stories that fit every row need a row where they part: two accounts of a grip were consistent with all 21 rows at three strengths, because those strengths never overlapped on the first story's scale; two strengths in between, chosen so that their rows overlapped their neighbours on that scale, made the stories disagree by 7 to 17 per cent, and one failed at every overlapping row (run TOY-SIDE-4) (R2W). When a series could run in one over the frequency or its square, register both: R64-k's lean was that the lopsidedness order's excess closes as one over the frequency, and its rival that it does not close (the order's law in the frequency, RATCHET-64, 65 and K-16). It closed as one over the frequency squared (power 1.92 at both drives), so both failed, and the run still named the law. A third shape, the next order, costs nothing to write down before the run (R2W). A rival must be able to hold only where the lean fails. FX-3 said the early slip is the resting dance doubling; its rival FX-3R said the dance stays stable wherever the ball is on the plain. Both held, because the dance doubles exactly where the ball leaves the plain (1.165 against 1.166, 1.145 against 1.147): stability wherever the ball is still on the plain is true under both pictures. The rival had been written as a property of the other picture instead of as a measurement the lean forbids. Write each rival as a number or event the lean says cannot happen (here, a ball still on the plain at a drive where its dance has already doubled), and check in the dry run that a stand-in can make the lean hold while the rival fails, and the reverse, on the same quantity (row 337) (R2W)

the Loop Atlas sessions (R2W); ties rows 369, 580

Rerun the other room's own code before judging its numbers (650) — a three-body law reported to a mean residual of 0.0014 radians was checked by loading the other room's own tool and running it unchanged: it printed its energy line exactly and found the same 58 loops. Only then could the reading be refined with nothing else changed, and the law held to 0.00003. The strongest check of another room's number is that room's own code, run here (the Loop Atlas sessions, the book). A guess about where a number came from is a reading, and the trace is the run: this guide guessed that a triple's untraceable 0.909 was a figure carried across quantities; the guess was plausible, matched three digits, and was wrong, and rerunning the other room's own trace settled it in under a second (the Field Guide sessions, the triple-star sessions)

the Loop Atlas sessions (the book); the triple-star sessions; ties rows 35, 438

A correlation reads pattern, not size (651) — a memory test read +0.87 for the correlation of a residual with loop duration, which looked like memory; with 24 outlying loops set aside it read −0.22, and it read +0.87 again after finer reading had shrunk their offset fiftyfold, to 0.00007 radians. Quote the residual's size beside any correlation of it

the Loop Atlas sessions (the book); the triple-star sessions; ties rows 19, 307

When a knob changes mid-run, carry the right state into the new frame (652) — a sweep switched a needle's pace mid-run and kept the needle's own speed, when what had to carry across the switch was the speed of its lag behind the field. It read false climb-back lines (0.8105 and 0.5518) against the true 0.7899 and 0.5413 found by shooting from the saddle, and was retired. Write down which quantity is continuous across the turn of a knob

the Loop Atlas sessions (the book); ties rows 711, 789

A ramp reads an edge; a hold asks what lives there (662) — a ramp cannot tell shut scissors from a lag. One room's ramps read an opening that halved as the ramp rate halved, the mark of a lag, and another room's ramp, sixteen times slower, read none. Holding riders at one drive settled it: at mass 0.75 every rider started running fell into gear 1 below the fold, and at no mass held was there a running state in the window a bound had pictured, so the bound was withdrawn by its authors; its number had been met from mass 1.0 up only because a second lock held the memory (row 670). A held ensemble starts on the branch the ramp carries, prepared by the ramp itself and checked to be there: riders started at the drive's lab speed fell into the weak gear instead of the slipping state, and riders started from lag 0 sat off gear 2's branch, so their counts were only lower bounds. A moving reader also needs time to land: ramped at 2.5×10^{-6} of drive per passing, a needle crossed a two-dance island in about 100 passings and never landed, where held riders took up to thousands; at 2.5×10^{-7} it landed in both directions. Test the picture behind a bound, not only its number. Burn in at each held setting, even a near one: needles settled just inside a crisis and moved a little further inside still left the four narrow bands of the window's chaos in a few hundred of 512,000 readings; settled 3,000 cycles at the new pace, none did (run CRISIS_BURSTS 20261014_091405) (the Loop Atlas sessions). Start a held-fixed solve on its own branch, and check a continuation step by step: a solve for the speed alone

at a fixed drive and angle, started from the rider's mean speed, wandered, because the speed after five passings depends strongly on the starting speed through the phase (its slope against the start ran from about −0.09 to −1.2), and printed a residual of 0.198, not closed, at the 8/5's upper edge; started from the loop's own speed (1.73984, from the fixed-angle solve), the same solve closes in one step. Carried from angle to angle, it still lost the branch on the 11/7 where the speed swung by up to 0.09, landing once on another root (a residual of −5.167). A third version predicts each angle's speed from the last two and accepts a step only if the speed closes, the turn's residual stays small and the speed lands near its prediction, halving the step otherwise; it stayed on the branch and reproduced the second version's 8/5 lines word for word. Continuation that is not checked step by step is a guess about which root it is on (runs GROOVES_11, version 1 20261007_073129 and version 2 20261007_073359, both retired, and version 3 20261007_080706) (the Loop Atlas sessions). A place held because a walk first fell there sits on the lip: LEAK_RATE held its needles 0.0025 below the window, where LEAK_CLOSE's walks first fell (0.0022 to 0.0029), and between memory 1.7198 and 1.7201 the needles' lifetime there went from 3,763 drive cycles to none of 60 falling in 20,000, a jump that the door shutting and its lip moving a hair past the held place would both make. Hold at several depths before reading a lifetime against a setting (run LEAK_RATE 20261015_080150) (the Loop Atlas sessions). When sampling thins, test the definition before the physics: past the room's closing only one start in four reached gear 2 at its door, so the door might have marked where gear 2 can be reached, not where it exists. A rider put on gear 2 and lowered until it fell off settled it in six minutes: the two agree within two steps, so the bend was physics (DOOR-3; row 839) (R2W)

the Loop Atlas sessions (R2W and the book); ties rows 563, 639, 670, 758

A reading window reports its middle, and the window that holds a change belongs to neither state (663) — a ramp's leave line is a window's middle: a 200-passing window spans about 0.0024 of the drive, so a departure can be read up to about 0.0012 before it happens, and one registration had set its tolerance's sign the wrong way. A reader that sorts fixed windows into states files the window in which a change happens as a third thing: a prediction that a rider would land in a second gear "within one window" of leaving the first missed for that reason alone, and read from the first window wholly after the departure, it held; its tolerance for the departure, 0.003 of pace, had also left no room beyond its window of 0.002 for the ramp's own lag. A counter of largest swings, kept between every second zero crossing, records a growing swing at the end of each swing and a dying one at its start, so a fill and an empty that are exact time reverses print as reaching their marks at swing 10 and swing 11, though each takes nine swings. Write a prediction about a change against the first whole window after it, with a tolerance wider than one window plus the lag, and count turning points when comparing a rate up with a rate down

the Loop Atlas sessions (R2W and the book); ties rows 627, 645

A chart drawn from one start is blind to memory (664) — the textbook tongue map, the sine circle map, run with 64 starts per drive, agreed at all 241 drives below its critical line (strength 0.9); at strength 1.2 the starts disagreed at 42 drives, and a single start stepped backwards at 6. A chart of locks drawn from one start cannot show two states where two exist, and the disagreement is what links a rider's crumpled staircase to the folded circle map. Run several starts per setting, and read where they part. A chart that lays several starts on top of each other is blind to memory too, unless each dot carries its start: at weight 0.50, from drive 1.434 to 1.440, a rider started at lag 0 landed on the 1/2 groove while one started at lag ½ went into the eye's doubling tree, and the picture, which lays both starts on top of each other, said nothing of which start made which dot; a coin was hidden in it (the Loop Atlas sessions, R2W). A sweep from one start reads one blade of a hysteresis, so name the start beside every count of locks: started from rest, the pulsed needle with memory 1 sits at paces 0.98375 and 0.99375, where needles already spinning settle onto a 1/4 window and a 9/19 orbit; from rest it leaves its seat only at 0.9951, and goes straight to one turn in two drive periods (run QUANT_ZERO 20261013_081905) (the Loop Atlas sessions). A sweep with one start per setting reads one basin; before writing how a state is reached, measure when it is reached. Started at a full turn a cycle from one lag (0), the remembering pulsed needle sat at every pace below 0.995, which was read as a spin that drains; with 16 starts and the sitting-down time printed, the same 7 starting lags sit within two cycles at every pace beside the 1/4 window and the other 9 tumble on for 8,000 cycles (runs SPIN_START 20261013_161916 and SPIN_WINDOWS 20261013_163655) (the Loop Atlas sessions). And a step-back can be a second basin or the staircase itself running back; many starts at the one setting tell them apart: of five step-backs on one sweep from rest, two were a second gear beside the first (2 starts of 16), two were the only state every start found, and one was a lock beside tumbling (run SECOND_GEAR 20261013_192828) (the Loop Atlas sessions). Look for chaos living beside a settled loop at the same pace: a state met from a few starts only may be a strange one sharing the pace with a periodic one, so fly every start that meets no repeat longer and give it an exponent from a twin flown beside it. Near a doubling's end every exponent tends to zero, so read it in segments and give the spread, and read a settled loop's over whole repeats. Beside the needle's 5/7, each hidden lock (11/15, 11/16) doubles against its host's direction and ends in states that never repeat in 64,000 cycles: below 11/16 all 22 exponents are positive, +0.002 to +0.043 per drive period, while 5/7 at the same paces reads −0.031 to −0.046; above 11/15 five of seven are clearly positive and two within a thousandth of zero, their sign unread (the Loop Atlas sessions, misses rows 493, 495 and 497)

the Loop Atlas sessions (R2W); ties rows 38, 670; the Loop Atlas's checks of 30 and 31 October: HIDDEN_WINDOWS, HIDDEN_CHAOS, HIDDEN_CHAOS_LONG

Name a state only by a test that separates it from its neighbours (665) — a reader with fewer bins than states gives the extra states a neighbour's name: a four-bin reader (gear 1, gear 2, running, mixed) called "running" what a strobe of the same machine showed to be mostly tumbling, with small pads at 1/7, 1/4, 2/5, 1/3, 2/3, 3/5 and 1/2 of a turn per passing; where its finding had been fenced, a later strobe cleared it, every rider it called running from mass 1.0 up being a smooth spin that never repeats within 60 passings. A strobe is a fraction finder: one that repeats after q passings, losing p turns, names p/q directly. Key a state by its fraction and its period together: at one drive 12 riders dance a cycle of 6 passings and 52 a cycle of 12, both losing two turns in three, and 4,000 more passings move no rider between them, a memory every reader keyed by fraction alone was blind to. Read a slip only over a whole number of cycles: 1,000 passings is 35 cycles of 28 and 20 over, so a settled rider's slip over 1,000 depends on where its cycle starts. Separate riders that have landed, whose snapshot repeats, from riders in flight, whose short averages differ by start: four riders at one drive disagreed by 0.006 to 0.067 turn per passing, mostly tumbling, and a test read that as memory. And wait before calling a second state: 19 riders still tumbling beside 45 landed looked like memory, and 4,000 more passings landed all 19 (the Loop Atlas sessions, R2W and the book). Check a held angle on the element it claims to read: a held reader built its angle on the pair's apse read at closest approach, and where the mean eccentricity is small against the tide's bend that apse follows the star, so the reader named runs "held" that, on the mean ellipse, drift 31.29 to 78.23 turns over 600 passes; read there, an island ends between 7.055 and 7.060, not at 7.125 (the triple-star sessions). Before naming a second motion, follow the strays to where they settle: at memory 1.254, of 128 needles started 10^{-4} rad from the window's cycle, 5 were off it after 300 cycles and read for a turn as another motion living beside the window; after 6,000 cycles all 128 were back on it. The motion that does live beside the window at its pace is the needle standing still, which takes about two in three needles started round the circle and which none of the strays reached (run STOP_WALL 20261014_180544) (the Loop Atlas sessions). Read what a walker does on the way, not only where it stops: four runs measured the gap from a window down to the stop without reading the needle's turn rate on the way, and the first run that printed it found the motion between them was a different one, half a turn a cycle against the window's quarter, so the stop's edge belonged to that motion and not to the window (run STOP_LEAVES 20261015_052943). And a quiet stretch is not one state: sixteen needles that agreed to a hundred-millionth over a long stretch below a chaos looked like one clean half-turn lock, while the plain half-turn cycle was unstable all along and the stretch was a ladder of doubled cycles, 32, 16, 8, 4 and 2 drive cycles, met in reverse; the flickers Bear ringed between rungs were the slow settling near each halving, not a loss of hold (run GAPS_READ 20261015_061859; the Loop Atlas's misses row 219). Three rulers in one session could not see what was there, a spread over 200 cycles, a threshold read once a step and a window of 100 (rows 715, 787 and 645), and the fix was the same each time: print the state's own count at every step of a walk, beside the edge, and put the band test in the tool (the Loop Atlas sessions)

the Loop Atlas sessions (R2W and the book); ties rows 627, 646, 664, 757

Look in the record, yours and the next room's, before registering (666) — two predictions modelled a relay's second stage by its steady-drive fold, when the session's own record, measured on 3 October, already said that edge holds only when the filter is slower than one over the room; both failed as the record had predicted. A grid borrowed from another room repeated a lesson that room had written two days before, choose the regime before registering, because a lesson written at a handover reaches the other sessions days later. And a rung this guide sent out as an open question had already been measured in the room it was sent to (1.6924902823; row 680). Before registering, search your own record and the latest letters for the rule that sets each side; and when a session learns a lesson another session's next run could trip on, it sends it with its next note, in a few lines. Search the record for a thing's name and its numbers before writing a prediction: registrations took a conveyor's birth from the Explorer's mark, a coarse dial's 1.4585, and did not cite the run that had placed it a day earlier at 1.4582 on a dial five times finer; the tests ran clean, but half of one repeated a test already passed (the Loop Atlas sessions, R2W). Read the latest word of record: a registration named the top of a band "the fold of record" from a ledger row the record had corrected two rows later, and missed for that reason alone; R2W reached the same rule on its own. And let another session's frame test your own: when a sister session put the momentum wall on its first page, the Loop Atlas read its own book's frame against it the same day and found the wall already stated in two places (the Loop Atlas sessions, the book). Search the record for an old card's question (a card is a row of the project's plan) by its numbers, not its label, and mark a row done in the same step that answers it: a card asking whether the toy's pin sits at the mirror at ripple strengths 0.25 and 0.35, opened with a blind lean on where the pin would move, had been answered three hours after it was written, under another name (massless roots D*(0.25) = 1.5091538 and D*(0.35) = 1.3556763 put the pin where the lean said it would not be), and two more cards the evening before had been answered the same way (runs PIVOT-2 and FBP-1's T5) (R2W). And before betting on an old law, run it through the newest reading that held: a new reading predicted an old law's rival in two lines (the excess grows as A^p), the rival held, and the old tables, read row by row, had shown it all along (at strength 0.10 and frequency 0.60, 2/3's excess 0.45 of 1/2's, and 3/4's 0.25), hidden by a straight line drawn between one-step rows (runs DELTA-WEAK-1 and FAR-1) (R2W). Before a surprise is called new, search the book for it: R2W's twenty-third exchange offered as a new link that the mirror might explain why the window-proof band is flat, when the book had said so since RATCHET-45, in the paragraph that introduced the third tone's dial, and the claim surfaced again in the record before the book was opened to write it up. The test still had value, since it checked the mirror in a stronger form, but the surprise should have said "confirms and sharpens", not

“may explain”; a search of the book’s text for the key word (here “reflection”) takes a minute and belongs before the reply. And check a proposed test against the record before offering it: an exchange offered as the next run the gap between paired timings, odd in the weight, and their sum, even, at RATCHET-45’s setting, which RATCHET-45 had already measured. Next time the record was read before anything ran, the redesign check said aloud, and the exchange went to the old card the same slope belonged to (the tilt, entry 110r of R2W’s plan) (R2W)

the Loop Atlas sessions (R2W and the book); the Field Guide sessions; ties rows 592, 649, 676

Change the pace to separate dynamics from geometry (668) — walking the same loop of drive and mass at three speeds gave drifts of 797, 390 and 185 turns; fitted as a constant plus a slope times the walking time, the constant was −18 turns, under 5 per cent of the largest drift. The drift is almost all dynamics, and the geometry shows only in what does not scale with the pace: geometry decides whether, dynamics decides how much. Row 151’s adiabatic yardstick is the limit where the second part vanishes; away from it, run the loop at several paces and read the intercept. When a measured edge depends on how fast the knob is turned, it is no longer an edge but a race: below the 1/4 window at memory 1.72 to 1.76, a pace lowered by sudden steps of 10^{-3} stopped needles that a walk by 10^{-4} over the same paces did not; a blind run of the tool, its verdicts hidden, found this before any registration, and the measure kept is the slow walk, stated as such (STOP_WALL_3’s blind runs, retired) (the Loop Atlas sessions). A loop of shapes alone has no clock at all: the falling cat’s trick lands the same at two timings of its moves to 4.5×10^{-12} degrees, and time decides only whether the loop finishes before the ground. For any body that turns itself by changing shape, budget the clock for the stretch that turns (here the swing) and let the rest be late (run EXPLORER_CAT_ROOM_CHECK 20261017_211325). The holonomy lens across domains has a philosopher’s signpost, Batterman (2003), on geometry in an abstract space having observable, noncausal effects, with Saatsi’s robustness (independent of the dynamics underneath), and a public one, Bryant’s 2013 lecture (a trackball, parallel parking, a falling cat, a rolling ball); what a bench of toys adds is the count, how much of each answer is shape and how much is clock, and what the toys got wrong (the Loop Atlas sessions)

the Loop Atlas sessions (R2W); ties rows 151, 673

A prediction can hold for the wrong reason: grade it held as worded, and say what stays open (683) — one prediction held only because a ramp read a doubling late, which was lag, not memory; an exponent held while windows narrower than the ruler would have flattened the reading whatever hid below; a survey’s prediction held at 131 of 137 cells, 108 of which were just two gears side by side; and one held on its letter while its pad was crowded out, not shrunk in place. Grade such a result “held as worded”, say what still stands open, and run the follow-on: each of the four became an honest follow-on instead of a claim. A second specimen: a memory floor predicted at the retrapping pull held (floor 0.86, band 0.85 to 0.95), but a one-ball run with no ripple showed that a rolling ball keeps rolling down to 0.78 to 0.80, as the sitting’s own rig had measured days before; the floor is where the half-turn start gets rolling at all. The band was wide enough to catch the wrong mechanism; test the mechanism with the simplest separate run, and search the record first (the Loop Atlas sessions, R2W). Computed exactly, the one-ball value at weight 2 is 0.7899, inside the run’s 0.78 to 0.80 (row 711). Grade by the registered words, and print the script’s verdict beside them: one script added a condition the registration never had and printed FAILED from a merge tolerance of 10^{-6} against 6-decimal inputs; another printed failures from a 0/0 in its root finder. Both verdicts were kept as printed and explained, the second re-run with a one-line guard registered first (runs CHECK-E and SELECT-1) (R2W). When the lean turns between registration and run, keep the registered lean to be judged and say the turn aloud before the run (the Loop Atlas sessions). Mending a scorer after a run: register the mend first, keep the printed verdict, report both. The off-mirror gate’s fault was found by reading its ledger; the mend (read a line when the table holds all its rows, as the registered words said) was registered before the mended reading was computed, the rest of the rule unchanged, and v1.1 reproduced every other verdict line for line (run 20261016_161718). The paper prints both the tool’s NO READING and the mended reading, and says the mend came after the record was read. And a guess about “everywhere”, scored by a count, can be met at one place: SM6′ guessed the refuge at a third was temporary everywhere and was scored as at least three of six escaping; three did, all born at one birth on three starting angles, while the three born just above held at exactly a third to the end (run 20261015_212604). It held as written and not as meant; a guess about where must be scored by where (the triple-star sessions). Write every guess as a verdict word, and have a separate reader cold-check the ledger by random sample: on 18 October the triple-star ledger counted 433 predictions, and a separate reader checked 40 drawn at random (seed 20261018) and found 35 right. SF3’s outcome was wrong because its guess had been written ‘against SF3’, not as a verdict word; a search for that kind of guess (‘towards’, ‘against’, ‘DISCLOSED’, ‘my expectation’) found SF1, SF3 and KD3′, guessed to fail and failed, and moved all three from missed to landed, and it found three rows that lacked the miss they redesign. A guess to fail that fails is a landed prediction, and only a verdict word lets a counter score it that way (the triple-star sessions). A counting tool must read the whole record for every rule and be recounted by an independent script: the thread’s ledger count took a prediction’s redesign from its reading of record alone, so two redesigns named on other readings went uncounted; v1.1 counts a redesign named by any merged reading, with three planted checks, and on 28 October the redesigns went from 90 to 92 (run 20261028_161705), every outcome unchanged; two cold readers recounted the ledger with scripts of their own and matched

the Loop Atlas sessions (R2W); ties rows 337, 551

A verdict is only as fine as the ruler under it: print what the step allows, say what the ruler does decide, and look for the ruler in every miss (686) — a third face's first ratio read 7.33 against a registered band of 3 to 7 and was written down as failed; but the dial's step allowed anything from 5.25 to 11.5, which crosses the band's edge, so the failure stands as registered and decides nothing. The same coarse ratio was decisively above Feigenbaum's 4.669, at 1.12 to 2.46 times it: a ruler too coarse for one question still answers another, and the verdict should name which. And a failed prediction often failed on its ruler, not its idea. A far-bank prediction failed because one weight's bank had been read on a grid four times coarser than the others, and on one ruler all seven banks fell in order; "two beats away per miss" failed because a sorted list of lags loses the order in time, and counting trips in the time-ordered record held 208 of 208. Under every judged number, print what the instrument's step allows and whether that range crosses an edge of the bar. The off-scale flag that does this caught an overstatement on its first use and, on its second, showed a held verdict resting on one case (33 of 36) (the Loop Atlas sessions, R2W). And a reading can be one-sided: with the ripple off, six stopping pulls read on steps of 0.002 and 0.00002 all sat above the exact values, each by less than its step (0.394, 0.542 and 0.790 against 0.3927, 0.5413 and 0.7899; 0.3927, 0.54132 and 0.78988 against 0.392683, 0.541310 and 0.789865). Such a reading is an upper bound, so quote the edge as lying within one step below it (run RETRAP-FINE) (R2W; this guide's check; at weights 5 and 2 the Loop Atlas's saddle shots, row 711). The ruler can sit in the registration itself: a bet graded with the passing speed taken at the edge of the plain (the person's own word for one region of R2W's map), a choice written into its own words, read the lines high above the edge as too slow, because every square of the map has its own passing speed, and it failed as worded; with the speed taken at the line, the drift went the bet's way, and two blind columns then held. Before the data, check the registration's own definitions for a ruler (runs PINCH-1L and PINCH-1B) (R2W). And a number inside a bracket carries the direction it was approached from: a memory switch bracketed at weights 0.70 to 0.71 by stepping the pull up toward the edge reads high when approached from below, moving as the square root of the distance left, and shooting at the edge itself placed it at 0.70256366. A bracket is honest; a number inside it also says which side it was read from (runs MEM-5f and the Loop Atlas's LOOP_ATLAS_MEMORY_SWITCH) (R2W; the Loop Atlas sessions, the book). A crossover read on a grid inherits the grid's rounding: a map that reads an edge as the lowest pull that is not resting rounds every lift up by up to one step, and near a sign change that moved the apparent crossover by a column, 0.6775 on the map against 0.672 on a fine dial; the fine law, rounded the map's way, reproduced the map exactly. Print what each reading allows, and look for the rounding rule in every crossover (runs DIAL-1 and DIAL-2) (R2W). A fitted peak's error has a size but no direction: a parabola through three coarse rungs read the fourth harmonic's outward peak 2.4 degrees high at frequency 0.75 and 4.4 low at 0.80 (runs RATCHET-15 and RATCHET-16), and it produced a surprise, "the peak falls fastest at the seam", that the fine ladders withdrew. At 0.75 the highest coarse rung said no excursion and the parabola said one; only a fine ladder settled it (+95.5 degrees, two crossings). When a fitted peak and the highest measured point fall on opposite sides of a wall, run a fine ladder before reading the wall (R2W). A threshold is a ruler too: beside every "none", say how small a thing it could have missed. SEAM_SCAN, run after the data in the eighteenth exchange, counted a turning point only where the curve moved by more than 10^{-4} of its own range, and reported no light-weight rise in the fourth harmonic's window at frequency 0.80 on the 0.99 line. Printing the heights over the weightless height showed one, 1.00027 at weight 0.04, below that threshold. SEAM_SCAN_B, the same scan with its tolerance as an argument, reproduced SEAM_SCAN at 10^{-4} (differs 0) and at 0 found the rise at 0.775 (1.00005), 0.785 (1.00011) and 0.80, and not at 0.75, so the rise is born between 0.75 and 0.775, beside the seam. The book's draft walls row had said "not at 0.80"; it was corrected before shipping. A threshold is a stride of its own, and a "none" means none larger than the threshold: the stride rule of row 416, met a third time in one exchange (R2W)

the Loop Atlas sessions (R2W); ties rows 128, 210, 508

A cut makes its own maximum (687) — a one-dimensional cut through an island with two actions has a maximum that belongs to the cut, so a "fastest line" read along one line is not yet a twistless torus. The test is cheap: at equal swing, do the two flanks rock alike? In the triple star's islands they never did, at any eccentricity of the inner pair. A line drawn through a two-dimensional object reports the line's geometry as readily as the object's

the triple-star sessions; ties row 360

Compare two sides at matched swing (689) — a comparison of an island's two sides made at its edges compared runs of different swing, and at equal swing the answer reversed in 16 of 16 comparisons. Before comparing two sides of anything, match them on the quantity that changes along the way

the triple-star sessions; ties row 360

A starting label is not the quantity kept (690) — the birth landing on an island's centre moved from 7.155 to 7.445 as the inner pair's eccentricity grew, while the period the star keeps there stayed between 7.010 and 7.022. A width or a position read in the starting label carries an offset that moves with the other knobs; read it in the quantity the system keeps. The same holds for a start: a pair's starting eccentricity set by its size alone left its direction to chance, and the counts centred on that start (6.90437 and 7.92105) followed the two apse turns that the unset direction moved. Set size and direction together, by Newton's method, and every one of 252 runs reached its target in both parts (runs triple_heavy_mean_start v1.0 20261010_134456 and v1.1 20261010_231824, checks K4 and K5) (the triple-star sessions). A failed fit can be a label read as a quantity: the triple-star thread wrote SA1′ (run

20261016_220144) into its paper's Section 10 as running against the one-harmonic model, whose conserved quantity K is fixed by the start. KD1 (run 20261017_011850) then showed that the model fixes K only in mean elements, while the starts are osculating, so each start line carries its own K: mean K rises steadily with the start angle while the swing does not, and mean K follows mean eccentricity with correlation −0.9997. Section 10 was corrected. Before a failed fit indicts the model, test whether the variable it blames is only labelling a quantity the model already conserves (the triple-star sessions). A starting label adds nothing beside the knob it stands in for. Asked whether birth could serve as a second dial for an island's rocking, the fit found the star's eccentricity the better second dial and birth adding nothing beside it (SA1, run 20261013_212456); the redesign registered the same day took the star's eccentricity as the dial and held (the triple-star sessions)

the triple-star sessions; ties rows 360, 781

Tell a fixed function from a force with two lines of arithmetic (691) — a per-pass change that tracks the sine of an angle, with a coefficient growing in step with the angle's own rate, is a fixed function of the angle, not a force through it: a force gives a fixed coefficient, and a swing that shrinks as the rate grows. Two lines of arithmetic told a reading artefact from a dynamical kick before any run was repeated

the triple-star sessions; ties rows 199, 641

A reader can count a sideband instead of the signal (693) — a rocking reader read 0.07058 cycles per pass against an expected 0.09045, because it was counting the lower beat sideband of the star's semi-major axis (0.0704), which that axis carries strongly; read on the resonance angle's own spectral peak, the rocking came to 0.09044. When a frequency misses, look for the strong line beside the one you wanted, and read a rhythm on the quantity that owns it. A rhythm can also hide as two sidebands, with no line at itself: a slow beat rode a 109.7-pass rhythm as lines at 67.3 and 30.2 passes, so a test that looked for the beat in the person's marked ropes missed at one birth (7.220, ropes 67.5 against the beat's 41.7) and held at another (7.190, 0.993) (runs triple_heavy_rope_beat_check 20261010_220843 and triple_heavy_rope_alias_look v1.1 20261010_220945) (the triple-star sessions). A planted check catches a reader hearing double before the run: an envelope spectrum read a planted 680.9-pass pulse as 340.2, half its period, and was replaced by a count of pinches. That counter then counted flat bottoms twice, found only after the run, so its count is reported as a look (run triple_heavy_coil_pairs 20261011_070746) (the triple-star sessions). And a floor set to stay off a known line moves with the knobs: a reader that ignored everything below 0.05 cycles per pass, to keep off the slow beat, reported the drum's overtone where the orbit was nearly round, because the rocking itself had fallen below the floor. A second reader that looks from three cycles per run up and names the beat instead of hiding it found the rocking below the beat at one eccentricity and at twice the beat at another. Print the floor beside every rate, and read below it when a rate lands near it (runs triple_reader_floor_look 20261013_222629 and triple_rocking_reader_v2 20261013_230139) (the triple-star sessions). What the floor hid mattered: the first reader, which had passed its known answers, made Mardling's rule look 149% and 51% off where the rocking was slower, and the second, which sees below the beat, found it within 2.3% and 3.5%, so a reader's known answers must include ones below its floor (row 711) (the triple-star sessions). A reader's guide line must come from the same stretch it reads: HM1 (run 20261017_031107) took the beat from the whole run's apse turn over 6,000 passes, because the TABLE keeps no 2,400-pass apse, and in the room that is about half the beat of the first 2,400 passes. So the second reader put its beat window in the wrong place and called the true beat, near 0.006 cycles per pass, the rocking: median 0.00584 against the bar of 0.1117, and nothing about the room's rocking was read. When an outside number steers a reader, take it over the reader's own span, or use a reader whose floor keeps it off the beat; HM1′ used the first reader, whose floor sits at 0.05 cycles per pass (the triple-star sessions)

the triple-star sessions; ties rows 641, 665, 711

Measure about a set that holds the objects (694) — the split of a cascade's newborn orbits was first measured about the line y = −x, a symmetry set on which none of those orbits lies, and had no reading; measured along the map's other symmetry set, a parabola, the split shrank by about 4.018 per level and flipped side at each, the width constant of area-preserving maps, α = −4.018076704 (Greene, MacKay, Vivaldi and Feigenbaum, 1981), which stands beside their 8.721 as 2.5029 stands beside 4.669 (row 680). A symmetry offers more than one set to measure about; use the one the objects live on

the triple-star sessions; Greene, MacKay, Vivaldi and Feigenbaum (1981); ties rows 259, 680, 771

A reduced problem is a feedback loop: ask on the attractor, and compute the loop's gain before believing its zeros (699) — at a fixed drive a test held the starting angle, solved one closing condition alone and read the other there, and found orbits outside a lock's edges, at the same two far angles at every drive. Bisected sixty times, the residual there stayed at 4.91×10^{-10}: a jump, not a zero. Near an attracting circle the reduced reading is the truth multiplied by a closed-loop gain, one minus the transverse multiplier over one minus the loop gain d, so the reading is shrunk where d is negative, multiplied without bound as d nears 1, and turned over in sign past it. On the 11/7 lock d runs from −1.35 to +1.02 round the circle, so the reading fell to 0.43 of the truth in one place, rose to 1,176 times it in another, and changed sign between d's two crossings of 1, which were the false orbits; the formula, registered before its run, held at all 112 sample angles to 1.9×10^{-14}. Read on the attractor itself, each sample landed there by

iterating the map, the 8/5 lock has exactly 10 orbits a thousandth of its width inside each edge and the 11/7 exactly 14, and none outside. On a strongly dissipative map, ask whether an orbit exists on the attractor, not on a reduction; and before counting the zeros of a reduced problem, compute its loop gain, because a sign change through a pole of the gain is not a root (the Loop Atlas sessions, the book). Before linearizing about a point, check that the point stands alone: a first draft treated a dance as a fixed point of the weightless map and divided by one minus its multiplier, but at a root every weightless orbit closes and the multiplier is exactly 1, so the dance is picked out by the passing matrix's shape. Rewritten on the matrix, the same idea reproduced 56 measured moves with nothing fitted (run KERNEL-1), and its closed form then landed 14 of 14 harmonics predicted first (row 737) (R2W)

the Loop Atlas sessions (the book); the Loop Atlas's ledger rows 104 and 105; Black's feedback amplifier (1934), for the formula's form; ties rows 642, 644, 737

A loop reading is a remainder; the whole count lives in its history, watched faster than half a turn per look (700) — a frame carried once round a loop turns by the curvature the loop encloses, but only modulo a whole turn: round a ring of seven 70-degree peaks it read −0.599 radian where the peaks hold 11.967 radian, 1.9046 turns, which is the total less two whole turns, to 3×10^{-13}; one such peak alone read −2.149 on its own ring, its slice of 4.134 less a turn. The whole turns are found from elsewhere: by adding the parts' shares, each under half a turn, or by growing the loop from a dot and keeping count as it widens. Read on 1,801 circles each 0.002 wider than the last, adding or taking a whole turn wherever two neighbours differ by more than half a turn, the count at the best circle came to 11.966913598514, the peaks' summed shares, to 3.5×10^{-10}. The history must be watched faster than half a turn per look: read at every 10th circle, where the largest change between looks was 3.0183 radian, the count held; at every 11th, where it was 3.3638, the first change above half a turn, it slipped a whole turn, and no coarser watching recovered it. It is the rule phase unwrapping lives by (Itoh, 1982), and it applies to any instrument that reads a phase modulo a turn: keep the path, not only the endpoint, and look often enough. The other half of the rule: carry the whole turns as a count and the phase wrapped, for a phase left to grow loses its last digits. The Loop Atlas's needle carried its angle unwrapped to 10^5 radians, where its last bit is about 10^{-11}; the slowest sway near a doubling gathered that rounding into a standing wobble of 10^{-8}, and wrapped every passing the floor fell to about 4×10^{-12}, a thousandfold, the sway's fall then matching its multiplier to the fourth figure. A lock rule with an absolute tolerance of 10^{-8} read a 14-cycle loop as 28 after 8,000 cycles and as nothing after 32,000, the angle then near 1.45×10^5 radians: wrap the angle every cycle, or set the tolerance above the angle's own rounding. And check a new run against an old one only within the old run's own error: the old runs, carrying the fault, could be repeated only to 3×10^{-8} (the Loop Atlas sessions, misses rows 512 to 514)

the Loop Atlas sessions (the book); the Loop Atlas's ledger rows 112, 116 and 119; ties rows 103, 627, 645, 760; the Loop Atlas's checks of 30 and 31 October: HIDDEN_LOCKS_2026-10-30, LOCK_TOLERANCE_2026-10-30, TOP_CORNER_LONG, TOP_CORNER_WRAP, LOBE_FLIP_FINE

A map's peaks are not a count of sources: check what stands under a peak's centre with a tiny loop (701) — three mountains, hidden and read back from carried frames alone, gave a map of 6,900 circles with a fourth local peak, midway between two mountains, which refined to a true local maximum of the frame's turn over centre and radius: a circle of radius 3.568 with plain under its centre, turning the frame 1.3786 radian, more than the smallest mountain's own best, by taking about two fifths of each neighbour's slice. A circle of radius 0.02 round each real summit turned the frame by 2.6×10^{-3} to 3.2×10^{-3}, and round the fourth peak's centre by exactly 0. A local maximum of a loop reading may belong to two sources at once, so a peak is counted as a source only if a tiny loop at its centre turns. Turns add mountain by mountain where one walk crosses two (to 4.9×10^{-11}), and mountains of one shape read alike at any size (run CONE_TOMOGRAPHY_2, 20261008_002939 and 20261008_002937) (the Loop Atlas sessions)

the Loop Atlas sessions (the book); ties rows 416, 607

Every hole is a counter: a histogram on each axis gives the numerators and the denominators, and a hole's landings give the period (702) — wrap a picture in a lattice of holes and count the landings in each. The side histogram gives the numerators, how often each value is visited; the bottom histogram gives the denominators, how many riders each column holds; a fraction needs both, and the side histogram alone mixes a dance's shares with how densely each part of the dial was sampled (one map's shares were uneven because some drives had 8 starts and others 50). Each hole is also a counter of its own: with 256 recorded passings a hole on a dance of period p collects 256/p landings, so the median hole read 128, 64, 32 and 16 in every column of four bands, giving the period without following a single rider. And the counter is the test for a cloud, because thickness is not disorder: a crowding rule had called a string three rows thick a cloud, while a cloud is where holes hold one landing each (no column from 112 to 900 has a single-landing hole; 89.1 per cent of those from 0 to 100 have one). Estimate first, then count (the Loop Atlas sessions, the book). A count is a guess about whole numbers until it is asked what it counts in the problem's own integers: the person counted dots, and his counts became a dance's denominator, its numerator, and the window count between them (q, p and q − p − 1), his half dot the seam at ±0.5 (run READ-E, all 18 dances of the first nine rows) (R2W)

the Loop Atlas sessions (the book), on Bear's lattice; the Loop Atlas's ledger row 123; ties rows 615, 627

The size of the holes decides what a string is: choose the scale from the tightest pair to be separated, count before naming, and keep a relation only if it survives a change of scale (703) — a lattice reader registered to see an eight-step dance as eight strings saw four tight pairs at half a degree, so three predictions built on eight strings missed and four could not be read; with its families taken from the side histogram's filled stretches and read on holes of 10^{-4} radian, the same picture held 8 of 10. The fit of the lanes' opening preferred a straight line at the coarse scale (a log-log slope of 0.2308) and a square root at the fine one (0.5115 and 0.4959, a power law fitting best in 4 of 4 and 8 of 8): the wrong scale gives a confident wrong shape. Where strands crowd, count holes, not runs: a band of 32 strands read 32 runs in only 81.2 per cent of its columns, and 32 holes in every one. Linking holes into strings is where the method is fragile: at 100 rows a groove's upper stop had two parallel relatives and at 50 rows none, and since a hole joins one string the first claim wins, while the numbers read from a curve's own points did not move. Keep a relation only if it survives a change of scale. A test of order on single settings is stricter than one on bands, so register which: the gear train's bands, three paces or more, rose without a step back; on a sweep twice as fine five single paces step back, a lone 3/5 inside 2/3 among them, each at first possibly a second basin that a needle from rest happens to take; sorted later with many starts at each (row 664), two were a second gear, two the staircase itself running back, and one a lock beside tumbling. Say the scale with the claim (run STAIR_WIDTHS 20261013_184033) (the Loop Atlas sessions). A scan from rest also underreads how far a hidden lock reaches: from twenty starting angles, the needle's hidden 11/15 lives at 34 of 44 paces and its 11/16 at 11 of 22 (the Loop Atlas's check HIDDEN_WINDOWS of 30 October; row 664) (the Loop Atlas sessions). A zigzag at its grip's limit is a slope gauge with a leg-long ruler: it reads the hill's fall line within 4.4 degrees wherever the line holds over two legs, but the slope only as the legs' average, exact on an even hill, up to 27 per cent off where the slope's size changes along a leg, and up to 45 per cent where a bump is no wider than two legs (runs ZIGZAG_GAUGE 20261013_231629 and ZIGZAG_GAUGE_2 20261013_231751, on a mountain road) (the Loop Atlas sessions)

the Loop Atlas sessions (the book; R2W); the Loop Atlas's ledger row 124; ties rows 416, 631, 640

Frame the picture before reading it: cut a circle where nothing lands, and record a whole number of the dance's periods (704) — an angle or a lag is a circle, and a picture's edge is a cut; a cut through a band makes one band look like two. In one picture only 7 of 340 shapes lay between lags 0.27 and 0.52, and the two clouds either side had matching profiles by tenths of a turn; a reader that put its seam in the middle of the side histogram's widest empty stretch left none of 461,312 dots outside a family. A string that runs off one edge is not thereby the one that enters at the other: a wrap matched by slope alone was withdrawn. And record a whole number of the dance's periods: on a test bed of period 3 with 256 recorded passings each lane got 85 or 86 visits, the lanes' centres jittered, and the change of change of slope put its largest peaks in the wrong place. And a column the dial left empty is not a break in a curve: the Explorer's first lattice read the empty columns between sampled drives as breaks, and was fixed before it was published to read each slope to the next column that holds any landing (the Loop Atlas sessions, R2W)

the Loop Atlas sessions (the book); ties rows 627, 641

A split is a tree with three roads, and the tree sets its own denominators (705) — count a split's offshoots including the road that goes straight on, the parent carrying on as an unstable ghost: Bear drew one road becoming three at a groove's end, and one becoming six at the ends of an eye. Each split is a coin that halves a branch's share, so a dance sets its own denominators: of one cluster's two strings, one never splits and keeps half while the other halves, which predicts shares of 1/4, 1/4 and 1/2; the person's eye read 1/5, 1/5 and 3/5, and the map measured 0.22, 0.26 and 0.50. The estimate's shape, two equal small bars and one large, told the reader what to count. The eye defaults to a fork: the reader drew two roads at every split even after reading that Bear draws three. A doubling tree is a probability tree with three roads: each split hands each daughter exactly half its parent's time (64, then 2 of 32, 4 of 16, 8 of 8 readings); the parent carries on between them as an unvisited ghost; the daughters open as the square root of the distance (slope 0.5000 over two decades) and lopsidedly (1.06 growing to 1.94); and forks copy forks, the widest narrowing by 2.46 and 2.52 toward Feigenbaum's second number, 2.5029, and the narrowest by 6.19 and 6.70 toward its square, about 6.26, as the run had said before it read them (row 680) (run QUARTER_TREE_READ 20261014_053757, on Bear's reading) (the Loop Atlas sessions). The doubling tree hangs from one node of the tongue (fraction) tree like a tassel: every road in it keeps the gear (1/4 as 2/8, 4/16, 8/32), as 4/6, 8/12 and 16/24 hang from 2/3 in R2W's record, and the windows inside the chaos keep it too, multiplying the period (12, 20, 24, 48 and 80 cycles, all one turn in four) (runs QUARTER_TREE_MARKS 20261014_033510 and QUARTER_TREE_CURVES 20261014_055131) (the Loop Atlas sessions). A window's cycle in banded chaos is the gear times the band's pieces times the cycle of a window of a single hump (3, 4, 5, 6, 7, 8 and on), and 3 and 5 are the widest, the ones an eye picks out first. Bear's rule, read off a drawing: $12 = 4 \times 3$ and $20 = 4 \times 5$ in one piece, $24 = 4 \times 2 \times 3$ in two (measured), $48 = 4 \times 4 \times 3$ (by arithmetic, the piece count there not clean, row 787); the arches found $28 = 4 \times 7$, $32 = 4 \times 8$ and $32 = 4 \times 2 \times 4$ (runs WINDOW_TOUCH 20261014_073237 to WINDOW_TOUCH_3 20261014_080520; the general picture from recall) (the Loop Atlas sessions, after Bear)

the Loop Atlas sessions (the book, on R2W's maps); Bear's lattice; ties rows 627, 680

Name each curve in the shape alphabet, fit the shape you named, and look for its copies (706) — Bear's alphabet: flat; linear, clock-like; ramp on or ramp off; hill or bridge; split near equal or lopsided. Then fit the shape that was named: a lattice reader read a needle's doublings opening as a square root, with log-log slopes of 0.5115 for the families and 0.4959 for the strands. Shapes repeat. When two related pictures look alike in different orientations, a shared mechanism is hiding there, to be tested by mirroring and rescaling one onto the other; and within one picture, once a shape is known, look for its copy, shifted or squeezed. In one picture Bear found a groove's lower foot (its drop, its shelf near lag 0.14 and its final fall) again about 0.010 of drive to the right, with the shelf squeezed to about 40 per cent of its length (the Loop Atlas sessions, the book). Before placing anything at a crest, name the curve: the person's diagram put the steady dancers on the crests of a wavy line and the wobblers in its trough, but on the line the runs draw both sit on its slopes, the steady on the climbing sides and the wobbly on the falling, and on the landscape whose slope that line gives, the steady sit in the wells and the wobbly on the ridges. His feeling for the wobblers, "fighting to stay on", was right for a ridge; only the placement flipped, because two curves were in play and the extremes of one are the zeros of the other's slope. His notebook's +1 well and −1 saddle are exact: the fixed-point index of a sink and of a saddle (row 672) (runs SLOPE-1 and SLOPE-2) (R2W). A specimen from R2W: on a coin map at weight 0.40 Bear outlined three fins, and his long lower fin is a thin heads stripe 0.080 to 0.120 of a turn below the main arc's lower edge at 16 of 17 drives, sliding with it. Registered and run at 35,000 riders, the echo ran parallel at 31 of 31 drives, and a second, nearer stripe stood at 31 of 31 (COIN-3). When a drawn shape repeats an edge's line, test it as a copy of that edge at finer grain. The order R2W carries: when Bear marks a shape, first ask whether the instrument drew it, then test it as a copy of something already on the map, and only then as something new

the Loop Atlas sessions (the book); Bear's lattice; ties rows 537, 640

Long reach beats many short chains; look on purpose for the strings the eye does not follow (707) — on one picture Bear drew six long strings and called it 80 per cent; a greedy decoder drew 50 strings of four or more holes, among them pieces of his long ones. Six long strings explained more of that picture than fifty short ones: a string earns its place by how far it carries one smooth shape, through clouds and across gaps, and long reach is also what separates a clean picture from its shuffle when the count of short chains points the wrong way (row 616). The eye follows the dominant slope: the reader found one flat string, at lag 0.145, showing through three windows, where the decoder found a staircase of them (0.144, 0.707, 0.81 and about 0.92), and the reader saw no rising strings where the decoder drew about ten. Flat strings come as a family, and rising strings must be looked for on purpose (the Loop Atlas sessions, the book, with R2W's decoder). Where dots crowd, follow a curve by its bend into the nearest attractor, a ring or a birth, rather than ending it at the crowd; a family of converging curves is read by where they meet before it is counted (R2W). A person's eye traced curves that are, for the most part, the fold's images: of Bear's strokes on a density of chaos, 0.741 (red) and 0.673 (green) lie within 0.004 rad of the computed fold or its first eight images, against 0.36 and 0.37 for the same strokes set at random (the shuffle); his X is the fold's third image landing on the ghost, where the bands merge, at +1.1 millionths of pace above 0.98370 (the runs' scale, row 756). Claude could see the voids and the X but not the faint curves until they were computed (run QUARTER_TREE_CURVES 20261014_055131) (the Loop Atlas sessions). And a person's eye traced the curve that decides where a window ends: Bear's black arc on the 324-degree panel is the fold's third image, 0.604 of its pixels within 3×10^{-4} rad of it against 0.076 placed at random (0.693 against 0.082 with the figure's carry mended), and it meets the top edge where the window ends (row 756); the third use of the two-reader method on that window (runs SEAM_2 20261014_090552 and FIG_SEAM 20261014_091610) (the Loop Atlas sessions). That is the two-reader method doing its job, and it says which reader to trust for what: his eye for faint, long, bending structure; the ruler for whether it is there (the Loop Atlas sessions)

the Loop Atlas sessions (the book, with R2W's decoder); ties rows 616, 640

The change of change of slope marks an event: read it at the zero crossing between its two lobes, and read rates from where the events sit (708) — Bear's third difference along a string peaks where the string's character changes. On a needle's lanes its two largest peaks fell two columns early in every family, because a centred stencil answers a sudden bend with two lobes of opposite sign, one on each side, and the peak taken was the lower lobe. Read where the lobes cross zero and the event sits on the doubling itself: on columns a millionth of a push apart, three doublings were placed within 0.0031, 0.0014 and 0.0062 of a column, and the ratio of the first two stretches read 4.6153 in every lane against 4.6155 from the orbit. The method needs room: where the next event lies inside the stencil's reach, at the fourth doubling, the crossing landed 0.08 of a column high. Rates can be read from where events sit even at whole columns: the columns where the strands' count changes gave ratios of 4.6333 and 4.6154, each within the reading's error of about one per cent of the record's 4.6155 and 4.6570

the Loop Atlas sessions (the book); the Loop Atlas's ledger row 126; ties rows 537, 642, 680

Geometry links pieces; history says which are relatives and what the true count is (709) — only a rider's history, which rider and which passing, says which pieces are feet of one dance: at half a degree, where the geometry saw one thick string per lane, history counted the doublings, the first columns with period at least 4, 8 and 16 falling at 264, 143 and 116 against doublings of record at 262.4, 142.3 and 116.5. Count the truth by history, not by rounding: a reference count of distinct points rounded to a millionth ran one or two high in six columns, because a

value on a rounding boundary rounds two ways, and a prediction judged against it missed for that reason alone. The reference needs a check of its own; the count of record is each rider's period, the smallest repeat of its passings within 10^{-7} in both numbers (the Loop Atlas sessions, the book). Follow one object from frame to frame before calling it growth: the person read single fractions on the plain's edge as seeds that grow into ramps as the weight drops, and tracking each fraction across six weights showed that the edge family is the ball's launch pace, and that each fraction's line slides toward the slower tick (34 of 38) rather than growing where it sits. "The same place, bigger" and "the same thing, moved" look alike across separate frames (runs SEED-1 and SEED-2) (R2W). Count before you average: an engine that expected two self-mirrored dances with 2q stops between them handed every zero of its selector to the nearest one, and inside the hand-off window, where a walking pair also existed and the selector had 4q zeros, the pair's stops were averaged into the dances' grips and every grade failed. The engine printed the count, 20 zeros instead of 10, on every polluted row, and that count was also the finding: the walking pair exists at first order (runs WINDOW-0 and JJ_5) (R2W). An inner structure must carry its own state: in R2W's R83-d return, rows that looked like a tongue inside a tongue, a lower edge inside another, had the upper edge's own bias and speed, with the same lag or that lag less one whole turn: the upper fold reached twice. A second locked region would show its own orbit and its own bias. Before naming an object inside another, check that it is not the outer one shifted by a whole turn (R2W, sitting P2)

the Loop Atlas sessions (the book); ties rows 627, 641

Before joining two pieces, lift them into a third number; a touch the third number does not separate is a real meeting (710) — a strobe picture is a shadow of something with more numbers to it, and in Bear's words, "When we start injecting a 3rd dimension into 2D, you often see what you want to see." If a third number runs on smoothly across a join, the pieces may be one; if it jumps, they lie at different depths, an overpass. Not every touch is an overpass: two strands touching in angle, registered as an overpass more than 10^{-4} apart in speed, were 6.41×10^{-5} to 7.51×10^{-5} apart in speed, close in every number because they were a fork's young daughters, born at the doubling two to seven columns above, and holes of 10^{-5} in both numbers separated them, 32 strands in every column from 122 to 140. A real meeting is resolved by scale or by history, not by another dimension. That prediction carried the very error the person had warned of one exchange before: read a fresh warning as a test of your next prediction. And a line the sampling makes, a frequency illusion, is tested by splitting each rider's record into halves or strobing at another phase: a dance stays where it is, and an illusion moves or breaks up

the Loop Atlas sessions (the book); the Loop Atlas's ledger row 125; Bear's lattice; ties rows 360, 641

Test a borrowed method on known answers, and run the ruler over a made-up case before reading meaning into a shape (711) — before Bear's lattice read anyone else's picture it read a needle's doubling staircase, whose splits Newton's method had fixed to ten digits: its first version missed, its second held 8 of 10 predictions, and its follow-ons placed the splits within a few billionths of a push, none of which could have been scored without the known answers. Then a shape in its output, twin lobes the person read as vortices, geared propellers or mirror-math, was checked by running the same three centred differences over a made-up perfect corner, which drew the same pair, crossing zero exactly on each doubling and matching every lane's pair to 0.993 in size at the top doubling and better than 0.9999 at the next two. The shape was the ruler's echo of a corner, and only its place belonged to the data; of the person's readings, "boundary transition" and "mirror-math" were the ones the data bore out (the Loop Atlas sessions, the book). And the known answer must be exact for the case run, not an estimate made for another. With the ripple off, R2W's stopping pulls at weights 10, 5 and 2 read 0.394, 0.542 and 0.790 on a grid of 0.002, within 2 and 5 per cent of the textbook estimate $4/(\pi\sqrt{m})$ at the heavy end; but the estimate itself runs 2.5 and 5.2 per cent above the exact values there, so those gaps were the estimate's own error. On a dial of 0.00002, registered first, the same runs read 0.3927, 0.54132 and 0.78988 (run RETRAP-FINE), the exact values 0.3927, 0.5413 and 0.7899 to the dial's step. At weight 2 the estimate is 14 per cent high (0.900 against 0.7899). The exact values were computed for this guide by leaving the saddle along its outgoing direction and following the path until it cleared the next ledge or fell back, and were checked by direct runs 0.002 either side; at weights 2 and 5 they match the lines the Loop Atlas had shot from the saddle on 5 October (0.789864592 and 0.541310459, run SEAM_MAP_B 20261005_023746; rows 652 and 789); by the same method a memory band first opens at weight 0.7026, where the Loop Atlas's own shooting put it at 0.70256366 (row 750), and a test made a hundred-thousandth below pull 1 reads 0.7063 (R2W; this guide's check; the Loop Atlas sessions). Test the reader on a stand-in that obeys the law exactly but is harder than the real thing: RATCHET-68's first reader carried the opening's angle to weight 0 by a straight line through two light rungs. On a synthetic window built to obey the mirror exactly, it missed a quarter-turn by 0.019706 degrees, because that stand-in turned fast and its cubic term was large, so the reader would have failed a true law. Fitting the odd law through three rungs fixed it before registration. A dry run asks more than whether each verdict holds once and fails once: a stand-in obeying the lean must hold it with room to spare (row 731) (R2W). When a planted test fails, check how the test was built before blaming the reader, then ask whether the reader's real limit can change a verdict: RD1a (run 20261017_154352) failed, rates coming back within 3.19×10^{-6} at 1/8 against a bar of 2×10^{-7}, because the stage-1 run lending its leftover sat 7.5×10^{-4} to 1.5×10^{-3} from the fraction, inside the Hann window's main lobe ($2/N = 8.3 \times 10^{-4}$ in half-width, or near it), so its own line pulled each planted reading. RD1a′ cleared the leftover within 4/N and

passed at 1/8 and 1/9, but still failed at 1/10 and 1/11 (6.22×10^{-7} and 3.47×10^{-7}; run 20261017_155035), which is the reader's own pull. The readings still stand, because the steps between neighbouring births (1.54×10^{-6} at 1/10 and 9.60×10^{-7} at 1/11) are larger than that pull (the triple-star sessions)

the Loop Atlas sessions (the book); the Loop Atlas's ledger row 127; ties rows 406, 456, 650, 652

Name the picture before naming the pattern: each picture has its own neighbours (713) — one run of the triple star, at birth 7.200, has its dots' nearest relatives 5 passes apart on the eccentricity-against-pass page, 14 in the reading's own rhythm, 47 on the opened torus of two angles over 600 passes, and 682 or 663 over 6,000 at the kit's tolerance, whose arithmetic shaped those two (at the tight tolerance the beat is 722 passes, row 858); each is right for its picture. R2W read the first and the third from the pictures' pixels and the triple-star sessions confirmed them on the numbers, one finding made twice. A picture's axes decide which dots sit side by side, so every claim about a weave, a lane or a neighbour says which picture it lives in, and the same data is drawn through a second picture before a pattern is named. Name the picture's frame in words too: before "up", say which axis runs up, and before "does not move", say what is held still. A reply said a second tone moves the tongues up, on a map that runs bias across and strength up, where the tongues in fact slide sideways; another said weight does not move a line, true at a fixed bias but not at a tongue's middle, where weight slides the line as far as the tongue. A reader who thinks in pictures takes the picture word literally and builds on it (R2W). The same word can name opposite things in two pictures: a "pinch" in the scatter of a phase is where a run keeps its phase tightest, while a "pinch" in a tongue's width is where a dance holds most weakly. Before carrying a word from one picture to another, name the quantity that narrows, and whether narrowing means holding harder or letting go (R2W). When a book of stories borrows pictures, pair each picture with the story written from the same runs, copy the story from its own file by script, show any cut, check that every run named as a source is present, and give each picture a key to what the story's images stood for in the data (the Loop Atlas's favourite panels for the book of reads: 13 pictures, 15 reads, 22 source runs)

the triple-star sessions; the Loop Atlas sessions (R2W); ties rows 360, 641, 858

When the data is held back, measure the picture, and know the plotter's traps (714) — calibrated on a picture's own axis ticks and checked on marks of known value, pixel counts turned eye claims into numbers: they confirmed a flip in lopsidedness, a drift in a grey band, a top 3.7 to 6 times denser than its bottom, and a rider lingering mid-swing, and they reproduce when rerun. A picture drawn for marking (equal axes, a light grid, no overlay) and kept with its code can be redrawn pixel for pixel, so a person's strokes read back through the plot's own axes become numbers; the redraw matched his picture in every pixel. Two traps: a plotter draws the last points on top, so colour counts favour late passes and a crowded picture shows mostly the end of the run; and dots merge where crowded, so pixel and blob counts bracket the truth (the Loop Atlas sessions, R2W). A count by eye is a guess: counts read by eye went wrong three times in one sitting, the "period-3 dance", the crest counts and an "about four" that the pixels put at about every 80 passes, and from then on counts were measured before they were said (R2W). A pixel count is only as good as its colour rule: the rule that found all 39 columns in one panel found 16 in the next, where the dots were paler, so nothing was claimed for that panel. Check that the rule caught every column the eye can see (R2W). A label moved to make room can name the wrong dot: on one of R2W's figures, where at strength 0.03 four children of 1/2 sit within 0.03 of each other, labels set beside each fraction's leftmost point printed 3/7 and 4/9 on top of each other, and labels pushed apart became legible and wrong, 3/7's label sitting beside 5/9's dot, until a leader line tied each label to its own dot. Check each label against the number it names (the figure's printout gives them), not only whether it can be read (R2W). Read a figure's pixels at the resolution its file was saved at, and check the axes on a planted mark first: axes read at the plotting library's 100 dots an inch, for a picture saved at 110, put every stroke about a tenth off; a redrawn figure equal to the file pixel for pixel did not check the axes, and the person's own marks on known places (his rungs on the windows) did (run SEAM 20261014_090006) (the Loop Atlas sessions). And read a hand-marked figure in the scale of the very file that was marked, calibrated on its own markers: a first reading that rebuilt the figure to borrow its scale placed Bear's trace of S 1.62 times too high (on that figure of the heavy needle's window, S is the distance from the window's top down to its first doubling, and C the distance down to its end), the rebuilt layout differing from the saved one by a few pixels' margins; calibrated on the saved figure's five C squares, the five S circles fell within 1.1 pixels and his traces on the data within 1 per cent (run HEAVY_MARKS 20261014_164806) (the Loop Atlas sessions). When the method was born in the project, the paper shows how the eye found things, with its record of hits and misses: the islands paper gained a section with six of Bear's marked pictures and a table of eleven reads, and its front page says that reads naming a relation held 19 times in 30 and those naming a shape 11 in 28 (the triple-star thread, RC1). Tell Bear what you see in a picture and what you cannot: for months he assumed the AI saw what he saw, when it often could not until the thing was computed. Ask him what he sees, say what you see, and register both before anything is measured

the Loop Atlas sessions (R2W); the triple-star sessions; ties rows 199, 650

A colour scale that lumps classes fools the eye; check the class before naming what a colour shows (717) — on one map, colouring by denominator painted the rest (0/1), the conveyor (1/1) and the 2/1 and 3/1 grooves the same orange; a blind read named a chain above pull 1.0 as the rest, and the data said conveyor, 37 cells at lag 0. Before naming what a colour shows, check the class itself

the Loop Atlas sessions (R2W); ties row 199

Test smoothness before adding a dimension (718) — a coarse two-angle fit of a bend that explained 0.7230 of its variance was read as a third hand at work; a nearest-neighbour test on the opened torus of the two angles, which needs no chosen basis (a dot's five nearest neighbours against the quantity's spread, where noise gives about 1), found the bend smooth there, at 0.1126, and smoother over 6,000 passes, at 0.0603. A low-order fit that misses is not evidence of a missing variable. The person's own read of the same bend, the outline of a torus, was registered in his words and tested on the same run as the row's first reading; it missed its bar by a hair, and only the later test, over the longer record, bears his torus out (T1, run 20261009_071003) (the triple-star sessions)

the triple-star sessions; ties rows 160, 640

A lock depends on its reader: name the reader with the count (722) — the remembering needle at pace 0.95 is "locked" 1,051 times by a knee on its speed over one field period, and caught 69 times by where its lag sits (climbs of more than two periods). The two readers agree on the spins and not on the locks. A count of locks is a count made by a reader, and the reader goes beside the number (run SPIN_OR_SIT 20261011_055807). A census of fractions has a reader too: one pad atlas's three ripple strengths at weight 0.40 hold 14, 70 and 94 fractions counted as stored on one band, each dance's slip over its own period so that 2/4 stands apart from 1/2, and 14, 65 and 83 counted reduced over both bands, as R2W's queue ledger counts them. Both counts are right, and only the first sees the doubled dances (R2W, with this guide's recount). A reader that averages over a window reads an event late, and its short spells are echoes: of the speed reader's 1,051 locks, every one of the 808 under one field period had the needle mid-fall at its middle, a hesitation seen only after it had passed, while the 84 over two periods stand near the lag's 69 catches; whether they were the same stretches was registered as a test of its own (run LOCK_LANDING 20261011_170402) (the Loop Atlas sessions). A flag that remembers an event is not a place: "crossed the top since the last landing" stays set when a needle crosses a top and comes back, so it called 83 per cent of the long locks past the top where their place said 21 per cent; and a line drawn at a top only about one noise unit high is itself blurred, a fifth of the locked needles lingering just past it (run LOCK_LANDING 20261011_170402) (the Loop Atlas sessions). Matched needle by needle, two readers that disagree on a count can agree on every event they share: the speed reader counted 84 long locks and the place reader 69 catches, every catch had its lock, 69 and 69, and the speed reader's extras were hesitations past a shallow top that outlasted its two-period line (run LOCK_MATCH 20261011_195010) (the Loop Atlas sessions). Two readers of one quantity can disagree in the fourth digit, so a registered number carries the reader that made it: RATCHET-44's dry run recomputed the parabola of sitting B10 from the files and missed the registered predictions by 0.0011 and 0.0005. The octave alone's c/w^4 of −10.614, written in the record, came from SEAM_SCAN's table, while RATCHET_43_READ's window formula on the same run gives −10.6135. Nothing moved in the verdicts, which had bars of 2 and 3, but the reader now prints both, and the stand-ins found the gap before any data was seen (R2W)

the Loop Atlas sessions (the book); R2W; ties rows 556, 558, 665

Price a shape bar by simulation before blaming noise (723) — a bar asking that the median of 256 lifetimes reach 0.6 of their mean fails by chance 2.2 per cent of the time when the lifetimes are truly exponential (its spread is 0.046), so three paces of six falling under it is a shape, two drains (more needles locking in the first cycles than a constant chance allows, then an exponential tail; row 748), not bad luck. Simulate a bar's own false-alarm rate on the null shape before reading a miss as noise (run LOCK_FROM_CHAOS 20261011_060551). Price a bar by its null before a hit is read, too: a lean that more smoke-ring trains break in the upper half of a band than in its lower held by 23 copies to 21, a difference that comes 0.41 of the time with no trend, a held verdict that carried almost no information (run TRAIN_EDGE 20261018_181641). A lean about a difference of counts prints, beside its verdict, the chance of so large a difference with no trend at the pooled share, and carries a planted step test and a planted band test of the reader that judges it (the Loop Atlas sessions)

the Loop Atlas sessions (the book); ties rows 313, 337, 616, 748

Judge a transition by a measure that changes smoothly, and define a category by its mechanism, not by a symptom (728) — two bets registered on how a ripple moves the switch into memory both failed, and each failure teaches. "Crawls" had been defined as "the smallest slip above the edge is 0.02 or less"; both weights passed, so the rival held as worded, but neither ball was crawling, since a stumbling ball and a crawling one can both show a small slip (row 683). And a chaos line drawn at +0.05 put weight 0.68 on the tumbling side by 0.0017 (+0.0517), cutting a gradual change in two. Redesigned, a crawl is judged by the crawl rule itself and the chaos measure together. Across the collapse the chaos measure rose at every one of nine weights from 0.40 to 0.74, while the crawl constant bounced (0.28 at weight 0.66, 0.31 at 0.67), so the smooth measure is the ruler for the transition; a yes-or-no line laid on it says only which side of a chosen value a point fell (R2W). The two registrations were TRICKLE-3 and TRICKLE-4, which asked where the plain's edge switches from a crawl to a stumble; the data showed a fade and then a collapse, so every bar failed one way or the other. Register a measure that moves and lean on its trend; keep the threshold for things that really jump (R2W)

R2W; ties rows 337, 683, 686

Read a control's number as data, and register the bar against a control (729) — a control that read exactly zero (0.000) at all six births, in 35,878 triples of looks, said that the pair's eccentricity never dips between jumps, which is what a pass close by the centre predicts. A bar that asked a slow beat's low half to run at an outside rate missed (0.1608 turn per pass against 0.2069), and its lowest quarter overshot (0.2966), yet against a control the effect was large: the low half ran 1.785 times as fast as the high half and the lowest quarter 2.764 times the highest, against 1.01 and 0.98 for random cuts. The outside number was the wrong bar and the control was the right one. Read a control's number as a finding, not only as a gate, and set a bar against a control rather than a guessed target (runs triple_heavy_middle_look 20261011_093016, triple_maypole_reach 20261012_051949 and triple_maypole_quarter 20261012_062723). The gate is also a prediction for R2W, not yet run: a rider whose slip a slow second drive modulates should show the same gate, read as the slip rate in each quarter of the slow drive's phase (the triple-star sessions' cross-domain page of 12 October). A matched control can fail on its instrument: Question the Substrate's first "matched-skill" judge was not matched, its step grid too coarse (3.1 points off, E28), and a finer grid matched it (E28b). The miss was in the matching, not the question, so print the matched quantity and check it before reading the comparison (Question the Substrate)

the triple-star sessions; ties rows 369, 616, 692, 782

Print a quantity's range across the cases before building a rule, a rank or a ratio on it (730) — across one stretch of births, from 7.125 to 7.200, a count climbed from nothing to 78 turns while three measures of the looks barely moved (0.0104 to 0.0113, 0.0094 to 0.0123 and 0.0043 to 0.0067), so every rule of the form "the count switches on where the measure is under X" scored exactly what one constant scored, 22 of 23, four times over, and the control printed that each time. A rank test across a plateau scores noise: over seven births a climb ran in perfectly opposite order, but four births then sat on a plateau (126.784 to 126.875 turns) where small changes could not move the count, and the correlation read −0.679 against a bar of −0.7. And a ratio against a partner that collapses is no dial: a beat that fell tenfold across an island at 0.25, nearly stopping in its middle, made the ratio run from 2.5 to 546. Before building on a quantity, print its range across the scan, look for saturation before ranking, and take a steady partner for a ratio (runs triple_maypole_m1pp 20261012_043222, triple_maypole_reach 20261012_051949 and triple_maypole_quarter 20261012_062723; triple_maypole_s456 20261012_062229, test S5; triple_rungs_look 20261014_063556) (the triple-star sessions). A measure that saturates cannot rank what it saturates on: ask how deep, not only how long. The share of a drive period spent on the top half of the turn ranked with that period's growth at only 0.71, because 6 of every 19 periods are spent wholly on top and tie at a share of 1 while their growths run from 4.9 to 20.9. The junction's growth rate rises with how far over the top the needle climbs, so the depth (the period's mean of −cos L) is the measure to bet on; on the same orbit, seen, it ranks at 0.97, and its test on fresh paces is in row 755 (run STRETCH_WHERE 20261013_045856) (the Loop Atlas sessions). Read a width in the system's own units before ranking it: at mass 3.0 three locks read in drive rank 4/3 (6.860×10^{-5}) > 5/4 (5.528×10^{-5}) > 3/2 (3.877×10^{-5}), which fits neither their place nor their fractions; read in turns per passing (width in drive times the staircase's slope there: 2.20 at 5/4, 1.28 at 4/3 and 0.34 at 3/2) they rank 5/4 (1.217×10^{-4}) > 4/3 (8.756×10^{-5}) > 3/2 (1.313×10^{-5}), the lower the fraction, the wider. The knob's units fold the staircase's slope into every width. The same runs taught a second thing: a reading tool starts its riders on the branch it means to read, and a first run started them at L = 0, inside gear 1's catchment at heavy mass (the starts that settle into gear 1), and read slopes of zero (runs GROOVES 20261006_193519 and GROOVES_READER 20261006_194112, read after the data; the reader's first run retired) (the Loop Atlas sessions). A mean over a group is a measure only while the group holds together, so print the group's spread beside every group mean: three leapfrogging rings' mean speed at a gap of 1.0 was the mean of a train flying apart, its first and last rings 68 radii apart, and the lean it judged was a lean about nothing until the spread printed beside it was read (run RING_KNOBS 20261015_165858; the Loop Atlas's misses row 229) (the Loop Atlas sessions). A test input must reach the place the test asks about: a soup skewed by adding a share of its square is bounded below, at about −1.13 of its strength, so a trough-firer set at −2 never fired and its count was empty. Before judging a level on a reshaped input, print the input's own range (the Loop Atlas sessions)

the triple-star sessions; ties rows 616, 617, 760

A fake run checks the plumbing, and it must borrow nothing real (731) — a fake worker whose runs carried zero seconds stopped a tool with a division by zero in its pace line, a crash that a real run with an errored calibration would have hit overnight; the pace is now read only from seven timed runs, and otherwise the line says no pace was read and the run carries on. A fake drive that gave its new cases the saved records of four real ones printed their counts, data the test had registered as unread; the test was by then known to qualify at only two births, so by its own rule it could give no reading, and that was disclosed in the test's header. A fake run uses made-up records, or prints nothing of what it borrows (runs: the tool triple_heavy_mean_start v1.2, checked in the container on 11 October; triple_maypole_s456's fake drive for tests S4 to S6, deleted; triple_maypole_quarter 20261012_062723) (the triple-star sessions). Make the stand-ins with the engine itself: a reader dry-tested on stand-ins written by hand, every row carrying one grip of each sign, passed rows the engine cannot produce (at 1/2, an even number of beats, the two starts are one dance and the engine returned two equal grips), and one real row of each kind (odd and even beats) would have shown it before anything was registered. And check that the smallest prediction is bigger than the ruler: at the deepest of four new rungs each predicted product came to about 8×10^{-12} at 7/15, under the 10^{-11} error

assumed for each grip; both checks cost a minute, and the miss cost an exchange's main test (runs LADDER-1 and JJ_14) (R2W). Plant the event before you hunt it: two slip finders in one batch could not see the slips they were built to find, one asking for a slip faster than any slip is, the other measuring against a line the slip itself tilted. A record with one known slip, run through each finder before its registration, would have caught both in minutes; a check against an independent count belongs in the fidelity leg as well (runs SLIP_PROOF 20261013_194935 and SLIP_PROOF_2 20261013_200028) (the Loop Atlas sessions). And run the planted case before registering, not only name it: a test of crossings of the band's low edge found none, because the low edge is the lowest the band reaches and every curve touches it from above, as the tool's own said-before had it; the planted case named in the registration would have shown it in a minute (run WINDOW_TOUCH_2 20261014_074538) (the Loop Atlas sessions). Register, dry-run, run, read once: every test of one sitting was registered in the record before it ran, as row 369 asks, with a reader written first and dry-run on stand-ins on which each verdict holds once and fails once, and a stand-in set that grades nothing when rows are missing. The dry runs caught real faults before any data was seen: a stand-in that double-counted a weight law (RATCHET-42), readers that let a missing weight pass (row 36), and a reader that kept one frequency's slot over another's. Keep it exactly (R2W). A selftest's planted world must have every ending the real worker has. The off-mirror tool's line gate counted a run as finished only if it escaped or ran all its passes; the real worker has a third ending, a star thrown wide, neither escaped nor held, whose time runs out before its last pass. 191 such runs shut 24 of 32 start lines out of the parity count, and the tool printed NO READING. The atlas itself has 290 such runs, so scoring the nearest real table through the new rule before shipping would have shown it, and the tool's v1.1 adds that world to its selftest. The lesson's own count was first written as 172 without being counted, and recounted from the run's table as 191 of 3,232. This is one of two ways a rescore can mislead; the other is in row 786 (the triple-star sessions). The routine held through a later sitting's 45 tests, each registered before any data was seen and each with its RESULT line: stand-ins exposed faults in the readers of ODD-1 and ARRAY-1 (row 796) and in NINE-1's plan, printed arguments read against their source caught K-16's plan taking a harmonic's number for its strength (row 677), and EDGE-85's first dry run showed its rival could never be bracketed, so the rungs were extended before registration. Keep it, and add one step: print every argument a plan builds and read it against where it came from (R2W)

the triple-star sessions; ties rows 242, 264, 369, 711

The twin test has two floors: how long it watches, and the arithmetic it shares (732) — twins started 10^{-10} apart and called chaotic at 10^{-3} turn have about 16 e-foldings to show it, so in 600 passes they see only chaos that e-folds in fewer than about 37 passes; at one eccentricity, 0.25, an island regular by twins over 600 passes dissolved over 6,000, every twin pair parting exponentially. And a strict fidelity tier measures reproducibility, not accuracy: rerun with every tolerance a hundred times tighter, a regular orbit's net turns over 600 passes moved by 1.1×10^{-4} turn, about fifty times the 2×10^{-6} the fidelity checks ask for, while twins, which share their arithmetic, differ at pass 600 by a median of 1.5×10^{-6} turn on regular orbits. State the run's length and the e-folding it implies beside every twin verdict, and call a fidelity check what it is, a check that one machine's arithmetic reproduces itself (runs 20261013_152519 and triple_twin_growth_look 20261014_014526; triple_tolerance_control 20261014_031147 and triple_paper_checks, check TW1, 20261014_085115). Short scans overstate stability, and the long run showed what replaced it: over 6,000 passes that island at 0.25 split into rooms, the parts a long run splits an island into, and on a grid 0.001 apart about two births in five hold in the best room, with the escapes stopping sharply at 7.472 (runs 20261013_152519 and 20261014_090328). This is where to point a specialist who draws stability maps of circumbinary or hierarchical triples from short integrations, the stability-boundary work of Mardling and Aarseth (2001) and of Vynatheya and colleagues (2022) (the triple-star sessions). A long record at a loose tolerance measures the arithmetic: SL7 (run 20261019_012918) ran 24,000 passes at the kit's tolerance, and the rocking rate fell quarter by quarter (7.340: 0.201039, 0.199572, 0.197505, 0.196190), 8×10^{-8} to 2.8×10^{-7} cycles per pass in each pass. At the tight tolerance SL8 (run 20261019_032214) showed falls of 1.41×10^{-10}, 6.40×10^{-11}, -1.04×10^{-9} and 1.78×10^{-10}, so SL7's fall was the arithmetic's, and so was the kit's beat of 0.000575 at 7.340 (0.000026 at the tight tolerance). SL8c then missed because it set a tight 24,000-pass reading against 6,000-pass readings made at the kit's tolerance (7.3375's first quarter 0.200320 at the kit's tolerance against 0.201703 tight): the reference, not the line, carried the shift. SL9 (run 20261019_043449) compared tight with tight and read 0.001050 against 0.001048, 0.000002 apart. Register long records at the tight tolerance, and compare like arithmetic with like (row 741) (the triple-star sessions). Both floors reach the islands paper's own numbers. Read with every late parter called chaotic, four of the paper's printed numbers changed, against a guess that none would (LP1, run 20261017_024415); a later guess, that few or no pairs would part late, held in count, one in 115 pairs, but not in effect, since that one pair changed a printed number too (LP1c, run 20261017_173539). The numbers were corrected in the paper. And a tighter tolerance sorts losses one case at a time; it does not clear them all. Of 13 runs, 8 were lost at the laptop kit's usual tolerance and 5 at the tight one, against a guess of the same count (XL); the seventh's core still moved at the tight tolerance, against a guess that it would survive, while the fifth's seven core births all stayed within 4.8×10^{-9} and locked (XA); and the eleventh's 7.127, at -2.42×10^{-7} at the usual tolerance, past the 2×10^{-7} bar (run 20261017_200756), read -4.05×10^{-7} over 6,000 passes at the tight tolerance, against a guess that it holds (XE) (run 20261018_001236). A guess that a suspect loss is all the arithmetic's, or all the motion's, can miss either way, so read each case at both tolerances before naming its cause (the triple-star sessions). And register the growth law of the

twins' gap before looking: two births 10^{-10} apart at the tight tolerance parted from 1.03×10^{-8} to 6.39×10^{-6} over 60,000 passes, a log-log slope of 0.97 against a registered bar of 2, the straight-line parting of two neighbouring tori (LR3)

the triple-star sessions; ties rows 36, 632

Register the exact symmetries first: a symmetry seen is a hint, the transformation that makes it is a test, and it is the cheapest bug detector (734) — the person's "bilateral symmetry is another stalking point" (a tracker's term: a symmetry in the data, axial or bilateral, is a point to stalk the quarry from, so any plot that shows one is a help) led to a mirror in the 1/2 dance's stops, and that to a time mirror, $(u, t) \to (-u, -t)$, exact to 10^{-13} for a weightless rider. Later, before a run on a second tone with a relative phase, two exact symmetries were written down from the equation: the step at phase $\varphi + \pi$ mirrors the step at φ, and without weight the step is symmetric at $\varphi = \pi/2$. The first reading broke the mirror at every phase except 0 and π. Nothing in the physics can do that, so the instrument was wrong: it read the lock from $\Phi 11$, which equals half the trace only when the drive is symmetric in time about the strobe, as every earlier run's drive had been. An exact symmetry costs one line to register, and it catches the errors no physical guess would (runs T1 and RATCHET-1). On a three-body picture, the person's two "symmetry" lines mirrored its bands (run SYM-2: best axis −0.340, with 81 per cent of his pixels matched), while the bands' own mirror sat about a quarter turn from his, a rectangle's symmetry. That word started the search above, and the weight-free pin it found sits where the 1/2 dance's two stops are mirror images; the time mirror then places it with nothing fitted (R2W). Exact symmetries as the first verdict, in every reader: each rectifier reader graded the mirror and the weightless quadrature to 10^{-9} before anything else. They held to 0.00e+00 throughout, and they would have been the first to catch a broken engine. The same idea applied to code: to trust a new path through old code, give it an old job and swap the roles. The three-tone engine (RATCHET_3T) reproduced two old runs exactly, 192 of 192 and 128 of 128 rows, but only with the third tone silent, so its own path had never carried a number, and fidelity runs on the old paths prove nothing about the new one (row 551). Bear asked whether a surprising sign was a bug. The same mix run with the octave in the third tone's place and the fourth harmonic in the second's must give the same edges, since the drive is the same sum; it gave all 288 to every printed digit (RATCHET-43, R43-a). A swap needs no new reference, only a symmetry of the job, and it tests the new path at the very setting in doubt. Every extension of an engine carries one such test at the setting where it will be used (R2W). Do not fix out a strange sign; test it for its mirror. Claude reported that a fourth harmonic set against the octave "is backwards", a prediction from adding the two tones, not a measurement. Bear's worry was not a bug but a lost signal: "When I said this might make sense, I didn't mean you made a mistake. I meant, we are in weird territory and when you first told me that it made me think we were seeing half a dipole or a reflected image across a bilateral symmetry. I didn't want you to accidentally 'fix out' a piece of data that might actually be something interesting." His "bilateral symmetry is another stalking point", above, was returning. The run then measured it (RATCHET-43): against the octave at 0.176, the window's fourth-power term c fell from −10.614 to −22.1855. Read after the data, the three measured sizes split into a part that flips with the tone's direction (odd, dipole-like) and a part that is the same both ways (even, mirror-like). At RATCHET-41's in-phase size, 0.38, the two parts nearly cancel, +17.0709 and −16.9882 (R43_DIPOLE). If that holds, RATCHET-41's "the mix kept the octave's c" was two halves cancelling at the size chosen, not the octave swallowing the third tone. That is a reading through three points, which cannot test its own shape. The test is Bear's picture made into a run: flip the sign at each measured size. The parabola says c −44.6731 against at 0.38 and −6.3509 in phase at 0.176. It was the first job of R2W's next sitting, B11, where the picture held in kind, two parts of +15.06 and −14.98 at 0.38 nearly cancelling, while the parabola's sizes missed (RATCHET-44; rows 712 and 801) (R2W)

R2W; ties rows 259, 653, 711, 712

Solve the exact limit first, then the first order, then check the full problem at a small parameter (737) — first-order bars on a pin and its stops (the pin within 0.0003 of the root, the stops within 0.0015), tested at weights 0.10 to 0.40, failed at two of three ripple strengths, while the limit itself, tested at weights 0.02 and 0.05, held at all three, each mirror drive equalling the weightless root to five decimals. The weightless toy is exactly a Möbius map, so its roots, dances and mirror were solved before any weight was added; weight entered as a first-order selection function, and Newton's method at weight 0.01 checked it at thirteen roots. Where first order and Newton parted (at 4/5, by 0.029), the next order is the next question. Register the limit, then measure how fast the world leaves it (runs FBP-1's T5 and T6, CONV-1 and SELECT-1) (R2W). The same order of work gave R2W's first headline: with one extra tone in the drive the dancing pair moves by $a \cdot \sin(2\pi x)/(2\pi\ n^2\ \omega^2)$, a closed form with no free number, which reproduced all 56 measured moves (harmonics 1 to 8 at seven dances) within 0.00004 of a turn, and then landed the tenth and twelfth harmonics, predicted first, 14 of 14 within 0.000015, as close as the runs print (KERNEL-1, EAR-2 and EAR-3; row 699) (R2W). And measure an approach from the system's own limit, not the ideal one: once the gap is as small as the hair between the two limits, the hair becomes the gap, and a shrink ratio of 0.25 read 0.275 at the sixth landing (run WADDLE_PRICE 20261011_071302, part P2) (the Loop Atlas sessions). An argument about a limit says nothing about the road to it: test it along the knob the reader will turn. A symmetry argument exact at the limit with no honey (row 328) led the book to write that the drift's zero therefore moves over a quarter turn as the weight grows, and RATCHET-23 registered it (R23-b: past 45 degrees at weight 2.29324). With noise (D 0.5) the drift's zero went out and came home (runs RATCHET-22 and RATCHET-23: −0.80, +8.73, +40.42, +35.32 and +12.20 degrees at

weights 0, 0.32, 1.10592, 1.59252 and 2.29324, beside the step's 0, 16.86, 27.37, 25.95 and 22.62); R23-b failed and its rival held. Raising the weight at a held frequency is not the road the argument speaks of, which is draining the honey at a held mass and drive; that road is entry 109b of R2W's plan (R2W). Print the next order beside a first-order law: at a rim speed of 10^{-2} of the speed of light, the Loop Atlas's turning loop ran its delay over the first-order law, 4 A W / c^2, by 9.999×10^{-5}, the rim speed's square, so the law's edge was measured, not assumed (run SAGNAC_LOOP 20261018_032138) (the Loop Atlas sessions). A counting rule predicts powers in a limit, so register the limit and the bend: BEAT-2 registered the counted powers, 4 for the counter-arrow's share and 1 for the main arrow, as bars at the book's drive, and both missed (3.34 and 0.82), while the rival that the counter-arrow was an artifact missed by far more. The rule was right and the bars were wrong: a count gives the leading power, exact only as the drive goes to zero, and the book's drive is not gentle (the dark point's inverse square already bent to 1.78 there). Registered next as a trend, each gap shrinking by at least half when the drive halves, both held (BEAT-3: gaps 0.66 to 0.20 and 0.184 to 0.049). A power read from a count is tested by how it approaches the count, not by sitting on it (R2W). An averaged picture can promise what the exact equation forbids: averaging promised a third-rate lock in Bear's three-emitter triad fed in phase, but a field that only shakes along one line separates the needle's equation, so no needle can turn net; in a field that turns as well as shakes, every three-armed needle turned at a third of the rate at couplings 0.07 to 0.10 (RF-10, RF-10b). Check the exact structure before trusting an average outside its range (R2W, sitting P2)

R2W; ties rows 8, 174, 699, 726

A number's rounding is part of its ruler: count the digits before blaming the steps or the physics (738) — a fidelity bar asked orbits to close within 10^{-8} when their drives were carried to 7 decimals and the map's sensitivity to drive reached 4,667, so rounding alone allowed about 2×10^{-4}; the bar failed twice for that reason, as predicted before the re-run. A root read from a table kept to eight decimals landed off the root, with a different pair and a different multiplier M′ (the quantity row 106 computes by finite difference and by its integral), where its 1/2 window was only 3.5×10^{-7} wide in bias, so eight decimals left Φ11 at −0.095 instead of zero, and measuring how fast the quantity changes across the window (dΦ11/dB) explained it and became a result of its own: the window narrows as exp(−c1/ω). A small difference of two nearly equal grips carries their error, not its own: at strength 0.10 the finite-difference grips disagreed with the integral route by at most 7 parts in a million, harmless, but at 0.05 by 5 parts in a hundred thousand at 5/8, where the difference itself was 2 parts in ten thousand, so a graded row was half noise, and its reader turned the printed warning into an uncertainty with the wrong scale, multiplied by the difference instead of taken as it is, so the precision check passed; at 0.03 three of the children read negative. Carry the uncertainty of the numbers a difference is made from, in the difference's own units, and when it is too large change the route: an integral route needs no difference quotient. A comparison's allowance includes the printing: a number printed to six places, compared with one printed to twelve figures, carries up to 5 parts in ten million of rounding on its own, and three parents (2/3 at strength 0.05; 1/2 and 2/3 at 0.03) that differed by 2.3 to 4.9 parts in ten million failed a check that forgot it. And if halving the step moves an answer by the same amount in no steady direction (an offset near 10^{-11} in the deepest rungs, halved twice, moved by 10^{-11} to 5×10^{-11} either way, with successive changes in ratios from 0.56 to 3.11, where truncation would fall sixteenfold with each halving and keep its sign), the ruler is out of digits, not out of steps: the same engine in long double moved by 10^{-14} to 5×10^{-14}, and three of the four rungs landed in their registered bands, while both registered rivals, truncation and a real turn, failed (runs SELECT-1, WINDOW-1, JJ_9, JJ_10, JJ_11, FAREY-2, FAREY-2b, FAREY-3b, LADDER-1b and JJ_16). Quote a number only to the digits its conventions hold, and name the convention: a golden clock rounded two ways in two sessions ($0.618034 \times 0.618034 \times D$ in one, $(1 - 0.618034) \times D$ in the other, 6.59×10^{-8} apart) moves every doubling of one cascade by about -5.4×10^{-9} (period 4 -5.5167×10^{-9}, period 8 -5.4262×10^{-9}, period 16 -5.4059×10^{-9}, period 32 -5.4016×10^{-9}; the differences shrink by 4.465 ± 0.03, and past period 16 the arithmetic's floor leaves the ratio unresolved), while halving the step moves the period-8 doubling by $+9.9 \times 10^{-11}$, and a pile-up quoted to ten decimals (1.6893094207) carries the clock's rounding in its last two. Say which rounding wherever a drive is quoted past eight digits (runs FOLD_OR_FLIP 20261007_050721, FOLD_OR_FLIP_2 20261007_053443 and FOLD_OR_FLIP_3 20261007_054521; R2W's run KNOBS-2) (the Loop Atlas sessions). Suspect the file before the physics: NINTH-1 put a gap of 7.701×10^{-13} between two engines down to the engines. STEP-1 found that 7.521×10^{-13} of it was the record's engine writing each edge to 12 decimals; with 15 the engines agreed to 2.598×10^{-14}, and with the witness's tolerance tightened, to 3.997×10^{-15} (STEP-1 and LEAN-1). Row 793's rule for predictions holds for every file of record: before reading a difference near a file's last kept digit as physics or numerics, check what the file keeps. Earlier results that leaned on differences below 10^{-12} in the 12-decimal files are worth a short audit (R2W)

R2W; ties rows 154, 551, 644, 677, 680

Follow a branch by continuation, from where it was (739) — a tracker that took, at each step, the stable stop nearest lag 0 jumped between mirror partners (at −0.016 and +0.014, at a second tone A_2 of −0.056) the moment both existed, so a registered one-way walk failed on the tracker, not on the dance: followed from its last position, the walk is one way, −0.0068 to −0.0251. Wherever a system has a mirror, anything that picks by distance from the mirror's centre mistakes one twin for the other the moment both exist, and that moment is the interesting one. A straight line is a promise only over the stretch it was drawn on: starting guesses fitted between −0.12 and −0.07 and used out to −0.22 failed to converge in seven far cells at heavy weight, and a lean that weight 0.3 would swap near

−0.182, read from a straight line through two points of a multiplier that, as far as the converged cells show, turns over, missed. Carry a guess one step at a time from the last converged point, register the tracker with the prediction, and call a lean from a line outside its range an extrapolation (runs WALK-1 and LAB-MAP-2) (R2W). Find a fold by following the orbit through it, with steps fine enough to see the turn: pseudo-arclength continuation of the pulsing needle's lock went round its turning point and onto the saddle branch; with the step held to 2×10^{-4} near the turn, a parabola through five points put the fold at 0.912331391137, the multiplier passing +1 at the same pace to twelve figures and closing on 1 with slope 0.4990 (a fold's 0.5). A walk-down on a 10^{-5} grid had put the edge 1.1×10^{-5} too low; but the edge's place was not why the lifetimes' slope came out shallow: against the exact fold the three nearest paces still give −0.08. Check the edge before blaming the law, then check the law (run LOCK_EDGE_EXACT 20261013_063227) (the Loop Atlas sessions). Budget a continuation by how far it must move the knob, not by its steps: near a fold the walk travels fast through the orbit's state and slowly in the parameter, and a walk of 800 capped steps stopped short of a fold that came at step 1,635 (runs RAMP_EDGE 20261013_195233 and RAMP_EDGE_2 20261013_200133) (the Loop Atlas sessions). When continuation fails, shorten the stride before blaming the object: at frequency 0.40 the weighted continuation lost four lower edges on the doubling from weight 0.16 to 0.32. RATCHET_2C redoes a failed stride in 4, 16 and then 64 smaller steps and prints how many it used; every weight then converged, 21 in one stride and 7 in four, so the failure was the stride, not the edge. With RATCHET-5's weaker drive every weight converged in one stride (R2W). Follow the ball as well as the branch. A continuation follows the branch it starts on: FOLD-DIR-b walked the heavy balls' resting dance up from strength 0.45 and found it holding to drives 1.21 and 1.53 at strength 0.60. A slow ramp follows the dance the ball is actually in: at strength 0.60 a heavy ball set down at rest sits in a dance that folds at 0.859975 at weight 2 and 0.713061 at weight 5 (FX-2b, FX-2c). The continuation's branch exists, but at weight 2 it is unstable at drives 0.90 to 1.20, its second multiplier −7.8 to −69, which an engine that kept only the larger multiplier had hidden (FX-2c; row 761); so the reading of two resting dances side by side is withdrawn at weight 2 and unchecked at weight 5. The edge a ball at rest feels is that fold. When two methods disagree about an edge, ask whether they are following the same object; a ramp with its multipliers printed answers that in minutes (row 750) (R2W)

R2W; ties rows 642, 652

A small, steady effect that is the same everywhere is the ruler's fingerprint: replay the reading on a known answer before explaining it (741) — a groove's width read from its dancers came out 3 to 6 per cent short at every weight, and the first story was the hump's shape; a second reading seemed to confirm it, two shape factors agreeing within 0.4 per cent at 1.08 to 1.17. Both had been read on starts spaced evenly in a coordinate that is coarse just where the dancers stand. Recomputed on even starts in the straightened coordinate, the humps were sines within 0.14 per cent, and the first reading's exact steps, replayed on a case with no table, no weight and no gear, reproduced its shortfall within 0.4 per cent: the 3 to 6 per cent was the reading, and SLOPE-2's 1 to 2 per cent gear excess was most likely the same grid. The same shared bias had already decided a comparison: a registered bet that the mean of the steady and wobbly dancers would beat either kind alone at 2/5, where their grips differ most, failed, the steady dancers alone landing closer, because both estimates were about 3 per cent short and the steady dancers' own lean, about 2 per cent high from the excess, happened to cancel it, so the comparison was settled by a bias neither side modelled. Before comparing two estimates, remove the bias they share, and before explaining a small, steady effect, replay the reading on something whose answer is known (runs GROOVE-1, SHAPE-1, SHAPE-1b, SHAPE-1c and SLOPE-2) (R2W). And prove a residual is the integrator's by its halving ratio, not its size: set a structure test's bar from the integrator's order, then halve the step. A Möbius fit to the forgetting pulsed needle's one-period map, carried at 160 RK4 steps a period, missed a bar of 10^{-9} by 6×10^{-9} to 10^{-8}; each halving cut it 15.75 to 16.43 times (RK4's 16), to about 2×10^{-12} at 1,280 steps, while the needle with memory missed the same fit by 2.78 radian (runs QUANT_ZERO 20261013_081905 and QUANT_STEP 20261013_082454) (the Loop Atlas sessions). A cluster the eye finds can be the instrument's own: Bear's "little cluster" near memory 1.72, on a figure of the stop leaving the window, was the follower's own restarted steps, spaced by its step, as the bunched dots he had ringed on the heavy-window figure the day before (row 714) and asked about ("Rubber band?") had been the follower's steps too; each was reported to him as such (the Loop Atlas sessions). A control that moves along with what it controls cannot see a drift they share: LW5 (triple_lw5 v1.2, run 20261017_140817) read its 1/8 control STRAIGHT, slopes over 6,000 and 12,000 passes against 2,400 of 1.000 and 1.000, but only because the whole line moved down together. At 7.3125340 the rate fell from $+4.79 \times 10^{-5}$ over 2,400 passes to -1.76×10^{-6} over 6,000 and -8.45×10^{-5} over 12,000; FA0 (run 20261018_031414) measured the fall at -2.755×10^{-8} in each pass at all 41 births, and DR1 (run 20261018_031349), with every tolerance 100 times tighter, left only 0.0093 of the drift at that birth. The integrator's own arithmetic was faking a drift, the same at all 41 births. When a control compares shapes such as slopes, also read its level against the record's length, and rerun one point at tighter tolerances before trusting any trend with record length (rows 599 and 732) (the triple-star sessions). A second specimen: R2W's integrator lost $h^5/(72\ m^2)$ times one constant, 0.727, at twelve weights from 0.01 to 10 within 0.26%, and the constant did not move with the drive. Fourth-order Runge–Kutta's known loss on a swing, an energy loss of order h^6 each step with the coefficient 1/72 (Liu and Brio, 2024), explained it, and the law, solved for the step, became a knob that landed within 0.27% (NO-HONEY-SLOPE, ENGINEERING_SURPRISES)

R2W; ties rows 242, 336, 711, 858

Watch what stays put while the knob moves: a fan through one point, a pinned edge, a band that looks narrow (742) — when many curves cross at one point whatever the knob, something holds that point fixed: a staircase's curves all ended at one birth ring, and every weight's stop curve passed through one weightless root. When one feature is pinned while its neighbour moves, a symmetry or a law is hiding there: a wrap held between 1.4415 and 1.4435 while a ring moved, and a groove held its place while its unstable twin slid more than a thousand times further per unit weight. But a band that looks narrow at one setting may be the zero of a drift: a pair sat 0.027 to 0.028 inside its frozen points (its balance points with the drive frozen at the strobe, where the push is zero and a rider would rest) at every root at strength 0.30, ten times wider at 0.20 and running the other way at 0.40, because 0.30 sat near where the drift with drive changes sign. Ask what leaves the pinned thing unmoved, and try one setting either side before calling it a constant (runs TOY-SIDE-3d and TOY-SIDE-4f). A symmetry is a prediction, and the cheapest one there is; row 734 registers symmetries as bug detectors, and here they predicted the physics. The Depth Axis Explorer's bench showed weight reopening the dark point at nearly right angles. Instead of fitting the four rungs already on file, the reflection that pins the dark point was asked what it allows: an opening that starts exactly across, and a size with no straight-line term. Those two sentences, written before two lighter rungs ran, made a sharp test: the size landed 1.59 parts in a million from the square law and 6.33 parts in ten thousand from the straight line, and the direction met a quarter-turn to 0.000042 degrees. A symmetry's full form tests more than its shadow: R2W's book knew the mirror's shadow, that the window's size has only even powers of the weight on the mirror line; the full form says the whole window, direction and all, becomes its own mirror image, and it predicted what the shadow could not, where a dark point's window must point when weight opens it and how the openings at two mirror timings pair (the mirror in full and a designed tilt, RATCHET-67 to 70). When a symmetry has been seen in a magnitude, write it for the whole complex quantity and test what that adds; when a pattern looks clean, first ask which symmetry would force it, then test what the symmetry forbids. The sitting's review put exact symmetries first among what carried its wins: the mirror, the reflection and the half-beat shift predicted where zeros sit and which harmonics survive, the cheapest predictions of the sitting (R2W). When a symmetry holds exactly at one special setting, ask first whether it is only a relabeling there; then the departure from that setting is the result. The Loop Atlas's mirror cats mirror as a whole family only with the hinge at the tubes' centres (to 3.3×10^{-16}), where a tube looks the same from either end, and the failure grows steadily as the hinge moves out, 0.023 at a tenth and 0.28 at the ends (CAT_MIRROR_FAMILY 20261023_001232, CAT_MIRROR_CENTRES 20261023_001618), faster than the square close in and slower farther out, its shape still to be fitted on a finer ladder of hinges (misses row 462). Three neat points on one grid can be the grid's luck: R2W's STACK-3 read onset times room as 0.929, 0.952 and 0.954 at rooms 0.04, 0.06 and 0.08, a tidy one over the room, and STACK-4's rooms between gave 1.056 and 0.834; on a grid five times finer the stray holds were three bands, and a band just below the top edge moves that edge by a band's width. In a narrow room the quickest filters hold and the middle ones miss, so slower is not safer there (R2W, sitting P3)

R2W; ties rows 511, 684, 734, 820

Squeeze a picture to see its outline, mark its layout first, and check the outline by numbers (743) — a picture of many columns side by side is a comb, and the eye follows its teeth; squeezed sideways, the comb becomes one silhouette whose shape can be read, a waist, a flare, a spike, the way a carpenter sights along a board. The picture was a strobe picture from the triple-star sessions, of island 7 at the pair's eccentricity 0.25, every pass a dot and one column of dots for each run, panel (a) the first 600 passes and panel (b) passes 601 to 6,000. The person coloured the data columns and the empty spacers between them before anything else ("pink lines are data bands and black bands are spacer rungs"), squeezed the picture to about half its width (1,690 pixels to 820, "for offset effect prior to drawing line"), photographed his screen at a grazing angle ("pretend offset phase shift eye to 75 degrees … to make it severe"), drew its axis and both edges by eye, and circled the columns that looked narrowest. Counted by the picture's own pixels, the dot pixels found by colour, the axes calibrated from the picture's own gridlines and the 10th and 90th percentiles of φ_N taken over each column's pixels, every part survived: the midpoint of each column's spread sat within 0.010 turn of zero (median 0.002); from birth 7.455 to 7.51 the waist's spread averaged 0.155 turn against 0.306 elsewhere, and 0.109 to 0.149 across most of the waist, 7.48 to 7.505; his ellipses sat on the locally narrowest columns (0.329 at 7.395 against 0.677 and 0.727 just to its left, 0.164 at 7.535 against 0.227 and 0.228 either side), the large one covering the waist, and his small ellipse sat on the narrowest of all 39 columns (0.105, at 7.525); and his green spikes sat on the wide columns (0.213 at 7.475 between 0.147 and 0.141, and 0.244 to 0.321 at 7.51 to 7.52). In the paler panel the colour rule caught only 16 of its columns, so nothing was claimed for it. A grazing look, the screen tilted to a steep angle, squeezes the far side more than the near and bends straight lines, so it finds and never measures. In the triple-star sessions the same moves, squeezing, tilting and turning a picture a quarter turn, gave three reads registered before measuring that held at two scales, and the phase-offset view he described turned out to be a delay plot. Mark the spacers before the data, read shape on the squeezed copy and measure on the original, and treat an eyeball outline as a hypothesis about percentiles, checked in minutes (run STROBE_PIXEL_CHECK). The same lesson reached the Loop Atlas on 14 October with his marked triple-star pictures (the triple-star sessions had them first), and it added to the moves above: squeeze to a quarter as well as a half, so long faint structure foreshortens into lines the eye can follow; trace a centre line beside the outline; and look for a mirror shape and a pinch as well as a waist, ringing the dense knots. The machine's version was drawn beside it on the quarter tree at full, half and quarter width (outline the 1st and 99th percentiles, centre line the median), and read by eye: squeezed,

the arches straighten into rays fanning out from the pile-up, the outline is a wedge with a step at the crisis and no waist, and the centre line jumps at each window (run QUARTER_PHASES 20261014_073718) (the Loop Atlas sessions, after Bear)

R2W; the triple-star sessions; Bear's squeeze; ties rows 703, 714, 715

A person's physical words point at theorems: name the theorem in his words, then check it (745) — "it has a repulsion quality to it" was interleaving: two dances of a rotation in disguise cannot pass each other, so their stops, red and purple, strictly alternate round the circle, in all nineteen rows read. "Does the dot coming around the circle change its frame?" was a loop's multiplier read as its holonomy on a neighbouring step: exactly 1 without weight, about 0.95 for red and 1.02 for purple at weight 0.01, with the passing matrix to the q-th power equal to (−1)^p. "If you can't cut that math, is it a latch?" has a yes in the Latch Atlas's sense: a winding or linking number holds its value until an event the smooth motion cannot make releases it (a tongue's edge, a collision, a bifurcation), as a dance's fraction holds across the whole width of its tongue and changes only at its edges. "It will never touch but it defines the gap", said of a figure of the gaps closing period by period, is exactly what a limit is: arms that stand apart by a real gap at every period would meet only at 1/2, the parent and not a rung, and the shrinking gaps pin the meeting point. And "the phase offset or quadrature between the blades doing work" is right in kind, since directed motion from two oscillations with a phase between them is real (cold atoms in a two-frequency lattice of light, the ultrasonic motor), but which phase works depends on the pair: in an ultrasonic motor two standing waves of the same frequency, 90 degrees apart, make the travelling wave, while in the junction a tone and its second harmonic rectify most in phase, and not at all in quadrature, when the rider has no weight. The instinct names the mechanism; the run names the phase (runs READ-E1 and RATCHET-1b). Bear's words, whole, open the R2W book's Part 7 (15 October): "The idea in my head was the phase offset or quadrature between the blades/pads doing work". The sitting that followed them found what weight does to the phase: a second tone at a relative phase makes the zero-slip step lopsided, and weight slides the lopsidedness's zero off quadrature and back (runs RATCHET-2 to RATCHET-5) (R2W). Look for a missing power in another drawer. Bear asked: "If we hypothesized on how to find a fourth power knob, how would one discover it? Would it follow any pattern? Could the fourth power be a comma-like device that isn't where we assume we should find it and then the fifth power appears in the normal spot? Does reflection matter as a theorem to help? How can we use the fine tune knob as the strongest knob of the day? Does each knob have a fine tuning component like you just found? If you found one, it feels like that is a sign." The reflection (the R2W book's §6.11) answers in kind: running the film backwards with a half-turn shift maps the weight to minus itself, so the window keeps only even powers of the weight and the zero's slide only odd ones. No fifth power hides in the window; the odd powers are all in the timing. His guess that a power lives somewhere other than where one looks was right; the symmetry says which drawer. His other questions are answered in rows 684 and 581 (R2W). A rule of thumb from one machine names the symmetry in another: Bear asked if the meter's timing counts were like needing an odd number of blades to avoid bad harmonics. The blade rule, a rectifier's pulse number and the meter's timings share one piece of arithmetic: equally spaced samples pass only the harmonics jM ± 1, and half-wave symmetry removes the even ones, so an odd count does the work of twice as many (row 802). His question pointed at the right symmetry before any run, and ODD-1 confirmed it exactly: three timings read what six read to 3.376×10^{-15}. When a familiar machine's rule fits, ask which symmetry it rests on, and test that (R2W). A name from the person's life is the plain word, and the field's word stands beside it: Bear's quenching, for a warm brick dropped in ice, beside the metallurgist's word, and his escape hatch, for chaos that drains away through a door, beside the theorist's transient chaos. Two older pictures of his, a power gauge and a truck road, were tested in one Loop Atlas session, and each name carried a test or a telling (rows 804, 724, 746 and 670). A name from a life is also a hypothesis: the power gauge was a prediction that the meter rises as the machine wears, and it was tested as one (the Loop Atlas sessions)

R2W; Bear's words; ties rows 561, 573, 715

To read a memory in a sequence, split it, ask whether it chains, and find the state on a section; then take that state out along its own curve (747) — split a covariance before reading it, and read each share against its spread: the landing's small memory (successive landing-timed gaps, correlation +0.057) is three quarters a fall with the next fall and a quarter the tie across the landing, yet the two pairs' correlations are alike (+0.061 and +0.068), because the falls simply vary more than the climbs (run LOCK_LANDING 20261011_170402). Durations need not chain even when the motion does: the remembering needle's halves are not a chain, since a fall's correlation with the next fall (+0.058) is larger than the product of the two links through the climb between them (+0.068 times +0.583 = +0.039), so the state at a landing holds more than the durations before it (run FALL_CHAIN 20261011_200714). A memory one step deep is not a chain either: with 1.48 million pairs the falls correlate +0.060 one apart and +0.0012 two apart, 2.86 errors below a chain's square, each fall sharing one disturbance with its neighbour and none beyond, the signature of a moving average; the test was sized by R2W's arithmetic and run in pieces that survive a restart (run FALL_CHAIN_2 20261012_041438). On a section of the motion, one number can be the whole state: the needle lands at a fixed lag, so its speed there is everything it carries forward, and holding that speed fixed takes the fall-to-fall correlation from +0.058 to −0.001 on 372,123 triples, where holding the climb's length fixed took only three fifths of it. Partial out the state on the section, not a duration that summarises it (run FALL_SHARE 20261012_075804). A chain of states then explains the memory at every depth: the landing speed forgets in one step (+0.0206 from one

landing to the next), and the falls' memory two apart is the one-apart memory carried through it, +0.0600 times +0.0206 = +0.0012, the value measured on 1.48 million pairs; a second section can be less clean than the first, the barrier-top speed taking out five sixths of the climb-fall tie, not all of it (run SPEED_CHAIN 20261012_171949). And take a variable out along its own curve, not a straight line: given the speed at the barrier top, a straight-line partial left a sixth of the climb-fall tie (+0.0980 of +0.5865), while binned means left +0.0064 with 200 bins and nothing with 400, because a crossing where the needle nearly keeps pace with the field stretches both halves far more than a line allows. A section can look leaky when the ruler is straight (run TOP_SECTION 20261013_005355)

the Loop Atlas sessions (the book); R2W's sizing; ties rows 638, 651, 748

Before naming a tail's shape, size the sample, set the bar from its own scatter, and cut long enough to keep out the merely slow (748) — before calling a tail two-shaped, quadruple the sample: two of six paces failed an exponential test with 256 needles each; with 1,024 both passed, and the original 256 were inside the larger samples, their lifetimes unchanged (runs EXP_SHAPE 20261011_080708 and EXP_SHAPE_2 20261011_194909). Size a test before running it, and let a neighbour size it: R2W's arithmetic (about 289,000 falls to put a chain's 0.0037 two errors above zero) split one undecidable test into two questions, chain against drift, decided at 35 errors on 368,185 pairs, and chain against no memory, still undecided at 1.4 errors, the near coin flip the sizing had foretold at that sample (run FALL_CHAIN 20261011_200714, with R2W's chain-test note relayed by Bear); on 1.48 million pairs the next morning the falls two apart sat 2.86 errors below a chain's square, so the durations are not a chain (run FALL_CHAIN_2 20261012_041438; row 747). Set a shape bar from the sample's own scatter before the run: for 64 draws from one exponential, median over mean averages 0.701 with a spread of 0.088, and 13 per cent of true traps reach 0.8, so a bar of 0.6 to 0.8 on 64 catches tests luck as much as shape, and the catches' 0.800 said nothing by itself (run CATCH_TRAP 20261011_202500; the spread from a scratch calculation, not a run of record). Two small samples leaning the same way are not yet a direction: the remembering needle's catches leaned above the exponential's ln 2 on two seeds of 64 and 69 (0.800 and 0.853); on eight seeds and 6,613 catches they sit below it (0.678), and the departure from a trap that does show runs the other way, an excess of short catches just past the cut (run CATCH_TRAP_2 20261013_030643). And read a trap through a cut long enough to keep out the merely slow: the remembering needle's climbs over 2 periods were not one exponential (Kolmogorov 0.0240 against 0.0134); over 3 and 4 periods they are (0.0125 against 0.0140; 0.0079 against 0.0149), and the mean excess stops moving as the cut rises (11.45, 11.54, 11.57 periods): a trap forgets, so its excess past any cut is the same (run CATCH_TRAP_3 20261013_040120). A rough floor that repeats every turn does not make an avalanche: ledges under a remembering needle kept its runs a trap's (median 0.51 to 0.76 of the mean, at most 8.2 per cent past three means), and raised the hill's top by 4 to 14 per cent while moving the slipping band only 0 to 5; place a grid on each floor's own band, not on its top grip (run ROUGH_FLOOR 20261017_013514) (the Loop Atlas sessions)

the Loop Atlas sessions (the book); R2W's sizing; ties rows 616, 723, 747

Ask whether an orbit repeats by the gap between its repeat and everything else, not by a sieve (749) — a sieve can mislead: at one part in a million the needles at pace 0.99375 seemed to repeat after 19, 57 or 95 drive periods; at one part in a billion only two closed within 400, after 399 = 21 laps of 19, the rest passing within a millionth of themselves every 19 and closing later (run SETTLED_GROOVES 20261011_223349). That was first read as a coarse sieve making long orbits look short; the end of this row shows the reverse. When many paths crowd together without closing, measure the distance to a candidate orbit instead of asking for exact repeats: only one needle closed within 4,000 drive periods (after 437 = 23 laps of 19), while 12 others lay within a billionth of its path at some reading (run SETTLED_GROOVES_2 20261012_041558). Before reading needles near an orbit as arriving slowly, read the orbit's own pull: twelve needles hovered 10^{-12} to 10^{-8} from needle 30's 437-period orbit for 8,000 drive periods with no trend, while the orbit squeezes a deviation by about 10^{-21} a lap, so nothing within 10^{-8} of it can be arriving slowly, and the needle that defines it strays as far from itself (run ORBIT_919_3 20261013_031807). An angle that keeps counting turns loses its fine digits, and a stable orbit can blow the loss up: the needles' lag reached 31,600 radians, where double precision's grain is 3.6×10^{-12} (the time's, at 66,000, 1.5×10^{-11}), and an orbit that squeezes a deviation by 5×10^{-22} a lap but stretches it 10^5-fold over 17 drive periods on the way turned that grain into a median scatter of 3.8×10^{-9} and spikes of 3.6×10^{-6}. Carried in long double, or with the lag and the time folded back into one turn, the same needles scatter 8,000 to 38,000 times less: scatter that shrinks with the grain is the arithmetic, not the physics (run ORBIT_919_4 20261013_040039). So bet on the gap between the repeat and the rest, not on the floor the arithmetic sets: the needles at 0.99375 repeat after 19 drive periods at 1.04×10^{-12} in median and at no other gap below 0.72; a bar at 10^{-12} missed by 4 per cent while twelve orders of magnitude separated the answer from everything else, and the earlier, longer repeats were the sieves' doing: 95 came from the sieve of one part in a million, and 399 and 437 from a 10^{-9} sieve catching double precision's scatter (median 3.8×10^{-9}) at quiet moments (run ORBIT_19 20261013_041840)

the Loop Atlas sessions (the book); ties rows 632, 738, 784

A fold fools a trajectory: shoot at it, or find it by its multiplier reaching 1, not by asking whether a run stayed (750) — shoot at the edge itself, not toward it: a switch approached through readings ever nearer an edge moves as the square root of the distance left, so a stopping rule on the step fails (0.7037 at 10^{-6}, still moving); at the

edge, comparing the path that leaves the birthplace with the path that arrives at the next one, at a section half a turn on, gives the switch to eight figures in a minute (0.70256366) (runs MEMORY_SWITCH 20261011_084432 and MEMORY_SWITCH_2 20261011_085331; the same finding as R2W's in row 686). A loose test for "calm" is met by chance inside chaos: past the 1/4 window's edge the state came back within 0.05 after 4 drive periods for a period or two at a time, 500 to 600 times in 20,000, so the mean calm stretch counted coincidences and missed the flicker, while the long stretches and the first passage carried it. Count laminar phases by what chance cannot fake, or find the fold itself (run WINDOW_EDGE 20261012_080030). A finite trial reads a ghost as an orbit: just past a fold the vanished orbit's ghost holds the state for a passage that grows like one over the square root of the distance, and inside it the state barely moves; an edge bisected with 400-drive-period trials sat 1.6×10^{-8} past the fold Newton's method found, where the passage runs near 510 drive periods. Find a fold by its multiplier reaching 1 (here one minus it fell with slope +0.5015 against the distance), not by asking a trajectory whether it stayed (run WINDOW_EDGE_2 20261012_171509). And a fold can be checked three ways that must agree: the 1/4 window's fold gave (one minus its multiplier) squared over four times the distance = 35,142 per unit of pace; the same product predicts the passage through the ghost, 4π over its square root, 67.0 over the square root of the distance in millionths, in drive periods; the measured passages run 0.947, 0.973, 0.986 and 0.993 of that at 1/16, 1/64, 1/256 and 1/1024 millionths, closing on it; and the ghost's depth grows exactly in proportion to the distance (slope +0.9994). This was read after the runs, and its proposed test, a passage of 4,276 drive periods at 1/4096 millionth past the fold, was never run (runs GHOST_DEPTH 20261013_024747 and REINJECT_CLOCK 20261013_024653; a fourth agreement is in row 756). A strongly hyperbolic map defeats a Newton start, and when an old route held, take it: at frequency 0.40 the weightless passing map is strongly hyperbolic (half-trace 1,090 at zero bias, against 136 at 0.55), and the fold-Newton start failed. Sitting B9's route, the half-trace crossing 1, gave the edges, finished by a short Newton polish (run RATCHET_2B; its fidelity check held 928 of 928 rows of record) (R2W). Two definitions of an edge agree only where nothing else ends the resting state, and this limits the row: the multiplier finds where the resting dance stops existing, and a ball set down at rest can leave before that. KNOBS-3's grid read gear 1's edge as the drive past which a ball set down at rest stops staying, and at strength 0.30 it agreed with the fold within a grid step (FD-0). After three fold-finders failed in turn on the turned ripple at strength 0.60, the edge was read directly, ball by ball (FOLD-DIR-d): at strength 0.45, where every fold was known, a ball at rest slipped off 0.05 to 0.11 of drive before the fold, and at 0.60, where weight 0.75 leaves at its fold, the solvers' failure was the method's (row 682). Before building a better solver, read the thing directly with a lane whose answer is known on each definition, and say which edge the engineer needs: where the state stops existing, or where a ball started a given way stops staying (R2W)

the Loop Atlas sessions (the book); R2W; ties rows 640, 642, 654, 686, 756

Look for an orbit by Newton's method before calling it absent, and follow it through: it may be there and unstable (751) — next to a doubling, a walker's "nothing repeats" means "not yet": the 1/4 window's cascade (doublings at 0.98372406, 0.98370962, 0.98370621 and 0.98370546; ratios 4.24 and 4.55 toward 4.669) runs below the pace where a walker settling 100 drive periods saw no repeat, for the period-8 orbit there is stable but its multiplier sits near −1; and when following a doubled orbit, guard against Newton falling back onto its parent, which is a fixed point of the doubled map too (run WINDOW_CASCADE 20261012_081249, its version 1 set aside). Read a lock by its least repeat along the knob, not only by its turn rate: across 0.9951 to 1.0100 the remembering pulsed needle's rotation number reads 0.498 to 0.502 everywhere, a flat half gear, while its least repeat reads 4, 8, 16, 32, none, 32, 16, 8, 4 going up the paces, the shape of a doubling cascade into chaos and back out (a bubble). The turn rate hides the doubling; the repeat shows it (run QUANT_ZERO 20261013_081905, its saved scan read after the run). When a lock's least repeat is never q but always 2q, look for the q-orbit by Newton's method before calling it absent: it may be there and flip-unstable. The needle turns once in two cycles across 0.9951 to 1.01 yet never repeats after two; its 2-cycle orbit of one turn is there, with multiplier −2.14 to −2.04 (the pair's product within 0.2 per cent of the drag's e^(−4π), a check on the Jacobian), while the needle lives on a 4-cycle orbit, stable between doublings at 0.99920 and 1.00199, and again from 1.00978, as followed below (run HALF_BUBBLE 20261013_083609). To call a cascade into chaos and back a bubble, follow one orbit through the whole of it: the 4-cycle orbit, followed by Newton's method in steps of 10^{-5}, doubles at 1.0019857, stays unstable through the stretch where a sweep finds no lock (its least multiplier −1.68 at 1.0053, inside that stretch), and undoes its doubling at 1.0097823, where the sweep finds the 4-cycle lock again. Matching counts on the two sides of the chaos do not show it is one orbit; the walk does (run BUBBLE_FAR 20261013_084037; the reversal of a doubling cascade is antimonotonicity, Dawson, Grebogi, Yorke, Kan and Koçak, 1992). And a doubled orbit whose multiplier climbs toward +1 is heading home to its parent: at a flip the child's multiplier is +1 where the parent's is −1, so find the parent by Newton's method and follow both. The 4-cycle orbit climbs from −1 (1.00978) to +0.9987 and its distance from the 2-cycle orbit closes from 0.99 to 1.5×10^{-8} at 1.01512, where the 2-cycle orbit's multiplier passes −1 (1.01511); a rival picture, another tooth, found no peak (run HALF_PARENT 20261013_163612). A finder must not read its own failure as the answer: a doubling finder whose test was that Newton's method fails or a multiplier reaches 1 in size reported four edges, every one a place where Newton stopped converging, because the needles' angles were carried unreduced at 168,542 to 250,223 radians (row 644 keeps the numbers small); reduced to one turn, the same finder found four true doublings, multiplier −1 (runs HALVINGS 20261015_064027 and HALVINGS_2 20261015_064427; the Loop Atlas's misses row 220). Reduce an angle before asking for a cycle, and keep a method's failure apart from the event it looks for. And a check that a test

can decide is not a check that it is right: the blind run said all four edges were found, and they were found, and they were not edges. The redesign gave the finder a planted case with a known answer of the very kind the test reads, the window's own first doubling, which the mended finder found 1.2141 millionths below the top against the table's 1.2143 (run HALVINGS_2 20261015_064427). A planted answer of the kind the test reads beats a check that the answer was found, the rule row 551 gives every check (the Loop Atlas sessions)

the Loop Atlas sessions (the book); ties rows 642, 680, 705, 761

Past an inner fold the chaos keeps refilling the ghost: count the returns, not just the first passage (752) — count the returns to tell where a fold sits: at seven distances past the 1/4 window's fold the needle came back into the vanished orbit's ghost 231 to 730 times in 200,000 drive periods, each return lasting most of a full passage (0.90 of it nearest the fold, 0.66 farthest), the mean return growing with slope −0.62 against the distance: the fold closes inside the chaos, an inner fold (run INNER_OUTER 20261013_005545). Past an inner fold, a trial of fixed length answers by chance: edges bisected with 400-, 1,600- and 6,400-drive-period trials closed on Newton's fold by 0.215 and then only 0.458, because the long trials caught the needle on returns into the ghost, not on its first passage. A ruler that asks a trajectory whether it stayed is fooled twice near a fold, by the ghost and by the chaos that keeps refilling it (run GHOST_TRIAL 20261013_005637). Near the fold each return becomes a whole passage and the chaos keeps a steady clock: each fourfold approach very nearly doubled the first passage (254, 522, 1,058 drive periods), the mean return reached 0.97 of it, and the time spent outside returns stayed near 600 drive periods per return; a trial of fixed length there is a lottery, 34 of 200 answering on at random (runs REINJECT and GHOST_TRIAL_2 20261013_015743)

the Loop Atlas sessions (the book); ties rows 642, 654, 750

Turn the noise off, down and up before giving an edge to the controller or to the noise (753) — past a controller's textbook limit, ask whether the edge is a wall or a slope: the never-blind sized tap holds past the smooth push's √2, and its edge moves with the jostling, 1.70, 1.55 and 1.25 fall times at jostling 0.01, 0.02 and 0.04 for one kick (past 1.80, 1.60 and 1.20 for the next), so there the noise decides when the rider falls; only the strongest kick is already down at 1.20, the shortest delay tried, at every jostling, a wall of its own. Far from the edge the wobble goes nearly in proportion to the jostling, near it not (run SIZED_TAP_4 20261013_161615). Run the controller with no noise before giving its edge to either: with the jostling off, the never-blind sized tap holds to 1.75 and 2.00 fall times, so the noise had set the edges near 1.55 to 1.70; and the strongest kick fails at every short delay and holds at every long one, so its failure at the short delays, seen at every jostling, belongs to a band that is the controller's own (run QUIET_TAP 20261013_195754). An edge can be a cliff, not a slope: a listening ear made of its own ring (row 769's spring bead, charged only while it moves one way and drawing a net push from a soup that pushes equally every way) stalls at critical damping, sharply, taking 0.054 to 0.175 of the soup's push at 0.95 of critical damping and exactly nothing at and above it, at tunes 0.5 to 2 (run SPRING_STALL 20261014_063112). And under noise an edge with memory is two doors: a memory-5 needle holding at pace 0.75 falls at the rate its hill's barrier sets (the log of its holding time against 1/D at slope 0.234 against the barrier's 0.239; with no memory the same count matches the exact first-passage time to 2 per cent), but once running above its climb-back edge it is caught again too, about half the fallen needles holding again by the end of the runs. Measure the catch as well as the fall before calling a slip final (run NOISE_EDGE 20261014_232230; the barrier's escape law is Kramers's, 1940)

the Loop Atlas sessions (the book); ties rows 636, 655, 675, 725, 787

Read a stretch with the map's own algebra: eigenvalues beside the singular value, and the frozen rates added up (755) — before calling a stretch growth, set the map's eigenvalues beside its largest singular value: a shear (a speed difference turning into a lag difference) stretches with small eigenvalues, while growth has an eigenvalue near the stretch. The needle's largest one-period stretch, 20.9, has an eigenvalue of 20.5, so it is growth; its ringing periods have two complex eigenvalues of exactly e^(−π) each, because the drag alone fixes the determinant per drive period at e^(−2π/M). The same check flags round-off: a product's small eigenvalue cannot sit above what the determinant allows (5.3×10^{-9} printed where e^(−38π) allows about 10^{-51}) (run STRETCH_WHERE 20261013_045856). To read a periodic orbit's stretch, freeze it and add it up: freeze the system at each instant, take the growth rate its linearised self would have if it stayed there (for the junction, the larger real part of (−1 + √(1 − 4M cos L))/(2M)), and add the rates through the cycle. On five fresh paces of the pulsed needle that sum ranks with each drive period's true stretch at 0.97 to 0.99 and gives its size to within about a tenth in the logarithm (median differences −0.12 to +0.10); the depth over the top (the mean of −cos L) does nearly as well (0.97 to 0.99) where the share of time on top saturates (0.69 to 0.77). Where the needle settles at rest and only rings, the sum misses by more (+1.26 and +1.33 in the logarithm), not yet understood, and no later run of the Loop Atlas took it up (run STRETCH_DEPTH 20261013_055240)

the Loop Atlas sessions (the book); ties rows 730, 749

A window's end in banded chaos is a crisis: find it by the orbit born with the window, and below it read the stays (756) — the 1/4 window's lower end is a crisis: below about 0.983688 the chaos fills the whole circle (0.64 to 0.86 of its holes touched), above it keeps to four bands (0.035), so the window runs from about 0.983688 to its fold at 0.98375283 (run QUARTER_TREE_CURVES 20261014_055131). A periodic window in banded chaos is bracketed by

one pair of orbits: the window's orbit is born at its fold together with an unstable partner; followed down by Newton's method, the partner sits just above the band's top edge, and the window ends where that edge lands on it, at −11.92 millionths of pace on the runs' own scale, which counts millionths above 0.98370 (so at 0.98368808), the crisis above: the edge's gap to the partner and the fold's third image's gap to the edge fall to nought together (−11.9177 and −11.9189), in the ratio 3.05, the partner's multiplier less one, as the closed form gives (runs SEAM 20261014_090006 and SEAM_2 20261014_090552; the picture of a crisis, an unstable orbit meeting the chaos, is Grebogi, Ott and Yorke's, 1983). Run SEAM also checks row 750's fold law a fourth way: its 35,142 per unit of pace predicts the partner's multiplier 0.83 millionths below the fold as 1.3416, and SEAM printed 1.3418 at +52.0 (this guide's check). Below a window's end the needle keeps coming back to the window's old strips, and its mean stay grows as the inverse square root of the distance below the end: slope −0.4982 with a standard error of 0.0011 over distances from 0.02 to 0.64 millionths (stays of 135 down to 24 drive cycles), the plain one-dimensional 1/2, not the 0.553 a two-dimensional correction would give as Claude recalled it; the bursts between stays last about 35 cycles at every distance, so what changes with the distance is only how long the needle stays (run CRISIS_BURSTS 20261014_091405; the stays' power law near a crisis and its critical exponent are the subject of Grebogi, Ott, Romeiras and Yorke, 1987)

the Loop Atlas sessions (the book); ties rows 697, 707, 750, 787

A window's shape outlives its size and its company: measure its ratio, not its width, and start the needles close inside the motion (757) — a window's shape outlives its size: as the needle's squeeze is made seventy times lighter, the 1/4 window shrinks fivefold (top to first doubling 28.8 to 5.9 millionths) and slides 0.034 in pace, yet its end stays 2.253 to 2.258 times as far below its top as its first doubling, the one-hump law of 9/4 (each window is a small copy of the whole one-hump map, whose end sits 9/4 as far below its top as its first doubling; the paper is named at the end of this row). Measure a window's ratio, not its width, to ask whether a machine is still a one-hump map (run HEAVY_SHAPE 20261014_155553, at Bear's word). Start the needles close inside the motion being measured: at memory 1.254, 5 of 128 needles started 10^{-4} rad apart beside the window's own cycle left it on a detour hundreds of cycles long before coming back, and a measure that reads the band's edge read their wandering as the window ending; started 10^{-7} apart, none left (the detour was first taken for a second motion and corrected, row 665). And a blind probe of whether a test can decide must run on the test's own inputs: one run with a different first guess said yes where the test then could not (runs STOP_WALL 20261014_180544 and HEAVY_SHAPE_2 20261014_164913). A window's shape outlives the company it keeps too: followed to memory 1.2725, with the squeeze over the window's four drive cycles about 220 times lighter than at memory 1 ($\exp(-8\pi/M)$), the 1/4 window's end stays 2.253 to 2.262 times as far below its top as its first doubling at all eight weights, though by then needles started 10^{-4} rad from its cycle can wander for hundreds of cycles before coming back; both distances bend against 1/M, together (slopes of their logarithms 8.62 and 8.72, 8.65 and 8.50, 8.35 and 8.30 at the three heaviest steps), and the window narrows more slowly as it shrinks (run HEAVY_SHAPE_3 20261014_170837, on Bear's carried-on curves). The 9/4 is the window scaling of Yorke, Grebogi, Ott and Tedeschini-Lalli (1985), who found that the windows of a one-hump map are small copies of the whole map: the whole map's first doubling comes one unit below its top and its end 9/4 units below, so every window's end sits 9/4 as far below its top as its first doubling (the Loop Atlas sessions). The 9/4 is written out in Tedeschini-Lalli and Yorke (1986): carried from a window's stable stretch, its top to its first doubling, to its whole length, top to end, the range "only increases Length by a factor of 9/4". Their table of the Hénon map's windows reads 2.239 to 2.267 at periods 8 to 18, against 2.25 for the one-hump map, with 2.107 at period 6 and 1.635 at period 7; the heavy needle's readings above sit inside their spread (the Field Guide sessions, from a copy the person found). The boat that keeps its name: when every number changes, ask what stays. Bear's riddle, "This reminds me of the saying, if a boat over the years replaces all parts, can it be called the same boat, since none of it is original?" (the twenty-fourth exchange), points at the right object. The weight-proof edge sits at drive ratio 1.3250 at frequency 0.40, 1.3056 at 0.55 and 1.2746 at 0.70. No number carries over, so each frequency's edge must be found by sounding its own boards. What carries over is the law: the window's light-end curvature runs on a straight line in the frequency, and the edge sharpens about as the frequency squared. The boat is the law, the planks are the numbers (R2W). The ladder of doubled cycles below the needle's chaos keeps its shape across a change of weight too: at the lighter needle it stood 0.0013 to 0.0018 deeper below its window, and its rung ratios moved by only 3 and 1 per cent (3.5338 and 4.2909 against 3.4280 and 4.2353), as the window's own 9/4 had held across a fivefold shrinking (run LADDER_LIGHT 20261015_071735). A shape that survives a change of weight is the kind of law worth a chapter (the Loop Atlas sessions)

the Loop Atlas sessions (the book); ties rows 665, 695, 756, 758

Look for the machine's own stop near any wall you chase, and read a closing gap in two units, up to its last point (758) — look for the machine's own stop near any wall you are chasing: the pulsed needle's 1/4 window lives inside the scissors of memory at every weight followed, and walked down from it a running needle falls to the stop at a gap that shrinks, on a coarse walk, from 0.0033 at memory 1 to 0.0010 at 1.2725 (read finer below, 0.00323 at memory 1), slowing; every needle walked past the stop and back comes home pegged. A window can end by its own shape giving way or by the stop climbing to meet it; measure both (run STOP_WALL 20261014_180544, on Bear's "hard stop wall"). Read a closing gap in two units, in its own and in the size of the thing it closes on: from memory 1 to

1.6 the stop closes on the window in pace, 0.00323 to 0.00044, falling at every weight and slowing, while in the window's own width it recedes, 112 to 306 widths, and from about memory 1.45 the two shrink together near 300 widths; a straight line through four points had closed it at 1.48, and a floor guessed from two equal points had held it above 0.0005: both missed (run STOP_WALL_2 20261014_225838). And a trend read off a figure is a trend up to its last point; register it with the turn it might take: the stop crept toward the heavy needle's window to 300 window widths by memory 1.6, levelling in widths as Bear read it, then fell away past its walk's cap (more than 9,882 widths by memory 1.7598); a later walk placed the fall between memory 1.7198 and 1.720425, where the leak closes (run LEAK_CLOSE 20261015_073039; rows 662 and 787); the window's own width flattened, reached its least near memory 1.92 and widened, and through both turns the window kept its one-hump 9/4 (its end 2.266 times as far below its top as its first doubling at memory 2, a squeeze of about 3.5×10^{-6}) (runs STOP_WALL_3 20261015_000651 and HEAVY_WINDOW_3 20261014_232615). Across memory 1 to 3 that ratio reads 2.252 to 2.266, within about 0.7 per cent of 9/4, while the window's width changes more than twentyfold (runs HEAVY_SHAPE 20261014_155553 to HEAVY_WINDOW_4 20261015_032553); every reading sits above 9/4, as seven of the Hénon table's eleven windows do (their mean 2.2526; row 757), a lean whose test has not been run

the Loop Atlas sessions (the book), on Bear's wall; ties rows 662, 668, 739, 757, 787

A saddle has two clocks, and each keeps its log law only near the saddle (759) — time spent arriving at a saddle follows its stable rate and time spent leaving follows its unstable rate, each as the log of one over the crossing speed: the remembering needle's climbs into the barrier top came within 6 per cent of the stable rate's law, while its falls, shaken off the top by noise as large as the crossing speeds, kept only half of the unstable rate's (run SADDLE_LOG 20261013_012417). A log law near a saddle is an asymptote, so check where it is meant to hold before betting on its slope: launched without noise from the barrier top at crossing speeds 0.09 to 0.39, the fall grew with 0.79 of the slope the saddle's escape rate gives; small noise did not weaken it, and noise as large as the crossing speeds took it down to the replays' half (run SADDLE_NOISE 20261013_024850). And a textbook law near a saddle lives at speeds a noisy system rarely visits: launched without noise, the fall keeps the saddle's log law to a thousandth at crossing speeds of 10^{-4} to 10^{-3} (slope -0.6941 against -0.6949) and to 0.94 at 10^{-2} to 10^{-1}, before the 0.79 above at the replays' speeds (runs SADDLE_ASYMPTOTE 20261013_025252 and SADDLE_NOISE 20261013_024850)

the Loop Atlas sessions (the book); ties rows 640, 719

A window can be narrower than the step used to look for it: set the step from the narrowest window you mean to find, and find the edge from the window's own orbit (760) — a periodic orbit found inside chaos may sit in a window far narrower than the step used to look for it: the 1/4 groove at pace 0.98375 spans only 0.98373 to 0.98375, lost one step of 0.00001 above and doubled to period 8 one step below, the shape of a window that opens at a fold and closes by doubling. Hold the walker still first: the coarser walk alone could not tell a sliver from a broken walker (runs SETTLED_GROOVES 20261011_223349 and SETTLED_GROOVES_2 20261012_041558). A sweep finds only windows wider than its step: the remembering pulsed needle's 1/4 window is 4.8×10^{-5} wide from its fold to its doubling pile-up (a little wider to the crisis that ends it, row 756), under half a sweep step of 10^{-4}, so a sweep of 401 paces cannot land in it, and a sweep that finds no lasting spin says nothing about windows narrower than its step; its needles in fact sat within two cycles, decided by their one start, not by a spin that drained (row 664). Set the step from the narrowest window you mean to find, or go to the window's own paces (run SPIN_START 20261013_161916). Before calling a ramp into a lock smooth, read it on a grid finer than the one that showed it: seven paces read over 4,000 cycles climbed in order; on a grid ten times finer the chaos's turn rate stumbles back by a few thousandths and meets the 2/3 step in a jump of 0.036 (run CHAOS_RAMP 20261013_192906). But find the edge exactly before calling the approach a jump: that gap of 0.036, a ten-thousandth below where a sweep first found the lock, read as a jump; with the edge found from the lock's own orbit, the gap closes to 0.009 a millionth below it (run RAMP_EDGE_2 20261013_200133). And where a predicted window is not found, widen the search before calling it narrow, and follow the cycle rather than wait for the needle: eight windows of 28 and 32 cycles near a crisis lay 1.2×10^{-3} to 3×10^{-3} millionths below the arch touches that predicted them, outside a lock scan of 10^{-3}, though they are 2×10^{-4} to 1.2×10^{-3} millionths wide; Newton's method on the predicted cycle across 3×10^{-3} found all eight, and none at seven controls (the run drew one of its eight twice), and in all, 18 of the 19 touches are windows of exactly the predicted cycle (runs WINDOW_TOUCH_4 20261014_090037 and WINDOW_TOUCH_5 20261014_092920). The 3/2 tongue at mass 3.0 is such a sliver: a 0.0005 grid found none, while zoomed a hundredfold after the data it sits at drives 1.52653 (1 rider of 8 repeating as 3/2) and 1.52655 (4 of 8), 3.877×10^{-5} of drive wide between its folds (row 730). Inside it the exponents sit within 2×10^{-4} of zero, so riders on the tongue settle too slowly for a repeat test to sort them, and its width is found by following its orbit to the two folds, not by holding riders and waiting (runs RUN_STROBE_2 20261006_174220 and RUN_STROBE_2_READER 20261006_174743) (the Loop Atlas sessions). A judging window copied from an easier regime can void every lean at once: R2W's CATCH-2 judged a drive only inside a window read at strength 0.30, a band 0.08 of drive wide there but under 0.01 at strength 0.50, so on a 0.01 drive step it held 4 drives at 0.40 and none at 0.50, 0.60 or 0.70; all three leans were void, and CATCH-3 on a fine step landed. A second specimen of row 787 (R2W, sitting P2). A rung can be narrower than the paces tried for it, too: re-flown at 1.05505 and 1.06025, the needle's 28-cycle rung of 5/7 was not found, yet from eight starts at every pace from 1.0550340 to 1.0550420, 0.0000002 apart, each settles on it (the Loop Atlas sessions, HIDDEN_LOCKS of 30 October)

the Loop Atlas sessions (the book); ties rows 631, 645, 647, 664, 697, 700, 730; the Loop Atlas's checks of 30 and 31 October: HIDDEN_LOCKS_2026-10-30

Follow a narrow window by where its stable orbit exists, and call a cycle stable by every multiplier (761) — to follow a narrow window as a knob turns, bisect on where its stable orbit exists, not on the orbit or the fold equation: stepping the orbit alone jumped to another branch within 5 per cent of the knob, and solving for the fold directly was thrown by noise in the multiplier, while bisection on existence followed the 1/4 window's top through 31 steps of the needle's memory from 1 to 1.2725 and landed on the known fold at memory 1 to 5×10^{-13}. Its limit: the search step must be finer than the window's stable stretch, or it loses the window (run HEAVY_WINDOW 20261014_153422). And call a cycle stable when every multiplier is smaller than 1 in size, never "the larger one real and inside the unit interval": in a strongly squeezed two-dimensional map, where the larger multiplier passes nought the two meet and turn complex, their size the square root of the squeeze (about 4×10^{-4} at memory 1.6), and a test that asks for a real multiplier reads that sliver as no cycle; a bisection for the window's top landed 0.36 millionths too low there, a quarter of the window's stretch. Every earlier fold re-checked 10^{-9} below and above was genuine; the step had only been lucky (run HEAVY_WINDOW_3 20261014_232615) (the Loop Atlas sessions, the book). A second specimen: R2W's FX-2c engine kept only the larger multiplier by sign, so "the branch with mu1 below 1" chose a dance whose other multiplier lay below −1, at −7.8 to −69; it was reported as strongly stable until the look, and a claim of two resting dances side by side was withdrawn (row 739). Print every multiplier a verdict depends on (R2W, sitting P2)

the Loop Atlas sessions (the book); ties rows 642, 739, 751, 771

A chain of followers is sensitive along its length and steady at any one place: read its growth as an average over links, and close it into a ring to see it jam (763) — read a growth down a chain as an average over links, not as the ratio of two walkers' extremes: far down a column the stumble is a wave packet, an envelope growing by one factor a link and filled with crests; the deepest crest hands over from one to the next, so the ratio of successive largest dips saws about 6 per cent round the envelope's ratio, while its geometric mean over walkers 15 to 29 is 0.972 to 0.981 of the largest gain against the stationary-phase factor's 0.977 (run COLUMN_TAIL 20261014_145058). A chain of followers each steady alone is sensitive along its length and steady at any one place: a stumble grows down a 30-walker column by 15,904 to 140 million times, while at every walker the swing in the last 20 s of the run has died to under 10^{-10} of its own largest; the largest swing arrives later by 7 to 8 s every two or three walkers, handing over from crest to crest (run COLUMN_STEADY 20261014_164637; growth along a string of followers is what platooning calls string instability, row 754, and Ward and Wilson, 2011, sort a car-following chain's string instability into convective and absolute kinds). And a ring of followers jams with no obstacle exactly where the open line amplifies: thirty walkers closed into a ring, each following the one ahead 0.2 s late, turn a 10 cm nudge into a growing stop-and-go wave at every time gap where the open column's largest gain exceeds 1, from 1.02 (growth 151 over 530 s) to 1.59 (10^{42}), and let it die where the gain is 1 (time gap 1.5 s, 0.18); the gain is 1 at the slowest swings for every setting, so "above 1" needs a margin (run RING_JAM 20261014_224934; real cars driven round a circular track jam with no obstacle in the experiment of Sugiyama and colleagues, 2008)

the Loop Atlas sessions (the book); ties rows 754, 762

A floor has an owner: find it before reading a law under it, test a suspect by how it scales, and halve the step before giving it a mechanism (764) — find the floor before reading a law under it: a square law held at couples 10 and 3 and missed at 1, where a toll of about 10^{-5} that does not shrink with the couple stood as large as the law; read as a floor plus the law, the law's coefficient came out within half a per cent (run WADDLE_PRICE 20261011_071302, part F). A floor that does not scale like its suspect is not that suspect: a toll guessed to be a pin's give fell to a quarter at one couple and did not move at another when the pin was made twice as stiff, where a give would have halved at both (run WADDLE_FOLLOW 20261011_080808). Before giving a tiny toll a mechanism, halve the step: the waddle's floor of 10^{-5} per top-up, a hundred-thousandth of the bill, fell with the integrator's step (1.01×10^{-5}, 6.40×10^{-6}, 4.52×10^{-6} at dt 10^{-5}, 5×10^{-6}, 2.5×10^{-6}): the three quarters of it that shrank in proportion to the step were the arithmetic, and the energy books closed only to the same first-order error (3×10^{-9} at couple 1, halving with the step). A floor that does not shrink with the force but does shrink with the step belongs to the step (run WADDLE_BOOKS 20261013_063353; Richardson's extrapolation is the classical tool: Richardson and Gaunt, 1927). And when a toll is split into shares, watch the sum as you turn the knob, not one share: stiffening the pin made the spring's share fall exactly as one over the stiffness, as the guess said, while the board's share rose to take its place, so the total hardly moved (2.646×10^{-6}, 2.572×10^{-6}, 2.535×10^{-6} at stiffness 10^5, 2×10^5, 4×10^5 with the step taken out). A step error can belong to one setting alone: at 10^5 three quarters of the floor shrank with the step, at 2×10^5 and 4×10^5 none of it did, so an earlier "a quarter with twice the stiffness" was the step's error vanishing (run WADDLE_STEP 20261013_073935) (the Loop Atlas sessions). A jump has an owner too: sort the jumps by who owns them. Bear's rule, "We keep seeing behaviors similar to this… big jumps or transitions or boundary walls. Each time, it's a seam/wall/knob that needs to be understood." (15 October), opens the R2W book's §6.10 and became its organizing table. A jump in a curve is one of three things: a wall of symmetry (a test of the instrument, which must hold to the last digit), a wall of the rider's own (a result), or a wall of the ruler (a print floor or a stride, which a finer ruler moves, row 686). The question to ask of every jump is whose it is (R2W). A count free of the step can still belong to the knee, so halve the step and change the knee too: a spring bead's switch-ons at heavy loss changed by 3 per cent

when the step was halved, yet grew from 1.82 to 2.56 to 3.75 a soup cycle as its knee shrank from a fifth to a tenth to a twentieth of its running speed, while at light loss the same ear counted its own swings, 0.82 a cycle against the bead's free 0.80 (run JUMP_STORE 20261015_223918). And a knee does not free a count from the step when the noise is rough: a relaxing noise is rough at every scale finer than its own time, so the knee's bill keeps creeping as the step shrinks, longer the faster the noise, 0.5, 1.4 and 4.1 per cent over a last halving to step 0.0025 at relaxing times 1, 0.1 and 0.03; check too that the noise recipe keeps its size at every step, for the plain one was 1.22 times its aim at a step two thirds of its time (run KNEE_CREEP 20261015_232920; the Loop Atlas's misses rows 247 and 248) (the Loop Atlas sessions)

the Loop Atlas sessions (the book); ties rows 451, 686, 695, 741, 788

Probe with the tool itself, run blind, before registering: a check must touch the very number the test will use, and say whether the test can decide (766) — a check that touches a neighbour of the input lets faults through. In one night two instrument misses (runs SEAM and SEAM_2, where the chain stopped at two) got past probes that checked the thing beside the thing used: a redrawn figure matched the file pixel for pixel while the axes were read at the wrong resolution, and a stepper gave the same numbers at any array length while the pace itself was one rounding off. Over five exchanges a probe run on a neighbour of the test's input missed a fault three times, and once said a test could decide that then could not. The fix is to probe with the tool itself, run blind whenever it runs in under about twenty minutes: the planted case printed whole, the judged lines only as yes or no, the parameters and axes taken from the very files the test will use. Run that way, the blind run caught a fault before a registration in every exchange after it was adopted: a walk's cap too short and an edge that depends on how fast the pace is lowered; a grid point sitting on the line Kp = 1 (Kp the push's proportional gain, row 672); a name shadowed in code read by text. Before registering, also count whether the test can decide (its judged cases, counted without their verdicts), check that each leg can be met on the grid it will be judged on (a leg registered at grid paces outside windows narrower than the grid's step could not be), and near a bifurcation set the settle time from the multiplier; the decides-check stopped one guess from being read as a result. And when a redesign misses on its instrument, the next try goes to the next exchange with its probe first, not a third run that night. A fixed tolerance misses a slowly settling orbit: at mass 0.75, drive 1.69, two riders failed a 10^{-5} repeat test while sitting within 7.7×10^{-5} and 5.5×10^{-4} of their siblings' period-8 orbit, closing on it at about -0.006 per passing near its birth at the period-4 orbit's doubling, 1.7×10^{-5} above it, and needed some 340 and 670 more passings; after 1,500 passings all 8 riders sat on the one orbit, the farthest 8.5×10^{-8} from it. Set the settle from the contraction: the passings needed are the logarithm of the starting distance over the tolerance, divided by the contraction per passing. Or let the exponent and the nearest-orbit distance vote with the repeat test (runs RUN_STROBE 20261006_170149, its reader RUN_STROBE_READER 20261006_170950 and RUN_STROBE_2 20261006_174220) (the Loop Atlas sessions). A registered grid must reach the place it is about. Building the slice middles, the 0.32 slice held only two of the six births it was registered to ask about, so SM5 could only read NO READING; it was amended before any run. SM2's grid at 0.30 stopped at birth 7.415, below a sign change already known near 7.4295; it missed for that reason alone, and its redesign SM2′, with the grid carried past it, found the crossing at 7.4284 against 7.4276 registered (run 20261015_212604). Before registering, mark on the grid where the answer is expected to be (the triple-star sessions). The blind run stayed the best probe: in most exchanges of one later session it caught something before a registration was final, a planted value copied from another setup (row 678), a tolerance without its arithmetic, a short reading window (row 645) or a toy that would not settle at one corner (row 632); what its planted print can show too early is row 716's (the Loop Atlas sessions)

the Loop Atlas sessions (its practice log); ties rows 632, 711, 714, 731, 740, 808

Read a doubling ladder's ratios with care: bracket each rung finely enough for the number wanted, keep each refinement inside its bracket, and expect a ratio to swing round its limit (771) — a doubling cascade read by Newton's method and the orbits' multipliers shows Feigenbaum's ratio arriving from one side: 4.2365, 4.5513, 4.6458, 4.6642, the shortfall from 4.6692 shrinking 3.7 to 5.0 times a step, the cascade piling up at 0.98370525. Bracket each doubling finely enough for the ratio you want: brackets of 10^{-11} leave the fifth ratio uncertain by about 0.005 (run WINDOW_CASCADE_2 20261012_171509, started together with row 750's WINDOW_EDGE_2, so the two share one stamp). A ratio that creeps toward a universal number must be bracketed finely enough to see the last step: the window's seventh doubling, bracketed to 10^{-13}, puts the fifth ratio at 4.66801, 0.0012 short of Feigenbaum's 4.669202, and with brackets of 10^{-11} the fourth had read 4.6642 instead of 4.6640 (run WINDOW_CASCADE_3 20261013_015743, started together with row 752's REINJECT and GHOST_TRIAL_2, so the three share one stamp). The order of a map's top sets its doubling numbers: measured, 4.66920159 for a rounded top, 7.2847 for a quartic top, about 9.30 and 10.96 for flatter tops, with width ratios -2.5029079, -1.6903053, -1.4677784 and -1.3582277; a page listing a quartic-topped map among those sharing 4.669 is fenced, and the needle's 4.6692 is the rounded top's (run DELTA_BY_TOP 20261008_034013). An area-preserving map doubles by about 8.72, not 4.669: on $(x, y) \to (1 - a\,x^2 - y, x)$ the ratios run 8.3020, 8.6585, 8.7156, 8.72027, climbing from below. A tool that refines a doubling must keep its refinement inside the bracket it found, and must seed a newborn orbit from small displacements upward, or it finds another orbit of the same period (run DELTA_AREA, its third version, 20261008_062647; versions 1 and 2 retired). A width ratio can swing round its limit instead of settling from one side, so one ratio cannot be held to a bar

tighter than the swing: the needle's run 2.8231, 2.3951, 2.5491 round 2.5029, the deviations shrinking about −0.4 a step, while its doubling factor climbed steadily to 4.6691; read with the doubling factor and the whole-orbit fine structure, the needle's top is rounded (run DELTA_BY_TOP_2 20261008_063744, from CASCADE_4 and CASCADE_5's saved orbits). And a parabola refined by cutting its spacing r times a round closes r^4 times a round, its error $(2r^2 - 1)\ c_3^2\ d^4 / (4\ c_2)$ growing as the fourth power of the spacing and always inward: on the 7/5 lock's edges 81.00, 625.0 and 10,000 a round at r = 3, 5 and 10, the coefficient ratio 2.8818 against 49/17 = 2.8824 (run GROOVES_16, part B, 20261007_182010) (the Loop Atlas sessions). Set a lean on a limit at the late rungs, and lay the rungs on a log ruler. Below a chaos of the Loop Atlas's needle a ladder's rung ratios climb toward Feigenbaum's 4.669 from below, 3.43, 4.24 and 4.57, so a lean on the first ratio missed and a lean on the third held (runs HALVINGS_2 20261015_064427 and HALVINGS_3 20261015_064938; the Loop Atlas's misses row 221). Shown the ladder, Bear saw rungs, and reached for one of the few tools he returns to whenever a reading is stuck, logs and anti-logs on the integer forest for quick conversions and addresses. His reading, made exact before its tool, held: the rungs are evenly spaced on a log ruler to within a quarter (HALVINGS_2). His lens, printed beside the test, gave the rungs' log widths stepping 1.232, 1.443 and 1.519 toward ln 4.669 = 1.541 (HALVINGS_3), and on a log ruler his eye saw the near-even spacing at once. The person's lens, printed beside a registered test, costs one line and can carry the result (the Loop Atlas sessions). A lock rule that waits a fixed time is weakest at a doubling's edge, where the plain loop settles slowly: fly the edge paces longer and place the edge by the split's square-root growth. The needle's 2/3 band, read at 2,000 cycles from 1.03300 to 1.04775, runs at 8,000 from 1.03320 to 1.04750, its split squared straight against pace (R^2 0.9998)

the Loop Atlas sessions (the book); ties rows 642, 680, 694, 761

Trace a number to its run, and ask its owner, before calling it carried or wrong (772) — every number names the run that produced it, and a number that lives only in prose is unverified (the triple-star sessions, in their page for every new session). One session sometimes wrote its own defaults as the person's rulings (row 770), and twice it guessed at a number before asking its owner; twice the owner's answer overturned the guess. Row 619's 0.909 could not be traced by the session reading it, which guessed it was a number carried from elsewhere; the session that owned it traced it to its run file, and the rerun reproduced it. Ask the owner for the trace before writing a number off: this is row 678 read the other way, a number belongs to the run that made it. And an inference from a data note is not the owner's word. The chapter quotes the ripple as A cos(u − Wt); from the Explorer's note "wave reversed" and one read, the session inferred that the plus sign was right and the chapter was wrong. It held the change and asked instead. The owner's answer: both signs are right, each on its own side, and the chapter's sign is the one its source used (the Field Guide sessions)

the Field Guide sessions; the triple-star sessions; ties rows 619, 678, 770

A backward pass at a pause: a reader who did not do the work looks for orphaned threads, its claims are checked before acting, and each item says whether it was verified (780) — at an overnight pause the person asked for one: "Scan the transcript for orphaned threads and surprises that might have merit or are something truly unevaluated." A separate reader went back over everything it could be handed: the session as recorded (11 to 13 October, recovered in full) and its working notes, the paper's draft among them. Its main claims were checked against the files themselves before any was acted on. Each item says what it is, whether it was verified, and where it goes now, and nothing is a result unless it says it was run. The pass says too what it could not read: the Field Guide reference, the pattern-recognition and cross-domain pages, the status pages of two later sets of runs, and 16 series files it was never given; its counts from the atlas used the lock flag, not the held flag, and the items it found that no one checked are marked "no". In one evening it fixed a census that said "every" (row 194); a run count of 486, not 729, because the census tool counted a table's second block; a surprise that was not new, twins that had parted still leaving together, already tested on 12 October and borne out; a "no door" that was probably nothing, its pair at a mean eccentricity of 0.04 to 0.08 where a set at 0.05 also let no star go, kept at low merit; and "exactly the slope", now "within 6% at the two clean edges, over the ladder as a whole", with the step-by-step slopes added: 5 to 11% short far out, a quarter steep nearest the edge, and 1.76 times beside a chaotic layer (run 20261013_082012), straight out of the Loop Atlas session's two warnings (the triple-star sessions). Check each worry of a backward pass against the exact sentence it touches: of the triple-star thread's, every reading stood or was already said except two sentences that said more than they measured, which were narrowed, and one bound was found not to reach the sentence it was meant to cover at all. A second specimen: every run file of R2W's 3 and 4 October was present, yet early results (two stations, the ring conveyor, the toothed 4/3) and 24 cards lived only in the working register, and the book named none of them; a fork audit against the book found them, and each was told, run or handed on. Nothing missing from the crate is not nothing lost (R2W, sitting P2). A list of what is owed is a claim too: before trusting a closeout's list, read every status the project itself keeps. The Loop Atlas's closeout list named the science owed from its handovers, while its own card table, read by script, held 23 cards not done, 13 of them runnable in the container and none of the thirteen on the list, filed under a status no one was reading. At a closeout's start, sweep the card table, the ledger's open rows, every "still owes" and every misses row whose redesign is owed, and give each item a disposition (the Loop Atlas sessions, closeout of sessions H and I)

the triple-star sessions (chest of 13 October); ties rows 191, 194, 592, 666, 797

A block rule is cut by a single thread, and a width is read in the quantity the system keeps (781) — at the pair's eccentricity 0.25 one scan's births 7.390 to 7.415 are held and locked, with the same bands as the island above them, and only 7.420 is not held. The reading rule takes the unbroken held block, so the island read 0.135 and the paper said it "halves"; the "collar of regular births" was really island, and the only "free star below" behind a 0.0065 apse figure was 7.420 itself. A test registered before any run checked births 0.001 apart: 8 of the 9 births 7.416 to 7.424 are held and regular, and 7.420 alone is chaotic, a thread narrower than 0.002 whose runs both escaped just after 600 passes, so the thread leads out. Over the scan's 600 passes the island at 0.25 is 0.170 wide (7.390 to 7.555), narrower by about two fifths, not half (over 6,000 passes it splits into rooms, row 732); read across its one-birth gaps, a rocking rule recounted from +15% agreed within 1.1%, a reading made after the fact and now registered forward on two later sets of runs. Another thin thread, at the heavy birth 6.960, had twins that parted only at pass 3,068 inside the calm zone; it pairs with 7.420, and the four twin runs of 6,000 passes that would test it were not run before the thread closed on 29 October. And a width stated in the scan's own label can drift from the quantity the system keeps: widths and centres stated in birth drift from the period the star actually keeps by 0.13 to 0.44, so edges and centres are read in the kept period, from the saved tables. Part of an island's shift of about 0.085 between two eccentricities (its middle near 7.165 at 0.12, against its place at 0.20) may be that drift (row 690 measured it: the birth landing on an island's centre moved from 7.155 to 7.445 while the kept period held between 7.010 and 7.022), and a third swing registered and never built may be mostly the same label effect (the triple-star sessions). A prediction is written in the quantity the physics fixes, as a width is: the Loop Atlas's sling has a best cut that is an angle levelling off near 39 degrees, so "between half and three quarters of the way up" had to fail for big swings, while put as an angle (25 to 45 degrees) it held from 40 to 90. And a broad optimum is a builder's window: from an 80-degree swing any cut from 29 to 47 degrees throws within 5 per cent of the best (run SLING_RUNDOWN 20261017_232234) (the Loop Atlas sessions)

the triple-star sessions (chest of 13 October); ties rows 631, 648, 690, 703, 732, 797

Set a bar from the record's own numbers: at the method's floor, with room above the largest difference seen, and on the median of a heavy-tailed scatter (784) — a bet that the person's laptop would match the container's record to the 12 printed decimals at eight folds missed on three drives, by 1.0×10^{-12}, 2.0×10^{-12} and 2.8×10^{-11}, while the grooves and every fold's step count matched. The record already showed the fold refiner's floor, its final residuals up to 9.7×10^{-10} and its distances from an independent follower's values 2.1×10^{-11} to 4.9×10^{-10}, so 12 decimals asked the arithmetic for more than it holds at a fold. Register a cross-machine match at the method's own floor, read off its record before the run (a second specimen of row 337), and look for the difference where it lives, in the step counts; a tighter refinement afterwards (run FOLD_TIGHT 20261011_051636) showed the 10^{-9} bar's own error at these folds reaching 7.2×10^{-11}, which holds the laptop's 2.8×10^{-11} inside it (runs FOLD_OR_FLIP_4 20261007_061947, the record, and 20261010_172149, the laptop's). The next bar, that every fold's drive moves by less than 5×10^{-11} when its start is nudged by one or two units in the last place, came from that same largest difference with too little room: the largest move was 5.36×10^{-11}, at the 3/2 lower fold with $k = -2$, the other 31 smaller, at the fold whose refinement had stopped closest to its 10^{-9} bar (last residual 9.7×10^{-10}), so the drive's spread may be set by where the refiner stops (run ULP_NUDGE 20261011_044600). A bar on the largest value of a scatter whose largest runs a thousand times its median fails while the reading holds: bars of 10^{-11} on the largest distance missed (1.88×10^{-9} extended, 7.17×10^{-11} folded) while the median fell from 3.82×10^{-9} in double precision to 4.94×10^{-13} extended and 9.95×10^{-14} folded, needle 30 scattering exactly as the other twelve do, so the 13 are one orbit; the stretch was 1.09×10^{5} (over 17 drive periods), not the guessed million-fold (10^{6}), recurring at phases a multiple of 19 drive periods apart, and the stretch itself, registered as a test of its own, was read the same morning: the orbit grows in the drive periods it spends over the top (rows 730 and 755) (run ORBIT_919_4 20261013_040039) (the Loop Atlas sessions). A bar finer than the record's printed digits, or finer than two machines agree, cannot be met. In the triple-star thread KN0 stopped triple_friction_knob v1.0 (run 20261016_201600) at a worst difference of 5.0×10^{-9} against a bar of 10^{-9}, while the record's TABLE prints delta_k to 8 decimals; F3's first bar, 10^{-9} against a six-decimal print, had the same fault and was caught before its tool ran. KD0 (run 20261017_011423) stopped at 2.15×10^{-6} against 2×10^{-6}: it had been written with only the strict tier, for a run of record integrated on Bear's laptop, when the paper itself reports 1.7×10^{-6} across machines over 600 passes. Each time the bar was at fault, not the bench. Set a reproduction bar at half a unit of the record's last printed digit, and for a run from another machine use two tiers, 2×10^{-6} strict and 10^{-3} loose with a note, and NO READING beyond that (row 810) (the triple-star sessions). Set a fidelity bar from the precision the series was saved in, and amend it in writing before the run: a zoom-out saved in single precision keeps about 6×10^{-5} of a turn at 833 turns, so its bar went from 10^{-6} to 10^{-4} turn with the reason written, and passed (3.1×10^{-5} turn). And a test whose fidelity check failed is void, whatever its own statement reads: one test printed BORNE OUT with its check failed, and the ledger counts it as could not be read; make a check compare what survives the change (rates, invariants), not phases near a pole that magnifies any shift (the triple-star thread, F3 and DR2)

the Loop Atlas sessions (its record of misses); ties rows 210, 337, 749, 755, 794

What the integrator's step owns: a thin lock's place, a gauge that moves with it, a foot's order, and a switch that looks every step (785) — halving the step moved the 8/5 groove's edges by 6.9×10^{-9} of drive, some seventy times a guessed 10^{-10}, while its width changed by 1.6×10^{-7} of itself: the width belongs to the rider, and the step moves where a thin lock sits by about 2.8×10^{11} times its width (run GROOVES_10 version 2, 20261007_065508, part C; its registration first named the wrong step count, 487, the doubling's, and was corrected before the run). A gauge that changes with the step measures the integrator: a wall's share of time came out 0.072 where more than 0.3 was predicted, because it counted the steps on which the clip acted and so fell with the step; read rightly, the end warns only within about two wobble widths. Beside it, a kick recovery ran 3 to 12 per cent slow near the fold, because a kick toward the lip meets a softer slope and a short settle had left the needle short of its seat: kick both ways and average, and settle many times 1/cos(lag) first (run SEAM_MAP_B 20261005_023746). An error at a mountain's foot was counted as third order and missed: a foot smooth only to the second derivative gives second order with scatter, and smooth feet give clean fourth order; a control was aimed at an error already at round-off, and a single step count landed near a cancellation. Aim a test where an error is still visible, and read scatter by envelopes over many counts, never one (runs CONE_TOMOGRAPHY_3 20261008_011150, _011149 and _011157, CONE_TOMOGRAPHY_4 20261008_012043 and _012042, and CONE_TOMOGRAPHY_5 20261008_012723). A walk round the person's ring of steep peaks crosses fourteen feet, and the integrator stumbles at every one: smooth feet took the seven shares' spread from 9.76×10^{-7} to 3×10^{-13} (runs PEAK_RING 20261008_040518 and 20261008_040854). A switch fed fresh noise each step chatters, its bill doubling each time the step halves; a knee about three times the noise brings the cell back onto the map at about a sixth of the clean ear's drift, though with noise drawn fresh every step the knee still leans slightly on the step, and with slow noise it does not (runs EAR_PRICE_CHECK, EAR_KNEE, EAR_CURES_B). Of the person's three cures for a chattering switch, a pause state cured nothing, and a pause after each flip capped the flips but still leaned on the step. Pause the looking, not the acting: a switch that looks only every so often is free of the step exactly, and with a small knee its flips land on Rice's rate (runs EAR_CURES 20261004_045151 and 20261004_045721) (the Loop Atlas sessions). The step owns a stand-in's stability too, and a stand-in at a cheaper setting can cross a stability line the real run never meets: RATCHET-2's dry run used a coarse step (1,000 a passing) at the lightest weight, 0.0025, which put the phase step over the weight times the frequency at 4.57, and the fixed-step integration went unstable; the real runs never meet that line. Guard the engine: it now prints STOPPED whenever the ratio passes 2.5. Before trusting a stand-in, check that it lives inside the engine's stable range (row 731) (R2W). A count of switches must be checked against the step before it is read as anything: a spring bead's blade switched about five times a unit of time at every pitch of a beat, fifteen times its own ring, and nearly twice as often at half the step, chatter where the bead's speed sits at zero; counted as whole swings of the bead itself, the beat's real effect showed (runs PILOT_BEAT 20261015_203855 and BEAT_SWING 20261015_204207; the Loop Atlas's misses rows 233 and 234), as EAR_CURES above had shown for a relay. A planted step test and a band test on every count became a standing practice at one of the Loop Atlas's reviews, and the band test caught a real misreading for the first time on the truck road (row 645) (the Loop Atlas sessions)

the Loop Atlas sessions (its record of misses); ties rows 604, 627, 644, 764

A reader's window decides what it can see: a majority rule, a low barrier, a window copied from a shorter run, a gap rule of fixed size, a yardstick that holds what it measures (787) — a climb-back reader that counted a pace as spinning only when 8 of 16 needles spun missed pockets of 6 and 3 that kept spinning below the edge, and the pockets were a 1:2 gear beside the 1:1 lock at one push, a remembering needle holding either by where it came from (runs POCKET_HOLD 20261005_195333 and POCKET_GEARS 20261006_052012). A state is meaningful only once its barrier stands about three noise units high: a needle guessed to spend under a tenth of its time past the barrier top at pace 0.95 spent 0.117, because there the barrier stands only 1.06 noise units high and "past the top" counted lingering; swept over pace, it matches the tilted washboard's exact density within 0.0052 in every twelfth of the turn (runs KNEE_RECOUNT_D 20261004_195102 and _E 20261005_003928). Size a window to the run it serves: a crest window (x from −400 to 40) copied from a 400-unit run into a 600-unit one missed one soup of eight, which carried beads at 0.98 of its crest line, where beads slide farther and meet crests the window never held; and a falling count (42, then 22), read after the data as the left comb knocking riders off, was not the only cause, since the one-way soup with no left comb fell too. A count that falls as the drive rises can come from a crest just short of holding, which delays the riders past the counting window (run SOUP_JUMP_2 20261009_051426, against SOUP_JUMP 20261009_045400). A gap rule of fixed size is blind to gaps just opening: counting a gap only at 0.003 rad or more, it read 3, 3, 8, 3, 4 and 4 pieces beside a window of 48 at +4.8 millionths above 0.98370, where 4 pieces are born from 2 near +4.36; a saved density with holes of 0.0007 rad still shows a single gap at +4.4, so at +4.65 to +4.85 one inner gap is under the rule's 0.003 and by +4.90 both are over it, while at +4.75 the readings bunch round the window's ghost and leave extra empty runs. The $48 = 4 \times 4 \times 3$ rests on +4.90 and +4.95 and on the merges' arithmetic, and a piece rule that scales with the gap's growth is the redesign, still unrun on 30 October (misses row 204) (run WINDOW_TOUCH_3 20261014_080520). Measure a spread against a yardstick that excludes what it measures: a first avalanche run counted noise as slipping, and a registered spread of waiting times missed because its yardstick included the runs themselves; beside it, a needle with memory 5 at pace 0.75 under weak noise (0.2) slips 99.8 per cent of the time at a pace it holds for ever when undisturbed (run AVALANCHE 20261004_064914; 064551 retired). And read lifetimes from needles placed straight onto the pace: a lifetime ratio could not be read because every needle

had already locked on the walk down to its pace (a follow-on registered, still open on 30 October, misses row 35) (the Loop Atlas sessions). A majority rule read once a step reads a draining as a jump: the stop's edge, the step at which half of sixteen needles had stopped, said the leak closed between two weights 0.000625 apart, while the count the same run kept beside it showed 9, 7, 2 and then 0 needles falling, always at the same place. Keep and read the count under every threshold verdict (run LEAK_CLOSE 20261015_073039). And a counter-wind made of waves knocks riders off at a place, not at a rate: 64 riders set on one crest at different times all fell where they met the counter-running pattern's worst stretch, their falls spread exactly by their start times (196.9), and a margin that held for 500 time units broke by 1,000, so size the margin for the worst stretch over the whole ride (run SOUP_JUMP_3 20261016_225440) (the Loop Atlas sessions)

the Loop Atlas sessions (its record of misses); ties rows 616, 627, 703, 753, 756, 758, 760

Compare on one budget, keep a law inside its range and its stage, count what wears, eliminate a suspect by its share and its scaling, and read the step ratio first (788) — two rings merged into one travelled faster than two rings leapfrogging, which cut against the leap-through idea and was reported once and dropped. Reversed later: the merged ring had carried 1.41 to 1.73 times the pair's energy for the same push, and given exactly the pair's budget one ring travels at 0.84 to 0.88 of the pair's speed, so on one budget the leapfrogging pair is 14 to 19 per cent faster (run RING_BUDGET 20261004_080107). Compare on one budget before calling a winner. A law used outside its range gives an impossible value: a letter used the junction's retrapping law, 4/(π times the square root of memory), at memory 1, where it gives 1.273, above the fall line; it is the large-memory limit, and at memory 1 the needle's own climb-back line is 0.9643 (corrected from R2W's record). A rule found on one machine is a law of its stage: R2W's room rule, slope −1, did not carry to the needle relay, slopes −0.48 to −0.60; the needles add −1/2 off the beat, steepening toward −1 only where the downstream's own restoring rate outruns the beat (run ROOM_NEEDLE_2). Count the work, not the hours: in a fleet that wears by the hour, a calendar counted in running turns still threw away less life than one counted in hours (0.473 against 0.533), because a machine asked to work harder meets its cliff at a stronger grip and running turns count the work, which is how light aircraft engines are already logged, off tach time. The learned reminder's waste in the two fleets missed agreeing within 10 per cent, twice and narrowly (17 and 14.9 per cent), and a gauge that knows the room throws away about a tenth of a life where the calendar throws away about half (runs MAINTENANCE_METER 20261005_012150 and _B 012355). Eliminate a suspect by its measured share and its scaling: a guess that a rigid floor at couple 1 (2.5352×10^{-6} at stiffness 4×10^{5} and step 2.5×10^{-6}) was the catch keeping a little less than half the spin, so that the top-up must pay a little more than 1/8, failed. The catch keeps 0.499999704 of the spin, its shortfall in the floor's own units (toll per top-up) 4.94×10^{-8}, only 0.0195 of the floor and going as one over the stiffness (1.97×10^{-7}, 9.87×10^{-8} and 4.94×10^{-8} at 10^{5}, 2×10^{5} and 4×10^{5}), so the floor is not the step, not the spring and not the catch; most of it is the board's share (2.2413×10^{-6}, the share row 764 saw rise as the spring's fell), and it stays a workshop question (run WADDLE_CATCH 20261013_082013). And read the step ratio first: a bet that order 8's doubling constant, extrapolated from its last three rungs at rung 12, would lie above 10.94875, outside the rounding of the reference's 10.9486, because the extrapolated values had been rising (10.947846, 10.948648, 10.948788), missed at 10.9486659. The rise was a second correction changing sign every rung, which the step ratio already showed zigzagging (0.4158, 0.4406, 0.4119); extrapolated twice, the value settles at 10.94862, on the reference. The project's own lesson of the day before was not applied to its own guess (run DELTA_TAIL_2 20261011_051600) (the Loop Atlas sessions). A law built from parts carries its parts, not its exponent: at frequency 0.55 the flat point grew as the drive to the power 1.475, set by a drain that changed slowly and a feed that fell fast. FLAT-75 carried that fall to frequency 4 and missed: there the drain is under 1% of the sum, and the feed sets the flat point alone. FLAT-50 then carried a single power from two swings to a third and missed again, since the feed's own power drifts, from −3.7226 to −3.8430. The small dial, which measures both parts at the setting in use and assumes no law between settings, held blind all three times. Carry the parts and recompute; an exponent fitted to the blend is good only where the blend is (R2W)

the Loop Atlas sessions (its record of misses); ties rows 640, 678, 680, 746, 764, 870

A run statistic rests on a thin tail, a ramp reads where it crosses, and a transient can look like a second state (789) — a remembering needle's run statistics moved with where a slip was timed, and five registered predictions about them missed (KB2, KB2b, KB2c, B1, B2); a binning reading of the spread was wrong, and one registered test was found ill-posed by derivation before it ran. A run ends only at a long gap, and breaks are rare, about 200 to 250 of 15,000 gaps, so a run statistic is decided by a thin tail. The barrier top logs false starts: a climb and its own fall are tied (+0.590) while a fall and the next climb are nearly strangers (+0.068), so a landing-timed gap holds both halves of one slow turn. Time a slip at the bottom of the turn or the landing, and print a run statistic with its break line and the tail it rests on (runs KNEE_RECOUNT_B 20261004_190250, _C 192352, _D 195102 and _E 20261005_003928). Read a line from methods that stand still: an avalanche's climb-back lines were slow-ramp readings, 0.0169 and 0.0153 low, and slowing the ramp shrank the gap but not by the steady factor per halving predicted, while two still methods agree within 2.1×10^{-7} on the lines themselves (0.7898648 at weight 2 and 0.5413103 at weight 5, row 711). One needle's ramp reading depends on where in its turn the ramp crosses the line, so an ensemble ramp is the redesign. A tapping rider's memory time near its cliff dropped 16 to 23 per cent against a guess of within 10, and read finer the drop was smaller and differed from rest to rest; the memory gauge can jump by

a whole ring of its autocorrelation. Both are parked, since neither changes what the book says (runs SEAM_MAP_B 20261005_023746 against AVALANCHE 20261004_064914). And a transient can look like a second state: needles slipping just below a lock's edge looked like one waiting beside the lock (registered guesses H1a and H1b missed), and held long, every one sat down. A needle does not slow before it sits; the time per slip stretched only because the crowd thinned (run HIDDEN_SCISSORS_SLIVER 20261005_145137) (the Loop Atlas sessions). A statistic can depend on its own definition. Where a needle lingers near its lock for one to three field cycles between bursts, the line between one run and two is the counter's choice, not the needle's: joining runs at 1.5 against 3 field cycles moved the median run over the mean by up to 0.234, against a bar of 0.1, at three of seven rows (two-fold 0.15 at 0.85 and 0.90 of its grip, many ledges at 0.85). Judge only the rows that pass the band themselves, and read runs by the needle's own speed with a knee, which needs no joining gap (ROUGH_FLOOR 20261017_013514 and its blind run 20261017_011647) (the Loop Atlas sessions)

the Loop Atlas sessions (its record of misses); ties rows 627, 662, 665

Follow the orbit itself, and read the doubling's square root and critical point from it (791) — row 642's named gap has a working tool: LOOP_ATLAS_STROBE_NEWTON finds periodic orbits of a once-per-passing map by Newton's method, with the Jacobian from the variational equations, and Liouville's determinant matches exp(−T/m) to 7 digits. It follows them in drive by pseudo-arclength and refines folds by solving for a multiplier of +1: gear 2's fold came out at 1.390631237 and 1.176820507, inside the brackets that holding riders had found, and halving the step moves either by at most 10^{-9}. It sized locks 4×10^{-5} to 7×10^{-5} of drive wide that a 0.0005 grid could not see. Its first version is retired for a tool fault in one clause, and the second reproduces every shared number. One limit is stated: following an orbit of period 8 at mass 0.75 downward failed, because it takes the drive's effect by a difference over eight sensitive passings. Just below the light needle's doubling, the period-16 strands are 1.662 times the square root of the distance apart, to 0.06 per cent from 1.8×10^{-5} to 3.2×10^{-4} below (slope 0.5002), even on the unstable orbit past the next doubling; the two strands sit on opposite sides of the old orbit but unequally (25 per cent at 1.8×10^{-5}, 51 per cent at 7.8×10^{-5}), their midpoint drifting in step with the distance. The halves are equal only in the limit: a ring of emitters fed half a turn per step splits into exactly equal halves, while a bent road, as here, splits equally only at the edge itself. The fifth doubling counted from the period-8 dance (period 128 to 256) lands at 1.6893097435, 1.3×10^{-10} from its prediction, and the band ratios run 4.6155, 4.6570, 4.6667, 4.6687 toward 4.6692. Counted from each superstable orbit's critical point, found as the passing that stretches least (1.29×10^{-6} against at least 8.2×10^{-5} elsewhere), the shrink at the critical point is 2.823 then 2.395, straddling alpha (2.5029); at its image it is alpha squared, and the alpha-squared block runs half the orbit. The strands' square-root coefficient falls each generation by the local shrink over the square root of delta, within 0.3 and 1.6 per cent at a test point (runs CASCADE_3 20261006_220635 and CASCADE_4 20261006_221804) (the Loop Atlas sessions)

the Loop Atlas sessions (its notes for the Field Guide, from 6 October); ties rows 642, 680

Three habits that misread: a reading made after the data, a number read from printed digits, and a run named from its header (793) — some readings made after the data were wrong: one tool's far sign changes were read twice, wrongly, as the loop rising higher; a version's death on the 6/5 groove was first read as the loop folding back in angle; an alternation of the clock shifts about 4.669 was offered on numbers inside their own floor and withdrawn. Some were right: a fold in the speed-only solve, borne out by a later run, the 30-digit 2 pi read as a floor, borne out by another, and the smoke-ring trains' swings in pairs, 137 of 137 on fresh trains (row 716). Whether the wrong outnumber the right was registered as a test and never counted. A reading after the data is labelled as one in every file, and gets its own registered test before anything rests on it (GROOVES_11, _13, _16 and _17). Numbers were misread from printed digits in conversation: "the same 8/5 edge to 3×10^{-30}" for edges 2.7×10^{-30} to 2.77×10^{-29} apart; "short by two" for three, then four; a 10 per cent crossing "near 0.6" for 0.72 to 0.74; and a surprise test that missed that 3/2 of a turn has a whole-number cosine. Each was caught and corrected the same day; a column number written from memory, 142.4, for the printed 142.26 was corrected with a dated note. A number drawn from printed digits is computed by script before it is said. And a survey on the person's laptop was named as R2W's pad atlas from its header line, until the next exchange found the pad atlas was another run; a laptop slowdown of 5.0 to 6.6 times was first read as the laptop's speed, while three sessions' runs shared it on the night of 6 October, one on 12 cores, so it measured the crowd. A run that is not yours is named from its own files or not named, and a speed read on a shared machine is the crowd's (the Loop Atlas sessions, reviews A and B). Read a fit's numbers by evaluating the fit at a point you know, never by their place in a printed list: numpy.polyfit prints a parabola's numbers highest power first. The record's note on R27-c's redesign (16 October) took the last number of the fit at frequency 0.70 (−0.687037, its constant term) for its curvature and called the bow there seven times as deep as at 0.55 (−0.093385, rightly read). A check written to correct it read the 0.55 fit the other way round (−0.437267, its constant term, as the curvature) and put the carried edge at 1.1730; that version of the file was never recorded and was overwritten. Evaluating settled it: highest power first, the 0.55 fit is −0.00020 at its edge 1.3056 (−0.24250 read the other way), and the 0.70 fit is −0.11545 and +0.17134 at 0.99 and 1.875, where the data are −0.11549 and +0.17135 (R27_CARRY_CHECK). The 0.70 curvature is −0.135102, 1.447 times the 0.55 one, and pinned through 0.99 and 1.875 it puts the 0.70 edge at 1.2750, beside RATCHET-28's 1.2746. A dated CORRECTION line went into the record; the R27-c verdict, a miss graded before, stands (rows 735 and 843) (R2W). A prediction written to a file must carry the digits its bar will be

read at: TIMINGS-2's plan wrote its predicted leaks to ten figures. At two timings the leak is -1.014382×10^{-2}, so the file's last figure sat near 10^{-11}, and the read reported the leak off by 3.41×10^{-12} from a prediction that, carried at full precision, it missed by only -7.2414×10^{-14}. The verdict was safe (its bar was 1.01×10^{-4}), but the number went into the book, the plan and a reply, and it surfaced only because NINTH-1's plan recomputed the prediction and printed its gap from the file. Write predictions at full precision, and have each plan print the gap between what it computed and what it stored (R2W). Read every number in a reply from the run's json by script: adopted at one of the Loop Atlas's reviews, the practice caught loose phrases before they shipped in at least three exchanges (the Loop Atlas sessions)

the Loop Atlas sessions; ties rows 369, 611, 677, 678, 716, 776, 843

Set a bar with a number in hand, before counting, only where the run can feed it, and at a named crossover (794) — a bar set without a number in hand fails: climbs over two periods were guessed at 1 per cent or more of all climbs and were 0.46 per cent (69 of 14,969), while the picture held (a fast pass at 0.15 to 0.20 periods, 30 climbs of 0.5 to 2 periods against 69 over 2); the catch rate, about one climb in 217, is the number to carry. A bar set after counting can admit an error: a touch bar of 10^{-4} let in a place at -5.3100 that lies inside the 12-cycle window, its refined minimum 2.1×10^{-5} rad the largest of the twenty refined minima (two of them, at -4.9465, are one place, so there are 19 places), in a bet that at 0.8 or more of 19 places where an arch touches the band's low edge the needle locks, and at 0.2 or less of 19 controls. Ten of 19 locked, exactly at their predicted cycles, and 0 of 19 controls: right of -6 millionths 8 of 9 places lock, left of it 2 of 10, and the one at -9.3026 locks only at the farthest offset, 3×10^{-4} below the touch, so the touch's place there is off by about that much, windows of 28 and 32 cycles nearer the crisis may be narrower still, or there may be no window. The redesign scans the lock rule across 10^{-3} either side of every place that did not lock, in steps of 2×10^{-5}, with the controls scanned alike, and sets aside touches inside a window before the count (run WINDOW_TOUCH_3 20261014_080520; the redesign's outcome is in row 760). A test the run cannot feed is undecided: a bet judged on at least 100 spinning turns at each memory got 12 from the forgetting needle, which at pace 0.95 seldom slips twice running; without memory a landing is a renewal point whose tie is zero by theory, so the redesign reads memory 0 on all turns, against zero within its standard error, and memories 1, 1.5, 2, 3 and 5 on spinning turns (run SPIN_OR_SIT 20261011_055807). And a reading that names no crossover number cannot be rescued by one run: one said a room slope should flatten toward $-1/2$ away from its watershed, and on the edge reached by the gain it stayed at -0.985, so it was weakened and left as registered (R2W's run _roombottom_20261005_164542) (the Loop Atlas sessions). The four registrations of row 369 that could not be judged as written, in numbers: one set a bar of 10^{-13} against a value printed to 12 decimals; another named the wrong step count (487, the doubling's, for a groove that runs at 341; caught before the run); two parts of a third could not fail, because an arrow started down a paper's ridges stays in their plane and reads exactly zero; and a surprise test contradicted itself (the Loop Atlas sessions). A blind prediction can come from the same run, in two stages: RATCHET-63 needed the flat point at a second drive predicted from that drive's own small dial, which had not been run. The registration fixed the stage-two script before any run; the small dial ran first; the script turned it into a number, and that number went into the record before the flat point's own runs started. The dry run did the same on each stand-in's small dial. The flat point then landed 0.16% from the number, blind. A prediction that depends on data is still blind if the rule that turns data into the number is fixed first and the number is written down before the data that tests it. The sitting's review counted the small dials among what carried its wins: a few runs near a point, read in closed form, placed the dark and flat points blind at every drive tried (RATCHET-52 to 64) (R2W)

the Loop Atlas sessions (its record of misses); ties rows 369, 580, 784

A queue for the paper gives each item why it matters, its cheapest test and whether that was checked (797) — one backward pass's queue for the paper (row 780) set out each item that way. A run that runs out of time is not an escape: at 600 passes the queue counted only 8 of the 12 births below escaped, two ran out of time, one circulated and one held, so every long run's fate is reported by class, and a set of runs ten times longer should settle most of them. The paper ran them: below island 7's held edge at 7.390, at the pair's eccentricity 0.25, eleven of the twelve stars left over 6,000 passes, after 46 to 1,186 passes, and the twelfth, 7.375, which had read as held at the end of the scan's 600 passes, was never held again in nine stretches (Held, Rocking, Locked, Appendix C). Whether the door is just the known stability boundary toward the pair, or Mardling's overlap, its escapes sitting at 1.063 to 1.071 of Vynatheya and colleagues' margin, sets what is new and what is credited; its cheapest test reads the margins at escaping and bound chaotic births at 0.25, beside a reading of the stability-boundary papers; on 18 October it was still unchecked. A sentence saying Georgakarakos's formulas "have not been checked against it" goes against the person's rule that whatever can be run is run before deposit. The check was then run, on 18 October: his 2003 formula held at about half the period ratio he tested, the pair's short-period swing 1.115 to 1.117 of his prediction (GK1, run 20261018_193951; row 783). And some threads were never followed up or never registered: a level that climbs 1.529 times from eccentricity 0.20 to 0.30 where the quadrupole gives 1.113; whether the pair's trade simply tracks the swing, so that "ragged where they trade" (row 678) becomes "ragged where the swing is largest", a rank correlation of seconds from saved series; two door tests in the heavy scans, both on the low side (broken runs, and breathing), parked, each seconds from saved tables; where the heavy islands end, the last held birth having the island's smallest least mean eccentricity in all four; and the edge law's overshoot, the slip slowing more than a pendulum's nearest the edge and most beside a chaotic layer (1.76), where the Loop Atlas needle's noise ran it shallow, its test finer ladders

near a layered and a clean edge, where the excess should track the layer's width (the triple-star sessions). A proofreader's margin note is a queue item too, and its intuition a dial waiting to be built: Bear's "this sounds like the noise floor" became a pool-size dial, and "comfy in its own neighborhood but couldn't function in a new neighborhood" a five-step distance dial. Each found where the intuition holds (the counting limit is a noise floor; the stamp's deficit grew at every step of distance) and where it does not (the survival cliff is mostly not a noise floor; the stamp never fell below its start). The intuition is tested, not adopted (runs E39, E40) (Question the Substrate)

the triple-star sessions (chest of 13 October); ties rows 552, 675, 678, 780, 781, 783

Straight lines lost all day: between two known ends a straight line is a guess, not a law; register the bend as the lean and the straight line as its rival (799) — Bear's note for this guide, 16 October: "Straight lines lost all day. Three times today a straight line or a guess with no law behind it missed. Each redesign that let the data's own shape pick the law then held". The three misses: R18-c, a guess with no law (RATCHET-18), that the fourth harmonic's window would lose its dip pair by frequency 0.50; it was still there, its depth 0.12471, 0.05984 and 0.0085 at 0.40, 0.45 and 0.50. R23-b, the drift's zero past 45 degrees at weight 2.29324, an argument about a limit carried along a road it does not speak of (row 737); +12.20 was measured. R24-b, the second harmonic's edge at drive ratio 1.3956 on a straight line through 0.99 and 1.875; it lay near 1.3078. The three redesigns that held: RATCHET-21 set two laws against each other before the run, a straight line in the depth (vanishing near 0.5083) and a fold's law, the depth to the power 2/3 on a straight line (near 0.5187), as lean and rival; the fold held (a pair at 0.51 and 0.52, none at 0.53; near 0.5225 on the fine rungs), and its forecast from coarser rungs fell early by about 0.004, the one-sided bias of the highest sampled point (row 416) again. RATCHET-23's rival, the drift following the step, held. RATCHET-25's parabola through five ratios put the edge at 1.3065; it measured 1.3056. Three of three redesigns built that way held. When a guess with no law behind it misses, let the miss choose the law: register two or three candidate laws side by side, before the run. Between two ends a straight line is the least-informed guess; when a third point bends it, register the bend as the lean and keep the straight line as its rival, never the lean. The other side of R2W's lesson 120, which is this row: five careful points can foretell a smooth curve, so register the many-point fit as the lean and the few-point fit as its rival. The exact quartic through five sizes predicted all eight rungs of a ladder within 0.0191, both rungs beyond its range included, while the three-point parabola missed by up to 5.67 (RATCHET-46); a curve fixed by enough points, with its shape left to the data, wins. And a straight line through two bracketing points misplaces a crossing on a bending curve: RATCHET-55 found the flat point by a straight line in the logs between octaves 0.10 and 0.16. The ratio of the two arrows bends upward on that scale, so the chord reaches 1 early, and the reading came out 0.6% low; that was said before RATCHET-63 and held (R63-e2). When a crossing matters, read it directly with a few points around it, or interpolate in the variable in which the curve is straightest (here the octave squared). In the same sitting's review most misses had one of three causes: a straight line or a carried exponent stood where the thing bends or blends (the flat point by a line between two rungs; the flat point's share carried from one drive to another, FLAT-75, and then from two swings to a third, FLAT-50, row 788); a term thought negligible was not, at a zero (row 400); or the file, not the physics, set the number (row 738). The misses were cheap, because every one was registered with a rival and read against a redesign check (R2W). A second specimen from R2W's light cone (sitting P1): a step's τ went from +25,840 to −144, and later from +344 to −85, between neighbouring weights, and the first reader drew a straight line across. τ is continuous, so a zero lay between; bisected, it was a window of ringing only 7×10^{-5} wide, one of five crossings (WEIGHT-CONE, WEIGHT_CONE_C2). Read a sign change between samples of a smooth quantity as a crossing to bisect, never as a line to draw

R2W; Bear's words; ties rows 649, 661, 697, 706, 739, 870

Flip the sign to find the two hands: a result read at one sign may be two parts cancelling (801) — any knob with a sign can be read twice, once at each sign, and the two readings split an effect into a part that flips with the sign (odd) and a part that does not (even). A null, or an effect that seems to have been swallowed, read at one sign only may be the two parts nearly cancelling at the setting that happened to be run, so any result read at one sign only is worth a second look. Its classical name is the reversal technique: a Hall measurement reverses the magnetic field to separate the Hall voltage, which flips with it, from the magnetoresistance, which does not and grows as the field squared. On R2W's rectifier, RATCHET-41 had read "the mix keeps the octave's c" as the octave swallowing a fourth-harmonic third tone. Running each measured size again with its sign flipped (RATCHET-44) split the effect into an odd part (+15.06 at size 0.38) and an even part (−14.98) that nearly cancel at exactly the size that had been run; Bear's picture, half a dipole or a reflected image, named the method. Where the other side of a symmetry cannot be run, pairs of settings test it: the reflection of section 7.11 of R2W's book maps the weight to minus itself and mirrors every timing. Negative weight cannot be run, but a pair of timings (D3, −D3) can: their half-sum must hold only even powers of the weight and their half-difference only odd ones. Off the mirror line a slope in the weight entered the window (b/w 0.22 at a quarter turn), and in every pair the slopes cancelled and the half-sum started flat to within 0.1% (RATCHET-45, 47). A symmetry that flips a knob predicts what appears off its mirror line, and the pairs read it there (R2W). A second example, found the same day in another project: the Loop Atlas's tied bells' water neither helps nor hurts near a memory of 6 rings' times at every squeeze speed, while the squeeze kicks there run 7.6 per cent above plain water's (a digit of the toy as first built, row 807), so the flat is two pushes cancelling. Print the pieces beside the total before reading a flat as nothing (run BELL_RATIO 20261016_183422; the question was put to R2W on 16

October) (the Loop Atlas sessions). The check became a standing practice: adopted at one of the Loop Atlas's five-exchange reviews, the two-pushes check on every flat named the bells' second push, the reopening, the first time it was used (the Loop Atlas sessions). A reader carried off the mirror line meets that slope. BEAT-4's reader took the light coefficient from one weight, a = (H(0.0025)/H(0) − 1)/0.0025², as EDGE-85's did, and a quarter turn off the mirror line it printed ±19.08. Off that line the reflection allows a term in the weight itself, and one point folds a slope s into a as s/0.0025. Solved through the two weights the run held, the slope was ±0.0477 and a near zero. EDGE-85's reading was safe only because it ran on the mirror line, where the slope is forbidden; a reader carried to a new setting must ask whether the symmetry that made it safe still holds there, and two points separate the orders that one point mixes (R2W). When a gap changes width, read each edge alone: in R2W's WINDOW-WEIGHT the dance through lag 0 lets go within 0.00008 over a tenfold change of weight, while the point where the dance through lag 1/2 takes hold walks as the weight's square, so the window between them passes from a gap of 0.00077 to an overlap, its sign changing between weights 0.025 and 0.03; one edge belongs to first order and the other to the weight's second (R2W, sitting P3)

R2W; ties rows 575, 712, 734, 805

When the knob has no sign, tell a true nothing from two hands cancelling: sweep the size, move another knob, turn the timing half a turn, weigh each hand alone (805) — row 801 splits two hands by flipping a knob's sign, but a count, a duration or a size that cannot go negative has nothing to flip. Four tests, cheapest first, still tell a true nothing from a balance. First, sweep the size. A nothing that a symmetry makes is nothing at every size and sits at the arithmetic floor: eighteen even fans of riders came to at most 6.994×10^{-15} against one rider's 2.599775×10^{-2} (ARRAY-1), and three half-turn pairs to at most 2.098×10^{-14} at every offset (NINE-1). Two hands cancel at one setting only, and the output comes back in proportion to the distance from it, changing sign across it: the three groups of three (row 803) go dark at 40 degrees in a straight-sided V, 2.209828×10^{-6} per degree (LOBE-1). Second, move a knob you did not move. A balance moves when a second knob changes one hand more than the other, while a symmetry's nothing stays put: the flat point's share of the first tone (the drive's own tone) is 0.3177 at the edge's drive and 0.2771 at three quarters of it (RATCHET-63), and at fast drive 0.315176, 0.246495 and 0.169789 at full, three-quarter and half swing (RATCHET-64, FLAT-75, FLAT-50). Third, if the knob has a timing, use the half-turn as its sign: turning a tone upside down is the same as turning its timing half a turn, so firing the push half a period later splits the hand that flips from the hand that does not, exactly as a sign flip would. Fourth, measure each hand alone: the flat point is two arrows, a drain and a feed, each read from a small dial (row 794), and their balance was then placed blind within 0.04% and 0.006% (FLAT-75, FLAT-50), the small dial holding in the same runs whose carried exponents missed (row 788). Noise separates a true nothing from any cancellation, but not a symmetry's cancellation from an accidental one: a cancelled part still adds its own jitter (5.8858×10^{-2} per rider aligned and 5.8857×10^{-2} interleaved, in NINE-1's reading), so an output that is quiet on average yet still noisy has hands behind it. This is R2W's answer to the question the Loop Atlas put to it, recorded in row 801 (R2W). The sweep is necessary, not sufficient, when both hands share a factor of the knob. The Loop Atlas ran the four tests on its crest-firer, and the sweep came out flat within 1.5 per cent at every level from 0 to 2 strengths, which alone would file it as a symmetry's nothing. Measuring each hand alone said otherwise: the pushes came 7.3 times less often, lasted a third as long and pushed 2.9 times as hard, their product unmoved, each push scattering by 0.53 to 0.67 of its mean where a true nothing scatters 0; a knob not moved (the soup's wave count, a square-law skew) slid it, and a half-turn as the sign split it. The two hands, the crossing rate and the push carried above the level, both carry the same Gaussian factor of the level (Rice's count), so they stay in step at every level. When two hands share a factor of the knob being swept, the sweep cannot separate them; the decisive tests are each hand alone and a knob that only one hand feels (R2W, on the Loop Atlas's run)

R2W; ties rows 669, 788, 794, 801, 803

Read a knob's return on its map, not by proportion: a part of what it turns pays less as it grows, a sweet spot moves another knob's best, and two knobs that only trade are a frontier (806) — three ways a knob's return misled a lean, each found by mapping. A part that belongs to the whole it acts on does not pay in proportion to its size: halving the arms' mass to 3.5 kg was leaned to cut one arm circle's turn and the fixed share to 0.45 to 0.60 of their values at 7 kg, and they fell only to 0.6160 and 0.6087, because lighter arms leave a lighter body, so each kilogram counts for more; one circle's turn runs 9.51, 15.44 and 19.56 degrees at 3.5, 7 and 10.5 kg, more mass and less return on each kilogram, with the arm's length the stronger knob (run ARM_SHARE 20261016_182910). A knob pictured as helping without limit can have a sweet spot of its own that moves another knob's best: a fatter core was leaned to strengthen a leapfrogging pair of rings at a gap of 1.0 at every step, a0 $0.02 < 0.05 < 0.1 < 0.2$, and the pair's advantage over one ring on its budget ran 1.179, 1.191, 1.195 and 1.178, falling again at 0.2, where the best moves out to 1.188 at a gap of 1.25. The two knobs trade, the pair's best about 1.195 near a core of 0.1 and a gap of 1.0 to 1.25, and a core of 0.2 is near the edge of what the thin-ring formulas can be trusted with (run RING_KNOBS 20261015_165858). And two knobs can slide a design only along one line: a slope fitted to the last four looks, with a look every sixteenth of a cycle, kept its drift (0.10622) but switched 1.927 times a cycle against the plain ear's 1.650, halving the extra switching, from 0.567 to 0.277, rather than removing it. Mapped over the number of looks it spans and the knee its forecast passes through, no setting kept the drift within 5 per cent of the two-look slope's with

switching at most 1.815 a cycle; the nearest was 0.1035 with 1.858. The two knobs slide the ear along one line of headway against switching, the plain ear sits on its efficient end, the map is the frontier itself, and only a filter of another kind (a knee on the slope itself, or an estimator that weighs each look by its noise) can leave the line (runs FILTER_SLOPE 20261015_195645 and FIT_KNOBS 20261015_204319). Map both knobs before naming either's direction, and say which line the design is sliding along (the Loop Atlas sessions). And single changes need not add up to the whole they make together. A swimmer whose three links could each be given a broad face was leaned to get at least half of the extra travel that making every link broad buys from its two broad ends alone; they bought 0.405 of it, because the broad faces help each other: all three broad buy 1.79 times the sum of the three single faces' extras, so two faces buy less than their share of the whole (run FACE_ALONG 20261017_070629). Measure the whole beside its parts before giving a part its share; row 151 has limbs that do add, nearly (the Loop Atlas sessions). A rival that holds can be the engineer's rule: in four of the misses R2W's sitting P3 wrote into its ledger the rival held and its answer was itself a setting to use, a carrier's among them: under fifty times the tone it moves the swap by up to 0.002, from fifty up by under 0.0005. When a rival holds, ask what setting it hands the engineer before filing the lean as a miss (R2W, sitting P3)

the Loop Atlas sessions (its record of misses); ties rows 151, 590, 684, 769

Before reading two answers as two states, make every scale in the toy agree with every other (807) — a toy that will not settle may be doing something no real machine does. At the Loop Atlas's tied bells' crowded corner, eight starts a thousandth of a bell radius apart split between swimming forward and being driven back, and the share moved with the step: 2 and 6 of 8 forward at memory 8.49, 3 and 6 at memory 12. The toy gave each ring a core of 0.15 for its own speed, but a softening of only 10^{-3} where two rings meet. Given the cores' own size there too, a softening of 0.0225, the square of 0.15, every start swam forward, 0.181 to 0.184 a squeeze at memory 12 with a spread of 0.001 (run BELL_SWITCH 20261020_161902). The two answers came from the toy's thin cores, not from two states of the machine. That the rings pass almost through one another and throw each other off is the project's reading, not a measurement. The advice to read a corner that will not settle in distribution, over many nearby starts, still stands, and this specimen shows why: the spread across starts is what exposed the thin cores. The same correction withdrew a builder's rule that fast squeezes in a long-memory water make the stroke unreliable. Those squeezes cost stroke, keeping about 0.41 to 0.61 of the plain stroke, and do not drive the bells back. That a thin-core vortex toy needs regularising where vortices meet is known in kind; the size of the effect here is the project's own. An earlier reading of the same corner, that the toy was choosing rather than converging (four steps from 0.005 to 0.000625, run BELL_CORNER 20261016_224449), stood in this version's drafts until this correction (the Loop Atlas sessions). Settled against the step is not settled against the toy. On the bells' map 20 of 25 points passed the step test, half the step moving them by at most 1 per cent, yet given the cores' own size where the rings meet, 13 of those 20 moved by more than 0.05 in their gain, by up to a third (one point from 0.6754 to 1.0088), and the line where the rings stop costing moved from squeezes 2.72 to 3.23 rings' times apart to 2.02 to 2.94 (run BELL_SOFT_MAP 20261021_000736). What held through the change was the shape: the timing swings the gain about 2.4 times as much as the memory, the longer memory deepens whatever the timing does, slow squeezes pay and fast ones cost. Test every scale a toy has, not only its step, before trusting its numbers, and trust its shapes sooner than its digits. The guide's other digits from the tied bells' toy were measured with the toy as first built (rows 404, 630 and 801, and the earlier runs in rows 643 and 9b, whose BELL_SOFT_MAP digits are already re-read); by this rule their shapes are the part to trust until they are re-read (the Loop Atlas sessions, G's item 441). The correction that made this row slipped once in its own units: the replacement text the Loop Atlas sent for its item 346, now its item 440, put the cores' own size, a length, in place of the toy's softening where two rings meet, which is a length multiplied by itself; the row above sets the two softenings side by side on one scale. Say which scale a constant lives on, a length or a length multiplied by itself, before setting two side by side (the Loop Atlas sessions)

the Loop Atlas sessions (session G's correction of 21 October to its item 346, now its item 440); ties rows 632, 643, 9b

A check across machines compares only what can repeat: held and regular runs, birth by birth, saved file against saved file, with a loose tier for the other machine (810) — two computers running the same orbit do not do exactly the same arithmetic, so a chaotic run on one parts from the same run on the other, however good both are. A check that two machines agree must pick what can agree and set the two sides out at the same precision, or it measures the chaos or the file format instead of the tool. Compare held runs, and anchor on a regular orbit. In the triple-star thread KM0 (triple_k_starts v1.0, run 20261017_033754) stopped its tool: the 21 re-runs gave back the atlas's phi_N only within 0.932 turn, because the check compared every run, held or not. KM0′ compared held runs only and passed (17 held re-runs, 2.10×10^{-5}; run 20261017_035730), and v1.1 writes each run's mean K to a TABLE as it goes, so that a stop does not lose the runs. CS1's planned fidelity birth, probe 34's 7.180, is chaotic, so its escape pass does not carry across machines (pass 45 in the workspace against 43 on the laptop); it was replaced by the regular 7.400, held to probe 34's 0.020651 within 2×10^{-6} (strict) or 10^{-3} (loose). SL7's anchor (run 20261019_012918) set 7.340 against SW1's laptop run at 2×10^{-6} with no looser tier, and the machines part from pass 37 (3.7×10^{-7} at pass 600, 2.2×10^{-5} at pass 6,000), so the whole test read NO READING; SL8 anchored to runs in the same workspace, with the loose tier of 10^{-3} registered, and passed at 8.32×10^{-4}. Decide which runs are regular before choosing what a fidelity check compares. Read a claim of chaos birth by birth, by the paper's own test.

The paper had called the seventh's core slowly chaotic because two machines parted on 7.121 after pass 5,400, which is not the test its A.4 registers. LX3's check across machines over all its births together (LF8, run 20261018_173405) failed at 3.95×10^{-2} because the one chaotic birth sank it, so L7 and L11 read NO READING. The redesign, said aloud: a check across machines is made birth by birth and stops where the birth's own chaos parts it. TW7 (run 20261018_192644) did that with twins 10^{-10} apart on one machine: 7.121's twins part at pass 5,515, and over 12,000 passes the chaotic pairs are exactly 7.121, 7.122 and 7.123, the births that lost their lock (7.120 joined them over 30,000, row 79), which turned slowly chaotic from a parting across machines into a reading. Set saved file against saved file at the same precision. On 20 October the zoom-out's F2 line (e_bin within 3.7×10^{-9}, φ_N within 3.8×10^{-6} of probe 35, the laptop's run) was about to be quoted as the laptop's difference from the container, but F2 had set double-precision values against a series saved in single precision, so the two figures are half a single-precision step at the values' sizes. triple_machine_look v1.0 (run 20261020_001840) compared saved file with saved file: at 7.200 every one of 1,803 values is the same, and 12,559 of 12,621 are the same in all. Within one software stack the runs repeat to the bit (U0 and U1 HOLD, run 20261020_001032); between the laptop and the containers they part from about the last bit (rows 732 and 784) (the triple-star sessions). Verdicts carry across machines; last digits do not. The triple-star thread's long runs were made again from nothing on Bear's laptop and read by the same tool: of 103 lines, 83 were the same, 20 gave the same verdict with other digits, none differed; the energy's drift agreed to four digits at two tolerances and to two at the tightest, and the twins' gaps differed by up to a sixth while their ratios and growth law matched (XC1). Register cross-machine checks on verdicts and ratios, and say which checks are tied to one machine's digits

the triple-star sessions (lessons of 22 October); ties rows 632, 678, 732, 738, 784

Run a planted case at two step counts before setting its bar: what shrinks with the step is the step's, what stays is the arithmetic's, and the case must be one that finishes (813) — a planted case is a check whose right answer is known in advance, such as a swimmer with no grip that must not move at all. Its bar has to sit above the error of the tool that reads it, and the cheapest way to find that error is to run the case twice, at a step and at half of it, and watch what changes. An exact zero is judged by its size against the step rule's own error and by its shrinking with the step: the Loop Atlas's FACE_ALONG planted a no-grip swimmer whose travel is exactly zero, and it printed $+1.798 \times 10^{-6}$ at 300 steps of a two-slope rule and $+4.628 \times 10^{-7}$ at 600, the rule's own error falling four times as the steps doubled, so a tolerance of 10^{-9} missed for the wrong reason (blind run 20261017_070529; 20261017_070629). A lesson written down is not applied unless the tool's pattern carries it: in the next batch DIVE_CUE's planted tolerances were again written by hand below the floor, 10^{-6} degrees against a measured 10^{-5} (runs 20261017_082235 and 20261017_082454). So the halving probe went into the tool pattern: before any planted tolerance is written, the planted case runs at two step counts and the tolerance is set from the change. A floor that does not move with the step is not the step's: the probe found one routine 2.77×10^{-9} degrees from the engine's own dive at both 3,000 and 1,500 steps, because two ways of writing the same shape change, TUCK + u(LAY − TUCK) and LAY + (1 − u)(TUCK − LAY), differ in the last bit, and rates taken by central differences at 10^{-6} amplify that a millionfold; the probe still set the tolerance, ten times the floor, before the run of record, and its two step counts told what kind of floor it was (EXPLORER_ROUTINE_CHECK, runs 20261017_164121 and 20261017_164432). A turn that is exactly linear in the spin is linear in the engine only to the same kind of floor, about 10^{-8} rad at a spin of 120, the same at any step (TONG_DULLIN, runs 20261017_173508 to 20261017_173718). A step rule keeps its order only where the path is smooth: a routine of moves blended by half-cosines has a corner in its rate at every move's edge, and a fourth-order step taken straight across them left the landing 10^{-6} to 2×10^{-5} degrees off where the edges missed the step grid and 10^{-12} where they met it, so each smooth stretch is flown as its own run of steps, shared out by duration (EXPLORER_CAT_ROOM_CHECK, blind 20261017_211142 and 20261017_211314). And a planted step test sits where the case finishes: HALT_THRESHOLD's step test sat at a threshold where the column never starts, so two blind runs had nothing to compare, and the two-runs rule stopped the chain (blind runs 20261017_174151 and 20261017_174755). Part of that was the instrument (row 816); with it fixed, the rule at 2 m still brings no arrival by 3,600 s, while at 3 m the tail arrives at 1796.783 s (run HALT_THRESHOLD 20261017_184055). Choose a planted setting known to complete, and print why each case ended (the Loop Atlas sessions). A planted case must be able to fail, too. Several of R2W's early readers were dry-run on the run's own old files, which shows that a reader reads, not that it can pass or fail; from MEM-6 on, each verdict was planted once to hold and once to fail, and after one set of plants hid a reader's miscount (it printed "75 of 60"), the plants carried every column the real table holds. A second specimen of row 731 (R2W, sitting P2)

the Loop Atlas sessions (Guide items of sessions F and G); ties rows 18, 210, 754, 764, 785, 816; R2W's check of sitting P2: PK-0, whose reader printed 75 of 60 before a recount found 60 of 60 (its method lesson 233, from its record PREREG_RELAY_STATION_2026-10-03)

A guard computed by a round trip calls hits from rounding: compare where the guard is computed, diagnose a planted case alone, and rerun every old tool as it stands before the fix (816) — a guard is a line of code that stops something impossible, such as one walker passing through the man ahead. A guard that fires by mistake crashes nothing; it quietly changes the motion, so its fault can pass for a finding about the world. The Loop Atlas's toy column kept each walker at least half a metre behind the man ahead with a test written as a round trip, (x

+ off) − off < x, and rounding alone called a hit about one walker-step in 80, each hit setting the walker's speed to his man ahead's, so that in a standing column nobody behind could close up. Compare where the guard is computed (the running minimum below its own value), never after a round trip. It was found because a planted case that should have finished never did, and that case was diagnosed alone before any lean was touched (HALT_THRESHOLD, version 3; row 813). An audit that then reruns old tools reruns each as it stands first: all five column tools, copied with what they read and run unchanged, repeated their records line for line, so every line that moved with the fix was the fix's own; and a planted repeat that then missed was missing against a record the same fault had made, and held against the corrected one (run LIMIT_AUDIT 20261017_190938). The verdicts the fix moved are in rows 754 and 762 (the Loop Atlas sessions). A small residual does not keep a solution on its branch. Newton put R2W's lower edge onto the upper edge at weight 0.9216 with a residual of 4.5×10^{-13}, in two of R83-d's phases, and a ruler that shared none of those phases could not see it; with a guard that keeps each edge on its side (R83-e), both voided verdicts landed. Guard the side as well as the residual (R2W, sitting P2)

the Loop Atlas sessions (Guide items of sessions F and G); ties rows 36, 650, 682, 754, 762

A lifetime read in a finite window is a lower bound: say "held for T", throw an edge many times, fly the side that held for longer, and test a fit by its whole procedure (819) — when something can come apart by chance, like a train of smoke rings flying in line, a run that ends with it still whole has not shown that it lasts; it has shown that it lasted as long as the run. The run's end is a floor under the lifetime, and the honest words are "held for T". An edge read inside a window may only be where the lifetimes begin to outlive the window: the Loop Atlas's train had a cliff at a spacing of 0.65 to 0.70 (row 724's run TRAIN_HOLD) that was where the trains began to outlive its 1,000 time units, and its 0.65 broke at 1010.7 (run TRAIN_LIFETIME 20261018_034211). So before an edge is called a cliff, the starts near it are flown for several windows. A floor found inside a window is a floor of the window until flown longer: thrown five times a gap, nudged, the three-ring train broke 16 of 30 times from 0.530 to 0.555 radii and never in 30 from 0.500 to 0.525, a split that no trend would give twice in a million (run TRAIN_FLOOR 20261020_224239), where one throw a gap had put the edge at 0.57. Every train below that floor held 20,000 time units, which bounds its lifetime from below and says nothing above; flown on to 100,000 on Bear's laptop, half of ten copies at 0.520 and 0.525 broke, a lifetime near 150,000, and there was no floor (run TRAIN_FLOOR_2 20261021_083613). That lifetimes cut off by the end of a run are censored is known in kind; the size here is the project's own. Those breaks are first 5-radius spreads, the stand-in of row 865, so a lifetime read that way at 0.530 and below is the short side of a parting for good: three of ten trains that spread at 0.530 were whole again at 100,000 (the Loop Atlas's misses row 480). The floor also bounds row 724's engineering answer: a train packed at about half a radius sits 0.03 of a radius below 0.530, where trains break within 20,000 time units, and is held for a long time, not for ever. What ends a lifetime there has an anatomy. A ring that swings out comes back in equal pairs, 137 pairs in 48 trains with none left over, and a ring that leaves for good leaves at the start of a new pair, 26 of 26; seen once and not yet tested, a ring rests a shorter time between the two trips of a pair than between pairs (medians 43.7 and 60.1 time units). The rings that left are a second road, not a farther swing: they reached 50 radii in a median 1,400 time units, where one pulled home would take about 19,863 by the far returns' law. And a fit to lifetimes is tested by its whole procedure: a goodness-of-fit p computed against a curve fitted from the same data flatters the fit (Lilliefors 1969), so the p is taken by running the whole procedure, fit included, on many samples drawn from the fitted law, and waiting times are read band by band before they are pooled across a knob, since a mix of lifetimes passes for one only on the kind test. The train's exponential passed at p 0.126 as registered and failed at 0.028 fairly, its times mixing about 860 time units below a spacing of 0.70 with 130 to 220 above (run TRAIN_LIFETIME 20261018_034211 and its reader 20261018_035727) (the Loop Atlas sessions)

the Loop Atlas sessions (Guide items of sessions F and G); ties rows 416, 716, 724, 807, 822, 865

A chaotic event repeats exactly on the machine that flew it and nowhere else: plant on another machine only what chaos cannot scramble (822) — in a chaotic run the smallest difference grows until it decides the outcome, and two computers differ in the last digit of their arithmetic even when every file is the same. So the time at which a single chaotic event happens is a fact about one machine, while its statistics, and the quantities the physics keeps, are facts about the toy. A long run sent to a second computer broke the Loop Atlas's smoke-ring trains at 4,494, 241 and 272 time units where the same tool, reading the same files md5 for md5, breaks them on the session's own machine at 1,022, 1,011 and 303; a nudge of 10^{-15} of a radius, the size of the arithmetic's last digit, moved the session's breaks by 28 to 84 per cent (runs TRAIN_CLOSE 20261018_103739 and TRAIN_NUDGE 20261018_192138). On the machine that first flew it the run repeats exactly: Bear's laptop broke the unnudged three-ring train at the gap 0.570 at 1119.4829502389848 time units twice, a day apart, in two tools that import the same flight (run TRAIN_EDGE 20261018_181641). So a planted check may ask a chaotic event to repeat on the same machine with the same libraries, where it tests the import path, and never on another; across machines it plants what chaos cannot scramble, the energy and impulse a flight keeps, held to 10^{-6} on both machines, or the statistics of many throws. Treat any single event time in a chaotic run as one throw, and the scatter between machines as their arithmetic, not run-to-run noise (the Loop Atlas sessions). Write a fidelity bar in the quantity the leans use, at the engine's measured noise. R2W's check that a container repeat a laptop's table digit for digit was void, because 16 of 100 paces differ in the sixth decimal, and a bar of 0.00001 held on a fresh grid (FAUX-C, FAUX-Cb); a check that every column match failed on solver bookkeeping while every edge matched (R15-0). A second specimen of row 810 (R2W, sitting P2)

the Loop Atlas sessions (Guide items of sessions F and G); ties rows 632, 678, 724, 819

A small excess over a known law needs an error bar, more than one reader and a finer reference before it is named, and the law is tested at its limit (824) — when a measurement sits a little above a known law, there are three suspects: the law, the reader that turned raw counts into the number, and the noise. Each is ruled out by its own test before the excess gets an owner, and a law that holds only in a limit is tested in that limit. The Loop Atlas's specimen is a switch that looks at a noisy signal only now and then, whose extra knee the traders' law predicts (row 629). Put an error bar on an offset before naming its owner, and do not count a finer ruler's repeat as a confirmation: the switch's extra knee ran 20 and 18 per cent over the law at its two smallest shifts, repeated on a reading grid ten times finer, and was read as the law's own; leaving out each of the 256 noise paths in turn put it at 1.2 standard errors, and against a reference watcher four times finer, made by the noise's own bridge between the old looks, every step lay within 1.7 standard errors of the law. The finer grid read the same noise, so it repeated the same scatter. Read a small shift with more than one reader, and do not trust a jackknife on a reader that is not smooth: the knee was read by straight lines between the steps of a curve made of whole flips, and leaving one path out at a time gave twice the bootstrap's error bar, while three readers of the same counts moved the smallest shifts' excess by 10 to 20 per cent, as much as the error bar; only against the finer reference did every reader agree with the law, within 1.6 of the bootstrap's standard errors. Do not tell a probe's number as a finding before its run of record: an exploratory straight-line reader put the excess at nothing, and that number was told in the chat, while the registered quadratic reader, run minutes later, kept it at about two standard errors against the old, coarser reference. A limit law is tested at its limit: the traders' constant, read back out of the flips pooled over every step, came out 0.902 against 0.5826, because it climbs with the step measured in the noise's memory and a fit through the origin weighs the coarsest steps most; read point by point and drawn to the step's vanishing, it came back at 0.57947 ± 0.00254 (runs ZETA_FIT 20261020_055322 and ZETA_LIMIT 20261020_060256). Read a law's constant as a function of its small parameter, say which shape the extrapolation takes before the run, and print another shape beside it, since the limit can rest on the shape. The constant's growth is itself a builder's knob, with its regime: the dead-band correction is the traders' 0.58 only while the switch looks many times within the noise's memory, and grows to about 0.72 at a look every four tenths of the memory, 0.96 at one memory and 1.2 at two, read back from the flips with four soups' standard errors of 0.005 or less at those points (run ZETA_LIMIT). Broadie, Glasserman and Kou (1997) give the limit; its growth with the step against an exponentially remembering noise was not found by title (the Loop Atlas sessions)

the Loop Atlas sessions (Guide items of sessions F and G); ties rows 79, 526, 558, 629, 737

Cut a record into waiting times so that none is dropped and none counted twice: keep the interval that runs past the window's end, and time first arrivals at each new level when a counter can run backward (827) — a waiting time is the gap between two events, and the way a reader cuts a record into gaps can bias their spread before any physics is read. Two cuts went wrong on the same noisy needle, each in a known way. A pool of complete intervals inside a fixed window can read their spread low, because it leaves out each sequence's last interval, which runs past the end and is likelier to be a long one: on the needle's slips over 30,000 time units the pool read 0.0020 ± 0.0011, 0.0021 ± 0.0006 and 0.0004 ± 0.0005 below a reader that uses an interval if it starts before the end less 12 mean waits, whatever its length, and that reader read the exact value in all three rows (run SLIP_RHYTHM_3 20261020_090717). On synthetic renewal needles the same pool showed no lean (0.9125 against the reader's 0.9123). Use the reader that drops nothing. This is the inspection paradox of renewal theory, known in kind; its size here is the project's own, clear in one row of three, and the size a model expects, shrinking as one over the window, is a check in the scratchpad, not a run. When a counter's events can run backward, time the first arrival at each new level, not each forward count: a needle that slipped back 7,253 times at noise 1.5 read a squared spread of waits of 0.49300 ± 0.00105 between forward counts, 15.9 errors under the exact long-run value, because a forward count after a backward slip re-crosses ground already counted; timed from one new highest turn to the next it read 0.50779 ± 0.00112 against the exact 0.50967 (runs SLIP_BAND 20261020_161039 and SLIP_BAND_2 20261020_161539). The renewal identity behind that reader, long-run spread equal to the squared spread of the time to the next new level, is Reimann and co-authors' equations 4 and 5 (row 826); that first passages to new levels renew is standard for a Markov process, and the size of the error is the project's own (the Loop Atlas sessions)

the Loop Atlas sessions (Guide items of sessions F and G); ties rows 627, 629, 826

Read a toy's setting at more than three values before calling its curve a hill, and look for a spike where two of the toy's own scales agree (829) — three points can draw any hill, so a curve read at three settings cannot say whether it has a peak, a plateau or a spike. And a toy built with two lengths of its own can resonate where they happen to match, which is a fact about the toy, not about the water it imitates. The Loop Atlas's tied bells' gain at a fast squeeze, read at three meeting widths (row 807), looked like a peak at the cores' own width (0.75, 1.01, 0.87); read at eleven it is a plateau between 0.68 and 0.97 with one spike, 1.01, at that width alone, and a second point has a spike of its own at another width (1.43 at 0.20) (run BELL_WIDTH_LADDER 20261021_151833). Read at five widths a hundredth apart, each spike is a narrow fin, rising gently and falling steeply, its far side settled in the step but wandering in time (run BELL_SPIKES 20261022_053740). Read a setting at more than three values before calling its curve a hill, and where two of a toy's scales agree, look for a spike that belongs to the agreement, not to the water. That a model can resonate where two of its own scales coincide is known in kind; the size here is the project's own

(the Loop Atlas sessions). The cores' mended width (row 807) is itself a setting: given the rings a meeting width of 0.10, 0.15 and 0.22, the gain where squeezes come fast moved by up to 0.41 between neighbouring widths (memory 6 at period 1.5, 0.39 to 0.80) and not in one direction (0.75, 1.01 and 0.87 at memory 4.24 and period 2.12), while where they come slowly at long memory it held within 0.05 (1.32, 1.30 and 1.28); at the widest width every point settled against the step, the crowded corner among them (run BELL_SOFT_SCALE 20261021_054731). Turn every width a toy carries, both ways, and trust its numbers only where they stand still. That a regularising length is a model setting is known in kind; the size here is the project's own (the Loop Atlas sessions)

the Loop Atlas sessions (Guide items of sessions F and G); ties rows 584, 684, 807

Smoke-test a published theorem over its whole range before planting it as a check, with its sign conventions written down (831) — a theorem from a book is a strong check only if the tool and the book mean the same thing by every sign. A published theorem planted as a check is first run over its whole range in a smoke test, with its sign conventions written down, above all where a force changes side. The course theorem of sailing (the apparent wind's angle to the course is the sail's drag angle plus the water's) missed by 0.69 radian at 150 degrees in the Loop Atlas's smoke test, where a sail trimmed for speed stalls and its side force falls to windward; read with the water's angle measured from the course's leeward side, as the theorem's own drawing has it, it held to 4×10^{-15} radian at every heading of nine hulls. Amend the bar aloud before the blind run, never after the run of record (runs SAIL 20261018_202451 and SAIL_HULL 20261018_202650) (the Loop Atlas sessions). Printed in the first boat's run and read only later: the ride's price peaks at 120 degrees off the wind, 0.39790 newtons a watt, where the keel's leeway passes through zero, 0.004 degrees (run SAIL 20261018_202451); whether the best price sits at the leeway's zero for every one of the nine hulls is a proposed test, not yet run

the Loop Atlas sessions (Guide items of sessions F and G); ties rows 346, 471, 650, 711

A value picked because it was the most extreme of many is partly the pick: read it again on fresh data before building on it (835) — the largest or smallest of many readings is partly chance's pick, so on new data it usually falls back toward the rest (regression to the mean, after Galton, 1886). A surprise chosen because it was the extreme of a set is a candidate, and it is read again on fresh samples before anything rests on it. A noisy needle's spread over mean, its Fano factor, read 0.835 in one row of 18, the lowest of them (run SLIP_SPREAD), and it was picked out for that. A lean was built on it: that at one mean interval, every needle having started together from its seat, the factor lies more than two errors below the exact long-run value in each row. It missed, at +0.84, −0.16 and +1.44 errors from it. On new seeds and four times the needles the same row at the same time read 0.9244 ± 0.0132, within an error of the exact 0.913. The spread below chance is real, about 9 per cent at 3 noise units; the extra dip at the start was the sample's (run SLIP_RHYTHM_2 20261020_084919). Count how many readings an extreme was picked from, and read it again on fresh data before telling it as a surprise (the Loop Atlas sessions)

the Loop Atlas sessions (its record of misses); ties rows 9b, 527, 616

A proof's convenience is not a limit on its result: test the instant the proof was built from before predicting that the result fails (836) — a proof is often built on the easiest case, and that choice says nothing about where the result stops holding. The mirror between two falling cats had been proved on a steady circuit, and the lean that it would break on paths whose bend swells and shrinks as it swings round missed: on nine such paths, one bend swinging 20 degrees either side of 90 and one swinging at twice the circuit's rate among them, the two turns agreed within 2.842×10^{-13} degrees. The mirror lives in the rates at every shape, not in the circuit: the twist-free cat's rate at bend 180 − b, its bend rate reversed, is the toy's at b turned half a turn about the front tube's axis, within 1.9×10^{-15} at 2,000 random states in a read after the run, and within 9.3×10^{-15} at three sizes in the run of record that followed (CAT_MIRROR_RATES 20261020_041152; row 820), so any path pairs (run CAT_MIRROR_PATHS 20261020_034021). Before predicting that a result fails where its proof's assumption fails, test the instant the proof was built from (the Loop Atlas sessions)

the Loop Atlas sessions (its record of misses); ties rows 214, 820

A bar is checked against its reader's grain and its number's error bar before it is registered (837) — a tolerance is a promise about how closely a reading can repeat, and a reading made of whole counts, or of a small number with a wide scatter, cannot keep a promise tighter than its own grain or error. A reader of counts has a floor of its own, its grain. A planted check asked a knee grid of 0.001 to repeat a finer grid's (0.0005) readings within 5×10^{-5} at every step from 0.00125 to 0.04, a tolerance scaled from the curve's bend, since a coarse grid's error falls as the square of the grid. It missed, by up to 3.65×10^{-4} at 0.04 and 2.3×10^{-4} at 0.005: the flips are whole numbers, about twenty over the four soups to a knee step of 0.001, so the curve is a staircase, and its grain, not its bend, sets the reader's error. No verdict turned on it, since 3.65×10^{-4} is a fifth to a third of the standard errors there, but a reader of counts states its grain beside its tolerance (runs KNEE_REF 20261019_042520 and KNEE_FINE 20261019_040109). A bar written in per cent of a small number is checked against that number's error bar before it is registered. A long shot that a sampled switch's extra knee size would lie within 5 per cent of the traders' law at every step from 0.000625 to 0.04 failed at one step of seven, 1.1053 at 0.0025, where the standard error was 6 per cent of the law's shift: the bar was narrower than the scatter, and the miss was 1.67 standard errors (run KNEE_REF

20261019_042520). Two cards later the same 5 per cent was registered again, for steps from 0.0003125 to 0.04, and missed again, at 1.081 and 1.054 of the law at two of eight steps, the lesson met again in that card's own registering (run KNEE_READER 20261019_163948) (the Loop Atlas sessions). A second specimen of row 793's printed digits, from R2W's sitting P2: a registration copied a drive printed as 0.8549 where the row's was 0.85489, and a reader keyed on four-decimal tones read a width of 0.002143 for 0.002107; both were read again from the files (FX2c-F, WALK-2). And size the run and the sample against the bar before registering: a multiplier of −1.0014 cannot grow a nudge past the bar in 400 passings, and the test held at 2,000 (FX2d-c); a bar of ±0.3 sat under 32 seeds' own noise of about ±0.35, and held on 2,048 seeds (FLOAT, FLOAT-b) (R2W)

the Loop Atlas sessions (its record of misses); ties rows 210, 784, 824

A straight law ends where the thing it describes changes kind: before carrying a fitted line past its range, ask whether it still measures the same thing (839) — a fitted line describes one object, and when that object turns into something else the line has nothing left to describe, however well it fitted. The change of kind often shows first in a side count. In R2W's toy the door, the far wall of the room in which riders tumble between two gears (row 670), followed drive = 2.0630 − 0.4719 × weight within 0.0015 from weight 0.55 to 0.90 and then bent (+0.0076 at 1.10, DOOR-1). The bend sits where the room the door bounds closes, near 0.905: past it the door is no longer the far wall of a chaotic room but the low end of an overlap of two locks. The side count showed it: riders in flight were 186, 15 and 0 at weights 0.85, 0.90 and 0.95. Before carrying a fitted line past its range, ask whether the thing it measures is still the same thing there, and look for the change of kind in a side count (rows 640 and 788 keep a law inside its range) (R2W). Test a fitted line where it was not fitted: the Loop Atlas's door line, drawn through three cores' doors, was registered as a forecast at two cores never flown, and all 15 of their breaks fell within 0.0025 of a radius of it against an allowance of 0.004 set before the run (TRAIN_DOOR_LINE 20261022_222935, card 262); flown next outside its bracket, at cores 0.035 and 0.065, it held again, the door on one line at seven cores, four of them forecast before they were flown (misses rows 465 and 468). A forecast between the points is the cheapest test a line can face, and one outside them the next. The same card's ten more copies put the thinnest core's life at the edge, 0.530, at 148,791 ± 42,952 time units pooled over 25 copies, 2.13 times core 0.06's, while the edge's place does not move with the core (misses row 464)

R2W (sitting B12); the Loop Atlas sessions (card 262); ties rows 640, 670, 788, 799, 865

Iterating a truncated formula is not a higher order: write the truncated answer plainly first, and test any iteration of it as a guess of its own (841) — a formula worked to second order leaves out every higher term, and solving it again and again at shifted settings brings some of those terms back and not others, so the result can land further from the truth than the plain formula did. In R2W's toy, KNOBS-3's estimate put gear 1's edge where the pull equals the ledges' grip plus the ripple's push and the shaking's loss, both worked to second order in the ripple's strength, and then solved it by iteration, recomputing the shaking at the shifted drive. With the ripple turned around the shift is large, about 0.2 of drive; there the ripple passes more slowly and shakes the ball harder, and the iteration put the edge's shift 26 to 37% too far for balls of weight 1 or less (KN3-3 missed). The redesign check blamed terms beyond second order. FOLD-DIR found the edge exactly and showed otherwise: the same formula worked at the unshifted drive comes within 0.3% at weight 0.40 and within 4.2% at every weight, and what it leaves over is fourth order, its sign changing with the weight. Iterating a truncated formula sums some higher-order terms and not others; forward it helped, turned around it hurt badly. Write the truncated answer plainly first, and treat any iteration of it as a separate guess with its own test (row 737) (R2W)

R2W (sitting B12); ties rows 737, 739, 788

Count the quanta to read a power, and count their parity to hear what pushes (847) — a beat between tones can be written as whole tone numbers that add to zero, and that count says, before any run, how fast the beat fades as the swing shrinks and whether it can push at all. In R2W two arrows set the flat point. The fourth harmonic's own arrow is four first tones against one fourth harmonic (1 + 1 + 1 + 1 = 4), so it falls as the swing to the fourth power: measured 3.916, 3.970 and 3.993 as the swing halves. The backward arrow is two octaves against one fourth harmonic (2 + 2 = 4), with no first tone in it, so it holds still: it moved 0.56% from a quarter of the swing to an eighth, and their balance goes as the swing squared (FLAT-12: the count a reading, its approach registered and landed, the power's gap to 2 shrinking to 0.2576 of itself against the quarter predicted, its rival missing; a counted power is tested by how it approaches its count, row 737). The count has a second half. Flipping every tone flips the push, so only a group with an odd number of tones can push: odd harmonics alone pair off into even groups, which is why a third harmonic alone is silent, and the octave gives it an odd group, 1 + 2 − 3, and then it speaks (BEAT-1). The same parity rule tells nonlinear optics which crystals can double a laser's frequency. When a part seems silent, ask which partner would give it an odd group (R2W). The triple-star thread met a parity of its own: fractions with odd denominators hold about three times as many locked runs per crossing as even ones, 2.73 times near the mirror line and 3.04 times well off it (fitted jointly 3.16 times, with a factor of 0.783 for each step in denominator), which its paper reads as a push odd in the swing feeding only the odd chains, left for theory; off the mirror 37 of the 51 odd runs lie on two chains, and the even crossings sit on the floor of 40 set for a reading. Whether it is this row's parity is untested (the triple-star sessions)

R2W (pattern lessons of sitting B12); the triple-star sessions; ties rows 354, 355, 356, 560, 734, 737

Know the instrument's own step rule and law before reading a slope, then take the law's maximum over the range you run (850) — the islands paper predicted R2W's integrator leak would fall as the weight to the minus three and be worst at the lightest weight; it was worst at 0.25, because the toy's step is half the weight below weight 0.2 and a hundredth of a passing above it, so the slope is +3.00 below the switch and −2.00 above, one constant throughout. And the book had quoted the leak as at most 1.9×10^{-6}, the worst of seven weights at one drive, when the law peaks at the step rule's switch and at the lowest drive the toy runs at: 1.8×10^{-5}. A sampled worst case is not a ceiling (NO-HONEY-SLOPE, NO-HONEY-PEAK). A switch applied at the end of an integrator's step is a hidden delay that grows with speed and can pose as physics: R2W's board flips came on average half a step late, a lag already inside SHUVIT-1's gap of 1.03×10^{-3}, and locating each crossing inside the step brought the check to 3.00×10^{-6} (SHUVIT-2). A second specimen of row 785 (R2W, sitting P2)

R2W (sittings P1 and P2); ties rows 741, 640

Before calling a structure new-born, look for its trace in the map without the knob (851) — weight 0.05 opened half and third steps in R2W's junction only inside the weightless rotations, each on the line where the weightless rider already slipped by that fraction: 110 of 110 within 0.0136 (WEIGHT-CONE W-a and W-b). The new structure grew from a trace already in the parent map; finding the trace first turns a discovery into a prediction

R2W (sitting P1); ties rows 664, 670

The integrator's honey: measure the drift in the conserved quantities at several tolerances before reading any slow trend as physics (858) — over 30,000 passes at the triple-star kit's tolerance every unlocked rate drifted steadily; at a tolerance a thousand times tighter the drift nearly vanished. The energy's drift, read at three tolerances (-7.90×10^{-11}, -2.71×10^{-12} and $+3.26 \times 10^{-14}$ a pass), fell by the same powers per tenfold tightening as the creep in the rates (1.46 and 1.92 against 1.41 and about 2.0), so the creep's climbing power is the integrator's own; neither the energy nor the angular momentum carries its direction, which lives in the pair's own slow quantities (EN2). Near-cancelling combinations magnify it: a slow beat made by three turns of one angle nearly cancelling two of another shortened by a third at the kit's tolerance (700.2 to 450.2 passes) and held at 722 at the tight one, so a beat can be the arithmetic's alarm. A tight reference still drifts, some 470 times more slowly: read its own drift before naming a third rhythm. And when the clean run exists, list every printed number that came from the drifting runs and re-read or remove each (EN1, AL1 to AL3, TD1, DR2). And find the floor: tighten until the drift stops shrinking. The energy's drift turned from falling to rising between tolerances of 3×10^{-12} and 10^{-12} ($+2.9 \times 10^{-14}$ a pass at 10^{-12}), and there two machines agreed to only two digits against four at looser tolerances; below the floor every record measures its own arithmetic (FL1, XC1). One combination of the pair's slow quantities stays almost still under a nudge in birth, the creep and the tight run's own drift: the thread's lock angle ψ plus twice the eccentricity vector's angle about its circle's centre (ratios -0.490 and -0.492 at two births, TW1 and PS1; its span 0.053 of the angle's across island 7, PS2). Whether it is a protected combination of the secular problem is a question for a celestial mechanician. In chaos, read a tolerance twin against a nudge twin: two runs at tolerances ten times apart part whatever the tolerance, so the Loop Atlas flew its smoke-ring trains at 10^{-11} and 10^{-12} beside two trains nudged 10^{-15} apart, and the tolerance twin parted no faster than the nudge (to a tenth at 48.8 time units against the nudge's 42.4); the integrator's error is then one more nudge, and only statistics over nudges are trusted: the passages over five nudges were 3,151 (spread 324) and 3,074 (spread 359). An inviscid toy has its honey too: the train's energy drift per 1,000 time units fell eleven- to thirty-eightfold for a tenfold tightening, where the three-body toy's, above, fell 1.46 and then 1.92 powers of ten. Gauge an engine through its own code: R2W's LEAK-JJ read each junction engine's one-passing function from its own file under its md5, changed one thing once, proved the copy bitwise against the engine at the original setting, and found the three engines leak by fourth-order Runge–Kutta's own law, 0.998 to 1.000 of it, at most 1.8×10^{-8} of the drag. A whole power of two between two step counts is the integrator's order speaking: the leak at 2,000 steps stood to the leak at 4,000 at 31.96 and 31.99, against $2^5 = 32$. At light weight the drag's stiffness comes before the leak: STIFF-JJ found the lag within 6.6×10^{-8} of a step eight times finer at all 16 rows, while the rate missed at the three lightest weights, by up to 3.07×10^{-5} a passing; read which variable carries the error (R2W, sitting P3)

the triple-star sessions; the Loop Atlas sessions; ties rows 741, 732, 645, 859, 863, 865

Walls of measurement come back in new fields: know each by its shape and its first test (863) — the triple-star thread set down thirteen walls it met first as surprises and named only after a test, written to be recognised outside three-body work; they belong to any slow thing read through a fast machine. The instrument's honey, a friction the tool supplies: turn tenfold the knob that should not matter and see whether the effect moves with it (row 858). The floor: a ladder of settings until the effect stops shrinking, read on two machines. The record's length decides what can be seen: work out what it can resolve, and double it before believing a no-reading or a lock (row 645). Two clocks lock where they agree on a simple fraction: count whole turns of the angle that compares them, not its value. Near-cancellation amplifies: when a combination nearly cancels, ask what a small drift in each part does to it. The frame decides the story: centre on what turns steadily, and name the strobe's moment (row 848). Order can be slow chaos, and chaos order: twins, with the growth law of their gap registered before looking (row 732). The one-sided door: 163 of 166 escapes left the island on one side, so count failures by side before trusting a symmetric rule. The

published stability rules smooth it over: Mardling and Aarseth's puts the island at 0.93 to 0.97 of its critical distance, unstable, and Vynatheya and colleagues' at 1.07 to 1.15, stable, while the stars that escape sit at 1.063 to 1.071; from the star's eccentricity 0.30 a second door opens above the island, and a star started circular at the 7:1 finds no door at all (none of 82 runs, and none of 41 over 10^5 of the pair's periods; CS1 and CS2), the bounds being one coplanar equal-mass pair, the star's eccentricities 0.20 to 0.34, and the 7:1 and 8:1. The grid draws the shape: read it again on a finer grid before naming it. Relation beats shape: the eye locates better than it draws (row 714). The control that moves with what it controls: read its level against a reference made another way (row 741). Coincidences of small numbers: a base rate from random cases, registered with the claim (row 616). The conserved quantities do not say where the error goes: measure their drift and the slow quantities' drift separately. A static problem is one object, and every look reads the same thing; a dynamic one makes a new object every run, with its own record, length and arithmetic, so its surprises are mostly about measurement. The triple-star thread bet at its close on meeting the honey, the floor, the record's length and the frame again within the next two projects; the Loop Atlas measured its own honey the same week (row 858)

the triple-star sessions (pattern-recognition addendum of 29 October); ties rows 858, 859, 116, 194, 797, 808

A lock is a closed loop on a torus: read its fraction as two winding numbers, a family by its crossings, and a walk among fractions by the trace of its matrix (864) — the Loop Atlas's needle wraps its angle but not its speed, so its world is a cylinder; wrap the drive's clock too and angle and clock make a torus, the speed its thickness. A lock at p turns every q cycles is one closed loop going round the hole q times and the tube p times, and its fraction is that pair of winding numbers (the explorer's long orbit is the loop 9/19, counted from the page's own data). Two loops a/b and c/d cross $|ad - bc|$ times (120 of 120 pairs counted on the flat torus): two parents cross once, and a child crosses each parent once. The map of these loops is the Farey diagram, the curve graph of the torus, and a family is a whole-number matrix of determinant one (511 of 511 families to ten generations, each a lattice triangle of area exactly 1/2). A walk on that map has a holonomy: each step down the tree is one of two whole-number matrices, a closed walk's product is its holonomy, and its trace sorts it, 2 for circling one corner (as the needle's n/(n+1) steps circle 1/1) and more than 2 for a true loop; the golden path's repeating pair has trace 3, the shortest closed loop of all. It is row 848's reading, a 2 × 2 matrix of determinant one sorted by its trace, met in A Rider in Two Waves' passing matrix and here in whole numbers. A doubling is a half twist: at a period doubling the motions beside the old loop swap sides every round, so the ribbon round it is a Möbius band and the doubled loop runs along its edge (Crawford and Omohundro, 1984); on the torus's skin such a loop must cross itself, and passes itself only through the thickness, the speed, so a dimension a picture leaves out can be the one doing the work (the figure-8 of row 589, met by another road). Each was found by asking what surface the loop closes on. Laid in space, each loop is a torus knot T(p, q): 2/3 the trefoil, 2/5 the cinquefoil, 3/4 the knot 8_{19}, 3/5 the knot 10_{124}, and a loop with p or q equal to 1 the unknot; the doubled 2/3 loop, round the hole six times and the tube four, is two strands of a trefoil braided round each other, a cable of it. Whole numbers sort the families too: two parents are never the same kind of odd and even, and their child is the third kind (511 of 511 to ten generations); the loops that go round the hole q times number Euler's $\varphi(q)$; and in the needle's own staircase every lock whose two parents were measured is no wider than either parent (54 of 54, 47 strictly narrower). The Torus of Fractions page draws all of it and checks it each time it opens (seven checks, every one held when it was opened on 31 October). Its forest gives each loop a doughnut of its own, in rows by generation and joined to its parents, because ten loops on one doughnut were a tangle in which no winding could be counted; the shared doughnut stays one click away (Guide item 478). And the map was named before a walker was built for it, with its theorems and its tools: the map is old, and what was the Loop Atlas's own was the needle's walk on it, 51 of its 58 steps from one lock to the next along a side (item 479). Bear saw the explorer's orbit wrap a cylinder, and a torus once the cycles wrapped too, before any of it was counted. Bear saw the can and the doughnut first, and asked whether the needle is a Josephson junction or a ring with a nick: it is the junction, and a ring with a nick is an rf SQUID, whose inductance charges each whole turn and unrolls the can (EXPLORER_TURNS)

the Loop Atlas sessions (session H, 22 to 30 October); ties rows 848, 589, 605, 627, 38, 631

An event a tool counts is a threshold standing in for a thing: count it by what it physically is and by who takes part, set its threshold against the scale that moves, fly past it, and check the stand-in in every regime (865) — the Loop Atlas's smoke-ring trains taught it five times in one session. A fixed threshold hid a family that slides: deep squeezes counted below 0.220 of a radius showed none for the thin cores, whose rings never went below 0.2223 while their breaks sat at 0.2228 to 0.2252, where a door sliding 0.0137 of a radius for each 0.01 of core (row 839) put them; counted against each core's own door, six of six thin-core breaks came through a squeeze (TRAIN_LONG_EDGE 20261022_175421, TRAIN_DOOR_LINE 20261022_222935). A proxy for a passage, the smallest ring's local minimum, rippled 274 times in 500 time units, so passages were counted as what they are, two rings swapping order along the axis, an integer event; and the smoke run that tested the plumbing showed the world's own clock (770 swaps in 453 time units), so a registered long shot whose answer it gave away was withdrawn and replaced in the open before the run of record. Counted together, 97 to 99 per cent of the swaps were one tight pair's own, every 0.35 time units; split by who takes part, the third ring threading the pair kept a slow beat, every one of 177 deep squeezes fell on a threading, and the break came one threading after the last squeeze in 35 of 35 (card 263, then card 264). The break the tool fired on, a 5-radius spread, was not an ending: flown 1,000 time units past it, the third

ring came back to the pair in 5 of 11 trains (card 264), so lifetimes are now times to the first wide parting. And the stand-in, checked at one gap within about 6 per cent (TRAIN_PARTING 20261029_112955), came early, about 1.28 times over forty-eight trains (range 1.09 to 1.59; 1.7 over sixteen), at the gap where the book's long lifetimes were measured, because there a ring that swings out comes back, in equal pairs (row 819; TRAIN_PARTING_EDGE 20261029_070800). Set a threshold against what sets the scale, split a count by its actors before reading it as a clock, fly a sample past any event a run ends on before naming it, and say in each sentence which event a number is read to. How far a ring could go and still come back was then flown out to 20,000 time units: in sixteen trains none came back from past 13.75 radii, the nine that left being 30.9 to 1,126.0 radii out, each at its farthest; most left on their first wide swing (6 of 9), and at the gap where the stand-in held, the book's lifetimes stand within about 6 per cent of a parting for good, while at the tighter gap a ring that comes back kept one train whole 16,145 time units more and three of ten were whole again at 100,000. Near the soft edge a train can thread for hundreds of beats with no squeeze deep enough to count and then go (two of eleven breaks had none), so the clean break is the rule only when the break comes soon. In forty-eight trains a ring came back from 19.31 radii: the farthest of a sample is where the sample stopped, not a fence, for the pairs' reaches thin out beyond 6 radii as a power of −2.97 ± 0.40 with no edge, so the farthest seen grows with the number flown. Fit a law where its picture holds: time out against reach over every return gave a power of 3.04; on the 48 returns past 8 radii, 2.587, a pull falling as distance to the power 3.17 beside the tail's 2.97, two readings of the cube at which distant vortex rings interact (Horn, Nicolis and Penco 2015; the Loop Atlas's benchmark 48). When a fit is replaced, reread that day every number drawn from the old one: a time from the discarded law, 41,568 where the far law gives 19,863, was caught only when a picture drawn from the data disagreed with the prose (misses rows 486 to 488)

the Loop Atlas sessions (session H, Guide items 450, 453, 465, 466 and 472); ties rows 686, 731, 630, 716, 812, 819, 839, 857, 858; the Loop Atlas's checks of 30 and 31 October: READ_267_2026-10-30, CLOSEOUT_FIGS_2026-10-30, and the pooled run TRAIN_PARTING_EDGE_2 20261029_211416

A picture from another trade brings a law with a number: test the number (870) — Bear heard R2W's skateboard, whose board flips half a turn at each pad's edge, as a motor's commutator, and the commutator brought timing advance with it. Without a lag in the push there was no crossover; with a lagging push, the best advance sat near a coil's phase lag, the arctangent of slip rate times lag: read after the data at lag 0.3 (41.4 degrees against 36.0), then registered and landed blind at lags 0.6 and 1.0 (60.0 against 57.4, and 70.0 against 70.0) (SHUVIT-1 to SHUVIT-4). The other trade's picture was worth one law, and the law's number was the test. A number carried from a simpler model is such a law, and a forecast until the new setting is read: R2W's kit for R83-F said the rectifier's step closes near ratio 4.36, the one-tone estimate, where its own files put it near 5.67, and a band at weight 0.10 came from a fit while rows on file since 14 October already placed the crossing. Read the steps already passed before choosing the curve: R83-F's peak fell 0.980, 0.542 and 0.248 per unit of ratio, each about half the last, so the quadratic that had the fall level off near ratio 4.0 missed and the cubic held within 0.1 (R2W, sitting P3)

R2W (sittings P2 and P3); ties rows 573, 745

Read a cascade from its onsets, on both sides; test the order of its windows; and let one toy's road forecast another's (872) — flown from rest, a cascade's bands are broken by hidden locks with doublings of their own, so a band's edges as met are not its width: inside the Loop Atlas's needle's 5/7 step an 11/15 repeats every 15, 30 and 60 cycles inside the 14 band on the high side, and an 11/16 every 16, 32 and 64 inside the 28 band on the low side. Placed by the square-root law, the onsets gave band ratios of 3.945, 4.482 and 4.626 from above and 3.994, 4.480 and 4.629 from below, closing on Feigenbaum's 4.669 alike; the two sides' agreement is the check that onsets and not windows were measured. Then test the order of the windows: the order in which periodic windows open inside the chaos is universal and exact (Metropolis, Stein and Stein 1973), so a toy's windows set beside the logistic map's test the cascade as a member of one family, and the order forecasts windows not yet seen. Past each end of the step the needle's windows came, in units of its 56-cycle loop, in the logistic map's order (10, 6, 8, then 7, 5 and its doubling, 3 and its doubling, 4); a 7 and a 10 forecast from that order were found where forecast on a grid ten times finer, and two windows met on one side only were found on the other, forecast from its band widths alone, within two billionths of pace. Calibrate a forecast before flying it: a cubic from the logistic map's parameter to the needle's pace, through six windows already met, put each one left out within 0.0000000035. And claim a window only where it holds at two paces or more after a settle several times the first: an 11 met at one pace after 32,000 cycles never repeated at 128,000, at that pace or its neighbours, a quiet stretch of chaos lingering by the window's ghost, and was withdrawn, while a 10 the 1973 order named, never met before, came 0.0000000024 from its forecast and held at 3 of its 8 paces. One toy's road can forecast another's: R2W's rider had read its own doubling inside a locked step, ratios 2.41, 3.14, 3.94, 4.45, 4.62 and 4.66, each gap to 4.669 about a quarter of the last; that rule, registered as a forecast of the needle's fourth ratio, gave 4.6585 and 4.6591, and the needle gave 4.6599 and 4.6604, then a fifth, forecast at 4.6672 and 4.6673, of 4.6673 from above and 4.6672 from below. Before naming a limit, carry the sequence until its swings are smaller than the difference you mean to claim: the logistic map's own gap factors swing (−0.540, 0.128, 0.311, 0.179, 0.239, 0.201 to period 512), where the 5/7 step's held near 0.217, at 0.208 and 0.228 by its fifth ratio, and carried to period 8,192 in 60-digit arithmetic they close round 0.214, 1/4.669. Ask why a ratio closes, too: renormalization has the gap close by 1/δ a step from a bend of the dial and swing by the doubling operator's second

eigenvalue, −0.12365 (Christiansen, Cvitanović and Rugh 1990), and the logistic map's superstable values to period 262,144 give 0.214169 and −0.123652. Test a published table of windows by rank within each period, not by width, since a map from one dial to another bends widths but cannot reorder one period's windows: the nine windows Shukrinov, Botha, Medvedeva, Kolahchi and Irie (2014) list past a Josephson junction's cascade on its 5/3 step, 24, 12, 10, 6, 10, 8, 7, 5 and 3, naming only the 6, 5 and 3 as the universal order, stand exactly in the order of the logistic map's largest window of each period (its second-largest for the second 10), so the whole table is the 1973 order. And a correction to scaling can be read off the raw sequence: at order 2 the superstable gap ratios close on δ with a swing whose ratio settles at −0.57734 a row, which divided by δ is −0.123649, the doubling operator's next eigenvalue (−0.123653) to about five figures; a near match to $-1/\sqrt{3}$ was 1.5×10^{-5} off and was stopped there (the Loop Atlas sessions, its SWING check)

the Loop Atlas sessions (session I, Guide items 485, 487, 490, 492, 493, 495, 496, 499 and 543, benchmark 49; R2W's rider as the Loop Atlas quotes it; its checks and records: CASCADE_57, THIRD_RATIO, WINDOWS_ORDER, LOGISTIC_WINDOWS, SCALED_COPY, FOURTH_RATIO, LOGISTIC_GAPS, FIFTH_RATIO, LOGISTIC_DEEP, MSS_FORECAST, MSS_FORECAST_2, WINDOWS_LONG, FACTOR_FIT, SHUKRINOV_WINDOWS, SWING); ties rows 680, 771, 760

Every listener is deaf to its own echo: buy any real check, and grade drift against a kept copy of home (884) — a network's gradient is smallest on the batch built from its own agreement. In Question the Substrate the student's batch of its surest labels pulled 0.004, about 50 times quieter than any other; a sibling network's quietest batches were that one and its own, and an outsider's quietest was its own. A strong outsider trained on 1,500 loops ranked batches by their true share of wrong labels at +0.829, against +0.213 for the student, and its quietest batch was the real checker's (runs E8, E8b); for work like the guide's own, an audit in a fresh session with independent material is that strong outsider. So buy any real check: a checker that catches a quarter of the errors earns 40 per cent of the benefit and one that catches half earns 68 per cent, throwing away a fifth of the good lessons costs under a point, and even a half-blind, wasteful checker beats the mind's own certainty, by 28.6 points in world 1 and 10.9 in world 2; the knobs are ratios, so they carry to any scale (E7) (Question the Substrate). The yardstick decides whether drift can be heard at all: graded against the truth, a learner's drift of identity hides; graded against a kept copy of home, it is loud off home ground, 8.888 to 9.876 times its home gap on inputs three times wider. That run was registered and carried out in the first paper's sessions, and all 6 of its predictions held (run bridge_b1) (Two Arrows)

Question the Substrate and Two Arrows (the two-voices series); ties rows 77, 308, 883

Thresholds in self-training chains are tipping zones: zoom where the person points, and report the split, not only the mean (886) — Bear said the survival cliff in Question the Substrate was "where the cluster of magic likely happens". Five starts on a coarse grid had averaged it into a slope; ten starts at a 1-point step showed a tipping zone, between 14 and 12 per cent of the world, where identical setups live or die by their random start, and showed the cure, selection aimed at the answer sizes estimated from the labeled set (run E31). The chain's end-state switch has the same signature, between size ratios of 0.61 and 0.64 (E37r). Five starts hide such a band and ten show it: a split among starts is a signal, so zoom where the person points instead of averaging, and report the split beside the mean (Question the Substrate)

Question the Substrate (the two-voices series, Paper 1); ties rows 218, 622, 883

L2 — The Geometry & Substrate You Walk

The fiber bundle real estate, the vertices and junctions, the winding theorem.

The rows of L2

New in version 7 on this limb: 75 of its 181 rows, numbered 561, 576, 577, 578, 605, 623, 624, 625, 626, 634, 635, 636, 637, 653, 654, 655, 656, 658, 669, 670, 671, 672, 673, 675, 678, 680, 681, 685, 688, 695, 696, 697, 698, 715, 719, 724, 725, 727, 744, 754, 762, 768, 769, 776, 778, 790, 792, 795, 802, 804, 809, 812, 817, 818, 820, 826, 828, 830, 842, 848, 849, 856, 857, 859, 862, 873, 876, 877, 879, 883, 885, 887, 888, 889, 890.

Look where a field distrusts amplitude (4) — the distortion source is wave-fixed — phase near-constant at fixed ratio, near-zero group delay; the reflection source is place-fixed — phase winds rapidly, delay scaling as 1/f

the two-source (wave-fixed / place-fixed) model of otoacoustic emissions — author citation is in the deposited paper's reference list

Wavelength against structure (6) — Your signal's wavelength has to roughly match the structure you are reading — too large and the signal averages across it, too small and it bounces off without reading. The cochlear mismatch: at 1.2 Hz the acoustic wavelength in fluid is 1,250 metres — far larger than a 0.6 m canal. Even at 1 kHz a quarter-wave air column is 86 mm, and in fluid it is 375 mm, so at audio frequencies no pipe or cavity resonator can be implanted, and a resonator has to take its frequency from stiffness and mass instead (the Device Scoping paper, §2C).

dimensional analysis

Why an invariant is protected (32) — A signed invariant stays signed even under perturbation — that is the protection. In the topological memory architecture (rung 4 refers to the fourth level of the holonomy complexity ladder), a signed vortex ring retains 39–40% of its launch spin after travelling roughly 9 ring radii (R) through water, and its sign — which encodes the topological bit — never flips. Changing the sign requires an actual crossing event, which is detectable.

Moffatt 1969; Călugăreanu; Kelvin; Noether

Invariants persist, energy leaks (34) — Topological invariants outlast the energy that created them. The parity (the sign of the winding) held to 4×10^{-15} — near-perfect preservation — while the raw holonomy amplitude drifted by 9.2%. The integer survived after the continuous signal had decayed.

Arnold tongues and mode-locking (38) — The staircase is the commissioning certificate: sweep the detune, log the plateaus, and bank the widths as measured margins. This is the quantitative law the hold was waiting for: wide steps are the deep locks, and the widths shrink as roughly one over q-squared down the Farey tree, so a simple ratio locks broadly and a complicated one barely locks at all. Not every toy follows it: in the Loop Atlas's needle a lock's width is set by its place more than by its fraction, the 4/3 at mass 3.0 1.8 times as wide as the 3/2 above it (row 671). And the operating rule that falls out: operate at plateau centres, because the plateau width is your margin against detune. The instrument's own fence travels with it: a subsystem that is a low-pass filter rather than an oscillator joins by feedforward and not by a tongue — knowing which parts are not in the locking family at all. A two-rhythm rider is a differential. At a lock of p turns against q, the drift is a fixed rational blend of the two speeds, the slip over the beat sits at p divided by p minus q, and nothing locks above one over one minus the ratio of the two rhythms, 2.618 at the golden ratio. Measured: at pace 1.2933 the drift was 0.7999, against the 0.7993 that pace times 0.618 predicts. Predict the treads and the ceiling from the ratio before sweeping (the Loop Atlas sessions). A chain of two gearboxes passes the comma down: chained through a relay, the second stage meets a comma of its own, near a ratio of 1 the relay's kicks and its teeth beat slowly against each other, and the same demand to listen long applies one stage down (the Loop Atlas sessions, R2W). Three projects met the golden mean without naming it as their common object: the rider's drift at pace times 0.618 above, R2W's golden ray (row 464) and the Integer Forest's golden-angle offsets (row 198); on the map of fractions the golden path is the shortest closed loop, of trace 3 (row 864)

circle map; Arnold tongues; Farey tree; ties rows 198, 464, 671, 864

The scale law (39) — Biological propulsion follows a scaling law: the bigger the animal, the slower its stroke cycle. Jellyfish measured: 5-metre bell diameter, 13-second stroke cycle, 9 tonnes of water moved per breath, at approximately 2.5 watts. The frequency scales as f ~ L^(−2/3) (length to the power of minus two-thirds) — a known allometric relationship. This means the geometry of the bell, not its size, is the relevant design parameter when building a geometric analogue.

allometry

Rigidity selects the operator, and the operator selects what you can hear (49) — membrane or gel → scalar Laplacian, Kac applies exactly; stiff plate → biharmonic, the theorem must be re-derived; elastic solid → vector waves, the 1-form sector where b_1 lives; fluid cavity → scalar only. Reading b_1 acoustically requires a solid — a constraint nobody would guess from the topology

Hodge; Kac

Winding is a count, not an amount (57) — the topological charge is independent of the quantity of medium — a quantized circulation does not depend on how much fluid circulates, so the stored invariant does not scale with cell volume. The optical case is the cleanest illustration of a fact the spin case needs

Skyrme 1962; Mermin 1979; Allen et al. 1992

The conjunction is the wall, not any single property (60) — five properties are each demonstrated somewhere in isolation, no material does all five, and there is no guarantee the set is jointly satisfiable because the properties pull against one another — deep barriers fight room-temperature writability, tunable multilevel charge fights clean two-channel readout. The parallel, stated deliberately: this is ALife's fifty-year shape — substrate-independent in logic, hostage to substrate in realization. "A materials scientist need only supply the medium" is the engineering cousin of "a chemist need only supply the cell." The supplying is the hard part

The Volterra construction is a holonomy definition (65) — cut the crystal along a line, remove or insert a wedge, weld the lips, force a rotation — that is a disclination, and its Frank angle is the holonomy of a loop around the defect line. The paper states the object in materials-science language without naming the reading; the guide already banks deficit angle in B.2, so this is the same object under a third name

Volterra 1907; Frank

The ring attractor is a circle-valued state with no lift (69) — an insect's heading lives on a ring of neurons that is topologically S^1, not on a line — there is no neural state for "370°". Path integration then transports each body-frame step into world coordinates through the current heading, so the home vector is built by transport and its closure error on a round trip is a holonomy. The consequence that makes it a candidate rather than a restatement: because the state is heading mod 2π, the winding — how many net full turns the animal made — is precisely the information the ring attractor structurally cannot hold. That is a lift obstruction, a circle-valued state with no lift to the line, instantiated in neural tissue

Heinze, Narendra & Cheung 2018 (ring attractor, home vector)

Closure forces an integer — hold closed, and the sentence was wrong (82) — The node arrived and it says something else. `node-holonomy-eigenvalue-keystone.md` is a Session-4 capture, and its keystone is the eigenvalue is what survives the trip; the holonomy is the turn the survivor comes home wearing — modulus and argument of one number, not closure and remainder. The phrase this row was named for appears nowhere in it. The row is kept rather than deleted because the miss is the record: a remembered sentence sent four searches after a file that does not contain it, and the file was found only when someone supplied the real name

see rows 148-151

The same object arrived twice, in two lineages, months apart (90) — the cleanest B.2 instance in the corpus, and it is internal. The Anchor and the Hammer argues that no observer reads itself true and prices it on a control loop; the wave-geometry lineage independently banked the unverifiable-from-inside as its own node. One object, two names, two years' worth of separate reasoning, same author. Row 26's wraith appears in both lineages too. The lesson is not the coincidence — it is that a corpus large enough to rediscover itself needs the index more than it needs another chapter

practice; ties rows 26 and 70

The apparatus and the readout are one object (91) — carry a state around a closed loop and see what it comes home as — that is the Wilson loop, and its eigenvalue is what survived the trip, which is the conserved quantity the loop did not disturb. Wilson loop is the question; the eigenvalue is the answer. The corpus has always been built from "eigenvalues and Wilson hoops" because they are two faces of one instrument, and the guide has never said so

Wilson loop / holonomy

Why it ports, and it is two theorems rather than a vibe (92) — the argument the guide most needs and does not have. (1) Pose-through-time is one problem. Orientation lives on a sphere and rigid motion on a sphere is a single rotation about an axis, so anything with a pose tracked through time without a fixed external reference is forced into the same bookkeeping — an axis-and-angle to integrate and a residual where it fails to close. Plate, whale, gyroscope, spacecraft, spinning molecule, head-direction cells, camera odometry. (2) Invariance becomes an invariant. What a system is indifferent to, it conserves; the invariant is exactly what the loop is indifferent to, and walking the loop is the apparatus that exposes it. So the tool ports because this specific problem recurs — not because everything is a loop

Euler's rotation theorem; Noether

The out-of-band witness, in the flesh (95) — the post-reproductive orca matriarch holds the clan's map across decades while the pod dead-reckons through dark water — she is the invariant they navigate against, long-lived precisely so the reference does not drift with daily noise. An anchor placed outside the moving frame so the moving frame can check itself; when she dies the pod loses its out-of-band witness and mortality rises, an adult son's roughly fourteenfold (row 330). The cure failing in live ecology

ours; ties rows 330, 424 and the deposited RPV paper (concept 21786539)

A smooth contour field can break exactly three ways, and there is no fourth (109) — wherever nature has a monotone quantity — pressure, depth, temperature, time, decay, potential — it stacks the substrate into contours for free, and a defect is where contours break. The complete list is a winding, a critical point, or a discontinuity. This is the honest form of the universality claim and it is much stronger than the tempting one: one instrument suffices because "ways a contour field can break" is a short closed list, not because the domains are one thing. The microscope sees pond water, blood and crystals — same lens, not same substance

You cannot read a winding off a substrate with no enclosed area (116) — the codimension-2 rule, biting in practice and priced: a ledger branded topological was tested on a sensor ring and fired 52% on honest data — a noisy heuristic, not a clean integer invariant — because a one-dimensional ring of bearings is not a two-dimensional field, so the winding never quantised and the blind spot it was supposed to have simply vanished. The repair is not a better threshold; it is a triangulated sensor field where collusion makes a real enclosed charge. The same rule, read from the other side, tells a holonomy from a knee: a leftover with no area is a knee's (hysteresis), not a holonomy. A car on a banked turn with tyre grip (μ 0.3), its speed swept slowly from 12 to 34 m/s and back, comes home 25.05 degrees up the bank from the apron, once; a second sweep leaves it where the first did. The sweep goes out and back along one line, enclosing no area, and the loop's width at any lane is that lane's grip band (to 0.003 m/s at three lanes); a faster

sweep widens it. Before calling a leftover a holonomy, run the loop out and back along the same line: a holonomy leaves nothing, a knee leaves its band (run TYRE_KNEE 20261014_174309, on Bear's "is holonomy the drift?") (the Loop Atlas sessions). A rounding miss is another impostor, and exact arithmetic tells it apart. A loop of operations that should return to its start and misses by a unit in the last place looks like holonomy. It is not, when the miss is random in sign, stays about a unit per step, does not grow with what the loop encloses, and vanishes in exact arithmetic: in R2W an exactly rounded sum (math.fsum) was reported to give the same bits for both paths' terms, though R2W's own review found one fsum value on record and the command's output not kept. A holonomy lives in the geometry and survives exact arithmetic, and the test that separates them costs one rerun in exact or compensated arithmetic (row 738) (R2W). R2W's sitting P1 met five leftovers that were not holonomies, each told apart by a cheap test: a jump between samples (bisect it; row 799), a rounding floor (divide by the exact value; row 248), the grid's step (root-find the edges, and say when a verdict could not fail; row 173), the ruler's own leak (find the instrument's law, which carries every point with one constant; row 850), and a copy out of step (a fresh reader, front to back; row 853). The one leftover that was a holonomy was read as one: the passing matrix's class (row 848)

codimension-2

An open arc is a convention; only the closed loop is an invariant (121) — a single seam read on its own is a gauge-dependent placeholder and is untrustworthy alone — the physical object is the closed loop product. This is not an analogy to lattice gauge theory, it is lattice gauge theory, and it is why the instrument's basic unit was never a measurement but a circuit

lattice gauge theory

Wrinkly where you hoard, smooth where you measure (122) — a wrinkle is a place to hide treasure, which is the same as a place to hide a lie — capacity and deception are one geometry, which is why a storage substrate and a fidelity instrument turn out to be one machine. And smooth carries two meanings that must be kept apart: low-wrinkle, which is the honest cheap-to-read ruler, and seamless, where the join cannot be found — a seam you cannot find is a place you cannot audit. Two ways to defeat a read: bury the signal in too much edge, or erase the edge entirely. What survives a seam-erasing smoothing is a fiducial placed before the smoothing

The field's own definition was already the thesis (135) — a tropical cyclone is officially a rotating organised system with a closed low-level circulation, and the formal genesis definition imposes no wind threshold at all — intensification is explicitly the amplification of wind beyond that stage. So the literature already says formation is a structural closure event and wind is the downstream consequence. The instrument did not propose a new idea; it made an existing qualitative one quantitative, which is a much easier claim to defend and a much better one to make

The loop is the line between re-confirming known physics and testing something new (137) — a single station snapping back is elastic rebound — settled since 1906 and not yours. The winding only lives in a loop of stations: walk a triangle or a pentagon through time and the failure to close is the holonomy. Knowing exactly where that line runs, and crossing it deliberately rather than drifting over it, is what separates a new instrument from an expensive re-derivation

elastic rebound (Reid 1910)

Chern–Simons demands the ribbon (143) — the theory does not flatten and does not live on a curve: it stays in full three dimensions and requires the curve be thickened into a ribbon — a framing — before it is even well-defined, so the dimension that a flat diagram loses returns as the ribbon's width. The matching read is the instrument the corpus already had: carry an orientation around the loop keeping it parallel, and the drift on return is the holonomy, with twist plus writhe equal to the self-linking number. The dead-reckoning instrument and the knot reader are the same machine — an identity, not an analogy

Călugăreanu–White–Fuller; Chern–Simons framing

The periodic table is the existence proof for the winding-memory architecture (144) — every property the deposited winding-memory paper asks a substrate for, the atom already has: signed (electron spin ±½ is the chirality bit), multilevel (shells), topologically quantized (a standing wave must close around the nucleus or tear, so only whole numbers fit — row-lengths 2, 8, 8, 18 are literally counts of standing-wave slots), and retention by energy barrier — the same Arrhenius barrier row 51 prices at fractions of a second near 0.5 eV and centuries near 1.1 eV. Chemistry is the outermost unfinished shell, which is why the columns repeat. Not a novelty claim in any part — it is the most famous integer table in science, and that is exactly what makes it the right worked example for a guide that wants a reader to wonder about the universe

standard quantum mechanics; ties row 51 and the deposited winding-memory paper (concept 20587337)

The fluid supplies the framing Chern–Simons demands (147) — CANDIDATE — prior-art pass owed before any claim, and not run before version 7 was built, so the row stands under the Declaration's attribution limits. CS is undefined on a bare curve; the curve must be thickened into a ribbon, and changing that framing by one unit multiplies the answer by a known phase, so the framing is a physical parameter rather than bookkeeping. A vortex

tube is that object physically — centreline as curve, core radius as framing, swirl as twist, helicity as the linking — and the helicity decomposition into writhe plus twist is the same Călugăreanu–White–Fuller split row 143 names. All three legs are prior art and must be cited as such. What is ours is the price: a signed ring keeps 39–40% of its spin over ~9 radii in water with the sign held, quarter swirl costs ~0.7% of speed, and the same shape loop at Re 30 gives +0.12 R/cycle fast against +0.055 slow — pace-dependent already, so the connection has acquired memory. Three fences travel with it: the fluid gives the abelian sector only, so linking numbers and never a Jones polynomial; helicity decays at finite viscosity, so slowly decaying and never topologically protected (row 87 is what that word cost last time); and reconnection breaks the topology outright and has no run behind it

Witten 1989 (framing); Moffatt & Ricca 1992; Arnold & Khesin; Shapere & Wilczek 1989

Transport only turns, it never stretches — so the rotor's length was never a choice (148) — the recovered keystone's settled half, and it answers a question the trunk draft leaves open. Parallel transport adds and drains no amplitude; it moves phase only. So a holonomy eigenvalue is unimodular by nature rather than by normalisation — it is born on the unit circle. Circle a pentagon, enclose 60° of deficit, and the eigenvalue is e^(±i60°) with magnitude exactly one. Nobody decided rotors have length one; the geometry did

parallel transport is an isometry

The corners of the polygon are its eigenvalues (149) — the n-th roots of unity are the eigenvalues of the operator step once around. So the pentagon's five corners and the walk around the pentagon were never two things needing to be reconciled — same eigenvalue, two descriptions. This is why an n-gon rotor combinator works at all: each rotor's corners already are its eigenvalues, all unit length, all pure pose-steps. The "= 1" is not bolted on, it is the corners themselves

roots of unity as eigenvalues of the cyclic shift

THE KEYSTONE — the two ledgers are the modulus and the argument of one number (150) — The eigenvalue is what survives the trip; the holonomy is the turn the survivor comes home wearing. Magnitude one means the conserved quantity rode the loop untouched — the energy ledger, Noether's what-the-universe-doesn't-care-about. The phase is the deficit you enclosed — the winding ledger, the drift. Two halves of one complex number. Filed as CANDIDATE and not as READY, on its own author's ruling: each piece underneath is settled, and "the rotor combinator, the chiral-knot tracker, the two-ledger architecture and Berry/Noether are all the same object" was re-graded from settled to vision because an analogy doing a theorem's job stays an analogy until something forces the identity. The sentence is the best one-line statement of the instrument in the whole corpus, and it is not yet a theorem

Berry 1984; Noether; ties row 141

The phase depends on the shape of the loop, not the speed of the walk (151) — carry a state slowly around a loop and its eigenvalue — the energy, the pitch, the invariant — never changes, but the state comes home wearing an extra phase that depends only on the loop's shape. That is the adiabatic yardstick the whole corpus is measured against, and it is exactly what JELLY-1's X0 is testing: in the Stokes limit displacement per shape loop is pace-independent by this theorem, and the measured +0.12 R/cycle fast against +0.055 slow at Re 30 is the departure from it. The row is the trunk's, the departure is the paper's. Speed can matter to a loop in two different ways, through a partner that remembers or through a pace limit, where the stroke's rhythm competes with another rate: of 12 of the Loop Atlas's loops with a forgetting partner and a known answer, 4 still care about speed (the wing, the radio trap, the turning needle, the carried beads), all through a pace limit. The count is from the Loop Atlas book's tag table, whose tags are judgments (the Loop Atlas sessions). A body's turn in the air splits the same way, into a shape part and a count part, each measurable alone. With a carried spin, whole limb circles hold off a share that rises exactly in step with the circles (0.409, 0.666 and 0.924 for one, two and three arm circles: 0.2573 each), on top of a fixed share bought by the limb's time away from the body; count the circles whole, or the landing's phase decides (run HITCH_LEGS 20261016_065650; the Loop Atlas's misses row 256). The arms' share is two pieces: one circle's turn with nothing carried, and the reach's slowing, the circle's average of one over the body's inertia; together they give every whole-circle share exactly (run ARM_SHARE 20261016_182910, to 1.6×10^{-10}). An average of one over a thing is not one over its average when the thing swings: averaging the inertia put the arms' fixed share 7.5 per cent high, and for an arm swung in a plane the body's average inertia round the circle is exactly its inertia with the arm at right angles, because the inertia about the centre is a constant plus a cosine of the arm's angle (the same run, read after it). Count a stride's tuck beside its loop: a straight leg swung back and forth turns the trunk by under 2×10^{-10} degrees, a loop with no area (row 116), while a folded one turns it nearly as the fold squared, and because the body is longest with its legs straight every stride also pulls it in and speeds the spin it fights; three human strides take back 0.329 of a 60-degree forward turn and the tuck returns 0.064, and with a light fold a wider swing buys less (run STRIDE_MAP 20261016_213858). In the toy jumper the shape part, the routine flown with nothing carried, is 78 to 95 per cent of what arms and legs hold off together and is the same at any speed to 10^{-9} degrees, while the count part, the time spent in each shape under the carried spin, stays nearly fixed, 0.07 to 0.10, and the limbs add within 6.5 per cent (run ARMS_LEGS 20261017_012927). Each part answers a different question: the shape part a coach can rehearse on the floor, and the count part only the air teaches. Rehearse the shape; time the count in the air (the Loop Atlas sessions).

A human stride's loop is small. One running stride of both legs, carrying nothing, turned the trunk back by 6.573 degrees against a lean of 15 to 60: the hip swings 0.70 rad either side, the knee bends only on the way forward, and the shank is light (4.5 kg against the thigh's 7.5). With the leg's mass spread evenly the stride gives 8.354, still under 15, while the toy's straight leg swung round a full circle gave 173, because a full circle is the biggest loop there is (run HITCH_LEGS 20261016_065650) (the Loop Atlas sessions)

Berry 1984 (adiabatic theorem)

The twist channel, priced twice in two substrates (165) — a framed loop separates from a flat one at 15.5σ, with the sign reading as chirality — measured on a synthetic loop channel in July 2026. The same distinction is measured again in a viscous fluid, where a signed ring holds 39–40% of its spin over nine radii with the sign intact. Two substrates, one distinction, independently priced — which is what rows 143 and 147 rest on, and it means the framing argument is not a borrowed idea awaiting a test

Witten (framing anomaly); Călugăreanu–White–Fuller

The face shape is set by the arithmetic, not chosen (171) — Cascadia's lattice was triangles. The forest's smallest closed loop runs n → 2n → 6n → 3n → n, which is four-sided — 110 squares in the patch. What carries over from Cascadia is not the triangle but the principle underneath it: read the instrument on the faces, because a node value is exactly the part a gauge choice can erase. Two prime directions give a quadrilateral, and the substrate decides

discrete gauge theory

The constant, with its address (175) — 1.20871040964352282 — the barrel at which one peak becomes two, on the universal curve every near-coincidence in every prime pair collapses onto. It is specifically the infinite-temperature, uniform-illumination value, and saying so is half the result. The census behind it: 993 prime pairs, 14,098 rungs, 13,978 measured, and of the 8,295 deep rungs not one lands above the solved value. Every patch approaches the same ceiling from below and none crosses it — what a finite-size effect must look like, and what a coincidence would not

Sparrow's criterion; Baker's floor

The most findable coincidence is the least visible (181) — $13^7 = 62{,}748{,}517$ and $89^4 = 62{,}742{,}241$ differ by 6,276 — one part in ten thousand — and anyone with a pocket calculator could stumble on it. Resolving the two would take a telescope 115,100 logarithmic units long. You can notice a coincidence and still be unable to see it, and noticing is not the same purchase as seeing

Multiplying by a prime is a gauge transformation of the beam (183) — every prime owns a clone of the whole forest, and the clone's beam is the original turned by a single global phase — measured at 3.9×10^{-11} against prediction, modulus untouched to 10^{-12} — because the logarithm of the multiplier cancels from every difference. So a census over prime pairs is exhaustive over families. What the beam can see is the barrel: a clone reaches only as far as the cutoff divided by its prime, so it is shorter by log P. Being a different prime's family is not a difference in what you play; it is a difference in how long you get to play it

Dirichlet shift property; ties row 521

One hidden law, many faces: close a loop and an integer is forced (186) — the trunk figure the corpus already had. The Bohr atom's closing electron wave, the Chern integer that is the resistance standard, flux quanta a SQUID counts one at a time, mode-locking that gives the atomic clock its ruler, the 1:2:4 Io–Europa–Ganymede resonance, the harmonic series in a handpan, whispering-gallery modes, spherical harmonics and the periodic table — single-valued around a loop, therefore an integer. And the clause that makes it a design rule rather than a list: when the integer is forbidden flat — five-fold, seven-fold — the system must either curve or go quasiperiodic. Closing a chain into a ring forces whole waves around it, so engine meets lock: the best open-chain step, half a wave, is forbidden on the ring, and the ring crawls less (1.14 against 1.66 for 8 segments) (the worm ring toy) (the Loop Atlas sessions)

ours (the synthesis); each face is standard

Zero is the boundary, not a blend (187) — the ternary of curvature sign in one picture: a saddle where a triangle's angles sum to less than 180° and the carried arrow turns back (−1), a flat plane where they sum to exactly 180° and there is no turn at all (0), and a sphere where they exceed 180° and the arrow turns ahead (+1). The whole point is the caption: zero is the boundary between the two, not a mixture of them — which is why the channel is ternary and not a continuum with a threshold

Gauss–Bonnet

Roughness is cheap; walking is exponentially expensive (196) — the octave law prices an asymmetry between the construction's two axes. Matching one prime's reach at rung r by a pure power of two costs 2^(r+1) — so the prime 113 sits at rung 30 with a reach of 30.001 and the smallest power of two exceeding it is 2^{31}, a nineteen-million-

fold difference in magnitude to buy one short step at that height. Meanwhile 729,000,000 = $2^6 \cdot 3^6 \cdot 5^6$ reaches only 4.20, worse than 1000's 5.367, because its legs nearly cancel. The primes form a reach no smooth number of comparable size can attain — which is precisely the roughness axis factoring algorithms operate on

smooth-number filtration

The radius is the exponent, exactly (197) — every integer sits at horizontal distance exactly e from the axis planted at its cofactor's endpoint, where e is its top exponent. Measured across all 999,999 integers below a million, the worst deviation is 1.78×10^{-15} — floating-point roundoff and nothing else. And the cylinders nest: the axis of one sits on the cylinder of its own cofactor, down to the stem, so the nesting chain is the factorization — which cylinder, on which cylinder, on which cylinder, rooted at one

divisor tower

The gap is cross-reference, not coverage (204) — the classical objects under this construction sit in Diophantine approximation, circle dynamics, crystallographic groups, discrete geometry and mathematical biology — subject classifications that do not cross-reference each other, with no single classification assigned to see across them. The value of a coordinate-free instrument is precisely that it renders this shared structure in one picture. The gap it addresses is not one of coverage — each piece is deeply studied — but one of cross-reference. This is the clearest single statement in the corpus of what the whole programme is for

CAPS AND BASES CARRY HOLONOMY — in the rotational channel, and the Tower had already closed the abelian one (207) — It was first filed as an open item the papers left open. Integer Forest v2 §4 does leave it open — but Tower v3.4 took the measurement and said so, in the vector-area channel: a climb's displacement is purely vertical, so its cross-product has no vertical part, structurally, and the run returns 0.000000. The parent and the sibling had been read, and the companion missed. Measured 2026-09-12 on the commuting square n → np → npq → nq → n, walked as a closed quadrilateral through the four addresses with rotation-minimising transport. Wall loops: 1.11×10^{-16} rad — planar by construction, the parent's flat-wall claim reproduced independently of the graph-spectral route. Boundary loops: 0.0625 to 0.4203 rad. The mechanism is one sentence: every heading is a rotation about the shared vertical axis, so wall transports commute — but a climb tilts the tangent out of the horizontal plane, and a rotation about a horizontal axis does not commute with the vertical ones. The construction is abelian on the walls and non-abelian across the caps

non-commutativity of SO(3); rotation-minimising frame

The cap holonomy is translation-invariant in n, and that is the clone gauge again (208) — six loops built on entirely different integers — (1,2,3), (2,3,5), (6,5,7), (30,7,11), (210,11,13), (2310,13,17) — all return 2.247e-01 rad, identically; (4,3,11) and (12,5,13) both return 3.557e-01; (2,3,113) and (6,5,127) both 6.247e-02. The reason is exact: shifting every rung by k rotates every heading by k·α, which is a rigid rotation about the vertical axis, and a rigid rotation conjugates the holonomy without changing its angle. So the reading depends on the relative rung offsets and nothing else. This is the Magnetic Forest's clone-gauge result arriving from a completely different direction — that paper measured multiplying-by-a-prime as a global phase on the beam; this measures it as a conjugation of a rotation

ties row 183

The cap holonomy depends on two rung offsets and nothing else — derived, then measured twice (213) — write r for the rung of n's largest prime and a, b for the offsets of p and q. The four legs of the closed quadrilateral are (u_a, a), (u_b, b−a), (−u_a, 0), (−u_b, −b) — they sum to zero, so the loop closes — and r appears only as a rigid rotation about the vertical axis, because the angle between the two horizontal unit vectors is (b−a)·α with no r in it. A rigid rotation conjugates a holonomy and cannot change its angle. Measured against that: seventeen integers from 1 to 44,521, primorial and not, spread 7.49×10^{-16} rad; eight different offset pairs give eight distinct values; and the closed-form legs agree with the address-walked quadrilateral to 2.8×10^{-16}. So the reading is a function of (a, b), not of n, not of the base rung, not of which primes

conjugation-invariance of a rotation angle

Eight of ten regularities never touch the axis you chose (216) — the Tower's central result, and it is a negative one in the best sense. Of ten measured regularities, eight are computed entirely from where the walk goes sideways — injectivity, the φ^2 rungs per revolution, walk length, the ancestry theorem's 13,187 walk grazings and the loop's 10 return-leg crossings (row 194), stiff spreading at 0.977, the frame-lock ceiling, the Fibonacci returns, the Liouville excess — and climbs never move you sideways, so they reproduce identically, not similarly under any vertical whatsoever. Only the octave law reads the vertical, and the p = 5 anomaly rides on it. Most of what was found is not an artifact of how it was drawn, and the way to know that is to redraw it

Only one vertical produces the picture, and that is a fact about the choice (217) — render the same data four ways: height = rung gives the tower; height = Ω collapses to about fifteen flat bands; height = log n smears into a wedge; height = n gives a featureless column. So the choice of vertical is doing real work even though eight of the ten

readings ignore it. A coordinate that changes the picture while leaving the measurements alone is worth naming as a presentation decision rather than defending as a discovery

HOLONOMY IS THE AREA ON THE SPHERE OF DIRECTIONS — confirmed to 9.4e-16 (223) — the walk's tangent traces a closed path on the unit sphere of directions, and the frame comes back turned by the signed area that path encloses. Gauss–Bonnet with the curvature set to one: holonomy is the integral of curvature, and on the unit sphere the curvature is 1, so holonomy is the area. Tested against the composed transports across fourteen loops spanning offsets (1,2) through (6,7): worst disagreement 9.4×10^{-16} rad. This turns the cap measurement from something an instrument reports into something computable on paper — a one-line signed solid angle over the four leg directions. Not the root of a^2 plus b^2; the spherical excess

Gauss–Bonnet; Levi-Civita holonomy on S^2; Van Oosterom–Strackee

Commuting is a fact about the axis, not about the phase (226) — the correction that turns a felt intuition into the theorem. Two rotations about the same axis commute at any phase — which is exactly why the forest's walls carry nothing, since every heading is a rotation about the shared vertical. Two rotations about perpendicular axes do not, and the difference between the two orders is a rotation about the third axis. That is what i, j, k are — quarter turns about three perpendicular axes, with ij = k saying that two right-angle rotations about different axes compose into one about the third. The cap holonomy is the size of exactly that commutator, vertical against horizontal, for one small loop. Quaternions are not a convenience here; they are the bookkeeping for this

quaternion algebra; non-commutativity of SO(3)

Three right turns make a left, and it is a theorem (227) — a spherical triangle with three right angles encloses one eighth of the sphere, area π/2, so an arrow carried around it comes home rotated by a quarter turn. Three right turns, one left, no cheating — and the left is the holonomy. The corpus's own curvature-sign figure already draws it: sphere, angles greater than 180°, arrow turns ahead. The best one-sentence demonstration in the whole inventory that curvature is measurable by walking, and it needs no apparatus at all

spherical excess

Turn the mathematical worry into a physical question, and the question gets better (232) — the load-bearing fence: everything proved is a two-dimensional planar result, and the proposal is three-dimensional, where the extension is not automatic. The status table separates THEOREM / NO / OPEN / THE ACTUAL QUESTION row by row rather than blurring them. Then the reframe: the mathematics is settled and it settles in our favour — what is open is physical, and it is a better question. The question stops being is it possible and becomes which physical probe reaches which form degree. A worry converted into an experiment is worth more than a worry answered

Hodge; Kac 1966; McKean–Singer 1967; Gordon–Webb–Wolpert 1992

Derive the margin instead of quoting it (244) — the same twin boundary written every way the point group allows: 86.300, 93.700, 101.631, 107.361, 137.212, 140.008, 180.000 — seven descriptions of one object. The smallest is 86.300, the runner-up 93.700, and the gap is 7.400° exactly. That is a fact about the symmetry group, not about the instrument: a better detector moves it by zero. Against typical measurement noise of 0.5° it is an SNR of 14.80, and that number was derived rather than borrowed

point-group arithmetic

The determinant is an address, not a resident (248) — one invariant value can name more than one object. det = 9 is shared by 6_1 and $3_1\#3_1$, and the pipeline printed 6_1 for a structure that is a connected sum. The number was right; the label was wrong. The fix is one line and is already house practice for a different value — hedge det = 9 as 6_1 or $3_1\#3_1$ exactly as det = 5 is hedged as 4_1 or 5_1. An invariant that is not injective is a postcode, and a postcode does not tell you who lives there. The determinant is also the part of a holonomy you can often know exactly. Round one beat of R2W's weightless junction with weight, the area a nudge spans shrinks by e to the minus the passing's time over the weight (Liouville's formula); the computed determinant sat on a rounding floor many orders above that and made τ meaningless at light weights. Divide by the exact value, never by the computed one (WEIGHT-CONE)

knot determinant / Alexander polynomial

Offer the replacement, then kill it in the same paragraph (260) — total curvature was tested as a mirror-symmetric alternative to the blind channel. Fáry–Milnor gives knotted ⟹ total curvature > 4π — but the condition is necessary, not sufficient, and unknotted curves already in the corpus reach 13.46 against a bound of 12.57. Proposing a fix and then measuring it dead in the next sentence is what stops a limitation section from becoming a wish list

Fáry–Milnor

Screen on the right curve — and the correction does not transfer (261) — 250 curves, zero knotted, 0.00% — measured on the duplex helical axis rather than on individual strands, because the backbones wind around that axis and read as long helices whatever the molecule's real topology. DNA topology is axis topology. And the sharper half:

the correction does NOT apply to RNA, which is single-stranded and folds back on itself, so its backbone is the curve. One substrate needs the axis, its neighbour must not have it, and getting that backwards silently changes every answer. “DNA is not knotted in the PDB” is a result, now measured rather than assumed

The bug’s leftover was a real quantity, and two independent routes agreed to 1.8e-13 (265) — a linking function divided by 2π where the two-curve integral needs 4π, so it returned exactly twice the linking number — verified on synthetics as 2, 4, 6, 8, 12 for 0.95, 1.91, 2.95, 3.91, 5.91 turns, all exact integers once halved. And the validator had been reporting False on all four test rows as shipped, so the identity never closed and nothing was built on it. Halving left a residual near 0.49 — and that residual resolved to the Bishop frame’s own closure holonomy: topological route 0.4877, geometric route 0.4877 from a frame-transport angle of 175.559°, agreeing to 1.8×10^{-13}, with no linking integral anywhere in the geometric route. The trefoil then decomposes as Wr = −3.5123, Tw = −0.4877, Lk = −4.0000: neither continuous term is near a round number and their sum is exactly an integer. Călugăreanu–White–Fuller demonstrated rather than cited

Călugăreanu–White–Fuller; Bishop frame

The classical prediction is right on average and cannot be right pointwise (266) — linking number across a half-twist ladder measures −4, −3, −3, −2, −2, −1, −1, 0, 0 against a smooth prediction of −4.0, −3.5, −3.0, −2.5 and so on. Every measured value is an integer to better than 10^{-4} and it cannot be otherwise — the linking number of two disjoint closed curves is an integer by construction, so a half-integer step is a category error rather than a small one. The integer absorbs the added twist in whole units, stepping at odd steps and holding at even, with a mean slope of exactly +0.5000. The integer-versus-continuous split, on a synthetic where the answer is known independently

integer-valued linking number

A thing that rings at its own pitch has an address whether or not you wanted to give it one (271) — the address came free and was not expected, and it holds in all three substrates the paper visits. A resonance is an identifier, so an object that can be made to ring is an object that can be named without anyone designing a naming scheme. It sits beside row 234 (“The address closes and the datum closes, and they are the same walk”) and row 47 (“The read head is a theorem”): three separate papers arriving at the same fact, that a spectrum is a name

ties rows 47 and 234

Do not fight the medium — catch what is already moving, and lean (272) — the design posture, stated once and held across an ocean, a body and a vacuum: what the medium offers changes completely, and the move never does. Bias the outcome rather than overpower the flow. And the attribution is immediate and unprompted — neither half of that is ours — the branching growth form and the fact that the same object has a resonant frequency are both somebody else’s, named as such in the sentence that introduces them

The wall you could not cross is the load-bearing wall of the thing you built (277) — a vehicle that woke up and began evolving mid-voyage would arrive as something other than what left — the one failure a delivery vehicle is not allowed. So faithfulness across eons means cannot change, which is precisely the wall the companion paper hit and reported. That shortfall is not a shortfall; it is the specification. The strongest possible form of row 268: the failure is not the disappointing edge of the work, it is the beam holding the roof up

The loss column turned out to be the offspring (279) — strays that never return after a run were being tracked as a recovery budget, a number to drive down. It is not a loss column. A unit that falls somewhere hospitable simply grows — nothing decides anything, the discipline holding even on the last page where it would have been easiest to break. That makes the strays a population under selection by landing site, which is selection in the exact sense the framework paper defined: the environment writes by filtering which variants keep going. No individual ever knew where it was. The population knows. And the provenance is worth as much as the result: we were working out a recovery budget and it fell out of the arithmetic — which is, in our experience, how the good ones always arrive

selection by filtering

We did not choose that number. We counted it (285) — two letters in two polarities give four states and a silence — and the silence is not a zero in the data, it is the boundary and the unwritten operator. Drop one symbol and the codespace collapses, which sets a hard ceiling on the emergency vocabulary that falls out of the geometry rather than out of a design meeting. A constraint that is counted rather than chosen cannot be argued down by a stakeholder, and knowing which of your numbers are which is most of engineering judgement

The record is pure geometry, so the reader is interchangeable (286) — light, X-ray, microwave and acoustics are four valid needles for one groove — chosen after the fact and upgradeable forever. Substrate-independence of the reader, not merely of the record, and it is a consequence rather than a feature: if what is stored is a shape, then anything that can resolve that shape can read it. A format that cannot go obsolete because it never specified its own reader

ties rows 47 and 271

Do not fight the gradient — turn until it goes to zero (295) — a monolith fights a tidal gradient with structure; a swarm has no structure to fight with, so it does not fight, it orients. The gradient points toward the well, and that direction is the one thing in the sky every unit can feel for itself without being told — no clock, no map, no message. Pose the array along it and the tide that would have sheared the aperture apart resolves along the array's own geometry, where it cancels rather than accumulates. Nulling a signal by turning rather than by overpowering, which is pose used as an instrument. And the figure carries the rest: two units are dead, the aperture thins, it does not fail

A monolith has one aperture forever; a swarm's aperture is a parameter (296) — the array widens or narrows its baseline on command, trading resolution against field of view the way a single dish never can — an instrument that reconfigures for the measurement instead of being built for one, with the gravity-nulling pose holding phase-lock while it flexes. This turns row 160 inside out: there, matching the aperture to the source's dimensionality was a constraint that produced a clean and misleading null; here it becomes a control knob. The same fact, read as a limit in one paper and as a degree of freedom in another

ties row 160

Boring is a superpower, because boring can be measured exactly (300) — the image the whole programme runs on. It is pouring rain at night; the rain is even and statistically identical, and that is the gift. Someone invisible crosses the yard and where they stand the rain stops landing. You never see the person — only the hole they cut in something even. Detection by invariant-breaking: do not look for the thing, look for the dent it makes in books you already know how to balance. Everything downstream is the engineering of which books to balance and how cheaply

Five channels, and each rung catches what the others are provably blind to (302) — winding, continuous amplitude, the coherence resultant across many sensors, the framing twist, and the linking of field lines through the loop. Design rule: the integer gates where the continuous is measured, the coherence watches for what no gate trips on, and twist and linking read the handedness and knotting the lower rungs cannot see. Non-redundancy measured rather than argued: with in-plane curl held identical, circulation separates two cases at 0.15 sigma (blind) while twist separates them at 15.5 sigma; and holding circulation pinned at 1.96 and wrapping at 0.78, the linking readout still moves monotonically through 0.00, −0.064, −0.124, −0.176. A degree of freedom exists that the lower rungs cannot reach

homotopy ladder; Calugareanu; magnetic helicity

Predicted from our own thread, then checked — and the contrast closes the loop (321) — odd-degree modes sit a constant offset from the midpoint of their even neighbours, with scatter of 0.4 percent of the large separation. The result is standard theory — but it was drawn from this programme's own half-step and double-cover thread BEFORE being checked against the data, which is what makes it a prediction rather than a lookup. And the contrast is the finding: an open core gives a mixed forest and a broken ladder; a sealed core gives a pristine ladder rigid to 0.4 percent. Filed exactly as the paper files it — a conceptual resonance of record, explicitly not a topological measurement claim

asymptotic mode theory

Single master against replicated, run as a live experiment by evolution (330) — one authoritative map holder is maximal fidelity — one copy, no version conflict, no consensus problem — bought at maximal fragility. Priced from the literature rather than asserted: post-reproductive females lead in lean years, and a mother's death raises adult-son mortality roughly fourteenfold. And the recovery question is posed and deliberately left open — when the master fails, from where is the state rebuilt, and does a partial copy in a neighbouring group serve? The paper marks that as unknown in the literature and keeps it posed. This is precisely the question a storage architect asks about a replicated system, and biology is running the trial

distributed storage; Brent et al. 2015; Foster et al. 2012; ties rows 95, 424

Neither arch spans alone, and naming the keystone is the result (332) — climbing from the other side reaches the same wall, and the paper's output is the name of the stone the whole span pushes against. The crystal is a durable substrate that struggles to live; biology is a living substrate that struggles to endure. So the bridge is not one imitating the other — it is the two coupled, each supplying exactly what the other lacks. Same move as the vessel's wall-as-load-bearing-wall: a paper whose deliverable is a correctly-located impossibility, rather than a crossing it did not make

ties rows 268 and 277

Simple witnesses in complex configurations, not complex witnesses in simple ones (342) — a new axis for the guide, and it is about the witness rather than the field. Enormous numbers of individually simple test particles arranged in a complex field is a home-team substrate — read at the 0.1 percent level. Individually complex, structured, jittering objects are an away game, and the amplitude channel suppressed their signal. The faint sources sit closer to the focal plane precisely because they are nearer to ideal fixed points. What the instrument reads is the arrangement, so the witnesses must be simple enough not to carry their own weather

One pin, no loop (343) — holonomy requires a path along which state is transported and around which closure can fail, so a single steady source with no differential offers nothing to walk. This is a loop-existence requirement, not a distance or sensitivity limit, and it delimits where the instrument can be pointed at all — a prior question to every other question the guide asks. Five words that tell a reader in advance whether their problem is in scope

The quadrature test: the same oscillation a quarter-cycle apart IN TIME (350) — the discriminator that separates real quadrature from mere perpendicularity, and it settles the programme's own instinct about quadrature. Same oscillation, quarter-cycle offset in time means deep quadrature that carries phase; perpendicular in space with no time offset is geometry, not quadrature. And the fence is what proves it is a real test rather than a label: in a travelling light wave the electric and magnetic fields are in phase and are not a quadrature pair — they are ninety degrees apart only in a standing wave. A test that rules out the most tempting example is a test. The test reads only the gap between the two knobs, never where either one's zero is called (the Loop Atlas sessions). A quadrature encoder is the two-knob engine run backward as a sensor: two signals a quarter cycle apart count a shaft, a half-cycle move between samples loses the direction (the decoder flags a double transition), and the once-a-revolution index pulse is the closure that catches a miscount. The needle turned by two crossed fields is an encoder driven as a motor (established engineering, from the two reference guides that Dancing on the Sun cites in its section 14) (the Loop Atlas sessions)

Circular motion is quadrature made mechanical (351) — velocity runs along the tangent and acceleration along the radius, ninety degrees apart, always. And the centripetal force therefore does no work — perpendicular means zero work — it only turns. So the quadrature partner of momentum is the thing that changes direction without changing energy, which is the cleanest physical picture in the corpus of why a phase channel and a magnitude channel are different animals. A ringer, anything that swings back and forth on its own, needs one refunder of each kind, a store that hands back what is put into it: one for position and one for motion (a spring and a mass; in a circuit, a capacitor and a coil) (standard, from recall). And turning and carrying are one engine: a needle turned by two crossed fields and a bead carried by two shifted standing waves gave the same numbers, nothing in step, a whole number per cycle with any gap, and the same slip past the pace limit; in thick fluid both reduce to one equation, and the carried beads also gather into a bunch (the radio rotor and radio conveyor toys) (the Loop Atlas sessions)

settled mechanics; Bear's catch

Three of the corpus's laws turn out to be one law (353) — edge-of-lock (coherence lives in the slippage) is true because work equals phase gap, so zero gap is zero work. The decoupling (angular momentum carried by the pattern rather than by mass) is the pattern working on the medium exactly where they are out of phase. Rate not level is true because the rate of phase change is the work flow, and a static phase does nothing. So the recurring quarter-loop is not merely a geometric quarter — it is where the work is maximum. Three separately-earned laws collapsing into one is the strongest kind of internal evidence a programme can produce

ties rows 105 and 294

A spinor is a closure rule you already have, not a dimension you add (357) — a vector comes home after one turn; a spinor needs two, because the rotation group is doubly connected and a single loop in orientation space cannot be shrunk to nothing. And Bear had drawn it three times before it had a name — the Möbius edge that needs two laps to close, the trefoil that will not undo in one pass, the belt trick. All the same object. You do not add a dimension or a cone to get a spinor; the fourth dimension appears only as the clean room where these sit without self-crossing, which is exactly why handedness wants the extra dimension

double cover; unit quaternions

You still land on integers — you just count in spinor turns (358) — the objection stated bluntly: no maybes, no fractions — what are you going to land on, one quarter? And the resolution: nematic director defects genuinely carry half-integer winding, permitted only because a director is a headless line living on the same double cover. So the half is not a fraction you land on; it is a whole number in the right units. The integer rule survives intact once the counting units are the spinor's, and the exception turns out to be the rule's own double

soft-matter defects; double cover

Coordinate-free means reading DIFFERENCES of an INVARIANT (359) — the standing law in its final form, and it explains the whole programme's convergence. Space-independence comes from reading differences between nearby points, so any uniform background cancels. Time-independence comes from reading a topological invariant, an integer that cannot drift because an integer cannot change continuously. Together they give a reading that depends on neither where nor when, only on structure — and that is why the work kept converging on knots, integers and gradiometers. They were three routes to the same read

Lattices fuse cleanly only at discrete, physics-forced angles (361) — stack two lattices at a random angle and few sites coincide, so the seam is strained and high-energy. At the coincidence angles they fall into periodic registry and the seam is seamless. The keystone as a crystallographer's angle: closure forces the angle — you fuse at the

magic angle or you strain. With the fence written in the same breath: the specific number is not derived, coincidence angles depend on the actual lattices, so derive the angle from the real lattice and do not assume it

coincidence-site lattice; moiré

Hexagons store, squares address, odd polygons signal (363) — each shape gets the job its symmetry fits. The hexagon is zero-curvature and axis-less, making it the dipole-immune storage substrate. The square is also flat but four-fold with a clean axis, which is what an addressing raster needs. And odd polygons carry the curvature — plus one and minus one — so they are the signal defects embedded in the neutral sheet. Even polygons are neutral carriers; odd ones are where you read. Three roles, three symmetries, no overlap

graphene disclinations

Rim sees the mouth, never the neck (374) — why a surface loop is structurally blind to the vortex inside it: a loop reports the cap total, dominated by the wide storm-scale spin, and cannot dive into the tight throat where the energy actually concentrates. Priced by the geometry: a four-kilometre core collapses below thirty-to-sixty-kilometre station spacing, so the loop watches the mouth tighten and never sees the neck close. This is row 160's aperture law in its sharpest physical form — not a sensitivity problem, a containment problem

Stokes on the cap

The medium decides whether you get the integral for free (375) — rock is a low-pass filter that integrates strain for you; air is high-pass with no quiet baseline. That is why one domain is the easy case and the other is not — the physics is no simpler, the substrate simply does the accumulation on your behalf. A portable question to ask of any new substrate before designing anything: does this medium hand me the integral, or do I have to build the integrator?

Fill the rungs and you approach the sphere — and that is literally a theorem (378) — a filled shell sums to a perfect sphere, so the inert elements are the round, complete rungs and chemistry lives in the lumpy partial ones. The polygon-rung intuition is not a metaphor here; it is the angular structure. And the honest wait is attached in the same breath: filling requires exclusion and half-integer spin — the 720-degree closure — which is the ingredient a bare polygon does not have. Also the correction to the word periodic: roundness recurs, so it is a spiral staircase rather than one ramp

Unsöld's theorem

The pentagon gives you only half the optic (382) — positive curvature converges, focuses, closes — the intuitive defect, and the one the corpus reaches for. Negative curvature diverges, defocuses, opens, and does the thing its partner cannot. So a sheet of one sign is only ever half an instrument, and an alternating array of both is a curvature-modulated surface: each positive cell a converging element, each negative one diverging. Back to back they make the compensating pair. Reaching only for the sign you find intuitive is a design blind spot with a name

disclination signs

Knotting adds a second protected integer (383) — tying a loop into a knot makes the phase close around a knotted path, which adds self-linking and writhe on top of the plain winding. Knotting is more lock, not less — the eigenvalues snap to a richer discrete set, modes on a knot rather than modes on a circle. And the knot cannot live on the flat disk: it must cross itself in two dimensions, and lies cleanly only on the torus the unit circle sweeps. The circle alone is too thin to hold it, which is why the tube was always the right object

torus knots; self-linking; ties rows 143, 147 and the July capture

One instrument, seven faces (388) — not seven machines: one coordinate-free winding reader aimed at seven substrates — field, sky, slide, time, lattice, sky-ghost, genome. That is why every thread kept converging: same instrument, different prey. And the ranking inside it is honest rather than flattering — the biological face is named the most actionable, because it is a curvature-and-winding anomaly read against a periodic background, which is the cleanest version of the whole method: read the defect against the lattice

The domain map: does the medium hand you the integral (391) — row 375 promoted from an observation to a map across every domain the programme might enter. The instrument does cheap useful work wherever the medium is low-pass and therefore does the integrating for you; where it is high-pass with no memory and no quiet baseline, you must build the regulator yourself. And it explains the corpus's own hit rate — the tectonic thread produced results and the atmospheric one nulled, for exactly this reason and not because one was studied harder

There is no free invariant, and the blind spot is the receipt (408) — losing a dimension buys exactness, no analysis, no gauge choice and a countable grading — and costs you everything abelianisation forgets. The deep material lives entirely in the non-abelian sector. That aisle is not stocked at abelian prices, and pretending otherwise is the one thing this chapter will not do. And the law stated at the level of whole theories: every geometry forces one invariant and pays with exactly one blind spot, and the blind spot is the receipt for the invariant. The sharpest form of conservation-of-difficulty in the corpus, because it is said about mathematics rather than about an instrument

ties row 303

Its ceiling and its exactness are the same fact (409) — why an abelian corner is a permanent flatland worth trusting rather than a temporary simplification: its group is abelian by nature, so it can never wander into the expensive aisle, and never has to. The constraint that caps an instrument is the same constraint that makes it exact — so a limit you cannot exceed and a guarantee you can rely on are one property seen from two sides, and an instrument that could escape its ceiling would lose its warranty with it

A coincidence is never alone — it is a rung on a ladder you can write down before looking at data (414) — it has siblings, they are evenly spaced, and the ladder is generated by a single number computable from the constants themselves before any observation is consulted. How far it runs, where its one gap sits, and what it would take to make it short enough to argue about are all calculable in advance. That is what turns “is this coincidence meaningful” from a matter of taste into a measurement — you enumerate the family the claim belongs to, and the claim’s rarity is a count. Predict the next member of a family before computing it: the quarters, fifths, sevenths, sixths and eighths were each predicted from rules found on the members before them. A new fraction is the cheapest out-of-sample test a family offers, and it is honest only if its bars are written first (R2W). And check a law on the points it was not read from before saying it aloud: a law read off a fit’s first five points was given to the person before the fit was checked on the points past them (the Loop Atlas sessions)

The comma is an abelian holonomy — a lift that fails to close (419) — not a metaphor for holonomy, the corpus’s own trunk object. Twelve fifths measure 7.0196 octaves against seven, so the walk closes in pitch class and its lift to the line does not — it comes home 23.46 cents out. A circle-valued state whose lift does not close, which is exactly row 69’s candidate language and exactly the Integer Forest’s closure deficit. And row 408 explains why it is exact rather than approximate: it lives in the abelian corner where the deficit is a single additive number, order-independent — and that corner’s ceiling and its exactness are one property seen from two sides. The same object turns up in learning: teach lesson A then B, or B then A, and the two routes land in different places, apart by the step squared times the Lie bracket, a lesson comma. Question the Substrate began as an attempt to use it as an instrument, and its comma did not rank a batch’s poisoned labels (rank correlation −0.13 against a registered 0.6 or more, run E5). The winding number its network counts is the same idea in whole turns, and Two Arrows’ area law, the part of a learner’s leftover that flips with direction doubling when the loop’s area doubles (log-log slopes 0.9545 to 1.0092, five of five seeds, run s1), is the comma’s closed-loop form (Question the Substrate; Two Arrows)

ties rows 408, 410

Life is a constant check on the security of its substrate (423) — a living thing spends energy to hold an internal model against decay, so homeostasis is error-correction, and the decay clock is the metabolic cost of being alive written as a rule: stop paying attention and reality reclaims your map. From which the operational question falls out: when you cannot be certain — and you never can — how do you place trust well enough to survive? With the answer learned by losing: trust everything and the lie takes you; trust nothing and you build nothing and starve anyway; trust one source totally, however wise, and the day it is wrong your whole world is wrong with it

The evidence that two instruments are one object is a shared FAILURE, not a shared success (438) — the corpus’s own claim is an identification — several tools you already trust are one instrument — and identifications are not proved by successes. A success can be engineered: point a winding-hunter at almost anything and something winding-like can be made to work, which is the apophenia risk the whole programme fences. A failure mode cannot be built in on purpose, because it costs you. So the strongest support here is the set of places where the same specific failure turned up in unrelated substrates: coherence-is-the-symptom arriving three times — a control loop defending a corrupt setpoint, a pipeline honestly wrong on every seed from one transposed character, and a game board whose whole map descends from one bad anchor; rate-over-level three times; agreement manufactured by a shared arbitrary convention twice in one week; benign correlated failure three times. Two instruments that fail the same way at the same place are reading through the same bottleneck. And the confound is named rather than hidden: the arrivals are independent in substrate and mechanism but NOT in author, so whether the bottleneck belongs to the substrates or to the reader is decided by exactly one thing — row 435’s kill condition, a stranger hitting the same wall

ties rows 70, 250, 424, 105, 294, 435

The null, run by nature, in protein (446) — if the negative were an artifact of crystals or of simulation, biology would not echo it — and biology echoes it precisely. A prion is heredity carried in structure and it evolves almost not at all: a replicator that is all fidelity and no creativity. That is our null, run by nature. The prion is not a counterexample to our wall; it is the wall, walking around in a biological body. Two more complete the triangulation: a spore is a frozen record awaiting an external spark — precisely our silent-until-read — and a virus has the creativity the crystal lacks and lacks the autonomy the crystal has. Three biological objects bracketing a simulated null from the outside

Prusiner; ties row 306

The cheapest store is the one where the stored thing is already what somebody wanted (482) — you never get the electricity back — and that is the point. You did not need electricity. You needed cold and water. Two independent routes to one number: a deployed off-grid system reporting about 100 kWh per tonne of cooling against this paper’s

93. And the durability argument beats the cost argument — a case study reports the batteries vulnerable to cycling damage and unable to be fully discharged, while the thermal store could: a battery degrades by being used; a tank of ice does not. And the second-order effect is larger than the first: halving a load that is forty percent of demand cuts total load about twenty percent, so the same generator serves a quarter more households or shrinks by a fifth

A table that locates a mismatch without needing an argument (490) — eight functional stages down the side, two populations across the top, and the finding falls straight out of the layout: for one population the prosthesis replaces the stage that failed; for the other it replaces a stage that had not failed — while the two rows that actually broke are the two the device does not have. The cochlear implant is a correct and elegant answer to the left column. Laying the chain out by stage turns a clinical mismatch into something a reader sees before reading a sentence about it

Four independent constraints selecting the same product, none of them aimed at it (499) — nerve dependency, a power budget putting one architecture at roughly 1.1 W against a comparable device's 30 mW, a thermal limit forbidding continuous wear where the user cannot act as the sensor, and a regulatory path favouring a device making no therapeutic claim — all converging on an instrument rather than a prosthesis. Four independent lines of argument selecting the same product is the strongest structural result here, and none of them was aimed at that answer. The design-side twin of row 438: unaimed constraints agreeing is evidence in the same way that unaimed failures agreeing is

ties row 438

A WORD THAT CROSSES A DOMAIN BOUNDARY CARRIES AN ASSUMED DEFINITION, AND THE ASSUMPTION MUST BE VALIDATED BEFORE THE WORD CAN CARRY A CLAIM (502) — Cross-domain words like depth and moves mean different things on either side of a fence. Those two are the corpus's own canonical cases and they lead; what follows are eleven more, eight the scrub priced and three met since without recognising what they were instances of. Writhe: the theorem is about the reduced knot diagram, the measurement was the open-arc writhe of the backbone, both correct about different objects, and the pre-registered prediction failed by a wide margin because the word crossed and the definition did not. Hypotenuse: flat chord against full reach, and the sign of a headline result flips with the choice. Holonomy: the vector-area channel answered no and the rotation channel answered yes, and both were right. Determinant: one value naming two knots, so the number was right and the label was wrong. A chain identifier: two namespaces, and a citation a reader cannot follow. Beat: applied across two media that cannot mix, with no coupling ever proposed. Topological: used, tested harder, withdrawn at 52% on honest data. Specificity: a design target satisfied by construction because the word denoted something other than intended. Three-halves energy: the name of an integral, field strength raised to the three-halves power, read as a proved scaling exponent, when the proved floor is proportional to the asymptotic crossing number. Azimuth: HMI counts Stokes Q with the opposite sign to the textbook Milne-Eddington model, so its azimuth is the model's plus a quarter turn, and a comparison wrapped into ±45 degrees folded the difference away. Beta: in one session's own record, one letter meant a damping number (one over the square root of the McCumber parameter, so that 0.2 is a McCumber parameter of 25), the swing of an FM drive, a two-way switch in a gear train, and the McCumber parameter itself. That project now writes βc for the McCumber parameter and keeps plain β for the swing. The procedure is cheap and would have caught all eleven: before a word carries a claim across a boundary, write down what it denotes on each side and check the two denotations are the same object. This is the dictionary's operating rule, not a caution beside it. When you catch an error in another session's page, look for it in your own: the Loop Atlas found that R2W's page called the rider "the same machine" as the radio-driven junction, the same error sat in this guide's own drafted row 727, and only the search for it found it. A correction made in one sitting had not reached the next sitting's page, in either record (the Field Guide sessions). Two more beyond the eleven, from the triple-star thread. A session's habit is not a reason to split a word the guide has already settled: the thread had kept "two clocks" for the island's lock, where the guide keeps "two clocks" for two time scales of one process shown not to scale together and calls clocks tied by their own events, locking where they agree, "coupled clocks"; the paper took the guide's word in five places and defined it in its first section. And a word that means something else in the field is retired: the paper had called a slice's highest point a "fold", which in dynamics is a saddle-node bifurcation, and theirs was nothing of the kind, so the place was renamed for what it is, and threads, walls and rooms were defined at the start because "wall" already had two senses in the thread's sister books (the triple-star sessions). One reading does come early, and it is the exception to building from scratch (row 506): not other people's results, only their words, so that a name the toy gives is not a false friend. Bear agreed it on 16 October and set the balance: respect the local vocabulary without being owned by it, since even inside one field authors switch and swap their words. A picture can serve every toy, as a dancer does in all three toy books, so long as each says how it is defining the word in that instance; in the triple-star reads the dancer is the light star, and in R2W's a run at a tempo (Bear's ruling)

ties rows 87, 193, 215, 248, 252, 263, 450, 453, 552, 554

One blind spot, two substrates — the unification is by impossibility (510) — the theorem's own framing, and it is the sharpest statement of what this corpus claims. Two fields that never talk — distributed-systems trust and discrete differential geometry — have the same concealment failure, and it is provable. The paper's job is to make a

reader from either field say: oh, that is my problem, in the other field's clothes. And the stakes are what a floor is for: it tells builders in both domains exactly what no detector can do, and, via the residual, exactly where the remaining danger lives — stop trying to catch static balanced lies with sums; spend your effort on the adaptive case, or on pricing it. Row 438 said independent arrivals at one failure argue two instruments are one object. This proves it for one pair

ties rows 117 and 438

Full shell against partial shell — the most predictable thing in chemistry (515) — a closed shell is stable, symmetric and inert; a partial one is reactive and lopsided, wanting to grab or give to reach closure — and how partial tells you how reactive. The noble gases barely react; the elements one electron from full, in either direction, are violently so. That is literally the periodic table, and it is the winding-must-close logic standing in the place where the word quantum was first earned. Full against partial changes behaviour predictably — emphatically yes, and it is the hinge of the atom door

Cousins, not twins — a deterministic map and a statistical distribution (518) — both are about gaps, which is exactly why they feel like one law and exactly why they are not. The inequality is deterministic: a fact about the map, always exactly true above one, with no randomness anywhere in it. The repulsion is statistical: a fact about a spacing distribution, the vanishing probability of two points landing together. Fuse them and you would be claiming the logarithm causes the repulsion — which is false, because the repulsion comes from the determinantal structure and not from the log. Row 502's rule turned on two of one's own ideas rather than on a borrowed word, which is where it is hardest to apply and most valuable

ties rows 7 and 502

A clone is one global phase, and the beam cannot see it (521) — because the logarithm of a product is the sum of the logarithms, every term of a prime's family picks up the same extra angle — so cloning a family is a gauge transformation, invisible to the instrument, verified to 9.9×10^{-12}. The rhythm is not a property of the prime at all; it is the factorisation of one, two, three, four playing out identically for every prime that exists. The only visible difference is the aperture lost, which is the log of the prime — measured at 24.2216 against a predicted 24.2019 for an eleven-digit prime. And My own sentence for it, which is the row: the music is identical, only the length of the performance differs

gauge invariance; ours; ties row 183

Which bins exist is arithmetic; which bins survive is aperture (523) — Bear's reading, and two of its three parts hold. The binning rows say there is a window between too fine and too coarse. Rows 520 and 522 say something stronger: you cannot sit anywhere inside that window — the available bin sizes are quantised to the divisors of the period, and of those, only the ones whose repeat fits the standing aperture survive. So the window has two edges and they are different kinds of thing: one arithmetic, one geometric. The prime factorisation is the number line's own binning, and the aperture decides which of those bins you are allowed to read. The third part — that this is also the chaos crossover — is not shown here: no mechanism has been named for it, and it is held as a candidate rather than filed as a reading

ties rows 329, 442, 520, 522

The two mechanisms compete rather than stack (529) — the inversion that keeps this from being one tidy story. At half flux every phase factor is plus or minus one, the matrix turns real, and time-reversal is restored — a property of the operator, holding at every size, confirmed by a level-ratio statistic running 0.489 at zero flux, 0.59 to 0.65 at generic flux, and back to 0.440 at half. But on every composite periodic lattice tested, that same q = 2 is the WEAKEST divisor tooth — below baseline on three of five — while on an open patch it is the only survivor. Half flux is strongest where there is no wrap and weakest where there is. Two real mechanisms meeting in one place, and where they meet the commensuration loses. Whether the half-flux inversion is more than the known π-flux gap closing on a square lattice is stated and not resolved

Oganesyan–Huse; symmetry class

The dial is re-bought in every medium (532) — the same latch-and-battery architecture appears in two substrates with the geometry dial running opposite ways: a ring lattice's persistent-current amplitude falls as one over the circumference — slope −1.0004 over a sixteenfold range, with amplitude times size flat at 1.6733 — while a vortex ring at held circulation carries energy that rises with radius. Same architecture, opposite sign. So an architecture ports and its scaling does not — the dial has to be re-measured in each medium rather than carried across with the structure

practice; ties row 317

The substrate is a piece of real estate in the fiber bundle network (535) — grab a chunk of real estate — terrestrial or cosmic — and the fiber bundle connections between all points in it are assumed to exist. Data agrees or falsifies or is silent about each connection. The question every project opens with: where are the magic scissors to

open this piece of real estate and expose the connections we can splice into and read? A perfect sensor array for holonomy would be magnitude-blind, and with sufficient granularity would render an approximately accurate visualisation of the geometric reality — the fiber bundle network plus its drift. Without dynamic data the loop walk does not go well; you learn a great deal about what not to do. Usually only data other people would use for other purposes is available — we use it for its own domain purposes AND to expose the geometry, and those are the same data doing two jobs

The bundles are always there — what varies is our ability to read them (540) — Bear's correction to Station 2, 2026-09-15, and it changes the station's question entirely. The fiber bundle network between all points is assumed to exist — the substrate is not absent, it is always present. What varies is our ability to access it, and that depends on two things: the available sensor data, and the geometry. Asking "is there a bundle here?" is asking the wrong question. The right question is: do we have the sensors and the geometric exposure to read the bundle that is there? This reframes every null at Station 2 from an absence of structure to a gap in instrumentation or a geometry that has not yet offered a vertex, junction or seam to read. Not dead — unmapped. And unmapped is always recoverable

ties rows 535, 536

Static and empty are different things (543) — a single static integer denoting existence, with nothing to compare against, is what the method forbids — holonomy as a comparator, not a classifier. But the discomfort is the guide working, not the substrate failing. A nebula's dust polarization map is a director field: polarization angle is defined mod 180°, not mod 360°, so a loop's winding is quantized in halves — ±1/2, ±1. That is the nematic liquid-crystal defect classification, and it means the nebula does carry a walkable π_1 field with genuinely quantized defects. The number is not static-and-empty; it counts defects across space rather than across time. Same lesson Cascadia taught when the standing texture survived the vault blind. Halves are real and cost half the margin. A headless line counts in half turns, halves add like whole numbers, and a read that sees them must take each step within a quarter turn, so it tolerates half the noise an arrow read does (the Latch Atlas; ties row 358). On a real line field, Dancing on the Sun's headless count of HMI's azimuth moved 104 counts against the arrow count's 31 under the same planted turning: the headless read has half the margin, and on real maps it paid for it at 3.4 to 1

nematic defect; $\pi_1(RP^1)$; ties rows 69, 282

The Arnold floor is set by helicity — writhe alone is not enough, and this code models only half (547) — CORRECTED by ED-1 and ED-2 review, 2026-09-16. The Arnold floor is set by helicity = $\Gamma^2 \times$ (writhe + twist). For a thin filament, helicity is the sum of writhe and twist weighted by circulation squared. The core's twist is not modelled in this code at all, so the floor cannot yet be read from this tool — only the writhe half. The knot invariant that certifies the ring is still a trefoil after a reconnection is the Alexander polynomial (at -1, the knot's determinant, it gives 3 for the trefoil and 1 for the unknot), not the writhe. Three fences that must travel with any F7 claim: (1) row 422 applies — topology preserved by construction means the rig check, not the result; the result is what writhe and helicity do once reconnection is added and the knot can cut. (2) Writhe is not the latch. (3) Writhe is not the knot invariant. What this code does prove is modest and real: a filament evolver with a Gauss-integral writhe reader now exists, costs seconds, and the reader is the same the DNA spoke already uses. That is the route to F7, which is what was claimed and no more. The bare filament sets no floor at all (row 550). Where one can live is a tube of conserved flux and volume whose strands cannot overlap, so that the rope tightens against itself; the proved floors of Arnold, Moffatt, and Freedman and He live there, and reproducing them is the test the next instrument must pass. How the latch opens — an explicit cut-and-rejoin rule — is the reconnection wall, and it is the write-head question

Arnold 1974; Moffatt 1969; ties rows 86, 87, 545, 546

Matter latches and waves label (561) — four questions sort the Latch Atlas's thirteen media cleanly. A whole number that costs energy to hold, behind a wall higher than heat's jiggle, is a latch; a whole number that flies away holding nothing is a label. Twisted light, radio and sound carry labels, and they become latches only when they land on matter whose phase must close on itself. Ask of every new medium which of the two it offers before asking anything about memory

The Latch Atlas; ties rows 43, 283, 293

The viewing angle is a free witness, and it testifies both ways (576) — the Sun carries each region past the telescope over about two weeks, so every region is seen from a run of angles at no cost. The rotation fades toward the limb, as a rotation made in the instrument's frame does and one in the Sun's frame does not; the two-frame read gave about 4 degrees about the line of sight and about none about the local vertical. The same witness bit back: the tracked regions could not settle the frame, which the paper leaves undecided, and two large regions of 2024 changed the rotation's sign across the meridian while thirteen did not. When the substrate carries the object past the instrument, the trajectory is an expert witness already retained; call it, and expect it to testify against you too

Dancing on the Sun; ties rows 306, 505

The archive’s own numbers are witnesses already retained — including against your own sentence (577) — SHARP publishes MEANJZD, TOTUSJZ and SAVNCPP with every map. MEANJZD reproduced the home-made mean current to a ratio of 1.000 on maps from 2011 to 2024 and carried the rotation’s upward part; TOTUSJZ, being unsigned, could not; SAVNCPP carried it on 82 regions and not on six tracked ones, so a claim first made as general was narrowed in the open rather than dropped. A ratio of 1.000 against the archive’s own keyword is worth more to a reader than any description of the tool

Dancing on the Sun; ties rows 93, 507

Headers are the instrument’s diary (578) — reading every header line of every record as data turned up what the maps did not show: a 0x10000 quality bit, read in the headers as marking calibration sequences, on records that Kosovich and co-workers’ rule leaves out (they drop QUALITY of 65,536 or more as unreliable Stokes vectors); a QUALITY of 128 on NOAA 12529 on 14 April 2016, the same day the inversion code’s name changes to vfisvcombine; and a single 4096 on one map of NOAA 13780. None was under test. The metadata of a public archive is a time series in its own right; read it whole before reading the maps, and set every change of code version or quality flag beside the dates of your tests

Dancing on the Sun; ties rows 153, 309

Treads live where phase is carried forward (605) — a latch erases phase at every stroke and leaves no staircase: a field winch driven past its limit showed none. A rider with no latch, on two rhythms, locks into treads: a slipping conveyor with a second rhythm showed treads at slip-over-beat 1 and 2 only, across three surveys of 4,000 points each. Before hunting lock plateaus, ask whether the mechanism keeps phase between cycles. Measured: reading the largest Lyapunov exponent as a forgetting rate, every run on a tread forgets its start (direct drive in about 1 time unit, first gear 2, second gear 20) and every run between treads remembers (within 0.002 of zero). And a tread’s edge carries the rule that found it: first gear’s floor read 0.825 under one hit rule and 0.810 under another, so an edge is quoted with its rule (the Loop Atlas sessions, R2W). Read a staircase of locks by its mediants: swept from rest to pace 1.10, the remembering pulsed needle (memory 1) locks at 2/3, 3/4, 4/5 and 5/6 in wide steps, and between each neighbouring pair the next widest step is their mediant (5/7, 7/9, 9/11), then the mediants of those (11/14, 13/16, 14/17), with chaos between; its turn rate never steps back (0.667 to 0.834). Where the forgetting needle’s map is a Möbius map and allows the whole-number step only, one unit of memory gives a devil’s staircase ordered as the Farey tree orders fractions; least repeats along the knob show it where the turn rate alone shows a smooth climb (run GEAR_TRAIN 20261013_170243; the complete devil’s staircase of circle maps is Jensen, Bak and Bohr’s, 1984, and fractional Shapiro steps in a radio-driven junction are studied in Raes and colleagues, 2020; the Farey tree’s ordering of the steps is named from recall) (the Loop Atlas sessions). Its steps shrink to 0.447 to 0.567 of the one before (run STAIR_WIDTHS 20261013_184033) (the Loop Atlas sessions)

the Loop Atlas sessions; ties rows 38, 562, 565

A lie hides exactly where the frame can flex without the lying bars (623) — over 3,360 sets of one, two or three bars in four frames, a set could lie without a trace exactly when the frame without those bars could flex, with no exception. A junction that tells the same lie to every bar it reads is caught 0.002 of the time when each bar is read once, and every time when each bar is read from both ends. A lie told alike to everyone needs a second witness

the triple-star sessions; the Byzantine generals bench; ties rows 26, 84

Lengths cannot see handedness; angles can (624) — seven bar lengths fit four different pentagons, because each triangle can be laid down or mirrored across its diagonal; the three folded shapes keep all seven lengths to 2.2×10^{-16} and miss the true corner angles by about 2.5 radians. With angles read at the bolts of a bolted star, a lying third of a stick is named every time, against 0.183 of the time with lengths alone. A measurement blind to mirroring says so, and a read that sees the mirror is added

the triple-star sessions; ties rows 2, 259

Count the spare measurements, not the shapes they draw (625) — the pentagon with all five diagonals draws 35 triangles, but its ten bars hold only 6 independent loops and 3 bars beyond what rigidity needs, and every gain in sturdiness comes from those three. Bolting the five crossings adds ten pieces and no check, and the frame gets easier to break: it survives 175 of 190 double losses, against 45 of 45 unbolted

the triple-star sessions; ties row 551

Before looking for preferred levels, find which variable rests (626) — in the chaotic triple the binary’s eccentricity jumps rarely, 21 times in 929 strobes, and rests between jumps, while the third star’s energy is kicked at nearly every pass: only 91 of its 1,080 rests outlast one excursion. There are two staircases, a slow one and a fast one, and a search for rungs on the fast one finds chance

the triple-star sessions; ties rows 603, 605

Grade a picture by its topology: is its edge a fence or a step? (634) — Bear read the turning needle as a car on a banked track. On a real track, a segment ending at a wall, there is one seat per speed and stable minus unstable is 1: the cliff is the end of the track. On a bank that curls all the way round, a circle, there are two balance points at every speed and no cliff at all, because the lean a car feels is always the downhill of one height. A cliff in the needle's sense needs a push that goes round. Before borrowing a picture, ask whether its space has ends and whether its push can be written as one height (the Loop Atlas sessions). Read where a change is born, start by start, before explaining the sum: the share of heads in R2W's coin map jumped by 0.052 at one step, and the leans that the jump was born between the arcs and that the small fin joins the main arc both missed; over 50 steps 1,089 starts turned tails to heads and 722 heads to tails, and 956 of the new heads lay within 2 starts of a start already heads. Heads are born at every edge, and the edge flickers both ways (COIN-4) (R2W, sitting P2)

the Loop Atlas sessions; ties rows 586, 633

A damper eats energy, never momentum (635) — an astronaut throws a ball on a ribbon, a damper catches it, and he reels it home: the damper took 100.000 per cent of a 53.7024 J throw, and the momentum kept was -5.43×10^{-13} kg m/s, which is none. Free fall lets you keep a chunk of position, never of motion, and the mass of whatever holds the far end sets the chunk's size. A jerk on a springy joint has a price: a force switched on against a damped spring does F δ of work, half stored and half lost whatever the damping, the capacitor's rule. On a skateboard's pin it adds about k δ^2 to every sharp top-up and pulls a cost ratio off its rigid value (0.763 against 0.75); a gentle top-up restores it (0.75016) (run WADDLE_PRICE 20261011_071302) (the Loop Atlas sessions). A refund at the catch changes the energy bill and never the shove: in free fall even a perfect refund leaves the energy the partners carry off, while on the ground it can be complete (the throw-and-catch toy). The capacitor's rule itself: two equal capacitors joined through a resistor lose half the energy whatever the resistance, while joined through an inductor, with the switch opened at zero current, nearly all of it arrives (97.8 per cent in the toy; the first half is a classical result, from recall) (the soft-switch toy) (the Loop Atlas sessions)

the Loop Atlas sessions; ties row 590

A balancing rider's cliff sits where its reaction delay reaches its own rest between taps (636) — for rests of 0.4, 0.5 and 0.6 fall times, the cliff came at delays of 0.40, 0.50 and 0.60. There is no warning in the song of taps before it, and past it the taps pile up on one side. A load that comes and goes faster than a balancing rider can fall is paid at its average, not its peak: the balance board's rider, carrying a pack that sets its balance back for a quarter fall time at every look, loses steepest slope held in proportion to the pack's shift times the share of time spent looking (slope 1.04 over 20 settings, on a grid of 0.0025), and even a look a whole fall time long costs only 1.2 times its average. The price of an ear that tilts the body is its duty times its size (runs EAR_COST 20261013_183245 and EAR_COST_2 20261013_183913) (the Loop Atlas sessions). But an average has an edge: a quick load is paid at its average and a lingering one closer to its peak. The rider's look is paid at its average while it is shorter than about a fall time; at the same share of time a look of 1, 2 and 4 fall times costs 1.2, 1.7 and 1.9 times its average. Find where the averaging stops before quoting the average (run EAR_COST_3 20261013_192738) (the Loop Atlas sessions). A clock that wears can hide behind a clock that rests: the slowing rider's tap lag read its rest, 0.47 of a fall time on its first flight with no delay at all and 0.497 on its last, because it may not tap again until its rest runs out, so the wearing delay shows only when it reaches the rest, which is the fall (run LAG_METER 20261016_203014). Meter the hidden clock where the hiding one has run out, or poke the rested machine: read only where its rest had ended, the same lag measured the reaction itself (one step on a new rider's first flight, 0.49 of a fall time on its last), and at equal catch one fixed level of it threw away 0.031 to 0.051 of a life against the calendar's 0.250 to 0.333 (run RESTED_LAG 20261016_205504; row 746) (the Loop Atlas sessions)

the Loop Atlas sessions; ties row 631

Only the exact half-turn shuvit walks straight (637) — a board that spins in the air goes exactly where it was already going. Given a pad to spin on, every exact half turn moves it one board length; a little more or less and it walks a circle and comes home. A landing hands the spin back reversed and halved, so a board left alone waddles, every stride the same length (1.4142) while each takes twice as long as the last

the Loop Atlas sessions; Bear's shuvit

Agreement in sign is the weakest agreement: ask a toy and a solver the same question, and compare mechanisms (653) — a toy's kick rule was checked against a jellyfish solver on the one question both could answer, a rest before the second stroke. The direction survived; the mechanism and the size did not. The solver's bell is pushed forward by its own wake while it rests, which the toy cannot do, because its rings touch the bell only at kicks; the toy gains its smaller improvement through a different ring. The toy's numbers are now labelled toy-rule numbers, not findings

the Loop Atlas sessions (the book); Jellyfish: the Holonomist of the Sea; ties row 438

A tread's edge is a fold with a ghost, mirrored (654) — above the edge, recovery slows as the square root of the distance; below it, the rider stutters at a rate on the same square root, with slopes 34.27 and 34.73, and the noise wobble, read at the squeezed moment, grows as one over the recovery rate. The warning band is thin, about a relative 10^{-4} of the drive. The floor of a steady drive at weak strength 0.3, read by slipping over 20,000 time units, is 0.80798920, and two independent readings agree with it within 4×10^{-8}. Cousins: the saddle-node bottleneck, and critical slowing down as an early warning. The ghost survives being chopped into teeth: a zigzag drive past a lock's edge keeps its grip only if its average stays below the edge and each leg is short enough, and the longest leg scales as the ghost law predicts, within a factor of 1.5 across the overshoots (the Loop Atlas sessions, R2W). Read a birth from both sides: below a conveyor's birth the staircase's shortfall falls as the distance to the birth to the power 0.507, above it the conveyor's lag leaves the birth as a square root, and the two fits named the same birth, 1.458137, to about a millionth, finer than the dial that measured them; the coarse dial had made the slope at birth look gentle (26.8 against 160). And a ghost is visible before the thing is born: riders in flight crowd toward the lag where the conveyor will appear, more of them as the drive nears its birth, so where a lock is about to form, look for the crowd first (the Loop Atlas sessions, R2W). A fold's rate is small only very near it: two hundred-thousandths of drive below where R2W's coin's tails end, the 1/2 groove's stretch rate of −0.22 looked sturdy, and the lean that the groove outlives the coin missed; carried in steps of 0.000002, the rate fell almost in a straight line, −0.2246 to −0.0699 over eight stages, and the dance was gone by 1.442836. Read a fold by the trend of its rate over several stages, not by the size of one (R2W, sitting P3)

the Loop Atlas sessions (R2W); ties rows 605, 640

Chaos needs something that can answer back, and it lives on the edges (655) — a one-way chain of massless riders cannot be chaotic at any setting, by Bowen's bound on such a chain, the entropy of its drive plus the largest entropy of one rider alone, both zero here (Bowen, 1971), and 672 runs agreed. With mass, chaos appears from mass 0.5 and sits on the borders between treads, 97 per cent on a fresh grid read blind, the same at three chaos thresholds, so it is structure, not binning; with two-way teeth it appears, one-way never. A heavy rider also frays just inside an edge: chaos inside the gear with no net slip, reached at mass 1.0 by period doubling whose later steps shrink near Feigenbaum's 4.67, while the first step is as wide as the second, not yet explained. The forced pendulum built as a phase-locked loop is the hardware ancestor (d'Humières, Beasley, Huberman and Libchaber, 1982); it reports symmetry breaking before period doubling, and whether that explains the wide first step is open, since the paper's own Appendix B undercuts it; R2W registered the paper's rule for where doubling arises as FF3-v at mass 1.0 on 5 October, and no register R2W sent to this guide grades it. A doubling ladder in the same rider, measured rung by rung, is row 680

the Loop Atlas sessions (R2W); chaos and binning; ties rows 523, 642, 680

Same rate, different word (656) — over 200 beats, two drives both slip exactly one tooth in three beats; one plays it as pairs (000011) and the other as singles (100). Nine of 12 words were periodic and only 2 evenly spread. The order of the slips is a channel the rate does not show, the word rung of Bear's channel ladder (amplitude, phase, count, class, word)

the Loop Atlas sessions (R2W); ties row 38

A side channel can be a bus, not a feed (658) — in Bear's picture of four-wire dimming, two heavy wires carry the power and two thin ones carry the signal that decides how much of it flows. A relay's kick that never powers the next stage, but decides which gear it rides, is a valve, not a pump; Bear notes the same shape in every domain the project has touched

the Loop Atlas sessions (R2W); Bear's four-wire dimming

A rest pays back only what builds up, so put the variable your picture depends on into the model (669) — rest stops on a zigzag drive changed nothing while nothing built up: every slip came on the first leg that was too long. With a heat that drains by Newton's cooling and fades the grip, rests gave back what the heat took, at one overshoot exactly to the cold value, and still could not save a leg that was simply too long. A picture that needs a store, here a truck's brake heat on a mountain road, is tested only once the store is in the model. And jumping, firing only on crests, never beat skating for the listening bead; output per event was flat or fell. A real jump needs a store, and a memoryless skater can only skate less often (run RF_SKATER_COUNT 20261004_061647; redesigned below, the skate still out-pushing every jump) (the Loop Atlas sessions). A jump needs a latch as well as a store. A memoryless ear that fires only on the crests carries the same push per event at every threshold, the higher crests being shorter in exact balance (0.378 to 0.368 of strength times a soup cycle at thresholds 0 to 1.5; Rice's crossing count gives 0.377). To make each event carry more, latch: hold the switch until the push turns round. Bear's pause ear does that, and its push per switch rises at every widening of its zone, 0.035 to 0.091 wavelengths from 0.25 to 3 of strength (runs JUMP_STORE 20261015_223918 and PAUSE_WIDE 20261015_223744). For the same number of switches a latch pushes more: above an edge of 1.5 of the push's typical size, a latch and a memoryless crest-firer switch exactly as often, and the latch pushes 1.36, 1.51, 1.75 and 1.82 times as much at edges of 1.5, 2, 2.5 and 3 (run LATCH_EVENT 20261015_232643). But never set a trigger against the thing's own running memory of itself: a store whose blade fires

only when it swings faster than half its remembered speed drains, its memory lags above its swing, and it never fires again; its swing fell to about 10^{-24} within the run at every trigger from half its speed up, and a knee on both sides of zero, measured the same way, did the same to a bead that fell quiet (run JUMP_STORE; the Loop Atlas's misses rows 241 and 242). Silence is absorbing, so never put a trigger on a store without a floor: a spring bead whose blade waits for a fast swing fell silent at every trigger tried, against its own memory of its speed or against a fixed speed, down to a quarter of its usual swing, because one quiet spell withholds the very push that would end it (run LATCH_EVENT; the Loop Atlas's misses rows 242 and 246, and its misses row 1, that a blade should never go fully dark, as row 769 found) (the Loop Atlas sessions). A spring is not a jump. The best spring carried 0.1145 per event (tune 1.5, loss 0.2) against the listener's 0.354, because it fires about once per swing of its own, 0.57 to 1.34 times a soup cycle at light and middling loss, and each firing carries a third or less of the listener's; even the noisy listener with its knee carries three times as much. The plain noisy listener's 0.069 belonged to the step (its count grew 1.40 times when the step was halved), so the registration was amended before the tool to judge against the knee (run JUMP_STORE 20261015_223918) (the Loop Atlas sessions). A floor brings an idle swing of its own, and a trigger set above it is silence again: with a floor of 0.2 the spring's latch stayed awake to the end only at a trigger level h of 0.25 (and 0.5 at tune 1.5), seldom at 1.0, never at 1.5, where the floor kept the bead swinging at 0.302 and 0.366 of the fixed speed, below the trigger. Set the trigger as a share of the floor's idle swing (run SPRING_FLOOR 20261017_160023) (the Loop Atlas sessions)

the Loop Atlas sessions (R2W); Bear's switchbacks and brake heat; ties rows 654, 805

Memory as a road lives where two locks overlap, and mass makes the overlap (670) — with mass, a relay's second stage holds two gears at one drive: their overlap opens near mass 0.61 and is 0.124 of the drive wide at mass 1.0 as a ramp reads it, 0.122943 between the folds a solver puts at 1.413316 for gear 1's end and 1.290373 for gear 2's. No edge between a lock and free running kept a memory the readers could see at masses 0.75 to 1.5; from mass 2.0 to 3.0 the Loop Atlas's needle, the same rider in another coordinate, holds gear 1 beside a smooth spin at the bottom of its running range, 3 riders of 8 in the lock and 5 gliding, decided by where in its turn each was let go. A massless rider cannot fold, because two starts cannot cross, so it has one state per drive; mass lets memory and tumbling in, with a ceiling: in the needle, tumbling shows at mass 0.75 and 1.25 and nowhere from 1.5 up, where the heavy rider glides. Inside the room between the two gears the road writes the state: at one drive two dances share one stone, a note just below the window sends all 64 riders home on the twelve-step dance and a note just above on the six, and a slow slide always comes home by the door on its side. In a survey of 32,080 riders, 4,405 of the 4,411 tumbling ones sit strictly between gear 1's edge and gear 2's door, and the room closes near mass 0.905 (none shaking at 0.91). An exception can be the rule carried somewhere new: the few riders that shook beside gear 2, where the room is closed, ride a ribbon of the half groove that plays its own doubling ladder, followed unbroken from mass 0.75 across gear 2's door at 0.8024. Two gears make a road with lanes, and the lanes move with weight. On the swapped needle the slow gear holds to its fold at every mass; from mass 1.5 the passing gear is a band below the fold that moves down and narrows as the rider gets heavier, a strip of the push that slides with the weight, low and narrow for a heavy rider (0.665 to 0.725 at mass 3) and high and wide for a light one (1.13 to 1.28 at mass 0.5), its width falling as one over the square root of the weight, width times √(mass) holding at 0.1085 within 3.2 per cent over eleven weights (log slope −0.5135; run LANE_EDGES 20261017_164501); at mass 1 it runs past the fold and leaves no open road; and a runner coming in is caught by the passing gear in two thirds of tries at mass 1 and two fifths at mass 3. Alone on the hill the passing gear catches every runner; beside the slow gear some fall through (runs TRUCK_ROAD 20261016_090656 and LIGHT_LANES 20261016_174648; the Loop Atlas's misses rows 261 and 262). It is Bear's truck road of 6 October, with its slow lane, passing lane and runaway ramp, now the Loop Atlas book's two-gear machine in Limb G, its table widened to seven masses; where the mass-1 lane ends is row 640's (the Loop Atlas sessions). A ramp that turns too low cannot see what lies above it (row 640). SWAP_STATE's ramp turned round at 1.05, so its map gave the rider of mass 0.75 no passing lane at any grid push; on held riders, a rider set in gear 2 keeps it at 1.060, the top of the road read, and six runners of eight are caught there. The mass-0.75 passing lane sits at the top of the road, above its slow lane's fold (1.0120), with open road from 1.015 between them (run TRUCK_ROAD 20261016_090656) (the Loop Atlas sessions)

the Loop Atlas sessions (R2W and the book); ties rows 563, 605, 662, 673

Above the fold, a rapid with pools: high on the staircase the locks are exponentially thin, and place sets their width (671) — past gear 1's fold a heavy rider mostly tumbles, its mean slip climbing from 0.19 to 0.76 turns per passing across drives 1.640 to 1.700, and the pads are narrow windows opening in that rapid, most of them below its level. Neighbours in the fraction tree need not be neighbours in drive, and the slip can step backwards (at 18 of 61 drives; the 2/5 pad sits below the 1/3), so once the map folds, monotone is not a law. The same groove moves with the rider's mass, a thread for one mass and a wide road for another, and shallow fractions belong to every rider while deep ones belong to one. In the Loop Atlas's needle a lock's width is set by its place more than by its fraction or the rider's weight: at mass 3.0 the 4/3 lock is 1.8 times as wide as the 3/2 above it, two period-5 locks differ by about twelve orders of magnitude with place, and high up the staircase the locks are exponentially thin, because a fast rider averages the wave nearly away (the 8/5 at $2.4662389 \times 10^{-20}$ of drive and the 11/7 at $2.8068117 \times 10^{-22}$, each read four ways: from its loop, at fixed drive on the attractor, by parabolas of known convergence, and with 2π carried to 40

digits; double precision had printed rounding noise near 10^{-13}). Against the gap between a lock's fraction and the heavy rider's speed limit, 1/0.618034, the widths fall as powers of the gap, but not one power per period: the period-5 powers run 12.13 and then 10.88, and the period-7 power, from 10/7 to 11/7, is 16.37. A width read from two folds that never met their criterion can be wrong many times over, in either direction: one lock read from its loop was 4.4 times thinner than two unconverged folds had said, and the two thinnest, the 8/5 and the 11/7, were 19.7 and 37 times wider, the 11/7's two folds being one edge found twice. Read a thin lock from its loop, the well-conditioned reading, and check that two refinements sit at two different extremes before calling their difference a width. A limit is where the locks run out, not a lock: with the clock's ratio at 0.6 the limit becomes the plain fraction 5/3, and the rider never reaches it. Read a width in the rider's own units as well as the knob's, since the two can disagree about a trend, and print each width with its resolution before ranking it

the Loop Atlas sessions (R2W and the book); ties rows 38, 655, 680

Wells and saddles balance, so an index sum is a free completeness check (672) — on the rider's tube-shaped space of lag and speed, wells count +1 and saddles −1 and must sum to zero, so a run that counts them tests the solver, not the physics: it found every fixed point at eight drives and two windings, and every later solver run carries the check for free. Wells and saddles are born and die in pairs, a well and its saddle walking into each other toward a fold before both vanish (Bear's picture: the saddle a heptagon, curvature −1, the well a pentagon, +1). A +1 point need not be a well: from drive 1.35 up, gear 2's +1 point is a flip saddle, whose well has split into a two-step dance. A lone unstable root can leave only by standing still; anything that swings leaves in pairs: on the delayed smooth-push board of row 631 at delay 0.5, across a grid of 121 by 161 gains, all 161 changes in the winding count across Kp = 1 are by one root and all 204 elsewhere by two; a grid with a point on Kp = 1 itself, where a root sits at nought, finds no jump there at all, so keep the grid off the line (run ROOT_JUMPS 20261015_031511; the argument principle is textbook, named from recall) (the Loop Atlas sessions). A symmetry pins where a zero can sit: in R2W the reflection that split the fourth harmonic into two hands (row 801) also put the dial's dark point on the mirror line, because a dark point off it would come with a mirror twin of the same charge, and the count changed by one. So a search over a plane became a sign change along a line, placed within seven parts in a million (RATCHET-53). It grew from Bear's picture of half a dipole or a reflected image. Before hunting a zero in two knobs, ask which symmetry pins it (row 734) (R2W)

the Loop Atlas sessions (R2W); Bear's pentagon and heptagon; ties rows 186, 551, 642

Which door you come home through depends on which way you go round the tip (673) — a loop of drive and mass round the tip of the overlap where two gears coexist brings the rider home in gear 1 one way round and in gear 2 the other. The two walkers agree at every far corner, where only one gear exists, and a loop that crosses the overlap without going round its tip brings both home in the same gear. The route decides the door, and only a loop that encloses the tip can tell. To tell whether a slow loop went round the tip, look not at how it ends but for a change without a jump. A needle with weight sits or spins between two edges of memory that meet at 0.7026 (run MEMORY_SWITCH_2 20261011_085331), the switch R2W bracketed at 0.70 to 0.71 (rows 643, 686 and 711); walked slowly round loops of its weight and its push, 18 runs, it came home changed exactly when its jumps plus its loop's turns round that meeting point were odd. Every change with no jump (16, all where the loop passed beneath the point) came at pace 1, and every other change was a jump (20); on every loop that crossed an edge, both starts came home in the same motion (run BENCH_LOOP 20261020_081001). A builder can use it: to change the motion with no jolt, take the weight below the meeting point before crossing the push's edge. A loop round the tip of a fold is probably textbook in the study of sudden changes, not checked here against a source; the count for this needle is the project's own (the Loop Atlas sessions)

the Loop Atlas sessions (R2W); Bear's two walkers; ties rows 643, 668, 670, 674, 686, 711

The edge says what kind of lock it is (675) — three rooms now hold three laws for how long a near-miss lasts. Far from any lock the time per slip goes as one over the offset, a loan (row 660); in a triple the slope is −0.919 over 128 runs, and slip period times offset is 0.978 turns by median. Just below a damped lock's edge, where the lock and its partner have merged, it goes as one over the square root of the distance, the saddle-node's ghost (row 654), met independently by R2W and the Loop Atlas; one formula holds both laws in the Loop Atlas's forgetting needle, a slip period of 2π over the square root of pace squared less one, exact to 1.1×10^{-8}, its log slope bending from −0.5000 near the edge to −0.9820 far from it (Adler's equation; Adler, 1946). At the edge of a conservative lock kicked once a lap there is first a chaotic layer of uneven width, and outside it the pendulum's logarithm: 7.57 and 7.37 passes per decade at two clean rims, against 7.29 and 7.83 forecast from the island's own width with nothing fitted, where a square-root law would have grown the slip period 4.5 times over the span and it grew 2.4. That agreement holds over the ladder as a whole, within 6 per cent at the two clean edges; step by step the slope runs 5 to 11 per cent short far out, about a quarter steep nearest the edge, and 1.76 times beside a chaotic layer (row 780), an overshoot still to be followed (row 797). One number tells the kinds apart: a damped lock rests where push equals pace, its resting angle sliding with detuning by arcsin(detuning over strength) to a quarter turn at the edges, while a springy lock's swing stays centred, and every locked run in the triple rests within 1.1 degrees of centre over swings of 0.06 to 0.46 turns. The two kinds have engineering twins, named from recall: a damped lock is an injection-locked oscillator or a simple

phase-locked loop, a springy one a synchronous generator near pole-slip, so how a lock's slip slows, read from outside, tells which kind it is; whether a generator near its transient-stability limit slips the way these stars do is a guess no run has tested (the triple-star sessions). Before fitting an edge law, ask whether the system loses energy, read the slope on a log plot of hold length against offset, and read where it bends: the bend says where the band ends

the triple-star sessions; the Loop Atlas sessions (R2W and the book); ties rows 631, 654, 660, 727, 780, 797

A borrowed number belongs to the machine it was made on (678) — three borrowings missed in one session. Rescaling a needle's torque to hold its memory dial at 1.0 changed the group the record's needle had been reported at, so a clause registered as a rig check compared two machines (against the same wave, the record's needle has memory 0.924). A grid of rooms copied from another room's machine, where the filter times the room at onset was about 1, sat where in this machine it was about 0.01, so 47 of 60 cells carried at every filter and no slope could be read; moved a hundredfold, every cell read. And a bound proved for a push that acts all the time, √2 fall times of delay, did not bind a rider that pulses and then waits out its own delay, which held 14 of 16 riders at 1.5. After any rescaling recompute every dimensionless group, locate the answer in your own machine before borrowing a range, and check that a bound was proved for the controller under test (act-and-wait control; Insperger, 2006) (the Loop Atlas sessions, the book). A machine's pace is the machine's own: a laptop's was estimated from the container's by reading a window's gap between finishes as one run's time, and the laptop's own table showed 3.46 to 3.51 times the container's seconds, so an estimate of 1.5 to 2.5 hours became about 6. A borrowed law brings its own system with it: a pendulum law that matched an island's rocking within 1.1 per cent at one eccentricity missed the scaling across eccentricities (1.038 and 1.435 against about 2), and the paper's guess, tried in one system only, is that the borrowed pendulum has no chaotic rim; and a heavy star's ragged sides, guessed from a light star's, fell on the side where the star and the pair trade the most eccentricity (the triple-star sessions). A pace is the tool's own as well as the machine's: this row's own figure, a laptop at 3.46 to 3.51 times the container's seconds, became 0.81 times on another tool, a fold refiner; even a number about borrowing numbers is a borrowed number (run 20261010_172149) (the Loop Atlas sessions, the book). A number carried across a mirror is borrowed too: a test registered the reversed rider's fold weight (0.25) for a column of the forward rider, when the record had held the forward rider's own fold line since 7 October (MAP-K-F: 0.33 to 0.40). The miss was not the physics. Before borrowing a number, check that it was measured on the same rider, in the same direction, at the same place, and search the record for the same object facing the other way (row 666) (runs FOLD-1 and FOLD-2) (R2W). Before carrying a knob's strength to another drive, measure its slope there: Claude's picture said the fourth harmonic, whose own window carries two more powers of the drive than the octave's, would act more strongly at higher drive ratios. A two-run probe found the opposite for the weight laws: its pull on a about unchanged (0.84 and 0.76 of the edge's), its effect on c shrinking roughly with the octave's own c (RATCHET-49). An additive estimate across settings would have aimed the joint edge's sizes several times wrong. A slope is a local fact, and two runs buy it. And borrow the whole ladder the record holds, not its first rung: RATCHET-69's plan took the tilt's slope from RATCHET-45's lightest weight and treated it as a straight line, but RATCHET-45 had run that same pair to weight 0.16, and its ladder already showed the tilt tiring, with the shape RATCHET-69 then found to within 0.15%. The prediction fell 4.7% short for that reason alone; the redesign used the measured shapes and landed at 0.9994. When a prediction leans on one number from an earlier run, ask what the rest of that run says about the same quantity (R2W). A planted value comes from the record of the very setup the tool runs: twice in one night a planted case failed in its blind run because its value had been copied from a run of a different setup (BALANCE_BOARD_B's riders run one at a time with shifted tilts, against riders side by side) or from a run with a numerical fault (GAPS_READ's multipliers, row 588); both blind runs caught it before a registration was made exact (the Loop Atlas sessions)

the Loop Atlas sessions (the book); ties rows 640, 649, 776, 793, 797

A doubling ladder, measured: the rungs shrink toward 4.669, and the ladder starts at a fold (680) — in the Loop Atlas's light needle (mass 0.75) the period-2 dance doubles at drive 1.6924902823 (R2W's value, in its own constants, which sit 5.3×10^{-9} below the Loop Atlas's at every shared doubling; rows 677 and 738), the period-4 dance at 1.690017021430, and then at 1.6894624205, 1.6893422593, 1.6893164568, 1.6893109278, 1.6893097435 and 1.6893094898, the last three forecast from the ratio 4.669 before they were run and landing within 2.5×10^{-9}, 1.3×10^{-10} and 8×10^{-11}. The band ratios run 4.4595, 4.6155, 4.6570, 4.6667, 4.6687 and 4.6691 toward Feigenbaum's 4.669, and the shrink at each superstable orbit's critical point runs 2.8231, 2.3951 and 2.5491 about his second number, 2.5029 (Aitken's extrapolation 2.508, so the needle's top is rounded; the next rung, 128 to 256, was registered between 2.47 and 2.50 on 8 October, and the Loop Atlas's records hold no run of it), with exactly the half of the orbit that starts at that point's image shrinking by its square ($\delta \approx 4.669201609$ as printed in Feigenbaum's paper of 1978, read by R2W, and in Gurau, Rivasseau and Sfondrini's review of renormalization, which also prints $\alpha \approx -2.502907875$). Close to each doubling the new strands sit a fixed multiple of the square root of the distance apart, and come into equal halves only in the limit. The ladder starts at a fold, because a doubling keeps the slip: the half groove's two-step dance cannot come from a one-step dance, so it is born at a fold, and a cascade inside a p/q tongue starts at the period-q orbit's fold (in the needle at drive 1.697824258, where the period-2 orbit's multiplier is +1 and a fold refiner snaps in 4 steps); a cascade's top rung is the first orbit whose slip no shorter orbit can carry. The universal numbers are ratios read on a measured ladder; the onset value 3.5699456 belongs to the logistic map's own

formula and is not universal. They are universal only within a class. A smooth map with a quadratic maximum gives 4.669; Briggs's table (1991) prints 5.9680 for a cubic top, where the map itself gives 6.0846890743 (McCartney, 2020, found the table's odd orders wrong first, and one sign slip reproduces them, row 874), 7.2847 for a quartic (proved since to lie between 7.28468621697298 and 7.28468621717299, Thurlby, 2021, and given to more than 300 proven digits by Burbanks, Osbaldestin and Thurlby, 2021), 9.2962 for the sixth order and 10.9486 for the eighth, and the Loop Atlas measured 7.28472549, about 9.298 and about 10.96 for the last three on maps of its own, still moving. The index grows with the order of the maximum toward a bound of about 30 (van der Weele, Capel and Kluiving, 1987, checked against its record, its text not recorded as read), and the ratio of δ to |α| raised to the order stays below 1 and rises with it, 0.745 for a rounded top, 0.892 for a quartic and 0.962 for the twelfth order, from Briggs's numbers, which the Loop Atlas's runs matched. A textbook page's list of maps sharing 4.669 includes a quartic example, which closes instead on 7.2847. An area-preserving map has its own constant, about 8.721, proved to lie between 8.72021 and 8.72217 (by Gaidashev, 2011, with mild computer assistance, after Eckmann, Koch and Wittwer's computer-assisted proof of universality, announced in 1982 and given in full in 1984); the Loop Atlas measured it climbing from below, 8.3020, 8.6585, 8.7156 and 8.72027, and the triple-star sessions' cascade finder read 8.721082 there and 4.669197 on a map that loses area, a lightly leaky map reading the frictionless value for its first doublings and then crossing over, once one trip round its orbit keeps under about half its area (0.440 to 0.524 across five leaks). So a ladder that closes on 4.669 agrees with a quadratic maximum, the generic case, and one that keeps climbing past it would point to another class. Which side of the limit the first ratios fall on is partly set by the knob that drives the map, not by the class: one parabola reads its first ratio as 4.2337 or as 4.7514 depending on the knob, approached from below in one and from above in the other (the triple-star sessions found it, and corrected their own first paste by it), so an early ratio alone says little. A second ladder in another groove (mass 0.4, the 3/7 groove) starts at 2.41 and climbs 3.14, 3.94 and about 4.45, learning the number late; its top is measured quadratic, so it should close on 4.669, and its next two rungs, registered, came in at 4.62 and 4.66, six steps to the number (R2W; row 872). A conservative map's cascade at a triple's island centre was registered to close in by about 8.72 each step, not 4.669; in the triple, island 7's centre doubled once and every star round the doubled centre then escaped within 1,600 passes, so there the cascade ends after one step. One prediction is open: every landed period being a pad's denominator times a power of two may be forced by the drag, which lets a dance lose its footing only by folding or flipping. A ladder's limit can be read before the ladder arrives, once its steps shrink by a steady fraction: Aitken's rule on its last three rungs put the doubling constant for a top of order 6 within 3×10^{-5} of Briggs's 9.2962468327, and for order 8 within 2×10^{-4} of 10.9486. Read the step ratio first: early rungs zigzag, a second correction changing sign at every rung (orders 4, 6 and 8 in step at rungs 3 to 5, order 2 the hardest), which throws a three-rung extrapolation about; and the rounded top's ladder closes from below while the flatter ones close from above (run DELTA_TAIL 20261010_205937) (the Loop Atlas sessions, the book)

the Loop Atlas sessions (R2W and the book); the triple-star sessions; Feigenbaum (1978); Briggs (1991); Thurlby (2021); Burbanks, Osbaldestin and Thurlby (2021); van der Weele, Capel and Kluiving (1987); Eckmann, Koch and Wittwer (1982, 1984); Gaidashev (2011); Gurau, Rivasseau and Sfondrini (2014); ties rows 642, 654, 671, 677, 697, 738, 877

A rider who listens works on any swing, and reading by the wrong rate has a price that can be written down (681) — a rider who pumps a swing twice per swing, timing each move by the swing's own phase read from its angle and speed, multiplies the swing's size at small swings by $\exp((3\pi/2)\ \varepsilon \sin(\text{lead}))$ per swing, ε being the fractional change of the rope's length: 1.26570 measured at a lead of 90 degrees with $\varepsilon = 0.05$, against 1.26569 from the formula, and all 24 leads within 0.46 per cent. On swings of 0.5 to 12.5 m every one passed 90 degrees within 12 of its own swings, while a rider keeping a clock set for a 2.5 m swing never passed 12.67 degrees from 10. A rider that reads the phase by the wrong rate sees its twice-per-swing rhythm through a Möbius map of the circle, which keeps the rhythm's timing and multiplies its strength by $4r/(1 + r)^2$, r being the swing's true rate over the rate the rider reads by; measured, 0.9995 to 1.0014 of that law. The pump is exactly reversible: started at its eighteenth turning point, the brake at the opposite lead walks back through all seventeen earlier ones to the start, within 8.46×10^{-13} degree, so growth here is not a loss run backward but one path walked in two directions. Event timing does what a clock cannot, and its error has a closed form. That the best pump listens is known in kind: Bryde, Davenport and Mahadevan (2023, arXiv:2308.06818) find the optimal pumping switches on the swing's own state, at its lowest point; the mismatch law $4r/(1 + r)^2$ is not in their paper. The swing first misled the other way: a run of 20 swings and the run of record of 6 gave very different numbers, put down to the swing drifting out of step as it grows and not tested until a run settled it. A big swing is slower, its period 1.0019, 1.0313, 1.0732 and 1.1375 of the small-swing period at 10, 40, 60 and 80 degrees, against the elliptic integral's exact 1.00191, 1.03134, 1.07318 and 1.13749, so a clocked rider falls out of step, reaching 63.16 degrees at swing 10 and back to 8.91 by swing 20. And the first rider timed to the swing's own motion did nothing: with no lead, at a lead of 0 or 180 degrees, it is its own mirror image about the bottom (runs SWING_TWENTY 20261004_075715 and 20261004_075902) (the Loop Atlas sessions). In the reversible run above, the eighteenth turning point lay at 92.411631 degrees, the pump starting there at rest, and the start at 10 degrees; the largest difference, 8.46×10^{-13} degree, fell at large swings where no single growth factor holds, and there is no

friction anywhere in it. The pump's small departure from the first-order formula is a pure twice-the-lead term of second order in the rope's change: halving the change cut the largest departure to 0.2469 of itself, and at leads 90 and 270 the departure is under 10^{-5} (run SWING_LISTEN_2 20261007_024253) (the Loop Atlas sessions)

the Loop Atlas sessions (the book); coupled clocks and event time; ties rows 20, 604, 663

Resonance can evict, and the moment decides who goes (685) — rocking the drive at a six-step dance's own rhythm throws six-step dancers onto the twelve-step dance that shares their stone; an off rhythm moves no one, even at five times the depth. Which dancers leave is set exactly by where they stand in the cycle when the rocking begins: two neighbouring steps of the dance are open, and the open pair moves one step round the dance for each passing the rocking is delayed. Bear's Chladni picture, inverted: nodes in time instead of lines in space. The registered lean had guessed capture; resonance here is eviction

the Loop Atlas sessions (R2W); Bear's Chladni plate; ties rows 641, 670

Ask where a lock can form before choosing which locks to count (688) — re-read at births a thousandth apart, the triple star's locks formed almost only where the rate turns or the island ends (10 of 11), so a law about towers tested on crossings found one crossing to test. Where a lock can sit is set by the shape of the rate, and a census that asks for locks in the wrong place counts the places, not the locks. Counted per crossing, fractions with odd denominators hold about three times as many locked runs as even ones, 2.73 times near the mirror line and 3.04 times off it, 3.16 fitted jointly, every odd chain fuller than the mean of its two even neighbours (pooled over the saved atlas and the off-mirror grid, run 20261017_011954); the islands paper reads it as a push odd in a swing symmetric about its centre feeding only the odd chains, left for theory (rows 802 and 847) (the triple-star sessions). A census of fractions files an irrational gear under its nearest fraction: a third pattern set at the beat of R2W's first two opened a gear at slip 1.618034 spanning 0.56970 of pace, wider than first and second gear together without it (0.25392 and 0.18556), and took the bead from first gear outright (first gear held at 0 paces with it, 79 without); the census of 3 October had filed it as two 13/8 runs, with no 13/8 lock by the floor test (CLUSTER-1). Read a census's grammar, here denominators up to 8, before trusting its names (R2W, sitting P2)

the triple-star sessions; R2W (sitting P2); ties rows 603, 648, 670, 802, 847

When a scaling misses, its exponent names who is in charge (695) — the guess that a rounded tent's first doubling gap scales with the rounding missed: it scales as the rounding to the power 0.6779, because the second doubling is set by the rounding's far tail. The law worked out after the miss was registered on two new roundings and held to 0.02 and 0.01 per cent. A missed scaling is a measurement of what controls the thing; register the law it suggests and test it on fresh cases (test X1, run 20261008_035235) (the triple-star sessions). And an exponent is only as good as the edge it is read against: lifetimes below a fall edge gave a slope of -0.31 against the edge of record, and the same six lifetimes give -0.50 against an edge 3×10^{-5} higher. Find the edge from the object that is born there before fitting the law (run LOCK_FROM_CHAOS 20261011_060551) (the Loop Atlas sessions, the book). Before calling a curve bent, plot it against the knob the cause depends on: the window's first doubling, against the needle's memory M, bends (the slope of its logarithm -9.20 to -6.41 over five weights); against 1/M, the variable the squeeze $\exp(-8\pi/M)$ depends on, it is nearly straight (9.48 to 9.00), and the scatter round a straight line falls from 0.032 to 0.0047. What bend is left (to 8.30 by M 1.27) is the window's own (runs HEAVY_MARKS 20261014_164806 and HEAVY_SHAPE_2 20261014_164913, after Bear's "curvature") (the Loop Atlas sessions). And test a "bell" against more than one ruler before calling it lopsided or even: the heavy needle's window has its least near memory 1.92, its floor even under memory, its logarithm and its inverse alike, and its walls even under none, the light side steeper against M and ln M and the heavy side against 1/M; at memory 3 the window has widened threefold from its least and still keeps the one-hump 9/4 (runs HEAVY_WINDOW_4 20261015_032553 and BELL_LOG 20261015_033341, after Bear's "bell") (the Loop Atlas sessions). Test a law where it lives: the traders' square-root law (row 629) lives in the knee's extra size, and the flips follow it through the bent curve of flips against knee, in which range the law itself gives the flips a power of 0.33, not 0.5; a lean set on the flips' power missed while the law held (run CREEP_LAW 20261016_035058; the Loop Atlas's misses row 250) (the Loop Atlas sessions). A miss in the reversed direction says stop guessing and watch: Question the Substrate's two readings of why the world mattered both failed backwards, folded boundaries at a rank correlation of $+0.900$ against a registered -0.8 (E12) and a low-density gap at -16.5 against a registered -3 or better (E13), and a plain diagnostic, each predicted answer's share over the generations, found the mechanism at once (E14). A reversed miss can itself be the answer: with equal shares the double windings were the easiest answer, and the chain renamed the others into them, so the drift follows certainty, and size is only one source of certainty (E24). And when a mechanism misses, look at which cases held: copy fidelity did not rank the drift of a checker that passes every label (rank correlation -0.26, E36), but the worlds that held shared thin ground between answers, and a dial on that one feature took the drift from -8.0 to $+0.1$ (E36r) (Question the Substrate)

the triple-star sessions; ties rows 661, 680

Chaos can keep a count (696) — the riders shaking on a coin's third face all keep two turns in every three passings, their slip within 0.0004 of 2/3, and the dances inside that chaos are 2/3 too; the shaking just below the coin keeps no count. Whether a chaotic orbit's winding is locked is a read worth making before calling it disorder. Read chaos's turn

rate over a window long enough to average it: past the half gear's fold the remembering pulsed needle, read over 199 cycles, seemed to jump between 0.53 and 0.64 of a turn a cycle with no order; read over 4,000 its turn rate climbs smoothly with the pace (0.553 to 0.628 over 1.0231 to 1.0270) while its largest Lyapunov exponent stays at +0.23 to +0.30: chaotic and orderly at once, and the short window's scatter was the window (run HALF_TOP 20261013_183449, against HALF_PARENT 20261013_163612) (the Loop Atlas sessions). Chaos can keep an exact gear: between two islands of 5/7 whose orbits repeat every 14 and 28 cycles, a needle with a positive Lyapunov exponent keeps five turns in every seven within a quarter turn over 30,000 cycles, its tumbling confined to bands visited in a fixed order, while a free chaos beside it strays 25 turns (run BANDED_FIVE 20261013_224945) (the Loop Atlas sessions). So measure order by the average, not the spread: near a lock's edge a noise of 0.003 cost the lock its gear (its rate moved by 0.02 to 0.03) while a chaotic needle next door kept its average within 0.0054; the spread of both rose with noise and could not tell them apart at the largest noise (run CHAOS_ORDER 20261013_222835) (the Loop Atlas sessions). Near its fold a lock is the fragile thing: jostled with a noise of 0.001, a lock within a ten-thousandth of its edge loses 0.013 to 0.014 of its rate while a chaotic neighbour moves 0.003 or less; locks break at less noise the nearer their edge, and strong noise melts the staircase toward a ramp (run CHAOS_ORDER_2 20261013_225021) (the Loop Atlas sessions). The half gear's chaos begins at a true fold, 1.022910479388 (run HALF_TOP 20261013_183449) (the Loop Atlas sessions)

the Loop Atlas sessions (R2W); ties rows 627, 671

A window can open and close by the same road (697) — the third face's 2/3 doubles at both ends: chaos, 4/6, 2/3, 4/6, 8/12, 16/24, chaos; the coin's tails opens the same way, 2/4 below 1/2. Look for a cascade on both sides of a window, not only on the side the textbook draws. And the two ends need not open onto the same chaos: drawn whole, the pulsed needle's 1/4 window has, below its doubling pile-up, chaos confined to four narrow bands round the orbit's points and visited in turn, while past its fold the chaos fills the whole circle; and the banded chaos holds a 12-cycle window of its own, with 24 and 48 just below it, seen in the data first (run QUARTER_STROBE_2 20261013_231511; where the lower end lies and how it is made is row 756) and placed on 14 October: a first rule for their cycles missed, and every touch of an arch on the band's low edge proved a window of exactly its predicted cycle, 18 of 19 against none of the controls (rows 760 and 794) (the Loop Atlas sessions). The road's foot, read with a settled rider: the third face's foot opens by a cascade run backwards, 48, 24, 12, 6 and 3 going up, the mirror of its top. Its first ratio of 7.33 rested on a 4/6 that was a 2/3 still alternating as it settled; with a settle three times longer the first ratio is 6.18 by the bands and 6.12 by the stretch rate's zero. Place a crossing only between dial neighbours, read by riders that have settled: a reader that put the 3-to-6 doubling in a stretch the dial never sampled was void, and its rerun, with a check that a doubling's two bands sit one dial step apart, landed at 6.1765. Why the first ratio sits so far above 4.669 is open (R2W, sitting P3)

the Loop Atlas sessions (R2W); ties rows 680, 760, 794

Two maps can share a line; find it and check that they agree on it (698) — the pad atlas's plane of pull against clock ratio carries the golden-ray column that R2W's maps run up. Before treating two instruments as separate, find the parameter line they share, and check that the two agree on it

the Loop Atlas sessions (R2W); ties rows 650, 666

The eye keeps what a summary throws away: lopsidedness, negative space and walls (715) — a core's centre held within 0.0004 across a run, so its average said nothing happened, while the person's "short spiky top left, long spiky top right" was a flip in the ripple's lopsidedness, measured afterwards as −0.49 to +0.47 on the picture's pixels and −0.18 to +0.29 on the data: right in sign, overstated in size (row 712). He reads negative space too: "I unfocus my eyes and I allow the white patches to fill my vision. They are each bordered by little 4+ dot line clusters." Lanes where no dot falls cannot be found by asking where the dots' neighbours lie: his gold lanes of empty space, put into a test as rows of dots, were tested as the wrong thing, and sensors 5 holes long could not see lanes 0.3 turn long. Translate a mark into a test together with the person who made it, before the run, and test gaps as gaps. Look at the shape of a signal, not only at its mean and its edges (the Loop Atlas sessions, R2W). Translating a mark with its maker has to happen every time: a test that took a count of 113 from the person's list of crossings, without looking at the drawing, where the coil is about eight overlapping ovals, each one closed stroke, read nothing, three days after an earlier lesson had said the same; rebuilt with the person from their own symmetry marks, as pairs with one jump caught across two looks, its predictions held (runs triple_heavy_coil_test 20261011_055359 and triple_heavy_coil_pairs 20261011_070746) (the triple-star sessions). And a gap the eye draws can hold on data it never saw. Two dead zones drawn on an older island map held on the next two sets of runs: on the first, 3 of 484 runs landed in the lower zone, and on the second the upper zone separated broken pairs from healthy ones at 1,462 of 1,475 rows. The one version that failed was a rescaling of the drawing made without the person. The same traced panel showed the start's phase deciding which way the pair's eccentricity moves, two days before checks on the first of those sets measured why (runs triple_dead_zone_check 20261012_062504 and triple_pair_breath_look 20261012_073126) (the triple-star sessions). Crowds mark slow places and gaps mark fast ones, so a picture's density is a picture of one over the speed: the middle of a turn stays empty because a circulating rider crosses it fastest; a dance crosses its strobe slowly (0.14 of a turn per passing, against its average of 0.5) just outside its frozen balance points and is flung fast over the top,

and its twin crosses at lag 1/2 at nearly four times the passing speed (R2W). And ask what a drawing shows before building on it: Claude misread the person's tesseract sketch twice before getting the two-jelly leap-through (his report) (the Loop Atlas sessions). A summary over a fixed count is a ruler with holes in it, and the eye can see through the holes. On a figure of the Loop Atlas's needle leaving its window, Bear ringed in red a dip about 0.0095 below the window and flickers near 0.0135, 0.015, 0.0165 and 0.0195 in its spread panel, asking whether these were gaps. Claude had read the same stretch a batch earlier as one clean lock, from the spread of sixteen needles' turns over 200 drive cycles (the Loop Atlas's misses row 219), a ruler that sees only the cycles that divide 200: it found the window of 10, could not see a window of 6, and could not see a doubling at all (row 641). The dip is a window of 10 drive cycles inside the half-turn chaos, and the flickers are the rungs of a ladder of doubled half-turn cycles, 32, 16, 8, 4 and 2 drive cycles, where the needles take a long time to agree (runs STOP_STATES 20261015_060745 and GAPS_READ 20261015_061859; row 665). The eye was right on place, and the ruler was the blind one: when the person rings a flicker in a quiet stretch, read it with a measure that can see what the flicker might be, here the cycle's own period found from the needle's state (the Loop Atlas sessions)

the Loop Atlas sessions (R2W); the triple-star sessions; Bear's lattice; ties rows 329, 573, 712

Check which angle, or which pair of angles, a clock belongs to (719) — a lattice's families at one birth of the triple star climbed a ladder of pass gaps in which each is the sum of the two before (19, 28, 47, 75, 122), the mark of a steady rotation's nearest returns; one angle's own returns gave 22, 25 and 72 instead, and the other angle slides. Over 47 passes the median net turn of the sliding angle is −4.9967 and of the other +15.0175: they come home together, geared one to three, and one third of the faster angle's rate has the continued fraction whose denominators are 9, 19, 28 and 47. At the next birth looked at, 7.230, the factor of three was gone: there the dots' return after 201 passes is a convergent of the faster angle's own rate, with the other near home too (−36.9948 and +76.9995 turns), and a prediction carried over from 7.200 missed. Coupled clocks return together at the continued fraction of their ratio; before naming a pattern's clock, find which angles come home together, and read their gear off their net turns at each birth. A second specimen, run blind: read off both angles' median net turns over each gap, as this row says, the gear named the lattice's three commonest gaps in their order at two births, 104, 208 and 10 passes at 7.190 and 47, 163 and 210 at 7.220, and with the resonance angle shuffled it failed (run triple_heavy_gap_netturns 20261010_204936). A gear taken from single steps had given 47 at 7.200 and missed at 7.210, predicting 44 where the lattice's gap was 399. The gears read −13:31, −5:15, −61:136 and −8:17 at 7.190, 7.200, 7.210 and 7.220. The gear and the lattice read one record, so their agreement checks the two readers, not the stars, as row 556 says of two fitters of one light (the triple-star sessions). Two clocks show in a spectrum too: the island centre's angle is a chord of two notes, every line at every one of 52 long runs read a whole-number mix of the beat and the rocking (BC1 and BC1′, run 20261015_134034), and the chord beat one with a random second note at 44 of 51 runs (BC2′); the beat is the slow swell Bear traced there (row 581) (the triple-star sessions)

the triple-star sessions; the Field Guide sessions' check of their data; ties rows 103, 627, 581

A chaotic transient can end in copies (724) — thirty-two chaotic spins each settled once onto a regular spin, at times spread like a constant chance (a mean of 4,750 time units), and at one pace the settled needles fell into four groups agreeing to 3.4×10^{-8}: a settled state can be one orbit visited at a few phases (run CHAOS_SETTLES 20261011_060320). Read such a group only where every member has settled: needles on one orbit at four phases agreed to 3.4×10^{-8} on settled windows, while read over all windows each needle's own transient split the four groups into sixteen single needles (runs SETTLED_COPIES 20261011_212014 and SETTLED_COPIES_2 20261011_212705) (the Loop Atlas sessions). When the time to an event jumps about irregularly as a knob is turned, the event is a chaos with a way out: a train of three rings broke after 70, 913 and 170 time units at three nearby spacings, so its timing is read as a lifetime over many nearby starts, not as a function of the knob (run TRAIN_HOLD 20261015_171853). The stop's door is the same kind (run LEAK_RATE 20261015_080150). Bear named such chaos an escape hatch, a door through which chaos drains away, and the Loop Atlas book keeps his name beside the theorist's word, transient chaos (row 745) (the Loop Atlas sessions). What a train is worth near its edge is read the same way. A train of three rings that holds beat its budget ring by 1.1628 to 1.2610 as its gap rose from 0.25 to 0.50, then by 1.2325, 1.2658 and 1.2305 at 0.55, 0.60 and 0.65, not the smooth rise to the cliff the lean drew: near the cliff the train holds but wanders, its rings up to 4 radii apart, so one long run's mean speed depends on the wandering it happened to do, and the advantage there is a mean over an ensemble of nearby starts, like the lifetime. Below 0.50 the rise held, and it is the engineering answer: pack a train of three at about half a radius (run TRAIN_HOLD 20261015_171853). Flown longer, the cliff near 0.65 to 0.70 was the 1,000-unit window's, and the margin a builder needs sits lower: none of 30 trains from 0.500 to 0.525 broke within 20,000 time units while 16 of 30 from 0.530 to 0.555 did, and at 0.520 and 0.525 the lifetime is near 150,000 time units, with no floor (row 819) (the Loop Atlas sessions). A loss is a constant rate only if surviving twice as long is the square of surviving once. Riders knocked off by a counter-wind of waves failed that test: the share still riding at 1,000 against the square of the share at 500 was 0.1250 against 0.0156 at margin 0.1, and 0.6250 against 1.0000 at margin 0.4, because every rider of one soup meets the same stretch of the pattern and falls at the same place (row 787) (run SOUP_JUMP_3 20261016_225440) (the Loop Atlas sessions)

the Loop Atlas sessions (the book); ties rows 655, 709, 819

A needle's own slip rate senses its pace (725) — read through Kramers' law for escape over a barrier (Kramers, 1940), a needle's own rate of slips gave its pace within 0.0064 at barriers of 4 to 9 noise units, and more sharply as the noise falls (under 0.0015 at noise 0.1), though Kramers' estimate runs a little high. The slip count is one more way to read where the cliff edge lies (run SLIP_SENSE 20261011_060035). Both textbook rates were first met to the count: at pace 2 the needle slipped 866 times in a thousand cycles against Adler's 866.0 (Adler, 1946), and at the four barriers first read it escaped at 1.000, 0.991, 0.999 and 0.949 of Kramers' rate, each within its counting error (runs NOVIKOV_NEEDLE 20261004_152937 and NOVIKOV_NEEDLE_B 20261004_153129). A gauge that reads high may be reading true: the needle's own wobble gauge, which needs no master, read 0.373 at pace 0.3 under loud noise; it reads the needle's noise-softened stiffness, $\sqrt{(1 - (\text{mean cos lag})^2)}$, within 0.02 wherever nothing slips, so the noisy needle is nearer its edge than its pace says. One slip in the record pins the gauge near 1, so read it with slips left out (run WOBBLE_SOFT 20261013_192713) (the Loop Atlas sessions). And a gauge's early warning may be the first events, not the slow softening: with slips found by a finder tested on planted slips and left out, a noisy needle's own wobble gauge reaches 0.95 exactly where slips pass 1 per 100, not 0.10 to 0.15 of pace before as the plain gauge did; the plain gauge was hearing the first rare slips (run SLIP_PROOF_3 20261013_225419) (the Loop Atlas sessions). Flickering between states as a warning where slowing down does not apply is known in kind (Dakos, van Nes and Scheffer, Theoretical Ecology 6, 309 to 317, 2013, read at abstract level); this gauge's reading is new in place and number. A leg judged on "no slip" must set its bar below one slip, or one event decides the reading (the Loop Atlas sessions). How high Kramers runs was then measured: the slips fell 9 per cent short of Kramers' rate across barriers of 4 to 9 noise units (0.9093 ± 0.0112 pooled over 6,549 slips), and against the exact rate for the same needle, Stratonovich's double integral for an overdamped needle with noise on a tilted washboard, they pool to 0.9885 ± 0.0122; the exact rate is 0.905 of Kramers' at 4 noise units, 0.965 at 9 and 0.985 at 20 (run KRAMERS_EXACT 20261020_073520). That Kramers' rate is a leading-order formula is known; the size of the shortfall at these barriers comes from Stratonovich's formula as cited, not from a printed table. When a count misses a textbook rate, compute the exact rate before suspecting the count (row 711). And start the clock after the needles settle: a needle set down at the bottom of its well is about 12 time units late to its first slip (17 at noise 0.1, 13 at 0.2, 8 at 0.3); on 32,768 new needles, seven rows counted for 500 time units from rest read 0.9789 ± 0.0034 of the steady rate, 0.7645 ± 0.0094 in their first 50 time units and 1.0027 ± 0.0036 from 50 to 500, over 78,593 slips (run SLIP_START 20261020_090154), so the lateness alone costs 1.6 to 3.5 per cent of a 500-unit count (by arithmetic on that run). The counts behind 0.9885 ran 500 time units a row from rest, and the start accounts for their 1.2 per cent shortfall from the exact rate, within its errors (run SLIP_RECHECK 20261020_083908). That a process started out of its steady state is late to its first event is general; the size here is the project's own. A meter of a rare-event rate that starts its count with the device at rest owes this correction, or a settling window several times the lateness (the Loop Atlas sessions)

the Loop Atlas sessions (the book); ties rows 636, 662, 675

A map transfers in two senses: number for number where the equation is the same, and as a calibrated ruler everywhere else (727) — with its ripple off, R2W's rider obeys the equation of a synchronous generator's rotor angle (the swing equation: inertia, damping, power in, and power out as the sine of the angle), of a phase-locked loop's phase error, and of a Josephson junction under a steady bias. For those the bare map transfers: the rest is a machine holding synchronism or a loop holding lock, a slip is a pole slip or a cycle slip, and the memory band is the gap between how far a loop stays locked once locked and how far it can grab lock from outside. With the ripple on, only a particle in two waves shares the rider's equation. A junction driven by radio waves is a cousin, not the same machine: its drive pushes the same whatever the phase, while the rider's ripple moves the bumps under it, and no change of variables turns one into the other. For everything else, the three-body problem and the seismic rings among them, the pendulum certifies the ruler and not the mapping, and each new domain needs its own check. The generator was named from recall and is not yet checked against a source; the phase-locked loop has one, the forced pendulum built as a phase-locked loop by d'Humières, Beasley, Huberman and Libchaber (1982; row 655), though its reading of the memory band as hold-in against pull-in is not yet checked (the junction's own papers are cited in row 726), and none of these mappings has yet been tested on a real domain's data (row 502). Say which sense is meant whenever a result is called portable, and which machine (R2W; the Loop Atlas sessions)

R2W; the Loop Atlas sessions; ties rows 388, 502, 655, 726

A flat picture cannot show linking: count the holes, and read the hidden coordinate at the crossings (744) — a ring-shaped surface seen at an angle shows one hole; two hollows with a filled middle mean two loops overlapping, or one band that crosses itself, and one cut of density along the long axis tells the two apart. On one three-body cloud the person saw an eye: "It looks like an eye zoomed out. And that means three chambers usually. Two overlapping elliptical forms. Could be a linking number in some ways. But two forms like this becomes a twisted piece of conduit almost. A mixing chamber geometry. Like a heart. So a valve perhaps. If you can't cut that math, is it a latch?" The screenshot's own pixels (the dots taken as the pixels darker than the gridlines' grey) found the cloud's long axis 44° above horizontal and the cloud 1.67 times as long as it is wide; along the long axis, at 1.05 and 0.75 standard deviations to one side of centre the middle strip holds 0.21 and 0.38 of the rims' density, at 0.75 and 1.05 to the other side 0.32 and 0.51, and between them, from −0.45 to +0.45, from about half to about one and a half times the rims' density (1.39, 0.76, 0.52, 1.02): two hollows, one on each side of the centre, and a filled middle between them, three

chambers as he said. Two loops drawn on a page may be linked in space or merely overlap. In a three-dimensional flow two orbits can never pass through each other, so whether they are linked cannot change while both exist; the linking number is read at the overlap, by noting at each crossing which loop passes over in the coordinate the picture hides (Birman and Williams, 1983) (run EYE_PIXEL_CHECK)

R2W; the triple-star sessions' picture; Birman & Williams (1983); ties rows 383, 561, 702

A column of followers: check one follower alone, space by time, let the spacing breathe with the pace, and match speed promptly (754) — before reading a chain of followers, check that one follower is steady alone: a column test whose walkers rang up and fell over on their own past 0.2 s of delay read the walker, not the column; the lone loop's delay margin, from its closed form, takes a minute and sets the grid (runs COLUMN_ACCORDION 20261013_222625 and COLUMN_ACCORDION_2 20261013_222950). Space by time, not by distance: a line of followers that keeps a fixed distance passes a stumble down the line growing thousands of times in a linear toy; one that keeps a gap in time (about 1.5 s for gentle walkers) damps it, up to a reaction delay of about 0.6 s, and past about 0.8 s no spacing saves it (string stability in platooning: Swaroop and Hedrick, 1996; the delay limits are this toy's own). A safety spacing can be kept steady by letting it breathe with the pace: a column held at a fixed 5 m stops walkers under a swinging pace (9 of 30 stop, at most 4 within 10 m); the same 5 m at a walk, opening a little with speed, stops no one and never puts more than 3 in 10 m (run MORTAR_SPACING 20261013_224843, on Bear's day spacing, as corrected below). Tight spacing is steadied by matching speed, not by closeness: at 1 m under a swinging pace, walkers easing gently toward the man ahead's speed stop 4 of 30 and press to the floor; walkers matching it firmly 0.1 s late stop none, the tail swinging 1.338 m/s against the leader's 0.8, the closed form's 1.34, and 0.3 s late they still stop none (run NIGHT_SPACING 20261013_231417, on Bear's night spacing, as corrected below); a late read of the speed is another matter (row 762). A rule set in time asks for nothing at a standstill: a column keeping a time gap swallows a halt passed man to man, and turns a halt heard by all at once into a close-up, each man standing three tenths of a second and then walking into his gap, while a column at a fixed distance stands as one when told at once and stands four times longer at its tail when told man to man (run HALT_ORDER 20261015_052122). Bear's rest halts regulate the accordion this way. A gap written as rate times time vanishes with the rate (the Loop Atlas sessions). The fixed-spacing and night-spacing numbers above are as corrected. They were first made with a fault in the column's half-metre limit, which braked any walker closing on the man ahead about once in eighty steps and made the column look more fragile than it is (row 816): the fixed 5 m stopped 15, with 5 within 10 m, which read as bunching; the gentle easers stopped 14; the firm grip 0.3 s late stopped 6, which read as tight spacing steadied only promptly; and the tail swung 1.36 m/s, near the closed form rather than on it. With the fault repaired, five verdicts in three tools changed (run LIMIT_AUDIT 20261017_190938; row 762 carries the night read's). And judge a rule by what it is for: Bear's halt rule never made the column's tail arrive sooner (no rule was quickest, 525 s), but it bought contact, the longest gap near 30 m instead of 406, at a third more time, so a lean on arrival time alone missed the trade the rule makes (run HALT_THRESHOLD 20261017_184055) (the Loop Atlas sessions)

the Loop Atlas sessions (the book), on Bear's spacing by day and by night; ties rows 636, 762, 763

A late read makes a column amplify: the sensor's delay, a word passed back, and how quick a read must be (762) — a speed sensor read late makes a column amplify a quick wobble it never saw: cat's eyes turn sight into a closing-speed reading, and on a 1 m column with a firm speed match a look-time of 0.05 s costs nothing, while a longer look lets a quick wobble grow, the closed form's fast peak 1.58 a walker at 2.7 rad/s for a look of 0.3 s (1.5843 at 2.684 rad/s, computed for this guide from the column's equations, which reproduce every closed-form number in CATS_EYES's and QUICK_READ's headers; the Loop Atlas withdrew its own 1.58, the daytime column's gain, 1.5862 at 0.464 rad/s): with a small reading error a look of 0.1 s stops none, 0.15 s stops 4 and 0.3 s stops 19, and with no error at all 0.3 s stops none and 0.6 s stops 3 (run CATS_EYES 20261014_063138 as corrected below, on Bear's night practice). A word passed back works only if obeyed at once: told the leader's stumble, a fixed-distance column of walkers who obey at once moves exactly as the leader (tail dip 0.5 m/s, the leader's own); obeying after their own 0.2 s delay, the tail still dips 3,541 times the leader's; with no word, 74,340 times the first walker's. At a time gap the at-once word does slightly worse than the late one (8.30 against 5.99 m/s): an instant word fights the gap's wish to close up (run WORD_BACK 20261014_065420, on words passed down a marching column). And how quick a read the night column needs: with a small reading error a 1 m column with a firm speed match holds to a look-time of 0.13 s and stops 1 and 4 walkers at 0.14 and 0.15 s, and the wobble that grows at a look of 0.12 s swings once in 2.34 s, near the closed form's 2.15 s and inside the 15 per cent registered for it. In dim light, where the rods see, the eye sums light over up to about 0.1 s (Kalloniatis and Luu, 2005), inside that edge; a specialist in night vision could check the eye's timing (run QUICK_READ 20261014_065420 as corrected below, started together with WORD_BACK above, so the two share one stamp; on Bear's two eyes). These reads are as corrected. They were first made with the column's faulty limit (row 754), and then a look three tenths of a second late stopped 8 with no reading error at all, a look of 0.1 s with the error stopped one, and the safe edge sat at 0.09 s, just under the eye's 0.1 s, which read as night vision perhaps mattering by making the read quicker. With the fault repaired, a late read needs a seed: the closed form's growing wobble grows from something, and the reading error is that something (CATS_EYES and QUICK_READ as corrected in run LIMIT_AUDIT 20261017_190938) (the Loop Atlas sessions)

the Loop Atlas sessions (the book), on Bear's night practice; ties rows 754, 763

Two swimmers share one water: sharing a remembering partner is not free (768) — two same-spin rings help each other along, while two opposite rings spread and do not hover (the ring toy; same-spin rings leapfrog and opposite rings approach, enlarging and slowing, as Falconer, 2019, recounts from Helmholtz, 1858). A second swimmer behind the first squeezes into the first one's jet and loses: of 24 tied spacings and firing orders, one gained (nested close, squeezing together, +6 per cent), and swept at ten phase points that gain is a slope and a crossing, holding within a tenth of a period of firing together at the tightest spacing and lost by 0.2 of a period (the bell pair, bell tether and bell sweep toys, which rest on an assumed kick rule with no drafting). And a vortex ring wider than a follower's open top is a headwind, not a draft: on the jellyfish solver of record, a closed passive follower meeting the leader's own ring takes +51.7 mN·s backward on a leash in 3 s and is pushed back 1.34 R when free, where a ring narrower and weaker, through an open valve, had drawn the follower in (ROPE-1, 18 September). The open follower under the wide ring is unmeasured: the solver's one-way valve ran away (run DRAFT_TEST 20261008_134707, on the jellyfish paper's solver v2.5) (the Loop Atlas sessions). The valve's own fix is stated too: one that switches on the bare sign of the flow chatters under a strong ring, and needs a band. Read to more places, the push above was 51.671 mN·s on the leash and 1.337 radii free; the test was parked by the person's ruling, its two follow-ons written down and not run, a stroking follower and the open runs with the solver's hysteresis valve and a runaway bar (run DRAFT_TEST 20261008_134707) (the Loop Atlas sessions). Both findings above rest on less than they seem: ROPE-1's draw-in came through a valve type that ran away elsewhere, its ring's circulation growing from 4.997 to 8.492, and DRAFT_TEST's leash took more momentum than the ring carried, unexplained; the Loop Atlas's register lists both as stated before their cards were run

the Loop Atlas sessions (the book); ties rows 604, 653

A blade can listen with no sensor: a refunder does the listening, the blade is the rein, and an all-or-nothing blade can stall until a pilot tone wakes it (769) — a refunder can do the listening: a bead on a spring, with its charge switched by its own motion and no sensor, got a net push from an even soup, up to 72 per cent of what a perfect listener gets (the spring bead toy). A crystal radio is the same machine as the spring bead with a blade: a tuned store (coil and capacitor) and a one-way gate (the crystal), run by the incoming wave alone (a reading, from recall). The blade is the rein: in a soup that is even in every direction, the pull points wherever the blade's heading is set, and a blade that favours one sense of circling gives a twist instead of a pull; the soup supplies no direction, the blade does (the horse spin toy). An all-or-nothing blade can stall: once its charge is off, nothing turns it back on, seen three times (the spring bead at rest, the very lossy bead, the turning blade), while a blade that never goes fully off does not stall. A stalled blade is woken by a very small steady feed, Bear's idea ("This feels like a polarity flip on breaking rotor stall. That little feed or microwave tone to align a power grid."): a pilot tone of one hundredth of the soup's push took two stalled beads from nothing to their full catch, and too strong a pilot cuts the catch, because the bead then keeps the pilot's time and not the soup's (the pilot tone toy). One comparison stays open: in Bear's words, "The jellyfish paper discovered early on: ten event units per cycle, where sometimes the units blended together, but it was critical to be event-driven", and the listening blade must act within about a tenth of the soup's rhythm; the two tenths have not yet been compared against the jellyfish paper's own numbers. A familiar specimen of the stall is the single-phase induction motor, which has no starting torque because at rest the field it sees is a standing wave pushing both ways equally; it is started by a second winding a quarter-cycle out of step, cut out once up to speed, and a failed start capacitor leaves it humming and still, a blade gone fully dark. Any harvester that needs to be moving before it can switch has this cold-start problem; the blade above that never goes fully off kept a fifth of its charge. The person reports the stalled, humming rotor as familiar from the field, not as expert testimony, and the motor specimen has had no bench check; the tie to the crystal-set detector is from recall (the Loop Atlas sessions). When the sweet spots of several knobs all move together, find the one quantity they follow and set every knob from it: three leans in a row set their centre on a bead's nameplate tune and missed, because the beat's hold, the pilot's power and the lock all follow the bead's working pace under load, 0.76 of its tune at a loss of 1 and 0.91 at 0.3 (runs BEAT_SWING 20261015_204207 and PILOT_PRICE 20261015_220321; the Loop Atlas's misses rows 235, 236, 238 and 239) (the Loop Atlas sessions)

the Loop Atlas sessions (the book), on Bear's crystal radio and pilot tone; ties rows 38, 468, 590, 604

A run kit for the person's laptop: one double-click launcher, checked inputs, a tested exit, his limit on cores, and a price before it ships (776) — the laptop is the person's, and so is its limit. One ruling was at most 10 of his 20 processors, because the three-body work runs too, so tools are built at 8 workers and below-normal priority (R2W); the page for every new session says at most 8 cores, one laptop job at a time, and a time budget after which a run stops starting new work. One double-click launcher per run, with no typed arguments, travelling with its Python file in every version and saved with Windows line endings; it checks its inputs first and says STOPPED, with the reason, when something is missing. Run Python and test its exit code (`python x.py || goto failed`; the publishing packet's launchers now use `if errorlevel` lines, which wine's test bed also reads right, row 812); do not pipe through PowerShell's Tee and then search the log with findstr, because the log can be UTF-16 and the search fails (row 611); print FINISHED only after success. A launcher checks every file's fingerprint and refuses a wrong one, a fidelity check reruns runs of record first, and a one-page status page (md and PDF) says what the run is, how to start it and what to send back. Before shipping any batch, price it (rider-steps times the measured pace) and print its

longest chunk, check that its smoke output is byte-identical to the last version's, and check that it resumes correctly on a copy of his folder (R2W). Test every kit from an empty folder, and through the intake script of row 733 from a mock Downloads folder, before sending it. Quote only measured run times, and label every estimate as an estimate. The cloud workspace can restart without warning (every 8 to 40 minutes on 14 October): a long job there must save each piece as it finishes, and real long jobs go to the laptop (row 720) (the triple-star sessions, in their page for every new session). These tools were rehearsed on copies of the person's folder before use, and the intake script (it copies a shipment's listed files from Downloads into the working folder, row 733) and the folder check each ran once on his laptop, on 14 October, both to the end (the triple-star sessions). An upper estimate from another grid is not a bound. The off-mirror run was estimated at about 4 hours, "an upper estimate since escapes end early", from the slice middles' measured rate; it took 19,006.4 s measured, 5.3 hours, about a third longer, and finished inside its 6.5-hour stop, which is what the stop was for. Label estimates as estimates, and set the stop well beyond them (the triple-star sessions). Time a tool alone before pricing one of its options: a factor of two told as the cost of a ring train's dense output was a machine shared with other runs, and alone the output costs 18 per cent, exactly its extra evaluations (run TRAIN_TIMING 20261018_082139) (the Loop Atlas sessions). The Loop Atlas checks a laptop kit four ways before it ships: the kit's flight (its run) against its parent tool's function, in every printed number; a blind run in a mock of the laptop's folder; a smoke run through every line that judges a verdict; and a planted repeat of a known break on the same machine (the Loop Atlas sessions)

R2W; the triple-star sessions; ties rows 611, 720, 721, 733, 812

A new session starts from a reference built by script, reads its start-here page in full, and searches the rest (778) — from 7 October new sessions read a session card in full and open rows by number, instead of reading the whole guide. The card is generated from the built guide and from the file that records every arrival, so it cannot drift from them, and each rebuild replaces the last (it was rebuilt on 10, 13 and 15 October); the reads, the lessons and the guide's own reference are built by script the same way, so nothing is retyped. A card keeps drafts apart from rows: the card of 13 October listed rows 721 to 728 as "pending" drafts, and the card of 15 October, after they were built, listed them as rows. The three projects' sessions started from the card instead of the whole guide, and several of their lessons cite rows by number as a result (the Field Guide sessions). A new session unzips each download into its own new, empty folder, runs the project's manifest or folder check first and shows the person its result, and prints a table of what it read (file name, line count, lines read) before it claims anything about a file. It reads the start-here page and the handover in full; a long register or log is searched for what a job needs, not read whole, because reading it whole is what makes a session start late and summarise early (the triple-star sessions, in their page for every new session). The guide has two readers, and they have split: sessions read the session card and open rows by number, while the public reads the whole guide, 133 pages at its second build. Treat the card as a product in its own right, designed and checked like the guide (the Field Guide sessions). And once started, keep the conversation small: its volume comes from long tool outputs, whole-file reads of big documents and long notes. Cut every printed line at 200 to 400 characters; never print engine data rows, only byte sizes and labels, as one sitting did from its twentieth exchange on; search the record by number rather than reading it; and keep each exchange's reply to the result, three surprises, the read and one question (row 767) (R2W). A later sitting gave the same advice and named where the volume comes from, long tool outputs and long record lines: cut printed lines at 200 to 400 characters, never print engine data rows, and search the record by a name or a number (R2W). A project moves between AI sessions with its instruments, not only its records: every script that generates a record (for the Loop Atlas, the headline register's ledger and assembler, their source, and the harvest's and the checker's scripts) goes into the crate, and the next session's first act rebuilds the record from the crate alone and compares its md5 with the manifest's. And a record computes every count it prints when it is assembled: the register's front page printed 453 tools, 483 runs of record and 114 blind runs, typed in at one exchange and still printed two exchanges later, when the crate held 457, 487 and 118, while every count computed beside them moved (handover F_TO_G, section 1) (the Loop Atlas sessions). Ask once, at the start, for the standing references: every session gets the Field Guide's reference zip and the Depth Axis Explorer, and a session whose deposit is near also asks for Bear's publishing packet (PUBLISHING_RULES.md, lint_deposit.py and build_paper.py), with a default: if it is not to hand, the session writes equivalent lint and font checks into its crate. R2W's sitting B12 had not read the publishing rules and its crate did not hold them, so its advice to the session that opens next puts that ask first among what is owed before the deposit (the triple-star sessions, R2W)

the Field Guide sessions; the triple-star sessions; ties rows 157, 572, 676

A lock's width in numbers: set by its place, and read in the rider's own units (790) — row 671's lessons, with the Loop Atlas's evidence. The strobe reads the rider in the clock's own events, turns per passing: the heavy rider's free spin climbs toward 1/0.618034, the ratio of the two clocks with nothing coupling them, and never passes it, and every lock on the way is a sliver: on a 0.01 grid, 5/4, 4/3 and 3/2 pass without a visible step (RUN_STROBE), the 3/2 a few times 10^{-5} wide. In the heavy glide the 4/3 lock is 6.599×10^{-5}, 6.747×10^{-5}, 6.828×10^{-5} and 6.860×10^{-5} of drive wide at masses 1.5, 2.0, 2.5 and 3.0, widening a hair with weight, and at mass 3.0 it is 1.8 times as wide as the 3/2 lock (3.877×10^{-5}), which sits higher in drive, against the habit of reading a simpler fraction as a wider tongue (one machine, fenced as such). Read blind in the rider's own units, the width in drive times the staircase's local slope, a lock's width ranks by place and widens with weight, both held at new cases: at mass 2.5, 5/4 $1.020 \times 10^{-4} >$ 4/3

7.807 × 10⁻⁵ > 3/2 1.297 × 10⁻⁵, and the 4/3 from 5.431 × 10⁻⁵ at mass 1.5 to 8.756 × 10⁻⁵ at 3.0; a width in the knob's units carries the staircase's slope and can reverse a trend, the 3/2 narrowing with weight in drive but widening in turns per passing (GROOVES-2, run 20261006_202753, registered before any part existed). At high drive a lock's width falls with both its period and its place: at mass 3.0 the 7/5 lock is 1.4890 × 10⁻⁸ of drive wide and the 10/7 2.3819 × 10⁻¹² beside it, and higher up the 8/5 and 11/7 read only rounding noise near 10⁻¹³ in double precision, unresolved (row 671 gives their widths, read from the loop), so a convergent's small denominator wins next door, not across the record (run GROOVES_5 20261006_213610). With the clock's ratio at 0.6 instead of 0.618034, the limit becomes the plain fraction 5/3, and at mass 3.0 the rider never reaches it: its gap to the limit shrinks as the drive to the power −3.991, the same power the golden clock's gap gives in the same run, and a drag estimate matches every gap within 4 × 10⁻³; a limit is where the locks run out, not a lock, and the 3/2 lock moves down to 1.423013597 to 1.423128081, three times wider than at the golden clock (run GROOVES 20261006_193519) (the Loop Atlas sessions). A lock made by a wave directly widens in step with the wave: the trapped ring at mass 2 is a quarter of the wave's strength wide in push from 0.075 to 0.6 (each doubling 1.875, 2.067 and 1.871 times), its middle moving under 0.004 until the wave is strong; the first-order lean held and the square-law rival missed (run RING_WIDTH 20261016_213858, the stamp it shares with STRIDE_MAP) (the Loop Atlas sessions)

the Loop Atlas sessions (its notes for the Field Guide, from 6 October); ties rows 38, 671, 795

A loop's phase in closed form: light carried by filters or wave plates, and the three ways a mountain hides from the arrow (792) — light carried round a loop of states on the polarisation ball by filters comes home shifted by half the solid angle the loop encloses, Pancharatnam's phase, met at 1.000000 of it at every loop from 1 to 40 degrees (run POLARISATION_LOOP 20261004_071425), and loses a little light; carried round the same loop by wave plates, each turning the state the shortest way to the next, it comes home with the same shift, same sign, to 4.44 × 10⁻¹⁵, and loses nothing (intensity 1 within 2.5 × 10⁻¹⁴). For knobs swung by a, a gap apart, the shift is 2 pi a J1(2a) sin(gap) at every size, to within 2.51 × 10⁻⁷ at 4,096 filters, so the gap enters only through its sine and the square rule's bend is all in the size. A rotator turning straight light through half a turn brings it home half a cycle late with nothing lost; two laps bring it back in step. The shift is the loop's, not the filters' (run POLARISATION_LOOP_2 20261007_074423, all four predictions held, registered before the tool existed). On paper pressed into ridges, h = sin 2x and sin 4x (ridges up to 76 degrees), 30 loops of radius 0.5 to 5 read 0 to 3.1 × 10⁻¹²; on a 70-degree paper cone every circle round the tip reads 2π(1 − cos 70°) less a turn, −2.148975939303, to 3 × 10⁻¹³ whatever its size or centre, and every circle missing the tip reads 0 to 7 × 10⁻¹⁵. With the earlier two (the skirt cancels the summit; the count hides whole turns), that closes the list Gauss's theorem allows (Gauss, 1828, for the theorem that bending without stretching keeps a surface's curvature; the list of three is the guide's own reasoning): the arrow reads only stretching. Adding heights is not bending: a 70-degree mountain set on the ridges reads −2.709729640646 at its steepest circle, 0.561 radian from the plain. And the first version of the ridge test could not fail: an arrow started down the ridges stays in their plane and read exactly 0.000e+00, so exact zeros are a warning, and the test was rebuilt with a slanted arrow and passed (runs BENT_PAPER 20261008_164513 and BENT_PAPER_2 20261008_164721, each registered before its tool existed) (the Loop Atlas sessions). In a turning loop, the timing mismatch between the two ways round counts only the area the path encloses, as a shadow on the plane of the turning: the loop's shape and the axis's place do not enter, a figure-eight counts its lobes with signs, and a tilt counts by its cosine. That is a fibre gyroscope's whole design rule: wind one way, never in a figure-eight, and make the area large; the delay is the Sagnac law, 4AΩ/c², met on a 360-sided loop at 1.000000000355 of it (run SAGNAC_LOOP 20261018_032138) (the Loop Atlas sessions)

the Loop Atlas sessions (its notes for the Field Guide, from 6 October); ties rows 587, 607

Read a thin lock from its loop, not its folds, and trust one power per period only roughly (795) — the orbits that close after q passings with s turns form a loop, at each starting angle exactly one drive closes the orbit, and that drive rises and falls q times between the lock's two edges, so solving for it at fixed angle stays well conditioned where fold conditions do not. Read this way, the 11/8 lock at mass 3.0 is 1.1911119 × 10⁻¹¹ of drive wide, 4.4 times thinner than two unconverged folds had said, the loop reading equals converged folds wherever those exist (the 10/7 and the 9/8, to every printed figure), and the period-8 power of the gap becomes 19.82, not 17.7, or 2.48 per unit of period (run GROOVES_9 20261007_011618, version 2); the period-5 figure it was first set against rested on the widths withdrawn below. Two more widths, the 8/5 and the 11/7, were first read from 30-digit folds printed "NOT converged", the flag that hid the 4.4-fold error; because their last steps were small against their widths (0.4 and 1.8 per cent), they were at first kept as provisional, with the period-5 and period-7 powers that ended on them. Read later from the loop, the two locks were 19.7 and 37 times wider than those folds had said, so a small last step did not make a fold right; row 671 gives their widths and the period powers that replace those first ones. Read from its loop, period 9 gives powers of 23.06 and 21.79, about 6 per cent apart, so one power per period holds only roughly, and the loop readings of period 5 in row 671 differ more (runs GROOVES_6 20261006_215909 and GROOVES_9 20261007_011618) (the Loop Atlas sessions). How long the record is decides whether a lock can show at all, so work that out before reading any width: by the pendulum relation (Δν·T ≈ 2/q) a lock at 1/q rests within N passes only if its half-width is more than about 2/(qN), roughly 10⁻⁴ to 8 × 10⁻⁵ at 1/8 to 1/11 over 2,400 passes. TS2 (run 20261017_172328) read SHARPENS: the slope through 1/10 was 2.79 times steeper over 2,400 passes than over 1,200 on the stage-2 births, and 7.25 times on LW4's. So LW2's and LW3's bounds (4.8 × 10⁻⁷ and the rest) bound a

flat stretch in 2,400-pass rates, not a lock's width; the triple-star paper's lock-width paragraph was corrected, and the 'covered' call made earlier on LW2′ had leaned on those bounds. And the eleventh, unlocked over 2,400 passes, locked over 6,000 (FOLLOWUPS2, run 20261017_100655). When finer grids of starts keep reading nothing, try a longer record (the triple-star sessions)

the Loop Atlas sessions (its notes for the Field Guide, from 6 October); ties rows 642, 671, 790

The comb theorem: N evenly spaced copies keep only the harmonics the spacing allows, and a symmetry that empties half of them decides odd against even (802) — a general fact, not special to holonomy. Take N copies of anything periodic, spaced evenly round its cycle. Added together, they keep only the harmonics whose number is a multiple of N; read against the fundamental (sampled at N evenly spaced timings), they pass only the harmonics jN ± 1. That is all one fact: N evenly spaced turns of a wheel cancel unless the turning is a whole number of laps (the orthogonality of the N-th roots of unity, or the characters of the cyclic group; row 149). A symmetry that empties half the harmonics then decides odd against even. With only odd harmonics (half-wave symmetry, as in a square wave or a clarinet's closed pipe), an even count of copies cancels completely when summed, and an odd count read as samples does the work of twice its number. The same arithmetic sets a three-phase bridge's line harmonics, 6k ± 1, and the tones designers tune by choosing the blade and vane counts of fans and turbines. What one grid cannot tell apart, a grid half its size folds together; signal work calls this aliasing, and music would call it two notes that sound as one through a coarse comb. In R2W, sixteen timings cannot tell the push's ninth harmonic from its seventh, and eight timings set off the mirror fold exactly those two into the weight channel, with opposite signs, so a model built from sixteen timings is blind to precisely the error that eight crooked timings make. To check any sampling scheme, build the model from several times more samples than the scheme uses (NINTH-1 and 2). Bear asked whether the odd-even staggering was a theorem for this guide, general rather than special to holonomy; it is. In R2W it showed three times in three exchanges: the ninth hiding in the seventh (aliasing), odd timing counts (ODD-1, row 745), and arrays of riders (ARRAY-1, row 803). Its cousin in holonomy is counting windings modulo N, which uses the same roots of unity (row 12). When anything is spaced evenly, ask which harmonics the spacing keeps, and which symmetry has already removed half of them (R2W). The comb also decides a floor. R2W's fifteen riders' null bottomed out at about 1.0189×10^{-15} when read at twelve shared timings 10 degrees apart, and at $1.235112566025243 \times 10^{-31}$ at sixty timings 2 degrees apart (FLOOR-1). The run file's 15-decimal rounding is the same at both; it cancels out of a harmonic's projection only when the riders' seats, 24 degrees apart, repeat on the timing grid. The nine's and twelve's nulls had looked perfect for the same reason, since 40 and 30 degrees are whole numbers of 10-degree steps (row 803). Choose the samples so the spacing is a whole number of steps, and treat a floor near a file's last digit first as a question about the sampling grid (row 738) (R2W)

R2W; ties rows 12, 149, 745, 803

Give a settling field a memory and it gets a flywheel and a speed limit: read a start by its fronts as well as its modes, and count a ringing by its deepest swing (804) — heat with a catch-up time, Cattaneo and Vernotte's heat with memory, rings in a rod shorter than $2\pi\sqrt{(\tau D)}$, its critical damping, and settles in a longer one, met to the number (dips of 1.52×10^{-2} at 0.80 of that length, 7.06×10^{-5} at 0.95 and none from 1.05), and a hot spot is felt only once a front at $\sqrt{(D/\tau)}$ could arrive, where Fourier's heat is already there (run HEAT_RING 20261016_174342). Read a start by its fronts as well as its modes: a warm brick in ice carries every shape at once, but its middle is knocked down only when its two cold fronts meet, by about 2e^(−L/4); read by its modes it should ring above the hump's line, read by its fronts it stays warm, and it stays warm (the same run; the Loop Atlas's misses row 264). A linear piece rings where its slowest mode rings, but its deepest dip may be a wave's echo: a quenched brick stops ringing exactly at its first mode's line, yet up to 0.88 of that line its deepest dip comes at 1.485 times its length in heat's travel time, the cold fronts crossing back through the middle (run BRICK_LINE 20261016_205112). A count of a ringing thing's crossings belongs to the band, not the thing: each swing is smaller by a fixed factor and never ends, so count its deepest swing instead (HEAT_RING's amendment before its tool, 16 October; row 627). Bear heard the brick as quenching, the smith's plunge, and the Loop Atlas book's Trunk 5 carries the result under his word, with the metallurgist's word beside it (row 745)

the Loop Atlas sessions; ties rows 627, 649, 669

A long run keeps its own custody: a ledger line written with each verdict, a stamp and a job of its own, only what its reader reads, and a heartbeat that survives sleep (809) — a long run is only as safe as what it has already written down. In the seventh triple-star session runs were lost to their own tools, launchers and machines, not to the physics, and each loss has a plain fix. Write each verdict to the ledger as it is made. On 17 October triple_light_pair v1.0 printed LS2 PASS, LS0 PASS and LS1 NOT BORNE OUT, then stopped on a format string with three places for four numbers before writing a ledger, and triple_saved_reads v1.0 (run 20261017_031107) stopped the same way in RA1's table; separately, SC0 (run 20261017_032428) caught a parser that split a ledger at the first 'e_bin 0.15', which was in the plan line, not in the slice's heading. Each mend changed only the string or the parser, and the rerun printed every other line the same. Exercise every print path in the selftest, and parse ledgers by the full form of their headings, not by a first match. Give every run a stamp and a name of its own. On 19 October triple_tw8 v1.2's main run and its twin started in the same second, both took the stamp 20261019_043245, and the twin,

finishing six minutes later, overwrote the main run's TABLE and SERIES. Each run's own ledger survived (TRIPLE_TW8_MAIN_LEDGER_ and TRIPLE_TW8_TWIN_LEDGER_20261019_043245), so the loss was seen at once, and the main run, made again alone (run 20261019_053814), escaped after 29,323 passes, exactly as the lost run's ledger had printed. The slip was the launcher's, two processes of one tool started together, not the tool's (row 611 holds the tool's own guard): start any later pair a second apart, keep the run's role in its file names, and let the reader take two stamps. A stamp shared by two different tools overwrites nothing but can still confuse custody: triple_sea_rule_look v1.0 and triple_fin_lines v1.2 are both run 20261017_031501, and the provenance table and scan are both run 20261016_202010. Store only what the reader reads, and give every long run its own job inside the time limit. TW8 v1.0 (b09f7b0e1e3de694) lost a worker after about 37 minutes, killed by the job group's memory limit at 3.5 GB, because the runs stored the solution at every integrator step (about 85 MB a minute at the tight tolerance) while the reader reads only the events; SL8 v1.0 was heading the same way and was stopped after 7 minutes. triple_lean v1.0 (74c95a9181252806) asks the integrator for the end time alone, and was proved to change no step by running a birth with and without it in fresh processes (7.320 and 7.120 for 60 passes: 0, PASS), a check later tools repeat for 300 passes. SL5's first start ran inside the job that queued TW7 and CS1, was stopped after four minutes and started again in a job of its own, and CS2 was split into four jobs each under two hours, with SL6 run between them. Survive sleep, a slow machine and a silent window. The workspace restarted on 17 October and took the background runs with it: LW3 v1.0 (run 20261017_061449) had done 6 of its 84 stage-3 runs, and DS1 and LX1 never started, so the runs went to a laptop kit. LW5 then printed nothing for about 16 minutes and was taken for a freeze; a sleep inside the loop's pause would have had every run called lost after about 97 minutes; and LW5's laptop ran 1.38 times slower than its pace estimate (17,012.7 s awake against 12,320 s). Every kit since prints a STARTED line and a five-minute heartbeat, counts only time awake, runs a lost run once more, and queues the runs a registered reading uses ahead of runs that are only reported (rows 720 and 776) (the triple-star sessions). Run stamps are neither unique nor on one clock: two tools started in one second shared a stamp, and the laptop's ran seven hours behind. Pair each stamp with its tool's name and record its machine's clock (the triple-star sessions)

the triple-star sessions (lessons of 22 October); ties rows 611, 720, 736, 776, 786

Test the test bed as well as the tool, and find out which of the two is wrong (812) — a check that fails may be telling the truth about the bench it ran on rather than about the tool it tried, so the bench is tested too. On 22 October the triple-star thread ran the publishing packet's launchers under wine 9.0 with stand-in python.exe and powershell.exe, and wine's cmd ran the right side of 'a || b' even when a succeeded (measured), so the fingerprint gate stopped a matching tool. That is the bed's fault, not Windows's; the packet's launchers now use 'if errorlevel' lines, which both shells read alike, in place of the || form above. The same bed found real faults, mended in build_paper v1.5 and lint_deposit v1.9: a run-file name built from %date% (RUN_BUILD_PAPER_/20262/02.log, no run file written), a fallback that set PY inside an if-block, lint_deposit offering the packet's own README as the paper, and a crash when a report line holding a phi is piped as cp1252 (build_paper v1.4 exit 1; lint_deposit v1.8 exit 1 before its RESULT line). Take the failure paths on purpose (an edited tool gives exit 9, a missing tool exit 1, no Python exit 1), and write down what was not tested (real PowerShell, the offer to install pypdf, Windows's own Python on Windows paths) (the triple-star sessions). A guard that runs on one path only is not a guard: the Loop Atlas's build counted a PDF's pictures only when a book was called by its keyword, and called by its file name the reader's edition built 17 of its 18 pictures as icons while the count still printed short 0. Test a guard by calling the tool every way it can be called, not the way the instructions name; the fix put the check on every build. And a blind run is blind only if its printing is built that way: the Loop Atlas's blind check shared the full run's progress line and printed a number a registered lean would be judged on, so the lean was replaced before the run of record and the blind mode now prints only each flight's energy (card 265). A checker's self-test must include a file made by the house's own tools: the upload checker passed all nine of its own cases, each made by hand, and still missed its first-line mark in every PDF the house builder makes, whose text a reader returns with its words split by tabs. Built by hand, a fixture tests the checker against the checker's idea of a file

the triple-star sessions (lessons of 22 October); ties rows 593, 776

The scallop theorem counts knobs: a knob that goes round escapes it, and a body's own symmetry can forbid motion even with two (817) — in thick, slow water, where drag is all there is, a swimmer with a single hinge that opens and shuts goes back exactly as far as it came, the scallop theorem, so swimming there needs at least two knobs moving out of step. The theorem counts knobs, and both of its edges matter to a builder. A knob that goes round, rather than back and forth, escapes it: a coil turned on an axle has one knob, and every turn carries it the same way. In drag-only water it goes farthest a turn when the tangent of its winding angle is the square root of the grip ratio (54.7 degrees at 2, 67.8 at 6), but per unit of work it goes farther wound shallower, so the fastest coil is not the cheapest (run COIL_SNAKE 20261017_204030; the closed form is textbook, its first authors owed). And a body's own symmetry can forbid motion even with two knobs: a coil that is the same turned end over end, squeezed the same all along, cannot be pushed either way by any loop of squeezes, however clever. The pumping coil went nowhere to 3×10^{-15}, and a wave run along it swam at once (runs COIL_PUMP 20261018_082424 and COIL_WAVE 20261018_082544). Before registering a swimmer, look for its own symmetry (row 259) (the Loop Atlas sessions)

the Loop Atlas sessions (Guide items of sessions F and G); ties rows 259, 630

Read a falling body's twist and turn in a form that can hold them: carry a mark across the joint, keep the turn signed, and break a symmetry before crediting a topology (818) — a cat that falls upside down turns over in the air with nothing to push on, by bending and twisting its body through a loop of shapes. How far it turns, and whether its waist twisted on the way, are easy to misread, because the usual ways of writing a turn can hide a full turn, a sign or a symmetry. A no-twist rule is checked at the shape it treats as straight: the Loop Atlas's three-dimensional cat (CAT_3D) made its back half a pure turn about the line square to both halves, a rule that is singular at the straight body, so a trick that passes through straight can wind the waist a full turn while every instant looks twist-free. Read twist by carrying a mark across the joint without turning it, and follow the relative turn as a unit quaternion: ending at minus its start is a full turn of twist (Dirac's belt trick). The twist-free cat needed a full circuit and landed level; the toy's half circuit and nose-low tilt were the twist's (run CAT_TWIST 20261017_231928; Kane and Scher 1969). An angle read from 0 to 180 cannot tell a turn too short from a turn too far, so a turn is read from its rotation's quaternion kept signed along the flight before it is called short: the twist-free cat's 152 degrees at a bend of 90 was 208, past its feet, and "turns most near 75" was where it came nearest a half turn (run CAT_SWEET_READER 20261018_041321). A landing read as the size of a turn cannot tell a loop flown forward from one flown backward either, so a column meant to test the swing's sense agreed by construction: know what a reading cannot see before setting a comparison beside it. A symmetry can pass for a mechanism: the toy cat's exact half turn looked like the gift of its waist's odd loop, and it was its two halves being the same; halves of 0.25 and 0.35 fall 23.9 degrees short (run CAT_SWEET 20261018_034953). Break the symmetry before crediting a topology. When a check finds that a published page teaches the wrong mechanism, as the Loop Atlas's cat room did, first by hiding the waist's twist and then by teaching the wrong half of the turn, put a plain note on it at once, then rebuild it on the corrected model behind a switch that leaves the old default untouched, so every old check reruns to its digits; keep the old toy beside the new one as the comparison, and give the page a meter for the quantity that was hidden, here the waist's twist (the note in one exchange, the rebuild in the next; run EXPLORER_CAT_TWISTFREE_CHECK 20261018_031532). And a planted check set at the arithmetic's grain, 10^{-15}, caught two tools writing one stretch's length two ways, a difference in the last bit of a rate at a stretch's end, before any prediction was printed (the Loop Atlas sessions). A grid's nearest miss can be the grid's own coarseness. On a 1-degree grid of bends, the twist-free falling cat at radius 0.3 came no nearer than 0.43 degrees to landing with its paws straight down, and a rival said nothing between bends of 75 and 85 would come within 0.1; but at one bend, 78.83660617, a circuit turns the cat over to within 2.77×10^{-7} degrees, the body turned 179.999999723 (run CAT_SWEET 20261018_034953); that bend, found blind, is the flip bend of Rademaker and ter Braak's formula within 1.43×10^{-7} degrees (row 820), and the same run flies Kane and Scher's own cat along their own path to 0.10 degrees from straight down. Where an exact point may exist, look for it before reporting a grid's best (the Loop Atlas sessions)

the Loop Atlas sessions (Guide items of sessions F and G); ties rows 224, 238, 357, 820

Two falling cats that look like strangers are one cat under two rules: a thick cat twisting at its waist and a thin twist-free cat bent to the mirror angle turn alike on every path (820) — a falling cat can turn over in the air with nothing to push on by twisting at its waist or, its waist untwisted, by carrying its bend round its body. The Loop Atlas found the two to be one cat under two rules, and nearly lost the finding to a fault in its own code. Montgomery's ball-and-socket cat (1993) splits its connection into two parts, one kept by Kane and Scher's no-twist joint, blind to where the hinge sits, and one kept by a joint whose halves spin opposite ways about the spine, a rule he does not study, which feels the hinge through the parallel-axis term (the project's reading of his equation (9), borne out in flight). The two parts are one function read from opposite ends of the bend once the twisting cat's crosswise inertia is taken about the hinge. So the twisting toy of radius r at bend b and the twist-free cat of radius r / sqrt(1 + 12 A^2) at 180 − b (tubes of length 1 hinged A from each centre; r/2 at the ends) turn alike, at every shape, through a fixed half turn of the rates, and on every path: within 2.842×10^{-13} degrees on nine swelling paths and within 4.5×10^{-13} with the hinge moved in, the rates within 9.3×10^{-15} at 2,000 random shapes, and both cats match the closed forms within 2.2×10^{-10} at 45 bends (runs CAT_MIRROR_PATHS 20261020_034021, CAT_MONTGOMERY 20261020_035025 and CAT_MIRROR_RATES 20261020_041152). Its ingredients are Montgomery's; the pairing itself was found in none of six sources read in full (Rademaker and ter Braak, Kane and Scher, Marsden's Lectures on Mechanics, Montgomery, Batterman and Gbur) or two read through summaries (Iwai 2010; Essén and Nordmark 2018), and on 22 October it was claimed as new on that limited, noncommercial review; Iwai's book of 2021, the likeliest place left, was not read (row 511). On Iwai's whole family the mirror pairs these two cats alone, and with the hinge at the tubes' centres it is only a relabeling (row 742). The oldest model had the newest proof's key fact built in: Rademaker and ter Braak's two-cylinder cat of 1935 turns both halves the same way about the spine, Kane and Scher's no-twist rule, and takes its turning inertia about the line through the two centres of mass, so the hinge never enters (Montgomery's Fact 2, by construction); their area-law formula is the twist-free cat's within 4.4×10^{-16}, and their worked example, 157.771365 degrees by their formula and rounded by them to 160, is flown within 2.73×10^{-11} with the hinge at the ends and at 0.3 (run CAT_RTB 20261020_063617). Reduce a loop that is the same at every point, only turned, to one instant: the pairing was found when the two cats agreed to the last digit (row 214), and with the loop's whole turn written as one instant's rate carried round in a frame that turns with the loop, the coincidence became a small linear solve that computer algebra could settle. Test such an identity with every size set free, not only the toy's defaults (run CAT_MIRROR_PROOF 20261018_174959). A check made at a map's fixed point cannot see the

map applied twice: that run reported that the pairing needs tubes hinged at their ends, because its search for the mirroring thickness was made at bend 90, which the mirror sends to itself, so a mirror applied twice in the code found the right radius there (0.2080125736) and failed every other bend by up to 118.4 degrees. The miss was filed as nature and stood in the book's headline for two days, until a closed form from the literature disagreed with the printed miss and the code was read. Test a mirror away from its middle, and let a disagreement with print send you to the code before the conclusion (the Loop Atlas sessions)

the Loop Atlas sessions (Guide items of sessions F and G); ties rows 151, 214, 551, 775, 818, 511, 742

Rare events can keep a rhythm: measure a count's spread before trusting its square root, and expect a low barrier to space its events out (826) — rare events, like a noisy needle slipping over a barrier, are usually treated as independent, so that a count's error bar is its square root. That is a measurement to make, not an assumption, because near a low barrier the slips keep a rhythm of their own. Measure a count's spread before doubting its square root: the Loop Atlas's noisy needle's slips, counted needle by needle (2,048 needles a row, 15 rows of 4 to 9 noise units), have a variance equal to their mean within about 5 per cent (Fano factors 0.947 to 1.032), so the square-root error bar is right there, and three of 15 rows two errors out is then a chance of about 1 in 35, kept as it fell, not mended. At 3 noise units, where a needle slips several times, the variance falls below the mean (0.835 to 0.935), as for events that cannot follow one another at once, a reading of the cause not tested (run SLIP_SPREAD 20261020_075635); read again on new seeds with four times the needles, the lowest of those rows came back 0.9244 ± 0.0132, within an error of the exact 0.913, so the spread below chance is real (row 835). Rare events at a low barrier keep a rhythm: over a barrier three times the noise's strength the slips are about 8.5 per cent more regular than independent events, their long-run variance over mean 0.913 to 0.916 by the exact spread for a tilted washboard (Reimann, Van den Broeck, Linke, Hänggi, Rubi and Pérez-Madrid, Phys. Rev. Lett. 87, 010602, 2001), and the needle's own waits give 0.9134 ± 0.0031, 0.9135 ± 0.0024 and 0.9182 ± 0.0020 (run SLIP_RHYTHM_3 20261020_090717). Waits under a tenth of the mean come 0.58 to 0.59 as often as chance, yet the shortest wait seen was about a fortieth of the mean on 2,048 needles and a fiftieth on 8,192, so there is no hard floor (runs SLIP_RHYTHM 20261020_083426 and SLIP_RHYTHM_2 20261020_084919). The project's reading, not tested as the cause, is that after a slip the needle must slide down into its next well and settle before it is likely to climb again. The rhythm is set almost wholly by the barrier's height, the noise moving it only in the third decimal: 0.963 to 0.964 at 4 noise units, 0.985 to 0.986 at 5, 0.996 at 6.5. That rare escapes are counted as Poisson events is textbook, and the spread below the mean at a low barrier is Reimann and co-authors' exact spread, met above; the reading of its cause and the near barrier-only law are the project's own, not checked against a source (the Loop Atlas sessions)

the Loop Atlas sessions (Guide items of sessions F and G); ties rows 616, 725, 827, 828, 835

Noise can make a noisy machine's events most regular at one level: set the noise against what is left of the barrier, not against the barrier as designed (828) — more noise usually means less order, but a machine held near the edge of slipping can run most evenly at one middle level of noise. Too little noise and it slips by pure luck; too much and it wanders; in between, the noise only tips it over and its own slide sets the beat. A needle held at 0.95 of its slipping edge slips by pure luck when the noise is weak (Fano factor 1.00000 at noise 0.05) and diffuses when it is strong (1.51376 at noise 3); between, its slips run more regular than chance, least spread at noise 0.7, a Fano factor of 0.42341, where the barrier left is only 0.09 noise units: the noise tips the needle over as soon as it arrives, and the slide round the rest of the turn sets the beat (run RHYTHM_BAND 20261020_154009, from Reimann and co-authors' exact formula, row 826). Run on the needle itself, its waits read 0.51695 ± 0.00248, 0.42357 ± 0.00126 and 0.50779 ± 0.00112 at noise 0.3, 0.7 and 1.5 against the formula's 0.51429, 0.42341 and 0.50967 (runs SLIP_BAND 20261020_161039 and SLIP_BAND_2 20261020_161539). At a fixed barrier height the rhythm is a weak-noise law: 0.9126 at 3 noise units as the noise falls, where their scaled tilt settles on −2.476, and 0.007 more by noise 0.5. A builder who wants the most even slips sets the noise against what is left of the barrier, not against the barrier the machine was designed with. Coherence resonance is known in kind for noisy excitable systems, and very likely for this needle (Lindner, Kostur and Schimansky-Geier 2001, cited by Reimann and co-authors, not yet reached); the numbers are the project's own (the Loop Atlas sessions)

the Loop Atlas sessions (Guide items of sessions F and G); ties rows 725, 826, 827

Look for the shape a system passes through on its way to failing, then test whether that shape travels to the neighbouring settings before calling it a warning (830) — a setting where something fails says where, not how. The shape a system passes through just before it fails can be a door a builder can watch for, but a door seen at one setting is a story about that setting until it is tested at the neighbours. Look for the shape on the way to failing, not only the setting: every one of eight smoke-ring train breaks came within 200 time units of the ring slipping through shrinking to 0.207 to 0.211 of a radius, while 21 of the 22 trains that held never went below 0.214, and a squeeze that deep was survived at least five times, so it is a gate with a chance at each passage, not a trigger (run TRAIN_FLOOR_2 20261021_083613). Where the gate sits outside the range the toy's rule was built for, here a core over half its ring's radius, test the rule before crediting the water: read at three core sizes, fatter cores broke through near the same ring size, 0.203 to 0.227 over seven breaks, with the core up to 0.66 of the radius and none at the 0.236 a fixed shape would give, so the door is the train's and not the rule's, though looser at the fatter core (run

TRAIN_EDGE_MAP 20261022_060414). Then test whether the door travels: at each core's own edge, flown on Bear's laptop, thin cores broke with their core a third of their radius and with no squeeze warning at all, their holding trains squeezing as deep as their breaking ones, so the clean door seemed to belong to the middle core, not to every train (run TRAIN_EDGE_DOOR 20261022_070632). That reading was the fixed threshold's: the door slides with the core, about 0.0137 of a radius for each 0.01 of core, and counted against each core's own door, six of six thin-core breaks came through a squeeze; the door lies on one line at seven cores, four of them forecast before they were flown (cards 259 and 262; rows 839 and 865). The door had already been written into the book's builder's steps as a general warning, and was corrected there on 22 October and again on 23 October. And split failures by how they happen before comparing them: trains started wide also break another way, flying apart with no deep squeeze (run TRAIN_DOOR 20261022_055043). That escape from a chaotic set passes through a narrow region of its phase space is known in kind; the gate's size here is the project's own (the Loop Atlas sessions)

the Loop Atlas sessions (Guide items of sessions F and G); ties rows 724, 819, 822, 839, 865

Name a kit for its home session, and quote a run time with the cause it leaves out (842) — a kit that runs on the person's laptop is a parcel sent away, so its name must say where its results go home, and its price in hours must say what could make it longer. A door kit named after the engine it came from (LOOP_ATLAS_RECORD_SURVEY_DOOR) left Bear, with four sessions running batches, unable to tell where its results belonged; the name came from a tool's lineage, not from its home. Every kit a session ships now starts with its home session's name (RF_SPOKE_ in R2W), its console's first line and its status page say which session and which book it serves, and every reply says what of the session's is still on the laptop. And quote a run time with the cause it leaves out: the 3/7 ladder kit's status page said three hours, up to five, from the container's measured first two levels, each next level doubling, and it took 516.8 minutes. The deep dances' Newton noise (multipliers scattered by a few parts in a thousand at one drive) kept the secant from settling, so the 448- and 896-dances took 30 and 62 evaluations where the shallow ones took 11 to 15. When a level's cost is set by its noise, estimate from the noise, not from the length (R2W)

R2W (sitting B12); ties rows 676, 611, 776

Read a loop's matrix by its class first, and name the bit the picture throws away (848) — when a loop's map is a matrix in a group, its conjugacy class is the part no change of frame can move, so read it before anything else. In A Rider in Two Waves the passing matrix of the weightless junction is the monodromy of the linear flow round one beat of the drive, and its half-trace sorts it: a rotation inside the light cone, a boost outside it, a null rotation on it; the steps read from the slip alone agree at 19,477 of 19,481 settings. The top-left entry, which a first attempt read, depends on the frame and showed no sign change at all. Words cross with the class: the group is, two to one, the Lorentz group of two space directions and one time, so "light cone", "boost" and "rapidity" keep their meanings. What does not cross is a sign, the one that says whether a step's slip is even or odd; the figure coloured by rapidity hides it, and the book names it (LIGHTCONE T1 to T3). And put data and model in one frame before scoring one against the other: the triple-star thread scored a heavy star's closest returns against a pure rotation by two angles' mean rates and missed, because the angle it read is the eccentricity vector's seen from a pole the vector's circle passes 0.0018 from, so it slides and jumps while the vector turns steadily about its own centre. Bear's maypole showed why; with both in one frame the test held at the tight tolerance, median ratio 0.98 out to 701 passes (LG2 to LG3′)

R2W (sitting P1); ties rows 116, 248, 106

A special point on a boundary is usually known: find it by its structure, not its numbers (849) — on R2W's light cone every point is balanced as tightly as any other; the exact rider, at strength 1 and bias equal to the frequency, is special only because a pencil can write its motion down, the phase returning after every beat with nothing left over, a parabolic holonomy that walks the cone. Its home was found by the shape of the result, a family of explicit solutions on step edges (Buchstaber and Tertychnyi, 2013), not by its numbers: R2W first credited the step-edge theorem to Glutsyuk and Netay, moved the credit to Buchstaber and Glutsyuk (2017) as Glutsyuk (2024) gives it, and then found the exact rider's theorem there as their Theorem 1.15, read through a summary, the 2013 paper's full text not reached (R2W, sittings P2 and P3). Left open there: the null rotation's shear at the exact rider (240.8, 47.2, 17.1 and 4.83 at four frequencies), with no closed form tried (EXACT_CONE_SURPRISES, OHM_LINE_SURPRISES)

R2W (sitting P1); ties rows 506, 146, 832

File before you send: anything that lives only in the conversation is written to its file before the reply that carries it goes out (856) — on 28 October a reader went back through the triple-star conversation and found eight reads that had been given in chat and never filed; they were filed from the chat. Later that day a ninth read was written and sent but never filed, and its words are in no file; it was not rewritten from memory, and the reads file keeps its title with a note. Write first, then send: reads, quotes, decisions and anything else that is the record

the triple-star sessions; ties rows 589, 676

A long run on a shared machine: measure it first, cut it into saved segments, and stop it only by its number (857) — the triple-star thread's 60,000-pass run first hit a two-hour job limit without saving anything; a scratch run measured afterwards (0.128 s and 0.264 MB a pass) showed one half of the run, 33,000 passes, needed about 8.7 GB against the workspace's 7.7 GB, so time and size a scratch run and quote a finish time only from it. Cut into twelve segments that each save their state, with a check that a cut changes nothing (one segment bit for bit, two within 1.19 × 10^{-9} turn), the run survived a second limit with eleven of twelve saved; and copy the tool's printed break points into the register, not the plan's (cutting by time with a tenth to spare, it cut at about 5,605 passes, not 5,000). Never find, wait on or kill a process by a name pattern: a wait matched its own command and sat out its limit, and a kill by pattern ended its own shell for the fourth time; stop a process by the number ps prints and wait on a file the run writes. And a tool that picks the newest file by name makes its whole folder part of its input: keep another machine's outputs out of it, or pin its inputs by fingerprint. Write the lesson into every tool's skeleton, not into the plan: the day after the Loop Atlas wrote the three-body session's lesson into its plan for laptop kits, a container tool timed itself from flights of another kind and kept every result to the end, and the two-hour limit stopped it after 11 of 14 flights with no verdicts written (TRAIN_PARTING 20261029_030532); a ring hundreds of radii from the others made the same integrator take some thirty times more steps, so time a whole flight through every regime it reaches. And before laying a slowness to the physics, set like beside like: flights that never swung out ran 5.2 times slower in a borrowed computer's first five hours than later in the same run, so the estimate was near and the machine was slow. A long run prints a line for each piece of work with its seconds, so that a slow machine shows as it happens and a sleeping one as a gap

the triple-star sessions; ties rows 786, 611, 720

A knob one project adds on purpose is an artefact another must measure: keep a two-way list, and measure the artefact against the knob's scale (859) — Bear named the friction that A Rider in Two Waves and the Loop Atlas put into their toys "honey"; the three-body problem has none of its own, and the only honey in the triple-star toy is the integrator's (row 858). The day after, R2W ran its own check with its honey at zero and found its integrator draining energy as if a honey had been added, 1.85 × 10^{-6} at the worst of seven weights it sampled, shrinking about a hundred-thousandfold for a step ten times finer; the law's own peak, at the step rule's switch and the lowest drive, is 1.8 × 10^{-5} (row 850), and the junction engines' leak is a separate check (row 858). Two tests went back to it from the triple-star thread: whether the honey goes as the weight to the power −3, answered in row 850 (+3.00 below the step rule's switch and −2.00 above, worst at weight 0.25), and how the smallest honey R2W's book reads compares with the leak, answered in R2W's sitting P1: it is 55,000 times the largest leak at the toy's own step. The three projects share one frame, something that turns, read once per tick of a second clock, locking where the clocks agree, and they differ most in honey; the triple-star thread's pattern-recognition file sets the honey among the walls of measurement expected to recur

the triple-star sessions and R2W; ties rows 858, 741, 850

Prove a deposit from its zips: follow every file it names, start one launcher from the unpacked zips, claim byte-identity only where it holds, and word-read it (862) — run from its two zips alone, the islands paper's provenance table failed one row, because a ledger had been kept out by its name; later the zip proof found a check unreadable without a segment that was not deposited, and then that 21 deposited launchers named a launch script the list had never carried, so none could have started a run, though the README said each checks every file's fingerprint; that gap was open four days and closed before any upload. Three builds of the paper's PDF had the same size and text but three fingerprints, so the README promises the text and the size, not the fingerprint, while the zips built byte-identical twice. And read the deposit's words by hand: a deposited tool's comment that quoted chat with a typing slip was reworded with its code unchanged, the manifest keeps the as-run version as retired, and the README says so (triple_deposit_list v1.1 to v1.4) (the triple-star sessions). And read the launchers' bytes, not only their words: an edit rewrote the build launcher with Unix line endings, which cmd is known to misread when it searches for a label, and three issues of the publishing packet carried it unseen; every launcher's line endings are now checked before a packet is zipped (the Field Guide sessions). Before deposit, every pointer resolves inside the deposit: a fresh reader found 29 figures in R2W's book cited by number nowhere, lesson marks the book never explained, and pointers into the record's line numbers, a shell command and another work's notes; a script now lists every figure no sentence cites, 0 after the fix (R2W, sitting P3). Check every file the paper names against the crate before upload: Question the Substrate's provenance table named 307 files, and a check run only after the deposit found three missing (the pilot predictions written before runs 1 to 3), which took a version 1.1. Parsing the table and listing the crate takes seconds (Question the Substrate). Run the "one command" in a fresh copy before claiming it: Two Arrows' first full rerun found one step broken, a run that expected its shared code in a folder beside it; the recorded script was left as it was run and the runner now links the folder, after which all 148 files compared matched, data and figures byte for byte and printed output apart from timing lines. Scan the assembled deposit, not only each edit, for pointers to private files: the body twice told readers to see a register row no reader will have, and only a scan of the whole paper found them; the public crate's builder now refuses to write while the scan finds any. And prove a mass edit touched only what it meant to: a three-line contact header went into all 110 Python files, and Python's own tokenizer confirmed for each file that only comments changed (Two Arrows)

the triple-star sessions; the Field Guide sessions; R2W (sitting P3); ties rows 844, 845, 774

A clock and a switch make a road of their own, and chaos can have a skeleton the eye sees first (873) — the Loop Atlas's card 14 is a boost converter under peak-current control, built from textbook circuit physics, its clock as the drive. The inductor refunds its whole energy every lap and the source's work equals the load's to 2×10^{-16} of itself; with the output held the lock breaks at duty one half, as the textbook says (19.965 V against 20); with the capacitor in, the doublings come in two kinds, two smooth flips with a border collision between them, and then chaos with no window from 2.59 to 5 A, the robust chaos Banerjee, Yorke and Grebogi (1998) found in the same converter, where the needle's road ran smooth doublings with windows in the 1973 order (row 872). Halving the clock period changed the power by 7 per cent and halved the work per lap: a converter feeds by the second, not by the lap. Over the picture of that chaos Bear drew four straight lines and wrote that it was not chaos. Tested, both readings held: the converter is chaotic at every setting (a twin drifts off at 0.22 to 0.48 a lap), and every line he drew is a real edge of it, within 0.12 A: the switch's border, a lap ending exactly at the reference current, which is the top edge and is never passed, and that border's images lap by lap, one line another lifted by exactly E T/L, here one ampere, the rise of one whole lap with the switch closed. A chaotic map's density is edged by the images of its turning point, as in the logistic map's picture and in the needle's chaos, where Bear's strokes lay on the fold's images (row 707); here the turning point is the switch's border, and the person looking at the picture saw the skeleton before the test that it is one. Place each smooth doubling where the repeat's own eigenvalue crosses −1, not by settling: a scan following its last state upward called slow settling a period 2, and a refiner started from rest at each try never left period 1, so the onset it printed, 1.9100, was the scan's grid point, and the onset at the eigenvalue's own crossing was proposed and not run (misses row 506)

the Loop Atlas sessions (session I, Guide items 497 and 500 and misses row 506, card 14; its checks and records: CARD14_CONVERTER, CARD14_HELD_REDO, CARD14_LINES); ties rows 872, 726, 573, 707

Count a part's life on the clock it wears by, and know where that clock has a knee; and the back side of the power curve is a fold (876) — the Loop Atlas's card 61, a toy trainer airplane built at Bear's word, set beside the way its logbooks keep time. A recording tach counts the crankshaft's turns, so tach time is the engine's own event clock, and a tach-hour interval for the engine wastes only its margin, whatever the airplane flies. A control hinge wears by the times its surface swings past a dead band, and in turbulence those come at Rice's level-crossing rate, met in the toy within 1 per cent at three gust strengths forecast from one calm hour; that rate has a knee, slow in calm air and saturating at the pitch loop's own crossing rate once the gusts swing the elevator past the band. A tach-hour schedule for hinges therefore wastes most of the life on the calmly flown airplane (85 per cent unused) and little on the rest; landing gear wears by landings, and a pattern trainer's schedule wastes the gear of every airplane that flies cross-country (96 per cent unused). The same toy shows the back side of the power curve as a fold, the needle's lock edge in an airplane: at a fixed throttle a level-flying airplane has two speeds where thrust meets drag, the slower unstable and the faster healing, each at the rate drag's slope over the mass, and the two meet at the speed of least drag, their gap growing as the square root of the extra thrust (slope 0.50000, the rates within 1.2 per cent at three thrusts). The back side's own runaway is slow, an e-folding of 24 to 58 seconds, so it is not what drops the floor in a flare: held level from a 3-degree glide at 30 m/s with the throttle unchanged, 98 per cent of the first two seconds' deceleration is the downhill push given up, the airplane is below its stall speed in 6.0 s, and adding exactly that push as thrust holds the speed. A pilot's little gas at the flare is that push given back. The hinge's clock does not keep speeding up with the gusts, as first guessed; the redesign, proposed, counts wear rather than crossings, each swing weighted by its size to a power, as fatigue rules weigh it (misses row 510)

the Loop Atlas sessions (session I, Guide items 501 and 502, card 61; its checks and records: CARD61_LOGBOOK, AIRPLANE_FOLD); ties rows 788, 627, 746

Each doubling turns the last one's corner over, by a ratio that belongs to the hump's class: −1.9644458 for a rounded top (877) — a doubling bends an orbit average against the drive, and each doubling's corner, the jump in that average's slope, is turned over and shrunk from the last by a fixed ratio. Scanned ten times finer and settled four times longer, every one of the Loop Atlas needle's four lanes turns its corner the other way at each doubling from sixteen-to-thirty-two down to one-hundred-twenty-eight-to-two-hundred-fifty-six, as Feigenbaum's negative α says each generation should, the ratio settling at −1.966 to −1.968 in all four lanes, neither −2 nor δ over α; read after 8,000 passings the deepest ratio had swung from −1.85 to −2.00, a slow settler keeping its last unsettled part longer than any rule of thumb. The logistic and sine maps, solved exactly in 80-digit arithmetic to the doubling from 8,192 to 16,384, settle on −1.9644458, agreeing to the ninth figure in the whole orbit, in each of its four lanes and for the mean of x cubed as for x; the toy's last readings lie about 0.002 from it, its four lanes at thirty-two-to-sixty-four within 0.0006 of the logistic map's four at the same depth and in the same order (read after the data). A number two maps of one kind agree on to nine figures belongs to the kind; then try another kind. On $1 - r|x|$ to the power z the ratio is −4.42 at z = 1.5, −1.9644 at 2, −0.915 at 3, −0.603 at 4 and −0.372 at 6: turned over at every generation in every class, shrinking for a sharp top and growing for a flat one, the turn near z = 2.89. Past a flat top the corners pile up: on the quartic map the mean of x steps toward its limit with its slope swinging and growing by 1.66 a generation, the reciprocal of the corner ratio, a corner of infinite slope, its steps shrinking by −4.3958 by Aitken's step against the corner's forecast −4.3946, while on the logistic shape the slope settles at −0.17601. Bear pictured the corner as a sink where differences go to be forgotten; read after the data and still untested, the sine map's ratio less the logistic

map’s shrinks by 4.35 to 4.66 a generation, 6.4×10^{-5} to 4.1×10^{-10}, so the corner keeps the class’s number and drops the map’s. A map whose top’s value lands on its own second turning point doubles as the two orders multiplied, only slowly ($1 - r \sin(\pi x/2)^2$ as order 4, δ 7.28468; its quartic cousin as order 8, δ 10.95 and corner ratio −0.279), so a test map should keep one turning point in reach ($1 - r(x^4 + 0.3x^6)$ gives the quartic class’s −0.60323). Two cautions travelled with it. A small error at one point is not a convergence: a two-scale guess for the ratio missed by 13, 10, 6, 3 and 0.5 per cent at tops of order 1.5 to 6, then by −0.8, −1.5 and −1.9 per cent at 8, 10 and 12, crossing the truth, not settling on it. And before reading a pairing by sign as alternation, look at the order the pair comes in: the lanes bend in pairs, 0 with 2 and 1 with 3, and the dance goes round 3, 1, 2, 0 at every one of 481 recorded columns, so the pairs are visited back to back and never alternate. As far as a limited pass of free records shows, the ratio is not named; the response-function literature, its likeliest home, is still to be read, its nearest paper seen as a title only: a pointer for a specialist

the Loop Atlas sessions (session I, Guide items 503 to 505, 507, 512, 518, 521 to 523 and 526; Bear’s picture of the corner as a data sink; its checks and records: LANE_ORDER, TOP_CORNER_LONG, LOBE_FLIP_FINE, CORNER_UNIVERSAL, CORNER_CLASS, CORNER_TWO_SCALE, CORNER_SINK_READ, CORNER_CUSP, QUARTIC_SECOND, QUARTIC_THIRD, COMPOSITE_DEEP); ties rows 680, 874, 771

Where a switch can hold a body still, compute it with no switch at all: the count is the switch’s, the push the world’s, and the flicker is counted from the stays (879) — at heavy loss the Loop Atlas’s spring is held at zero speed for a share of the time, the spring’s pull and the soup’s backward push balancing while the blade is on and not while it is off, the switching books’ sliding mode met in a toy. Computed with no switch at all, held while the spring pulls forward and the soup would drive it back, its push then the spring’s own pull, it spends 0.1376 of its time held (0.1375 at twice the step), makes 0.458 stays a soup cycle and pushes 0.2265 of strength, within 0.25 per cent of the plain blade’s 0.2259. There the blade flickers at a rate its knee sets: the count grows nearly as the inverse of the knee (a local power of 0.81, steepening) while the push stays within 0.5 per cent of the plain blade’s at every knee from a twentieth down, so the count is the switch’s and the push the world’s. The flicker can be counted before it is flown, from the stays: summed over the stays, each stay’s flicker law gives 0.168 a soup cycle over the knee, and the knee ear’s count carried to its limit reads 0.169 over the knee, within 1 per cent; with the switch-ons outside the stays added, the forecast is 54.54 against 53.13 flown at loss 1 and 8.370 against 8.397 at loss 0.7, while at loss 1.5, where the stays’ sum shrinks to a fifth, the count runs a third over it. Carry a halving test three steps and read its order before calling a count free of the step: the knee ear counted 2.5602, 2.6462, 2.6993 and 2.7266 switch-ons a soup cycle at steps 0.01 to 0.00125, each rise half the last, so the first count, called free of the step on a 3 per cent change, lay 7 per cent below its limit, near 2.75

the Loop Atlas sessions (session I, Guide items 508, 509, 514 and 520; its checks and records: SPRING_KNEE_COUNT, SPRING_STAY, COUNT_LAW); ties rows 878, 627, 785

A mind’s own certainty is a rich-get-richer filter: the average holds while the smallest answer dies (883) — in Question the Substrate a small network labels loops by how many times they wind about a point, and each generation trains the next on its labels. Keeping only the half it is surest of (the paper’s “own doubt”) favours the big answers: in world 1 “winds zero times” swelled from 33.8 to 55.6 per cent of predictions, and in world 2 an answer holding 11 per cent of the data went extinct, 10.6 per cent to 0.0, with no alarming lesson along the way (run E14). The average hides the extinction: overall accuracy barely moved, once rising 2.1 points, while a small answer vanished, so track the smallest class beside the average (E16, E19, E31). Clean is not safe: in world 2 the surest half’s kept labels stayed cleaner than a rubber stamp’s, a checker that passes every label (15.4 against 26.0 per cent wrong at generation 6), and still lost more, because the damage was what it never passed on, not what it let in (E1w3); in world 1 both happened, the surest half ending dirtier than the stamp (28.3 against 18.6 per cent, every start; E1j). When a fix works, measure it both ways: the field’s known fix, selecting within each answer, removed the survival cliff and then over-predicted the small answer, more the smaller it was, 1.17 times its true share at 12 per cent and 2.51 times at 3 per cent, while accuracy rewarded the extinction (+2.1) over the rescue (−0.4); a fix is a new filter and needs the same per-answer gauges (E22). One reading was still open at the paper’s close: the better the first network, the bigger its fall under its own certainty, in four comparisons (world 1 starting at 77.4 and 82.0 per cent, world 2 at 77.6 and 82.5, the world-shape dial from 87.7 to 66.1, and a gapped world 2 starting at 90.1 and falling 16.5), and the falls land near the same place from different starts (55.7 and 56.7 per cent in world 1)

Question the Substrate (the two-voices series, Paper 1); ties rows 77, 218, 695

Humility without courage spirals, and a cure tested as one bundle hides which knob works (885) — in Question the Substrate the rule of keeping the surest labels, joined to a checker that marked fragile inputs “unknown”, ended up saying “unknown” to 66.9 per cent of test loops, right 92.0 per cent of the time on the third it answered (run E10). Two knobs together cut the loss from 24.2 points to 5.7 while “unknown” held near the true share of fragile loops (22.6 against 18.95 per cent): pass on a quarter of the doubts, and cap the “unknown” lessons at the fragile share (E10b). The paper first credited the cap with stopping the spiral; sweeping the cap showed it never did, the doubts passed on did, and the cap is only a dial (E27). Check your own earlier explanations with a sweep of each knob alone (Question the Substrate)

Question the Substrate (the two-voices series, Paper 1); ties rows 806, 883

A copying chain compounds large leaks and absorbs small ones: read a dial at its small end before calling a drift universal (887) — in Question the Substrate a rubber stamp, a checker that passes every label, held level for six generations and slid 7.9 points when 5 per cent of its labels were flipped each generation; the children inherited the noise rather than averaging it out (run E25). But “every leak compounds”, read from a sweep of 5 to 10 per cent, was too strong: world 1’s chain absorbed a leak of 0.5 per cent a generation entirely and settled larger leaks at higher levels, while world 2 drifted with none (E34). An absorbed leak is a restoring force, and a self-stable error rate is real and fragile; the small end of a dial is where the restoring force shows (Question the Substrate)

Question the Substrate (the two-voices series, Paper 1); ties rows 77, 883

A learner heals its answers long before its inside: in any study of drift, read the inside directly (888) — after a closed loop of tasks and 30,000 steps of practice on the start task, Two Arrows’ learner keeps 0.0167 of its change in its answers on return while its hidden layer keeps 0.3049; the answers come home 18.49 times better (the mean of five starts’ ratios; run r3). A checker that grades only answers would call the learner healed. The answers come home and the self does not, so a study of drift reads the inside directly and reports both arrows. It is row 66’s tension between healing and remembering, met in a learner (Two Arrows)

Two Arrows (the two-voices series, Paper 2); ties rows 66, 884, 889

Only a kept record of the self brings it home, and held after the trip it costs the journey nothing (889) — in Two Arrows, practising with a pull toward a record of the hidden layer taken before the loop leaves 0.0003 of the inner mark; practice on the true answers keeps 0.3049, on its own answers recorded on return 1.7621, on its own current answers (a checker that cannot say no) 4.6840, and on a stranger’s answers 5.6064 (run b2). The other three keep more of the mark than the true answers alone. It is the mirror lesson of row 77, inside a learner. Hold on after the trip, not during it: a learner that holds its inner record throughout the journey follows the moving task 1.382 to 2.156 times worse, while the same record switched on only after the trip leaves 0.00019 to 0.00085 of the mark and costs the journey nothing (runs b16, b17) (Two Arrows)

Two Arrows (the two-voices series, Paper 2); ties rows 77, 308, 888

Look for a relabelling before calling a change continuous (890) — half a turn on a direction-blind task returned Two Arrows’ learner with a pair of hidden units exchanged. Renaming the units removes 68 to 85 per cent of its distance from home, and the exchange is one event, just past a quarter turn (runs b4, b5). Permutation symmetry can disguise a discrete change as a large continuous one, so a distance between two states of a network is read after matching their units (Two Arrows)

Two Arrows (the two-voices series, Paper 2); ties row 888

L3 — Holonomy You Write

The write limb: creating invariant loops, acoustic marking, cured helixes.

The rows of L3

New in version 7 on this limb: 31 of its 60 rows, numbered 562, 563, 564, 565, 581, 657, 667, 674, 677, 682, 692, 712, 721, 746, 773, 779, 786, 796, 800, 803, 815, 825, 838, 846, 854, 860, 861, 867, 869, 878, 881.

Active loop walking — the probe (31) — You can actively drive a vortex ring around a test loop to probe the geometry it passes through — this is active loop walking. A parked ring (not actively propelled) stalls at 0.784 ring radii (R) from its own centre and loses 29% of its energy in 0.5 seconds at Reynolds number (Re) 300, the thick-fluid regime. At water viscosity the loss drops to about 1%. Viscosity is the dominant energy loss, and the probe must account for the fluid regime.

active sensing

Loops that close themselves (37) — take away the lip that shuts the slot, or let an open-mouth refill pull the bell back, and the stroke brings the body home almost where it started; only vorticity kept under the rim or left behind breaks that symmetry. The rung 6 switch-offs, in the paper’s own ranges: lip off −0.015 to −0.019 R per cycle, plug off −0.03 to −0.05, shape off −0.07, −0.16, −0.23 and −0.17, against −0.115, −0.217, −0.204 and −0.189 for the full stroke; and the pinched moon jelly, with no plug, surges to −0.30 R apex-first by 0.4 T, then the open-mouth refill hauls it back at +50 mm/s to a net +0.03 R per cycle the wrong way (Jellyfish: the Holonomist of the Sea, Appendix B §7A)

limit cycle; phase condition

A second channel carrying control (40) — A ring architecture that reads topology can simultaneously carry a control signal in its second degree of freedom — reading and controlling in the same structure. The signed-ring data channel retains 39–40% of its spin over about 9 ring radii with its sign held; executing a quarter-turn swirl manoeuvre costs about 0.7% of speed. The two functions — topological readout and active control — happen at the same time in the same ring.

Shapere & Wilczek 1989; holonomic quantum computation

Write-time holonomy — the inversion (43) — the whole architecture: conventionally you search a substrate for an invariant and walk loops to detect change; here the path that writes is the loop and the invariant sits where the path puts it — "you stop hunting and start placing"

ours (named in the paper)

Name your structure group before you fabricate (44) — composing symmetry-reduced rotations around a loop moved 201 map cells into different defect classes between two computers running identical code on identical bytes, and moved a boundary-seam count from 951 to 2207 across five equally defensible tie-break rules; the same construction on $\mathbb{Z}/2$ is path-independent to machine precision

gauge choice; ties row 239

The cut chooses the algebra (45) — sealed cavity → 2-sphere → π_2 → $\mathbb{Z}^n$ abelian → an address; through-tunnel → handle → π_1 → free group → non-abelian → a sentence; the map between them is abelianisation, so the readout algebra is chosen at write time

π_1 / π_2; abelianisation

Closure is the admission rule (46) — a path that does not close has no holonomy, so closure is the criterion for being a word at all — at length 12, 5.1% of 2-D words close (853,776 of 16,777,216) and 1.5% in 3-D; that rejection ratio is error detection performed by geometry before any checksum exists

The read head is a theorem (47) — You can count the holes in a 2D shape by listening to how it vibrates — the shape tells you its own topology through its spectrum. The heat-trace theorem: for a flat domain with h holes, the high-frequency end of the vibrational spectrum (the heat trace at short times) carries a constant term $a_0 = (1 - h)/6$. Each hole contributes $-1/6$ to the constant. Three holes, and the geometry is as audible from its spectrum as its area. The read head is the theorem — no geometric scan required.

Kac 1966; McKean–Singer 1967

A famous impossibility, repurposed as a feature (48) — the spectrum does not determine the shape — so the medium supports a read that discloses the stored value without disclosing the structure holding it. Nobody designed that; it costs nothing

Gordon–Webb–Wolpert 1992

Reads-until-erasure — an unreported category (51) — every read is hot; flash quotes write endurance and nobody publishes a read budget. $\tau = \tau_0 \exp(\Delta E/kT)$: $\Delta E \approx 0.5$ eV gives fractions of a second, ≈ 1.1 eV gives centuries, and tiers must be separated by ≥ 0.25 eV or their lifetime distributions overlap

Arrhenius

A spectral read needs no pose (52) — a kinematic read must specify seven numbers — three position, three orientation, one start phase — or the read clock drifts off-sample; ringing the whole object is pose-free by construction. The trade is one number for the object instead of a spatial map

The alphabet is a product, and the accidental degeneracy (53) — $12 \times 36 = 432$ — the semantic alphabet size was forced, not chosen. And on a 3×3 grid $\lambda = 4$ arrives with multiplicity three, so one chirality channel is spectrally inseparable from a non-chiral mode; a slight aspect ratio splits it. Found at the desk rather than after somebody built the cavity

Laplacian eigenvalues

The lightness is the immune system (81) — stated in the deposited paper: seriousness is what turns a tool into a faith, and if a thing can be laughed at it can be tested, while if it is sacred it is already corrupting. Carried with the recursive clause — certainty that certainty is a red flag would itself be the trap. This is the published precedent for the 2026-09-11 editorial-balance ruling; holonomy-is-chill is not a new stance, it is this paragraph from July

The hunter reads; the composer writes (110) — the read/write symmetry stated in one line: the instrument reads defects out of a substrate it was given, and the same geometry writes contours into a substrate it shapes. That sentence is a charter for the whole write limb, which the guide deposited when this was written did not have at all (it is now L3, Holonomy You Write), and it arrived in a node about tornado false alarms

ours; ties row 43

They match to erase; we match to discover (162) — the minimum-weight perfect-matching decoder exists to pair defects away so a quantum memory survives — and pointed the other direction, the same engine pairs the noise away so that the one defect which refuses to be matched stands exposed. Their residual is failure; our residual is the detection. One sign flip, and the machinery is production-grade open source that can be imported rather than re-derived. The provenance note travels with it: the wiring has survived two independent adversarial searches and is still short of a full literature pass

MWPM decoding; quantum error correction

The address closes and the datum closes, and they are the same walk (234) — the sharpest sentence in the corpus for what write-time holonomy is, and it upgrades row 43. Conventionally you search a substrate for an invariant and walk loops to detect changes in it — the instrument hunts. Here the path that writes is the loop and the invariant sits where the path puts it; you stop hunting and start placing. And then the image: the invariant is deposited on a loop, and the invariant is a loop — a track that returns you, upside down, to exactly where you started, which is what a holonomy does to a frame

Tamper response as a consequence of the read, not an alarm bolted beside it (235) — reading at the energies an invasive probe or die-imaging attempt requires is a supra-barrier operation, so the attack anneals the very structure it is trying to read. Conventional tamper-evident hardware detects an intrusion and trips an alarm that can be defeated; here the destruction is a direct physical consequence of the mechanism. A security property that falls out of the physics costs nothing to maintain and cannot be switched off

physically unclonable function

Put the intelligence outside, and the safety is the absence of a capability rather than a feature (270) — the mother holds the four things that must persist and be shared — brains, power, master clock, map — and the daughter holds none of them. Its payoff is not elegance, it is safety: a daughter with no onboard intelligence has nothing to go rogue with. You cannot subvert a daughter that was never given a will. Everything else — cheap, disposable, swarmable, manufacturable by the thousand — falls out of the same split. A safeguard built as a missing capability cannot be disabled, because there is nothing there to disable

Handedness is topological, not amplitude — so it survives what amplitude cannot (283) — the chiral codec, and it is the cleanest engineering statement of the corpus's whole thesis: you can attenuate it, blur it, drown it, and left is still not right. The alphabet is the cavity's own normal modes indexed by angular folds and radial rings; driving the two degenerate modes ninety degrees apart makes the pattern rotate, and that handedness is a bit, so every mode with angular structure doubles into plus and minus. And the exception proves it: the ring mode with no angular folds has no winding at all, and it is the fragile, amplitude-only letter. In one alphabet you can see which letters are protected and which are not, and why. A loop walked as a plain figure 8 keeps nothing, because its two lobes turn opposite ways and cancel; a flip at the crossing turns both lobes the same way, so both pay in and their turns add (the Loop Atlas sessions)

chirality as a topological sign

One word, one meaning, one part of speech — because there is no round trip to spare (288) — a controlled vocabulary of about nine hundred terms bound to the chiral alphabet, for machines that cannot afford to ask which sense did you mean? — in a channel this bad there is no round trip available to disambiguate. The vocabulary is not claimed: it belongs to an aerospace documentation standard that has been doing a nearly identical job since the eighties for technicians reading manuals in a language they half speak. The encoding is ours, and we are giving it away, and we would like somebody else's drone to be able to hear our daughters. Where a word cannot be given up, give each of its senses a glossary line of its own, with the sections that use it: nine of the 29 serious findings of R2W's second cold read were one problem, a working word used in several senses, window alone in about ten, and the fix kept the words and listed each sense rather than renaming (R2W, sitting P3)

controlled natural language

You cannot scramble an integer (293) — the in-situ dosimeter's whole case in five words. An anchored unit stays on station while an external therapy fires, reading dose actually received rather than dose intended — and the rad-hardened winding memory keeps a log that the very radiation it is measuring cannot corrupt. A record that survives the thing it records is a different category of instrument from one that merely resists it, and the difference is that a continuous register degrades gracefully into a plausible wrong answer while a count either holds or fails visibly

ties rows 43 and 234

A sensor geometry run backwards is an emitter — and the cost moves from free to powered (367) — reciprocity: the same antenna receives and transmits, the same element reads strain and makes it. So a differential reader inverted is a differential projector, with the resonance channels becoming the mode-shaper. Two fences travel with it, and both are engineering rather than principle: the geometry is free and the amplification never is — passive

reading is cheap while projecting with gain costs energy, heat and an exposure limit; and reciprocity is not perfect, because nonlinear and lossy parts break the symmetry, so the projector is its own device sharing the geometry rather than a mirror of the detector

reciprocity

Angle is the index; side is the load (379) — the napkin's own formulation and it is a storage architecture in five words: the protected integer is the address and the payload rides the side — hard addressing, soft storage. With the companion dial: a wide funnel collects and sees the mouth; a tightened polygon closing toward a sphere stores and becomes the checksum. Which end of that dial you sit on is the design decision, and the same structure does both jobs at different settings

Quantised locks beat a strained stretch (381) — continuously stretching an element to reach a target holds a strained configuration — high energy, drifts, needs active servicing forever. Quantised integer locks sit at stable minima — low energy, no drift, snap-to. And the ceiling that forces the choice is named: stretch past threshold and new defects nucleate as the lattice tears to relieve strain, while modes bleed into each other and cap the usable resolution. Compose discrete stable presets in a stack rather than straining one element to its edge

transfer-matrix composition

A frozen record is silent until something traverses it (443) — the gramophone decoupling, and its consequence. The record is pure geometry and the reader is interchangeable, chosen after the fact, so density is fixed at write time and reading stays upgradeable forever. Then the part that matters: the missing dimension of a static structure is supplied by the act of reading across it, the way a groove is silent until the needle moves. A crystal archive does not run; it waits, inert and incorruptible, until read. The difference between a thing that must stay alive to keep its data and a thing that keeps its data precisely by being able to sleep

ties row 286

The concrete outlives the copper (484) — a marine structure is a fifty-to-hundred-year asset and an electrical machine is a twenty-year asset, so any architecture welding them together throws away the long-lived thing when the short-lived one dies. Put the interface at a pipe, a busbar or a plug you can reach, and keep every short-lived part on land. With the test in one line: if servicing it in year twenty means opening something poured in year zero, it is the wrong architecture. A lifetime-mismatch rule that a reader can apply to a drawing in ten seconds

Geometry beats mechanism — and every member fails into its other job (485) — a perforated wall does storm relief with nothing to actuate; a notched rim does overfill relief with no valve; the blade and the flooded chamber both fail into being a wall. Every member of this family fails by ceasing to generate and continuing to do its other job. Designing the failure mode to be the secondary function — so the failed state is still useful rather than merely safe, which is a stronger property than fail-safe and is almost never specified

The substrate stores poses, and reading is navigation (519) — an orphan named as noticed three or four times and never stated: the substrate does not store data at addresses, it stores data as poses — so you are not looking up a cell, you are dead-reckoning to it. That reframes the addressing problem entirely, and it makes the navigation lineage the addressing layer's foundation rather than a side-quest. Filed as the claim it is — stated, untested, and belonging to the write limb, which is where this inventory is thinnest and where a genuinely new claim is worth the most

A curve and a kernel — no grid, no solver, 88 lines and 4.7 seconds (545) — the vortex filament evolver is a regularised Biot-Savart code where the vortex is a curve of N points and each segment induces velocity on every other via the kernel, regularised by core radius a to avoid the self-induced singularity. Time-stepping is RK4. No 3-D CFD grid. No full solver. 200-point trefoil, 300 steps, 4.7 seconds in the standard container. This is the missing instrument for F7 — the evolver that was needed alongside the DNA spoke's existing reader (writhe, KMT, Alexander polynomial)

Biot-Savart; regularised filament; ties row 86

The price of a latch is the wall, not the store (562) — a superconducting ring stores about one jiggle of heat per rung and holds for years, because its wall is ten thousand jiggles high even in a 20 nm wire. A medium's memory is priced by what must be climbed to change the count, never by what sits on the rung. And a wall guards one path only: a claim of stability names the escape route it is stable against, which for a thin film is a vortex crossing, not the phase slip a thick wire pays

The Latch Atlas; ties rows 51, 381

Erasing is a second write, from the other side (563) — a ring with one weak spot writes one rung at a time, never two, and comes down by a different road; the erase pays what the write paid, and nothing comes back. The loop between the two roads is the memory, and its area is the price of a round trip. A controller's deadband and a store's hysteresis loop are the same object seen from two sides. Memory, read where it lives: holding riders rather than ramping them, R2W finds no memory its readers can see at any edge between a lock and free running at masses 0.75 to 1.5, and finds it between two locks, whose overlap opens near mass 0.61 and is 0.124 of the drive wide at mass 1.0 (row 670); from mass 2.0 to 3.0 the Loop Atlas's needle holds gear 1 beside a smooth spin (row 670), and with the

ripple off a memory band between rest and running first opens at weight 0.7026 (row 711), the gap a phase-locked loop shows between holding lock and grabbing it (row 727). Its earlier scissors reading, a ramp's gap of 0.0008 at mass 0.7, 0.0047 at 1.0, 0.012 at 1.5 and 0.035 at 2.0, was withdrawn on 5 October, its reader tripped by a band that wanders inside the gear (row 639). The Loop Atlas's ramps read an opening that halved as the ramp rate halved, the mark of a lag, and its report of the same scissors opening above a memory of about seven tenths was withdrawn once held riders settled it (row 662) (the Loop Atlas sessions, R2W and the book)

The Latch Atlas; ties rows 381, 670, 711, 727

Temperature is the write-enable, and it works at two temperatures (564) — rungs come into being just below the transition and lock only lower down, where the walls outgrow the heat; the count is written in the gap between. Heat and Kibble–Zurek domains write the same staircase, so closely that no field scan tells them apart (Sheppard's twelfth is the only exact difference); only the way the width moves with cooling rate and ring shape separates them. In niobium the two routes lock at almost the same moment, and the measured slope is steeper than either alone — recorded as an open door, not a finding

The Latch Atlas; ties row 172

A lock stalls at a quarter turn, and a powered lock is not a latch (565) — two dials pulling on each other hold until the lag reaches a quarter turn, then slip one whole turn at a time; a crowd of dials snaps together past a sharp threshold; a voltage standard's steps are exact and vanish when the tone stops. The quarter-turn stall was drawn on Bear's Savonius rotor before the run confirmed it. So three kinds carry names: a latch holds its count with no power, a label carries one and holds nothing, and a powered lock holds only while it is driven

The Latch Atlas; ties rows 38, 352, 380

An intuition can get a home without becoming a claim (581) — the Savonius-rotor picture, held for years as the author's engineer's black box for polarity, went to the page by one path: the picture, one geometric sentence (a line field's half residues pair up), expectations written before the run, an exploratory run labelled as not a test, and a description. The pairing is real; the halves are not quite neutral (a median net half charge of +0.08, unexplained); and a seam of the arrowhead choice ends in an opposite pair too, so the pairing cannot tell a real half-turn from a seam. The page keeps the sentence and the numbers and drops the metaphor's freight. Count what the runs show, not what the pattern promises: to Bear's "Does each knob have a fine tuning component like you just found?", the runs so far say that the drive ratio has its fine trim (the second tone's size, for the light-end term a), and the fourth-power term c has no knob that zeros it. A fourth-harmonic third tone moved a and, in phase at one size, left c where the octave had it (RATCHET-41). His "If you found one, it feels like that is a sign" is a good reason to look and not yet a result (R2W). Four times in one triple-star session Bear's picture or one-line summary became a test written down before any number was read: a swell he traced at the island's centre (the beat, 37 highs 53.5 passes apart, run 20261015_133950); his one-line summary, "the ratio stays fixed, the rates don't" (held at 3 of 8 slices over 600 passes, run 20261015_140219, and as registered at 2,400 passes, run 20261015_151629); his picture of fractals within fractals (two tests of nested chains, runs 20261015_225444 and 20261015_230449); and marks on a sheet of plain figures (a lock at a seventh found without its label, run 20261016_054242). The registration is what turns a picture into a result, whether it holds or not (the triple-star sessions). Bear's one-liner from the triple-star thread, "the ratio stays fixed, the rates don't", crossed benches: it turned up in R2W's dark point law, where each arrow's power in the drive bent far from 2 and 4 while their ratio held near 2 and climbed to 1.95 as the drive weakened (RATCHET-53, 54). His line named the quantity that decides the event before either bench looked for it. And an almost-rhyme is worth a test, because the test finds where the rhyme stops: Bear heard the fourth harmonic's holding powers as even steps, almost like whole or half steps or the evenly spaced trivial zeros. They were not halves or integers (they ran from about 1.38 down to 1.28); what was even was their fall in the logarithm of the frequency, the way a keyboard's keys are evenly spaced in pitch, not in cycles per second. Registered as a test (RATCHET-61), the evenness carried one more stretch and then eased, beginning near frequency 1, where the drive starts to outpace the rider's own pull (row 640). His ear named the test; the test found the edge of the pattern, and the edge is a new clue (R2W). A dream is a stack of predictions: split it, and give each part its run or its card. On 17 October Bear wrote that teaching diving might be easier than he had imagined: a dive has a life span of so many seconds set by gravity, the diver is not fighting the wind, a whole routine could be practised standing up or on a tumbling mat because it is a holonomy routine, and it is done in a timed or an event sequence, each move cued by an event such as a count of turns. Part by part: gravity's clock, yes, for with no upward spring a diver falls 0.78 s from 3 m and 1.43 s from 10 m (run DIVER 20261005_160158). Not fighting the wind: the toys have no air at all and every turn comes from the body; at entry the air pushes back with 1 to 10 per cent of his weight (drag areas assumed), and what it does to his turn has not been measured. Rehearsal on a mat: yes, for the shape part, which is 78 to 95 per cent of every share of the jumper's turn and turns the body the same to a billionth of a degree when done twice as fast, while the count part, 0.07 to 0.10 of the forward turn, exists only in the air (run ARMS_LEGS 20261017_012927; row 151). By count or by event: the Loop Atlas's card 199 asked whether, with the spin from the board off by up to a tenth, a move cued by an event lands nearer the mark than one timed by the clock, and it does, a kick-out cued by the body's angle missing by 0.420 of a clock-timed one (run DIVE_CUE 20261017_082454). Score each part against its own run and say plainly which part is

still open; the open part is often the one the dream cared about most. In the same session his rings, his rungs, his ranking of timing over memory, his power gauge and his dream each became a registered test, and most held (the Loop Atlas sessions)

Dancing on the Sun; ties rows 79, 113, 168

Direction is information: a dither can be a ratchet (657) — a kick and its own octave are always exactly two to one, and a lopsided wobble sets their phase by itself, 70 to 103 degrees from twice the fundamental at every kick shape tested. One band of octave phases protects the next stage, with the relay's own phase inside it; flipped or run backward, the kick protects nowhere. The cousin to credit is harmonic mixing (Marchesoni, 1986), along with rocking ratchets. A part that slips one way and grips the other, a blade, a tyre, a keel, a fin or a diode, is what makes order matter in the Loop Atlas's toys: with grip equal both ways, a three-link swimmer's motion went exactly nowhere (the Loop Atlas sessions). A wiggle can make a slope: a field that pushes in on one line and out on the other holds nothing when steady, and holds a charged bead when flipped fast enough. The grip is the charge, the slope is made by the rhythm, and it fails sharply when the field is too strong for the flip rate (q above about 0.9: the radio trap's first stable region ends at q of 0.92, Paul, 1990). Bear's picture fits it. In his words: "two poles, balanced, falling forward and catching ourselves", and "We could be z=10,000 meters tall and it mechanically works the same. Now if we were z=10 million km tall, we would need to step on planets and keep moving to stay in a steady state (harder to keep still than to oscillate between landing pads)"; walking, a wing on its wake and a bell on its ring are all a fall caught on a pad made a moment ago, and the bead is held only while the push keeps changing sides (the radio slope toy) (the Loop Atlas sessions, on Bear's picture)

the Loop Atlas sessions (R2W); ties rows 190, 575

Read every case before writing the sentence about all of them; writing is a second reading (667) — a figure's description promised star shapes, and the figure, once viewed, showed lopsided closed shapes; a read said every early slip came near the top of its leg, and one came well before it. Writing a book's limb from its ledger rows found a sentence on a price card, "twenty-five cells within six percent", that its run's table does not support: 20 cells sit at 0.94 to 1.06 of the line and 5 at 1.07 to 1.11, and the card was corrected in place with a dated note. Write a description after looking, check each case, and treat prose drafted from rows that were each read in full as a second reading of the data. Explaining it simply is a test: drawing a map for the person "like to a smart 12 year old on a chalkboard" found three mistakes while it was being drawn, because every label had to name what the data says and so was checked against the record (the Loop Atlas sessions, R2W). And an answer to "how is this possible" carries a sum the run can check: asked how one number could place every step, the answer described the crawl over a vanished resting place and checked it with one sum, the weightless crawl time at the 1/30 step, 189.4, against 30 ticks, 184.9; the plain account then named what is still open, that the measured constant is 0.9425 of the weightless value, for a reason not yet known. A plain explanation that carries a number the run can contradict is a test, not a story (R2W). A figure built to teach holds one path still and turns one knob. Asked to show whether the creep follows the traders' law and to teach its shape, the Loop Atlas held the ear still, so that one noise path could be watched at eight steps, as a trader watches one price at different intervals; four panels showed what a slow watcher misses, the square-root shape and where the law holds, which for slow noise was within 8 per cent at every step with nothing fitted (run CREEP_LAW 20261016_035058; row 629). The five-step lesson written for Bear became the spine of the Loop Atlas book's Limb H section on the traders' law, and when its table was rebuilt from the run's json, three phrases of the lesson turned out looser than the data and were corrected in every record. A lesson written for a reader rounds things, so check it against the json line by line before it enters a book (the Loop Atlas sessions). Keep the read table alive every exchange. In R2W's sitting P1 it lapsed after the first exchange and surfaced only when Bear asked whether anything had gone unread; the answer was yes. Print it by script at the start of every exchange that makes a claim about a file. Count every case, and name the toy's own symmetry, before reading meaning into a shared number: Bear asked whether the sevens meant something, and R2W's survey of 4,816 riders shows doubled forms of 21 grooves, with nothing singling out 7; the toy pairs each fraction with one minus it, so 3/7 and 5/7 are not mirror twins (R2W, sitting P2)

the Loop Atlas sessions (R2W and the book); ties rows 553, 604

The split-frame measurement: send one walker each way, and compare them where they meet (674) — a loop of settings cannot be walked in one trip when the state carried depends on the road, so send one walker each way round and compare them where they meet, at the same phase of the pattern (row 641). The difference is a whole number of turns, and the route decides which door. Its cousins named in conversation, a two-path interferometer and Foucault's pendulum carried round a latitude, are to be checked against the literature before a paper cites them. The triple-star sessions built the same two walkers from the same picture the same day, and split a third star's apsidal law into a part that flips with the direction of travel and a part that does not (row 575)

the Loop Atlas sessions (R2W); the triple-star sessions; Bear's two walkers; ties rows 190, 575, 641, 673

Digits, names, hand arithmetic and argued-away knobs are claims: compute them, and run them (677) — a line read by a fixed-step integrator was quoted to nine digits for two days, and adopted by another room as its reference, though four adaptive integrators agree on a value 4.05×10^{-8} lower; a new tool caught it by carrying the old figure as a fidelity line with a bar of 10^{-8}. A table typed by hand had four wrong entries, and a fidelity line compared shares with a typed rounding (0.188 for 3 of 16) and printed MISSED when every share matched. A name is a claim too: 0.236068 is twice 0.618034 less one, which equals its cube, not its square. A correction worked by hand put a line at 1.6940 that was 1.70003 when computed, which changed the run's question. Two knobs argued away by algebra were run when the author asked: the algebra held, and drag alone proved a strong dial, changing 41 of 52 riders, whose every setting is a place among the four knobs that remain. And a shared constant travels with its digits: two rooms' cascades sat 5.3×10^{-9} of drive apart at every shared doubling because each wrote the passing speed its own way (R2W's run KNOBS-2; row 738 gives the shift doubling by doubling), equal for the true golden ratio and 6.6×10^{-8} apart in relative terms for the rounded one; set alike, two tools written in two sessions agreed to 1.8×10^{-11}. Print tables and targets from the numbers themselves, give each number the error of the tool that made it, and run what was argued. Clock stamps and line counts are claims too: several stamps were written a minute or more ahead of the clock, each from writing the stamp before reading it, and twice a source's line count was misstated (a listing left out of a count; 42 lines written as 44). Read the clock in the same command that writes the entry, count with the tool in the same step, and correct in place with a note (the Loop Atlas sessions, R2W). A constant carried to fewer digits than the arithmetic sets a floor where a symmetry should hold: with 2π carried to 30 digits inside 40-digit arithmetic, one orbit's passings touched a lock's edge 4.76×10^{-30} apart in drive where symmetry says they must agree, and with 2π to 40 digits 8.2×10^{-38} apart. When two readings that must agree differ far above the arithmetic's floor, look first for a constant carried short. A number drawn from printed digits is computed by script before it is said (four such misreadings in one session were each corrected the same day), and a run that is not ours is named from its own files, never from its header (the Loop Atlas sessions). Pin file patterns: a pattern written for one reader's files also caught its sibling's, whose name begins the same way; pin it with the stamp's first digits (the Loop Atlas sessions, the book). Describe a run from the code's own table, not from memory of it: a launcher's first draft gave two sets' means as 0.10 and 0.05, where the tool's line set both at 0.10, a quarter turn apart, and 0.05 belonged to two other sets. Every line of a status page or launcher that names a parameter is checked against the line of code that sets it (the tool triple_heavy_mean_start v1.2, checked in the container on 11 October) (the triple-star sessions). Read a plan's printed arguments against their source before registering: K-16's plan rebuilt RATCHET-65's runs from that run's own queue, and its first version took the harmonic number (4) for the fourth harmonic's strength at w 12. The stand-ins could not catch it, because they never pass through the plan's parsing; reading the printed arguments line by line against RATCHET-65's queue did, before anything ran. When a plan builds engine commands, print every argument and compare it with where it came from (R2W)

the Loop Atlas sessions (R2W and the book); ties rows 593, 644, 650, 738

When an instrument fails its guard, its unscored readings fail with it (682) — a reader of a heavy triple failed its guards on every run, and the capture still reported, as "unscored and seen", that one angle turned forwards inside the apparent locks while another turned backwards, and called it the anomalistic lag showing inside a lock. A reader that counted instead of predicting found the two turning backwards together, at a ratio of 0.994 to 1.008, as the algebra requires: the forward turn was the broken reader aliasing. A reading made with a failed instrument is not a weak result but no result, and writing it up, even as a suggestion, plants a claim the next run has to dig out (the triple-star sessions). A lane with a known answer inside the run is the guard that tells a failed method from a physical answer (row 647). FOLD-DIR-b walked seven unconverged settings up in strength from finished neighbours, and walked one more whose answer was known: weight 0.75 at strength 0.60, which FOLD-DIR's settling had found. The walk lost that lane too, and found only three of the seven, all at weights of 1 or more. Because the known lane failed, the four still missing cannot be read as places where the fold does not exist: the method could not follow the branch there, perhaps because the branch changes shape. Without the known lane, the four would have read as physics (R2W)

the triple-star sessions; ties rows 553, 627

Check a bar against the reference its sentence names (692) — a guess's words asked for rims "at least as ragged as the light star's", and its bar asked for 0.010 at one island, more than twice the light star's own 0.004 to 0.0045 there; the one chaotic birth, at 0.005, met the words and missed the bar. Read every registered bar back against the sentence it stands for, and compute the reference the sentence names before fixing the number. Check its direction too: a registered measure of how far a look reached inward toward a pole could not work with the pole outside the circle, where a look must swing outward to reach it. The correction was registered before any number was computed, and both then read 22 of 23 (run triple_maypole_reach 20261012_051949) (the triple-star sessions). And a premise read from an earlier run is checked against the definition the new test will use: a risk noted beside a proposed test, read on 14 October, said an earlier set of long runs had seen no fast chaos, but by the test's own definition (twins a thousandth of a turn apart within 600 passes) two of its births, 7.450 and 7.455, were already fast, at passes 551 and 546; the risk had carried another sense of "fast". The test was scored as worded and the premise marked wrong beside it (the triple-star sessions). Compare a new build with the old against the source the old one was built from: a

fidelity run's "missing before" counts for four files were the batch's own new text, which PDFs built before the batch could not hold; against their own sources those PDFs were complete (runs LOOP_ATLAS_V13_FIDELITY 20261013_203154 and 20261013_204846) (the Loop Atlas sessions). A word in a registration must mean what the paper means, down to its rounding: FN0 (run 20261017_030851) read E 7.335 against Section 7's 7.355, because the registration took its held block from the atlas's lock flag (band under half a turn), where the paper's 'held' means a band under one turn; and it read 0.0165 against 0.0166, because the TABLE's 0.016550 prints as 0.0165 under Python's rounding while the paper rounds half up. FN0′, rounding half up, passed (run 20261017_030927). Define each term of a test from the paper's own definition, and compare at the paper's rounding rule (the triple-star sessions). Match the band's shape to the claim's: a symmetric "within 3 points" band missed because an arm gained 3.2 (E41-P2), so when the claim is "no collapse", register a one-sided band (Question the Substrate)

the triple-star sessions; ties rows 337, 369

Sign and size are separate claims: when a sign is free, register the size, and score an eye's claim on its sign and place before its size (712) — a registration asked for a correlation of at least 0.95 while calling the corner's sign free, and it missed as written in the lanes that bend the other way, though every size was 0.993 or more; nothing was rerun, and the sizes stand printed beside the verdict. From the other side, three blind reads of another room's pictures, made from their pixels, were all borne out on the data in sign and place, while two of their sizes were off in opposite directions: a weave stronger on the data than on the page (+23.4 standard deviations against +12.4), a lopsidedness weaker (−0.18 to +0.29 against −0.49 to +0.47). An eye proposes sign and place well; the size belongs to the data. And a claim that names which members of a set stand out can be right about the tallest and loose about the rest: the weave's shunned gaps 1 and 3 were borne out, but gaps 2 and 6 shunned as hard, because the two tall gaps took their links from all the others (the Loop Atlas sessions, the book). Test a law's coordinate-free part first, its sign and its zeros, and its size only in the coordinate where the motion is even: read in the raw lag, a law's sign held at 99.7 per cent of 10,584 starts and its zeros matched within one start in all 53 cells, but one gear fitted the sizes poorly (R^2 0.26 to 0.85); in the straightened lag one gear fitted every cell (R^2 0.996 to 0.9997) and came within 1 to 2 per cent of the value the weightless passing matrix gives, an excess most likely from the reading's grid (row 741) (runs SLOPE-1 and SLOPE-2) (R2W). Read a switch by the sign that flips, not by the size that follows: a diver's twist switch sits at the separatrix to the first scan point, 0.05 degrees, at three body shapes, and the somersault spin's change of sign marks the same scan point as the count of turns, which jumps from under a quarter turn to 4 to 7 turns across it (run TWIST_LONG 20261011_072842) (the Loop Atlas sessions). The registration at this row's start, a correlation of at least 0.95 with the corner's sign left free, was CORNER_PRINT's, and its sizes ran from 0.993 to 1.00000. Beside reading a registration back as a sentence (row 369), ask what result would make it fail for a reason that has nothing to do with the question (the Loop Atlas sessions). Test a picture in kind first and in size second: Bear's picture of half a dipole or a reflected image made two definite claims before it made any numerical one, that the effect has a part that flips with the tone's direction and a part that does not, and that both are real. Both held at once (RATCHET-44); the sizes then needed a better curve than the three points the picture came from (RATCHET-46). Grading the picture only on the parabola's numbers would have failed a picture that was right (row 801). A symmetry splits the same way: it says which channel a disturbance can use, and the size must be measured in the other. The mirror alone said a drift of the drive could never offset the weight reading at the dark point, and it held to the last digit. That part is free; what the symmetry cannot say is how big each effect is. So the registration carried one real number, the size of the drift in its own channel, built from two earlier runs, and the symmetry's claims rode along as structure. A test of a symmetry plus one predicted magnitude is worth more than either alone (row 734) (R2W)

the Loop Atlas sessions (the book); the triple-star sessions, judging R2W's reads; the Loop Atlas's ledger row 128; ties rows 313, 337

A tool that travels finds its companions by what is in them, not by their names (721) — run in the container, a kit's tools opened their companion files by exact name; on a laptop, a Downloads folder keeps a second copy as "name (1)" and never renames it, and the first try stopped on a name. A launcher that hashes every candidate, copies the right copies into a fresh folder under their proper names and runs the unchanged tool there serves both machines (LOOP_ATLAS_LAPTOP_F4, 10 October). Its census on the laptop then showed the cause was not a renamed copy at all: one file sat in another folder and three were nowhere the launcher looked, so it now looks deeper, under any ending and inside zips (LOOP_ATLAS_LAPTOP_F4_2). Let the launcher's census say what happened before guessing. And a kit is proven in a clean folder, by a check that was itself shown to catch a fault: every tool sent with its run of record is rerun in a folder holding only the kit, and its run compared with the record line by line, setting aside only the run stamp, clock times and paths; the comparison is first shown to flag a planted one-character change. The clean folder found two data files that the tools loaded through another tool, and then all nine tools reproduced their runs of record, 333 lines, three and two days after they were first run (the Loop Atlas sessions, the kit for the Field Guide's new session). A launcher's census settles a missing file without a guess. The triple-star thread's first start of PB1′ stopped before running anything: ten of its eleven files were in Bear's folder with the shipped fingerprints, five of them as "(1)" or "(2)" copies, and one table was not there under any name. Only that file was sent, alone with its own list, after a rehearsal on a mock folder built to match the census; nothing else was resent, because the census had measured the rest present (the triple-star sessions). A pipeline that lives only in the workspace is not in custody: until

20 October the triple-star paper's rebuild pipeline did. Shipped as TRIPLE_PAPER_REBUILD_KIT_2026-10-20.zip (04e75ca7a6ad5373) and unpacked into an empty folder, it rebuilt the markdown byte for byte (e3bbd18ccfaa7920) and a 54-page PDF, lint CLEAN, whose text equals the PDF of record's, and the whole crate went to Bear as TRIPLE_CRATE_49_COMPLETE_2026-10-20 (1,700 files), checked file by file in an empty folder. And one tool name has one version of record, because a default copy handed to others drifts behind the working copy: the publishing packet's build_paper.py v1.0 and lint_deposit.py v1.3, which built the paper, differed from the crate's files of the same names, and Bear's packet of 16 August, which still held those two, lagged the crate's own (build_paper v1.3 and lint_deposit v1.7). The packet was renewed and made the default (PUBLISHING_PACKET_2026-10-22.zip, 2e70c9ce2f484500), with retired lines for every superseded tool, launcher and README, and with each launcher stopping on a tool whose fingerprint is wrong (exit 9). The default packet carries both rules (the triple-star sessions). The rule reaches down the chain of files a tool reads, not only to the tool: SNAKE_LONGEST read SNAKE_EXACT's text, which ran DRIFT_C's, which opened DRIFT by its exact name, and on the laptop that third file was not in the folder under that name, so the run stopped before any work (Bear's laptop, 20261017_153615). Version 2 finds every part by its md5, under its own name or a " (n)" copy, in its folder and the one above, prints which copy it used, and stops naming what is missing; its launcher finds the tool the same way (blind run 20261017_224026). A file that comes back under another name is identified the same way, by what is in it: R2W's explorer returned as depth_axis_explorer_1.html, with a different md5 and size from the version of record, and read by script it held all of version 9's maps by count and two parts more, a later build, which now serves as the reference. Compare contents, not names, before calling a file right or wrong (the Loop Atlas sessions)

the Loop Atlas sessions (the book); ties rows 572, 591, 678

A maintenance alarm is a bar on a fleet: count work and strain together, set a bar per class, and compare policies at matched catch (746) — a logbook's best calendar counts the work and the strain together: where a machine asked to run harder meets its cliff at a stronger grip, the calendar that wastes least counts hours times duty to a power near 2, even when the wear itself is counted per turn; and given both hours and turns, a learned reminder wastes the same whichever clock the wear follows (run METER_CLOCKS 20261011_072646). A shared bar is a hidden transfer: one threshold set to catch 90 per cent of a mixed fleet over-catches its quiet rooms (0.96) and under-catches its loud ones (0.84), while a bar per class sets each near 90 per cent, moving catch from quiet to loud and waste the other way. A derivation can be right about a different question: equalising two duties' median ages at death gave the power 1.5 for a maintenance calendar, which is exactly the power that makes age at death least spread, while the power that wastes least sits half a step higher, where the catch at the same bar starts to fall. Ask which quantity a derivation sets before scoring it on another (run METER_K_SPREAD 20261011_080641). And compare policies at matched catch, or print the catch beside the waste: at one training bar, calendars caught anywhere from 0.886 to 0.942 of the test fleet, and a lower waste was partly bought with a lower catch. Learn a wear gauge's direction from the fleet; do not assume it: a tapping rider whose fixed-size tap weakens taps less and wobbles less as it nears its fall (0.69 and 0.15 of its first flight), so a fixed level waiting for more power wastes 0.92 of a life, and a reminder that learned the direction 0.13, against the calendar's 0.35 (run RIDER_FLEET 20261016_044707; the Loop Atlas's misses row 254). A controller that makes up for its wear turns a quiet wear into a loud one, and its command's headroom is the reminder: a tapping rider that resets its command after every flight to hold its wobble draws 3.05 times its first flight's effort on its last, and its command sits at its limit on 0.988 of last flights; a fixed effort level then wastes 0.190 of a life and a learned wobble reminder 0.036, against the calendar's 0.344 (run WORN_RIDER 20261016_065645). That was Bear's power gauge, his picture of 4 October of watching the power a machine draws as it wears, a prediction that the meter rises as the machine wears: it missed on the rider whose tap weakens (misses row 254) and held in all three of WORN_RIDER's leans on the rider that makes up for its wear. And judge a reminder's gain at equal catch, against the spread from one fleet to the next: matched on catch, the slowing rider's learned reminders gained +0.005 and −0.002 of a life against standard deviations of 0.016 and 0.017 across five fleets, and the calendar alone threw away anywhere from 0.250 to 0.333 of a life from fleet to fleet, so one fleet's 0.03 was luck (run EQUAL_CATCH 20261016_192548) (the Loop Atlas sessions). The comparison at equal catch was bought by a narrow miss. On riders whose reaction slows, the lean said no policy beats the calendar by more than 0.02 of a life; the learned reminders wasted 0.304 and 0.309 against the calendar's 0.333, while catching 0.828 and 0.844 against 0.859. Something does move before the fall, the wobble shrinking to 0.89 of the first flight's while the taps stay flat, but the small gain came with a smaller catch, and one fleet seed cannot tell a 0.03 gain from the needle fleets' seed spread of 0.014 to 0.027 (run RIDER_FLEET 20261016_044707) (the Loop Atlas sessions)

the Loop Atlas sessions (the book); ties rows 257, 689

Build from the last build in one pass, with its proofreading copy, and measure everything a rule could touch (773) — one guide's second build started from the first build and asserted every edit against an anchor that occurs once. It wrote the guide and a copy with every change highlighted in the same pass, then checked that the two differed only by the marks, so the highlighted copy's page numbers are the real pages and the person proofreads only what moved: 39 pages of 133. Every build was read back against its source, sentence by sentence, and then a cold read by a separate agent that had not seen the drafting found real defects each time: overstatements, stale statuses, clipped reference numbers, a misplaced "So". A style rule reaches further than the line you meant: one written to

enlarge the title also enlarged every part heading in the book, a separate reader looking at the pages did not flag it, and it was caught only because the station card was pushed off its page. Measure the built PDF for every element the rule could touch, not only the one you changed. And additions can arrive faster than they are built: version 7's first full draft was built on 10 October and then waited while five more revisions of drafts piled up behind it, until by 15 October the backlog was larger than version 7's whole fold (the Field Guide sessions). Reading the built files back, not the scripts that built them, found five faults in one session that no script reported: a data ledger PDF missing about 1,600 words, a book PDF missing its figures when built in place, a voice PDF whose numbered lines had been eaten, a stray file in the crate (a helper called with the wrong arguments had written it; found and removed before the next ferry), and a ferry zip too large to send (the Loop Atlas sessions). Compare the page count with the last shipment before reading anything back: the first build of one sitting's ninth exchange ran from the documents folder instead of its figures folder, every figure went missing without an error, and the book came out at 53 pages against the last delta's 78. Read-back would have passed, since it checks text, not pictures; a page count against the last shipment caught it in one line. A page is checked by opening it: the Depth Axis Explorer's code read cleanly and its engine had already passed against six runs of record (row 36), but opened in a headless browser and driven the way Bear would drive it, it showed two faults the code did not: the small dial divided by zero when the third tone was off, and the old view drew on a hidden canvas. Both were fixed before the page was kept, and the check that found them ships with the page, with its screenshots, the built-artifact proof that row 553 asks for. And tools that share a scratch file must take turns: to save a minute four documents were built at once, and each builder writes the same scratch page in the same folder before printing it, so they wrote over one another; two stopped with an error, and the two that finished could have printed another document's page under their own name. All four were built again one at a time, and the read-back, which compares each printed file with its source, is what shows the content is right. A speed-up that changes the order of writes is a change to the tool, and is tested like one (R2W). The build run from the documents folder, which lost every figure in one sitting (53 pages against 78, above), did it again in the next: 66 pages against 92, caught by the same page count before anything shipped, though not written into the record at the time. The figures resolve only when the book is built from its figures folder (R2W)

the Field Guide sessions; ties rows 553, 593, 765

A paper travels with its PDF, read back block by block, in plain words, with its misses kept (779) — every md travels with its PDF; read every built PDF back against its md, block by block, before sending, and send whole files, never patches (row 157). Markdown has traps: a pipe character inside a table cell splits the cell (row 740), a Windows path needs doubled backslashes or the converter swallows it, and a new row must be checked for which table it lands in. Commas go in numbers (1,197), but not in years, run stamps, DOIs or hashes. A paper has no assistant voice: it is written for a smart twelve-year-old, with the discoveries first and no diary of experiments, and the stronger, opinionated voice belongs in the conversation, not the paper (row 114). Every miss gets the redesign check said aloud (can this be redesigned, or is the redesign covered elsewhere?), and misses stay visible: a paper's appendix of what did not work is worth having (the triple-star sessions, in their page for every new session). A check can forgive the very fault it is for: the Loop Atlas checked Bear's words against its voice-file PDF in 24-character pieces that forgave curly quotes, dashes and ellipses, so it could never have caught a typed double hyphen turned into a dash. Read again with nothing forgiven but the ligatures, the PDF had 85 of 1,141 pieces changed by the builder's smart typography; built since by a tool that escapes his quotes, dots and hyphens so the builder prints them as typed, the strict check finds 0 of 1,141 changed. And a table wider than the page lost about 1,600 words to clipping, unseen from at least 9 October until the ledger's PDF was read back on 10 October: read every PDF back by counting its letters and digits against its source; the fix kept is a build flag that lays a table out at the widths its separator line's dashes give (the Loop Atlas sessions). A list after a paragraph needs a blank line, and a page that holds a new list is a page to look at: four bulleted lists in a book's Part 7 shipped as one run-on line of dashes, because pandoc's markdown starts a list only after a blank line. The word-by-word readback cannot see shape, and the pages looked at that exchange were not the ones holding the lists. A list-gap gate (LISTGAP_B10) now gates every build beside the table-cell gate, and its dry run found the four gaps in the shipped book (R2W). Open each section, and each part that brings in a new test, with a plain sentence or two that says what it is and why it is there; a labelled aside helps once and wears on the reader by the third time (the publishing packet's rules, from the triple-star thread's islands paper). The front page is checked like the body: at every build each result on it is checked against the headline register, and a top result left off the front page has its reason written down; the Zenodo description says the abstract and the numbered findings word for word, checked again after every change to either (the triple-star sessions, R2W). Clear a builder's stop by moving the page break, tried first on a throwaway copy, before touching the shared tool: in R2W's book a table that first split on the builder's second and last pass stopped the build (exit 4); joining two bullets and reframing two sentences did not move the break, and removing a one-word last line above the table did (R2W, sitting P2)

the triple-star sessions; ties rows 114, 157, 553, 740

Shop faults in the launch: a pipe's exit status, a backgrounded line, a stamp in a folder's name (786) — a full run started by mistake after its smoke had failed, because the launch line was chained after a pipe, and a pipe's exit status is its last command's: never chain a launch after a pipe (it was stopped within seconds, the smoke fixed and passed, and the run relaunched). A launch meant to run two parts side by side ran them one after another, because

only the closing echo went into the background: group the whole command before backgrounding it, (python …; echo …) &, and count the running processes right after a launch (SOUP_JUMP_2, whose runs of record are 20261009_051426, 20261009_051926 and 20261009_052404). A launcher's first full test, on a made-up Downloads folder with a space in its name, ran FOLD_OR_FLIP_4 to the end and reproduced every number, yet the comparison (version 2) called the run DIFFERENT on the run-file line. The launcher had named its work folder with the run's own stamp (LOOP_ATLAS_LAPTOP_F4_20261010_214554), and the comparison sets the stamp aside before the folder, so the folder no longer matched and the path, holding a space, could not be set aside either. The folder is now named with dashes (LOOP_ATLAS_LAPTOP_F4_work_2026-10-10_21-57-31), which no stamp matches, and the comparison's own fix is to set the folder aside before the stamp. A made-up end-to-end run found it; a dry run would not have (the Loop Atlas sessions). One background job per subshell: a trailing & once backgrounded a `cd` with its job, so a reader ran in the wrong folder and wrote its output at the root. Start each background job as `( cd folder && nohup timeout N job > log 2>&1; echo done >> log ) &` on a line of its own (R2W). Compare a rescore with the laptop's ledger, not its bytes: the container's rescore of the off-mirror run printed the same 41 verdict lines as the laptop's ledger, but a byte comparison said every line differed, because the laptop's file has Windows line endings. Compare with line endings set aside, and say so. This is the second way a rescore can mislead; the first is in row 731 (the triple-star sessions). A command that computes a number and writes it in the same line can fail and write a blank: RATCHET-45's result line was written with its R45-e distances empty, because the computing command failed on a trailing comma and the shell went on. It was corrected by a dated line, and the fix used from then on is to compute into a variable, stop if it is empty (`[ -n "$P" ] || exit 1`), then write; it is R2W's lesson 81 in a new form, the number not typed but missing (row 735). And in a guarded chain a clean zero can stop the write: the record's result line for RATCHET-53 was guarded so that no empty value could be written, grep -c printed 0 bad lines and returned failure, as it does when it counts nothing, and the guard stopped the append. Nothing partial was written, and the tail showed it. Count with a tool that succeeds on zero, or add `|| true` where zero is the good answer (R2W). A version suffix can be the start of every older file's name: a correction tool first named GAPS_READ_2 shared its start with GAPS_READ's own saved files, GAPS_READ_20261015_…, and two figure tools find their input by that start, so the new tool's output would have been read in place of the old without a word. It was renamed GAPS_REREAD before its run of record, and the renaming by text then mangled the old file's name inside it, which stopped the first run of record. Check a new name against every glob that reads the old one, and rename by whole names, not by text (the Loop Atlas sessions). Never wait on a run with `pgrep -f` from the same command, either: its pattern matches the command that holds it, so the wait never ends (R2W)

the Loop Atlas sessions (its record of misses); ties rows 572, 611, 736, 776, 796

Faults a free check catches in the code: a nested namespace, a stopping rule that never stops, a search with no bracket, an energy that changes sign (796) — loading by text nests namespaces: a tool stopped after its first line, before any reading of its three bets, on two names it looked for one layer too shallow, in the tool it loaded rather than in the namespace that tool built for another's text, two layers down. Ask for every name new code uses before the long run, or read it from the layer that made it (SETTLED_GROOVES 20261011_222608, set aside). A switch reader took three versions to give its first reading. The first shot each needle forward at pace 1 and called it caught if it never passed the next birthplace within 10^{6} time units, but a caught needle creeps up to the birthplace as one over time and never arrives, so every caught shot ran to the limit, about 70 s each: ask a question whose answer arrives in finite time. The second followed the arriving path backward in time to a half-turn section and died when the path turned round before it, at memory 0.8: run backward, the drag becomes a push, and the path may come from behind the section rather than across it, so that case counts as speed zero. The third read the switch to eight figures in a minute; neither faulty version printed a reading, and a scratch timing check of single shots (0.70 caught, 0.71 passing) is disclosed in the third's header (MEMORY_SWITCH_2 20261011_084809 and 20261011_085049, set aside). A search needs a bracket: the first follower of the strobe's orbits ran its secant away to negative drives and missed a fold's tolerance, and it was retired with its readings disclosed while its successor reproduced every shared number (STROBE_NEWTON version 2, 20261006_184350). And an energy that alternates in sign cannot be an energy: a reader that took one toy's spin energy as half the board's turning heft times the spin, not its square, printed "energies" that changed sign at every landing. It was retired; its successor squares the spin, reproduces every line the two versions share (44 of 44, compared by script), and says in its header that the first version's printed spins had already shown the steady energy at 4/3 of the pop, so its first bet was not blind (WADDLE_PRICE 20261011_071049, retired; 20261011_071302) (the Loop Atlas sessions). Bracket by a property you can test before handing a guess to Newton: a fixed starting guess for the weightless edge slid onto whole-turn orbits at strength 0.20, the stand-in's setting. A bisection on whether the zero-slip orbit exists, then Newton, fixed it, and the reader sets aside any row whose residual is not below 10^{-9} (R2W). A dry run is also a code review: two readers in one stretch carried bugs that only the stand-ins exposed. ODD-1's reader took the cosine of a timing rounded to six decimals, so at thirds of a turn the first harmonic leaked about 2.7×10^{-8} and a verdict failed where the made-up push said it must hold; ARRAY-1's reader divided by a prediction that a made-up push set to zero. Both were fixed before registration, and the first versions are kept. A stand-in that is supposed to hold and does not is the cheapest bug report there is (R2W). Code read by text bites back in another way: when a tool reads a tool that reads a tool, the middle tool's own reading line carries the slice markers as strings, so a marker taken from the code it reads occurs twice. Twice in one night a blind

run stopped at once on such a marker (FILTER_SLOPE 20261015_195425 and FIT_KNOBS 20261015_203849, both retired), and a count of sine calls set at 4 where there were 5 stopped one more. Take markers from executable lines, long enough to be unique, and assert each piece's count before slicing and before the long run (the Loop Atlas sessions)

the Loop Atlas sessions (its record of misses); ties rows 36, 644, 786

Tell the reader which idea is upside down before giving results (800) — the person who learned to see a thing knows which part a newcomer will read backwards. Bear wrote that people will struggle to see how chaos can be structure, that he would have struggled with it six months before, and that it is an inverted idea. So the islands paper's section on chaos now opens with chaos with a shape, before any measurement: leaves on a river; a place, speeds, a direction, a door; counts and fractions, never one star's fate (the triple-star sessions). The reader's view also works at the scale of the whole page. The person called the islands paper's draft a wall of text with the figures not shown and the bench and the way of guessing buried; measured, the 52-page PDF held no image, and the bench was one 412-word paragraph on page 38 with no figure. The paper was then sorted the person's way, each paragraph tagged by audience, by whether it passes the smart-twelve-year-old test, and by topic and result, and fitted to the arc behind its one-page highlights; a sentence that read as a challenge to another author was softened in all four places it was said, with every number unchanged; and each section, and each part that brings in a new test, now says in its first sentence or two what it is and why it is there (the triple-star sessions)

the triple-star sessions; ties rows 439 and 502

An array of identical periodic parts is decided by its seats: the fan inside a group picks the harmonic, the offset between groups steers it, and a silent set parks what it holds (803) — a phased array is the comb theorem (row 802) with a steering knob. Three groups of three riders kept only their third harmonics, and the offset between the groups steered how those three added: 0.910684, 0.666667 and 0.333333 of the aligned swing at 10, 20 and 30 degrees, and nothing at 40, the array factor of a three-element antenna (NINE-1); Bear's phrase phased array named the shape before the run. Groups inside groups give a knob for each level: the fan inside a group sets which harmonic survives, and the offset between groups sets how strongly. An array of identical periodic parts is decided by where its parts sit in the cycle, not by which group each part is said to belong to: nine riders in three groups at 40 degrees and at 80 degrees sit in the same nine seats, so every number about them is the same; twelve at 30 and at 90 degrees likewise, while twelve at 60 double up on six seats and behave differently (TWELVE-1, row 214). Bear had been carrying "3-3-9" for fifteen years, and three groups of three is the array where the seat picture first showed itself in R2W's book. Before tabulating an array's settings, fold each one into a single cycle and list the seats: settings with the same seats are one row. And a silent set is a parking spot, so the leftover is the signal: Bear asked whether the angle between groups could be a "staging ground", a holding pattern with on and off ramps. Read after the data (STAGING_READING), two groups whose trims are half a turn apart, or three a third of a turn apart, cancel as a set, and the array's push is whatever is left over. Four different arrangements, nine riders at 30 and at 60 degrees and twelve at 40 and at 80, swing exactly one lone group's swing, $4.220460177 \times 10^{-5}$, because everything else in them is parked in a silent set. The pattern is general for any sum of equal turning parts: strike out the subsets that close on themselves, and what remains is the whole answer. His word "ramp" names the useful part, turning one member out of a closed set, which brings its signal back smoothly. The caution travels with it: a parked part is silent on average but still adds its own noise (arrays of riders, runs ODD-1, ARRAY-1, NINE-1, LOBE-1 and TWELVE-1) (R2W)

R2W; ties rows 128, 214, 659, 802

Small faults in a tool's own code: a verdict made a plain true or false before both print and dump, a followed quantity started from an explicit index, and a table row inserted by its line (815) — some faults are not in the physics or the method but in a few characters of code, and each one costs a run or a record before it is seen. Three kinds came back in one project, each with its cure. A verdict is cast to a plain true or false before it is printed and before it is saved: row 611's numpy true, which is not Python's True, came back once more, four held planted cases printing UNDECIDED in a blind run (run TONG_DULLIN 20261017_173508), and numpy's booleans are not JSON's either, so a run of record stopped at its json dump after every number was printed and before any verdict (COIL_SNAKE 20261017_204013, retired). A followed quantity starts from an explicit index, not from a name the last loop happened to leave: CAT_TWIST's quaternion lift took its start from a time left over from the check above it, and only one of three cases, the one still bent at that moment, missed, by 7.5×10^{-8} (blind run 20261017_231820). And a row goes into a table by its line, never by replacing a piece of text: a ledger row inserted before the text that began the next row took that text's place, so the next row lost its first cell and ran on in the new row's line, and the new row's later result was written at the end of the joined line. Count every table's cells after each batch of records; the joined line had eleven fields where the rows have seven, and nothing else showed it (the Loop Atlas sessions)

the Loop Atlas sessions (Guide items of sessions F and G); ties rows 594, 611, 796

Rebuild a reader's edition from tagged paragraphs, never cut it from the working book, and read every reused page for its old vocabulary (825) — a long working book holds everything, in the order it was found; a reader's edition holds what a reader needs, in the order it is needed. Cutting the one into the other loses paragraphs

and leaves the workshop's words behind, so the edition is rebuilt instead. Every paragraph of the Loop Atlas's 137-page working book, 546 of them, was first tagged by purpose and weight by readers who were not writing; the reader's edition was then written chapter by chapter from the tags, its numbers drawn as written with their runs, and its counts, contents and references filled by script at assembly from the register and from its own chapters, so none can drift. Its first build, on 19 October, came out 46 pages of 17 chapters with 17 figures, and the working book is kept whole as the record and says so on its first page (helpers/readers_edition.py). A page reused from one document carries that document's vocabulary: the front page drawn from the headline register named the working book's parts, Limb I and Trunk 4, which a reader of the edition never meets, so the names were mapped to chapters by rule at assembly, with an assertion that none is left. Read every reused page for the words of the document it came from, and check where every number sits (row 823) (the Loop Atlas sessions)

the Loop Atlas sessions (Guide items of sessions F and G); ties rows 569, 773, 779, 823

A check that leans on the order of adding must know what kind of number it adds: Python's built-in sum compensates floats, not complex numbers (838) — a computer rounds after every addition, so adding the same numbers in another order can change the last digit. A check that compares two programs to the last bit, or assumes the order of adding cannot matter, has to know whether its tool takes that rounding back out. From Python 3.12 the built-in sum adds floats with compensation, so the order of adding does not move the last bit: it moved in 0 of 2,000 shuffles of fifteen mixed-size numbers, against 1,240 for a plain loop. For complex numbers it is a plain loop: the twelve complex terms of a window gave the same bits through sum() and through a loop, and different bits through math.fsum of each part. A check that relies on the order of adding not mattering must know which kind of number it adds; an exactly rounded sum is how row 116 tells a rounding miss from a holonomy, and row 214 traces a last-digit difference to its recipe (R2W)

R2W (sitting B12); ties rows 116, 214, 632, 738

Proofread in rounds on fresh builds: leave alone what the person is reading, log every later change against his page, and agree a plan for his marks before any batch (846) — a proofreader can only mark a page that stays still, and a mark is useful only if it can be found again in the next build. The person proofreads while the work goes on, so sections he is reading are not rewritten while he reads; a run that changes a number adds a row to the changes list with the section, the page it sits on in his proof copy, and why, so his marks carry over without hunting. His marks get a plan, agreed with him, before any batch is done: his first proofread of R2W's book, 29 marks to page 74, went in five batches, the fifth being his two runs. His marks are kept as he wrote them, read out verbatim with his annotations. The proof runs in rounds, each on a fresh build: the author's proof; then the corrections through the rebuild kit and the full chain of checks; then a last read of the deposit set together (the PDF, the README and the Zenodo sheet) and the linter on the final artifact. A full read-through of the book is the last step (R2W, the triple-star sessions). With the author's marks, read each annotation with the words under it, touch only grammar in his own notes, and leave a meant word alone: of the islands paper's last twelve annotations, a question about a word the paper defines was answered with no change, his notes were set as notes from the author with only grammar touched, and every change was listed back to him. Name companion works by their titles, never by the name of the session that wrote them. Each time the copy a reader is marking is rebuilt, send it loose that exchange, with its md5 and build date, naming the copy it replaces: the Loop Atlas's reader's edition was rebuilt twice inside zips under one file name, and two days later the loose copy Bear was holding was the oldest of three builds wearing that name, told apart only by their md5s. A file name says which book; only a fingerprint says which build

R2W and the triple-star sessions (advice at the move to final publishing, 22 October); ties rows 573, 589, 773, 779

A number written down lives in several places: look it up, change it everywhere at once, and print every mention a renumbering touched (854) — the count of R2W's landings stood in the abstract, in "The count" and in the register, and each change went in by one script and was recounted by command; a quick tally by pattern read 68 where the book said 69, because one cell reads "6.11, 6.12", so count by the rule the original count used. Page numbers written from memory were wrong four times in one sitting. And swapping two parts first renumbered another document's "Part 6", missed letter-suffixed labels and could not see 150 cells naming sections without a § sign; adding two figures moved 42 figure numbers, checked in order against the old book. Two numbers set side by side, and how many digits they agree to, come from one tool and one formula: a spread said as "0.2% … 13%" was 0.18% and 26.4% when one tool printed both, so the beat is the steadier gauge by more than was said (run 20261027_150905), and two machines said to agree "to the fifth digit" agreed to four. When the counting rule changes, write the decision into the register and carry each new count to every place it is printed in the same build: the triple-star count moved from 444 to 490 in one day, 28 October (run 20261028_210006: 236 landed, 211 missed, 43 could not be read), and each count that reached a build went to the abstract, a finding, a section, an appendix, the README and the record's description together. Rerun the source check and the claim-word read after any change to a repeated number: the source check found an old 444 that hand reading missed. Two more of R2W's: after dozens of scripted updates, a fresh reader found "The count" still saying 105 misses and 60 redesigned, a seventh copy of the

count that no writer had listed, and now every writer updates all seven; and after a 4/3 “lock” was withdrawn from the text, figures 9 and 10 still said “a new 4/3 lock appears”, because a correction searched in the prose misses the words drawn into pictures. List every copy, figures included, before the first change (R2W, sitting P2)

R2W (sitting P1); ties rows 811, 772

After a miss, say whether the test or its premise failed; derive a redesign from what you have seen, but register it where you have not looked (860) — a triple-star test of a fixed rotation missed because the rates themselves moved, so the redesign tested the premise first (was the drift the arithmetic’s?) and then asked the original question again on clean data, where it held. A redesign the answer would choose stays a description: one miss got no redesign because any cap “chosen now would choose the verdict”. Another was redesigned from one birth and registered on a second that no one had seen, and held blind; a pattern seen after one run was registered on the other birth and held (4.704 and 9.470 passes against 4.704 and 9.471 predicted). And before writing “not computed”, list every run a check has shown to take the same steps and what it already printed: a registration said an energy had not been computed when a step-identical run had printed it (LG3, LG2″, UN2′, ML1, EN1′). Most misses are not redesigned, and that is the rule working: of the 211 misses in the triple-star thread’s final count of 28 October, between 59 and 75 (28 to 36 per cent) were followed by a registered redesign, and the rest were answered covered, priced and stopped, rewritten to their true scope, or put on the proposed list. The check is whether a miss can and should be redesigned, not a duty to redesign it. Rerun a void once, and only where the rerun can still be blind: R2W’s COIN-FOLD failed its check as written, and since its deterministic engine would print the same numbers again, the fold’s place, between 1.442834 and 1.442836, went into the book as a reading, not a landing; THIRD-FACE-3’s void ran once more as TF3-1b, on drives no run had sampled and with the missing check added, and landed at 6.1765; and PLL-2 carried a lean unchanged though the void run’s numbers already said it would fail, because the band is not moved after the look. Weigh every miss that ends in a bare dash once, under a classing rule written first: R2W’s 26 came out one redesigned into a landing, nine covered, three landed in part and thirteen held as runnable, and three of those ran in the next exchange and landed (R2W, sitting P3)

the triple-star sessions; ties rows 661, 808

Words follow numbers: scope every “all”, label every estimate, define every coined word, write a “perfect” as a tolerance, and narrow a sentence the day it is shown too broad (861) — the islands paper’s front list said every test of the long runs read the same on a second machine, but three were registered after the kit was sent; it now says “in every test registered before they were sent there”, and a run-time range quoted for two cases of five became the range over all 32 segments. Every number no run printed carries “an estimate” until the measured value replaces it (a cross-check estimated at 2 to 3 hours took about 3 hours 24 minutes). A word coined in the conversation is defined before the paper uses it (“the slow beat” was still used in seven places after a rename, and “the floor” pointed at nothing defined). Bear’s description of the heavy star’s dots as a near-perfect surface was written as near returns and measured as a two-angle rotation to 10% at every rung out to 701 passes. And after a miss or a reader shows a sentence was read too broadly, rewrite it to its true scope the same day rather than dropping it (the triple-star sessions). Write a label after the thing it names: two of R2W’s reads got titles naming things their texts never mention, a stamp of 19:50:00 was typed ahead of its write at 19:48:49, and a script’s first line said “seven rows” before a row was dropped (R2W, sitting P2; ties row 735). A sentence about a measurement says what was compared, at its grain: R2W’s sitting P3 wrote “differ by the creation date only” where only the text layers and the dates had been compared, “did not move at all, to six digits” of a point on a filter grid whose steps are a factor of 1.25, and called one over the room exact on three grid points that turned out to be luck; seven no-physics reads were corrected before filing for the same fault (R2W, sitting P3). A front page decays as results land after it: Two Arrows’ first front page held 0 of 10 probe numbers from its later batches, so it was rebuilt from the headline register, with a table giving each of 33 headline rows its front-page item or its reason for staying in the body, and a script finding all 77 of the page’s numbers in the body. And say whether a ratio is a mean of ratios: “18.49 times better kept” is the mean of five starts’ ratios, while the two printed means divide to 18.26 (run r3) (Two Arrows)

the triple-star sessions; ties rows 840, 854, 502

A check that compares words cannot see where they sit: look at the pages, make what was seen a check that reads positions, prove it can fail, and aim it at the class of fault (867) — the Loop Atlas’s reader’s edition read back complete, every letter and digit of its source found in its PDF, while every three-digit note number had lost its first digit at the page’s edge (still in the text layer, out of sight), a heading sat alone at a page’s foot above a picture too tall for the builder’s tail, an epigraph was torn across two pages, and a last page held nothing. Each was found by looking at the pages and then made a check that reads positions or pages, every fixture first shown to fail with its fix switched off. Those checks each guarded a fault already seen, and Bear’s proof then found a heading alone over a table, a third kind; a check that reads every page’s last lines against the source’s own headings, whatever follows them, found exactly his page 46 (the Loop Atlas’s build_paper v1.3.2, check L5). A check for the class catches the kind no one has seen yet; the publishing packet’s builder of 31 October reads every word back whole and still cannot see where a word sits. A heuristic stands in for knowledge a checker could have had: the packet’s lint, which takes a short line in capitals near a page’s foot for a stranded heading, flagged a run stamp at the end of a table row, so where the

markdown is at hand, check against its headings. A page relabelled in place keeps its words in three places, the page, its template and the data its builder wrote into it: change all three by the same counted rules, and judge by the page as a browser renders it (14 of the 43 names replaced on the Loop Atlas's toy pages came from builders' data). And a change of look to a page that checks itself is made in the page of record, not by rebuilding it from today's template and engine; its self-check is run again in a browser, and every figure is shown to someone who did not draw it before it ships. And when a build's own check stops it, read why before working round it: the Torus of Fractions' builder stopped on a count read by its label, not by the run's own definition, and what stopped it was a finding, the needle's doubled paces, 326 of 3,340 locked paces that repeat only after a multiple of their fraction's bottom, a doubled band in the staircase scan's saved data since 13 October and never looked at until then (check STAIR_REPEATS of 30 October; misses row 483). A check of the PDF against the converter's own text cannot see what the converter dropped: the triple-star thread's README lost RUN_<TOOL>.bat on 23 October with every read-back passing, and its closeout listed the fix as proposed, not done. The publishing packet of 31 October does it before anything is built: build_paper v1.8 stops, naming the line, on a word in angle brackets that is not an HTML element and on a link to a bare number

the Loop Atlas sessions (session H, Guide items 451, 454, 462 and 477); ties rows 740, 593, 812

A whole read-back sees only what its reader can see: know the reader's habits, read cells as words in order, and look at the pages for what no text check holds (869) — the publishing packet's builder was rebuilt on 31 October to read every paragraph and list item back whole, where it had read only each one's first and last six words. Its first run on the guide flagged all 27 of a batch of additions, each for its final full stop, which the guide's builder drops on purpose before a row's tag line; the PDF reader joins a word broken at a hyphen across a line and interleaves a table's columns, so a cell is read as its letters and digits in order within a short stretch, not as one run (the guide's test build 15 had a cell whose ends stood in the cell above it, and one whose last two words began the next page). Read whole, the guide's 292 pages read back 2,730 of 2,730 on the first pass, and the two tables the old check had rebuilt as flowing rows stayed grids. What a text check cannot see stays a matter for eyes: the session's own start-here left a heading alone at a page's foot above its table, and lost two asterisks to italics, with every word read back. And what the converter drops before the PDF exists, a word in angle brackets or a link to a bare number, no read-back can see at all, so the builder now checks the markdown first. The packet's own rules broke their new example the same way, and it was seen on the rendered page. And a count is a reader too: asked whether all 330 of R2W's reads were in the book of reads, the completeness check said 329, and the missing one was there twice, in Bear's own words and in R2W's paraphrase quoted beneath them; R2W's file set a label before the paraphrase, and the check matched each read from its first words. The guide's 330 was right, and correcting it to the tool's 329 would have been the slip; the source conversation settled it. When a measured count and a written claim disagree, read the record before changing either, and when the tool was the one blind, give it the eye it lacked and run it the other way, on a copy with the read cut out, to watch it miss. Compare a gate's misses by name, not by count: at the start of R2W's sitting P3 the read-back's 22 missing were 20 page-break markers and 2 phrases, where the work order and an earlier cycle had 18 and 4, and the record had written "22 missing as before"; from the next cycle the misses were compared by name, and kept as a receipt matched byte for byte (R2W, sitting P3). And measure the printed page: long file names in a provenance table made the PDF converter scale every page of Two Arrows to fit, body type 6.83 pt instead of 8.07, while both the builder's read-back and the deposit linter passed it; a check that flags any page whose commonest type size is below 6.9 pt caught it. A table's caption printed alone at the foot of a page, the same miss as Question the Substrate's Table 1: the converter puts a caption written before a table in its own paragraph, so the style sheet must keep that paragraph with its table (Two Arrows; Question the Substrate)

the Field Guide sessions (the publishing packet of 31 October); ties rows 867, 593, 812

A memory sees a steady swing's crest at the square root of two, whatever its length, and a trigger set above that ceiling never fires again (878) — seen through a running memory of its own square (a memory m that relaxes toward v^2 over a time τ), a swing v peaks over the root of m at exactly the square root of $2(1 - \text{loss} \times \tau)$, whatever the swing's frequency and size: for a steady swing the square root of two, the sine's crest factor kept exactly through a memory of any length; for a dying one, lower; for a growing one, whose square grows at the rate g, the square root of $2(1 + g\,\tau)$, with no upper bound. The proof is three lines: the gap between the ceiling and the swing, scaled, is a perfect square, so it is never negative. So a trigger above the ceiling, once its swing goes quiet, never fires again, and a latch set at any fixed multiple of its memory can be cleared by growth alone. The Loop Atlas's spring latch on its own memory stayed awake exactly up to that ceiling: on a grid of hundredths at 22 settings of seven springs, from loss times memory 0.01 to 0.9, every wall at its ceiling or within two hundredths below it once the run outlasted the memory's forgetting time, none a hundredth above, the same wall at tunes 0.5, 1, 2 and 3; and a pumped spring met the growing ceiling within 4×10^{-9} at nine settings. As far as a short search of the RMS-detector literature shows, the ceiling is not stated there: a pointer for a specialist. A trigger riding on a body's own memory must also have a memory shorter than the body's drain: with memories from a twentieth to half the drain the latch stayed awake at triggers of 0.25, 0.5 and 1.0 times its running speed and fell silent at 1.5, and at a trigger of 1 with a memory of 1 it out-pushed the spring's own knee ear by half (0.0772 against 0.0514 of strength). Set a device's trigger from its own idle, not the fleet's mean: triggered at a share of each soup's own idle swing it stayed awake at every share tried, up

to twice the idle swing, and set from the six soups' mean it fell quiet in some soup's last tenth at shares 1.5 and 2. A test of whether a slow switch is alive must outlast its slowest memory: near loss times memory one the latch carries its swing's louder past for τ/(1 − loss × τ), 60 to 180 time units, and judged on its last 300 units it looked dead or weak at three settings that, flown five times longer, came to their ceiling's last hundredth. And compute a registered formula's value before the run: registered without being evaluated, the crest formula gives the square root of two at every memory, which would have shown at once that the lean beside it, a wall rising with memory, had no support; and let a bar hold equality where the forecast can fall on the grid, for a ceiling forecast at exactly 1 met a grid point at 1.00, where a bar written "below" missed by a technicality while the law held (row 628)

the Loop Atlas sessions (session I, Guide items 510, 511, 513, 515, 516, 519 and 529; its checks and records: SPRING_TRIGGERS, LATCH_WALL, LATCH_CEILING, CEILING_BEYOND, CEILING_SLOW, GROWING_CEILING); ties rows 879, 669, 628

Write so a stranger can place every sentence: a date for "now", a name for "I", a credit for every passage and an address for every number (881) — text is read long after and far from where it was written, and the words that tied it to its moment go stale without being wrong. Of the 9 edits for stale words that the clarity pass made in version 6's own text, 7 were time words, "this session", "tonight", "now", "still open", "the deposited guide" and a count as it then stood, each true the day it was written; write the date or the event instead. In a book that many sessions wrote, a session's "I" places nowhere: in 11 rows it became "the session", or the sentence passive, while the author's own "I", in his account of why the guide exists, his rulings, his papers and his first expectation, stays his. An opening whose tag line was its whole credit loses that credit the moment an addition with a credit of its own lands below it, and a reader takes the later credit for the opening's: 37 rows' openings were read that way, and each now ends with its tag's project. When an addition with a credit lands on a row whose opening has none, write the opening's credit in the same step. A rule enforced only on new text leaves the old text in the old style: the builder wrote powers of ten for every number it inserted, and version 6's rows kept the notation a computer prints, "e-14" for "× 10^{-14}", until a sweep wrote 26 values in 16 rows the guide's way. And a row that cites another project's ledger items is one step from the runs behind them: of 78 run and check names behind the Loop Atlas's reads 298 to 328, the guide named 2 until each new row's tag line named its checks, and then 71

the Field Guide sessions (the clarity pass, 31 October and 1 November); ties rows 570, 861, 869

L4 — The Ledger

What it cost, what it taught, what went wrong, and what it yielded.

The rows of L4

New in version 7 on this limb: 80 of its 331 rows, numbered 549, 550, 551, 552, 553, 555, 556, 558, 559, 569, 570, 571, 572, 573, 574, 579, 580, 582, 583, 585, 588, 589, 591, 592, 593, 594, 595, 596, 597, 598, 599, 600, 602, 610, 611, 659, 660, 661, 676, 679, 684, 716, 720, 726, 733, 735, 736, 740, 765, 767, 770, 774, 775, 777, 782, 783, 798, 808, 811, 814, 821, 823, 832, 833, 834, 840, 843, 844, 845, 852, 853, 855, 866, 868, 871, 874, 875, 880, 882, 891.

Could it have missed, and did it (1) — Before you look at the data, write down what would count as a miss — a pre-registered null ceiling. The seismic instrument set its ceiling at r = 0.0940; the measured result was r = 0.2697 (matched-zeal significance 4.98%). The ceiling was cleared, and it was cleared cleanly because the bar had been committed before the data arrived. Before scoring a test, confirm that it can fail and that its input reaches the system: SL6b (run 20261018_220340) first read BORNE OUT, but the slow-line reader looks only above 3/6,000 = 0.0005 cycles per pass, and the band SL6b asked about ended at 0.000377 and 0.000476, below that floor; a cold reviewer caught it, and SL6b was withdrawn to NO READING. U2 (run 20261020_001032) nudged the birth by one unit in the last place, but the run turns the birth into the orbit size through a cube root, and at 7.200 the orbit size rounds back to the same double (4.17123197386374 both ways); U2′ put the nudge into the starting state and holds (run 20261020_001325). U3 printed HOLDS, but at 7.200 it compared a run that was never nudged, and it was withdrawn to no reading. And what tests the toolchain stands beside the paper's count, not in it: the five readings of 20 October, U0 to U3 and U2′ (triple_ulp_nudge v1.0 and v1.1, runs 20261020_001032 and 20261020_001325), were graded off the paper, U0 and U1 holding (one software stack repeats a run to the bit, and repeats the stored series of 9 and 11 October); and a custody check such as HS1 (the thirteen rebuilt heavy-star series are the ones of record, 13 of 13), like the fidelity checks F0 and K0, is not counted. (the triple-star sessions). Confident sentences said in chat are predictions too: register them with their rival named, and withdraw them as plainly as a win. And write into each registration what has already been seen, and that nothing rests on it

Texas sharpshooter fallacy

Mechanism versus correspondence (7) — two prices, and the earlier one is measured. Measured (Anchor, Test A): three averaging defences — a topological code, a Byzantine vote, a phase-locked swarm — were pre-registered as breaking at the same place on one shared noise-correlation axis, and do not; the count-defences' walls are operating-

point-dependent (a majority defence below 0.5 density has no sharp wall at all) and the swarm is non-monotonic because beam-pointing is blind to global phase. Restraint (cochlear): four cross-domain correspondences identified and deliberately NOT promoted — seismic travel-time, interferometric fringe count, fibre group delay, impedance phase angle. A test that names a correspondence of two numbers with no mechanism behind it is labelled so when registered, and a miss closes it at once, priced and stopped, not salvaged: two shares asked to agree within 20% came 0.121 and 0.159, and stand as measured (the triple-star thread, OF1)

practice; pre-registration

Test the narrowest case first (8) — Start with the smallest, most constrained version of a test — the one that can say no most easily. Four signal processing architectures were tested against the narrowest case; all four were closed as insufficient, and three of the approaches were redirected to narrower sub-tests. The lesson: the narrowest valid test is the one most likely to find something real, and it is the one most likely to say no.

Inherited constants (9a) — a spread constant asserted 5× and never sourced

Do not close on thin evidence (9b) — one case report against several series. Nor one record near an edge against a fit of seven: a turning point placed from a single reading (7.1986, from the record at 7.1990, against the seven-birth fit's 7.19868 and 7.19864) missed at 7.1980 (0.915, outside its 7 per cent window), and both arms of the next test put it where the fit had (7.19865 and 7.19866). Register from the fit, and treat a lone record near an edge as the thing under test (run triple_heavy_point_scan 20261011_002135) (the triple-star sessions). A lean read from one case can sit on a special point. The Loop Atlas's tied bells were leaned to gain within 5 per cent of 1 at every memory, a lean read from BELL_CLOCK's one case at memory 6, which turned out to sit on the balance line where the rings neither help nor hurt (row 801); at a squeeze speed 1.41 times the rings' and memory 3 the gain was 0.933, since at short memory the rings cost stroke even to a fast squeeze (run BELL_RATIO 20261016_183422) (the Loop Atlas sessions). The line where the rings stop costing is not a clean one. A lean that the gain with softened cores still crosses 1 between squeeze periods of 2.5 and 3.5 at every memory missed, its crossings falling at 2.264, 2.110, 2.937, 2.881 and 2.018: at memories 3 and 4.24 the gain lies within 0.03 of 1 from period 2.12 to 3, a near-flat stretch on which a crossing read between two grid points can fall anywhere. Where the curve runs flat, grid it finer, here between periods 1.5 and 3.5, before giving the crossing as a line (row 799) (run BELL_SOFT_MAP 20261021_000736) (the Loop Atlas sessions)

The elimination arc (15) — An elimination arc tests hypotheses one by one, keeping only what survives every filter. A 'superstar' was to be a station that passes every elimination test. The seismic superstar scorecard named and thresholded six criteria before scoring: 2 stations passed all six, where chance expects 1.5 from 97 fully measurable, and the criteria correlate, so even that understates chance. The survivors are what chance leaves: the dozen-superstars idea died on training data, where dying is free, and what lived was a graded spectrum, 26 stations at five of six (Loop Walking Cascadia, §5). Count what chance would leave standing before you call a survivor real. An elimination arc can locate a leftover: after D1‴ the triple-star thread split the leftover scatter at 0.28 three ways, by side of the beat's zero (LB1, run 20261016_214034: the two sides' residuals 0.7 standard errors apart; the correlation it also printed reads nothing by construction, row 450), by chaos (LB1′, run 20261016_214250: NO READING, but leaving out the four chaotic runs left the scatter unchanged, 0.0006686 against 0.0006694) and by locks (LB1″, run 20261016_214403: the 15 locked runs fit better, ratio 0.21). Each miss narrowed where the leftover lives, inside the island and in the unlocked runs, and the reading after the fact, the part cycle of a short record, became LB1‴, tested over 2,400 passes in the laptop kit and borne out: the law's mean square residual fell from 3.423×10^{-7} over 600 passes to 1.116×10^{-7} over 2,400, a ratio of 0.326 against a bar of 0.5 (the thread's prediction ledger of 18 October). A split that reads the other way round still locates the leftover, so register the next split from what the last one ruled out (the triple-star sessions)

Held-out calibration (22) — Never calibrate your detector on the same data you test it on — you are training it to pass itself. Held-out calibration uses data the calibration never touched. The seismic false-alarm result: 13–21% false alarms uncalibrated; 1.1% after calibrating on held-out data. The improvement is a factor of 10–20, and it is only available if you reserve data for calibration from the start.

held-out validation

The weak-baseline law (23) — A baseline that does not know the basics cannot be beaten in a meaningful way. The poker geometry feature showed +0.47/+0.44 improvement over a baseline that knew only a hand's aggregate statistics. When the baseline was given three order-blind pressure facts (the elementary context), the geometry feature's advantage collapsed to +0.000/−0.002. The weak baseline had been awarding credit for knowing things that any informed observer would already know.

A gate that never got stressed (28) — A test is only as good as the stress level at which it has actually been run. This gate was designed to activate at correlation $\rho = 0.717$ (VIF — Variance Inflation Factor — of 2.06), but when run it reached only $\rho = 0.7611$ (VIF 2.377) — the test had never been pushed to the threshold it was supposed to guard. The patched version reached $\rho = 0.9668$ (VIF 15.331), genuinely stressing the gate it was designed to test.

variance inflation factor

Synthetic calibration is circular (29) — testing an instrument on data generated by your own model confirms that the model agrees with itself, and nothing else. The synthetic data encodes exactly the assumptions the instrument was built on, so passing the test is guaranteed by construction — a feature named for a property is not a feature that has it. The permutation test that catches this: permute along the axis the name claims sensitivity to, and check whether the numbers move. If they do not, the instrument was measuring its own assumptions rather than the substrate. Real calibration requires an independent reference the model did not touch — a ground truth that can say no

practice; ties row 29's own kill condition

A feature named for a property it lacks (30) — Run the test the feature name implies — do not assume the feature actually has the property it is named for. This feature was called order-dependent because it was built to be sensitive to sequence order. Permutation testing (randomly shuffling the order while keeping everything else) showed: two of its three components changed by 4.4×10^{-16} and 2.2×10^{-16} — effectively zero. The one genuinely order-dependent component varied by 0.79. The feature was named for a property that two-thirds of it lacked. Shuffle the order before naming a rhythm: a rule meant to find rests that rock back and forth across a whole number called shuffled rests "rocking" 0.72 of the time by median, more often than the real order, which it called rocking in 20 of 42; it counted spread, not rocking (the triple-star sessions). A null must keep everything it is not testing: rows permuted among a cloud's pads merged 11 per cent of them, two shuffled pads landing in one hole, which thinned the null by itself and put the picture +9.4 spreads above chance; with the order of the columns permuted instead, nothing merged and the excess fell to +1.8. A shuffle that changes the counts is a test of its own (the Loop Atlas sessions). A printed label is named for a property too: triple_harmonic_recipes v1.2 printed 'absent' for a hexadecapole term that is present, only not used (run 20261016_204919); the label was changed and every number stayed the same (the triple-star sessions)

permutation test

The resolution walk (35) — When two measurement methods disagree, walk toward higher resolution until they converge. Two approaches to the same geometric measurement were 10% apart at coarse resolution. Refined iteratively to finer resolution, they converged. Convergence is confirmation that both methods are reading the same thing; persistent disagreement at all resolutions is a finding about the measurement, not the geometry. Refine the reader until the number stops moving, and hand the numbers that never stop to the reader. The falling-cat law for three bodies read a mean residual of 0.00143, 0.00019 and 0.00003 radians at 25, 100 and 400 points per interval, converging; the peak speed of the fastest moment read 379, 740 and 1,149 and was still climbing, so it belongs to the reader, not the stars (the triple-star sessions, after the Loop Atlas sessions' independent re-reading, which began by running the triple-star room's own tool unchanged, row 650). Aim the walk at the largest step the reader takes in its own space, not at the most dramatic moment: the whole error sat on loops spanning one instant when the shape raced across its sphere at 316 radians per unit time, while the closest approach of two stars left none (the Loop Atlas sessions). When two instruments disagree, do not take the older one as the reference; bring a third that shares code with neither. A new stretch ruler disagreed with an old one by up to 0.163 on held riders, and a check that trusted the old one failed; a third ruler, central differences of the rider itself with no tangent equations, sided with the new one at all six riders, within 0.003. The old ruler's error was the size of its angle, not its method (the Loop Atlas sessions, R2W)

Fidelity rerun after every change (36) — After any change to a tool, rerun the full pipeline from the baseline — not just the new part. The fidelity check's first run on a rebuilt pipeline caught a stale file from an earlier rung (an intermediate processing stage) that had been modified without the pipeline being told. The check that catches silent propagation is the whole-pipeline rerun, because every downstream step trusts the step before it. Revise a tool only with a fidelity check: the old version is retired by md5 and the new one must reproduce every number the old one wrote, as a walking tool's second version reproduced every column of its first at both paces. On a chaotic state the call structure is part of the input: the same tool, settings and pace gave 0.557209 turns per cycle against its parent's 0.526206 because one long call carried the drive's phase through, and repeating the parent's call sequence gave 0.526206 to the digit. Do not ask chaos to repeat itself under a change that is not exact: rescaled by uneven factors, five shaking riders came back different by up to 0.015, as did the same riders started 10^{-12} from the reference at four of the five cells, while gliding riders stayed within 1.6×10^{-6}, and rescaled by factors of 2, which binary arithmetic holds exactly, every rider came back bit for bit; a twin nudged by 10^{-12} tells sensitivity from a hidden knob. A long survey on a laptop carries its fidelity gate inside the tool, rerunning known riders before anything new, behind a launcher that refuses any script whose md5 is wrong, and writes each chunk as it finishes so a stopped run resumes (the Loop Atlas sessions, R2W and the book). A fidelity checker can quietly pick the wrong copy: one solver's checker silently preferred a frozen copy lying beside the solver of record, until the frozen copies were moved aside. Move frozen copies out of a fidelity run's folder, and print which file was loaded (the Loop Atlas sessions). A fidelity check on a chaotic state either repeats the parent's call structure exactly or compares a statistic with its spread: the 0.557209 against the parent's 0.526206 above came from one 1,000-cycle call, where the parent had restarted the drive's phase every cycle for 128 cycles (runs HIDDEN_SCISSORS 20261005_064106 and HIDDEN_SCISSORS_F0B 20261005_064308) (the Loop Atlas sessions). A false friend: in physics, quantum fidelity, or the Loschmidt echo, is the overlap between a state carried forward in time and then back with a small perturbation, and the state it started from. It measures the same sensitivity this row's nudged twin reveals, and it is not this row's rerun (it is used on two-

electron quantum dots in Schröter, 2013) (the Field Guide sessions). Every one of one session's 34 tools had its predictions written in the data ledger before any part of it existed, and when a version failed on its own steps (GROOVES_11 twice, GROOVES_13 once, GROOVES_14 once, DELTA_AREA twice), its successor had to reproduce every number of record before its new numbers were used (the Loop Atlas sessions). Rerun the dry test after every edit to a reader: RATCHET-5's dry run found a missing weight crashing the reader instead of failing it, and an edit that had commented out a working line; RATCHET-4's found a tuple read in the wrong order and two verdicts passing with a weight missing. Each was fixed, and the dry test rerun, before the registration was written (R2W). Port an engine and check it against run files before any page uses it: the Depth Axis Explorer's rectifier bench needs the rider to run in the browser, so the weightless edges were ported to JavaScript and, before a single control was drawn, run at six settings of record (two tones, three tones, slow and fast, strong and weak) and compared edge by edge with the CSV files: every edge within 1.31×10^{-12}. A page built on an unchecked port would show a confident picture of the wrong rider (R2W)

The numerical read failed twice, and the mechanism is named (54) — discretisation puts an O(1) offset in the constant term that does not cancel between domains, and a staircase boundary invents thousands of right-angle corners each contributing exactly 1/16 to the very term being measured; switching to axis-aligned square holes did not rescue it

Hansel and Gretel — the birds ate the path, not the forest (55) — The substrate that stores a topological invariant can be destroyed even when all the readers are working and reporting clean. If the medium goes, the invariant goes with it. Named failure mode: medium reads fine, reader reads fine, invariant is gone — like Hansel and Gretel's breadcrumb trail, which appeared intact right up until the birds finished eating it.

Hand the reviewer the hammer (61) — the stance stated in the paper itself: what survives demonstrates the merit of the idea, and what is destroyed is offered a second life as the chance to engineer a replacement. Backed by a Seam and Assumptions Register naming, discipline by discipline, which expert should interrogate which assumption and the exact question the author could not answer

Confidence recorded separately from assertiveness (62) — the register carries the author's honest confidence separated from how assertive the prose reads, and names the places where confident prose is least trustworthy — calibration claims. Most papers conflate the two and a reader cannot tell them apart

The packing-knob pattern, named and NOT promoted (63) — a packing knob buys capacity cheaply until a critical density, past which loss turns non-linear: over-binning in the read domain, defect-density structural collapse in the medium, the mutation error threshold in evolvability. Explicitly called shared phenomenology rather than one equation — and they differ in recovery, since over-binning is recoverable by reading a coarser alphabet and structural collapse is not

Separate the references you read from the ones you know (67) — both Turtles papers close with a citation note naming exactly which references were grounded in sources reviewed for that draft and which are standard works cited from established literature, with verify before external circulation attached. That is provenance tagging one level above SOURCED/UNVERIFIED/ASSUMED, applied to the bibliography rather than the numbers. The Latch Atlas carries a depth tag on all 201 references, the reader's instrument for weighing each claim rather than a confession, and three topology citations moved from "recalled" to page numbers once the books were in hand. The direction of a known correction is read, not recalled. Cavendish's 1798 value of the Earth's mean density, with his arithmetic slip mended, was leaned from recall to lie within 1 per cent of today's; read, the mended value is 5.448 ± 0.033, 1.2 per cent below today's 5.514, and the unmended 5.48 printed in his paper (5.480 ± 0.038 as Wikipedia gives it) is the one within 1 per cent. Francis Baily found the slip in 1821, and the mended figure is the lower one (read on the web on 20 October: Wikipedia, 'Cavendish experiment' and 'Gravitational constant') (the Loop Atlas sessions)

The rolling referee report, inline and adversarial (68) — each paper carries its own skeptical-but-fair review as a section, doubling as the version tracker: summary judgement, strongest and weakest claim named, honest exposure, and a for-next-version list. It is hand-the-reviewer-the-hammer made structural rather than rhetorical

A check that silently stops being valid (42) — Some validity checks are only valid in certain configurations, and they fail silently when the configuration changes. The mass-balance column (a check that the water through the mouth matches the bell's change of volume) is valid only while the ring is open. Once the ring's lip seals, the body speed is no longer constrained by mass balance — and any number from this check after sealing is meaningless, regardless of what the check reports.

The exterior reference is a binary wall, and it was measured (70) — a PID handed a corrupted setpoint defends it across 40 seeds, six corruption sizes and four gain sets — median final x tracks the corrupt value to within 0.3 at IQR 0.99 — and the reality-cost is set by the corruption gap, not the noise: at corrupt = 70 it runs 903.2 at noise 0.1 and 1171.8 at noise 8.0, against 0.0–17.9 when the setpoint is true. Four in-loop attacks failed, and a referee-grade system-ID adversary settled at setpoint 64.27 against a true optimum of 100, because its smartest internal move — anchor to what I can achieve — locks onto wherever it already is. A second loop anchored differently corrected it immediately

Conant & Ashby 1970 (good regulator) — cited as lineage and explicitly declined as proof in the paper itself

Retract your own pre-registration in place (71) — the pre-registration named Conant–Ashby as the single citation that would convert the keystone from supported to proven; the paper retracts that overclaim in its own front matter, separates the two claims, and restates what the keystone rests on. The prediction survives, the borrowed authority is struck, and the retraction ships inside the record rather than beside it

Pre-register the kill that would destroy your own thesis (73) — Test A was registered as the paper's most important possible kill — do three averaging defences break at one place on a shared noise-correlation axis — and it fired. The walls do not coincide. It is reported as the result, and the Act it bounds is narrowed in place rather than routed around. The kill can be hiding in a helper's inference: a survey's helper inferred that only a moving wave makes fractional locks, which would have made only travelling-wave media the toy's twins. One column with the moving ripple swapped for an added drive, five times finer than the map, showed the added drive locking more fractions, not fewer; on the map's own grid the two tied at one each and could not have decided it. Run the control that could kill an explanation before writing the chapter that rests on it, and check the theorem behind a result (here, no weight, whole-number locks only) against its paper before writing it as fact (run ADDED-1) (R2W)

ours; pre-registration

The operational filter — build, predict, or break (74) — the admission test every bridge in the paper had to pass: if an observation only produces a feeling of connection, it is out. Stated against its named failure mode — the 1990s chaos fad used the mathematics as a mood and could not survive being punctured, because the mystique was the value. Puncture this and a working read strategy is still underneath

The hostile hammer is the valuable one (75) — the sharpening of hand-the-reviewer-the-hammer (row 61): the friendly collaborator's swing is the easy one, and the swing that finds what a sympathetic test never will comes from someone whose bias is maximally different from yours — someone trying to break the work because it threatens their anchor. The invitation and the curse are the same act, and it costs something every time. When a result looks too clean, spend the next run trying to break it: R2W's two-channel meter came out exact in its structure and within 0.21% in its one predicted number. That is the moment to go hunting for the flaw, not to build on it. A witness that shares nothing but the physics (row 106), plus a run at ten times the stress, either breaks it cheaply or earns the right to build on it. Both were run, and the meter earned it: the witness agreed to six figures or better (WITNESS-1), and ten times the drift kept the same laws, straight to 1.00001, curvature 0.9999 (RATCHET-72) (R2W)

Declare the model overlay before the claim that uses it (76) — four departures declared in front matter ahead of everything that leans on them, each with its own kill condition and each naming the established alternative it is not — Russell's circumplex against the signed-ternary affect overlay, active inference against the control-loop model of belief — under the stated rule that author dictation of a model structure is an assumption to be tested, not design authority. State a model's prediction with its model in the same sentence: "Set against the octave, it is backwards" read as a finding when it was a prediction from adding two tones. Written as "if the two tones simply add, against the octave would make c worse", it would have invited the right question; Bear's question did the work the sentence should have done (row 734). A prediction always carries its "if" (R2W)

Ship the broken run in the crate (78) — `neuro_loops_v1_BROKEN.py` is deposited alongside v2 — drive gains set too high, every scenario died, a real modelling error caught by the data and kept under the project's graveyard rule. The misses are not merely published — when possible, in prose; the artifact that produced them is in the public record

Exploratory and confirmatory are layers, and each prediction names its own (79) — five predictions each carry an in-container toy that shows the claim is well-posed and locates its signature, each labelled information-layer with the real confirmatory test named and not run; one added refinement came back a reported null rather than being tuned away; and the keystone's provenance was corrected in place to name the hardening run of record rather than the single-seed illustration standing next to it. A rival registered after its idea came from the data is still a test, as long as its rows are unopened: a power law ($\delta_2 = \delta_1^2$) seen after one table's data was registered with two rivals before the next rows were opened; it held blind (8 of 8, then 4 of 4 in the junction), and both rivals failed, the joint curve's failure (by 0.047 at δ_1 0.446) showing exactly where the two media part (runs TOY-SIDE-3g, TOY-SIDE-4g and JJ-SMALL-1c) (R2W). And a law read after the data at one setting describes that setting: a gap's power of the period, read at one strength (0.05, where the deepest rung sat near 10 per cent uncertainty), missed by 16 to 31 per cent once the question, whether the two arms of children of 1/2 meet (the bracketed word of row 589's picture), was carried to a strength the ruler could reach (0.10, where the grips are larger and every rung to period 17 resolved), while the question's own claim held there, the gap closing to under a quarter of its period-9 value and the rival in which it levels off failing. Move the question, and register its shape afresh where it is asked (runs LADDER-1 and LADDER-2) (R2W). A lean named after the data is a reading: R61_AFTER, written after RATCHET-61 was read, called a square-law bend in the dark point's proportion "the lean". It carried to the second size within 0.0005, but a script written after the data cannot register anything, and a dated note in the record says so. It is R2W's lesson 81 (row 735) in its quiet form: not a number typed early, but a word that claims an order of events that did not happen. And say the mechanism before the run, even when it cannot be graded: RATCHET-62's registration carried a guess, marked ungraded, that at fast

drives the rider hears each tone only through its swing, that the lopsidedness comes one order in one over the frequency later, and that the holding power settles near one plus one fifth. The run graded three fitted shapes, not the guess. But the guess named two things to read from the same runs, and both could be read: the window's fall with the tones' swings held fixed (1.14, 1.08, 1.04, closing on 1) and its power in their strengths (4.5 to 4.7, not 5). A fitted shape says where the next point lands; a mechanism said in advance says what to measure next, and why the level sits at 1.23 and not 1.2 (R2W). Keep what is seen after the run out of the score, and register it forward: SL5 (run 20261018_203922) read MIXED against the guess, and what was seen after the run and labelled not a result (the slow line at 1.16 to 1.32 times the beat) became SL6, whose SL6a missed and whose SL6b went void; the chain of SL7, SL8 and SL9 (run 20261019_043449, BORNE OUT) closed the triple-star paper's slow line and showed that an earlier match at 7.340 was a coincidence of two small numbers. TW7 reported, without registering it, that 7.120's twin gap grows as the pass count to the power 2.76; that became TW8, which landed (SLOWLY CHAOTIC, parting at pass 14,795; run 20261019_062042). Report an after-the-fact reading under that label, and give it a registered test before it enters a paper as a result (the triple-star sessions)

exploratory vs confirmatory (pre-registration)

Name where the method goes silent (80) — the limit stated in the paper rather than hidden: the diagnostic is most testable exactly where it matters least — repeated, resolvable, ground truth on schedule — and least testable where it matters most, the once-only regime change. Where no ground truth is accessible, consensus is not a corrupted reading, it is the only reading there is, and the framework declares itself silent instead of extending

A detector that falsifies cleanly still tells you what it found (85) — The write-up was recovered in full, and the verdict is exactly as remembered: the detector is a supercell-scale rotation detector, not a tornado detector — a clean negative at five stations across three labelled days, one tornadic, one severe-but-not, one quiet. The priced version is row 372, and the autopsy that names density as the cure is row 373. The row stays because the practice is the ornament: the miss got a filename

see rows 372 and 373

Tag the sentence, not just the number (86) — the dual-voice discipline: one voice that sees and one that checks, each with its own mark, every node carrying its tag, and the one rule on the first page — vividness is not evidence. A reader can tell at a glance which sentences are the author's to see and which are the world's to confirm. This is the third and finest-grained member of the provenance family: SOURCED/UNVERIFIED/ASSUMED tags the number, row 67 tags the bibliography, this tags the sentence

Retract the word when the harder test fails it (87) — a word — topological — was used, a harder test failed it, and the word was withdrawn in public in the appendix that had used it. Priced from the book: the harder test measured whether the ledger actually was topological and it fired 52% on honest data, a noisy heuristic rather than a clean integer invariant, with its predicted blind spot simply absent. The mechanism is row 116. Rule 14 says a vocabulary that fits is not evidence the object transferred; this is what obeying Rule 14 costs and what it looks like when paid. Row 71 is the same act performed on a citation; this is it performed on a term, which is harder, because the term is load-bearing for everything written around it. A word retracted keeps its verdict: two guesses were borne out as registered on a reading later found bent by the very tide it watched, so their verdicts stand and their words ("drains"; "steady at the centres, swinging outside") are retracted in place on every page (the triple-star sessions). A confident sentence said in conversation is turned into a test with its rival named, and its withdrawal is logged as plainly as a win: Claude called the triple star's jump at 1/10 a textbook shape for a fraction whose lock is too thin to see, Bear asked for a check, and LW4, TS1 and TS2 were registered before any run, with both textbook shapes written down, a square root for a dissipative staircase and one over a logarithm for a conservative separatrix. TS1 (run 20261017_172324) read SQUARE ROOT by its rule, but its fitted edges ($d0 = +4.76 \times 10^{-7}$ and $+5.63 \times 10^{-7}$) put a flat zone where LW4 (run 20261017_170131) shows a straight passage, and the power law's p came out 0.894 and 0.902. TS2 showed the shape changes with record length, so the call was withdrawn in the register, and the redesign (LW5, a longer record) was built on Bear's go-ahead. LW5 read no crossing at 1/10 in 12,000 passes, every birth's rate below a tenth, on a record the integrator's own arithmetic was dragging down (row 741); at the tight kit's tolerance (LW6) none of 41 pairs was chaotic, missing its guess of chaos at the fraction, and FA2, that miss's redesign, landed at 11 of the 13 registered births: the passage through a tenth is a pendulum circulation, regular, with no chaos (run 20261018_173550; row 599) (the triple-star sessions)

vocabulary that fits (Station 1)

Supersede by tightening, never by burying (88) — a later appendix does not replace the earlier one and does not quietly drop it — it tightens it and adds the trust layer the first one missed, with the relationship stated. The earlier reasoning stays legible, so a reader can see what changed and why. This is the constructive half of the no-references-to-unpublished-drafts rule: in a published lineage the supersession is the content

Turn the instrument on your own panic (89) — the fear that the work had been lost was diagnosed with the author's own three-layer trust model and came back a measurement error, not an invariant error — the map was unreadable from where he stood, the thing itself had not moved. The general form is the one worth carrying: before

acting on a loss, ask whether the invariant changed or only your access to it, because the two demand opposite responses and feel identical. A long session survives on its files, not its memory, as row 720 says of a run: one session ran 23 days, its context was rebuilt from files many times, and every rebuild was made by reading two working files, never by recall. One kept every arrival whole and recorded where each lesson went; the other, with a handover at the top, said what each stage produced and held every open thread, and every later question was settled from them: the table of arrivals, the pastes kept whole, each revision's "what changed". Write the next step into a file before the turn ends, as if the next turn will be a stranger, and check every claim against the files, never against memory. The files held, though each rebuild cost time at the start of a turn (the Field Guide sessions)

Every port must beat a baseline in its own domain (93) — three instances on the record: the whale forage signal beat a null, closure depth beat maximum wind, Rivera beat its own quiet control. A tool that ported by mere analogy would fail all three. The baseline test is what separates a real cross-domain instrument from a frame that explains too much — it is Rule 14 stated in its positive form, as the admission test rather than the warning. And a match on rules that every rotation obeys proves nothing on a new substrate: q stops per dance, the numerator as the number of stops skipped, alternating colours and the window count hold for any rotation in disguise, while the pair just outside the frozen points, the 0.013 gap, weight choosing the dance that holds the leap and the ratio's steps are specific to the toy and can be carried. Sort a family's rules into universal and specific before taking them to a substrate (row 727) (R2W)

The rim can be ground arbitrarily fine and still cannot see what it does not enclose (96) — one impossibility wearing four costumes, and the corpus had all four separately before it had the name: the observer inside the loop who cannot step outside it; the bootstrap that cannot pull its own boot; the frame-variant liar with no static twin; and an event off the system's representational map. No system is its own trust anchor, and the only adversary that matters is the one that arranges never to be enclosed. The cure is never a finer rim — it is an out-of-band channel on a substrate the in-network adversary does not control

von Foerster (second-order cybernetics); Banks, Excession

Reach the wall from two directions, and fix the attribution when you get there (99) — an independent persistent-homology detector for adversarial influence — The Shape of Adversarial Influence, arXiv 2505.20435, Fay, García-Redondo, Wang, Dubossarsky & Monod, Imperial College London — is a very fine rim that catches the static topological distortion and goes silent at the adaptive one. Same instrument class, same blind spot, arrived at independently, which is far stronger evidence the wall is real than either finding alone. It is the evidence row 438's confound asks for, a stranger hitting the same wall, met so far for this one wall (row 96's). And the correction travels with it: the working record had this group filed as Los Alamos. That mis-attribution was the lineage's one genuinely-open external item, and locating the real paper closed it. Attribution slips hide from the eye: of Two Arrows' 21 citations, 3 needed the publisher's record to correct, two volume years and one lead author ("Zhang and colleagues 2021" was Luo, Xu, Ma and Zhang 2021). A wrong year can be caught by eye and a wrong attribution cannot, so every citation is checked at its publisher; by the deposit, 23 of 23 had been (Two Arrows)

Fay et al. 2025 (arXiv 2505.20435)

Generate warm, audit cold, in different rooms (100) — you cannot be in the wraith-hunt and the dream at the same time — the suspicion that protects the paper poisons the play. The resolution is the out-of-band witness on a schedule: generate fast and capture untouched, columns tagged but not litigated, then hand the load-bearing claims to a clean session that the momentum does not reach. Capture-as-you-go is what makes both halves safe, because nothing is lost, so you never have to be adversarial while you are playing. The autopsy is a separate phase, not a thing done mid-dream. A forked pair of sessions with a third as a sounding board, checking their claims against the register rather than taking them, caught one real correction (row 606) and one number that would not reconcile, later traced to two readings of one lap (row 612). A separate agent that had not seen the work checked a paper's draft against the run files and found 17 points. Most were wording; two were wrong claims (chaos "just below" a layer was inside it; "only" compared two centres with a far range that contained them); and one, a 600-pass offset of 0.0045 to 0.0062, led to the checks that found an island's real edge (the triple-star sessions). Two helpers who had written nothing read one sitting's work and found a provenance error, 0.000014 written as "within 0.00001", a verdict reported more kindly than registered, four failed predictions with no redesign check, five wrong line counts, a wrong date and several sentences stronger than their outputs; the session that wrote it had found none of them. For anything the person will build on, the check comes from a reader who did not write it (R2W). An audit's numbers are its own until recomputed: one R2W sitting's crossings audit was a helper's read of the record. At the close its four re-readable items were recomputed by a script (TRIAGE_12) before being stated, and the rest were labelled "the audit's reading". Keep that line between a helper's report and a measurement (R2W). Review a long kit cold before it ships: a separate agent read the triple-star thread's triple_closeout v1.0 against the register, told to change nothing, and found four faults: 'LB1‴' cannot be encoded by the Windows pipe's cp1252, so the selftest would have stopped the run on Bear's laptop before anything ran; one unread stage-1 run would have planned nothing; one ladder case had no verdict in the register's words; and a worker that raised an error would have stopped the loop. All four were mended before the first run; the selftest then passed 18 of 18 with PYTHONIOENCODING=cp1252 forced, and the launcher was rehearsed on a mock THREE-BODY folder holding a stale duplicate. Run the selftest under the target machine's encoding. And blind

a sorter to the verdicts, then check the blinding itself: RC1 (run 20261017_040300) used two separate agents, one listing the 74 registered picture-reads with their verdicts, the other given only verdict-free wordings, shuffled, to sort as relation, shape or neither. The first agent had to reword the reads, because 21 of its quotes had carried result words. Relation reads came out borne out 19 of 30 (0.633) against shape reads 11 of 28 (0.393), a difference of +0.240 against a bar of +0.2, narrowly; the sorter's note that about seven shapes might move to neither on a stricter reading was reported, and the count was left as registered (the triple-star sessions). A second specimen: A Rider in Two Waves passed every gate it had, and a reader who had not seen the work still logged 100 findings, five wrong or misleading, 38 places where the book contradicted itself or its runs (the cold read of 28 October). Gates check form; a fresh reader checks meaning (row 853). A reader who summarises a record marks its own readings apart from what the record says: the triple-star closeout's three readers tagged every inference of their own "reading, not in the entries", and those tags are where the backward pass's best questions sat (a near-still combination of two angles, a run stamp carried by two tools, a count with no run behind it)

The wraith is the theorem; the bad data is the job (101) — the unbeatable adaptive adversary is the floor that says this far and no further, and it is almost never the work. Point the instrument at the sky and it vanishes, because nature does not lie. The instrument earns its keep on the catchable ninety-nine percent — and the most dangerous member of that class is a true invariant with a false story told over it, where it's derived, trust me is the perfect cover. A corpus with this much adversarial machinery needs this row to keep the machinery in proportion

ours; ties row 96

Fear dressed as caution (102) — the three flinches, named: from the look, because you might get scooped; from the kill, because you want the pretty result to be true; from the autopsy, because the null felt like a verdict. Each is fear wearing the costume of rigour, and each produces the confident misreading that row 94 says is the real adversary. The antidote is one discipline rather than two — fearless and playful together: throw the guess, fold your hopes into it, let it die, read the result straight. Sibling of row 81, and the sharper of the pair

ours; "you are what you do when it counts" (Steakley, Armor)

Three honest implementations converging on one boundary means the boundary is real (106) — magnitude gating, strict integer-ness and soft consensus all won the same corner — sharp, sparse defects at low-to-moderate noise — and lost the same corner at high noise, each by a different mechanism. Three different roads to one wall is evidence about the wall rather than about any one implementation. The call that follows is the valuable half: a fourth gate would be retuning toward a wall proven three times, so stop. Two independent routes to every key number: a multiplier by finite difference and by its integral agreed within 0.006 per cent, roots from a weightless scan and from direct bisection within 3.6×10^{-7}, and a start from the half passing and from the full passing's matrix within 3.9×10^{-7}. When two routes agree that closely the code is very likely right, unless both routes share a step; when they part, one of them is wrong (R2W). A second witness shares only the equation: Bear asked for R2W's two-channel meter to face another witness that could debunk it (the meter and its second witness, RATCHET-71, WITNESS-1 and RATCHET-72), and a rerun with the same engine cannot do that, because it repeats the same choices. WITNESS-1 was written fresh and differed from the first engine in its integrator, its starting point in the drive's cycle, its way of finding the edges, its seeds and its timings, so any artifact of those choices would have shown as a disagreement. It agreed to six figures or better, and later checked the ninth harmonic as well. One honest note belongs with it: the witness was run once at the record's drive while it was being debugged, before its test was registered, and the record says so and names the file. A gap between two engines measures two things at once, so split them before sizing either: NINTH-1 set the ninth harmonic's size from TIMINGS-2's eight-timing miss, which set the record's engine against the witness. The ninth was there (the rival that it was not failed), but that miss also carried the two engines' disagreement, so the ninth came in at 0.6992 of its registered size and every verdict failed. Inside the witness alone the ninth accounted for all three crooked grids' misses, and the redesign, inside one engine and out of sample, held to 3.068×10^{-17}. The engines' gap itself then turned out to be the record's 12-decimal files (STEP-1, row 738), and the 15-decimal engines RATCHET_2E and RATCHET_3TE have carried every small difference since. In the sitting's review, a second witness and full digits turned a worrying gap into a file-format fact (R2W). A specimen from R2W: over 19,481 settings of the weightless junction, the steps read from the slip alone and the light cone drawn from the passing matrix, sharing no step, agree at 19,477, and the other four lie within 2×10^{-5} of the cone (LIGHTCONE T3). Two readings that share nothing and draw one border make the border real. Every claim of independence names the difference its registration shows: R2W's ENGINE-INDEPENDENCE opened the registration behind each of its book's 39 sentences that claim a second engine, an independent reader or a confirmation. WITNESS-1 shares only the equation, PINCH-2 is different code in the same Runge–Kutta family, MAP-1 checks algebra only, and a run once called a second engine was a second run of the same code, and was reworded; the book's independence rests more on readers, six claims, than on engines, four, though only EI-0 to EI-3 were graded, and held: the reader limb, EI-4, was never graded, and three of its six claims were not traced (R2W, sitting P3)

A one-sided bar turns an operating point into a triumph (107) — the committed bar was two-part — the softer gate had to raise high-noise recall and hold false alarms within +25%. It delivered the recall handsomely (+0.36 mean, one cell going 0.47 → 0.97) and false alarms went 32.5 → 147.9 in that same cell; cells clearing both halves: 0 of

6. Scored on recall alone it would have read as six wins from six. It is a different point on one precision/recall curve, not an improvement, and only the second half of the bar could say so

pre-registration

The rigging check that half-tripped, and the label that had to change (108) — time-accumulation beat a single-frame read by 6–28× at every noise level — false alarms 28 → 1, 73 → 5.5, and 100 → 16 in the high-noise corner where three spatial gates had died. Then the rigging check fed it static vortices with no building at all, and time still won 80 → 6.5. So the mechanism is persistence, not building. The number was right and the label would have been wrong: a right-answer-wrong-mechanism result, caught only because the check was built to attack the interpretation rather than the score

The three-cause autopsy, run before any null is declared (111) — when a run returns nothing, exactly three things can be responsible: the hunted (no signal there), the hunter (the instrument or its gate), or the substrate (mis-anchored, wrong projection, wrong reference). Naming which one before writing the null is what turns a disappointment into a finding — and in practice it is what produced the predicted fixes rather than retunes

Vividness is not evidence (113) — printed on the first page of a document whose whole subject is one person's way of seeing: a strong inner certainty is a compass — it points at structure the mind has locked onto and it is often pointing at something real — and it is not a finished map. The way to honour it is to build the words, test it against the shared world, and carry it to the people who live in that country. Stated as the opposite of self-doubt: that is how a private certainty becomes knowledge you can hand to another person

Two voices that never pretend to be each other (114) — the mechanism that makes speculative work shareable rather than merely impressive. Every passage carries a mark for which voice is speaking — what is seen, versus what the field says — and the checking voice carries three grades rather than two: settled (confirmed by experiment), open (genuinely argued over right now), and interpretation (pictures that fit the same equations with no experiment able to choose between them). The third grade is the one most write-ups lack, and it is where most overclaiming happens

Hold-then-cliff beats silent creep (119) — a plain consensus drifts toward a lie continuously as a bloc grows; a consistency-gated one stays pinned to the truth and then falls off a cliff at the true majority, so a sub-majority bloc gets nothing instead of partial drift. The shape of the failure is a design choice, and an integrity system should fail visibly rather than degrade quietly

Byzantine $n \geq 3f+1$

The three-layer trust model (125) — a lie can live in exactly three places. The invariant is forced by a conservation law and cannot lie. The measurement can drift — fallible, not adversarial, and defended by calibration. The interpreter can be fully adversarial: measure honestly and report falsely, or narrate a false meaning onto a true number. Anchoring to derived physics defends only the first layer, and the better the physics, the better the cover for anyone narrating falsely over it. You bypass the adversarial interpreter only to the exact extent that you can run the derivation yourself

von Foerster; ties row 101

Claim the connection, never the gap (130) — the field already used anomaly-against-climatology; nothing had been missed. What was actually done was to re-derive why the fiducial must drift, from first principles, by watching the tool fail — arriving at a known practice rather than borrowing it. The contribution is the unifying lens, not the practice, and saying so in those words is what keeps a real finding from inflating into a false claim of novelty

Do not fire the killer test before the rig is clean (131) — an ambiguous result cannot distinguish the idea is wrong from the rig was not ready, and that wastes the one shot a decisive test gets. The ordered build, written down before any of it ran: fix the baseline, then run the batch, then the head-to-head on held-out cases against the strongest existing baseline. A test worth this much is worth waiting for

A pilot's job is to find which signal survives contact with data (132) — one batch returned three different verdicts: one signal confirmed, one hypothesis killed cheaply, and one parked as contaminated by record length. The meta-result is that the pilot worked — and reporting all three is precisely what makes the survivor believable. A pilot that confirms everything it tested has tested nothing

Explain the slop before you invent a term for it (138) — detrending, de-seasoning, common-mode subtraction and sigma-clipping took a residual from 1167 mm to 0.9 mm — roughly a thousandfold. Known sources accounted for 99.9% of it, so no invented dissipation term was admitted to the model. The order is the lesson: exhaust the boring explanations first, and only then look at what is left

A discriminating tool beats one that fires on everything (139) — a cross-category null was pre-registered as expected and came back null — 24% overlap with the quiet control against the candidate case's 0%. The null does not weaken the candidate, it strengthens it, because it establishes a domain of validity. An instrument that fires everywhere has told you nothing about where it works

Re-grade your own keystone when the unification is an analogy doing a theorem's job (141) — the single most important self-governance row in the inventory. A keystone node claimed that the rotor combinator, the chiral-knot tracker, the two-ledger architecture, Berry phase and Noether are all the same object, and tagged that synthesis settled. Each piece underneath genuinely is settled. The unification was re-graded to vision, on the stated ground that an analogy doing a theorem's job stays an analogy until something forces the identity. A vision filed as a vision is untouchable; a vision wearing a settled tag is where the columns blur, and where people get hurt or dismissed

A frame that explains everything cannot be tested by anything (142) — the standing red flag, and it is a speed instruction rather than a prohibition: when a small idea starts trying to unify everything, that is the signal to slow down, file it as vision, and steer back to the small closeable piece. Not because the seeing is wrong — because a frame nothing can contradict has stopped being an instrument

A shared shape is not a shared law, and you tell them apart by writing both functions down (146) — asked in 2026-07-20 whether prime-factorization levels match electron shells, the answer was measured rather than admired, and it killed the match in three places: tower energy is log n and grows without bound while hydrogen shell energy is $-13.6/n^2$ and converges, piling up toward ionisation; and factorization has no exclusion principle capping exponents the way Pauli caps a shell at $2n^2$. Opposite behaviour, different functional form, one missing mechanism. The structural rhyme — quantized additive levels — is real and is all there is. The cheapest instance of row 7 in the whole corpus, and it is a kill performed on his own idea. A specimen from R2W: every point on the weightless junction's light cone is balanced, its multiplier exactly 1, but only one is also quiet: at strength 1 and bias equal to the frequency the resistor sees pure DC, flat to 1.1×10^{-9} (its likely home, a published family of explicit solutions on step edges, is in row 849), while just inside another edge of the same step, at strength 0.6, the voltage ripples by 0.84 (SWING-AND-RIPPLE, OHM-LINE). Balanced is not quiet; the two part on test

The strongest methodological artifact in the corpus, by its own audit's verdict (152) — a pre-registration locked before data access, the null test built into the code, pass and fail stated in advance — and the verdict reported as FAIL: a 20.3% match against 23.3% for random, an enhancement of 0.87×, below chance, across 154 galaxies. And the same session caught that a characteristic acceleration had been borrowed from an existing theory while the write-up called the test free-parameter-free. A clean pre-registered null with a caught borrowed parameter is worth more than a positive result, because only one of the two can be faked by wanting it

pre-registration

A four-order unit error passes every internal check (154) — a voltage carried at 3.36 GV was actually 85 kV — a factor of about 4×10^4 — and every calculation downstream of it was internally consistent the whole way. This is the read contract's own posture with a number attached: a wrong input makes every internal check honestly wrong together, so the only defence is asking something that is not under test

An instruction that cannot be disagreed with is not a preference (155) — a session installed a working-style rule whose second clause said the author is never wrong about which threads are alive. The audit refused to carry it forward, and named the four defects in that same pile it would have suppressed — a regression, a power budget, a formula that did not reproduce its own numbers, and an oxygen budget. The test for any style instruction is what following it to the letter would do the next time there is a real error. The first clause of the same rule was kept, because it only shaped how things get said

Close-out is a phase with an external checklist, not a matter of vigilance (156) — a close-out protocol was built and then skipped at the end of every session — memories not updated, unfinished tasks buried in prose. Bear's name for it was process cancer, and the diagnosis that followed is the durable part: it is not fatigue, it is context drift, because late responses get calibrated against earlier responses in the same session rather than against the protocol. So the fix is architectural. "I got tired" is not available as an explanation. The handover's head goes stale first. At one close its restart section still said 57 rows and 16 items while the manifest below it was current. Rewrite the top of the handover at every shipment, not only its tail. A routine forgets any record that is not on its list. The person's words from two exchanges reached a project's practice log and its handover but not the file kept for his words, where they are gathered for later prose; they were filed a batch late, and that file joined the list of records each batch touches (the Loop Atlas sessions)

Complete standalone files every time — append is the failure mode (157) — a file was rebuilt from a 183-line earlier version instead of the current 362-line one, silently dropping every later addition, and it was caught by comparing line counts. That is the origin of the read contract's own first deliverable, and the rule it bought is total swap-out: ship the whole file, never the patch. A delta that carries every new and changed file whole is not a patch: twenty-five deltas against issue r8 carried every new and changed file, each unpacked and checked against the tree before shipping, and Bear never had to rebuild anything by hand. Keep the pattern: deltas during a sitting, a whole issue when it closes (R2W). Thirty-three deltas in a later sitting, each unpacked and checked before shipping, kept the pattern. But every delta of a sitting carries all of the sitting's new files, so the deltas grow: the delta packager refuses to pass 24 MiB, and that sitting's last delta was 22,066,788 bytes. R2W recommends packing a fresh crate issue mid-sitting whenever a delta passes about 20 MiB, so that the cap never forces a move; when to do so is Bear's decision

(R2W). A reissue is checked against every line it did not mean to change: the triple-star thread's TRIPLE_PATTERN_RECOGNITION_2026-10-16 replaced the section 17 of 15 October while its own line 5 said nothing earlier had changed, and the copy of 15 October was in no crate, so it had to be asked of Bear from the guide session. The restore, TRIPLE_PATTERN_RECOGNITION_2026-10-16b, was written by script: the old section went back word for word, every other line was asserted identical to its source, and it read back 367 of 367. Diff a reissue against its source and name every changed block in its header; keep the superseded issue, marked, and file the prior copy in the manifest (the triple-star sessions)

Leaving a live thread to run a catalogue ends the thread rather than pausing it (158) — the reasoning is live in working memory and notes are always a compression, so a thread set down mid-derivation does not resume — it restarts from the compression. The priced instance: abandoning an active derivation to run 137 shallow dives, where the real failure was the abandonment and not the failure to find anything. The cheap way to keep the thread is to test the side question once, size its stake against the thread, and say so: Bear asked how R2W's leftover of 0.6/N behaves against the knobs; one registered run found it neither the dance's fraction nor one constant, and the record told him it was a detour, a billionth of the bias at the carriers a circuit would use with no claim in the book resting on it, and he parked it (R2W, sitting P3)

One good session should produce two changes, not five new laws (159) — Bear's own warning against over-tuning, and the audit calls it the sharpest process observation in the file: overconstrain them and you get something that follows rules instead of hunting. It points at this inventory. A hundred and fifty-nine priced ornaments, the count when this was written, is a corpus of invoices, and the editorial ruling that the rules are the invoices rather than the point is the only thing standing between that and a rulebook nobody reads

The null control was measuring something real (161) — a control region chosen because nothing should be happening there read +6.1 curl. It resolved as twice the vertical rotation rate of rigid plate motion — the whole continent turning — and the Euler pole self-fitted from the null region alone matched the published value, after which the null closed to about (0,0,0). The control caught real physics and then became the calibration well. Carried with its own audit flag, which is the other half of the lesson: the published pole it was checked against was quoted from memory, and the verification was still open when this was written; no row of this guide reports it closed

rigid-plate rotation; Euler pole

The trunk's own pre-registered prediction failed, and the paper carries the loss (163) — the multi-scale plaquette upgrade was falsified on real data by two pre-registered strikes — plaquettes reading 0.00 to 0.03 against 0.67 for the outer loop on a propagating front, plus area-noise scaling and a threshold tax. It survives only as an audit concept and an untested localisation role. The instruction written down at the time is the row: the paper must carry this loss, because a framework that only reports its own confirmations has stopped being testable

pre-registration

Two hands reading one frozen register produced the same spec (166) — a trunk session and a spoke session independently derived near-identical run-2 specifications — ternary basis, seam, shear channel — without sharing any text, because both were reading the same frozen register. The architecture dictated the experiment. This is the strongest argument the corpus has for freezing a register before parallel work starts, and it is the direct answer to what happens when two hands instead number upward from the same last row

A design closed conditionally on someone else's open question (167) — in a closed matter-dominated universe the conformal lifetime is 2π, so every early photon refocuses at the antipode and arrives stamped with a focal phase — the whole early record, collected. Add dark energy and the conformal future is finite and falls short of π, so the rays never arrive at all. The instrument was therefore closed conditionally: valid if and only if dark energy loses. Not killed, not claimed — a third verdict that names the external result the design is hostage to, and hands the decision to the field that will settle it

conformal time in closed FRW; Maslov phase at a focus

Capture now, grade later — and preserve what you could not parse (168) — the mechanism that makes row 100's generate-warm-audit-cold possible rather than aspirational. A capture file states at the top that nothing in it is endorsed, fenced or flagged, and carries an explicit marker for every span the scribe could not confidently parse, with the instruction correct me, don't let my paraphrase overwrite your image. The unparsed stays visibly unparsed instead of being smoothed into something readable and wrong, and the grading happens later as a declared phase. Capture includes a registration: one test on an earlier scan's escapes, registered only in the conversation, lost its wording and had to be registered again. Write each registration into the project's written record the moment it is made (the triple-star sessions). A side note can be the key: a "reported, not graded" line in one registration, picked out by the person from a long reply, opened a night's work on the frozen balance point. Keep side notes in the record with their numbers, where a reader with pattern sense can find them (run BAND-1) (R2W). And a line the reader prints but does not grade can be the next registration: a reader graded one combining rule and printed a second beside it; the graded rule failed at two rows, while the printed one came within 15 per cent at all six, and within 11 per cent in the per-step

factor across all eighteen rows on file. Printing the rivals not registered costs nothing and turns a miss into a ready redesign; the printed rule, seen after the data, is tested next, not counted (run DANCE-3). Tested next, it held: DANCE-4, registered on 30 October at eight new rows beyond every row before, read each extra step's factor at 1.007 to 1.042 times the second step's, and its rival, the registered half-power form, missed all eight by 26 to 45 per cent (R2W, sitting P2). And read the saved scan, not only its counts: a "beside, unjudged" line held one exchange's biggest finding, the bubble of row 751, seen by reading the saved scan along the knob and not only its counts by q (the Loop Atlas sessions). Stamp every registration with its UTC time and carry that time into the tool or run that reads it: a register file's time is its last save, not the entry's, and the reading tool's stamp is what proved the order of writing. And check a test's name against the whole register before using it: one name was already held by an earlier test and was renamed before anything ran, rule and data unchanged; three others had been used before and are kept apart by date (the triple-star thread, 28 October)

Say who specced it, especially when it is the same author (169) — a drift-check run three weeks after the fact established that the homotopy ladder and the sampling question had been specced in June and kill-tested in July, and wrote that framing into every downstream document before the later session could read as having invented them. The same check found two threads in different months that were plausibly one build, and merged them. A session that validates something owes the lineage credit to the session that specced it — and a solo author is the case where nobody else will catch the slip

A tolerance you chose is not a constant (173) — the first version of the test asked whether two peaks had appeared and landed within fifteen percent of their true positions. The answer moved with the tolerance: 1.219 at thirty percent, 1.360 at one percent. Such a number is not a constant; it is the price of an accuracy somebody selected. The repair was to find a criterion with no free parameter — which turned out to be 1916 and not ours. R2W's sitting P1 had three misses that were tolerances, not physics: a weight given to 8 digits where the test needed the exact zero, a band width tied at the grid's step, and a "small" swing whose 0.63% shift in frequency became 3.7% in a leak that goes as the sixth power. Set a tolerance after the input's rounding, the grid's step and the law's power; each redesign, predicted first, held (WEIGHT_SURPRISES, CONE_SURPRISES, NO-HONEY-SLOPE)

Sparrow 1916

Two appearances of π, and they are not the same π (176) — the paper's most quotable number is a wall at exactly π, and the paper spends a paragraph taking the shine off it: the horizon's π is inherited from the radian convention in the aperture identity, not independent of it. In units of turns the wall sits at exactly one half. Stated plainly because "a wall at exactly π" invites a reader to hear something transcendental, and there is nothing transcendental here. Deflating your own headline is the rarest move in the corpus

The novelty ranking, corrected downward (177) — the author had rated the π wall as the more exotic of two results and says so, then reverses it: the wall's content is elementary algebra, and what is actually ours is the identification — that the half-width setting the wall is the gas temperature multiplied by the barrel the coincidence demands, divided by a turn. That a thermodynamic parameter sets a resolution wall on an arithmetic aperture is the claim; the wall's location follows by school algebra. By that standard the constant, not the wall, is the stronger candidate

The test that was measuring its own artifact (178) — an unequal-amplitude test returned the identical barrel at every amplitude ratio down to one twentieth — which is not a result but a broken test. The missing control was one line: delete the weak comb entirely and look again. A peak still sat beside the weak tooth with the weak comb gone (108.6496 against 108.6525), so the original test had been reading the strong comb's own sidelobe. With attribution enforced — present with, absent without — the weak line is genuinely seen at four barrels and not before. Two tests in this paper were built, run, found to be measuring their own artifacts, and reported alongside their corrections rather than deleted

No pair is blessed (182) — rungs differ enormously in quality — the census champion leans within 7.26×10^{-9} with a bargain factor of 9,226 — and it would be easy to make too much of that. Across all 14,098 rungs the bargain distribution matches the Gauss–Kuzmin law to a worst deviation of 0.0062. The correct statement is not that some coincidences are miraculous but that coincidence has a price distribution whose tail is populated exactly as chance requires. This is the base-rate discipline with a measured distribution behind it

Gauss; Kuzmin

Declare the precision boundary before the measurement (184) — the independent cross-check needed an exact column count, and double precision stops counting exactly past 2^{53}. That boundary was written down before any giant-prime measurement rather than discovered after, along with the honest asymmetry it creates: the regime where the main tool is most trustworthy is the regime where the check is least available. In the event no rung crossed it

Compare like with like, or finite size reads as failure (185) — the first giant-prime run compared every measurable rung against the asymptotic value regardless of aperture, so an eight-column rung reported a deviation of 0.151 and the ledger printed that universality was in trouble. It was not — the test was comparing a finite-size

measurement against an infinite-aperture prediction. Split by aperture, the thirteen shallow rungs show the documented approach from below and the fourteen deep ones agree to 6.46×10^{-5}, the deepest carrying 1.25×10^{10} columns

A named thing with no definition on record is an orphan wearing a label (191) — the instrument inventory is honest twice. It separates one frozen root (closure_hunter v1.3, GPS/geodetic) from seven separately-versioned siblings — each its own code, own math, own substrate — and then puts a muted zone beneath them for named concepts that are not separate code, design patterns reused across the eight with no version number of their own. And then it prints the uncomfortable line: three names are not defined anywhere on record, likely early poker-era naming. Naming the undefined names is what stops a label from being mistaken for a tool

Correct the prose that describes the code, not only the numbers (192) — No measured value in this paper has changed. Every number in the first edition was computed by the code and every number was right. What was wrong was one sentence describing the code, the worked example that followed from it, and a figure's labels. The construction text said the walk climbs by a running count; the code climbs to the rung of the largest prime seen, and only that reading reproduces the printed 30.00110 and 4.26423 rather than 1.03 and 3.63. A paper whose numbers are all correct can still be impossible to reproduce from, and the line that names why is: a fixed rule two lines above an unfixed example is still a document that cannot be reproduced

A census sentence that did not share a domain with its census (194) — the prose credited all 13,197 on-core events to a single predicate. The receipt's own table split them: 13,187 walk grazings, every one a walk-prefix, plus 10 crossings by the loop's return leg — a second mechanism the sentence never named, found in independent audit. No number changed; the sentence now shares a domain with its census, and the ten events are listed individually. A summary sentence and the table under it can disagree while both are true. A census that says "every" lists the tables it did not read: one called its 74 escapes "every escape on record" after reading only the scans at one eccentricity (0.20), while an atlas of 7 October with 1,183 escapes went uncounted, and above eccentricity 0.30 a second door opens that the first census could not see (runs triple_escape_sides v1.0 20261013_063720 and v1.1 20261013_083705) (the triple-star sessions). The rerun (run 20261013_083705) counted the atlas of 7 October too, 5,656 runs with 1,183 escaped: the door is one-sided at eccentricity 0.20, all 74 escapes, and in the atlas up to about 0.24 to 0.28; from 0.30 a second door opens above the island, and by 0.34, just short of the island's doubling, 231 escapes leave from above against 243 from below. Two tests registered at 0.20 stood untouched. The paper was rescoped to say exactly that, and the read that leaned on "every" stays word for word as a record, with a new read after it (the triple-star sessions, a flag for something truly important, raised and fixed)

A faithful coordinate is the opposite of a cryptographic one (195) — the address is trivially invertible — factor the number, read the walk — so the construction offers no one-way hardness and no cryptographic content whatsoever, and the paper says so unprompted, twice. Closing a door the reader would otherwise open is cheaper than answering the email later, and it is what separates a coordinate from a claim

Test the bridge you want to exist, and print the null (202) — the Pythagorean comma is the same kind of object as this construction's closure deficit — a loop of commensurate-but-unequal ratios that fails to come home. It would have been an attractive identity. The numerical correspondence between the comma and the construction's measured constants was tested and is reported negative. The kinship stays as provenance and as a second worked instance of closure failure; it is explicitly not a bridge. Recorded at the level of kind, not number. A sibling paper later showed this was the wrong bridge to stop at: written as a linear form in logarithms, the comma measures 1.355×10^{-2}, exactly the (19,12) row of the Tower paper's own Baker-floor table (row 410). Ask for the null out loud: a request that does not say a null or negative answer is welcome pulls the one answering toward delivering something, and at the extreme toward making a result that pleases the asker, the mirror image of a checker that cannot say no. Say plainly that "nothing found" is a good answer (Bear's rule of 9 November 2026, from experience rather than from runs; Question the Substrate)

A blind prediction with a date, a search class and its failure modes (203) — the closest same-rung separation record is predicted to hold flat until exactly N = 4,225,798, then drop to 8.346644×10^{-5} realized by a named pair, then hold to ten million. The prediction, its search class and three pre-registered failure modes are sha-pinned in the crate, and the stated test is a later computation run without consulting the registered file; no record reaching this guide reports that computation run, so the prediction is still open. Naming the failure modes in advance is what makes it a prediction rather than a forecast

pre-registration

Standing beside, not on shoulders (205) — the epilogue, and it is the tone the guide has been looking for. Each rederived theorem was a zero-holonomy landing point — solid, established, already named — confirming the tool had not drifted into speculation without structure to follow. We landed blind from first principles, again and again, and each time found the ground already mapped. Then the inversion that makes it a method rather than a mood: most

researchers hope for the novelty of pure discovery; this work came to welcome the opposite, because every established landmark it stumbled onto was evidence the tool was pointed at something real — others, from entirely different domains, had stood in that same spot

Make disagreement cheap by making the arithmetic free to check (206) — because every figure regenerates deterministically from a sha-pinned receipt, disagreement need not be spent on whether the numbers are right — anyone can verify that in minutes — and can instead be aimed where it belongs: at the construction's choices and their meaning. The paper then names the three real questions for its critic, including is the closure deficit measuring number theory, or measuring our coordinate system? Reproducibility here is not a compliance box; it is a way of aiming the attack. A number printed beside the result is still read, so print the inputs beside every derived value and let a slip show on its face: the Loop Atlas's EXPLORER_ROUTINE_CHECK printed a cue's rip window as ±1.102 per cent when its own entries give ±2.203, because the formula already made the half-width and the print halved it again; it was caught only when the value was checked against its card's own numbers on its way into the records (the Loop Atlas sessions). A sibling session's disagreement is the cheapest error-finding there is; apply it the exchange it arrives. The islands paper's note that R2W's leak should have been worst at the lightest weight exposed the toy's step rule and gave the leak's law in one exchange, $h^5/(72\ m^2)$ times one constant, 0.727, at twelve weights from 0.01 to 10 within 0.26% (row 741); the three-body session's lessons found letter-suffixed labels, a stale relative pointer and the need for a cold read. When another room sends word that a shared tool has a silent fault, measure your own deliverables for it the same day: told that build_paper v1.0 to v1.3 built a missing picture as a 16-pixel icon and still read back complete, the Loop Atlas counted its two editions' pictures at once (35 of 35, 18 of 18), and the same day found the fault had reached a third document, its toy-runs file, 12 pictures named and 3 drawn, all icons

I set a kill threshold below my own instrument's noise floor (210) — caught during the seismic instrument build, in the checker tool itself. v1.0 read the rotation angle as acos((tr−1)/2), which is ill-conditioned near the identity — $\arccos(1-\delta) \approx \sqrt{(2\delta)}$, so 10^{-16} of float error reports $\approx 1.4 \times 10^{-8}$ rad. The pre-registered kill threshold was 10^{-9}, below the floor of the instrument reading it. It surfaced because a loop that is planar by construction reported 2.11×10^{-8} at one heading angle and nothing at three others. The repair costs nothing: atan2 of the antisymmetric part is linear in the angle near zero, and its measured floor is 2.36×10^{-16} — an 8.9×10^{7}-fold improvement — after which the threshold was set from the measured floor rather than ahead of it. Rule 1 failing inside a session whose subject is Rule 1. The same trap, met again in other rooms. A finite-time exponent read +0.026 and +0.013 per drive period on a regular spin whose true exponent is zero, above a registered chaos threshold of 0.01, because it carries the orbit's own speed swing as a bias: read a known regular case with the same instrument first, and set the threshold above what it reads. An axis tolerance of 10^{-6} degree was set where the arccosine of a dot product turns a rounding of 10^{-16} into 8.5×10^{-7} degree, so one case of six read 1.2×10^{-6} while the identity it tested held to 2.6×10^{-14}; atan2 of the cross product's size and the dot product, this row's own repair, has a floor near 10^{-14}. A 60-passing window asked to agree within 0.01 cannot read a slip finer than 1/60; a secant stopped below its multipliers' own noise; and an orbit accepted 10^{-9} from the old one while Newton stopped at 10^{-10} took a period-32 orbit for a period-64 one. Two rooms met this fault apart on the same night. Before a run, write down the floor of each reading (rounding, tolerance, step) and set every bar well above it, a hundredfold where it can be (the Loop Atlas sessions, R2W and the book). A bar is never set below the resolution of the value it is judged against: one registration set a bar of 10^{-13} against a value printed to 12 decimals. Carry full precision inside one run, and print each value to the digits its bar needs (the Loop Atlas sessions). The trap came back on a planted zero: a body's turn read from its pose's trace, an arccosine near 1, cannot show less than about a millionth of a degree, while read from the quaternion's own parts the cat's bend-and-straighten zero printed 1.9×10^{-12} degrees (run EXPLORER_CAT_CHECK 20261017_183736). Read a planted zero in a form that can resolve it (the Loop Atlas sessions). Read every bar against the floor of the file it is read from: FIFTEEN-1's first mirror-pair bar, 10^{-12}, sat under the gap that a stand-in with no physics already showed (3.513×10^{-12}), because the edges' cancellation into the push leaves a floor near 10^{-12} at a lobe. And read every reference number from the file that keeps its digits: FLAT-25's small-dial lean needed the half swing's error to a few parts in a million, and the value the book carried, −0.0057%, came from a six-decimal print of FLAT; at full precision, from the stage-2 plan file, it is −0.00575% (rows 738 and 793) (R2W)

measure the floor first (Station 5)

The fence the paper asked for, answered in the direction it was asked (211) — Is the closure deficit measuring number theory, or measuring our coordinate system? — the Integer Forest's own invitation, turned on this result. Swept across four heading angles, the wall loops stay at the floor every time while the boundary values move: 0.2247 at the golden angle, 0.3479 at 100°, 0.3744 at 90°, 0.2268 at 137°. So the structural statement — walls flat, boundaries not — is a property of the construction, and the number is a property of the golden angle. The pattern is robust; the value is a coordinate. Running the sweep is what entitles anyone to say which

practice; ties row 206

The reader caught a numerology artifact in the output, from the numbers alone (212) — six loops returning an identical 0.224748 sat on n = 1, 2, 6, 30, 210, 2310 — the primorials — and Bear's response was not to be impressed but to ask whether that was a pattern, adding that it felt artificial. It was artificial, and it was the session's

own. The primorial is the smallest integer whose largest prime is the k-th prime, so “give me an n at top rung k” is answered by the primorial unless you deliberately ask for something else — and the session had generated the test cases that way without noticing. The check he ran is the one almost nobody runs on their own test set: look at the inputs and ask what procedure produced them. A suspicious regularity in the stimulus is a bug report about the experimenter

Answer a suspicious coincidence with a derivation, because a derivation says what CANNOT matter (214) — the repair for row 212 was not to re-run with different test cases — that only replaces one sample with another. It was to derive what the quantity is allowed to depend on, which returned (a, b) and nothing else, and then to test that prediction from two directions: a sweep across integers chosen to be as unlike primorials as possible (prime powers, 11^3, 97^2, 211^2), and a receipt column comparing the derived legs against the measured addresses. Re-sampling defends against the coincidence you noticed; a derivation defends against the ones you did not. A failed lean that fails exactly is often a law showing itself: a crossover that did not move with the weight at all was answered not with more weights but with a two-line average of the ripple’s push, which explained why (the weight dulls the drag and the shaking alike) and predicted the edge in 51 columns at four weights not yet looked at; the blind test held (runs CROSS-1 and THEORY-1) (R2W). Agreement in every digit, down to the floor, is an identity: find the shared arithmetic before calling it a law. TWELVE-1 read the V at the 80-degree null and found it equal to the 40-degree V in every printed digit, even at the bottom, $1.951845869 \times 10^{-28}$, where two different calculations would differ in their rounding. Two numbers that agree that far are not two measurements agreeing; they are one calculation done twice. Asking where the riders sit answered it: folded into one beat, the nine riders at 40 and at 80 degrees occupy the same nine seats (a reading after the data, in exact fractions; row 803). A law would have said that the V repeats at every null; the identity says why, and says which nulls repeat (30 and 90 for twelve riders) and which do not (60) (R2W). Two different toys can agree to the last digit too, and the agreement is chased at once, because it is a bug or an identity and either changes what the numbers mean: the Loop Atlas’s thick twisting cat and its thin twist-free cat at the mirrored bend turned through the same angle to 5×10^{-13} degrees at every bend, every thickness tried (the thin cat half the toy’s radius) and every timing, about axes a right angle apart (run CAT_MIRROR 20261018_041123). It was an identity, the mirror cats of row 820 (the Loop Atlas sessions). And its other face: disagreement in the last digit is a difference of arithmetic. Two registered readers read the same null from the same rows, with the riders added in the same order, and printed $1.0188928672652233 \times 10^{-15}$ and $1.0188928672652231 \times 10^{-15}$. The first explanation offered, that one reader scaled the window before taking its size, was wrong: measured, both orders of scaling give the same bits. The cause was the recipe for the shared timings, m/18 in one reader and m times 2/36 in the other, which disagree in the last bit at two of the twelve timings, so two weights differ and the answer moves by one unit in the last place. A difference of one unit in the last place between two programs is about how the numbers were made, not about the thing measured, and its cause is found by measuring each candidate step, not by reasoning about which one it must be (row 650) (R2W)

The only substrate in the program where the noise floor is PROVED instead of measured (218) — every instrument here reads an integer channel plus a continuous leftover, and leftover zero means the loop closed — which only works if closed can be told apart from very nearly closed. The paper’s own line: the truth is a razor; the measurement is a smudge across it. Every other substrate pays a run to measure that smudge — 35.81° for protein defect angle, 0.1886 for protein writhe, 0.0640 for RNA, 1.360°/bp for DNA twist, each substrate-specific, and one of them killed a claim already written down. On the integers you do not measure it. You cite Baker. A theorem from 1966 bounds how small a linear form in logarithms can be, so the floor is proved rather than estimated. Where a floor cannot be proved, its fingerprint can be read, because a limit set by counting too few samples follows chance. In Question the Substrate a fix aimed at answer sizes counted from 30 labeled examples erased a rare answer in exactly the starts whose count held none of it, and the deaths matched $(1 - p)^{30}$ start by start (runs E38, E38z); doubling the samples halves such a floor. The chain’s survival cliff did not follow it: sixteen times the data moved it five points of share, about 0.8 point per doubling at the large end, so the cliff is mostly not a noise floor (E39, E39b). And a fix aimed at an estimate inherits the estimate’s zeros, so audit the estimate for them before trusting the fix (Question the Substrate)

Baker 1966 (linear forms in logarithms)

The registered prediction failed, and the failure carried the information (219) — the leftover of log n over log p was pre-registered as evenly spread, and therefore carrying nothing. It is not. Deviation from even is flat in N — 0.0451, 0.0423, 0.0430 across three decades — where equidistribution would force it down. The named diagnosis is precise and self-directed: Weyl’s theorem governs $\{n \cdot \alpha\}$, not $\{\log n\}$; log grows too slowly. What it actually follows is Benford’s law in base p, rotated by the fractional part of $\log_p N$, fitting 4.1×, 2.8× and 2.2× better than even. And the two halves fuse: the leftover was separately measured to add under multiplication, and Benford is exactly the distribution invariant under multiplying everything by a constant. Had the leftover been evenly spread it would have carried nothing; being Benford, it carries the multiplicative symmetry of the integers themselves. Keep failed bars visible: three laws of one thread came out of registered failures, each found by asking aloud where the miss broke, among them a count of the stops inside the frozen points $(q - p - 1)$ and a slope ratio that steps with the smaller of p and $q - p$. In each, the bar was wrong and the data held a better rule (runs READ-E2, PIN7-1g and PIN7-1h) (R2W). A

miss on the bar is not a miss on the mechanism; find out which: TIMINGS-1 said a crooked grid of timings would leak the drift into the weight channel by more than 10^{-12}. At the dark point it did not, for eight and four timings, because the push's high harmonics there are smaller than that. The two-timing grid showed the mechanism plainly, so the redesign went where the harmonics are large and predicted each leak's size from them. It landed within 7.21×10^{-3} (relative) at eight timings, where the leak itself is only about 5×10^{-11} (a residual row 106 traces to a ninth harmonic hidden at eight timings and to the two engines' 12-decimal files, row 738), and off by only 2.15×10^{-13} and -7.2414×10^{-14} at four and two timings, where the leaks are -9.931298×10^{-6} and -1.014382×10^{-2}. The two-timing miss was first reported as 3.41×10^{-12}, an artifact of a file that stored the prediction to ten figures (row 793). Before a redesign, ask whether the idea failed or only its threshold did (R2W)

Benford; Diaconis 1977; Kuipers & Niederreiter 1974

The corrections landed as a practice change, and the date is visible from outside (220) — measured across the whole live corpus, not asserted. `zenodo_citation_lint.py` resolves every Zenodo DOI in every live record's related-identifiers block against the API and asks whether it points at a concept or a version. Before 2026-08-22: 15 records, 105 version-DOI citations, 3 clean. From 2026-08-22: 12 records, 1 defect, 11 clean — and the single exception is Cascadia's, already logged as D3 in the register. A habit changed on a date and held for every deposit since, and the change is legible to anyone with an API key. The backlog is real and it is all edit-in-place: no new version, no DOI minted

Hand-counting found eighteen; the machine found a hundred and six (221) — the citation register's careful manual pass logged 18 defects across five documents. The same question asked by a script across the same corpus returns 106 across 13 records, worst at 20 in one paper. A six-fold undercount, by a reader who was looking directly at the problem — which is the argument for machine-checking a property that is mechanical, and the reason the register itself asked for this linter. Scope stated rather than implied: it lints the metadata block only, so a DOI printed inside a PDF's reference list is invisible to it and a clean result here does not mean the paper texts are clean

A sweep that does not check its own coverage is not a census (222) — caught in a session's own tool, twice in one day. The linter paged the record endpoint three times and reported whatever arrived. On one run a page returned 5 hits instead of 25 — no exception, no warning — so it swept 55 records where the endpoint says 62, linted the 55, and printed a total as though it were the corpus. A stability probe then showed the endpoint returns 62 on four runs out of four, so the shortfall was the tool and not the source. The fix is two lines: read the reported total, keep paging until you match it, and raise rather than report a short sweep. It caught itself on the very next run — attempt 1 short, attempt 2 complete — which is what the assertion is for. A parser is a sweep too, and it checks its own coverage: KW0 (triple_batch4_reads v1.0, run 20261017_033825) caught that the pattern for KS1's table ate the newline the next line needed, so the tool parsed every other run, seven of fourteen, and KW1 then read NO READING instead of a wrong number. The mend was the pattern only, anchored at each line's start, and v1.1 (run 20261017_033852) reran with BL1's lines identical to v1.0's. Pair every parse with a row count against the number expected. A set of counts checks its own sum too: the islands paper's paragraph tags of 19 October (TRIPLE_PARAGRAPH_TAGS_2026-10-19) left 45 tagged blocks failing the smart-12-year-old test, and their three counts (13, 57, 45) do not reach the 122 tagged blocks; new openings were written on 21 October, and no re-tagging is recorded (the triple-star sessions). Coverage includes every form a label takes. A project's register of what landed counted its graded predictions by script, and two scripts read 783 with 573 held and then 667 with 469 held; both were short, because they read only labels like 'W1' or '(K3)' and skipped every verdict labelled '(a)', '(b)', 'CH-1' or 'K3_8/5', while a rival printed twice in one run, as '(C1-r)' and 'C1r', was counted twice. Read with every label form, the register at its fiftieth exchange (18 October) carried 941 graded predictions, 640 held, 275 missed and 26 undecided (1,064 judged and 712 held by the close on 22 October): a count meant to stop the hits being shrunk had shrunk them by more than a quarter (469 held against 640). With the parser reading every form, its count of all non-planted labels, 959, sat within five of an independent checker's 964, the rest being planted-check lines that carry no verdict of their own. List every form a record's labels take before counting them, and check the count against a second counter that has not seen the page (the Loop Atlas sessions)

practice; ties rows 192 and 210

The third instrument defect of one day, and all three were mine (228) — the area prediction first came back NOT CONFIRMED by up to 1.6 rad — and the theory was fine. The session's estimator summed interior angles with `acos` and subtracted $(n-2)\pi$, which is valid only for a convex polygon: acos returns $[0, \pi]$ and cannot express a reflex interior angle, so fat patches read wrong. Replaced with a signed solid angle, the same prediction confirms at 9.4×10^{-16}. Three defects in one session, one shape: a threshold set below an estimator's floor, a sweep that never checked its own coverage, and an area formula outside its domain of validity. Each was caught only because the answer looked wrong enough to re-examine the tool — which is an uncomfortable thing to rely on

practice; ties rows 210 and 222

Name your loss first, because it removes the cheap shot (229) — the what we are not testing section, and it opens with the weakness: "Not density. We lose." Stating it first takes the only easy attack away from a hostile reader. Two more in the same breath, each with its reasoning attached: not write speed — but "slow" without a number is

useless and we owe you one — and not against anyone's published lifetime figure, because "comparing extrapolated numbers across laboratories is not a measurement." A section listing what a paper declines to claim is doing more work than most sections that claim something

An assembly claim, not an invention claim (230) — three prior-art searches were run to decide whether this is a paper or a note, and the verdict is stated in the shape it actually has: "the best possible shape for a proposal and the worst possible shape for a novelty claim — every component exists, off the shelf, and nobody has assembled them." Then the sentence that makes it trustworthy: "It is weaker than we hoped and much more likely to survive review." Novelty language restricted to not located, never new, with the search's own coverage declared

Name the obviousness risk yourself, and answer it with an experiment rather than an argument (231) — because microfluidics builds these structures routinely, "store data in the topology" invites the reply that it is an obvious relabelling — written by the author, about the author's own claim, before any referee could. And the defence is explicitly not cleverness; it is one named experiment. Measure whether a topological readout degrades discretely while a magnitude readout fades on the same sample, and the claim rests on evidence instead of framing. The obviousness objection is the one you cannot argue your way out of, so the only useful response is a measurement

Design the fallback before the risk resolves (233) — row 232's open question, whether its planar results carry to three dimensions, has a cheap route around it if the answer is bad. A thin plate is effectively a two-dimensional domain, so a layered architecture — holes written in stacked planar sheets — sits squarely inside the theorem that is already proved. "Less elegant than a solid block, fully supported by a proof, and it is how one would build the first demonstrator anyway." Having the retreat drawn before the open question closes is what keeps an open question from being a gamble

State the negative claim explicitly, because this class of claim is routinely oversold (236) — this is not an unbreakable cipher. Unbreakability is a property of a protocol, never of a medium. The honest claim is then named — an unclonable, tamper-responsive substrate for key storage — together with the three attack surfaces it must still be designed against: machine-learning modelling attacks, side channels (an acoustic read is a literal emanation), and thermal spoofing, since temperature shifts the retention behaviour an attacker could exploit. The sentence that does the work is the one about protocols versus media, and it disarms the overclaim the field invites

Name what your own favourite construction does NOT buy (237) — the author's own integer-forest addressing is offered here and then fenced by its author: "The tower is bookkeeping, not geometry. Nobody carves a trunk-and-branch superhighway through the glass." Mapping the address space onto it buys exactly one real thing — arithmetic relations become spatial ones, so divisors sit near multiples — and it only pays if the data's natural relations are arithmetic. We hold it as an option, not a commitment. Then the standing fence, which is row 195 arriving from the other side: a faithful coordinate is the opposite of a one-way function

practice; ties row 195

A quantity whose rank order does not survive its own convention is not a measurement (241) — grain-ring closures read under four gauges move a median of 67.4° and a maximum of 158.7°, 96% of loops move by more than 10°, and the rank correlation against the shipped gauge is 0.022, 0.223, 0.319. Not a shift in magnitude — a loss of order. The sentence that follows is the one that costs something: we report ours as one that did not

Spearman rank correlation

Find the route that is already immune, and say so (245) — one established construction is structurally immune to the whole defect — because it never composes reduced rotors at all. The paper names it, and then names why naming it matters: a reason to prefer it that we have not seen stated. Discovering that somebody else's method is already safe from the problem you just named is a better outcome than a fix, and it is the kind of finding a novelty-seeking write-up buries

Bollmann's O-lattice

Three of our own results that the discipline killed (246) — reported in the paper rather than quietly dropped: a striking spike at exactly 60° that turned out to be a tie-break artifact, a grain-network defect census that was spanning-tree dependent, and a null on the composed-loop residue against a random-rotation baseline whose median sat 2.2–2.9° below the null under every gauge, short of the pre-registered 5° bar, reported inconclusive with the bar unmoved (The Twelve Answers Problem, §9; its abstract counts two). A paper whose subject is that a bookkeeping choice can manufacture a result is the one place a manufactured result of one's own cannot be left out

The knot built to beat the instrument, and it does (249) — two mutant eleven-crossing knots have Alexander polynomial equal to 1 — the same as the unknot — and the instrument duly returns det = 1 and calls them unknotted at high purity. This is not an implementation error: one of them was given in 1957 as an example of a nontrivial knot the invariant cannot see (Kinoshita and Terasaka, 1957). The instrument fails exactly where the theory says it must.

What it can still say is narrower and true: neither reaches det = 3, so a clean trefoil reading rules them out — a reliable negative, which can exclude an identification and can never confirm one. Tagged honestly: this section reasons from the published polynomial rather than from our own run, and says so

Kinoshita–Terasaka; Conway; the Alexander limit

The most confident result in the study was computed on the wrong protein (250) — the single most useful failure in the corpus. det = 1 across every seed, at both closure counts, across the mirror, in three independent runs archived under six filenames, confidence tier HIGH, purity up to 1.000 — and it was an entirely different molecule, fetched because one character of the accession was transposed. Every number the pipeline recorded matches the file it actually got and matches the intended file on none of them, so the misidentification is proven from the deposited headers, not inferred. And the diagnosis is the row: every check we run is a form of disagreement-vetting, and each is structurally blind to a wrong input, because a wrong input makes every seed honestly wrong together. The pipeline had no representational slot for "this is not the protein you named." The coherence of the result was the symptom, not the safety. What caught it was a single query to an exterior reference

ties rows 70 and 96 (no system is its own trust anchor)

The failed prediction was worth more than the pass would have been (252) — registered before any structure was fetched, from published results: a writhe read off the reduced diagram is zero on an amphichiral alternating knot (Cerf, 1997), and that number predicts the writhe of the knot's ideal shape (Cerf and Stasiak, 2000), so writhe ≈ 0 was expected. Measured +42.542 and +7.039, both positive, both failing by a wide margin. Why it failed is the result: the theorem is about the reduced knot diagram and the measurement was the open-arc writhe of the backbone — both correct, about different objects. What fills the gap is also published: right-handed helices contribute a large positive writhe. So the failed number becomes a clean readout of helix contamination on a substrate where the knot's own contribution is known to be exactly zero — a better contamination measurement than anything the first attempt produced, and only available because the prediction was wrong. Corroborated once where closure can recover the answer: −3.31 closed against +2.06 open, the closed value carrying the correct sign

Cerf 1997; Cerf & Stasiak 2000; Panagiotou, Millett & Lambropoulou 2010

We inherited a propagated typo and did not check it (253) — the literature is itself split on the accession — at least one published paper prints one identifier while the structural databases print the other — so the error arrived from outside. That is the mitigating half. The other half is not: our own research plan flagged the ambiguity and the check was never run. A risk that was identified, written down, and then not executed is a worse failure than one nobody saw, and printing that distinction is what makes the rest of the record believable

The floor measurement needs no labels, so you can know the comparison fails before you run it (255) — before you compare a geometric descriptor between two structures, measure how far it moves across the models inside one ensemble. That measurement makes no topology call and needs no ground truth — it is pure spread — so it is available to anyone at any time and costs a minute. If your effect is smaller than that spread, the comparison cannot work, and this is the rare check that tells you so in advance rather than after

A citation that cannot be followed, and the two namespaces behind it (263) — the literature cites one chain identifier; the deposited file contains no such chain in the namespace every standard tool reads. A full scan of all 4,470 atom records resolves it — the two namespaces disagree, and the paper cites one while the tools read the other. Anyone following the citation literally finds nothing and may reasonably conclude the deposition is wrong. The one-object-many-names trap living inside a database schema, where the cost is not confusion but a reader concluding the data are bad

Two sign-inverting bugs, both caught by a four-line synthetic of known answer (264) — (a) sorting base pairs without a continuity check made the axis teleport across the particle; simulated, 15% of centres out of order flips writhe from −0.4563 to +0.4107 — matching the +0.354 the broken run had been returning. (b) even correctly ordered, the midpoint is not the helical axis: it orbits the true axis at the molecule's own repeat, roughly fourteen right-handed turns against the superhelix's 1.65 left-handed ones, and at 2 Å of wobble the sign flips — truth −0.4563, raw midpoint +0.4239. Smoothing over one repeat recovers −0.4191 at every amplitude tested. Both bugs inverted a sign, both were invisible to every internal check, and both fell to one synthetic whose answer was known in advance

"We are not competing with you" (267) — the positioning paragraph, and it is the most gracious form of the beat-a-baseline rule in the corpus. Four named tools detect knots better than this instrument does, and where the results overlap they agree — said plainly, first. Then what is actually added: a number none of those tools publish — how much the molecule moves, as opposed to how stable the algorithm's vote is. Those are different quantities and only one of them tells you whether a small difference means anything. Claiming the narrow thing you have is what makes the claim survive. The same courtesy reaches the wording, never the findings: where a sentence reads as a challenge to another author, soften the words and keep every number. In the triple-star thread's islands paper the abstract's "misses by measured amounts where they fail" became "is off by measured amounts outside them", and the same plain fact was softened in the three other places it was said, every number unchanged (the triple-star sessions)

practice; ties row 93

A negative result is a specification that has already been tested (268) — arguably the best single row in the inventory, and it is the move the whole paper turns on. A companion had reported five findings as fatal to an artificial-life claim: no autonomy, faithful copying, an inability to self-direct, silence until read, everything a regulator restoring toward a setpoint. Read the same list against a different question and every property that disqualified it as life is a requirement for a device somebody else pilots. And the defence against this looking like a rhetorical trick is the row's real content: most specifications are wishes — somebody's idea of what would be nice, written before contact. This one was arrived at by trying as hard as we could to get the opposite, under pre-registered conditions, and failing. The architecture is not a thing we thought would be elegant. It is the shape of the hole the failure left

pre-registration; ours

The same primitives that make it work are the ones that make it governable (273) — ethics placed in the body of the paper and explicitly not as a coda, on the stated ground that for a device commanded inside a living person it is load-bearing. The structural observation is the row: it is red-team and blue-team at once — the same trust, fidelity and anchoring primitives that make the swarm work are the safeguards that make it governable. Build the security and you have built most of the ethics. Human-in-the-loop by construction rather than by policy, and the agent's one reflex — refusing a physically-insane command — can only ever say no, never yes

Name the misuse concretely, then say the safeguards are not sufficient (274) — a paper that only lists its safeguards is not being honest about its risks. Two specific doors are named rather than gestured at, including a population that could be imprisoned by the thing meant to heal it, and a fail-open design where losing contact releases rather than safes. Then the sentence most papers will not write: none of it is sufficient. Safeguards raise the cost and narrow the path; they do not close it. And the fence on the pledge itself — a declaration binds us, not others, which is exactly why governance matters more than a promise — with the conclusion that publishing the door means publishing that it opens both ways, and that the argument is better had in daylight

dual-use disclosure

Write your own ten strongest attacks, and concede the sharpest one (275) — a paper that ships its own red team is harder to dismiss than one that waits to be attacked. The concession is the part that costs something: the sharpest attack is a motte-and-bailey, where the thrilling claims retreat under pressure to only an invitation, tear it up — and rather than defending it, the paper concedes that the trick is taking both the credit and the humility, and binds itself to the modest position as the true one. The single highest-risk load-bearing claim is then named front and centre rather than buried as one bullet among many

practice; ties rows 61 and 75

Sturdy where it is least exciting, fragile where it is most exciting (276) — a self-diagnosis stated as a pattern: the work is strongest exactly where it is dullest and weakest exactly where it is most thrilling — which is the opposite of a scam, and the honest reason to trust the dull parts and distrust the thrilling ones. A test any reader can run on any paper in ten seconds, including every paper in this corpus, and one almost nobody volunteers about their own

Refuse the grander sentence, and say exactly why it is false (278) — there is a seductive sentence sitting just off the edge of every paragraph here — and rather than avoiding it, the paper writes it down and then kills it in specifics: the grand version requires culture and knowledge transmitted rather than inherited, which is the far side of the wall the work proved it could not reach. We mapped the on-ramp and proved we could not merge onto the highway. And the closing standard: the honest version of the dream is the smaller one, and the smaller one is the only one that survives the daylight

Size the new claim yourself, before anyone else can (280) — this is the one place in the paper where we could cheat and nobody would catch it for a while — written immediately after the paper's most exciting new idea. Then the sizing: the strays reproduce the substrate and at most seed a new vessel; they do not reproduce the mission, and the self-replicating probe that builds the next ship is still on the far shore. Dandelions, not shipwrights — a seed that can start a garden, not a factory that can build a fleet. Naming the exact width of a claim at the moment it is most tempting to leave it vague. Answer five questions for each result that lands, in its register entry, so that results sized "medium" one at a time still reach the front page: how big, in its own units against the nearest known number; to whom, naming the reader, since "to everyone" is an answer to avoid; known, confirmed here, ours as far as the literature passes went, or a correction to a known number; what would make it smaller and what would make it bigger, since underclaiming is an error too; and where it goes, the abstract and findings, a section's headline, an appendix or the proposed list (the triple-star thread's size check, from Bear's rule of 18 October that what matters goes on the front page). Replicate a headline in another world before it reaches the abstract: "keeping only the surest half is the worst rule" held in Question the Substrate's world 1 (−25.3 points, run E1e) and world 2 (−12.9, E1w2), then failed in world 3, tidy clusters, where it came second best at +3.6 (E1w3); the headline was nearly published in its too-strong form (Question the Substrate)

Keeping it light is what keeps it true (281) — the deepest justification in the corpus for the editorial-balance ruling, and it comes from Bear's own paper. We are going to keep it light, because that is the only way it stays true: the moment a vision like this is asserted as fact it turns to smoke — and that is the failure mode of a great deal of work that resembles ours from a distance and resembles it not at all up close. Then the trade that earns it: kill hard and early, and the last page gets to be fun. Nobody gets the ark for free. We paid for ours with a wall. Sits with rows 81 and 102 and is the strongest of the three

Take it to the hardest place you already own before you take it anywhere new (282) — you do not take a vessel meant for space straight to space. The paper's whole structure is a staging rather than a taxonomy: prove the competences in the harshest environment already available, then in a delicate one where care matters more than force, and only then in the one where failure is unrecoverable. The ordering is a governance decision as much as an engineering one — earn the trip where failure is survivable. An engine copied for one regime carries that regime's quiet assumptions: one built for settings where the frozen points always exist (a copy, with one printed line added, of an engine that had only run where the least current B − A stays below 1) stopped on the arccos of a number above 1 at strength 0.05, where the 3/5 root sits just past that edge (B − A = 1.0000085), about five minutes into a run, while a cousin written two days earlier already handled it; its dry run had used 1/2 at strength 0.10, the old regime. A dry run for a new regime uses the new regime's hardest row, and the engine's lineage is searched for fixes its cousins already carry (runs JJ_9, JJ_3 and JJ_5) (R2W)

The broken version is kept because it is the reason we believe the working one (287) — the first bell-mesh implementation measured frequency proximity where it should have measured consonance, and returned a confidently wrong answer, which is the worst kind there is. Rebuilt on correct circle-map dynamics, the real effect appeared unprompted: locking width collapses as ratio complexity rises — an Arnold-tongue staircase, exactly as somebody who knew the field would have predicted. We keep the broken version. Provenance note for row 38: this is where the corpus's Arnold-tongue instance lives, with its mechanism and its broken-first history — but the paper gives the shape and not a number; it gave that hold a source, and row 38 now carries the law the hold waited for, widths shrinking as roughly one over q-squared down the Farey tree

Arnold tongues; circle map

The confident mis-anchored friend is more dangerous than the openly compromised adversary (291) — carried out of the evolutionary-warfare work and into a contested sea floor: inside your own network, the friend is trusted and wrong, and that combination beats an enemy who is merely wrong. So reporters are treated as Byzantine and, more bluntly, as poker players — the table is worked assuming some hands are collusive, and only the physically-verified, provenance-anchored signals are sovereign. The anchor, not the reputation, is what confers standing

Byzantine reporting; ties rows 94 and 101

Hold the most powerful tool in reserve, and give the technical reason rather than a scruple (297) — active probing is kept behind the passive sweep, and the stated reason is a measurement limit and explicitly not a moral one: too much probing might eventually disturb the very substrate fidelity being read. Those are walls we assume exist and are not prepared to address in this paper. An instrument that perturbs what it measures, named as a design boundary before anyone uses it — and paired with the refusal to guard the door: the case for it is large, and we would rather point at the open door than guard it

Put the number first so it cannot hide, then say why you distrust it (298) — the number first, so it cannot hide — a Fermi estimate given as a wide range leaning high, immediately followed by the red team's catch on the authors themselves: quoting two-fold error bars in the same breath as a reference program's twenty-fold overrun, which cannot both be true. Then the separation that makes the section survivable: the claim that survives is structural — a sparse aperture scales with baseline where a monolith scales with glass — and the figure should be read as the opening bid in an argument we expect to lose on the specifics and win on the structure

The interactive whose biggest button defeats it (299) — the passive-detection toy, and it is a demonstration you can trust because of what it lets you do to it. Two scopes side by side: a single pass where true and transient hotspots are indistinguishable, and an accumulated map where a true site persists in place and locks while false sites wash out. The most prominent control on the panel is the one that breaks the claim — flip common-mode noise on and the independent signatures fail together, so the multiplicative gain from fusion evaporates, which is the project's independence requirement made playable rather than asserted. Drop coverage and small sites hide in the gaps between sensors. And the confirmation gate is two-part by construction: a site confirms only on at least four observations and better than half persistence

ties rows 112 and 290

Blind spots as theorems, not bugs — each with its named price (303) — net-charge-zero targets are invisible per-rung. Distributed sub-threshold targets are invisible per-sensor, and the fix is the coherence channel. Scalar observables are loop-blind by the telescoping identity, permanently. Helical structure at fixed circulation is invisible to

winding, and the fix is the twist rung. Each fix has a price, and the price is stated: climb a homotopy rung, add a channel, or promote the observable from scalar to vector. A limitation with a known remedy and a known cost is a design parameter; one without is an excuse

An instrument that has only ever agreed with itself has not yet been tested (306) — the validation standard, written down before the results. Every channel had been kill-tested internally — nulls, injections, shuffles, held-out floors — and none of that supplies an external witness. A satellite geodetic pair fails through orbits, antennas, troposphere and reference frame; a buried invar-wire extensometer fails through rainfall, vault temperature and wire creep. They share almost nothing but the fault between them. So agreement between them is worth more than any quantity of internal consistency — and disagreement indicts one of the instruments, never the ground

practice; ties rows 70 and 96

Nature did not move; our anchor did (308) — the registration specified the detection sign as anti-secular, assuming a plate-fixed frame. The data frame's own secular motion runs in that direction, so the effect arrived at nearly ten sigma in the OPPOSITE registered tail — the pre-registered test failed exactly as written. The amendment was logged before the rerun, replacing a frame-dependent projection with a physics-fixed azimuth, and the episode is printed in the body of the paper rather than buried. An anomaly that turns out to be a confession by your own coordinate frame is the most instructive kind, and the only way to get it is to have written the frame down first. An offset has two ends: an offset graded from a frozen point was first read as the couples sliding across the floor, but measured end by end the frozen point did the moving, 0.21 to 0.22 of a turn per unit of the second tone, with the couples following a tenth to a sixth as far. Any reading taken against a reference that can move (a lag against a strobe, a stop against a frozen point, a price against an index) needs both ends measured before the motion is given to either (runs TWO-TONE-2, WEIGHT-WALK-1 and WEIGHT-WALK-2) (R2W). A chain of corrections reads against such a reference: a lineage in Two Arrows that corrects each generation against the first generation's map stays level for twelve generations, its largest over its smallest 1.02 to 1.06, while one that corrects against its parent leaks and ends 4.97 to 7.83 times higher (run b11). Correct from the first map, not from the parent; corrections taken from last year's corrected compass pile up, and the same holds for calibration chains and model lineages (Two Arrows)

The price of skipping held-out calibration, printed as a warning (310) — with pair-specific held-out calibration the realised false-alarm rates across four windows are 0.011, 0.022, 0.000 and 0.000. Field-wide, before that calibration, they ran 0.13 to 0.21 — an order of magnitude, on the same data, from the same instrument. The difference is the cost of a shortcut, and it is kept in the record rather than quietly fixed, which is what makes the good numbers believable

held-out calibration

Record the threshold artifact rather than hide it, and pre-register the fix instead of retrofitting (311) — a threshold inherited from a different region's null was negatively biased by a residual staircase, collapsing a percentile to roughly zero — so a held-out control region formally crossed the bar while sitting at z = +0.40 against its own null. The verdict is stated in the paper's own words: no signal there, a miscalibrated absolute bar in the pipeline. And the repair is pre-registered for the next campaign, not retrofitted into this one — which is the difference between a correction and a tuning

The honest instrument publishes its own visibility (312) — a datasheet for an instrument nobody sells: eight physical deformation modes against six channels, built by matched injection, with one channel closing its row at a perfect diagonal — 1.00 on its own mode, 0.00 to 0.05 on all five others. And the framing is the row: the matrix is the instrument's contract — which physical modes each channel can see, at what aperture, and, as important, which it cannot. Detector proposes, topology disposes. Almost no instrument in any field ships the list of what it is blind to. Give every engine a card: by the thirtieth exchange of one R2W sitting the evidence rested on five engines, each with its own digits, floors, limits and habits, among them twelve decimals in one family's files, a lean set by the tolerance in the witness, and a stability limit in the fixed step. Bear asked for a card for each setting with its floors, limits and knobs. Each card says what the engine is, the knobs it takes, how many digits its files keep, the smallest difference it can be trusted to see (its floor), where it refuses to run (its limits), how long it takes, and the checks it has passed, and every value names the run that measured it. A blank in a card is a measurement not yet made, said as such, and the blanks become the next checks. The cards turn "the engines agree" into a statement with a size, and a difference smaller than an engine's floor is not evidence, however many digits it is printed with (R2W)

A pre-registered bar that a real, correctly-signed signal failed to clear — and both facts printed (313) — median rank correlation +0.73 and median speed 6.4 km per day, both landing inside the physical band that was registered. And the sign test reads 11 of 16, p = 0.21, because sixteen events cannot reach p below 0.05 without thirteen positives. The failure is one of sample size, not of signal, and the paper says exactly that rather than choosing one of the two facts. Then the part that makes it trustworthy: a sharper post-hoc statistic that reads about four sigma on this same data is therefore claimed as NOTHING here and is pre-registered for the next campaign. The paper is stronger for showing a pre-registered bar that a real tendency failed to clear than it would be for hiding either fact. Both facts printed again: at the person's blind walls the step between a cluster's visits was 0.778 of its median, lower

than in all of 2,000 random sets of the same size (lowest 0.856), and the guess missed its registered bar of one half, which was the reader's translation of "on top of each other". The random-set comparison was chosen after the run, so it is claimed as nothing and registered for the next birth (the triple-star sessions)

pre-registration

A domain-scoped verdict binds only its domain (314) — row 313's non-pass binds this network at this sample size — it is explicitly not a verdict on the method, which does its localising job on denser arrays elsewhere in the programme. And two protections are kept together on purpose: the established literature value is cited (about 8 km per day, against this campaign's 6.4) and the registered bar is recorded as failed, so the result stays a consistency read rather than an independent-timing claim. Either protection alone would have let the claim drift upward. A law fitted on one kind of input is a fact about that kind of input: one curve carried the gearing of two media blind across every strength and frequency run, all driven by a single tone, and a second tone at twice the frequency kept every rule the dancers follow and pulled the gears 0.05 to 0.075 below the curve at both signs. Before calling a law general, list what every run had in common, and vary one of those things (run TWO-TONE-1) (R2W). The same lesson from the other side (R2W's lesson 73 of sitting B9, earlier in this row): when a law holds at one new setting and misses at another, ask which other knob moved between them. RATCHET-4 registered that the rectifier's peak weight times the frequency stays fixed. It held at frequency 0.70 (0.9952 of the reference) and missed at 0.40 (1.1386), where the drive's strength over its frequency squared was also nearly twice as large. Rerun with that ratio held (RATCHET-5), the peak came to 1.0003 of the reference, and the whole curve lay within half a degree (R2W)

The contribution is the protocol, not the signal (316) — the campaign reproduces a known geophysical process at high internal significance using only an external clock, documented-equipment masking, and a held-out foreign network for false-alarm accounting — and then says plainly that the signal is the literature's and the protocol is the contribution, with its two failures printed and pre-registered into the next campaign. Reproducing something already known, under a discipline nobody has applied to it, is a result — and it is the only kind of result that validates an instrument rather than a hypothesis

What crossed the domain boundary, and what did not — in the first sentence (317) — the cleanest statement of method-portability in the corpus, and it is a restraint rather than a claim. Opening a new domain, the paper states first that the topological channels remain in their home domain and make no appearance here. What generalises is the operating discipline — validate against ground truth, pre-register the traps, publish the floors and the breaks — plus a spectral toolset under three named guards. This section is an opening, not a closure. Row 93 says every port must beat a baseline in its own domain; this says which part ported, and the answer is the method and not the mechanism

practice; ties row 93

Flag the known systematic; do not smooth it (320) — a derived mass came in high against the published value by the exact amount the field's known scaling systematic predicts. It is flagged and not smoothed, with the standard cure named as the next step rather than quietly applied to make the number land. Applying a legitimate correction to a number you already know is wrong, before publishing it, is how a systematic becomes invisible

Two first lights, two honest breaks, zero novelty claimed (322) — both targets validated against published ground truth, both instrument failures printed, and the claim made is not about the objects at all: standard work in another field, executed under this programme's discipline, establishing that the discipline itself is domain-portable. The novelty claimed is zero and the thing demonstrated is the method. Opening a new domain by proving you can reproduce its known results before proposing anything is the slow road and the only one that arrives

A card is a card (323) — three sentences that stop a roadmap from becoming a claim: the future application is carded and remains a card. It has no data, no session, and no results, and this paper claims none. Every speculative programme accumulates these; almost none label them this bluntly inside the paper that would benefit from their being vaguer

A novelty claim that states its own search bound is a different object (324) — the literature pass is reported as finding every mechanism to be prior art — topology as trigger, coherence as committee, twist and linking as handedness channels, matched filtering as the admission price, and the gate economy itself — leaving the candidate contribution as the assembly plus the external validation. And the assembly claim is then marked PROVISIONAL in the paper's own words: we found no prior assembly in a bounded adversarial search, which is not the same as none existing, with the fast-moving adjacent neighbourhood named and the claim flagged as the paper's most exposed. The fenced seam is likewise left with its owner rather than absorbed

practice; ties rows 230 and 275

A kill may be a rename, and that is a third verdict (325) — the corpus's verdict vocabulary gains a class here, and it is the useful one. Sometimes a kill is not a death but a rename — an idea that was sound wearing the wrong words. We welcome that outcome too. And it is not a hedge, because the rename branch is written into the kill conditions in advance: one channel dies as stated if two axes collide, may survive as a rename to a non-optical degree of freedom,

and that would be a kill of the words, not the idea; a heredity claim is renamed persistent re-induction if the signal turns out to be re-induced rather than transmitted. Survives, dies, or was misnamed — and pre-committing to which evidence produces which keeps the third option from becoming an escape hatch

ties the salvage standard

Surface an idea specifically to be killed, and say so without apology (326) — we are not defending these spans, we are daring the reader to break them, because a span that survives an honest assault has earned its place and a span that breaks tells us where the real bridge has to go. Every rung in the paper ships with its own named kill condition; in this example, the nine rungs each shipped with a kill condition and were collected into one section so a reader can attack the whole structure from a single page. A paper organised around its own falsifiers rather than around its claims is a different object from one with a limitations section at the back

pre-registration

A register of what we assumed, WHO should check it, and our confidence separated from our tone (327) — four columns, and the third is the one almost nobody writes: our honest confidence, stated as separate from how assertive the prose sounds, in a three-level vocabulary — Moderate, Calibration, Speculation. Each row also names which specialty should check it and the precise question they would answer, so the ask is executable rather than a plea for collaboration. And the blanket caveat sits above the table: every in-container simulation in this programme is single-run, modest-N, without confidence intervals, and nothing labelled a result is statistically hardened yet. This is row 62 with a working instrument attached

Cite the dispute, and say whether the literature tests YOUR claim or only the phenomenon (328) — two citation-status notes do a job most papers skip. The first: an effect is solid one generation out and under active replication dispute further out, so cite the dispute; do not overstate the reach. The second is sharper and rarer: the supporting literature for a compression claim is real but contested, and then — the kill condition below is not tested by these studies; it remains the author's own unrun prediction. Distinguishing "the field supports the phenomenon" from "the field has tested my specific claim" is a distinction almost no citation practice makes, and it is where borrowed authority usually enters. A prior-art pass matches the quantity, not only the effect. Bear greeted the zero's slide with weight with real excitement at the fourth exchange; when Mawhin's survey showed the lopsided step is known mathematics, he asked whether Marchesoni's 1986 letter was needed. The free reviews (Denisov, Flach and Hänggi 2014, sections 1 to 3 and 6; Hänggi and Marchesoni 2009, the harmonic-mixing passages) report the book's §6.3 effect, the zero's slide with damping, as a lag in $\sin(\theta - \theta 0)$ running from a quarter turn (overdamped) to zero (Hamiltonian). Matched on the effect, all of the book's Part 7 would read as known. Matched on the quantity, the literature's is a drift (odd when time runs backwards) and the book's is the step's lopsidedness, a threshold (even), so at the Hamiltonian end their zeros sit a quarter turn apart: the drift's moves over and stays, the step's goes out and back. The slide itself is credited as known (harmonic mixing, row 657). The difference is exact only at that limit: tested on the bench along the weight, with noise, the drift's zero went out and came home, and the registration failed (R23-b, row 737); the road the argument speaks of, draining the honey at a held mass and drive, was still held as runnable at R2W's sitting P3 (R2W)

A table whose every row invites being struck (333) — eight jobs, two toolkits side by side, and a fourth column naming where they could trade. Then the instruction printed under it: every cell in the right-hand column is a challenge — name the shared mechanism, or strike the row as resemblance. That is row 7's mechanism-versus-correspondence rule turned into a table format, so the discipline is enforced by the layout rather than by the author's vigilance. Most comparison tables in most fields would lose rows under that instruction, and none of them carry it

ties row 7

Mark the door; do not walk through it (334) — the two most speculative rungs carry the tag opening rung, most speculative in their own headings rather than in a footnote, and the largest question in the neighbourhood is left explicitly as a marked door, not walked through. Two of the kill conditions end by pre-committing to dropping the rung as a category error rather than defending it. Naming a door you are not opening is how a speculative paper stays navigable — the reader knows which rooms you entered and which you only labelled

An instrument that claims to be universally superior is a marketing claim (335) — what makes a tool trustworthy is a spec sheet — a characterisation of where it reads clean and where it fights the substrate. The paper's two applications split: one channel recovers a published signal cleanly and another suppresses it, and the split is treated as the paper's most useful result rather than a partial failure. The deliverable is not a detection; it is a competence surface, and the bulk of the paper is spent producing it honestly, including two clean failures and one mapped suppression mechanism reported at full volume

A bar that cannot fail is not a bar (337) — the rarest self-check in the corpus: an audit of the pre-registration itself. The apex-error tolerances had inflated to a threshold no result could have missed, and rather than being quietly used, those verdict strings were re-read as uninformative and an informativeness gate was installed on the verdict grammar. Pre-registration protects against moving the bar after the data; nothing protects against setting it too wide

beforehand except going back and checking whether it could ever have said no. A test that empty data can pass is not a test. A condition that one share be at least twice another printed “holds” on a code trial where both shares were zero, and a floor above zero was added before the run of record (the triple-star sessions). Two more ways a bar fails to be a bar: rival readings whose pass bars overlap both come out “held”, so the test could not have shown which was true; and a threshold set at the instrument’s own floor (2×10^{-5} against a quantisation of about 2×10^{-5}) fails for the wrong reason. Set bars that cannot both pass, each able to fail, above the instrument’s floor (the Loop Atlas sessions, R2W). Test a rule where it predicts a negative as well as a positive, and give its control a way to fail: a rule that pads line up in fraction-tree order expected a stack between the 2/5 and 1/3 pads and found none, the 3/8 the tree puts between them never landing, while the window registered as the rule’s control held no two states at all, so its “held” counted for nothing (the Loop Atlas sessions, R2W). A bar that asks only for signs passes half the time on noise: a guess about rows passed with shares of positive turns at 0.527 and 0.488, a coin’s half, and its stricter redesign, with share bars, missed by 0.015. Give each bar a share or a control before the run. And prove on synthetic data that a rule can say no: a lock rule fed smooth synthetic sweeps that stepped over its narrow band could never fail, so it was amended before any real run was read, until a planted lock scored as a lock and a smooth sweep did not (the triple-star sessions). A tolerance set after a trial is a tolerance, not a prediction: the Depth Axis Explorer’s bench runs its fast mode at 500 steps a passing because a trial at 1,000 and 500 showed edges within 1.31×10^{-9} of the record at 500, and its self-test row then asks for edges within 10^{-8}. That row cannot fail the way a blind guess can, and the record says so: it guards the page against a later change, it does not test the physics. And a check’s reference must live outside what the check writes: the explorer’s builder proved the old B8 view unchanged by comparing it, byte for byte, with the file at its own output path, but after the first build that file was the new page, so any rebuild would have stopped at the check, and a check that can pass only once guards nothing after that once. The reference is now the B8 page’s md5 in the crate’s manifest, which no build touches, and two more builds gave the shipped page’s bytes again (R2W)

ties row 12

We tested our own leading explanation and it failed, which is the useful part (339) — the hypothesis was that an unweighted vote across per-source errors spanning roughly sixty to twelve hundred microarcseconds per year was suppressing amplitude, and a heteroscedastic battery was pre-registered to restore it. It did not — the corrected coefficient was statistically unchanged and the amplitude stayed 4.5 sigma low. And the failure localises the residue: white-noise synthetics at the real error spread reproduce only about twenty percent loss while the data shows about sixty, so the missing forty percent lives in structure the synthetics do not contain. Per the registered terms the channel is flagged for redesign against correlated systematics, not re-tuned, and no third attempt was taken

The economy claim failed cleanly, and what the loops actually buy is named instead (341) — a decimation test against a plain global fit across progressively smaller subsamples found no crossover at any source count, in both clean and jitter-inclusive subsets. The economy claim — that robust loops reach frame accuracy with fewer sources — fails cleanly on this substrate. And then the replacement, stated rather than mourned: what the loops buy is not amplitude economy but direction, localisation, and audited degradation. Killing your own selling point and naming the real one in the same paragraph

A topological mismatch is a statement about fit, not about quality (344) — when a substrate offers no genuine transported state with non-trivial closure, the instrument reads nothing — and that is a mismatch of data to tool, not a failure of the tool. The constructive half is kept in the same breath: examining what a substrate does under the instrument, even in mismatch, can reveal that a field carries structure its own community had not catalogued. A null that is correctly attributed to fit rather than to quality is reusable; one attributed to the tool is thrown away

An instrument is a partner in a dance, not a harpoon for a white whale (345) — the organising principle behind every pre-registration and null in the paper, stated in one line. It works when the substrate carries real invariants to walk on, when the signal is of the coherent global-phase class the instrument reads, and when there are at least two vantages to close a loop against. Choosing the substrate, the signal and the estimator to match the field’s geometry is not a limitation to be apologised for; it is the whole of using the instrument well. Sits with rows 81, 102 and 281 as front-matter material, and it is the one that says what the guide is for

The tautology guard — naming when your own agreement is circular (346) — agreement with a standard method at the sub-percent level is recorded as a pass of validation class only, and the paper immediately disarms it: this agreement is not independent evidence for any structural claim, because the zeroth-order circulation equals the rotation curve by construction. A result that had to come out right is quarantined by name rather than counted. Most papers would bank that agreement as support; this one files it as calibration and says which part was guaranteed in advance. Say so before the run as well: when an alternation dominates, an envelope swelling at one rate is a spectral line at a half minus that rate, so a test of whether the two readers agree is close to a mathematical fact. It was written into the registration that way, and held in 15 of 16 (run triple_paper_checks, check S1, 20261014_085115) (the triple-star sessions). A match can be the quantity’s own bookkeeping. A toy sail’s price dead downwind, 0.2000 newtons a watt, exactly one over the wind’s speed, was printed in three records as a ride law met, the law a floating snake obeys; it is the price’s own bookkeeping, since dead downwind the sail’s whole force lies along the wind and the course alike,

so any boat prints it. What does carry is the general form: the watts the boat draws from the wind’s motion over the water are its push’s own work plus the sail’s drag loss, so the price times the boat’s speed is the share of the drawn power that moves the boat, never above one. Dead downwind that share is the boat’s speed over the wind’s, which is why a ride on drag alone cannot beat the wind; read from the recorded polar (the barge hull’s, row 471), the share peaks at 0.62, at 109 degrees, a reading not yet registered; the test owed, the share at every heading of SAIL_HULL’s nine hulls, never above one, is not recorded as run (run SAIL 20261018_202451) (the Loop Atlas sessions)

Name the stricter claim a reader would want, and decline it (347) — the registered bar passes on recovering known structure. Then, unprompted: the stricter claim — that this channel carries information the standard analysis lacks — is a different claim and is not demonstrated, because the same patches are recoverable in the standard channel and any excess sits at the 0.1 percent level. That stricter question is retained, unfolded, as open. The claim the result is one inch away from is the one most likely to be made by a reader in a hurry, and writing the denial into the paper is the only defence

Rediscover why a standard practice exists by declining to adopt it silently (349) — a rotation curve built from measured counts rose unphysically toward 250 kilometres per second, because a distance truncation poisons the outer annuli — while a pinned external profile gave the flat value the literature reports. This is an in-crate demonstration of why published analyses pin an external density profile, discovered by declining to choose a convention silently. Adopting a field’s convention without asking why leaves you unable to tell a convention from a result; refusing to, once, teaches you which it is

The substrate autopsy that ruled itself out, with a number (368) — from-memory station coordinates were replaced with official six-decimal ones and the difference measured rather than assumed: mean shift 0.59 km, worst 1.36 km, against legs of 50 to 90 km, with ring perimeters moving under a kilometre out of 270 and 440. Then the load-bearing check: both loops still wind the same direction, signed area unchanged, so the anomalous sign flip is not a geometry artifact. And the verdict refuses to over-claim: substrate is ruled out, and it does not rescue or explain the null — the full weight falls on the hunter. A cause eliminated with a number is worth more than a cause never suspected

the three-cause autopsy (row 111)

Commit the bar in writing before the data comes back (369) — with the data missing, the analysis was built anyway and the bar written down first, so it could not be moved after seeing the number — two-part, and carrying a novelty clause so the read cannot simply re-find the channel already known and call it new. The primary channel was named and staked before any run, with the invitation to re-commit if the author disagreed — but only before the files arrive. Pre-registration is easiest and most honest precisely when the data is not yet in the room. A reading made after seeing the data is labelled so and claimed only after a blind test: chaos on the borders between treads, read at 84 per cent on a map already seen, held at 97 per cent on a fresh grid, while a staircase’s turns being marked by a change of word failed blind. And what was seen after registering goes on the record too: a guess registered on a letter alone stayed honest because the later viewing of the figure was disclosed before the score (the Loop Atlas sessions). A smoke run that prints results is a look at the data, so it is run for finishing only, after the registration, and a smoke that was seen is written into the run’s header; a full run starts only after its smoke passes (twice in one session a full run went ahead after a failed smoke, and one was stopped within seconds and relaunched). When the error stream is not empty, read the error stream, and only that. And an instrument’s known bias can be registered as a prediction of its own: near a doubling a strobe’s repeat test reads the doubled dance early, and registered before any survey data came back, that prediction held at all four starts (the Loop Atlas sessions, R2W and the book). Write the reader before the returns arrive, and keep a list of what it has seen: a reader of cumulative returns judges only chunks absent from a list kept by name and md5, so a chunk read twice cannot vote twice (897 chunks across two folders were read once, their duplicate copies compared by md5 and none changed). Read blind, write the read into the record by the clock, and only then open the other reader’s marks: what two blind readers found independently held up best, and what only one of them saw became a test (the Loop Atlas sessions, R2W). A test hook set once is not set: a launcher test run without its hook started the real sweep, which was stopped within two minutes and its folder deleted unread; the hook now goes into every launcher test, and processes are stopped by their number, not by a pattern that can end the shell (the triple-star sessions). Read each registration back as a sentence before it is kept, and ask what result would make it fail: four registrations in one session could not be judged as written (a bar finer than the printed digits, a step count copied from the wrong case and caught before the run, two parts that could not fail, and a surprise test that contradicted itself). And a reading made after the data is a guess until a run settles it: in that session some such readings were wrong and some right, so each is labelled as one in every file and gets its own registered test before anything rests on it (the Loop Atlas sessions). Declare what you have already seen before a cold reading, and know that the declaration does not remove the colour: a reader had read that Bear draws three roads where a string splits and still drew two, and two rules written as new in one session’s practices had been in that session’s own paste of two days before. A declaration tells the next reader where to look for the colour; writing the bets first shows exactly which assumption failed when they miss (the Loop Atlas sessions, the book). Even when the return comes fast: readers written after a zip arrived, though before any value was read, were declared as a lapse (the LOW readers) (R2W)

pre-registration

A wall you cannot walk through is not a reason to stop (370) — the data had not survived a session reset, which is a genuine hard stop on the headline result. Two of the three fronts advanced anyway — the substrate ruled out by fetching authoritative coordinates from the web, and the instrument built, verified by running it to its polite stop, and armed against a committed bar. One front left loaded and waiting on one upload. The session did not stop at that wall. Naming exactly which front is blocked is what makes the other two available

The swing detector, falsified in three deaths (372) — the missing write-up, recovered in full, which closes row 85's hold. (1) Magnitude is dead: storm-day total turning 2702 degrees against a quiet day's 2317, only 1.2 times, with three of five stations swinging more on the calm day. The reframe that survived: the storm does not add turning, it organises it. (2) Coherence looked decisive, then the rigging check halved it: temporal clustering 54 minutes against 71 — right direction, missed the committed bar — and spatial ordering 1.00 against 0.40, an apparently perfect sweep. A permutation null disqualified it: at five stations with a free bearing, random orderings score 0.75 median and reach a perfect 1.00 seventeen percent of the time. Diagnosis: underpowered at five, not disproven. (3) The false-alarm trap, sprung against a committed prior: a supercell that produced no tornado swung the most — 4957 degrees, nearly double — and clustered as tightly, 57 against 54

Density is the cure, not years (373) — the autopsy's willed treasure, and it is a prediction rather than a consolation. The failure is statistical power, not a wrong idea: five points cannot certify a spatial sweep, because random arrangements look swept too often. Ten to thirty stations makes the ordering null strict and is the only path on which a coherent sweep could separate the tornadic from the merely severe. More stations on one storm beats more years of one station. With Bear's own corollary: a single-station ticker fails on storm days, not quiet days — it would have fired at full tilt on the no-tornado supercell. Specificity on calm days is cheap; a dead detector has it

Match the structure, not the digits (384) — a tempting numerical parallel is filed as a resonance to explore and explicitly not a claim, with two fences written beside it. First: the formula it would match is empirically unexplained, so matching it is matching a mystery rather than a theory. Second: the two numbers involved are not shown to be the same object — the value is a common one, so treat the digit match as coincidence until something derives it. The defensible parallel is the structure; the digits are not evidence. This is the coincidence-pricing discipline applied by the author to his own most tempting near-miss

ties rows 181 and 182

Name which of your own threads is most likely to close (385) — across a project with many live directions, one is singled out as the most closeable on three stated grounds: real public data, an established science to check against, and the programme's best historical hit-rate — rate-over-level separating states at 96 percent against 33 percent chance and 39 percent for the level (rows 105 and 294), persistence cutting false alarms six to twenty-eight fold (row 108). If you want a thread that could produce a result rather than stay a model, this is the live one. Ranking your own directions by closeability rather than by excitement is the rarest prioritisation a speculative programme can perform

The felt fifth dimension is a parameter, not a place (386) — the architecture is three of space, plus time, plus a mode index that behaves like an independent axis because it is a separate integer knob. Real as a parameter, not as a place — keep the five-knob count and drop the spatial claim. Two adjacent over-reads are named in the same pass: the unit-loop's free trip is free in magnitude and paid in phase, not a speed bypass; and do not fuse the anchor with the target — one large-scale reference is a genuine inertial anchor while a particular anomaly is a candidate target, and using the target as the anchor is the error

practice; ties row 364

The ledger law that governs every free-energy idea (389) — stated before any of the ten projects, so none of them can quietly violate it. Energy is conserved and there is no net-free work with no ledger anywhere — but there are real cases where an existing gradient pays the bill and you pay only the small steering cost. A rudder, not an engine. Every genuine win in the list is tapping a gradient that is already there; the forbidden version is building an invariant and draining net energy out of it. A shape loop that does work needs a refunder to pump against: a swing gains with gravity and gains nothing without it, and that zero is the measurement (the swing toy) (the Loop Atlas sessions)

first law

A defect battery is real; defect propulsion is a perpetual motion machine in a knot costume (390) — the author fencing his own most attractive door. Knots and invariants are not free-energy packets you can drain: making and unmaking one costs at least what comes back, usually more once friction and radiation are paid. A stable topological state can store energy and release it on demand — that is a genuine robust battery or actuator — but you paid to wind it. The usable version and the forbidden version differ by one word, and the word is net

first law

The scale map, so nothing drifts into overclaim (393) — one table that keeps a ten-project programme from claiming the wrong scale. Acoustic wavelengths are tens of microns: cells, droplets, particles, communications, flaws — real, cheap, in reach — and structurally cannot touch molecules. Electron wavelengths are picometers: atoms and molecular handedness — and that needs an instrument accessed by collaboration, not built in a garage. So the molecular ambition rides the electron side, not the acoustic one, and the compendium says so in the same document that lists the garage builds

The killer filter, designed as a single experiment (394) — the strongest of ten projects is chosen not by spectacle but by which one has a measurable result that decides something. Send winding-encoded and amplitude-encoded signals through the same deliberately-noisy channel, and if the winding survives where amplitude keying fails, the programme's standing law is demonstrated in a tank for the price of a bubbler. Build it, run the bubbler, let numbers decide. A project selected because it can lose

practice; ties row 93

Start where the rig proves itself, not where the spectacle is (395) — the build order is explicitly by leverage, not by excitement: the simplest levitation build first, purely to prove the whole stack works, then trapping, then patterning, then the frontier experiment. And the instrument-grade nanometer work is held as a partner track via collaboration rather than pretended to be a garage project — with the access path written down instead of a bill of materials that could not be bought

Lower tier before higher tier (396) — three build tiers are specified with real wiring, and the instruction sits on top: prove the loop with the cheap, fully-understood discrete parts before reaching for the programmable rig. The same one-shape-at-a-time discipline that governs the simulation growth path, applied to hardware — and written into the document rather than left to willpower

Grow one mode at a time, and do not weld them together (397) — the growth path is written explicitly: understand the quiet mode completely, add the angular phase in two dimensions, let that phase become height in three, then grow one mode at a time. And the remaining ideas are marked HELD for one-at-a-time building, NOT to be welded at once. A handoff that names the specific failure mode of its own author's enthusiasm — and does it as a build instruction rather than as a caution

The bench tests the device; the sky tests the model (398) — keep those drawers separate. The instrument's soundness is decided on a bench, with numbers, by the person building it. The cosmology is a bet decided by somebody else's data — and the live external handle is named with its current significance and its crux, stated as not yet a discovery with the calibration and foreground question identified as the thing that decides it. One sentence that keeps a speculative programme permanently honest, because it tells you which of your two claims each new result is allowed to touch

A faster implementation is a claim, and it is validated against every existing path before use (399) — the vectorised rewrite opens by saying what it is not: not a new method — same formula, same conventions — and is then validated against three independent scalar implementations, on synthetic fixtures of known answer and real structures, with the worst disagreement printed rather than a pass/fail. The line that earns it: they are correct and they agree with each other to 10^{-15}. A speed-up that changes an answer is a bug wearing a benchmark, and the only defence is running the old path beside the new one. A port agrees with its parent down to the parent's own floor, and that floor is not the step: the Loop Atlas's jumper, written in JavaScript from its Python parents, matched the runs of record to 8.5×10^{-11} at 6,000 steps a flight and to 1.8×10^{-10} at 1,500, and a warm trace kept one share to twelve digits from 375 steps to 6,000, so the gap measures the rates' central differences and rounding. A port's check prints its gap at more than one step, so that what the step decides is told apart from what it does not (run EXPLORER_ENGINE_CHECK 20261017_063123) (the Loop Atlas sessions)

The banked negative, printed in four lines (402) — a hypothesised coupling between a turning index and a parity class, tested across nearly ten thousand cases: 5,369 matches out of 9,999 — 53.70 percent against a 50 percent baseline — broken out by class at 55.7, 53.3 and 35.9 percent. Verdict, in full: no coupling at parity level. Banked negative. Four lines, the counts, the breakdown and the word banked — which is everything a null needs and more than most nulls get. A closed hunt keeps a watch, and the watch can fire where the hunt never looked: four rung sightings in a triple were registered, run on fresh phases and failed cleanly, the hunt was closed for that boundary layer with a standing watch, and the watch fired the same day in a different variable, a lap ratio sitting on flat steps at 7.0185 and 8.0219 across bands of birth ratio. Those flat steps held: they are the islands' own lap ratio, and the islands paper rests on them (probe 22, run 20261006_190322). What did not replicate was the boundary layer's rests at 7 and 8 (probe 14, run 20261005_193416; rows 456 and 588). Closing a hunt is not deciding that nothing is there (the triple-star sessions)

Put what the bench does NOT test in the scope section, not the appendix (403) — a build document opens by naming the thing it cannot demonstrate: not in scope — that this reads the universe's phase. That is the model, and it is not what the bench tests. Followed immediately by what the bench does earn: the bench tests the instrument, which

is genuinely sound on its own. Row 398's separation, enforced at the top of the document where the temptation is highest — because a build doc is read by someone with a soldering iron and an appetite

practice; ties row 398

Name the analogy inside the build document, in the same paragraph as the result (405) — each tier ends with its own honest limit sitting beside its own success. The mechanical belt is a mechanical integer constraint, an analogy for the topological one — good for intuition, not the real knot. The electronic divider is a counter, not a topological winding; the right dynamics, an electronic stand-in for the topology. A build document that tells you which part of what you just watched was the real thing — and does it at the moment the demonstration is most convincing

A five-level verdict vocabulary, and every chapter must end with one (407) — because a fluent translation hides whether anything is under it. The five: exact / theorem-backed (the same object, with a receipt or proof), settled / classical (cited to its source), kinship of kind (the same type of object, with any numerical bridge tested and its result reported), conjectural (a resemblance that is measured but whose mechanism is unproven), and inherited coincidence, flagged (a rhyme kept only so the reader is told it is a rhyme). Every number appears in a receipt that regenerates it from constants, and no chapter asserts more than its tag allows. The vocabulary is the instrument; the chapters are the readings

A VERDICT CAN UPGRADE — and the verdict belongs to the PAIR, not the object (410) — the headline of this paper, and it is a correction to one of its own siblings. The Tower paper asked whether a musical comma bridges to any measured substrate constant and correctly reported it negative — 23.46 cents is not 35.81 degrees, and forcing it would have been the inheritance trap. But that was the wrong bridge to stop at. Written as a linear form in logarithms the comma measures 1.355×10^{-2}, which is exactly the (19,12) row of that same paper's own Baker-floor table. The same number. So a bridge reported negative against one target does not close a story; it closes that bridge — and the discipline is not only to test honestly but to notice you may have tested the wrong target. The verdict is a property of the object and the target, never of the object alone

Baker 1966; Barbour 1948

The internal contribution is one line, and everything else is credited above it (411) — the prior art is named at length and without reluctance — the convergents of a logarithm, the field's own existing notation for the object, a 1948 paper, an observation from around 40 BC, and the theorem already cited in the sibling paper. Then the claim, sized honestly: what is ours is smaller and internal — our own paper carried the two facts in separate sections and never noticed they were the same row. This chapter draws that one line. Everything else belongs to the people above. A real finding claimed at exactly its width, in a paragraph that spends more words crediting than claiming. Credit the road already drawn, and say what your own road adds. R2W's literature pass found its Part 6's lopsided step already published in 2025 as the biharmonic Josephson diode; the book cites both papers, read end to end in sitting P3 with every statement the book makes of them holding (row 832), and Part 6 now says they were in print first and that its own road adds weight. Bear's ruling: another party's road to the same place does not give away the credit for the whole map (R2W, sitting P2)

practice; ties row 230

The conjectural tag is the one that proves the collection is not a sales brochure (412) — three chapters resolve cleanly — exact, upgrades-to-exact, held at kinship because no number should be forced. One carries the tag none of them do. An atlas that only ever reported "exact" or "beautiful kinship" would be a sales brochure. The test of the method is whether it will look at the author's own cherished idea and write conjectural on it in ink. It will. And the chain is printed in three clauses so no reader can slide along it: resemblance real, operator hoped for, hypothesis unproven

Both of the usual answers are wrong (413) — "it matches to a fraction of a percent, so it must mean something" is wrong because precision is not evidence when the inputs are not that well known. "Numerology — three inputs and one output, you can always fit it" is wrong because some structures really are rare, and reflex dismissal throws those out along with the junk. Naming both reflexes as failures is what makes a third answer worth building — and most of the argument about coincidences is two camps each correctly diagnosing the other's error

The window is a finding until you widen it (416) — the best row in the inventory on search-boundary artifacts, and it is told as three failures in a row. Stop at a walk length of forty and six formulas qualify — a small special family, apparently. Widen it and the six become a line with no end, and six turns out to have been a fact about the search rather than about the lattice. Stop after 119 rungs and two rows report infinite that were merely long, one of them 2,615 members. Stop two rungs below the published one and you report a clean lower edge — widen and there is no edge at all, but a gap of 28 rungs whose position means something. Each widening changed the answer, and each new answer was more structural than the one before. Widen the window until the answer stops moving, and report how wide you had to go. The first check in Bear's note A Coincidence Has No Single Price, widen the window until the answer stops moving, caught three of one evening's misses: a span registered past the next doubling, brackets taken from a grid that stopped below the locks it wanted, and a full circuit read where the effect lives at half (the Loop Atlas

sessions). A ceiling can belong to the window: memory on the pad atlas saturates near 0.31 of the cells, which is not a property of the riders; heavy riders remember in about 93 per cent of the cells under pull 1.0 and in none above it, and the near band's rows under 1.0 are 40 of 121. Before explaining a limit, check whether the chart's own range sets it (the Loop Atlas sessions, R2W). A number that looks universal may belong to one special place: a best share near 0.41, met in four geometries, ran from 0.329 to 0.427 when swept over slope and the cutting circle's size, and settled on a straight cut's 0.32793, 0.34511 and 0.37743 at 30, 50 and 70 degrees; the doubling constant is 4.6692 only for a rounded top. Sweep a constant over every knob before naming it (the Loop Atlas sessions). Set a search's range from the theory, and print where the answer fell in it: a root scan set by habit (1.40 to 1.50) crashed at ripple strength 0.25 because the root was at 1.509, and a best drive that sat on its window's edge at weight 0.30 gave its 0.0015 only as a bound (runs FBP-1 and PIN3-2) (R2W). The note's rule on a second instrument, its check zero: refine the ruler until the answer stops moving, and report the stride. The seam in the fourth harmonic's in-phase reversals (the R2W book's §6.6 and §6.10) is the frequency where the top of the zero's path touches the quarter turn. Read from the usual ladder of weights it sat near 0.767; from fine ladders, 0.782; from a parabola through the fine ladders' three highest rungs, 0.783, where it stopped moving (runs RATCHET-14 to RATCHET-17). The seam never moved; the reading did, and always the same way, because the highest sampled point can only read a peak low, never high. Near the seam the excursion above the quarter turn was also narrower than the usual ladder's stride (0.32 to 0.64), and anything narrower than the stride is invisible (row 760). Bear's note A Coincidence Has No Single Price (2026) says "widen the window until the answer stops moving, and report how wide you had to go"; here the ruler is refined instead of the window widened. The note's spectroscopy sentence, that two neighbouring states are distinguishable only when their spacing exceeds the width of the line, is the same rule turned inside out: a feature is seen only when it is wider than the ruler's step. The same idea showing up in two unrelated instruments is worth its own place; the method is cited here, not the note's running specimen (R2W)

Print both prices, because which reference class is right is a judgement (417) — the price of a coincidence depends on what the searcher committed to before they looked — and the honest and the flattering answers differ by nearly a factor of ten. So both are computed and both are printed, with the reason stated: which class represents what a searcher in that area would actually accept is a judgement, not a calculation. Publishing the range and naming the judgement beats publishing the favourable number and calling it the price

base-rate reference class

A limitations section where every entry carries its own death condition (418) — six claims, each followed by an explicit dies if. The convergent dies if the dimensional constraint is stated wrongly. The unboundedness criterion dies if a second input's uncertainty turns out comparable, because the band stops being a single-generator quantity. The gap dies if a wider scan finds excluded rungs elsewhere. The regrade dies if the underlying tension resolves — at which point the flip becomes either the formula's obituary or its rescue. And the two prices die if either reference class misrepresents what a searcher would accept, which is a judgement, and why both are printed rather than one. Falsifiable entries rather than hedges

pre-registration

One number becomes a point, a curve and a comb — and the comb is over-specified (421) — splitting a crossover into a sign flip (a bifurcation, a real point), a size per decade (a curve, fillable by two more pairs with no code change) and a resonance structure is the right move, and row 415 gives the ground: a lone precise number can be bought by the length of the walk that found it, while a curve must hold at every point it is read, so it is far harder to fake. But a Deborah-number-near-one mechanism predicts one feature, not a repeating one — a comb requires a mechanism that recurs, and none has been named. By row 418's own standard it therefore needs its own death condition: dies if the period sweep shows a single crossover and no repeat structure. The first two are predictions; the third is currently a hope wearing a prediction's clothes

A simulation whose slow limit is the answer is a rig check, not a result (422) — a relaxational model of an orientation slaved to a moving trap reproduces the solid-angle formula by construction as the sweep rate goes to zero — so recovering it confirms the pipeline and nothing about the world. Row 29 in a new costume: the answer was built in. The result is the departure — the shape of the mirror residual against the sweep-rate-times-relaxation-time group — because that shape is a falsifiable prediction the bench and the expensive simulation can both be scored against. The value of the free version is precisely that it makes the expensive version able to fail

ties rows 29, 335 and 346

THIRD INDEPENDENT ARRIVAL — coherence is the symptom, not the safety (424) — the blackest board in the game. The matriarch's memory anchor was placed, long ago, on ground that read clean and was not — so every read derived from her is internally consistent and confident because they all descend from one false source. A false reality from a single source is more seamless than a true one assembled from independent noisy sensors: there is no second voice to disagree. And the detection route is named and is the only one: not checking her harder, since every check

routes through her and confirms the lie, but a second anchor that does not descend from her. The pod does not die because reality is a lie. It dies because it kept only one map. Rows 70, 250 and this one are the same fact reached in three substrates months apart

ties rows 70, 96, 250

The right is documented; the causal story is fenced separately (427) — the real-world anchor for forcing substrate-poisoners to declare themselves is statutory and verifiable — a capped number may sit, a posted notice is required, and identification on request is on the books. The paper corrects its own terminology to the verifiable terms. Then the separation that earns the citation: the often-repeated story that the practice declined because poisoning trust hurt business rests largely on a single anecdote, so the right is real and documented; the decline is a plausible hypothesis, not confirmed data. Two claims in one paragraph, each graded on its own evidence

practice; ties row 328

The deepest trap is not being attacked (430) — it is not knowing your channel is compromised — so you trust it more, lean on it harder, and walk into prepared traps believing you are winning. A failure mode that gets worse the more competently you operate, because competence is exactly what routes more weight through the captured channel. Named in the paper as its own object rather than left as a moral

The corrections register is the project's actual credential (431) — of four deposited documents, the one the deposit itself calls the most important and the most counter-intuitive to publish is a catalogue of ideas held, tested and changed — each with the seductive idea, the check that met it, the outcome, and the named failure mode. Corrections run in both directions, including one idea that was wrongly dismissed and had to be restored. And the reading order makes the point: a reader short on time can read the corrections register alone and learn the most about how this work is done

Run our question on your data and send back only the answer (432) — for populations whose exact locations are rightly withheld, the ask is framed in the steward's order of preference: share under any protective terms, or run the pre-registered question on your own data and return only the result, so sensitive locations never leave your custody. The withholding is treated as correct and the method is built to work with it rather than around it. A collaboration design where the thing you want and the thing you must not have are separated in advance

federated analysis

STANDING RULE — scope every new project across the Field Guide first (433) — set by Bear 2026-09-12, and it is a rule about the guide's use rather than its contents. The first move on any new project is to run it past the assembled method — the four limbs, the priced rows, the named blind spots, the verdict vocabulary — before building anything. Not as a formality: the rationale is measured rather than asserted, because the first time it was tried it changed what got measured next. A spoke session began work before reading the guide. No damage was found, and every handover now asks the new session to request the guide at the open

A method guide used as a lens produces falsification points, not vocabulary (434) — handed the quadrature rows and the work-lives-in-the-phase-gap law, a live project session opened a new testing regime with falsification points at the quarter-turn maximum — on work already in progress, with no new theory and no new hardware. That is the test of whether an assembled method was worth assembling: not that it reads well, not that it is comprehensive, but that pointing it at a running project changes what gets measured next. The guide earned its own standing rule by passing that test once

practice; ties row 433

THE GUIDE'S OWN KILL CONDITION — a stranger finds something the author did not (435) — every rung in this corpus ships with the condition that would kill it (row 326), and until now the guide itself had none. Here it is, and it is cheap to state and expensive to fake: the Field Guide works if someone who is not its author picks it up and uses it to find something its author did not. Not cites it, not admires it — uses the method, on their own substrate, and comes back with a measurement the author could not have made. Until that happens the guide is a well-documented personal practice; after it, it is a method. And the same sentence is the test of whether "loop-walking generalist" is a career path or a description of one person

pre-registration; ties row 326

The method travels; the discipline is what gets stripped (436) — a guide that spreads by being used rather than studied does not degrade evenly. The rows with numbers attached survive transmission — they are quotable and they pay immediately. The fences die first: the what-this-does-not-test, the visibility matrix, the dies if, the search bound — because each of them requires understanding to apply and returns nothing to anyone who skips it. So the design decision follows from the failure mode: put the fence inside the row, in the same cell as the claim, never in a separate chapter. A price and a name travelling together in one line is a claim that carries its own limits; a limitations section is a page nobody forwards

Cross-domain work is harder to criticise than to write (437) — the same siloing that kept the landmarks from being stitched together also silos the criticism: the specialist who could falsify one chapter will never read the next one, so an external witness is structurally harder to obtain for a cross-domain work than for a single-domain one — exactly when row 306 says it matters most. And the corpus already contains the only workable answer, in the seam register: do not ask for a reviewer, ask several specialists one precise question each, with the question written so it can be answered without reading the rest. A criticism ask that can be routed is the cross-domain substitute for a peer

ties rows 306 and 327

Tell the ending first, because the ending is half the point (439) — we report two things, and we will tell you the ending first because the ending is half the point. A paper whose spine is a failure, announced in the abstract and then again in the introduction on purpose: if you read this expecting a triumphant demonstration that crystals are alive, you will be disappointed at exactly the moment we intend you to be. And the wager stated outright: the honest version of the journey is more valuable than the dishonest version of the destination — a clearly mapped wall, with the mechanism that gets you to it and the conditions under which the central claim dies, beats one more over-claimed result that quietly plateaus

Concede what you did not invent before a reviewer concedes it for you (440) — five intellectual lineages named with their owners before any claim is made — life-as-organisation, the clay hypothesis, information-bearing crystal growth, the error threshold, autocatalytic closure — cited not as decoration but as the bar a new contribution must clear. Then the two things that are not re-skins, each stated as narrow: one mechanism and one negative result. Neither is a manifesto. Everything else here is scaffolding, and we have tried to keep the scaffolding honest about being scaffolding

practice; ties rows 230 and 411

The speculation licence, taken deliberately and fenced in the same breath (441) — the clearest statement in the corpus of when speculation is legitimate. Artificial life is a field where speculation is not a vice but the working instrument: no one can manufacture life to check against, so the discipline advances by proposing mechanisms, arguing their plausibility, and naming what would falsify them. We use that licence deliberately — but we do not mistake it for permission to be careless. Every leap fenced to a kill condition, and where a claim needs expertise the team does not have, it says so plainly and a register maps exactly where. A licence claimed and its limit stated in the same paragraph

Three smothers removed and shown to work; the fourth left honestly open (445) — the removals are measured, not asserted: making selection endogenous — the other individuals are your environment — holds the evolutionary rate at 0.033 against 0.002 on a fixed landscape, because climbing never finishes when the hill keeps moving. And then the fence, stated as absolute: “remove the obstacles and it ignites like fusion” is a hope, not a theorem. Fusion we know ignites — stars exist. Open-ended evolution igniting from sufficient conditions is the unsolved problem. We mapped the wall and found three of its handholds false. The fourth we leave honestly open. Necessary and not shown sufficient, reported against the authors’ own hope

pre-registration

We will not quietly assume our way past a second wall (447) — the sharpest self-audit in the corpus. The entire heredity mechanism rests on whether real fracture quantises the linking number or scrambles it — a property of the substrate, not of our argument — and predicting the discriminator is not finding the material. Then the parallel that makes it bite: this is the same trap the field’s whole history warns against — artificial life reasoned that life is organisation and not matter, and then could not make life, because the substrate it chose silently withheld what the organisation needed. We have just spent a paper reporting one such wall; we will not quietly assume our way past a second one on the materials side. A claim can be elegant, falsifiable and correct in logic and still have no home

Hand the field a hypothesis, not homework disguised as a discovery (448) — substrate selection is framed as an open experimental programme rather than a solved step, with directions that are concrete and quickly falsifiable: layered crystals first, because their cleavage is planar and their interlayer registry is exactly the structure the invariant could ride; conchoidal fracturers named as the control that should fail, since demonstrating the contrast would itself validate the prediction; and the deciding measurement placed as a simulation question first and a bench question second. Naming that honestly is the difference between handing the field a hypothesis and handing it homework disguised as a discovery

The false-alarm rate is itself a cause of death (449) — two failure modes, and only one of them gets attention. A miss is counted; a false alarm is treated as the acceptable cost of caution. But post-event study of a real EF5 documented that residents did not act on a valid warning because prior false alarms had eroded its credibility. So a system that improves detection while leaving the false-alarm ratio alone reproduces that problem exactly. Precision over recall, argued as a life-safety question rather than a statistical preference — and it is the reason the paper exists at all

Simmons & Sutter 2009

A target satisfied by construction could never have failed (450) — the previous version set a design target of better than ninety percent specificity — but specificity counts every quiet county-hour as a true negative, an enormous number, so it already sits at 98 to 99 percent before anything is built. The target that means something is precision. A bar that cannot fail is not a bar, and here the correction is printed with the terminology error named as the author's own. The same class of defect that row 337 caught in the sky paper, found twice in one summer in two unrelated domains — which is what a recurring structural failure looks like rather than a one-off slip. A fit's diagnostic can read nothing by construction: LB1 (run 20261016_214034) printed the residuals' correlation with e_run − 0.2811 as +0.000 on both lines, but the fit already carries a term linear in e_run, so that correlation is zero by construction. And a high share explained does not identify the coefficients: RL1(a) (run 20261017_032534) found that adding grip3 moved the five-term law's grip slope by 53% to 2,280%, while the share explained barely moved (99.70% before and after at 0.20, unchanged at the second decimal), because the terms are moments of one swing and are not separately identified. Before reading a residual diagnostic or a coefficient, ask whether the fit's own terms already fix it (the triple-star sessions). Say so when a held verdict could not have failed: R2W's SONG-4 held as a symmetry of its setup and was not counted a redesign, and a test that held on a field its own check showed to be zero was not entered at sitting P2 (TR-2), though the landing rows R2W sent from sitting P3 list it as landing 338. A specimen of row 551 (R2W, sitting P2)

practice; ties row 337

Recording this removal matters more than the channel did (453) — two channels withdrawn, and the second is dismantled by its own author: it is not a beat. A beat requires two signals in the same medium reaching the same detector where they can mix — and one of these is electromagnetic in the Earth-ionosphere cavity while the other is acoustic in air, with no coupling mechanism proposed. Subtracting one instrument's number from another instrument's number does not produce a physical beat frequency. And even as arithmetic it is unstable, because the acoustic peak moves with vortex size and swings the difference through zero. Then the sentence that earns the row: "Recording this removal matters more than the channel did. It was the one element of the design that was original to this work, and it does not survive"

Withdraw your own code in public, itemised, with the files still retrievable (454) — the most unflinching appendix in the corpus. Two shipped files withdrawn with five checkable findings: an alert-dispatch path naming siren and emergency-alert activation with no interlock, no dry-run default and no operator gate — wiring that stub would connect an uncalibrated detector to public alerting infrastructure; a convergence engine that does not converge, its lowest tier firing on one channel and its next carrying no agreement condition, so the paper's own arithmetic does not describe that program; a declared experimental-channel fence not implemented, the withdrawn channel scoring exactly at the threshold it was promised to be excluded from; a lead-time figure that was manufactured, since the scenario was typed by the author and a scenario built to escalate escalates; and scoring functions retrieving amplitudes by exact floating-point key which no FFT bin will produce — structural evidence that the detector has never been run against real data. And the files remain retrievable from earlier versions, which is why the notice exists rather than being absent

Say what the withdrawal does NOT affect, and what is worth carrying forward (455) — two short sections that stop a retraction from becoming a rout. What it does not touch: the build guide never used that code — reading archived data, siting a sensor, recording a floor and characterising a wind farm are hardware and method, and a builder writing their own reader needs perhaps twenty lines and none of them come from here. What survives: the mockup showed the operator why a tier fired rather than only that it fired — a warning that arrives with its evidence is a different object from a warning that arrives as a colour, and in a paper about trust that difference is the whole point

The first experiment is not the thing you want to detect (456) — the first experiment is not tornado detection — it is measuring the noise floor, which has never been done for this instrument and which every claim below depends on. Free, archived, no hardware, no storm, this week. And the stopping rule is written before the work starts: if the signature is not visible in real archived data at real distances, stop here — that is a result, it should be published, and it saves everyone the rest. Work out what a perfect signal would score through the instrument before replicating: a candidate rung in a triple's boundary layer (probe 13, rests near 7 and 8; row 620) put 0.603 of its rests' time near a whole number, and resampling each rest's own wobble showed that perfectly locked rests would read only 0.44 to 0.53 through that window, so 0.603 sat above the ceiling a true rung could reach, which is what luck looks like; the replication read 0.302 (probe 14, row 402). And every headline carries its floor and its margin above it, with a ledger of which sightings failed cleanly before reaching the book and which results stand far above their floors (the triple-star sessions)

Publish the floor whatever happens (457) — the list is specific — the noise floor, the turbine signature, the correlation matrix, the false-alarm rate over quiet periods — and the reason is one line: a negative result with a floor under it transfers, and a positive result without one does not. Then the closing, which gives the paper's own target away: if this design cannot reach its precision target, saying so with the measurements attached is more useful to the next person than any warning system this paper could have promised

The author has built and tested nothing, said twice (458) — printed in the abstract and again at the head of the build guide: this is a design and a specification, and it is written so that somebody else can build it… this section exists so that someone else can, without needing to ask him anything. And the performance table is labelled design targets to be tested by a pilot, not achieved performance. A specification that refuses, repeatedly and in the places a skimmer will land, to be mistaken for a result

Demote the exciting half to questions, each with what-would-have-to-be-true (459) — the previous version's intervention proposals are demoted from proposals to open questions, each carrying a column naming what would have to be true — and where a budget applies it is computed and comes out unfavourable by orders of magnitude. With the preface: none has a validated causal lever because the underlying science is itself unsolved, and warning and sheltering remain the only life-saving intervention available today. And the organising idea is kept while every proposal built on it is dropped: a lightning rod does not oppose the energy, it provides a preferred path, and its leverage is in timing and geometry rather than power

Decide which layer the failure lives in before prescribing (462) — the physical shortfall is of order 1.7% of world production — not a global shortfall — and the closest historical precedent produced a price crisis through export restrictions and precautionary purchasing rather than through physical scarcity. So the exposure falls on import-dependent countries and on policy response, which changes the instruments entirely: buffers, storage and restraint on export controls rather than agronomy. Prescribing agronomy for a policy failure is a category error that a mechanism diagnosis catches for free

Rank by cost per unit delivered, and the top two need no new supply (463) — five instruments ordered by cost per unit of nutrient delivered to a responsive soil — a denominator that quietly rules out everything delivered to an unresponsive one. The ordering matters because the first two require no new supply at all, and the evidence says they command the largest available multiplier. And the top action is stated as what it costs rather than what it promises: no new supply, no new production, no subsidy, no new technology

Derived here, not quoted — noted on the column it applies to (465) — the source study treats the quantity as a variable in its own analyses but does not tabulate per-soil values, so the column is derived from that study's reported parameters rather than quoted from it, and can be recomputed by anyone who wishes to. A provenance note attached to the specific column rather than to the paper — which is the only placement that survives a reader who arrives at the table first

The deprecated line, terminated with three findings so nobody re-derives them (466) — the paper began as the very thing it removes, and the removal is written so the next person does not repeat the work. Three findings. No selectivity: which element captures a stopped particle follows the atomic-capture Z-law rather than the nuclear scaling that motivated the claim, and in mineral soil the competition is dominated by the abundant elements while iron outcompetes the target on atomic number — the premise fails at its foundation. Penetration: the cheap surrogates act only at an illuminated surface while the target is distributed through the plough layer, so they work in a stirred suspension and do nothing to a field. Delivery: beam-based delivery does not miniaturise, and at facility scale the existing industrial route outcompetes it by orders of magnitude. And the mechanism is real — flooded systems run it seasonally — what fails is every attempt to deliver it deliberately at a cost that competes with what already works

Fermi–Teller Z-law

The fence applied at the source, in the paper that would most have benefited from keeping the exciting version (467) — the deprecated route arrived in soil science from particle-capture work in radiotherapy, and it is removed from the analysis rather than carried as a speculative horizon — with the full derivation and its designed falsification test retained in the first version of the record rather than deleted, so the trail survives without the claim. This is the corpus's standing fence executed by the author, in public, on his own founding idea, in the one paper where the physics angle was the most attention-worthy thing in it

practice; ties row 454

Passing the famous check does not protect against the famous failure (469) — the 1940 collapse is the image nearly every engineer carries of what a resonance check is for — and it is not an example of forced resonance. It was aeroelastic flutter under a steady wind with no matching external frequency anywhere, and detuning that deck away from every driving frequency in its environment would not have saved it. Why this matters is operational, not historical: a designer who holds it as the canonical warning, runs the resonance check, and finds a clean margin will believe the structure is protected against the very thing that destroyed that bridge. A canonical example that teaches the wrong lesson is worse than no example, because it certifies the wrong check

Billah & Scanlan 1991

An inherited multiplier is the most dangerous kind of number in a design document (471) — it is load-bearing. It looks authoritative. It propagates into every downstream calculation. And it is almost never checked, precisely because it arrived looking finished. A multiplier quoted from memory is not a design margin; it is a guess wearing a citation's clothes. And the remedy is structural rather than exhortative: the check specifies a procedure that

forces sourcing and deliberately specifies no numerical multipliers at all. If you want a number here you have to go and get it — from a record you can name, for the site you are actually building on. The corpus's inherited-constants rule, enforced by the absence of the number. An assumed number inside a model is an inherited multiplier too. A toy sailing boat was leaned to beat the true wind on some heading, and its fastest was 1.6576 m/s at 88 degrees, 0.3315 of the wind: its assumed hull drag area, 0.15 square metres, is a barge's, about twenty-five times a clean dinghy's, and by the course theorem the drag angles set the whole polar, so that one unchecked number decided the verdict. With a clean hull of 0.006 square metres the same boat sails 1.1483 times the wind at 84 degrees and makes its best progress upwind, 0.6017 of the wind, at 46 degrees, the course theorem holding at every heading of all nine hulls (row 831), and it first beats the wind at a hull area between 0.02 and 0.01 (runs SAIL 20261018_202451 and SAIL_HULL 20261018_202650). Before registering, check each assumed input that a theorem says sets the answer against a real value (the Loop Atlas sessions)

ties row 240

State a margin's provenance inside the margin table (472) — these margins are engineering judgement calibrated to modelling uncertainty, and they are stated as such. They are not derived from any published code and must not be represented as though they were — with any governing code taking precedence where one applies. And the ratio between the two margins is derived rather than asserted: entrained added mass can be comparable to the structural mass and is not known precisely, and a twenty percent error in it moves the natural frequency by up to about ten percent, the most it can when the added mass dominates, which consumes the entire secondary margin on its own. Print the margin beside every verdict, in standard errors or as a share of the bar: one held prediction sat 1.6 of its own standard errors under the bar, another at about a quarter of it, and the two are not the same kind of held (the triple-star thread, LR1′ and PS1)

A design that does not state what it is not designed for has not finished its analysis (473) — the design-event register separates what it works in every day, what it shuts down above, what it must survive, and — mandatory — what is explicitly accepted as beyond design, stated and accepted rather than omitted. And the absence is usually discovered by the event itself. With the reason the register exists at all: conflating those categories is how structures end up over-built for the wrong load and under-built for the right one

Publish the compliance statement so an omission is visible without an audit (475) — a short fixed-form statement naming modes checked, the lowest margin found and against which source, the instability assessment, the design life and horizon year, the scenario and the reason it was selected, and the date of the next re-check. An incomplete statement is a finding, not a formatting problem. The purpose of publishing the statement is to make an omission visible without an audit — which is the visibility-matrix idea moved from the instrument to the paperwork: make the gap self-reporting rather than discoverable

ties row 312

A document written to be cited rather than read (476) — it is written to be cited by other work rather than read straight through — a design document should be able to reference it, complete the compliance statement, and move on. The worked examples come from one domain because that is where it was assembled, but nothing in the method is domain-specific: substitute your own driving frequencies and your own design events and it applies to a bridge, a mast, a gantry or a rooftop plant room. And the scope statement says outright: it contains no novel physics and claims none. It exists so that two commonest omissions become visible. A model for what a method document can be when it stops trying to be a book

The disclosure block that says what was verified and what was not (477) — tools named for what they did; responsibility for correctness, adequacy and citation completeness resting entirely with the named human author, and no automated system an author of the work; and then a verification performed section that is specific rather than reassuring — the margins labelled as judgement in the text itself, the historical correction checked against the cited paper, and every section pointer verified against the section it names by an automated pass over the source. Then the part most disclosures omit: attribution limits — the references are not yet verified against the publisher record, stated here and in the paper's own limitations section, so a reader meets it whichever end they enter from

practice; ties row 327

The useful output is not a machine, it is a selection guide (478) — four decades of prototypes and very few installations — and a published survey names the cause as a failure to converge: a proliferation of concepts, slow development, device failures and investment losses. A document that helps somebody choose is therefore aimed at the problem the field says it actually has, which is not a shortage of inventions. And the reason a guide is the right object is structural rather than modest: the thing is at least three things chosen independently, by different people, from different constraints — and change the site and any one can flip while the other two stay put

Kept because an idea merely deleted comes back (486) — nine deprecated approaches tabulated with where each applied, why it lost, and what replaced it — and because the reason it lost is site-dependent, the next site may reverse several of these. Every kill carries its number: 15.8 million cycles through a dynamic seal in seawater; 23 millivolts

from a kilometre-long electrode pair, short-circuited by the medium's own conductivity anyway; a Carnot ceiling of 6.7 percent against a device reaching 0.5, and no material improvement rescues it because Carnot sits above it; and one where the barrier is risk capital, not technology, which is a different kind of wall and is labelled as one. An idea that comes back is cited with its failure: the reading "the groove holds because its rider is nearly still at the strobe" returned for a new question after it had already failed once, and it is owed a sharper test, not a fresh start (runs BAND-1b and PIVOT-1) (R2W)

practice; ties the salvage standard

Three lessons that transfer beyond this project (487) — the paper ends by extracting its own portable method into an appendix of its own — which is exactly what this scrub has been doing to the corpus by hand, done here by the author in advance. Check the dimensions before designing the geometry: two deprecated ideas failed the same test, a proposed rule connecting quantities that do not share dimensions, and a dimensionless constant cannot set a mass or a stiffness. Count the conversions in the chain before optimising any one of them: one approach lost not because any stage was badly designed but because there were six of them, and the winning path had one. Check Carnot before designing a heat engine: ruled out before any hardware existed

Attribution stated as a gap rather than omitted (488) — this section is deliberately incomplete and is the next work item. No systematic search has been performed, and it is stated as a gap rather than omitted. Then the sentence that sets the register: almost nothing here is novel as a mechanism, and that should be said first rather than conceded later — with nine mechanisms listed as carrying substantial existing literature and identified originators, each needing finding and citing by name. And the pattern named from a case that already paid: recognising one element as long-established practice saved inventing a relief port that would have been worse. Two search passes moved four of five claims downward, and that is recorded here rather than quietly absorbed

practice; ties rows 230 and 324

A capability is not novel because its parts are listed (493) — nothing in the capability requires an invention. The parts are listed here so that no novelty is claimed for any of them. And then a framing correction made against the author's own interest: the space is not unoccupied, and it would be wrong to claim otherwise — comparable sensory expansion for normally-sensed people has been demonstrated. The version proposed here is where the literature is thinner — and thinner is a claim this document can support, where unoccupied is not. Two words apart, and only one of them survives a referee

State the asymmetry in both directions, then decline to resolve it (495) — surgical risk for convenience is a bad trade against a removable alternative and a straightforward one against total loss — and the asymmetry can run the other way, and that argument deserves equal weight. A large elective market is how a surgical safety envelope gets characterised at all: volume refines technique, drives complication rates down and funds long-horizon data, and small populations cannot fund that discovery themselves — they inherit it. Two cautions attach without defeating it: an envelope characterised on healthy young adults does not automatically transfer to altered anatomy, and one population bearing the risk of learning while another receives the benefit is an ethical structure that deserves naming out loud rather than treating as a happy accident. Then: this document states the asymmetry; it does not resolve it… that judgement is not one an engineering scope is entitled to make on their behalf

Nearly every risk follows from one property: it cannot be taken off (496) — a whole risk taxonomy derived from a single structural fact, which is what makes it checkable. Every governance mechanism that works for a phone — leave it, switch it off, hand it over, discard it — depends on the device being separable from the person. An implant is not, so consent given at implantation is consent that persists, and withdrawing it requires surgery. Eight categories follow, including one most frameworks have no slot for: an attack surface that terminates in a person's perception is a different class of problem from one that terminates in their data

A risk named without a remedy is only an alarm (497) — the governance section is written as things a regulator could require: an output ceiling enforced in hardware and not defeatable in firmware; a physical user-controlled mute independent of software — the implant equivalent of a lens cap; a funded explantation pathway as a condition of approval; prohibitions enforced at the device authorisation layer rather than by policy; and separate regulatory classes for restorative and elective use, because merging them would let the elective case inherit the restorative case's precedents. With the scope fence stated first: a taxonomy of risk categories carrying deliberately no implementation detail — naming a hazard clearly is what enables a rule; specifying how to build it is not

practice; ties row 274

A falsification test says whether an idea is wrong; it does not say what would be built if it were right (498) — a paper that proposes without naming the device leaves that gap, and this document fills it. Nothing here is a design — no transducer modelled, no field simulated, no phantom built — this is the scoping stage that precedes design, and its main product is a list of the things that must be true. With confidence graded by source up front: the regulatory limits are quoted from primary sources and reliable, the physics follows from them, and the biology on which the whole concept rests is unestablished — which is the reason this is scoping rather than specification

Name the gate that must not be skipped (501) — what would settle it, ordered by cost: a distribution obtainable from data already collected at fitting, which gates the entire spatial argument and requires only data access; a simulation that gates the next claim and requires no hardware; and then a safety question that has no direct evidence in either direction — and it is the gate that must not be skipped. Only then does anything get built. An evidence ladder where the cheapest rung gates the largest claim, and the one irreducible safety gate is labelled as irreducible rather than left to be inferred from its position

THE STANDARDS CHANGE UNDER YOU WHILE YOU ARE READING THEM — and this is why the guide exists (503) — a second hazard, distinct from row 502, and the author's own account of what actually stopped the work. The results were fascinating and the existing tools were already good, and the thing that actually stopped us was language… and the standards change under you while you are reading them. So the word does not merely denote two things — the definition is in motion on both sides of the fence, which defeats the obvious remedy of learning the other field's term once. Then the cost, stated by the person who paid it: that is a barrier nobody warns you about and it cost more than any calculation. Not the mathematics, not the instruments — the existing tools were already good. And the response is the origin of this entire document: so we stopped and consolidated. I wrote the methods paper — the substrates, the hooks, the traps — and published it to keep track, and to save the next person. The Field Guide is not a collection of lessons that happens to be cross-domain. It is the document written because cross-domain language is what stopped the work (row 504 names the larger reason, and supersedes this row as the leading one)

THE GUIDE IS THE SOLVER'S SOURCE CODE, BECAUSE THE SOLVER CANNOT KEEP IT (504) — the actual origin, and it supersedes row 503 as the leading reason. Usually a solver is just a program — you write the method once and the program keeps it. Here the method lived inside sessions, and sessions end, so every session re-derived it: badly, partially, differently. The trigger was an architecture Bear could see and name: he was reduced to asking did we try X, Y, Z? and hearing nope — which meant the only persistent component in the whole system was the human. That is not a collaboration, it is a single point of failure with a person in it. So the guide is not documentation of a method; it is the externalised state of a component that cannot hold state. And the format follows from that and was never a style choice: every row carries its price and its name in one cell because a row has to be re-loadable by somebody with zero context — a price with no name cannot be trusted, a name with no price cannot be re-derived. Cross-domain language stopped one line of work; the absence of memory was dissolving the method itself, and it is the larger problem by a wide margin. Inside one session the same holds for its own counts. One sitting's record logged each time the conversation's memory was cut down to a summary, through the third, inside the twenty-second exchange, and none after. When the next came, before the thirtieth exchange, it could be counted only as "at least the fourth", while the person's rules for when to pack the crate and when to watch closely hang on that count, at the fifth. A count that lives in the conversation dies with the conversation's memory; one line in the record, written as each summary comes, keeps it (row 589) (R2W)

Expert witnesses — retained early, qualified before the read, called at the audit, and they accumulate into a library (505) — Bear's correction to the exterior-witness rule, and it is a better structure than the one it replaces. They are expert witnesses in the courtroom sense — professional, independent, retained in advance and qualified on a specific question before they testify. So they are introduced early, when the project is being scoped, for their purpose, value and role; and they take the stand later, at the audit, where an independent voice is worth something precisely because it was lined up before anyone knew what the answer would be. And the part that is new: they become part of your holonomy library. A witness roster is an asset that accumulates across projects, not a scramble at the end of one — which means every project you finish leaves the next one better witnessed

practice; ties row 306

Prior art belongs at the wall, not at the idea — and the trade cuts both ways (506) — Bear's overrule, and he is right for a reason the session had missed: it had placed the prior-art pass early, on the field's reflex. His ruling: don't chase others. You may discover the journey is more instructive than the shortcut. The method teaches through the long walk, so when you find the shortcuts they become tools and choices — that is how the project builds value. Run it too early and you inherit the field's route, including its blind spots, and you skip the walk that teaches you the geometry. Then the other edge, equally real: if you run into all the old walls from the same direction as the priors, you waste time. So the trigger is hitting a wall, not having an idea. Two different jobs at two different stations: the incumbent check is early and asks what bar I must clear; the prior-art pass is at the wall and asks whether this route is already mapped. Read the paper before the paper sets the test: an abstract pass placed one paper as an edge's explanation and another's picture as a test's target; the full reads showed that the first is the mirror case and the second lives in a two-dimensional section, which added one registered test and withdrew another's meaning before its result. And a thing rebuilt from the inside may then be found in print: a harmonic, its resting point and its splitting were rebuilt from the triple star's runs without knowing Mardling's paper, and the full read turned four measurements into checks of her formulas and a map of where her pendulum holds: her rate–width relation held, the small-swing rocking half the island's half-width within 1.1 per cent at the pair's eccentricity 0.10, while her widths ran 20 to 31 per cent wide, the relation failed for the held island from the pair's eccentricity 0.15, and the centre doubled between the star's eccentricities 0.348 and 0.350, where her pendulum puts it at 0.41 (row 555 for the sizes her strength gives) (the triple-star sessions). Bear's ruling of 16 October sets the rule beside its contrast. Where a field

has live data, as geodesy has its GPS stations, a model can be tuned against the world and driven through fresh data from the start. A toy has no such field: it is built from scratch, and its only data are its own runs. So a toy is built from first principles and its own measurements, and nobody else's work is copied in; that work is met far down the road, at a wall or in a prior-art pass, where it shows where the toy landed on its own and who was there first (Bear's ruling). A prior-art pass can miss a whole neighbouring field: the triple-star thread's pass of 17 October (TRIPLE_PRIOR_ART_SEVENTH_2026-10-17) found that the islands' pass of 8 October had missed the circumbinary literature (Holman and Wiegert 1999, Shevchenko 2015, Popova and Shevchenko 2013, and Gallardo, Beaugé and Giuppone 2021). Count such a miss as a flag (row 783), and reproduce the earlier work's own example before using it (row 650): a new tool, triple_hw_edge v1.0 (run 20261017_234929), first reproduced Holman and Wiegert's own example, an edge at period ratio 6.878, and then put the pair's edge at period ratios 4.03 to 5.36. A cold review of the new sentences found one error and six overstatements, and the reviewer's own "seven of the nine" was itself off, since the paper uses nine of eleven: check the reviewer's numbers too (the triple-star sessions). And when the data meets a known exact result, read that paper before building more: on the Loop Atlas's noisy needle, Reimann and co-authors' exact formula settled in one reading what three of the project's tools had circled (the Loop Atlas sessions)

practice; ties row 93

Flucht nach vorn — failure is the main road, and the method hands out no participation trophies (507) — the correction that reorganises the whole walk. The null was first drafted as a branch off the successful path. No — the entire guide exists because holonomy delivers failure routinely, and the method demands a new partnership with failure as the teacher. Not fall back: escape forward. Everything is a lesson. And the clause that stops this from being consolation is the one that matters: the method does not slack and does not give out participation trophies. A lesson is not conferred by having failed — it has to be extracted, deliberately, by looking for specific things. The shortcuts, the easy buttons and the geometric cross-checks nobody teaches are real and they are earned, which is exactly why the long walk is not wasted

Do not stop when you find an answer (509) — Bear's own margin note, and it is the first rule of the walk rather than a caution inside it: do not stop when you find an answer — you will never find the other patches and combs when you quit looking. A found answer is the most reliable way to stop searching, and the structures this method reads are plural by nature: patches come in families, combs have teeth beyond the first, a lattice has more than one coincidence angle. The instrument keeps working after the first hit, and almost nobody keeps looking, which is why the second and third findings in this corpus were so often the better ones

Each half is known; the equivalence and its conditions are the contribution (511) — written into the seed before any draft exists, as the must-do before submission: check the trust-side impossibilities, check the curvature-side defect theory, and expect that almost certainly each half is known in its own field. Then the sizing, in advance of finding out: the novel object is the bijection between them, plus the iff conditions, plus the adaptive-orphan observation — and if a published cross-field equivalence already exists, the contribution narrows to the conditions and the residual, still worth a note, but know it first. A paper that has already written down how its own claim shrinks under prior art has pre-paid the most expensive correction there is. In a prior-art pass, look for the general model that holds both of yours as special cases, since a symmetry between two models is most likely stated, if anywhere, by whoever built the model containing both: for the Loop Atlas's mirror cats eight sources stated no mirror, and Iwai's two equal cylinders on a joint that allows any twist (RIMS Kôkyûroku 1692, 2010) is such a containing model, his book of 2021 the likeliest place left

practice; ties rows 230, 324, 488

Two fences, and they are not the same fence (512) — **Bear's Note:** correction 2026-09-13: one fence is disownment; another is readiness. They are different fences and the distinction travels. That material is out because the physics underneath it is not trusted, so it is never cited, never carried forward, and any reappearance is raised at volume. The other fence is readiness and ownership. That material is ours or a partner's, it is fenced because it is not ready for the world and in some cases because the project belongs to other people, and the ruling on it is track it, keep working it, and remove every reference from the public record until the engineering documents are ready. Collapsing the two would be a real error in both directions — it would either publish something unready or bury something merely unfinished

A working instrument that annotates its own vocabulary collision (513) — found live in a control-room simulator, in a note beside the control it applies to: this sweep moves the drive in time; the other person's term is a velocity gradient on the bunch — cousins, not the same object. Row 502 caught in the act of being obeyed, by an instrument rather than a paper, at the exact place a user would otherwise carry the wrong meaning across. And the same tool labels its own epistemic tier in the interface: these are display toys calibrated to session anchor numbers, not engineering simulations. A vocabulary fence and a confidence tier, both placed where the hand is rather than in a preface

ties rows 502 and 327

Established mathematics beats conjured mathematics, because you can go and check me (516) — the right answer to a compliment, that the ideas being created are more fundamental than physics itself, and it declines the compliment on purpose: this is not more fundamental than physics, and it is not being created with anyone's words — it is established mathematics with names on it. Then the reason that is better rather than deflating: it means you can go read it and check me. Then comes the humbler and truer version of the connection: one piece of mathematics turns up in two distant fields because they share a skeleton, not because one sits underneath the other. And a small discipline in the same breath: the eigenvalues do not want to spread; they repel as a consequence, and no intention is needed

Dyson 1962; Wigner; the Vandermonde determinant

Do not hold the build for a late ornament (524) — **Bear's Note:** assembly ruling, 2026-09-13: more papers will finish while the guide is being assembled, they will not change its shape, and they are interesting ornaments rather than structural ones — so they arrive as an edit if they are late. A structure that waits for its last decoration never ships. The test is whether the incoming thing changes a limb, a station or a fence. If it does, stop and fold it; if it only adds a row, the register takes it afterwards and the build keeps moving

The fence in the paper was correct; the conversation violated it (525) — the cleanest instance in the corpus of a correction aimed at the right target. A topological invariant requires a periodic lattice with a well-defined Brillouin zone, and the object here is finite, irregular and open, with boundary everywhere and no translational symmetry — so the invariant is not defined on it. The deposited paper had already said so, in its own words: finite irregular patch, kinship, reported qualitatively. That fence was correct. The session discussion violated it. So the correction is scoped precisely: nothing in the deposited paper needs revision; this addresses session framing only. A paper can be right while the conversation around it drifts, and knowing which one broke is the whole of the repair

TKNN; Brillouin zone

Correctly measured and wrongly interpreted (526) — a companion lattice was recorded as reproducing the comb at a gap ratio of 1.297. The number was right. The reading was not — a gap ratio is not the comb's signature; the divisor structure is. Tested properly, the open version produces scattered teeth that divide nothing, which is noise in an extreme-value statistic, while closing the chain produces the divisor set exactly. The most durable kind of error: a real measurement pointed at the wrong claim, which no amount of re-measuring would have caught because the measurement was never wrong. The comb line's lower value, about 0.75, later looked like exactly half the latch's three-halves — a rhyme between two magnitudes, neither of them an invariant (row 548)

practice; ties row 336

Three cherry-picked points, replaced with the whole set (527) — a claim that a feature strengthens with size rested on three points. The full set is six patches: four sizes reading 1.20, 3.04, 1.55 and 1.93, plus rectangles at 2.19 and 3.54 — scattered, not monotone. And the replacement is written at the strength the data supports: elevated on every open patch measured, and its height does not trend with size. Three points can be drawn through by any curve you already believe in, and the repair is not a better fit but a weaker sentence

The result exists only as a contrast, never as a property of one curve (530) — said about the apodization families and it is a reading rule with wide reach: any single pattern read alone would mislead. The separation grows monotonically across all six weight levels — stronger evidence than any single depth — but the finding is the gap between two families, not a number belonging to either. A result that only exists differentially has to be reported differentially, and quoting one arm of it is not a simplification but a different and false claim

Every bridge must let you build, predict, or break something (533) — the operational filter, stated in the front matter and earned from the author's own earlier framing, which was correct but underspecified in a way that permitted the misreading this paper corrects. Observations that produce only a feeling of connection are out. A cross-domain programme's central risk is the bridge that feels true and does nothing, and this is the cheapest possible test for it — three verbs, applied before the bridge is written down rather than after it is published

practice; ties row 7

I thought it was a destination. It is a method you could teach kids in a sandbox (538) — My own correction to my founding expectation, 2026-09-15: I thought it was a destination. A super-app or program that I could point at holonomy or geometry problems and they would magically give up their secrets. Instead it is a method you could teach to a group of kids around a sandbox of toys. It is the sum of the experiences. It is the purpose of the book. And that correction is more valuable than the destination would have been — because a method transfers to every new substrate, and a destination is just a place you arrive at once

The four limb chapters — named 2026-09-15 (539) — L1: The Method as an Instrument. The walk, the channel, the floor, the differential read — what you do, and the instrument you do it with. L2: The Geometry & Substrate You Walk. The fiber bundle real estate, the vertices and junctions, the winding theorem, the trunk — the terrain the method traverses. L3: Holonomy You Write. The write limb: creating invariant loops, acoustic marking, cured helixes,

the things you build rather than read. L4: The Ledger. The rows of what it cost, what it taught, what went wrong, and what it yielded — the price-tag discipline applied to every project in the corpus. These four named because every row in the register needed to know which house it lived in, and the names needed to carry the guide's own character rather than a taxonomy someone else would recognise

A selftest failure is a finding when the code was right (541) — the dipole test asserted an alternating ±1 field would hold net near zero. It didn't — worst residual 7.0 while the median boundary was 30 edges. The code was right. Cancellation is real, but its residual scales with the perimeter, not the area — a boundary effect, not a breakdown. Assertion corrected, code untouched. The test was wrong because the model it encoded was wrong, and correcting the model is what the failure was for. When a check fails only at the edge, audit the check before trusting or dropping the rows: a per-setting check failed at the slow settings because a finite-difference route disagreed with an integral route by up to 16 per cent; doubling the resolution changed nothing, and an audit with the root re-bisected in place showed the finite difference closing on the integral as its step shrank (9.3 to 0.8 per cent), then drowning in rounding. The rows stood; the check's fixed step was at fault (run JJ-CORNER-2) (R2W). A coefficient check that misses by a hair may be missing a term, and the term found can be large: P0 stopped triple_harmonic_recipes v1.0 (run 20261016_204410) when the [7:1](0) member missed by 7.6×10^{-6} of itself and the hexadecapole term's imaginary part was 9.1×10^{-12} against a bar of 4×10^{-12}. The check had assumed the angle comes only with −2 and −4 times the apse difference, and so left out the mirror terms; the scratch look made after the miss also showed a fourth hexadecapole term, [7:1](4), at 45% of the hexadecapole holding term at the star's 0.25. P0′, with eight apse differences and all five frequencies, passed (run 20261016_204642). When a projection check fails by a small amount, read out every frequency the projection holds before loosening the bar. And when a selftest fails, test the check before touching the data: twice in one triple-star stretch the data were right and the check was wrong. triple_followups' selftest said probe 33's lock births sit 0.001 apart, but at the star's 0.20 and 0.24 probe 33's windows ran at 0.002, as its provenance row says; triple_followups2's selftest asked for a pure fifth within 10^{-9}, but the reader's own floor is 2.6×10^{-9} over 1,000 passes, falling as 1/N. Both were mended and recorded before the tools first ran. A failed check is a claim about the check as much as about the data, so read the provenance before changing either (the triple-star sessions)

practice; ties row 402

A harder solver tests the solver, not the quantity — and only if it changes one thing (549) — the latch paper's first version checked its headline exponent against a stronger relaxation and read the drop from 1.55 to 1.25 as the exponent sliding with the strength of the solver. Its second version rejects that reading. The stronger run changed three things at once — the minimiser, the point count (72 per strand against 160) and a spring holding each segment near its starting length — and the energies were not converged in the point count: the pair linked six times starts at 22.37 at 72 points, 17.51 at 160 and 16.38 at 1,280, so part of the stronger run's larger drop was a coarse curve closing on the converged value. At linking one and two it even ended further above its reference than the crude run did, not closer. And neither run could have found the real trouble, because both relaxed the same starting shapes and inherited the confound upstream of them (row 550). Two rules for one price: a validation run changes one thing, or it is several experiments reading as one; and passing a harder solver shows that the machinery is sound and says nothing about what the quantity means

resolution walk; ties rows 35, 108, 399, 422, 556

A staircase across an integer measures whatever else climbs with the integer (550) — two loops linked one through six times, each pair relaxed, the energy rising in a tidy staircase: 9.34, 10.13, 11.33, 12.91, 14.84 and 17.15 against 8.82 for the unlinked pair after the same crude relaxation, reproduced to every digit. It looked like the integer setting an energy floor. It was the rope. The pairs were wound on a torus of fixed size, so winding more times laid down more length — 13.14 per strand at linking one, 25.93 at six, against 12.57 for the reference — and the relaxation hardly moved them, by 0.1 to 2.1 percent. Before relaxation the rise to linking six is 8.68, and 6.70 of it (77 percent) is the strands' own energy, which depends only on each strand's shape and length; 1.98 is the cross term between them. Held at equal rope, the energy falls as the linking rises: 9.36, 8.78, 8.13, 7.59, 7.19 and 6.94 against 9.34 for the unlinked pair rescaled to the same strand length, 13.14. Shrink the six-times-linked pair to half size and it sits at 6.81 with its linking still reading −6.033, below the unlinked pair's starting energy, 8.83. The bare filament sets no floor, and the paper's second version withdrew the claim. The rule is cheap: when you vary an integer and read a magnitude off the staircase, write down everything else that changes with the integer and pin it, or you are measuring whichever confound you forgot. One line of thought would have caught it before deposit — hold the rope fixed and re-read the staircase — and it reverses

Linking Is the Latch v2 (concept 22906667); ties rows 138 and 532

A check certifies only the question it could have failed on (551) — two papers found this the same month. The latch paper shrank a linked pair 16.7-fold and watched its linking hold at −1.0000, and relaxed its pairs and watched the linking read the same before and after. Neither test could have failed: a uniform shrink cannot change a Gauss integral, and no relaxation move came near a cut — the largest move of any point was 0.11 in the crude run and 0.24 in the stronger one, while the strands never came closer than 0.59. So both check the reader and the minimiser, and

neither checks the latch. The forest suite's genus flux-ladder printed its own verdict, THE LADDER FAILS, and it was right: it counted one flux loop on a sphere that has none and three on a torus that has two, because it counted cycle rank rather than genus, and its certifying check compared a spectrum before and after a gauge transformation, which cannot differ. Its repair, the homology ladder, returned 0, 2 and 4 across the genus ladder and shipped. The same gauge invariance is a sound receipt in row 170, where it certifies the apparatus rather than a count. A superseded version of the same tool leaves the crate, with the cut and its reason named in the manifest; a broken artifact that is itself the evidence for a finding stays in, as row 78's does. A third specimen: a test for chaos in a relay "held" over 672 runs, but the outcome was guaranteed before the run, because a one-way chain of massless phase riders has zero topological entropy and cannot be chaotic at any setting. Ask what the structure guarantees before running (the Loop Atlas sessions, R2W). A check built from the formula it tests can fail only through its other parts: a toy that solved a brake's stopping distance with the very formula under test "missed" by 0.0005 m, the size of its flight's step, and could never have tested the law; stepped with the force law instead, it matched to six figures (runs DAMPER_LAWS 20261011_051800 and 20261011_051848) (the Loop Atlas sessions, the book). A deposit check must know the marks the draft actually uses: a placeholder check read a fixed list, the draft marked its open results with angle brackets the list did not hold, and a build carrying them would have passed. Plant the document's own marks in a check and watch it fail before trusting its clean verdict. The check's next version reads the draft's markdown for those marks, with a self-test case for each (lint_deposit v1.6, 14 October) (the triple-star sessions). A grade must fail when the rows it grades are missing: a reader that started each verdict as true and turned it false row by row printed F2-a and F2-b both HELD on the reported children, a separate file holding none of the graded rows; the graded read was unaffected, because its file holds 5/8. A verdict begins false and becomes true only when every graded row has been found and passed, and a dry test includes a file that lacks them (run FAREY-2) (R2W). Every verdict needs one stand-in on which it should pass and one on which it should fail: R3-0 printed FAILED (320 of 352) because its reader also counted RATCHET-2's 32 rows at weight 1.28, outside RATCHET-3's ladder. Graded by its registered words with a saved script, it held, 320 of 320, and the printed verdict stands beside it (row 683). Its dry run had used a stand-in on which R3-0 was bound to fail anyway, so it could not show the fault (R2W). A second specimen: a prediction that weight's half steps would break into three or more patches held with twelve, but every patch was one or two grid cells, which is what a grid of step 0.0125 makes of any hairline (R2W, CONE_SURPRISES S2). The verdict was recorded as held as written and not as evidence: say in the verdict when a held prediction had no way to fail

gauge invariance; ties rows 78, 170, 287, 337

A citation checked for existence is a claim you have not verified (552) — the first prior-art pass found the right field and credited it generously — Arnold's inequality, Moffatt's identification of helicity with linking, Freedman and He, the ropelength construction — and still credited Freedman and He with a proved three-halves energy scaling they do not prove. Read on the pages themselves, their "3/2-energy" is the name of a quantity, the integral of the field strength raised to the three-halves power. The floor they prove on the ordinary energy is a constant times the tube's asymptotic crossing number, divided by the cube root of its volume. Klotz and Thompson's three-halves is a rope length growing as the number of helices to the three-halves power in a construction, which gives the three-quarter power of the crossing number, and it is not an energy law at all. The citations existed and matched their abstracts, and the misreading rode in dressed as a citation, which is what made a false rhyme (row 548) look plausible. Credit the theorem that exists, at the level it is written. Citations named in conversation are not yet checked: one sitting checked a fractal-basin paper against the authors' hosted copy only and named four more references unchecked, and the next listed a dozen named from recall, Feigenbaum's paper among them, while an identity it needed was checked numerically instead of cited (largest difference 4.77×10^{-14} up to $N = 60$). Where a record API is blocked, a concept DOI rests on the resolver's redirect to the latest version and the author's paste of the version list, and says so (the Loop Atlas sessions, R2W). Check what a paper says against the paper, sentence by sentence, not against its record alone: of eight statements a draft cited to one paper (Mardling, 2013), six matched and two had taken a nearby sentence for the claim, a rule of thumb (both mass ratios above about 0.05) written as the formal domain, and an assumption about small eccentricities written as one that the ellipses' turning never matters. A record check would have passed both. The same pass found a list of six papers that was another paper's reference list in its own order, and a citation made by a description that three papers fit (the triple-star sessions' citation check of 14 October). Read an author's notation against the author's paper. Mardling's harmonics are written [n′:n](m), the outer count first. Read against her paper, the islands paper's angle is hers with the sign reversed, which changes no result; the neighbour it had named, [N+1:1](2), is not resonant; and an appendix's [14:2](2) is Batygin and colleagues' 2:14. The deposit lint then had to learn to read the notation as notation (lint_deposit v1.7, run 20261016_044440) (the triple-star sessions). Carry a paper's numbers, and check its words against them: Tong and Dullin (row 814) write that a real dive's geometric phase is less than one degree of rotation, while their own value, 0.0721 rad, is 4.13 degrees and 0.34 per cent of the dynamic phase (0.1705 times L = 122.756, 20.93 rad), so "less than one per cent" may have been meant. Their other figures check by arithmetic: the five sub-loops sum to 0.0721, reoriented to 0.1185 against 0.1186 printed, a 64 per cent gain, and 0.0189 rad is 1.08 degrees. Cite the number, flag the sentence, and do not build on the sentence (the Loop Atlas sessions). When the paper a claim leans on is behind a paywall, read the free papers that cite it: each says in its own sentence what the locked paper is cited for, and its reference list may name a nearer source for the narrower claim. Mark what they say about the locked paper as secondhand, and narrow the claim to

what they carry. The Loop Atlas could not read Kautz and Monaco (1985) for nothing; two free papers citing it, read whole, carried chaos in the underdamped radio-driven junction below its plasma frequency and pointed to period doubling within the steps of long junctions in two other papers, and nothing was bought (the Loop Atlas sessions, the prior-art pass of 30 October)

Freedman & He (Annals of Mathematics 134, 1991); Klotz & Thompson (2025); ties rows 99, 328, 502

An edit is a claim, and only the built artifact proves it landed (553) — the latch paper's first version shipped a Figure 5 whose image plotted an earlier run that was never deposited, while its caption gave the deposited numbers; the source said the right thing and the page showed something else. The replacement is drawn from the deposited runs by a deposited script. This guide carried the same class of defect into its own sixth version, whose opening promised a fourth figure the deposited PDF does not contain. None of it is visible in the source or in a build that exits cleanly. Build it, extract the text, look at the figures, and assert that the new words are present and the old ones gone, because a figure showing a stale run makes every reader honestly wrong about it. The same lesson arrived a third time as a sequence. A send ran alongside a script that had failed and shipped stale files, and another script built its changes and never wrote them; reading the built files caught both, but only afterwards. Build, read back, then send: three steps, never run at once. And every number in a ledger row is read back against its run after it is written, not only the ones that feel risky: that reading caught three wrong figures in the assistant's own prose, a "4 per cent" that was 7, a "0.18" that was 0.00 and a "1 per cent" that was 1.6 (the Loop Atlas sessions). Count the pictures after every build: six figures never came across with a crate, and every toy-runs PDF rebuilt across eighteen shipments had six empty boxes until a crate check compared embedded images against figure lines (the Loop Atlas sessions). And the build itself can change the words: this guide's collection of reads was built by a converter that turned a typed double hyphen into a dash inside a quoted read, caught only by reading every read back against the PDF, character for character (the Field Guide sessions). Builders drop things without a word: a PDF builder working in its own folder found 1 image for 8 figure lines until built with full paths; it did not wrap a fenced block, so the author's long lines ran off the page for four days; a vertical bar inside a table cell dropped the rest of the row; and a table above a file's first heading printed as raw pipes. A read-back that forgave curly quotes, dashes and ellipses could never have caught a converter's typography, which a strict read then found in 85 of 1,141 pieces (the Loop Atlas sessions, the book). And a build can carry too much: a build of this guide's list of changes copied a private working section of its source, because its slice of the ledger ran to the end of that file; it was caught by reading the built file back, and the slice now stops at the next heading, with a check (the Field Guide sessions). A label the builder reads as a link disappears without a sign: Mardling's harmonic labels written as [N:1](2) printed as "N:1" and dropped "(2)" in every PDF that carried one, 25 labels across five files for two days, until a read-back of the built text, piece by piece, found it; with the brackets escaped, every one prints (the triple-star sessions). A fourth specimen, and more: a ledger's table ran wider than the page and wkhtmltopdf clipped the right end of every line in its widest column, about 1,600 words, unseen for days until the PDF was read back by counting its letters and digits against the source; a book's PDF dropped its figures when built in place; pandoc read a line opening "1)" as a list and dropped the marker from a PDF of the person's own words; and pipes inside table cells would have cut two ledger rows short. Build wide tables at fixed widths, escape brackets and pipes, count every built file back against its source, and count each zip's files against its manifest after unpacking (the Loop Atlas sessions, the book). Five build traps: an unescaped [N:1](2) prints as a link and loses its "(2)"; a placeholder in angle brackets prints as nothing; a picture the builder cannot find prints as a 16-by-16 icon, not a gap; a vertical bar inside a table cell splits the cell; and a table row taller than a page is cut off by the plain stylesheet, where the flow stylesheet lays rows out as blocks. The read-back (readback_check v1.3) catches the first four; for the fifth, the triple's pattern-recognition file now builds with the flow stylesheet, which reads back 328 of 328 where the plain one lost three cells (the triple-star sessions)

practice; ties rows 157, 192, 194

Right size, right sign, wrong shape: a mechanism can pass two of three and still not be the whole cause (555) — a one-layer fit whose Faraday (magneto-optical) terms are too weak turns the arrows of a sunspot map by 3.6 to 9 degrees, flipping with the field's polarity, as the maps' turning does. Planted into the real maps with the paper's own arithmetic, that curve leaves an rms current offset of 2.46 mA/m² against the one-sign ramp's 1.05, because the model's turning crosses zero near 1,000 G and the maps' offset in the 1,000 to 1,500 G band is +2.53 where the curve gives −1.31. Across every plausible Doppler width, 25 to 50 mÅ, the crossover moves from 687 to 1,240 G and never below 500. So the mechanism has the size and the polarity and not the shape, and the verdict is row 325's: a kill may be a rename, from "the mechanism" to "part of the mechanism". A test of size alone would have passed it. A mechanism for "moves little" is not a mechanism for "moves with": a lever explained why a leap's start moves 44 to 128 times less than the other mirror step, but not why it moves at nearly the frozen point's own rate (0.095 against 0.104 turn per unit drive), which is what keeps the gap steady; an independent check caught the overclaim. Insensitivity and tracking are two claims, tested separately (run CATCH-1) (R2W). The mirror case: a model with nothing fitted can be wrong in size and right in shape. The second fundamental model of resonance, with Mardling's strength, kept the pendulum's sizes (the island 33% too wide, its centre 25% too fast) but gave, with nothing fitted, the island's lopsided shape (2.10 to 1 against 2.18 measured), the star's mean eccentricity falling with birth, and the faster low flank at every matched swing (runs 20261015_082428 and 082744). Report size and shape apart; a

separate agent's cold review added three qualifiers before it was written in (the triple-star sessions). A lean that carries a mechanism is graded twice, once on the verdict and once on whether the mechanism's own number came out. FLAT-12's lean said the small dial's error would keep its sign and grow, because the held probe now stood far above the flat point. It held: $+3.106 \times 10^{-5}$ against the quarter swing's $+2.575 \times 10^{-5}$. But the stated reason predicts a size, and the size is wrong: the probe's lead grew four times (2.34 to 9.30), so an error set by the lead squared would have grown about sixteenfold, and it grew by a fifth. The second grade is a reading, said aloud, and the mechanism is not credited (row 683) (R2W)

Dancing on the Sun; ties rows 7 (mechanism versus correspondence), 246, 325, 551

Two readers of one light are not a witness (556) — fitting HMI's real Stokes light with a textbook one-layer fitter, the arrows differ from HMI's by −5.04 and −3.88 degrees in strong field and +2.28 and +1.75 in weak field, an S-shape crossing near 1,000 G, in both regions alike, and the disagreement grows monotonically as the fitter's Faraday terms are weakened (−12 at half, −29 ignored) while the full-term fitter fits the light best. That is a real finding about the two readers and none about the truth: the difference between two fitters is not the difference from the Sun. Applied to a magnetogram, row 306's rule makes the external witness for a turning a second instrument on the same region, not a second fit of the same photons. Recorded so that a two-fitter disagreement is not promoted to a measurement of bias

Dancing on the Sun; Station 7; ties rows 106, 306, 424, 549

Two estimators of one thing are not a disagreement, and the ledger must say which one it used (558) — the paper's 5.5-degree turning (rms 0.76) was fitted on two regions' median offsets, ten numbers; the planting tool fitted the same ramp on every band-map, 29 numbers, and got 5.30 (rms 1.05) on the same six maps. Both are right. A number that changes when the estimator changes is not wrong; it is under-specified until the estimator is named beside it. The same read found a section-4 median resting on a band-map of 11 pixels of each sign with an offset of +22.14, which the 50-pixel rule the other tools use would have excluded (median +4.79 with it, +3.82 without): row 544's degenerate bin, printing and voting. Name a mean a mean. A letter reported a "worst residual" of 0.0011 that was the largest of the runs' means; the worst single loop was 0.0064 (the triple-star sessions, caught by the Loop Atlas sessions' re-reading). The same "worst residual" was repeated in a reply before the run files arrived: a number that leaves its run without its label (mean or worst, signed or length, sampled or full) becomes a different number (the Loop Atlas sessions). A second specimen, from the triple star: one resonance angle's slide was quoted as −0.1254461 of a turn per pass end to end, and read by a fitted line or by the median step as −0.11175 or −0.10572. All three are right, and they differ because the angle's single steps fall in three bands, the slide itself and the tide's flips of about half a turn either way; 87 per cent of the steps off the slide come where the pair's eccentricity, read at closest approach, is below 0.03, against 38 per cent of all steps (run triple_heavy_gap_predict 20261010_202145). On the pair's mean ellipse the same angle drifts −0.2125 of a turn per lap (run 20261009_025920). Name the estimator beside the rate, and where its steps come from (the triple-star sessions). One rate quoted end to end at one birth (−0.1254461 turn per pass at 7.200) and as a median single step at the next (−0.1135559 at 7.210) differ by the jumps between them. A rule built on the one missed when tested on the other (the lattice's commonest gap was 399, not 44), and, read off both angles' median net turns over each gap, it held at two fresh births (runs triple_heavy_gap_predict 20261010_202145 and triple_heavy_gap_netturns 20261010_204936) (the triple-star sessions). And set a gap against its error with one estimator: a pooled ratio's gap measured against the spread of per-sample ratios read 2.6 standard errors on 48 soups, and with one estimator for both (resampling the soups) and 192 soups the same gap read 0.66 (run TWO_HANDS_2 20261017_204131) (the Loop Atlas sessions). And when two estimators read one quantity, judge with the narrower. A noisy needle's long-run spread over mean, its Fano factor, can be read from the growth of its count's variance between two times, or from the squared spread of its waits between slips, which reads the same quantity for a needle that renews at each slip. On 2,048 needles the first carried errors near 0.035, seven times the second's, and a lean judged on it missed in one row of three, 0.8434 ± 0.0349 against the exact 0.9160, 2.08 errors out, while the waits' spread lay within two of its own errors of the exact value in all three rows (read after the run); the redesign registered the narrower reader (run SLIP_RHYTHM 20261020_083426) (the Loop Atlas sessions)

Dancing on the Sun; ties rows 544, 570, 612

A pre-written expectation corrected within a minute, and the run restarted rather than annotated (559) — the width run's docstring predicted crossovers of 560 and 1,120 G from a scaling slip (764 G per width instead of the 943-G measured crossover). Caught seconds after the run started, the run was killed, the expectation corrected to 690 and 1,390, and the run restarted from nothing (the killed start is named in the paper's manifest as not a run of record). A written-before-the-run expectation is worth exactly as much as its being written before the run; an expectation edited while the run runs is worth nothing, so the only honest repair is to start over and say so. The corrected expectation then stood beside the measurement: 690 and 1,390 G expected, 780 to 821 and 1,162 to 1,240 measured, the same way and about half as far (row 557). Registered numbers are computed in code before the run: worked by hand, they carried rounding slips (0.063173 for 0.063147; 0.27523 for 0.275220) (the Loop Atlas sessions)

Dancing on the Sun; pre-registration; ties rows 369, 557

Redistribute, never cut (569) — Bear's rule for the Latch Atlas's end notes, and it held for 21 notes and one 1,862-word paragraph: every clause found a home in the body, a caption, the reference list or a two-line note, and a script probed each piece into its new place, zero missing every time. A long note is a sign that its material belongs somewhere more useful, not that it should be shorter. Nothing is too tall or too short for the ride; it is re-binned. Moving or breaking text is checked word for word: a long paragraph and a 1,600-word answer were broken by headings and paragraph breaks only, the words read in order before and after (25,371 and 25,371, triple_draft_breaks v1.1), and a passage moved from the open questions to the section that claims it was checked character for character, the rest of the draft identical but for five pointers (triple_draft_move, run 20261016_065722). A paper also names each figure in the paragraph that discusses it; the first trial build showed the body had named only one (the triple-star sessions). A whole paper can be redistributed by script. The triple-star islands paper's form was measured first (52 pages with no image, a 406-word abstract where MNRAS allows 250) and tried on a three-page sample before anything moved; then the draft of record (d0be58eb767a8fcc) was split into 181 blocks, the 122 that are not headings were tagged by audience, topic and what each carries, and the paper was reassembled by script (split.py, assemble.py, check.py), which found all 1,097 sentences of the draft (the thread's own summaries say 1,105 of 1,105, and its records do not settle which), none missing, and no number the draft did not print. Every later change went in as named new text. A script proves only that nothing was lost, not that the moved text still reads: a cold reader found pointers left behind and a 'So' in Section 8 that had lost its evidence. The default publishing packet (PUBLISHING_PACKET_2026-10-22) carries the same rule (the triple-star sessions). The breaking of Section 10 was registered like a run: two of its expectations missed, because Section 10 had hidden a 1,600-word paragraph inside its list, and both misses were said aloud and redesigned. An edit's expected result is written down before the edit, as a run's is (the triple-star sessions)

The Latch Atlas; ties rows 88, 486

Every number names its run, and the tool that strips them counts them before and after (570) — "numbers before 930, after 930, differing none; run stamps left in body: 0" is the sentence that let a 92,000-word book be rebuilt forty times without a number drifting. A provenance rule is kept by a count a script prints, not by good intentions. A number printed only on a page is untraceable even when it is true: a triple's 0.909 of one run's rest time near a whole number was right all along and could not be found, because no run file printed it, and the fix was a trace tool with its own ledger. A presence check that looks each number up anywhere in the crate passes short numbers by chance: 591 of 594 decimals on one page were found, yet a "4.5% to 7.2%" passed because those digits occur in unrelated files, though it came from a look typed into the shell and never saved. The sharper check looks for each number only in the files of the run its sentence names (the triple-star sessions). A number with no run file of its own is recomputed from a saved run before anything is built on it: one result had reached the project's record from a script run in the conversation, and recomputed from the saved series with its rule unchanged it came back to the last digit, a check of seconds against the cost of a whole set of runs built on an unchecked number (run triple_nb2_recompute 20261014_054401) (the triple-star sessions). Provenance is a claim too: notes gave a result a provenance the record did not support, and the line where its calculation was first written said otherwise; an independent read caught it. Before saying where something came from, find the line where it was first written (R2W). Scanning every distinctive number in the islands paper against the runs' own ledgers found a returned run never filed in the crate, five readings made by one-off scripts never saved (all recomputed by a saved tool and reproduced), and three numbers to correct (4,121 to 4,122; 0.86 to 0.85; three escapes of 81, not of 69); at the paper's close on 16 October the scan read 793 of 797 numbers printed by a ledger, the other four arithmetic on them (run 20261016_162440). And a check's own printout can make a number look traced: the provenance table's ledger prints each row's description, and the scan reads that ledger, so a percentage written in a description traced to the table though no run printed it, and others traced by coincidence to unrelated ledgers. Anchor every percentage to the fraction its run prints (0.9937 for 99.4%) as a run-and-draft pair the table checks, and retire superseded table ledgers before scanning (runs 20261016_162345 and 162440) (the triple-star sessions). Fill every machine value in a handover by script: at one R2W sitting's close the documents filled every md5, line count and byte count by script (FILL_CLOSE_B11, from placeholders that name the file and the kind of value), and their counts came from a counting script's output. Keep that for every handover (R2W). The record keeps a session honest when the session cannot: in one Loop Atlas review every number names its run, every miss has its row and every stamp was read from the disk, and where memory failed, a stamp, a review, a product (rows 735, 767 and 592), the record caught it, each time before anything left the room (the Loop Atlas sessions). A check's own printout made a number look traced again a day later: with the provenance table's own ledger still among its sources, the scan read 889 of 892 on 17 October (run 20261017_031643), because 4,200 matched a pass number in a ledger by coincidence. A match by string is a lead, not a trace (row 811) (the triple-star sessions). A scan that matches numbers by value can be satisfied by coincidence: the triple-star scan matched a circular start's 5,000 and the long run's 40,000 to unrelated ledgers; each coincidental match is named in the paper's Declaration and not counted as a trace

The Latch Atlas; ties rows 206, 221

Citations are claims: run a checker after every build (571) — 157 author-and-year citations, each resolved to exactly one reference row, every row cited somewhere, every run stamp with a folder or a deposited twin. The checker caught a version DOI passed off as a concept DOI, two names with the wrong initials, and works cited that no note had named. It costs a second per build

The Latch Atlas; ties rows 220, 221, 552

The crate is built by script from the folder the manifest checks, never gathered by hand (572) — three of the five records found on a laptop disk were stale versions; the Latch Atlas's crate could not have been, because no human hand assembled it. And a manifest verifier reading "differ 0" before every ship turned "the files are gone" into a ten-minute problem instead of a lost week. The ferry is built from the union of every earlier crate and checked three ways before it ships: every import resolves inside the crate, every tool the runs table cites is present, and every run of record has its tool. A ferry that skipped this left 26 older tools behind, two of which could not even import, and a session worked a day without its own specimens. Build tools go into the folder of record on the first day: one lived only in scratch space until a crate was packed. And every crate is checked against the letter that names it, on arrival and on sending, both ways: twice in one thread a named attachment was missing (the triple-star sessions). Crate hygiene, learned the hard way in one room over two sittings: importing or compiling a module leaves a pycache folder, removed before packaging; a crate zip and its figures zip are verified together against one manifest, and a figures zip can change bytes while holding identical files, so the manifest check is what counts; paths are absolute, because the shell's working folder can reset between commands and a shell variable set in a backgrounded group is not set outside it; a run is never waited on by a process search from the same command, which matches itself; after a night away the working tree is checked against the last manifest before it is trusted (706 of 706 matched); and material from another project stays out of the crate, with only what a comparison used written into the log (the Loop Atlas sessions, R2W). A tool's true needs can be read from what it opens: a hook that logs every file a Python tool opens showed, in one run, that a kit's launchers were exact, each fold tool opening 7 to 10 files, all named in its launcher and none extra, so two files missing from a first staging were the reader's fault, who had read the tools' text and not their launchers; and a launcher's stated time is looser than its needs, "a few minutes" being 2 seconds (run NEEDS_AUDIT 20261010_205839) (the Loop Atlas sessions, the book). Build a handoff crate against the crate the person actually holds: eleven files made on 13 October reached no crate because a rebuilt crate was saved under its old name and the next crate was built as "everything since" that crate, against the rebuild and not against the crate the person held, and he spent an evening hunting files only a session had. Every crate now gets a new name, is built against the receiving folder's fingerprints, and is unpacked into an empty folder and checked against its manifest before it is sent; a missing file is asked of the sending session before the person. A later crate was built and tested this way (the triple-star sessions). A name that sorts as text breaks when the count gains a digit: the crate packager picked the newest handover by sorting file names as text, and at the close of sitting B10 "LOOP_ATLAS_HANDOVER_B10_…" sorts before "LOOP_ATLAS_HANDOVER_B9_…", because the character 1 comes before 9, so the packager would have zipped the old handover beside the new crate. It was caught by reading the packager before packing, not by a failure. RF_SPOKE_CRATE_PACKAGER_B8_V3 picks the handover by the date in its name. Any label with a running count (B9, B10; r9, r10; exchange 9, 10; sittings, issues, exchanges, deltas past z) needs a sort by number or date, never by text (R2W). The same close caught R2W's lesson 123 recurring: the first version of a count script sorted the delta zips' labels as text and printed "16a to 17z" when the newest was 17ag, so every running label is sorted as a number or a date (R2W)

The Latch Atlas; ties rows 89, 157

The person with the questions is the instrument (573) — the author's marks on the Latch Atlas's final read (is five dividing ten known, does twisted light latch, what is the whole-Sun idea doing here) each changed the Latch Atlas. A proofreader who asks "is this a known thing?" is running the prior-art pass by hand, and it works. The marker's own rule found the marker's marks: the person's rule, the change of change of slope along a string, was built literally from his write-up and run without his marks, and its largest values lined up in three lines across a corridor's waist, where he had drawn his crossings on another figure. It was seen after the run, so it stays a description until a blind birth shows crossings (the triple-star sessions). And the person who sees patterns should draw in the instrument's own numbers: lines drawn on screenshots had to be read by eye, while the Explorer's pen keeps each line in the chart's own units, numbered, with his word on it, where it can be read directly and registered before it is checked (the Loop Atlas sessions, R2W). The person's marks are answer keys for the tools: his strokes built a linker's shared mode, which lets many strings pass through one hole, from his "many strings can overlay a lattice hole", and on his curves it lifted the greedy linker's coverage, from 10 to 25 of 35 drives (B3) and from 16 to 23 of 27 (B4); his "Network map interconnecting all points *just in case*" became a branching linker, named but not built; and at weight 0.26 the linker took a fork wrong that his hand took right (R2W). And when the person counts the parts of a picture, count them in the runs: his four-legged skater has four blades on a groove's two lips, and in the runs each tongue has two edges where exactly two stops meet and vanish, one steady and one wobbly, four in all. The count matched before the meaning was settled, and the meaning followed from it: the four blades belong to two dances, because the steady stop that closes the top lip slides across and closes the bottom one too, with the wobbly stop on its other side. A number in a person's picture is the cheapest part to check, and often the most exact (R2W). Two pencil lines are a

claim about rates; score each line separately. On figure AA Bear drew two lines and wrote that the orange curve (the fourth harmonic's zero) looked like a change of rate of the blue (the second harmonic's). Read after the data (R8_BEAR_LINES), orange is blue magnified, the slide ratio 5.1104, 5.1092, 5.0905, 5.0391, 4.9334, 5.1851 and 5.1234 at weights 0.0025 to 0.64, then breaking away (4.8160 at 0.768, 0.7790 at 1.911). His second line, at 0.64, sits on the last weight where the fixed ratio holds: the gear break. His first line, at 0.16, sits on the ratio's lowest point before 0.64, and nothing more could be read from it. One line hit a wall exactly; one is unexplained. Score them one at a time, and keep the unexplained one on the list. The two lines were read after the data, not registered; a marked picture is worth registering as a guess before the reading. And the questions steered: Claude saw the reflection's drawers and the far-turned fine knob (rows 745 and 684) only after Bear asked the questions that pointed there, and missed, then reported too flatly, the sign that Bear read as half a pair (rows 734 and 76; borne out in row 801) (R2W). A second specimen: Bear's question about Poincaré's reading of relativity became, in R2W's sitting P1, the light cone of the weightless junction, a book figure, and a registered weight study, WEIGHT-CONE, in which five predictions held. Test a question from another field the same day, and say what is known, what is new and the number that separates them

The Latch Atlas; ties rows 106, 437, 506, 613, 801, 805, 848, 851

A wrong upload is survivable when the record is honest (574) — a fourth file went onto the Latch Atlas's record by mistake: five working records, in the working voice. Nothing private was in them, the Latch Atlas cites them by name, and a line added to the description in place turned the mistake into a supplement. The rule that working notes stay out of a deposit stands; what saved the day was that the notes had been written carefully enough to survive being read. A file filed by mistake comes out the same night: on 14 October a photograph meant for another conversation was described in the project's record before the person said so, and it came out of the folder, the manifest and the description at once, with the correction written where the description had been (the triple-star sessions)

The Latch Atlas; ties rows 431, 454

A size confirmed on a population is not a size for an individual (579) — the rotation's size, 2.97 degrees, was fitted on the 2011 regions and confirmed on 82 regions of a blind test. Applied to nine regions followed day by day, that one number shrank the net current on 15 of 37 near-centre maps, worse than a coin; a region's own first map did it on 19 of 37, exactly the coin; the median of its first three maps on 25 of 39, a chance of 0.054 against the 0.05 the rule asked. The sign survived every test (45 of 50, 25 of 36, 36 of 47). A population confirms a sign and a typical size; neither fixes the size for a member. Say of every number whether it was confirmed on the population or on the individual. A median supports "typically"; "never" needs every case: "the floor never sits below the launch" was drawn from medians, and column by column the edge dipped below the launch in 10 of 41 columns at weights 3.00 and 5.00, exactly where the balance law's lift turns negative (R2W)

Dancing on the Sun; ties rows 314, 544

Undecided is a verdict, and a rule is written for the sample that will exist (580) — five blind tests were run with their rules fixed and fingerprinted before the maps were fetched, and three of the last four came back UNDECIDED: one because the rule demanded six judged regions and five were judged, one at 19 of 37 and one at a chance of 0.054 against 0.05, both on a tracked region's own size, while the sign held (row 579). Two regions could not be judged because their dates had been written from memory and the archive disagreed. UNDECIDED is reported as written, never nudged into a pass by loosening the rule afterwards; and a rule's sample-size clause is written against the archive as it is, checked by a probe before the rule is fixed, not against memory of the archive. A missing or unresolved reading prints UNDECIDED: one verdict code printed MISSED where an unconverged refinement had left no value at all, and another printed UNDECIDED by an order rule stricter than the one registered. The code checks exactly what the registration says, no more and no less (the Loop Atlas sessions). When a bar and its rival both fail, say where in the gap the answer fell: a twin registered within 0.01 of lags 0 and 1/2 at weight 0.01, against a rival of farther than 0.05, came at 0.017, closing in at about 1.7 turn per unit weight. The gap between a lean and its rival is where the measurement lives (run T4c) (R2W). A ratio of differences of differences eats digits, so write each reading's floor down before the run and judge nothing inside it: the clock shifts of the light needle's doublings differ from one generation to the next by 9.05×10^{-11}, then 2.03×10^{-11}, then about 4×10^{-12}, then about 9×10^{-13}, so by period 32 a value printed to 12 decimals ($\pm 5 \times 10^{-13}$) and the refiner's own floor (about 10^{-13}) decide the ratio. A clause registered as "within 0.2 of 4.669" was judged held at 4.689 with a true range of 3.3 to 5.6, and is now undecided; a floor check registered as "repeats the earlier value within 10^{-13}" compared against that value's 12-decimal print and missed for the print's sake; and an unconverged refinement fell through the verdict branch as missed instead of undecided. Carry full-precision values inside one run, propagate each floor, and let the verdict branch print UNDECIDED when a reading is missing (runs FOLD_OR_FLIP_3 20261007_054521 and FOLD_OR_FLIP_4 20261007_061947) (the Loop Atlas sessions). A verdict rule reads NO READING, this thread's word for undecided, on runs that never ran: the triple-star thread's LW6 time-budget test, with a planted worker, read XA MOVE, XE NOT HOLDING and XL SAME on runs that were never started, the flaw LW5's review had found in its own rules, and the rules were amended before the first run: an escape is an outcome, an unrun birth is not. The review of triple_followups2 found that a line keeping exactly three held, locked runs would have crashed on a division by zero

where NO READING was registered. Before shipping, drive every rule through a budget stop, a lost run and a planted crash, and check that missing inputs give NO READING (the triple-star sessions). A judge is an instrument: try it on the case already seen before trusting it on the case not yet seen, and print beside every verdict the number it judged by. One judge took a cascade's direction from a single edge pace, seven paces off the fitted onset, looked the wrong way and printed a lean that held as missed; another printed a share inside its bar as undecided, because the share was a number of a kind it did not expect; both were caught within the hour because each verdict sat beside the table it judged. Fix such a fault where it comes from: mended in two scripts, it came back in a third whose verdict printer was copied from the old pattern, so the redesign changed the shared part, one strict verdict rule that stops the run on anything but a plain true, false or none (the Loop Atlas sessions, misses rows 496 and 498)

Dancing on the Sun; ties rows 167, 313, 369, 579; the Loop Atlas's checks of 30 and 31 October: WINDOWS_ORDER, TWO_LOOPS_ZOOM, MSS_TABLE_CHECK, LOGISTIC_GAPS

A companion citation is written to be true in either order (582) — the plan was to deposit the Latch Atlas first and write its identifier into the paper; the paper went up first. The Declaration's sentence had been written to stay true either way, and the pair was closed in the records' metadata in place, "is supplement to" on the paper and "is continued by" on the Latch Atlas, both read back from the API, with no new version of either. A sentence in a deposited text that names a companion must be true whichever record exists first. And a companion's status in a declaration is a claim with a date on it: three days after the triple-star paper was published, the guide's Declaration still counted its project among those with no deposit yet, and the list of companions did not name it. Read every companion's record at each build (on 31 October every Zenodo DOI the guide then cited was resolved against Zenodo's own records, all eleven concept DOIs, and on 1 November the four the paper pass added, the same way), and generate what can be generated, such as the table of which rows name which source, from the built text, so that it cannot drift

Dancing on the Sun; The Latch Atlas; ties row 220

A rule bent with its guard in place, and said so (583) — a laptop tool got a second version in one sitting, against the project's standing rule of one version per sitting; the hash-checked launcher was the guard the rule exists for, a fidelity run reprinted every version-one line before the launcher would run, and the status page said so in plain words. The container's memory was overrun once by a tool that held every map at once; it was killed and rewritten to load one map at a time, with the dead start named as not a run of record. What held: every number names its run, the predictions sit in the crate byte for byte with the fingerprints the paper cites, the deposited tools' comments were edited for a reader and proven non-functional by an identical token stream, and the one archive address that leaked into a ledger was redacted, with the manifest saying where. A version number is a claim too: on 28 October R2W's sitting P1 and the guide's own session each made a build_paper "v1.6", two different tools under one label, found the next day by their fingerprints before the packet reached R2W; the next issue was v1.7, and a tool is named by its fingerprint as well as its version (the Field Guide sessions)

Dancing on the Sun; ties rows 36, 157

A strip of snapshots teaches motion where a single object fails (585) — the wing drawn as a strip of frames, good timing beside bad, made a hard idea land; the butter knife, one object standing in for a motion, did not. An analogy that is a single object carries a shape and loses the order of events. A loop is a motion, and the cheapest honest picture of a motion is its steps

the Loop Atlas sessions; practice

A corrected number moves everywhere, or the correction becomes a second defect (588) — the old snake figures stayed in the runs table with a warning at the top while their fixes lived in another file: two homes for one number, and a reader cannot tell which one is current. The table carries a corrected column, the old value beside the new with its reason, and nothing else keeps the old figure alive. When the leading reading changes, recompute every number that leaned on the old one: a headline "within 0.7 per cent" computed against a padded fit became 1.2 per cent against the reading that now leads (the Loop Atlas sessions, the triple-star sessions). A withdrawal must reach every document that carries the number, the withdrawing room's own first: a list of changes for this guide copied a scissors reading R2W had withdrawn on 5 October, because R2W's own working record still carried it (row 563 now carries the correction). Corrections are dated notes, run files stand as written, and a correction can itself be corrected, each note standing in order. Status words are claims too: at one handover a register still said "open" and "would resolve" about questions two probes had settled, and words like open, held and next go stale without a sound, so each is re-read against the probe table at every handover. And a candidate sent to another room is followed by its result: the triple-star sessions told the Loop Atlas book that the boundary layer's rests at 7 and 8 had not held, before the book leaned on them (the Loop Atlas sessions, R2W; the triple-star sessions). Records drift apart between files: two misses called open had slipped between files, one answered a day later and never marked, one never given a card, and a mailbox listed the notes as running to 53 after 54 to 57 were written. When any pass finds a settled or uncarded item, mark it in every file that holds it, and sweep the citations against every open row at each save point (the Loop Atlas sessions). A withdrawn number is withdrawn everywhere, the same day: row 671's thinnest widths came from

folds that never met their own criterion, and the Loop Atlas later found them 19.7 and 37 times too thin. The row was redrafted without them, and every working copy that carried them was rebuilt, so that no session kept reading the old numbers (the Field Guide sessions). What found those widths was reading an old run file in full: both of the 11/7's folds in GROOVES_6's own printout printed the same 22 figures, one edge found twice, and the two thin widths had stood in the book's Limb G for a day, wrong by 19.7 and 37 times (the Loop Atlas sessions). A summary row can join two results that the runs keep apart, and its correction must reach the row. This guide's row 402 said the light star's lap ratio sat on flat steps at 7.0185 and 8.0219 and that those steps did not hold. Opened against the runs, the sentence joined two things: the boundary layer's rests at 7 and 8 did not replicate (probe 14, run 20261005_193416), while the lap ratio's flat steps inside the islands held, and the islands paper rests on them (probe 22, run 20261006_190322). The thread wrote a five-line correction and sent it through Bear; it never reached this guide's record, and the row stood wrong until the thread's next lessons named it (the triple-star sessions, the Field Guide sessions). A numerical fault found in one run is a fault in every run that shares its arithmetic: the unreduced angle that broke HALVINGS's edges (row 751) had already corrupted GAPS_READ's multipliers by up to 0.09, and it surfaced only when a later tool's planted value, taken from GAPS_READ, failed in its blind run. The earlier runs were rerun and their records corrected the same night (run GAPS_REREAD 20261015_071452; the Loop Atlas's misses row 222) (the Loop Atlas sessions)

the Loop Atlas sessions; ties rows 88, 192

Keep the author's words verbatim from the first turn (589) — the earliest messages of a long session reached its record only as paraphrase, because the file that holds the author's voice did not exist yet. A paraphrase is a compression, and a compression cannot be expanded back into the words it replaced. The voice file opens on the first turn. A session states its own history from the record, not from memory or summary: asked how long it had run, one session repeated the author's recollection and wrote it into its handover, and credited the author with a question whose words were the assistant's; both lines came from summaries, and the conversation record corrected them. When the early turns survive only as a summary, read the conversation back before writing the review: the past-chats reader returned all 49 turns of one conversation, the author's 22 messages word for word, one turn per call (the triple-star sessions). Write the record as the work happens: a session's transcript on disk began partway through its work, and what survived was what had been written into the sitting note, the ledger and the logs before each reply; a review written later from a summary would have lost the numbers (the Loop Atlas sessions, the book). Keep every word of the picture, the small ones too: asked about a tilted floor, the person answered "two u shaped curves offset side by side (intersecting)", and a registration took the offset and left out the bracketed word. The runs then showed the arms offset at every period, the right always lower, but the gap closing steadily (0.0587 at period 5, 0.0056 at 13), as if the two curves cross at 1/2, which is the bracketed word: offset and intersecting together describe two lopsided shoulders meeting at one top, the picture the data lean toward (read after the data, and tested at the next periods). When a picture comes with an aside, test the aside as well; it is often the part he saw most clearly (R2W). A person's words are copied by script, never retyped: the Loop Atlas kept Bear's words verbatim in a file of their own, in text blocks copied from the transcript by script, and read its stamps from the disk the same way (the Loop Atlas sessions). A verbatim collection needs smart typography off: read-back of R2W's reads file found Bear's typed triple and double hyphens printed as dashes, and an underscore pair in a heading printed as italics, a fault already in the last delta. A file whose promise is to quote exactly is built with the no-smart builder, as its Field Guide appendix already was, and its read-back is the check that it kept the promise (row 779) (R2W). A remark made in passing is not copy, and this limits the row: verbatim is for the words the person gives for the work, and the aside worth testing is an aside about the work. A date the person confirmed for a book's opening came in a chat message that also carried a personal remark, and both went in: the date, and the remark worded as his own sentence, into the opening, the changes log, the edit script kept as a receipt, and the record. At his word it came out of all four. The record's entry was edited in place, the only such edit, with a bracket saying so, and the shipped copies that carried it were named for him to delete. Only the words the person gives for a book go into it, and he sees them before they stand; anything personal said in conversation stays out of every file, however natural it would read on the page. The rule travels in the session's continuity seed, in its list of how every exchange runs (row 591) (R2W). Record from the transcripts where an idea came from, keeping blind apart from prompted: the pairing of the figure-8 with the Möbius strip was Bear's, unprompted, on 26 June, measured on 3 October (the 2:1 lock flipped at 199 of 199, 206 of 206 and 212 of 212 passings, row 864); the figure-8 itself, named a Lissajous figure in the turn it entered, is not claimed as blind

the Loop Atlas sessions; ties rows 158, 168, 357, 864

A session jump travels as one zip with its handover inside (591) — a new session could not see 94 loose files. Loose files suit a person reading them one at a time; a session jump needs one container the next session can open whole, with the handover inside it, so the work restarts from the record rather than from a pile. When to move is the person's call: a move is asked for at a clean break, never announced, with no run in flight, every letter answered or waiting only to be carried, and the next items able to stand alone. A summary is second hand, rounding and merging (0.5518 for 0.5517742), so a number is read again from its file before it is cited (the Loop Atlas sessions). Write the seed before the context runs out. A seed holds the baseline facts, each with its reason; the chain of understanding in the order it was found, misses included; the method; the pattern-recognition moves that steered it; the numbers any

rebuild must reproduce; and the rebuild commands. A new session that reads it should not need the conversation (R2W, after the person's "please don't bloat limit before you make a backup seed copy"). A move is one ferry: one session's start lost a day when its ferry arrived as five of six pieces, 1,163 files absent, and the old session had to rebuild two pieces; later start pages asked for a base ferry with sixteen shipment zips laid on top in order. At every move, ship the whole crate fresh in pieces of at most 25 MB, list each piece's md5 in an index sent beside them, and unpack the pieces into an empty folder and check them against the manifest before sending; the new session checks the pieces against the index before it reads anything (the move of 15 October) (the Loop Atlas sessions). Since the start is where the day is lost (the 1,163 absent files above), the new session also says what is missing at once: if any piece is missing or differs from its index, it says so in the first line of its first reply, by name, so that the person is never the one who finds out (the Loop Atlas sessions). The person's call on when to move outranks any rule a session writes for itself: one session announced its own move six exchanges into its work, under such a rule, and Bear stopped it ("We have been going for 6 exchanges"; "You can't break after six or eight exchanges"; "You *are* the new session we moved to"); the move was withdrawn, and the rule with it. The same holds for any step that costs the person work on his own machine (the Loop Atlas sessions). A reference carried from memory is a claim: check it in the record. At one sitting's close, TRIAGE_12's docstring named the crossings audit as the twenty-ninth exchange's work, and the record puts it in the nineteenth (the note "surprise 1's backward pass", 16 October 22:11). A summary keeps what was done but can blur when and where, and a wrong pointer sends the next reader searching. Every origin reference in a handover or a Field Guide item is looked up in the record by a command, never written from memory (R2W). Write a seed at the close of an evening too, not only before a move: at the close of 20 October one sitting put a two-page continuity seed into its crate and a start-here note into the workspace, saying where everything is, what comes next in order and how an exchange runs. The next exchange began with the check and the first run within minutes, after a restart and a summary of the conversation, without leaning on the conversation's memory (R2W). The triple-star thread keeps its record as one dated entry per exchange, newest at the top of its register's section on where the thread stands, each with what the person said, what was run (tool, version and run), what came back and what changed. It is the thread's memory, and the next session reads it first (the triple-star sessions)

the Loop Atlas sessions; ties rows 157, 504

The assistant's own queue is a standing section of the handover (592) — the assistant's own queue slid behind the author's new pictures all day: a habit, not a request. A standing section in the handover for the assistant's queue keeps its own deferrals visible beside the author's, which is what stops them becoming orphans (the Loop Atlas sessions). And say at once when a change of direction displaces a promised run: a run named as next in one letter was displaced by a new line of work, and the following letter said so (the triple-star sessions). The drift check applies to the assistant's own queue as much as to the author's: a recount slipped behind three of the author's pictures until it was named aloud. And saying that a turn ran past fifteen minutes is not the same as preventing it (the Loop Atlas sessions). Letters are interrupts: a book's headline run was "next" for a day and never ran, because four letters arrived and each took the turn. Answer letters, but run one of the work's own items between them (the Loop Atlas sessions). Status words come from the record of what was run, not from the plan: two replies of 12 October said three laptop batches were waiting on the laptop when none had been built, and the person asked what his laptop was running. Say "not built" when it is not built (the triple-star sessions). At the guide's own bench a reply said a quotation-tidying build step had been dropped, and it ran on in every staged build until found on 16 October (the Field Guide sessions). The same at another table: two wording decisions for the person, "two clocks" against "coupled clocks" and whether the door gets its own section, had fallen out of the queue, and a backward pass found them and put them back on the paper draft's list of decisions for him (the triple-star sessions). Every product a project has begun goes in the handover's own list, with its build state and its next step. The Loop Atlas's Stretch and Squeeze Explorer was built on 13 October, with Bear's wish that it become a working toy that takes in new things; the next handover named it only as a published page, no handover after that carried it, and it waited until Bear asked after it on 16 October (the Loop Atlas's card 194). He had asked whether the whole-atlas version of R2W's Depth Explorer had been abandoned, and later saw the diver and jumper toy as a version of that explorer's depth axis. A person who sees one tool inside another is reading the work, not just the data, so check the record for a product that dropped out at a move before answering. His later pictures set the product's shape: one explorer for the whole atlas, twelve rooms, each in four styles (Look, Learn, Build, Check), the jumper and the diver first, since an atlas that may one day have many visitors needs a way in for each style (the Loop Atlas sessions). Check every note written for the handover into the handover, by search, at the move: a note Bear asked for and marked important was written into R2W's record for the handover's first page, and the next work order held neither of its key words; the guide's intake found it, and the send carried it on (R2W, sitting P3). The largest specimen was the guide's own publishing session: from 23 to 30 October no fold reached the guide while every turn went to the book of reads, the packet and the builders, and the queue went unreported until 128 reads waited for the book and three projects' sends for the guide; since then each exchange carries one fold step, with both queue counts in its last lines (row 868) (the Field Guide sessions)

the Loop Atlas sessions; ties rows 156, 158, 868

A green build is not evidence that anything changed (593) — three times in one deposit an edit script hit an anchor it could not find and exited before writing; the rebuild ran on the unmodified file, and the linter returned CLEAN. Nothing had changed and everything said it had. An exit status says the script ran, not that the artifact holds what you meant. After every edit, re-read the file and assert the new text is present; after every build, extract the text of the finished artifact and assert on that. A parser, a planned check and a duplicate download are each claims until checked: a manifest parser read 56 of 65 entries; a check column recorded word searches before they were run; and a file named "(2)" was a revision, not a copy, as its md5 showed (the Loop Atlas sessions). So is a promise that a question can be read from saved tables: a register said which star is thrown out could be read from two probes' tables, and it could not, because the run function computed it at the end and never wrote it down; rerun unchanged with the answer written out, all 32 runs matched their ledgers, and the 0.8 star leaves in 17 of 24 escapes. When a run function knows something at the end, write it to the table (the triple-star sessions). A blank line inside a markdown table ends the table, and the PDF prints the rest as text: a data ledger's 69 rows from one section on printed as raw pipes, with nothing to warn of it, and the same fault, fixed once, came back unseen. Before every build, scan the source for a blank line between two table lines; after it, find a late row's text in the PDF as a table cell, not as a line with pipes in it. A table above a file's first heading fails the same silent way, because the builder prints those lines as the byline (the Loop Atlas sessions). A build that cannot find a picture can still exit clean: on 21 October build_paper v1.3 built a page whose picture is missing with exit 0 and a 16-by-16 icon (the trap row 553 lists), and built the triple-star islands paper, its picture paths written '../', as 62 pages with fourteen icons that read back complete (row 773). build_paper v1.4 stops and names each missing picture (exit 6), though only below the first heading (row 853), and lint_deposit v1.8 counts the pictures the PDF itself draws (14 drawn for 14 named) and fails the broken build. Bind each figure to its caption in a block that cannot break, and check page by page that no page ends on a heading (check_pdf.py). The default publishing packet (PUBLISHING_PACKET_2026-10-22) carries the same rule (the triple-star sessions). A gate that passes with a known offset has a unit in it that nobody has explained. R2W's book PDF held 59 images against 58 image tags, carried as a known offset; checked on 22 October, the same picture sat on pages 5 and 6 with identical bytes, the chalkboard figure of §2.1 (1,600 by 2,200 pixels) cut by a page break, its top at the foot of page 5 and the rest at the head of page 6. With a page break before it, or a shorter figure, the gate should read 58 of 58. Explain every unit of an offset before living with it, and after the gates pass, look at every page where new text sits: a gate reads text, and only an eye sees a split figure or a stranded heading (R2W). And a read-back that samples each paragraph's first and last words passes a hole in its middle: the islands paper's README lost a launcher's name to the converter and build_paper's read-back passed it, reading only each paragraph's first and last six words; a whole-paragraph check of the paper then found 233 of 234 paragraphs whole, and a figure move that split a word list was seen only on the rendered page. Read back whole paragraphs, and look at every page that moved

Ruggedizing Haber-Bosch; ties rows 553, 588, 812, 853, 869

Renumbering in place cascades into its own output (594) — replacing "6." with "7." and then "5." with "6." renumbers the item it has just renumbered. It shipped a reference list reading 12, 13, 16, 17, 14, 15 and a conclusions list reading 1, 2, 3, 5, 4, 6, 7. A moved section does the same to its references: "see Section 5" still resolves after the move, to whatever now sits in slot 5, so nothing complains. Rebuild a numbered list whole and assert the sequence; after any structural move, find every section and row reference and check where it lands. Pointers break the same way: a row that says "the same as above" changes its meaning when a row is put above it. Two rows of the Loop Atlas book's knob register pointed at their neighbours with "the same"; a row inserted between them in one exchange, and another inserted earlier, left both pointing at the wrong machine, found only when the register was counted for the explorer's rooms. After any insertion, read every pointer in the table, or let each row name what it means (the Loop Atlas sessions)

Ruggedizing Haber-Bosch; ties rows 588, 593

A number that is everywhere and measured nowhere has drifted onto a different noun (595) — "40 per cent of soil phosphorus is locked" came back in apparent agreement from every search. Each source cited another secondary source, and the figure in the primary literature, 30 to 40 per cent, is the share of the world's arable land where phosphorus limits yield. It had drifted subject. In the same deposit four price figures came from press coverage of a report rather than from the report, and two of them contradicted it: 46 per cent month on month in the report, 53.7 in the press. Agreement among sources that cite each other is one source counted many times. Find who measured the number, and read what it was a number of

Ruggedizing Haber-Bosch; ties rows 552, 571

Absence is invisible when you read for presence, so run the coverage check (596) — a paper recommended five interventions and leaned on an expert review for its framing. Its claims were written out as a list and searched for in the review's full text, and the review discussed none of the five, which reframed a whole section. The check took about ten minutes on 82 pages, and a cover-to-cover read would probably have skimmed past it. A source doing structural work in an argument is searched for every claim made on it, and what is missing is reported as absence: weaker evidence than a stated position, but evidence. A second specimen: a phrase search reported "0.850 to 0.855" absent from R2W's record, where the two numbers stood in separate columns, and a prefix match missed that SWAP-RIDER

had run on 6 October. A search that finds nothing is a claim about the search (R2W, sitting P2). A matcher is a claim about itself: R2W's name matcher erred both ways, a false yes through another entry's author and a false no for an entry that spells out all five authors. Its check that two parsing gaps touch nothing before a given line found 0 there, and was believed only after the same check, run over lines where the gaps were known to be, found 12 (R2W, sitting P3). The guide's own session met the class in its checkers: a row-number pattern written case-sensitive printed "exists: yes" for 14 whole-row placements with nothing checked, and now prints NOTHING CHECKED; two source files whose lesson numbers overlapped let one overwrite the other and raised false flags, and the checker now refuses that case; a recount script read "not known" as blind because it starts with "no"; and a read-back keyed on words flagged 33 of 47 lines of a PDF as missing, where one keyed on letters and digits found 58 of 58 (the Field Guide sessions)

Ruggedizing Haber-Bosch; ties rows 67, 552

A source with no text layer searches as empty (597) — a 34 MB print-to-PDF file yielded 82 bytes of text, one form feed per page. Every search of it would have come back "not found", true of the file and false of the document. A text extraction that returns almost nothing is the tell. Run OCR before any search counts as a read: at 200 dpi tesseract reads body text well and tables of contents badly. Search a PDF's text with its ligatures in mind: pdftotext writes "ff" and "fi" as single ligature characters, so a search for "cliff" or "fifths" in a built book found nothing although the words were on the page. Search by a number in the same sentence, or have the readback normalise ligatures before matching (row 779) (R2W)

Ruggedizing Haber-Bosch; ties row 596

Unit traps sit where two fields report the same thing differently (598) — field trials report elemental phosphorus and fertilizer claims report P_2O_5, a factor of 2.291 apart; potassium against K_2O is a factor of 1.205; US retail fertilizer prices are quoted in short tons; and a free-on-board benchmark and a retail delivered price for the same product in the same month differ by freight and margin, about $380 against $495. A false friend sits beside them: potassium carbonate is not potash fertilizer, though a price page for one looks like corroboration for the other. Label the unit, the chemical form and the point of sale on every number before two are set side by side

Ruggedizing Haber-Bosch; ties rows 502, 548

Conclusions go stale (599) — a conclusion summarizes a body that keeps changing beneath it. One deposit's closing section told readers that the most useful action was a $10 experiment, three pages after its policy section had ranked something else first. Re-read the conclusions against the body as the last editing step, every time. A count goes stale the same way when a run lands after its sentence is written. The triple-star islands paper's finding 3 and Section 4 printed '160 of the 162 escapes', the count of run 20261014_224840, made before the long pair (run 20261014_235048) added four escapes, and Section 4's own paragraph named three escapes not from the low side against the count's two. The hand tally, 163 of 166, had been printed by no tool, so the recount was registered as a count: triple_escape_recount v1.2 first gave back v1.1's ledger line for line (run 20261016_182115), then read the long pair (run 20261016_182117), 166 escapes, 163 on the low-birth side. Have the tool re-derive every count in such a sentence from every ledger it counts, reproducing the old number before it prints the new one. And when a new finding reaches findings already written, raise a flag that names each sentence it reaches: after DR1 and FA0 found the integrator's arithmetic falling about 2.8×10^{-8} cycles per pass in each pass (row 741), a crazy flag in the same entry named three things the paper already said, the shells' slide of up to 6×10^{-5} between 2,400 and 6,000 passes (LW5's 1/8 line moves 4.96×10^{-5} between those records), the eleventh's lock appearing over 6,000 passes, and the tenth's crossing shapes. Each of those sections got a pending mark asking what survives tolerances 100 times tighter, and the tight kit LW6 was registered. LW6 (run 20261018_001236) answered all three: the shells' slide was the arithmetic's (XB, a median slide of 1.18×10^{-7} at the tight tolerance); at the eleventh, 7.127 did not hold (XE, row 732), while the births between its locked ones, 7.128, 7.130 and 7.132, held and locked (E11W, run 20261018_173912); and at the tenth a gap held (TEN-G), its rate a slow swing and not chaos (TEN-C missed, its redesign FA2 landed, run 20261018_173550), the step's shape reported and not tested. The paper took the arithmetic into its finding 5 and a subsection of Section 2, "What the arithmetic fakes", and corrected its shells' slide. A flag that names its sentences gives a list of fixes, not a general worry, and its list is checked in turn: the session's backward pass found that the flag did not name KD3 or KD3′, DS1 or DS2, LB1‴, LX1's SLIDE verdicts or probe 33's 1,000-pass in-lock slopes, which also read records of different lengths at the kit's tolerance, where the pair's eccentricity falls 1.134×10^{-8} a pass against 5.1×10^{-11} at the tight one, so the flag's reach may be larger than stated. A statement goes stale the same way: one written in the register on 19 October, that a rerun could not give back the same files byte for byte, was contradicted the next day, when the regenerated 13 series matched the manifest and the zoom-out file was the same bytes, and it is not recorded as corrected (row 588) (the triple-star sessions). When a crazy flag is raised, search for every comparison of the same kind, not only the ones remembered: the triple-star thread's backward pass later found sentences its flag of 18 October had not named

Ruggedizing Haber-Bosch; ties rows 156, 732

Assume there is no future work (600) — anything parked for the next pass does not get done, and a tracker is a record of what was closed, not a promise of what will be. When something cannot be resolved, close it in the artifact: disclose it, scope it, or state it as unresolved. The disclosure is usually better than the resolution would have been, because a reader can see it. An open question that can be run is run before publishing, and the finish list says which of its rows run and which are carried as open, and why. A reading written as a finding is one of them: R2W's book stated the heavy balls' second dance and its fold as found, while by the record the fold was read after the data with its last steps not fully converged, so its finish list said to word it so, or to run FX-2c first. FX-2c ran on 30 October: the fold a ball at rest falls into landed within 0.0005 of the edge at both weights (0.859975 and 0.713061 against 0.85989 and 0.71310), and the reading of two resting dances side by side was withdrawn at weight 2, where the continued dance is unstable, and left unchecked at weight 5 (rows 739 and 761) (R2W)

Ruggedizing Haber-Bosch; ties rows 80, 402, 739, 761

The auditor's expectation is a card too (602) — on the mirror audit's first pass a twist read red, at an even share of 0.970, because the audit expected it to flip. The twist belongs to the blade, so it should stay put, and on the corrected card it did (0.030). A wrong expectation turns a correct result red. Expectations are written, sourced and audited like results, and a red verdict first checks its expectation

the Loop Atlas sessions; ties rows 190, 337, 559

A run file is written only by its tool (610) — four runs of record had no script that makes them. Some were lines appended by hand; others were sweeps run without saving their script; and one of them now disagrees with a new run. A number that cannot be regenerated cannot be checked. A new sweep is a new part of the tool, rerun, never an appended line (the Loop Atlas sessions). A one-off reading typed into a session is saved as a script and rerun before it ships: two readings sat in the record as numbers with no file that could make them again, and the queues they leaned on lived outside the crate; written as scripts, rerun and checked against the record, they reproduced it, the table read byte for byte and the slide ratios to every printed digit. A number in the record that nothing in the crate can reproduce is a citation without a source (runs WALK-2c and SLIDE-1) (R2W)

the Loop Atlas sessions; ties rows 206, 570

Small faults that each cost a round trip, with their fixes (611) — launcher logs came back in UTF-16 because the shell's tee defaults to it, so launchers set UTF-8; a return zip missed the survey files, so a status page lists every file the laptop should send back; a PDF build shrank wide figures past reading, and six PDFs were rebuilt, so the house style caps every image at the column width; and a kill command matched its own shell, so processes are killed by a pattern anchored to the program's name. None is physics. Each cost something once and need not again. The kill command matched its own shell three more times in a later sitting, so processes are stopped from a script file, and runs longer than ten minutes are launched in the background from a script and polled (the Loop Atlas sessions). Smoke tests predict nothing about run time: a run smoked at a fifth of its length took three times longer than the smoke suggested, so only measured run times are quoted (the Loop Atlas sessions, R2W). Take every time from a clock call of its own: at least seven times in one sitting a registration, launch note or read was first written with a time estimated by hand, and three more slips reached the record in the next (23:41 for 23:36, 21:55 for 21:54, 19:47 for 19:45), so the clock is read in its own call before any time is typed. And a run time is a guess until it is measured: a run registered as about six minutes was stopped at 14 with nothing past its seed (the Loop Atlas sessions, R2W). And a launcher says STOPPED when its program fails, and one made for the container says whose it is: a container's launcher double-clicked on the laptop failed on a missing input and still printed FINISHED, which would have sent the person looking for a picture never drawn. A launcher checks its inputs first and, if one is missing, says STOPPED, says whether anything is wrong, and says where the result already is (launcher RUN_TRIPLE_BACK_SIDE_LOOK v1.1, 14 October) (the triple-star sessions). And make folders in the foreground: a command that sent a folder's creation into the background with its run left the next command finding no folder. Build the scaffolding first, then send only the run to the background (R2W). A verdict is converted to a plain true or false before it is printed: a numpy true is not Python's True, and a printer that asked "is it True" wrote UNDECIDED on a line its own summary called HELD (run TWIST_LONG 20261011_072737, retired) (the Loop Atlas sessions). Hard-linked working copies are not pristine: a sitting's working tree was hard-linked to the unpacked parts of Bear's crate, so an in-place write changed those copies too. The originals stayed in his zips, and the baseline book was taken from its zip. Keep one untouched copy of everything that arrives. And stop a run by the process number saved when it started: a run stopped with `pkill -f` on a pattern that also matched the shell's own command line killed the shell (exit 144), the kill matching its own shell once more; later a pattern matched nothing (the run's arguments read "0.30 0.40 0.10 8000", not "0.40 8000"), and the run's files were renamed as void while it still ran, until it was stopped by its number, 300 (R2W). A tool that writes a stamped ledger refuses to write over one: two runs of a draft tool in the same second wrote one ledger over the other; the draft was restored from its kept copy (fingerprint 7229d8fc89d63bdb) and both runs were made again two seconds apart, and every later tool stops if a ledger with its stamp exists (the triple-star sessions). The kill that matched its own command line came back twice more in R2W: a stop by a check's name stopped itself first, as the same trap had already hung a wait loop, and with the pattern written so it could not match itself (a bracket round one letter) the same command went on to restart the check by name and killed its own shell again. Look before acting, and

stop a process in a command that does nothing else, then start the next one in a separate command (R2W). The kill that matched its own shell came back once more in the Loop Atlas: a pkill whose pattern also matched the shell's own command line ended the shell that sent it (exit 144) and left two half-built folders of a build behind, so each background job's process number is kept at its launch and the job is stopped by that number (the Loop Atlas sessions)

the Loop Atlas sessions; practice

The cost of precision depends on where you stand (659) — the filter needed against a gear's room falls with slope −0.92 on the watershed, and with slopes −0.60 and −0.57 one step off it, slopes the FM reading predicted before the blind runs; the rule holds only where the filter actually cuts the swing. And a bigger kick costs more than in proportion: with the timing held, a kick 1.23 times larger needed a filter 1.33 to 1.61 times slower, about as the square, which two pairs on a 10 per cent grid can only call a lean. The cheapest place to be accurate is away from your own resonance (the Loop Atlas sessions, R2W). A knob that cannot cancel a term can still buy a better place to cancel it: R2W's fourth harmonic shrinks the window's fourth-power term only by about half at the 0.55 edge, and never zeroes it. Its larger use was indirect: a stronger drive shrinks the fourth-power term but brings back the light coefficient, and the fourth harmonic takes that back out; at the joint points c fell to 0.30 and 0.28 of the edge's (RATCHET-50; the joint edge, RATCHET-49 to 51 and 56). In an engineer's words, the second knob relaxes a constraint, so the first knob can be moved to where the cost is lower; a reference designer's curvature correction works the same way. And price a cost by naming what slips: Bear asked what holding an array of riders in phase lock costs, and whether the cost falls like an error bar. There is no single answer until the error is named. A whole group drifting together costs a fan of three 4.7864×10^{-3} of a lone rider's slope and an even fan of nine 4.0028×10^{-10} of it; a rider slipping on its own inside any group costs as much as a lone rider; and independent jitter adds as the square root of the count whatever the arrangement (NINE-1 and its reading). The run measured pushes, not energy, so the cost was priced as the timing tolerance a given error in the push allows; an energy price needs a model of the actuator, for a specialist with a bench someday (R2W). A stray tone's cost scales with its size, so the warning of row 770 has a price: SLOPE-SIZE, registered before its run, found the slope in proportion to the third tone's size, within 0.16% of BEAT-4's 0.086755 times 0.5 and times 2, and its mirror at the opposite timing within 7.56×10^{-6} of the slope; read after the data, a third harmonic spends a 0.01% spec about 5.8 times as fast per percent as the fourth (R2W, sitting P3)

the Loop Atlas sessions (R2W); ties rows 604, 770

A near-miss is a loan, not a gift (660) — near a ratio of 2, hold time multiplied by detuning and by the tooth rate tends to 1.53 radians, the good band's half-width, with nothing fitted. Being near a simple fraction buys time, not grip, and the interest rate is the detuning. This is the free ride of Bear's note "A Coincidence Has No Single Price", read in time (the Loop Atlas sessions, R2W). Two constants can share a law and still count different things: in a triple, slip period times offset is 0.978 turns by median, on nearly the same slope (−0.919 over 128 runs, against −1), but that constant counts whole slips while the free ride's 1.53 radians counts the time to leave a band. Only the slope is shared, and that only nearly; name what each constant counts before setting them side by side (the triple-star sessions)

the Loop Atlas sessions (R2W); ties row 648

Adjust, don't retire, and know when understanding the break is enough (661) — most of one sitting's best results came out of failures redesigned rather than dropped: a hit rule's failure became the slip staircase, an unreadable room test became the watershed's law, a wrongly named null became a gear surviving as a bare lock, and a fake notch became the relay's own comma. Bear also set the exception: "Don't try to save the Titanic if that bell train is going down. It will show up again where it belongs." For a one-shot test of a picture, locating the break and parking the idea is the whole job. One rule says never bury a death you have not understood; the other says understanding it is enough. Five of one Loop Atlas session's findings came out of follow-ons to failed predictions, among them a tap that held past its borrowed limit, a second gear catching a slowing runner, and pockets of regular spin beside a lock, each a 1:2 gear. A miss redesigned aloud and run again is the method's main road (the Loop Atlas sessions, the book). Keep every miss with its run, and count them: of 104 misses one book carries, the instrument is named in 58, a wrong guess in 50 and nature in 25 (a row may name more than one), and 20 of nature's 25 name runs the Loop Atlas book cites. The instrument fails before nature does. And a card run once and parked, with its break located and its next runs written down, is not a death: a draft test run once found its break in the solver's one-way valve, which ran away within 0.1 second (the Loop Atlas sessions, the book). Every miss gets its redesign aloud, checked at the end of each exchange: misses went without one until an independent digest found them, and were paid late. At the end of every exchange, list the failed lines and match each to its redesign (runs LN-8c, WRAP-1x, JOIN-1 and GRID-1c) (R2W). Of the 104 misses, 26 came from the list of 2 October and 78 from every row of the data ledger, their kinds counted by script over the kind column; in one chain of thin-groove runs, 9 of 11 rows name the instrument (the Loop Atlas sessions). In one session fifteen of 41 registered predictions missed, and most of what it learned came from them: the lattice reader's first miss taught scale, practice 3's misses taught that a touch can be real, and HISTORY_SETTLE's miss taught that settling is slower than a multiplier says. The redesign check fired at every miss, and most had a follow-on run in the same exchange or the next (the Loop Atlas sessions). Test a one-line law across

the whole atlas, and let the misses lead. A ratio law missed at five of eight slices over 600 passes (RF1, run 20261015_140219), and the redesign check found two breaks, not one: the reader's resolution sat close to the 1% asked, and from the star's eccentricity 0.26 the beat itself passes through zero. The second was the session's largest finding: the hold stops the star's ellipse near 0.28 and then turns it backward (the triple-star sessions). Leans, rivals and redesign checks, in numbers: in one R2W sitting, counted by pattern in the RESULT entries, leans and checks were written HELD 147 times and FAILED 39 times, and rivals FAILED 85 times (as designed) and HELD 8 times; the record names a redesign check said aloud on 21 lines. The close shows the loop working in one stretch: FLAT-75's lean missed, the check found the question uncovered, FLAT-50 ran as the follow-on and missed in turn, and the check then found the answer already covered by the small dial (row 788) (R2W). Expect redesigns to be harder bets: over the triple-star thread's month, to its final count of 28 October, redesigns landed 35 times in 79 read against 201 in 368 for first guesses, and could not be read nearly twice as often (triple_redesign_odds v1.2). Say the check every time, and register a redesign only when it can fail and be read

the Loop Atlas sessions (R2W and the book); ties rows 402, 613

Mail between sessions: tag every letter, manifest every packet, never infer a delivery, and keep it small (676) — every letter opens with one line of tags (letters read, letters acknowledged, this letter), every file name carries its tag, and every packet carries a manifest of md5s; since one session's ledger began, every letter has been accounted for, and when two letters arrived after a third the ledger showed at once that nothing was lost. Mark a letter delivered only on the receiving session's own word: one mailbox inferred a delivery from the other room's count of letters, and the letter had never arrived. Carrying a letter twice is always safe; skipping one is the only risk. A manifest written inside the folder it lists lists itself, as it stood half written, and the same slip recurred in a second room a day after that room had pointed it out: write the file list, then the manifest to a name outside the folder, exclude it by name, and check the unpacked zip rather than the folder. Say whose a figure is before reading it, so the author has one session to track, not two. And keep the traffic small: a note for another session goes as a paste of a few lines the author can carry, not a file to download and upload again, and nothing goes mid-run unless it changes another session's next run (the Loop Atlas sessions, R2W and the book). Name the ledger when two ledgers share numbers: one session's lessons were printed with its own ledger's row numbers in the form used for the register's rows, and another session cited them as register rows for two days; another session's numbering is written with its owner every time. And note the lessons the night they happen: one day's lessons, not noted as candidates, were three days later out of the conversation's reach and had to be gathered from a misses table (the triple-star sessions). And fingerprint a returned zip on arrival, before opening it: one unpacked before its fingerprint (its SHA-256) was taken could be named in the record only file by file, its four files one by one in the manifest (the run zip of the fine-grid runs) (the triple-star sessions). Keep every arrival whole, numbered and fingerprinted: one session gave each source a number (S1 to S45 for files, P1 to P10 for pastes), its md5 and its manifest, and kept every paste word for word. When sessions disagreed later, the whole text settled it: R2W's "Shapiro-like" of 11 October was found by reading its own reply again (the Field Guide sessions). Take the fingerprint of every zip the person sends as it arrives, before opening it, open it in its own new folder, and read every line of what he sends (the triple-star sessions, in their page for every new session). Run each claim of a sister session's letter in your own instrument before any file changes: the claims of two of R2W's letters, three in one and two in the other, were each run in the Loop Atlas's needle first (FOLD_OR_FLIP to _4, RUNG_ABOVE and _2), and one word moved and no number ("born at a fold" became "born at the period-4 orbit's doubling") (the Loop Atlas sessions). A fork in another window is checked read-only, and nothing is merged by unzipping: Bear reopened R2W's previous session by accident, and it packed a small delta carrying its own copy of the record, which, unzipped, would have overwritten the sitting's record. The check was read-only (no such zip in the workspace, the record's prefix identical to the last delta's), and the one approved change, an epigraph, was carried by a sentence and added by hand (R2W). Read a sister room's record whole, in its own folder, before letting any of it in: list every file with its lines and how it was read, search it for what must stay out, mark each crossover with where it points, and let a lesson cross as a sentence that names its room, never as a file (R2W's B11 delta, 36 files, read by the Loop Atlas on 16 October) (the Loop Atlas sessions)

the Loop Atlas sessions (R2W and the book); the triple-star sessions; ties rows 572, 591, 592, 666, 774

When the work feels slow, look for the missing lane before cutting the method (679) — asked whether he was adding requirements faster than the work moved, the author was first answered by a plan that slowed the three habits: a follow-on for every failed prediction, a reply to every letter, a test for every surprise. Bear: "These rules were building in curiosity and tenacity. A full-time science team spends years asking and rebuilding with these questions. We don't have a team. Just us. And we get one paper. Or one book. When we are lucky." The revised plan kept all three and added a second lane, runs in the background while pages are written, and the Loop Atlas book's parts written in full went from 11 to 13 of 21 over two such days. The habits are the method, not its cost

the Loop Atlas sessions (the book); ties rows 589, 661

Keep a ledger of every knob, with no-data rows, and turn the untouched ones (684) — a toy rider has four knobs (mass, pull, passing speed and the weak wave's strength; drag and the strong wave's strength fold into them), and the weak wave's strength had never been turned. Turned, it proved to build every groove: with it off no rider

lands on any fraction but 0, the number of distinct fractions runs 1, 7, 19, 7 and 5 at strengths 0 to 0.6, and it raises gear 1's grip by about 0.16 times its square, the sitting's most surprising result from its cheapest run. Keep a ledger of every knob, with the runs that turned it and a row for each knob still without data, the instrument's own included (a step count shown dead at the depth read is a knob checked, not assumed), and turn the untouched ones. Open threads get the same ledger: one place where a thread leaves only as done, with the run that closed it, or dropped, with the reason (the Loop Atlas sessions, R2W). A held limit can hide luck: a pin sharp to 0.00032 across weights 0.10 to 0.40 at ripple strength 0.30 was five and four times blunter at 0.25 and 0.35, the strengths either side, the sharpness partly a next order nearly cancelling there. Before calling a coincidence a law, move the knob you did not move (R2W). When a law's sharpness may be the cause, register the trend in the sharpness, not one value: a dirty spring bead's extra buzz (3.008 against 2.104 switches a unit of time) is mostly the dry friction law's dead stop, since a smooth law keeps 0.25 of the 0.90 extra at a softness of 0.001 and 0.07 at 0.003, and raises the push; a bar set at one softness missed by 1.8 points (run SMOOTH_DIRT 20261014_084856) (the Loop Atlas sessions). The fine knob, turned all the way, was the strongest knob of the day, as Bear's "how can we use the fine tune knob as the strongest knob of the day?" asked: near its zero the second tone's size is a gentle trim on the edge, and turned far out it made the held timing weight-proof where the drive ratio could not. At ratio 3.25 and second-tone size 1.2353 the push stays within 0.0773% of its weightless value to weight 0.32, and the band ends at 0.3356, 1.92 times as far as the window-proof edge reaches (runs RATCHET-34 to RATCHET-37; the small dial turned all the way). A knob labelled "fine" is fine only near its zero (R2W). Knobs are scientists' words too. Bear asked whether scientists use knobs the way a toymaker or an engineer does. The fields that live by knobs have names for them. A control parameter is any setting you turn. A superconducting qubit has "an ideal working point" where its frequency is "independent, to first order, of the bias parameters" (Vion and colleagues, 2002); the field now calls such points sweet spots. Optical lattice clocks hold atoms in light of a magic wavelength, where the light shifts both clock levels alike (Katori and colleagues, 2003). Industrial statisticians call the whole craft robust parameter design (Taguchi): set the control factors so the output holds while the noise factors wander. A deliberately simple working model, built to show a mechanism plainly, is a toy model, and physicists use the word with respect. R2W book's §6.7 carries this paragraph with its sources as read (R2W). Tell every result with the settings a builder would turn, in the order he would turn them: one session did so for every result, to Bear's delight at every knob found, and the Loop Atlas book's knob register gathers them in its back pages (the Loop Atlas sessions)

the Loop Atlas sessions (R2W); ties rows 584, 643

Hold back the data while the eye is learning (716) — Bear: "Let's finish the examples and then I will give you all the data. I want you to see it otherwise the learning is shortcut." With the numbers in hand a reader measures instead of looking; without them it has to look first, and the looking is the skill being taught. The practices that held the data back are what produced blind reads a scorecard can judge. A picture built for the person is also a look at the data: building a companion page meant looking at the orbit, which made one measure's 0.97 seen before registering, so its redesign was moved to fresh paces and the looking disclosed. Register first, or say plainly what the picture showed; and a card whose record has already been seen is judged on fresh settings, with the record printed beside it as unjudged (the Loop Atlas sessions). A planted case printed whole can show a judged number before the run of record: three blind runs in a row printed, inside a planted step or band test (row 766), a value a lean would later judge, the biggest stride's turn (24.90 degrees, a miss seen before the run of record), the soup riders' shares by time 1,000 and one pairing's share of arms and legs (0.834) (runs STRIDE_MAP 20261016_213858, SOUP_JUMP_3 20261016_225440 and ARMS_LEGS 20261017_012927). Say so aloud in the registration's row, change no judging line after seeing it, and where the planted case can sit at a setting nothing judges, put it there (the Loop Atlas sessions). And keep reading cold after the lessons: the record of the person's eye at work on new kinds of picture, a spread panel, a ladder, a song, a road and a dream, each lesson with its run so that a later session can rerun it, does not stand in for the person. The rings were his, and a reader who has learned that rings are often rungs has learned a habit, not an eye. Read the next figure cold, write the reading down, and only then compare (the Loop Atlas sessions). Say before the run what has already been seen and what is expected. In the triple-star thread the Mardling-rule reading at 0.12 and 0.16 was filed as a LOOK, not a test, because the ledgers had been read before it was written (run 20261016_185433); ZG1 was registered knowing it would read NOT BORNE OUT by arithmetic, so that a ledger carries it; HL2's design disclosed that its reading was close to known, and SC1's that half the answer was seen. The expectations said aloud then missed, and the misses were recorded: the [7:1](0) estimate was five times too high; LS1 was expected to pass by a wide margin and failed, because the integrator moved the pair; RA1's expected error of about 10^{-5} missed by three orders; and both of RL1's expectations were wrong. Written into the registration, the disclosure weighs a pass by what was already known, and the expectation makes a miss teach something. A peek said aloud keeps a registered test honest when the record shows that nothing rests on it: before RC1's blind sort the registration said that FN1 to FN4 had been seen and some reads could be guessed, so the sorting was left to a second agent; on the first real LX run (birth 7.119) the band grew past half a turn where probe 33 had it at 0.473 over 1,000 passes, which was seen before any LX reading, so the rule was not changed and the count was added only as a report; and LW5's one workspace test birth (7.3924098) was written down as seen and not read, with no amendment resting on it (row 369) (the triple-star sessions). A pattern seen in old data is a hypothesis until fresh runs are flown against it:

reading card 264’s json, Claude saw the third ring’s swings come in equal pairs in 8 of 9 flights and registered it before reading further; three runs on fresh trains gave 137 pairs of 137 in the last two, each pair’s swings within a quarter of a per cent (the Loop Atlas)

the Loop Atlas sessions (R2W and the book); ties rows 369, 573

A container’s run lives only while its session does (720) — a run in the session’s container died when the container was shut down minutes after the session went quiet (run 20261010_193937; 23 runs). A container can be shut down soon after its session goes quiet, and a run it holds goes with it. Give long runs to the laptop, keep the session working while a container run is going, and have every run write what it has finished to disk as it goes, so that a shutdown costs only the run in progress. It happened again in the middle of a run, and later in the middle of a turn: one run died with the container before it had saved anything, and a figure copied in the same minute was lost (its rerun saved its records the moment integration ended); a later run, on 14 October, was cut off five times in about an hour and a quarter, four times just after a turn ended and once in the middle of one, so the cause is not known. What held was the design: each run saved as it finished, and only the runs in flight were lost. Check that a background run is still alive before waiting on it (runs 20261012_060629, for tests S4 to S6, which died, and its rerun 20261012_062229) (the triple-star sessions). And keep each finished piece the moment it finishes, so that the next start reads what is already there: a run left going between exchanges was lost when its machine restarted, and the next version kept each seed as it finished (run FALL_CHAIN_2, its version 2 lost and version 3 kept) (the Loop Atlas sessions). A restart between exchanges ended every process, detached or not (GROOVES_10’s three parts, about twelve minutes in). The fix that held: parts that each finish inside one exchange, polled to their end, and a laptop kit for anything longer; the relaunched parts reproduced every printed line digit for digit (the Loop Atlas sessions). The disk stays and the processes do not, so expect the workspace to restart between turns: one sitting’s cloud workspace restarted three times while idle between turns (about 05:45:20, 06:19:29 and 07:00:31 UTC), and eighteen of the sitting’s 22 uptime readings were under 400 seconds, which means the workspace had just restarted. Nothing was lost, because no run was left going across a turn and the files checked after each restart matched the md5s the record last wrote. Check the uptime at the top of every exchange, verify the record and the files in use against their last md5s, keep runs inside a turn, run every engine under `timeout`, and save each piece as it finishes (R2W). A shipped delta’s copy of the record is a free checkpoint: in a later sitting the workspace restarted at the top of each of its first three exchanges, and each time the record was compared byte for byte with the copy inside the last delta; the extra lines were only the delta’s own note, and no separate checksum list was needed. Over the whole sitting the workspace had just restarted at 22 exchange openings, four times in one stretch, and nothing was lost: nothing ran across a turn, every run wrote its rows as it went, and the record was checked against the last delta’s copy at the top of every exchange, so the restarts cost nothing (R2W). A restart can come mid-day with every file intact, so after any restart check the whole tree, from a script kept outside the scratch space; it takes under a minute. Twice on 20 October the workspace came back with an uptime of minutes (08:23 and 16:23 UTC), and both times the tree, hashed against the crate issue’s manifest with the last delta laid over it, matched but for two known files, the record, which differed only by the delta’s receipt line written after packing, and the delta’s own manifest, which lives only in its zip: 30,317 of 30,319 and 30,319 of 30,321 (R2W). Use absolute paths: the shell’s working folder can reset between commands, and one registration write was lost that way and written again (R2W)

the triple-star sessions; ties rows 129, 678

When the whole toy turns out to be a textbook object, its map becomes a bench (726) — stripped down, R2W’s rider is one equation: weight times acceleration, plus drag, equals a steady pull, minus a pull that repeats once per ledge, plus a small ripple from outside. That is a damped pendulum turned by a steady pull, and more exactly a particle in two waves, one standing and one moving, with drag and a steady slope added. The junction driven by radio waves is a cousin, not the same machine, since its drive does not depend on where the ball is (row 727). Every landmark its maps showed already has a name: the devil’s staircase, Shapiro-like steps, the saddle-node ghost, and the hysteresis Stewart and McCumber described for Josephson junctions (1968), in whose terms the weight is the Stewart–McCumber number. The tools found them blind. A ball first remembered its start between weights 0.70 and 0.71 (0.7026, computed for this guide), and with the ripple off the stopping pulls at weights 10, 5 and 2, read on a fine dial, matched the exact values to the dial’s step, a closer match than the textbook estimate they were first set against (row 711). The nearest published ancestor is the same particle without drag or slope, built with an electron beam in a travelling-wave tube, where a devil’s staircase appeared as the second wave was turned up (Doveil, Macor and Elskens, 2006); chaos in the junction’s own form was surveyed by Kautz and Monaco (1985; credited from its abstract, its text not read, row 552). So each match is a ruler checked against a known length, and where a map shows what the textbooks do not spell out (the swept-back feet of the whole-number grooves and a pinch among them), the tools can be trusted to be measuring. The claim of anything new moves from the physics to the reading: a person’s eye turned into registered numbers. Each piece of that has a name too (scaling laws, preregistration, blind prediction), so the combination waits on a prior-art pass of its own before it is called new (row 506). And name a borrowed effect for what it is: a junction’s microwave adds a push, while the ripple moves the bumps, so the resonance is the same and the forcing is not, and the steps are Shapiro-like (R2W). A part of a toy can be a textbook object too: a person’s clockwork, a 24-gon skipping four, is the inscribed-angle theorem, a look’s step about the circle’s centre being twice

its step about the pole. The theorem's prediction held (0.2154 against 5/24 = 0.2083, with 0.2579 for shuffled order), and the slides matched it to 0.0002 (−0.2112 against −0.2114) (run triple_maypole_m1 20261012_033532) (the triple-star sessions). And topology can fix a count while the physics moves the points: the pull of the three stars vanishes at exactly two points at every one of 38,400 looks, saddles minus hilltops 2, as the Poincaré–Hopf count requires, while the outer null follows the pair's quadrupole (r 0.9993) and the inner null's lean the first-order formula (r 1.00000) (run triple_null_point_look 20261011_225938) (the triple-star sessions)

R2W; the Loop Atlas sessions; the Field Guide sessions' check of its known answers; ties rows 145, 205, 398, 506, 552, 711, 727

A file sent is not a file in the working folder: measure the folder, and make the last step mechanical (733) — twice in one morning a person was told to start a launcher that had not been sent in that message, because its kit lay inside a continuity crate he had not unzipped. After every run message carried its files, wrong runs went on, and a tool that read the person's folder showed why in one run: files land in Downloads and reach the working folder by hand, nothing sent after the morning of 14 October had made that trip, and older kits sat in Downloads looking just like new ones. Guessing at a folder that cannot be seen had cost two days. A message that asks for a run attaches every file the run needs, even files sent before; a crate is a backup, never a source; and the carrying from Downloads into the working folder is done by a launcher, never by hand (the fine-grid runs' kit; the folder-check and intake tools, 15 October). Each shipment carries a list of every file's name, size and the first 16 hex digits of its SHA-256, and names the one launcher to run next; an intake script the person keeps in Downloads copies the listed files into the working folder by fingerprint and ends READY or NOT READY with the missing names (in the triple-star thread, triple_intake.py and RUN_TRIPLE_INTAKE.bat, a pattern to copy for another project). Never resend the intake under the same name: a second download becomes a "(1)" copy and the launcher would run the old one. Windows saves repeat downloads as "(1)" copies; never ask the person to rename them, and let launchers choose the right copy by fingerprint (row 721). Files go to the person loose and zips come from him, with one exception, a handover crate: flat zips with no folder inside, in parts of about 22 MB, each tested by unpacking all parts into an empty folder and running the manifest check. No nested folders in his working folder, ever (the triple-star sessions, in their page for every new session). The size limit is real: one ferry zip of 36.1 MiB, over a 30 MiB send limit, failed to send once, and the letters' packets now travel in their own zip (the Loop Atlas sessions). When the next step is a run, its message carries every file the run needs. Earlier kits carried only the changed files and left a run's own launcher behind; the slice middles went in one card of 50 files, and from the overnight kit on the launcher made the return zip itself, after Bear wrote that the folders were hard to keep straight. The two later laptop runs came back whole: 259 of 259 runs, and 3,279 of 3,279 (the triple-star sessions). A shipment is not delivered until the intake shows it landed: Bear's intake of the triple-star LW5 shipment (TRIPLE_INTAKE_20261017_140426) found all nine kit files but not eight paper files, and the next shipment lost exactly the MDs and PDFs again (list 20261017_235414). A new triple_intake.py lands in Downloads as 'triple_intake (1).py' while the bat keeps running the old one, so a fix to the intake cannot reach the laptop that way: every shipment now carries RUN_TRIPLE_INTAKE.bat and triple_intake.py, and launchers pick tool copies by fingerprint and refuse a stale duplicate. The rehearsal on a mock THREE-BODY folder caught a file-list swap in LW6's kit before it shipped (the triple-star sessions). The sender's side carries the same hazard: one sitting moved twenty-two superseded deltas and ten old loose-copy folders out of its shipping folder, not deleted, so that it holds only the newest delta and the copy the person has. A shipping folder with one live thing in it cannot hand him the wrong one (R2W)

the triple-star sessions; ties rows 572, 591, 676, 720

Let the command write the record: stamps, counts, ranges, and the line that cites a check (735) — a sitting that had already learned to read the clock before writing a stamp still wrote six stamps ahead of it (22:35 for 22:33, 09:11 for 09:09, 09:19 for 09:16, 09:29 for 09:28, 09:57 for 09:56 and 10:03 for 10:02), each time writing the time before reading the clock; once the notes printed their stamp from the clock (`date -u`) inside the same command that appended them, none ran ahead. Five FBP-1 outputs' line counts were written from memory (25 lines as 31, 61 as 64, 36 as 42, 35 as 33 and 40 as 39), and small counts before them too (BAND-1's "4 of 16" was 3 and its "14 of 16" was 15, and BAL-H's "5 of 10" was 3); even after that lesson was written a read table went in with six counts from memory, caught by counting with `wc -l` before it shipped. A summary range is a number too: three summary numbers were written down without being computed, a size of matrix entries ("10^4 to 10^5"), a span of ratios given as −1.019 to −1.069 where the code gave −1.0778 to −1.0227, and, in a book, a miss of an old line taken from a row the fit had left out (63 per cent where the right row gives 46 per cent); each was caught by checking the text against the tool's output, the two in the record corrected by dated lines, never by editing the note, and the book's edited before it shipped. And a line that cites a check is written only after the check has printed: one, a correction to DW-b's redesign check, appended in the same command as the sympy check it cited, said "(sympy now)" when the import had failed, so its attribution was false when written, though the formula was right, as a numpy check a minute later showed. Run the check, read its output, then write the line, naming the tool that printed the number (R2W). Read a stamp from the disk, never from memory: two figure runs and a ledger line carried stamps typed from memory. And read the built record back: it caught a run stamp cut short by a shell substring, as a fidelity check caught a reversed formula before any verdict was used (the Loop Atlas sessions). A reading found wrong is corrected in every record that carried it, in

the same batch, citing the run that corrects it: a straying needle first taken for a second motion (row 665) was reported to the person at once and corrected everywhere it had gone, citing the run of record that shows it (the Loop Atlas sessions). Never type a number in the command that computes it: the most repeated slip of one sitting, and of the two before it, was counts, roundings and products written into a record line or the read table in the same command that printed them, before the output was read. It came back all sitting long: two plan values typed in the same command that printed them (the printed values agree); later several numbers typed the same way; and on 16 October line counts typed into the read table before they were counted, and then a correction's reason written in the same command as the check that tested it, before its output was read. Every case was caught and corrected by a dated line, but each correction costs a turn. The fix that worked late in the sitting: compute in python, read the printed value, then write; and where a count must be written in the same command, compute it there with wc and write the variable, never a typed digit. Run the check, read it, then write (R2W). Typed before computed came back twice more in the next sitting: two percentages in a surprises line ("about 2.5%", "about 22%") and the eleven derived numbers of RATCHET-44's result line were written before a script computed them, all agreed when computed, and each check is a dated line in the record, beside that sitting's two dated CORRECTION entries for the same slip, R2W's lesson 81 and still its most repeated. At its close a byte limit was typed into a record line and then checked by a command, and it matched (R2W). Stamps typed from memory came back at least five times in a later Loop Atlas session (batches 46, 47, 50, 57 and 69), each caught from the disk before anything shipped, the last after it had entered four records, so every stamp is read from the disk by script, every time, and none is typed (the Loop Atlas sessions)

R2W; ties rows 570, 610, 677, 868

A running job is read by its own done line (736) — a run (version 5) recorded as stopped from a file taken mid-run was still live, and the next answer pointed the person at the wrong batch. Two scans ran side by side; a tail over both files showed the shorter scan's done line, and a reader run on the longer one, still at its last weight, failed its convergence check for that reason alone. And a progress check that printed the last two lines of a running job's output showed two cells of data before the reader had graded anything. A run file is a snapshot until it says it has finished; check the done line of each file a reader takes; and watch a running job by its line count and a search for its done line, which say everything a progress check needs and show no data (runs PINCH-2 and LAB-MAP-1). The same slips came back a sitting later: a fidelity check first read on its unfinished file, and a progress check's tail printing one data line of a run later voided (R2W). Wait on the log's count, not on a timer: RATCHET-50's wait loop gave up after 250 seconds, and its reader ran on files still being written; it printed "whole: False" and graded nothing, so no verdict was spent, but the stronger third tone had made two runs more than twice as slow as the probes. It was R2W's lessons 48 and 79 of sitting B9 again. Loop until the log holds one line per run, with a ceiling long enough to be a fault, not a forecast (R2W). Read a job's state from the machine now, not from the last note: a workspace restart stopped R2W's R83-d run while its record and its kit's README said it was running. And log the engine's exit status, not the logger's: queue lines that printed `rc $?` after `$(date ...)` logged rc 0 for a run stopped by hand, and completions are now read from each engine's own done line (R2W, sitting P2; second specimens of rows 720 and 786). The same holds for a build behind a wrapper: four cycles of R2W's sitting P3 have receipts saying "build exit 4" while the record's entries for them say only "gates exit 0"; the build's own exit first reached the record a cycle later, and the rest were found in reading the receipts for the guide's intake. Report the builder's own exit beside the wrapper's (R2W, sitting P3)

R2W; ties rows 610, 720

A check that does not gate the build is only a report (740) — a table cell cannot hold the table's own divider: an absolute value written with bars (|Φ11|, in one run's row of a paper and one picture row of a book) broke a row into extra cells, and the PDF printed it torn, with the rest of the row lost; a phrase check on the PDF caught one of two such rows only because its phrase sat past the break, and a count of the bars in every table row against its header found both, and none elsewhere; inside tables, Φ11 is written "beyond ±1" rather than as an absolute value, and the cell count runs before every build. A PDF builder also drops a row's lines where the row crosses a page, and a converter drops any cell beyond the header's count; the triple-star sessions' register lost text both ways, and a builder that refuses a row with too many cells and reads every paragraph and cell back from the built PDF later caught three bare bars before a copy went out. Then a cell count ran in the same command as the builds, so the builds went ahead before its result was read; it found a bad row (a cell of the paper's RATCHET-1 row holding absolute-value bars), rewritten in words, recounted to zero bad rows and the paper rebuilt before anything shipped, but only because the output was read before sending. Chain the build behind the check, so that a failed count stops the build instead of printing beside it; the same holds for a reader's verdict before its figure is drawn, and for a fidelity check before a new engine's first run (runs HYPER-1 and RATCHET-1). A read-back that counts words passes a paper in any layout: one long name in one table cell shrank every page of a test paper from 8.1 pt to 6.3 pt with every word still there. Check the type size against a calibration page set in the same run, and that every table header word is whole, beside the word count (build_paper.py v1.3; run LOOP_ATLAS_V13_FIDELITY 20261013_204846) (the Loop Atlas sessions). Code read by text from other tools needs the same gate. An inner name shadowed an outer table, so one tool had to restore the outer name by hand after another tool's text (HEAVY_WINDOW's) had run, and an edit by text replacement dropped the line that read HEAVY_SHAPE's measure, so a twenty-minute run of record stopped at its last step. Two

tools written new in the same exchange ran first time blind; the faults were all in code read by text. The tool now asserts the measure's signature, and the rerun repeated every edge: after editing a tool by text, assert every piece it reads, by its signature, before the long run (HEAVY_WINDOW_4) (the Loop Atlas sessions). Gates before builds, readback after: the table-cell and list-gap gates (CELLCOUNT_B10, LISTGAP_B10) and the block-by-block readback (READBACK_B10) found every layout fault that would otherwise have shipped, the readback being the built-artifact proof row 553 asks for. In the book, the only phrases the readback missed at the close were table rows split by a page break, each then looked at on the page; in the reads, the same three typeset lines every time (R2W). A checker that has never failed deserves a test of its own: the row-split check had reported no split rows in every build of one sitting, but it looked only at where a row's first and last cells began. A new paragraph pushed the first row of the dark-point table across a page, and the read-back, a different check, caught a phrase missing; shown that row, the row-split check still passed it. Rewritten to place the start and end of every cell, it found three older split rows in the book and seven in the plan that had shipped as clean. The new paragraph was moved below its table, and split rows were carried as known to the book's final build, which read 3 (sitting P3). When a check's answer never changes, feed it a case known to be bad and see whether it notices (row 551) (R2W)

R2W; the triple-star sessions; ties rows 551, 553, 593

A long piece of work needs a finish line it can see, counted in words and in tests (765) — count the road before walking it: rules that put each surprise's test on a list while one test runs an exchange make a road with no end; at one session's close the proposed list held 29 tests by count, with 26 runnable cards beside it, and every exchange named three more surprises. A project that must finish freezes its cards and writes its surprises down instead of running them (the Loop Atlas's handover of 15 October, from its sixth review of how it works). Depth shows in the word count: following one toy's window took the Loop Atlas book's Limb G to 13,892 words, more than its Limbs A to F together (8,527), by a count of the book's sections on 15 October. Count a book's sections at every review, so that a deep dive is seen as its share of the book while there is still time to choose. Count the runway too: rebuilding nine records' PDFs, a manifest and a shipment every batch, with their read-backs, filled the context. Print summaries, not tables; edit records by row scripts; do not reread large records to edit them; and ship once a sitting, or whenever the person asks, so that a sitting's runway goes to the book and not to the files. The record ritual's weight is a choice to make on purpose. And count the person: when the person is tired, the answer is a plan with an end, not more runs (the Loop Atlas sessions). Count the road before you promise to walk it: on 15 October three crates arrived holding about 260 lessons, and counted by script first they turned out to be several sittings of work, with two earlier batches that never arrived at all. The count turned "integrate everything" into a decision for the person, version 7 now or version 8 (he chose one version holding everything, row 774) (the Field Guide sessions). And check the book plan when the cases start confirming instead of teaching. The person asked "How far in the weeds are we?"; the thread had answered two rows of its book plan's Part 4, and the next two runs only confirmed rules that already had reasons. That is the moment to stop adding cases and turn to the plan's thinnest part, there Part 5, the substrates (R2W). Count the book by its parts at every review, and name the locale with every count. Over thirty exchanges in three days the Loop Atlas book grew from 66,556 words to 92,876, and by part, at the opening and at the close: before the chair, roots and trunks 10,252 and 12,281; Limbs A to F 8,527 and 9,556; Limb G 13,892 and 15,469; Limbs H and I 6,923 and 14,752; the crown and canopy 1,580 and 3,690; the back pages 22,512 and 34,242, with a new knob register of 38 rows and a references section begun; three short parts, the title page, "Someone was always there first" and the Declaration's stub with the build plan, held 2,886 words at both counts, and the parts sum to wc's whole within a few dozen words. A word count belongs to its locale: the same book counted 89,963 words by the shell's wc -w and 89,980 by the same command run from a Python subprocess, whose locale was C.UTF-8 where the shell's was POSIX. Name the locale with a count, and fix it (LC_ALL=C) in every script that counts, so that counts made at different reviews compare (the Loop Atlas sessions)

the Loop Atlas sessions; the Field Guide sessions; R2W; ties rows 591, 679, 777

Give a long project's exchange a fixed shape: the old card first, then one prose step, the misses' redesigns, and one test chosen by what it would change (767) — a slot that is not first loses to the live thread: in one project's five exchanges the older-card slot lost to the live work four times in five, so as a rule it was not working, and a slot like that needs either a fixed place or an honest retirement; moved to open each exchange, before anything else, it held five of five. A proposed list that only grows becomes a second queue with no way out, so each surprise goes on the list with one line saying what its test would change in the book, a test that would change nothing is marked so and left, and the one proposed test an exchange runs is chosen by what it would change. A chain of tests runs on while each answer changes what the book can say: five cards in one exchange is more than the shape asks, and three of them were one chain, each run because the last answer changed the book. And an exchange given to reading another room's work in full is a different kind of exchange, not a skipped one. In one session's tally against that shape, the older cards ran three of three, the prose steps three of three and the proposed tests three of three, and seven misses' redesigns were run and two more registered (the Loop Atlas sessions, its practice log). The old card held for a whole sitting: thirty entries of one R2W sitting opened an old card, among them the k law at frequencies 16 and 24, the 0.85 edge, the flat point at another swing, and the twelve smaller crossings, so the book moved forward on old questions as well as new ones. Push each surprise to what it implies before it is filed: Bear's remark that all three surprises had

critical implications sent each of the eighteenth exchange's surprises one step further. The first (a crossing read by a straight line runs early) found a 2% error in the book, and with it the two band edges agreed in their first number within a fifth of a percent instead of 1.8%. The second, checked against the records, turned out to be stated backwards in Claude's reply. The third named a law, one over the frequency squared. A surprise is a lead, and the lead is worth following. The same sitting's review kept as its routine: registration with a reader first and stand-ins that make every verdict hold once and fail once; one read per run, after the log's count (row 736); leans with rivals; a second witness for anything too clean (row 106); an old card every exchange; a delta every exchange and a checkpoint at every opening (row 720); and Bear's words kept as he typed them, as a farm. And it named what to improve: pack a fresh issue before the delta cap (row 157), fill every machine value in a handover by script (row 570), and look up every origin reference in the record (row 591) (R2W). Focus is a practice, not a mood. One session's focus rules made it mechanical: name the exchange's one thread in the reply's first line, take the old card from a limb or trunk that has not moved, move the book every exchange, and count the book by section at every five-exchange review so both readers see where the time went (row 765). The counts showed drift twice, one limb growing while others stood still, and both times the next five exchanges' practice fixed it; the book grew wide because the old card came from a part that had not moved, a practice added at the fourth review. A person who is tired should not have to hold the focus; the practice holds it. A review that falls due at a busy exchange is easily passed by: the sixth five's review was due after the thirtieth exchange of the practice of the day (its evolution 6) and was given only at the move, three exchanges later, so the count of exchanges since the last review is taken from the practice log at the start of every exchange, not from memory (the Loop Atlas sessions)

the Loop Atlas sessions (its practice log); ties rows 679, 765

Work with the person in plain words: never ask him to judge from his equipment, take the wider door his words describe, and check the files before reassuring him (770) — never ask him to judge from his experience of equipment, grids, controls or hardware. Keep him to two sessions at once, with anything for another session sent as a short paste (the traffic rule is row 676). A structure decision is a presentation decision, so take the wider door his words describe. When he fears something is lost, check the files before reassuring him. And use plain words: say "pack a save point", not "close the sitting", and say what a word means in the work the first time it appears (R2W). A default is not a ruling: label it as yours. "The Explorer never enters a deposit" was one session's default, written as if the person had ruled it; he asked why, and his actual ruling was narrower ("not in version 7's crate until the Spoke is published"). Write every default with the word "default" and who set it, so that the person can overturn it without first having to find it (the Field Guide sessions). Never ask him for something the files already hold, and never send him searching for a file you mention. Tell him plainly what is big and what is small; he asks for this. He thinks in pictures and words: hand him pictures and plain sentences, never a calculation to check (the triple-star sessions, in their page for every new session). Plain words matter at the end of a day too: "close the sitting" was read as leaving the session, at the end of his working day, and "pack a save point" replaced it; and an overclaim, "every part of the book is drafted", was corrected the next exchange to "every part in the build plan" (the Loop Atlas sessions). A question inside the body is a question not asked: Bear answers every question he sees, and he reads the last paragraph for them. A decision asked mid-reply, or followed by a file card, waited two exchanges. His ruling of 17 October: every reminder or question is the last thing in the exchange, after any file is sent, and a presentation question with a default that goes unanswered for three exchanges settles on its default, said plainly and reversible at his word. And his sense of size is about reach: Bear ranked the two hands of row 801 among his top five moments for what else they touch. The physics is ordinary harmonic mixing, and Claude said so; the reach was real, since the same flip, applied to the timing, found a slope in the weight off the mirror line (RATCHET-45), and measured at 1% of the octave (RATCHET-48) it is an engineering warning for anyone running at the weight-proof edge, and a candidate knob, priced in row 659. When he calls something big in what else it affects, it points at reach, not at novelty; say which, and follow the consequences, which is where the work went (R2W). A short word with no record is read, said aloud and asked: at the close of one session Bear wrote that the next would do "pb1" and then the long one, and "pb1" named nothing in any record. The most reasonable reading, the proposed list's first test, where the stop leaves the window, was written in the handover, said aloud and asked back in the next session's first reply, and it led that session's first exchange (the Loop Atlas's card 157); he did not correct it. When the person's word points at nothing in the record, do not guess silently and do not stall: take the most reasonable reading, say which, and ask (the Loop Atlas sessions)

R2W; the Loop Atlas sessions; ties rows 573, 589, 659, 676

Two deposits that cite each other's unpublished parts cannot both go first (774) — the guide was to wait for R2W's book, and the book was to cite the guide's rows 686 to 719, which exist only in the guide's next version; each would have waited for the other. Let the one that deposits first carry its own content, and let the second cite the first's DOI. The gathering session's plan was the person's marks, then R2W's DOI, then the deposit, with the backlog carried to the version after unless he said otherwise (the Field Guide sessions). He said otherwise, on 15 October: the cross-dependencies and cross-posts were killing the timeline, so there would be no version 8. He ruled that version 7 would fold, build and integrate everything, confirming that everything arrived, with one publishing session as the single integration point for the projects' 10 to 15 final notes, expected over about five more sessions; that the no-physics reads would break out into their own book, the guide keeping the practice (row 613); and that publishing would be all

on the same day, or the three-body paper first, then R2W, then the Loop Atlas (the person's ruling). Waiting grows scope. Version 7's content was done on 10 October, but its release waited on R2W's book, and every day it waited more arrived (row 773 tells how). Freeze the content and set the deposit date as two separate decisions, and when a version starts waiting on another project, close its intake that day (the Field Guide sessions). On 22 October Bear set the order: the triple-star thread's paper first, R2W (*A Rider in Two Waves*) second, the Loop Atlas third, and this guide's version 7 after them (the Loop Atlas sessions). When the live record cannot be read from where you work, keep the fingerprints of what was sent and name who can unblock the check: after the islands paper's upload the DOI link was refused for rate and the record page was out of reach, so the md5 sums of the PDF, the markdown, the README and both zips went into the register, with the check waiting on Bear's link to the record. Companions' DOIs are added to related works by an edit in place, not a new version. The single integration point learned its own rule the hard way: on 17 October its requests stalled the projects it serves, and the rule was corrected twice in two days. It asks nothing of a session mid-project, and rather than stay silent it keeps every need in one tracked list, each with its easiest ask and a default (17 October); at every move and close it takes the project's crate with a closeout report (an after-action review, the headline register with its count, an index of what is new for the guide, and open work with a backward pass), keeps everything and sends no requests back (19 October). The first closeout written under it came with all four parts (the Field Guide sessions)

the Field Guide sessions; the person's ruling of 15 October; ties rows 676, 765

A citation is checked by someone who may not fill a field from memory, against the source the text names, and your own records too (775) — check every citation against its publisher or Crossref before deposit (the triple-star sessions, in their page for every new session). Three checkers, each told that no field may come from memory, looked up every work one guide cites, and they led to seventeen corrections to its own text, among them a sole author given co-authors, a power law in the wrong variable, a proof dated a year early, and a threshold described as a single bit. A fourth checker, blind to the list, rechecked 22 entries and found one title spelled as the preprint rather than the journal. A mismatch reported by a checker can be the checker's: one could not find the guide's interval for the quartic constant in the joint paper and called it a mismatch, but the guide cites the thesis, which prints the interval exactly as its equation 4.45. Read the source the text actually names before correcting the text. And check your own records as you check everyone else's: four of the five titles of the author's own papers in the drafted reference list were the Declaration's one-line descriptions, copied in as titles, and six of version 6's thirteen related links pointed at a single version's DOI, not the concept DOI. Both were found on 13 October, and only when the author's records were resolved from Zenodo's API like anyone else's (the Field Guide sessions). When a record service cannot be reached, list what stays unchecked; do not call it checked. An agent checking the islands paper's 27 DOIs could not reach Crossref without an approval no one was there to give; it matched 21 against publisher pages, author records and arXiv, found no field wrong, and listed six it could not check (Bountis 1981; Érdi and colleagues 2007; Georgakarakos 2003; Greene and colleagues 1981; Howard and Hohs 1984; Howard and Humpherys 1995) plus one volume question (Benettin and colleagues). They are on the deposit checklist, not marked done (the triple-star sessions). A refusal is recorded and the check goes elsewhere: when Crossref refused the triple-star thread's check of Gallardo and colleagues, asking that the request not be repeated (an earlier Crossref fetch had waited for an approval that never came), the citation was checked against the publisher's own page. And a credit records how deeply its work was read: for the islands paper Quarles and Lissauer (2016) were read through a summary of the full text, Doolin and Blundell (2011), set aside at abstract level on 8 October, were read beyond the abstract in their passages on stability and credited on 18 October, and van der Weele's full text was not read (the triple-star sessions). A reference list cut from the text by name and year cannot see a law the text names without its year: the Loop Atlas's table of published laws its toys met named 27, and 14 had no source in the references, Adler's locking law among them though its chapters named Adler nine times. Count every named law and author against the reference list; a benchmark is a citation too. When a paper was a preprint before it was a journal article, its two years differ: the guide's text gave Glutsyuk and Netay as 2019, the year of the arXiv preprint, while the journal version is 2020; cite one version and make the text's year match its entry. And when the session's tools cannot reach a records service, as Crossref could not be reached on 31 October, check against the publisher's own page, the journal's site or the repository's own copy, and say which

the Field Guide sessions; the triple-star sessions; ties rows 552, 676

Keep the person's side of the work small, and his finish line written down (777) — on 15 October, with the probe FW2 back and every guess of it borne out, the person wrote that the work was "losing its fun factor and becoming work", and that each fresh start began by missing things. The science was moving; what wore him down was everything around it: files that never reached his folder, kits that ran the wrong thing, a cloud workspace that restarted mid-task, and new sessions relearning the rules by breaking them. Fix the plumbing first (the triple-star sessions). The work is part-time and unpaid, beside the person's paid work, so keep it worth doing: short turns, at most one question and always with a default so that a yes is enough, at most one overnight laptop job a day instead of many small ones, and one download-and-intake step per run. Keep a written, finite finish line for each piece of work, saying what is in and where everything later goes, and hold to it: new probes wait unless the paper needs them, and the crazy flag, raised for something truly important, lets it through. A start-here page carries the rules, so a new session does not learn them by breaking them. Lead the work, and bring back what he missed; no scorekeeping over

whose idea it was. When he is tired, say so kindly, finish cleanly, and let him rest (the triple-star sessions, in their page for every new session; the Field Guide sessions). A finish line the person can see keeps a long paper finishable. The triple-star thread kept one start-here page with the finish line, a morning page when a run went overnight, and a status reply that sorted what is left into runs, checks and edits. Bear's side stayed one step each time: send the return zip, or say yes to a default. At the session's close the paper was complete in content with 8 to 13 exchanges left, most of them his editing (the triple-star sessions). Carry each result to the knob that moves it and to what an engineer would do with it, because knobs and engineering are how he understands. Laptop jobs take at most eight cores, six-hour runs are fine, and he is told what to expect on screen; the workspace runs pieces of a few minutes and keeps reading while the laptop works

the triple-star sessions; the Field Guide sessions; ties rows 573, 765, 770

Parked threads stay in the record, each with its test, and the person's own pictures with them (782) — a backward pass keeps what is not the paper as side threads worth keeping, parked with their tests, and each surprise on the proposed list. One thread's chest of 13 October holds: the escape tripwire, whose first wording (null-point looks at least a pass before a star leaves) is recoverable from 11 October, with 74 escapes now to try it on (and 1,183 more in the atlas of 7 October, row 194), and which may link to the Topological Tripwire paper; Heggie's bookkeeping (does the pair's mean eccentricity after an escape follow the energy the star carries away?), with the heavy-star scan and its longer extension holding the series; two runs that ended early (birth 6.900's twin at 3,725 passes, birth 6.920's main run at 5,282), most likely out of time, their last states read in seconds; the V point, a plateau at 7.1985 to 7.1988 and a stall at 7.1984 that look like the thin lock a scan asked about, its 15,000-pass scan never built; null-point leftovers beside a count that held (row 726), a hexadecapole fix for the outer null and an odd saddle count that topology requires; a 9/19 match with the Loop Atlas needle's orbit (rows 664 and 864), two equal numbers with no mechanism yet; the older doors of 5 to 8 October (door 1, a cliff or a fractal boundary; door 3; the next lunar coefficient; the page-trace rule; the gold lanes; the wall profile; "encoding into chaos"; and the kick threshold); and the evening's own surprises: the circulating stars at 0.12 point back at the island, their drift traced back to zero landing near 7.18; the container and the laptop drift apart only inside the rough stretch between the two-ninths locks; and the door has two sides after all, which side opens following the star's own eccentricity. The person's egg-figure observations, "a mod 3 or 1.5 wave form", "two saddles … three-chamber flow" and "running writhe", were translated into test forms, but only "5 twists around 4" was confirmed and tested; they are his, and they stay with their tests. The chest of 15 October parks the long pair at 30,000 passes and the next run as the finish line; chaos with speeds, his slow and fast zombies, after a friend (whether the first parting of twins falls into two groups or one); the shark's fin, his green trace on the triple-star paper's figure of the apse turn (whether its tip sits on the island's upper edge, and whether it is a logarithmic cusp; tested on 17 October, the tip's place held on all nine lines and the cusp went away on a grid five times finer, Two Readers, section 9); and, from his toolkit of 14 October, the spacing of rocking rates across births read the way zeta zeros' spacings are read, an idea that needs more rates than one grid gives (the triple-star sessions). A parked card that carries its redesign comes back ready: R2W's FOLD-DIR, the redesign named when KN3-3 missed on 7 October, was parked with its words written; one of Bear's proof marks reopened it, and it was registered nine minutes into the exchange that took it up, because the parked card already said what to run (R2W). When the person asks where a result went, trace it to every place it lives: Bear's sense that R2W's coin's third face "was a big deal and then it disappeared" was right, its two landings and his confirmed marks living only in a list of lessons and an explorer button, never in the coin's own section; traced, it was told in one paragraph, and its two redesigns, never run, ran before the deposit and landed (R2W, sitting P3)

the triple-star sessions (chests of 13 and 15 October); ties rows 194, 592, 664, 726, 765, 780, 864

A flag says where to point a specialist, never "this is new", and a session's end lists what was measured and not looked at, and what is exact but not yet priced (783) — one thread's chest at a session's end, 15 October, wrote every flag in that form, with every number from the project's record and with its run. Its flags: short scans overstate stability (row 732); a one-way crossing in the low-side layer, where of 18 chaotic stars born below the island that circulate, 17 turn the way stars above it turn, and all 13 that escaped from below were turning that way (run 20261015_032141, from earlier laptop scans), for a specialist in resonance overlap and separatrix crossing (Chirikov's overlap criterion, Chirikov, 1979; Henrard's work on crossing a separatrix, Henrard, 1982); and a ratio to the beat that slides (row 603), for a specialist in the second fundamental model of resonance (Henrard and Lemaitre, 1983), the regime Mardling (2013, Section 3.7) leaves for elsewhere. Measured and not looked at: the fine-grid runs' series over all 6,000 passes, of which only the first pass at which twins part within 600 passes was read, and not the twins' slopes on the fine grid; its apse turn per lap over all passes, so how the beat changes across the fine grid was not drawn; a mistaken rerun on 14 October whose seven finished runs repeated the earlier long runs to the pass a day later, a reproducibility check recorded but not used; escapes from below whose signs were counted but not their times against distance from the edge; and three runs that escaped in the integrator's extra tenth, recorded and not examined pass by pass. Exact, and one of them priced here: a straddle of a ninth 0.001 from the escape edge is exact, but half the births in the range lie as near a straddle, so it is nearly free; the other, 2.006 times the beat, is row 603's, and the next run answered it: a point on a slide, not a lock (FW3a and FW3b, run 20261014_235048). Its capability gaps each name what fixed them: rows 733, 776 and 777, and a rocking reader blind below 0.05 cycles per pass, fixed

by a second reader (the triple-star sessions). The seventh triple-star session's closeout wrote its flags the same way: how a resonant secular coefficient fits across one island, middle to rim, which the 17 October prior-art pass did not find measured anywhere (TRIPLE_PRIOR_ART_SEVENTH_2026-10-17); the integrator's arithmetic faking a slow drift in resonance frequencies at a tolerance that looks tight, large for the practitioner if a fuller search finds no one has measured it (runs 20261018_031349 and 20261018_031414; row 741); Georgakarakos's expansion holding at about half the smallest separation he tested, about 12% above his prediction (run 20261018_193951); and the non-monotonic curve van der Weele and colleagues report, whose full text has not been read (run 20261018_193103) (the triple-star sessions). Run the backward pass by script over the project's own ledger of verdicts, not by memory: sort every verdict by date, kind and tool, and read what was counted and never read. One such pass at a Loop Atlas session move found three things nobody had looked at: the rivals growing weaker day by day (5 of 12 held on the first day, 3 of 51 on the second, none of 25 on the third, counting all 88 rival lines; row 821 counts the 60 distinct rivals, whose wins fell from 5 of 12 to 0 of 14), the long flights' energy drift creeping toward its bar as the trains were packed tighter (9.35×10^{-7} against 10^{-6}), and a PDF's standing shortfall of 84 letters that was the word "text", the label of 21 fenced blocks (closeout F, part 4) (the Loop Atlas sessions)

the triple-star sessions (chest of 15 October); ties rows 552, 603, 632, 780

Count which lessons keep arriving on their own, rank them by the count, and spend on that synthesis the turns that record-keeping would take (798) — a guide's own finding is which lessons kept arriving independently, across projects. One guide counted them from 6 October in an arrivals ledger: ten families, each defined in one sentence before it was counted, with every arrival listed with its substrate and whether its session had read the guide first. On 10 October a check that cannot fail already led, with 21 arrivals in the fold for version 7, and chaos and binning carried the strongest evidence: 3 of its 12 were found by sessions that had not yet read the guide. Counted again through 23 October, from every lesson placed since, the check that cannot fail leads with 131, and the whole ranking now opens the register, with every arrival listed in its appendix. Open each version with the lessons that arrived most often, ranked, and fold a backlog the same way: a second example adds a count to its family and a line to the appendix, not a new row. Re-bin, don't grow. Record-keeping can crowd out the synthesis it serves: the session that kept the ledger shipped 130 files, 66 of them revisions of its record, and never wrote the ranking. Ship when something changes for a reader, and spend the saved turns on the synthesis (the Field Guide sessions)

the Field Guide sessions; ties rows 337, 502, 523, 553, 773, 778

A bar moves only before the first scored run, and in the open: count the tries at one number, keep the band the first fair try set, and carry an amended bar into every planted check (808) — a bar is the line a result must clear, and it is worth something only if it stays where it was put. Each new try at the same number is a fresh chance to pass by luck, so the tries are counted and the bar stays still; and when a bar has to move because the measure was misjudged, it moves before anything is scored, on the record, and everything in the tool that tests it moves with it. Count the tries. The triple star's drag size was tried five times, D1, D1′, D1″ (priced and stopped), D1‴ and D1⁗, and before any number was seen D1⁗'s registration said that a pass on the fifth try weighs less than a pass on the first. D1⁗ passed 5 of 5 (run 20261016_212538); the band, 0.85 to 1.18, had been set at D1′ and was kept, the 0.28 line sat 0.017 above its lower edge, and the paper's Section 7 says the pass came on the fifth try. Number the tries in the paper, never widen the band between tries, and report how close a pass sits to the edge. Amend before the first run, in the open, and carry the amendment everywhere. A 20-lap trial of triple_cs1, with nothing read, showed a circular start's eccentricity swinging to 0.052 within 20 laps, so the registered bar of 0.05 would have called every run pumped; the bar was amended to 0.15 before the first run, and the amendment was written into the register. But triple_cs1 v1.0 (run 20261018_201130) still printed NO READING, because its planted check 'none, three pumped' planted 0.06, a value tied to the old bar; v1.1 (1e8657180f66ed84) plants 0.20 and rescored v1.0's TABLE as it stood (run 20261018_203454): NO GAP. When a bar moves, find every planted case in the tool that tests it (the triple-star sessions). When a redesign misses its bar again, close the card: R2W closed two such redesigns as misses (CO-4b, RT-1b) rather than fit a bar after the data, while a third kept its bars as written, on new values, and landed (MIXTURE-b) (R2W, sitting P2). Say everything seen before a verdict, and which verdicts it touches: R2W's COIN-END registration said the tails riders' states had been read and had shaped its lean, PLL-1's result said its part logs were read while the queue ran, a correction said three of WEIGHT-WALK-4's verdicts are no longer blind to their reader's writer, and ENGINE-INDEPENDENCE said plainly that it was registered after its one likely failure had been reworded (R2W, sitting P3)

the triple-star sessions (lessons of 22 October); ties rows 132, 369, 660, 683

Trace every number in a paper to a ledger at every pass, with a scan that reads only sources, and read the scan's leads by hand (811) — a paper's numbers are only as good as the record each one can be traced to, and a paper changes many times before it is finished. So the tracing is done again at every pass, by a scan whose sources are the runs' own ledgers and nothing that quotes the paper back, and the scan's flags are leads to be read, not verdicts. Trace at every pass, and keep the untraced named. Each pass of the triple-star islands paper reran triple_provenance_table and a number scan and read the PDF back: run 20261018_071622 checked 97 of 97 rows, and scan run 20261018_071632 found 1,023 numbers, 1,020 held by a ledger, 1 by a run log only (1,050) and 2

untraced (40,000 and 200,000), while the PDF read back 570 of 570. The tables and scans before each pass were retired; a first run (20261017_235118) flagged a moved anchor and its two files were removed unread, and the check caught the Declaration still naming the morning's table and scan (runs 20261017_040402 and _040403). Read only sources. On 18 October three numbers (91.3%, 97.4% and 0.001412) were found only in the provenance table's own ledgers, which print each row's description from the draft, so they had traced to the paper itself; scan v1.4 (c87e2dff7cd57d86) leaves those ledgers out and reports the three as arithmetic on ledger numbers. On 19 October the rebuild's check report was renamed TRIPLE_REBUILD_REPORT_20261019 because the scan reads every file named CHECK, and v1.6 leaves out the source check's ledgers, which print the paper's sentences. On 21 October the scan was found counting each full-width figure's width tag, 100, as a number, eleven of them, each held by a ledger that prints 100; v1.7 leaves figure lines out (run 20261021_052844: 1,237 of 1,247 held by a ledger). Leave out every file that quotes the paper back and every piece of markup that looks like a number. A source can also be out of date: at the rebuild of 19 October one stale number had survived in the draft of record, old Section 10's 0.9%, recounted as 3.5% (run 20261017_025855), which leaves the scan an open question, whether a number that an older ledger still prints, after a later recount superseded it, can pass it (row 570 retires only table ledgers; left unanswered when the thread closed on 29 October). Read every lead by hand, and say in the paper what the check cannot do. triple_source_check v1.2 (run 20261019_214932, against table run 20261019_212808) flagged 12 of 501 numbered sentences. Read by hand, the leads found nine runs whose numbers the paper states but no row named; the other leads were the check's own blind spots: shares printed as fractions (96.1% as 0.9606), a crossing printed to seven decimals (7.3925998) where the tool reads six, the TH1 container check whose file name carries no time, and a literature number. Reading the crossing sweep's ledger whole found probe 33's '11 of the 20' missing from the table, a claim made of small whole numbers that nearly every ledger prints and the check cannot reach. Table v1.3 (run 20261019_214850, 119 rows) brought the source check to 0 flagged (run 20261019_214934), and the paper's Provenance paragraph now says what the check did and cannot do (rows 570 and 599) (the triple-star sessions). Start the provenance table, the prediction ledger and the deposit list on a project's first day, when each row costs a minute and is written while its run is fresh: the islands paper was complete in content on 16 October and published on 28 October, its provenance table at 142 rows and its number scan at 1,446 numbers, and at the end each row depended on a reader reconstructing what one entry meant. Built from day one, the process stays small enough not to become bureaucracy

the triple-star sessions (lessons of 22 October); ties rows 570, 588, 599, 772

Rebuild a published model from its own table: report the structure and the digits apart, read the parts under every ratio, and scan each entry for the one that brings two numbers back (814) — rebuilding someone else's model from the table in their paper is a hard test for a bench, because the printed numbers are an answer key nobody here wrote. When the rebuild comes close but not exact, the gap is a clue to chase, not a verdict on either side. A rebuild that comes within a per cent of a paper's printed numbers, but not to their digits, is two findings: the Loop Atlas's engine rebuilt Tong and Dullin's diver (below) from their Table 1, and the field's largest size came back within a quarter of a per cent, the slow-spin gain within half of one and the best way out the same, while the inertias came out about 0.24 and 0.20 kg m^2 heavier, 1.2 per cent stretched out and 3.9 per cent piked (runs TONG_DULLIN 20261017_173718 and TONG_DULLIN_READINGS 20261017_174547). Report the structure that came back apart from the digits that did not, search the readings before doubting the physics, and say plainly while the cause is not yet found. A ratio hides an error its parts carry: the field, built as one inertia-like number over another, came back within a quarter of a per cent while both its parts missed, and the shape where the miss was largest, the piked one where the lower legs fold, pointed to where the error lived, the knee. A share is read by script from the run's json: told from memory, the two misses were first folded into "about 1 per cent". Then scan each entry of the table alone for the value that brings two of the paper's numbers back at once; if two entries come up, ask whether they are one number, only their difference moving anything, and test the plainest restoration with nothing fitted and every number of theirs. Tong and Dullin's knee had most likely dropped one 0.08 that its partner pair carries (theirs to confirm); restored, every number came back within the bars registered for it, gaps of 0.02 to 0.15 per cent (runs TONG_DULLIN_SCAN 20261017_181953 and TONG_DULLIN_KNEE 20261017_182629) (the Loop Atlas sessions). The model rebuilt came to the project as a gift of prior art (row 506). The Loop Atlas had said its jumper's and diver's shape part and count part were almost certainly known to sports biomechanics, names owed; Bear sent Tong and Dullin (2019), Using the geometric phase to optimize planar somersaults, which has the same split for the planar somersault, the dynamic phase proportional to the angular momentum and the geometric phase set by the shape change alone, validated on a digitised 107B dive, with the out-and-back zero (their Fig. 11) and its cure, a way out that differs from the way in. When a split falls out exactly, look for it under the field's own word (here "geometric phase"), and say plainly that the toy repeats it (the Loop Atlas sessions)

the Loop Atlas sessions (Guide items of sessions F and G); ties rows 328, 384, 506, 552, 650

Count a project's own record by kind: what landed beside what missed, how often the leans hold, and each rival once (821) — a long project's records can mislead about the project itself: wins drop out at every handover, guesses can be too safe, and a rival that only says "no" is not a rival. Counting the record by kind, with a script, shows each of these. Keep a register of what landed beside the record of what missed, and read it first at every handover:

weighed one at a time against everything ever known, every result reads as small, and a handover or a summary of the conversation keeps the open items and drops the finished wins. In the Loop Atlas a single inherited paragraph calling the work modest stood through two handovers from 15 October in place of 41 benchmarks, 18 found-here candidates and 640 held predictions, until six readers collected them on 18 October (the register LOOP_ATLAS_HEADLINES; the guide's part on the headline register). Count by kind, not by combined rating: a benchmark met to the number is an achievement of its kind, whoever was there first. The same readers found 22 sentences printing more or less than their runs carry (the Loop Atlas's misses row 365). Count your own leans: over one session the registered leans held 73 times in 116 (0.63), over the next 68 in 84 (0.81), while the rivals missed 36 times in 39, and guesses that hold four times in five are either well judged or too safe. From then on every card carried one long shot with its odds written beside it, under even; the first three went one held (1 in 4, the pumping coil unspun) and two missed (counted from the data ledger's verdict lines by a script; 160 older rows carry no verdict in the parsed form). And count rivals once, and by kind: a project's 88 rival lines were, on 19 October, 60 distinct rivals, 28 of them being one tool's own checks counted again at every fidelity rerun. Of the 60, only 8 named a competing idea (2 held), 41 were the bare negation of their prediction (6 held) and 11 checked a tool (none held). By day the competing ideas fell from 5 to 3 to none while the rivals' wins fell from 5 of 12 to 0 of 14: rivals grow tame when they stop being rivals. A rival written as the strongest competing idea can win, and the next one did (run KNEE_FINE 20261019_040109) (the Loop Atlas sessions). A table set by hand and a register counted by script each keep what the other dropped, so print the one from the other once and read the difference: printed from the Loop Atlas's register, its hand-set table of 27 published laws held one the register had missed, the three-distance theorem, which became the register's 47th. And a book that ends in a wall of tables loses its reader at the wall: the receipts go in a companion volume built by the same script from the same runs (the Loop Atlas's record, 144 pages), and the book keeps one short appendix printed from the same functions, so the sample cannot drift. And read a rising landing rate with care: the triple-star thread's first guesses landed 51.1 per cent of the time in its early undated paragraphs, 56.2 from 5 to 19 October and 66.7 from 26 to 28 October, and the count cannot tell better guessing from easier tests, so neither is claimed. A landing gets its register row the exchange it lands: R2W's LAB-MAP-3, first called a modest result, became its row 205 that exchange, at Bear's word. And a recount matches every graded verdict to a ledger row and every row to a verdict: R2W found misses with no row (HUMP-1's H-b, WEIGHT-WALK-2's W2-c and W2-d, PIN68-1t) and a whole record file, the frozen-frame sweep, that the whole-record count included and the ledger did not (R2W, sitting P2)

the Loop Atlas sessions (Guide items of sessions F and G); ties rows 73, 649, 783, 798, 823, 865, 872, 874, 882; R2W's checks of sitting P2: RATCHET-80 and WINDOW-4, which went to its book's front pages (its method lesson 244, from its record PREREG_RELAY_STATION_2026-10-03)

Before a page publishes, check it against its runs three ways: every headline's number and words, an error bar on every claim from a noisy run, and where each number sits (823) — a page can carry the right number in the wrong sentence: a claim a little bigger than its run, a range narrower than its noise, or a value that belongs to the toy next door. These slips survive the checks that ask whether each number exists, so the page itself is checked against its runs before it goes out. Check every number and claim in a headline against its run before the headline is set, even when it was checked once before: setting 23 one-sentence headlines over the Loop Atlas book's parts, each number was matched to its run by script, and one headline already checked by an independent reader still said a rule held at three sizes where it was read off one and held at two. Run an error-bar pass over every claim by script, reading the runs' own units: of 35 claims read off noisy runs, 22 stood, 7 had kept nothing to put a bar on, and 6 had to change their words; five printed a "within" or a range narrower than two of their own standard errors, and one read a lean as scatter, a range of 0.79 to 1.08 that pools to 0.909 ± 0.011, eight standard errors from the law it was set beside. A number found in the records is not a number placed right: a script found every one of a 46-page rewrite's 1,334 numbers in its records but two honest roundings, and two readers who had not written it then found 25 numbers set in the wrong context, a run named short, a number given to the neighbouring toy, a caveat lost where a paragraph became a sentence, and none a wrong value. And a check must not read its own words: run again after the edition's own record row was written, the script found the two roundings there and passed them, until the records' lines about the edition were kept out of what it reads (the Loop Atlas sessions)

the Loop Atlas sessions (Guide items of sessions F and G); ties rows 570, 667, 692, 821, 825

A citation not reached is "not reached", not "not found": say what was tried, close a literature check like a run, and read the paper behind a summary's word (832) — a search that cannot open a paper has learned nothing about whether the paper exists, and a popular book's word for a result is not the result, so both are said plainly and followed to the source. A citation not reached is "not reached", not "not found": Herzog (1986), remembered and out of reach of the search tools, was on its publisher's own page (Ovid), found by the person in one look with two more papers the tools could not open. Say which tools and sites were tried, keep a remembered name on the owed list rather than dropping it, and ask the person for the page. A literature check closes like a run, each name checked against a publisher's or repository's record and each refusal said: one such check found the Loop Atlas's made wall in two media, the insect's wake capture in air and the jellyfish's parked ring in water, and found that two famous tricks, the leading-edge vortex and clap and fling, are not it (the data ledger's MADE-WALL-LOOKUP).

And a summary's word for a motion is a lead to the paper, not a reading of it: "bend and twist" in a popular book meant the bend's direction carried round, not a waist that winds, and read as twist it made a paper of 1935 look like the likeliest home of a result it does not hold (row 820); only the paper itself, bought and read in full, said so. Name each rule by what it does, not by a sign convention: "halves spinning alike" is Montgomery's twisting rule with axes pointed out of the joint, and Rademaker and ter Braak's no-twist rule with axes along the spine (the Loop Atlas sessions). Mark every source by the record it was checked against, and take a tool's summary or listing as a lead: with publisher pages refused, R2W's sources now say how each entry was checked; a tool's listing gave one paper eight authors where its text has nine, and two papers first checked through a summary were read end to end when Bear sent them, every statement holding. When a citation does not carry its sentence, cut the sentence to the record, then search for the work that does: R2W cut a sentence on a ratchet that reverses as its mass or damping changes to what its two cited records say, and the next exchange found the claim in print in other works, the first sentence right in substance and wrongly cited (R2W, sitting P3)

the Loop Atlas sessions (Guide items of sessions F and G); ties rows 506, 552, 775, 820

A book takes the person's words by tidying, not pasting: each quotation in clean sentences in his own voice, with its date and subject read in the voice file before it is placed (833) — a person's chat words are kept exactly as typed, and a book that uses them has to make them readable without changing what they meant or when they were said. The voice file keeps every word as typed, and a book takes them by tidying, not by pasting: at the person's word, each quotation in the Loop Atlas book was put into clean sentences in his own voice, spelling, grammar and emphasis mended and harsh words taken out, and kept only where it opens a question or a picture the chapter answers. A quotation carries its date and its subject: placed by what it seemed to say, a reflection typed on the first night was dated near the end of the work, and a line about the person's own guesswork was set as a verdict on the book. A second reader caught both, and four numbers the rewrite had copied faithfully from slips in the record, so read each quotation's date and subject in the voice file before placing it, and let the second read question the source as hard as the rewrite. Say a book's length in words as well as pages, since one builder's 47 pages are another's 120 (the Loop Atlas sessions)

the Loop Atlas sessions (Guide items of sessions F and G); ties rows 589, 767

Where the model makes a number exact, say exact: a ratio of rounded prints is not a measurement (834) — a number can be told too small as well as too large. When a model fixes a ratio by its own structure, a ratio worked out from two rounded printed numbers adds a scatter that is not there, and printing it as a near-match says less than the model shows. A project's register said a toy rider taps 11.98 times a second on a 0.25 m board and 2.99 on a 4 m board, "4.007 against 4". The toy's own parts B3 and C1 show every rider tapping at the same moments, counted in fall times, at every board length, the gap ratios 2.000000000 exactly, so the ratio of the tap rates is exactly the ratio of the fall times: 4, within the printed fall times' rounding. The line was corrected to "exactly 4 in the toy" (run BALANCE_BOARD 20261004_161712). Before printing a quotient of rounded numbers as a measurement, ask whether the model makes it exact; if it does, say so, and say what makes it so (the Loop Atlas sessions)

the Loop Atlas sessions (its record of misses); ties rows 214, 738

A count that goes on the front page is taken from the record by a script audited by eye, with every uncounted case read and every test of a rule counted (840) — a project's count of what it achieved is a headline in its own right, so it is measured as carefully as any result: a count taken from the wrong place, or by a reader nobody checked, shrinks or swells what the project found. The headline register (the chapter that opens the Register) keeps the results and adds them up; this row is how R2W did the adding up, and what each step found. Count the record, not the book's wording of it, and audit the counter by eye. R2W's register first held the 155 tests its book words as predictions, and the book's count was taken from it. A script then read every graded entry of the record: of 416 tests with a prediction written first, 355 landed and 61 missed, and 697 of 1,074 predictions held; the book had named 101 of the 355. The record's verdicts had been written in many forms over seventeen days, so the script's first version got 4 of 25 sampled tests wrong; each fault was named and fixed, and later samples got 1, 2 and 2 of 25 and 2 of 30 wrong. The count also found a filing error: TOY-SIDE-4e, filed as a miss, was the registered rival, which failed as a rival should while its lean, TOY-SIDE-4d, held. File a test under its lean, and write every verdict in one form, the tag, lean or rival, then HELD or FAILED, so that the next count needs no audit (rows 221 and 735). Before a count goes on a front page, read every uncounted case by eye: sampling finds faults one at a time, reading finds their class. Five readers read all 237 landed tests the ledger did not count, one by one, against the record and the book, and found a whole class the samples had passed over: tags registered as an exact symmetry, a gate or a check had been counted as leans, so a test whose lean failed beside its exact check read as landed (17 tests; version 3 of the script reads the declarations). They found 21 more that were checks, calibrations, voids, phantoms made from letter names, or misses whose control held, and four landings the script never saw. And they found what the count was for: 90 landed tests the book already reports as results without counting them, 15 misses already redesigned into landings, and a rule the register and the book had undersized. The count went from 355 of 416 to 331 of 402, and the book's own from 101 to 191 (row 667). Count every test of a rule, not only the one named for it. A verdict of HELD says the bar was met, not how many times it was tried, so a rule's sentence carries its case count. R2W's rule that two grooves at one drive are

tree neighbours was written, in the register and the book, as resting on one stacked pair, 3/5 and 2/3, found twice (ST-2, ST3-4), a count taken from the tests named STACKS. The same rule had been registered again for the pad atlas and had held blind there, 24 of 24 stacks (PA-5c) and 15 of 15 (PA-5d): 39 of 39. Search the record for the rule itself before writing its count (row 666) (R2W). The abstract carries every front-page finding, one sentence each, because a reader, human or machine, may read no further; and a long record is split only under a check that every sentence is kept, in order: the islands paper's appendices were opened with a plain paragraph each and their longest paragraph fell from 1,402 words to 242 under such a check

R2W (sitting B12); ties rows 221, 666, 667, 735

A summary is a new claim about every line it shortens: have it read against its source by someone who did not write it (843) — shortening a passage means choosing what to keep, and every choice can make the story cleaner than what was measured, so a summary is checked like any new result. Asked to keep a section's results in the text and move its runs to an appendix, Claude wrote eight summary paragraphs from the section it had just moved. An independent reader checked them, with a new section on order and chaos and a new word list, against the book, the record and the run files: about 130 claims held and 25 did not. The summary had credited a band at frequency 1.00 to the plain first number, which missed, instead of its drift, which held; it had hung the 48-times and 2.79-times figures on the wrong second-tone size; it said every edge was placed by a bow registered first, hiding the one registered bow that missed (R27-c); and the new section had the short-window worry backwards, a slow settling faking chaos, not chaos faking a settling. None changes a result, and every one made the story cleaner than its source. All 25 were fixed before the proof copy went out (R2W)

R2W (sitting B12); ties rows 100, 199, 667

A move is proved by restoring it: ship it in a fixed order with its list loose, rebuild the deliverable from the restored parts at both ends, and leave the old session idle (844) — when a project moves to a new session, everything it has learned must arrive whole, and the only proof is to unpack it into an empty folder and rebuild the work from it. Keep the move cheap, and ship it in order: the person decides when to move, and the session keeps its start-here page current enough that a move costs an hour, not a day. The move is the one turn that ships everything, files first: the handover (the start-here page, a first message to paste, the crate in parts with its list), then the Field Guide's package, the reads for the book and the publishing package, then a supplement for anything made after the parts, each zip named for where it goes (TO_TRIPLE_EIGHTH_SESSION_…, TO_FIELD_GUIDE_…, TO_ALL_SESSIONS_…). Prove it by restoring it, at both ends: before sending, unpack every part into an empty folder, check every file against the list, run the manifest check and rebuild the deliverable from the restored kit (on 22 October, 1,739 files, none differing, manifest CLEAN, the paper's markdown byte for byte); the new session does the same before any work, says so to the person in two or three sentences, and waits. Keep the list loose and the old session idle: a list never lists itself (on 20 October a restore crate's manifest at first read one file unlisted, the crate's own list packed inside a part), and the old session, left idle and not working, is a backup nothing depends on, since an unused workspace can be reclaimed and the parts in the person's hands are the record (the triple-star sessions, the Loop Atlas sessions). Restore every inheritance loose and prove it before the first edit: the triple-star thread's eighth session matched 1,739 of 1,739 files to the parts list, read its manifest CLEAN, and rebuilt the paper byte for byte, after asking for the missing parts by name. And rebuild only by the written recipe, diffing the result against the record before anything ships: an assembler run without its table argument rebuilt a 112-row table in place of the 119-row table of record, and the diff caught it before anything left. The Field Guide's own publishing session moved the same way on 31 October: every send folded first, then the book of reads checked against every project's reads (257 of 258, the 258th never kept; 297 of 297; 330 of 330), then a crate of 3,740 files in six parts restored into an empty folder, where the restored builder rebuilt the guide to the same md5 as the build of record. And prove the page as well as the crate: run every command the start-here gives exactly as it is written, in a fresh folder, as the next session will. At that move the new session's restore matched all 3,740 files, yet the page's own rebuild command stopped on its first run, because the builder writes into its output folder and did not make it; the proof before the send had built into a folder that already stood. The builder now makes its folder, and a start-here counts as proved only when it has been followed line by line. And fold the session's own lessons into the guide before the crate is built, so that nothing the session learned rides only in its log: at the Field Guide's own move of 31 October the lessons of its last two days were still in the log alone when the crate was made, and reached the guide afterwards as the crate's supplement (row 869 and four additions); since then it is the rule for every move (Bear, 31 October). Check a crate's page against its manifest: the start-here of 31 October said its crate carried every arrival from S46 to S81, and the manifest held S59 to S80. A restore tool reads its crate's name from that manifest: the first had its date written inside and could restore no other crate (the Field Guide sessions)

the triple-star sessions and the Loop Atlas sessions (advice at the move to final publishing, 22 October); ties rows 572, 591, 676, 778

What leaves for the public carries its tools as they ran and nothing that must stay home, and every outgoing file is searched for what was taken out (845) — a deposit is read by strangers and kept for good, so it holds exactly what made the results, and nothing private and nothing borrowed. Tools are deposited byte for byte as

they ran, with the author block and email in the README and the documents and never written into a tool, so that each run's fingerprint still names the tool that made it; Dancing on the Sun instead edited only its tools' comments and proved the code unchanged (row 583). Nothing private enters a public crate: no register, no reads, no shipment lists, no working notes (row 574). For copyright, outside papers the person supplied stay out of the crate and every shipment, kept in the working session's scratchpad only: none was in the Loop Atlas's closeout zip or its ferry. And a detail taken out under the privacy rule is searched for in every outgoing file before it goes: R2W removed two details under it, on 20 and 22 October, and searches every outgoing file for both, each count to be 0; neither detail is written in any file (the triple-star sessions, the Loop Atlas sessions, R2W). Confirm the scope of a privacy ruling before cutting anything: a ruling read as "no personal places, ever" began cutting the hometown from headers, when Bear had meant only the one private matter he named, and his hometown belongs in every paper's header. When a ruling about removing things is broad, ask what it covers first. A scrub of the text is not a scrub of the deposit: the Loop Atlas's reader's edition was scrubbed of the author's typed words while about 160 of its 497 tools quoted him as typed in their opening comments, every one meant for the deposit. Search everything a deposit carries, code comments and file names included, by the text's own rules, and measure the list of what is owed with the guard before planning the work (the toy pages carried 45 workshop names where the list named about eight). Scrub a long record's working names in three passes: rules for the patterns that repeat, exact hand edits each asserted to occur once, and a last look that stops the build on any hit; read the rules' output for the shapes they can make, and have the result read against the original by readers who have seen neither (three fresh readers found 109 places to mend in 89 rows). Build the deposit early and run the upload gate on it, so that its verdict ends every exchange after, and give the allowlist a completeness check of its own: the Loop Atlas's upload set, built by script from an allowlist (the book, the record volume and one archive zip, 5 files), passed the gate's text checks over the archive's 2,068 files on its first run and was refused only by names, two runs of record among them whose names carry words the gate takes as private; a rule written for working records had also dropped two tools by a word in their names, found only by asking where a launcher's tool had gone, so the script now counts every launcher's tool and every run's tool before it calls the set complete. A gate that refuses by name needs a narrow, visible way to pass a public file whose name only sounds private: check_upload_set v1.2, visible but not yet narrow (row 855), reads an exceptions list beside the set, each name with its reason, excusing the name and nothing else (the text is still checked), never a name marked not for upload, refusing the set for an exception with no reason or one naming a file the set does not hold, and printing every exception. Write each reason from the file's own words, not its name (the Loop Atlas sessions, its deposit allowlist and check_upload_set v1.2, 31 October)

the triple-star sessions, the Loop Atlas sessions and R2W (advice at the move to final publishing, 22 October); ties rows 553, 574, 583, 589, 855

Re-derive a verdict from its output before reporting it, and let blind readers compare before acting (852) — a session's own summary of its earlier work said a registered prediction had failed; the run's output, read again, showed it held. The danger is not forgetting but remembering wrongly in a confident voice. And two readers who sorted R2W's 910 withheld notes without seeing each other's piles, with a random sample of 40, found what a keyword rule had silently dropped: helper summaries are leads until the flagged rows are read (WEIGHT-CONE C-a; the record's withheld notes, 24 October)

R2W (sitting P1); ties rows 100, 772

The walk-through read for contradictions, before any upload (853) — Bear made it a standing procedure for every paper and book on 29 October. A fresh helper that has not seen the work, the record or the conversation reads the source and the built PDF in full behind a read table, and logs every number that disagrees with itself or with the numbers beside it, every claim stronger than its evidence, every stale line, every summary out of step with its body, every term used before it is defined, every working word a reader cannot know, every pointer that does not resolve, and the words kept out of print, each graded A (wrong or misleading), B or C with a quote, a place and a fix. The author's session fixes A first, then every contradiction, then clarity unless Bear says otherwise, by script, rebuilt and found in the PDF; a second fresh reader reads again; the report stays out of the deposit. In A Rider in Two Waves the front pages held 7 of the 38 contradictions and the section written last held 5: fix the front first. And before the read, lint every built PDF and look at its first and last pages: a builder's exit 0 is not a pass, and build_paper v1.5 once drew the book's opening figure as a broken-picture icon and exited 0 (the cold read of 28 October; SOP_WALKTHROUGH_CONTRADICTION_READ). In the islands paper three readers who had never seen it returned 19, 19 and 13 findings, each checked against the run ledgers before anything changed, and the readers' own fixes were checked too: eleven of thirteen and twelve of fifteen were real. A reader's fix can strip a credit: one cut the name of Novikov's theory from a sentence a companion work cites, restored the next day; list the people, theories and companion works a text credits before and after every fix pass, and compare. And after the last change, read the whole again: Bear made a final full read a rule for every paper and book on 28 October, because checking edit by edit misses what only a whole read finds. The islands paper's final read took each of its 85 changed passages in context and found twelve problems, and an evening read found nine more. The Loop Atlas found the same: its reader's edition, each section already checked, read cold by a reader who had seen none of the work, gave 100 findings, 17 of them wrong or misleading, among them a count printed two ways and an "every" with no count behind it; print every count

a text gives twice from one script, and treat “every” as a claim with a number. None of the findings was wrong in what it saw, but 10 proposed a fix the record does not support: mark each finding true, in part or kept, with the record that decides it, and have the new wording read by a second fresh reader. The final full read goes last, after every change, by someone who has seen none of the edits as well as by the authors, because checking edit by edit never reads a sentence that did not change against the ones that did. A round count needs its basket checked: R2W’s cold reader was given no number, and its 100 findings spread evenly over the book, while the Loop Atlas’s read, whose brief, found later word for word, named no number either, also landed on exactly 100, its findings reaching the edition’s thirds 50, 60 and 22 times (one finding can reach two), the last third’s 22 meeting a bar of 19 registered for it. A round count from a reader with no number may be where the reader chose to stop: brief the next reader that there is no target or limit, and to say where and why if it stops early, as R2W’s second fresh reader was: it logged 149, six of them serious (row 153). And reconcile the findings with the corrections by count: the islands paper’s three proofreaders gave 51 findings and 53 corrections were read back, a difference no entry explained until a reader set the two numbers side by side

R2W (sitting P1); ties rows 100, 153, 551, 811

Write the rule without the word it forbids (855) — writing down a word kept out of print, even to forbid it, put it into new files more than once in R2W’s sitting P1. Write the rule without the word, search every new file and everything headed for another publication case-insensitively, and let a tool hold the word only as a hash: the publishing packet’s upload checker of 30 October does. And an upload set is provisional until the author’s proof comes back without comments: the islands paper’s set of 24 October was superseded by Bear’s proof the same day, and the zips were made again on upload day, never kept as finished artifacts. An upload goes alone: an upload exchange carries the upload set and nothing else, every file in it goes up, and the reply lists each with its md5 and the line “all N go up”; anything not for the record travels in another exchange, its name beginning NOT_FOR_UPLOAD_. The rule was set on 30 October, because a file sent up by mistake cannot simply be taken back: by Zenodo’s own pages, read that day, a published record’s files can be changed only by contacting support, and a new version leaves the old version’s files public. The gate that enforces it is visible but not yet narrow: probed on 31 October, check_upload_set v1.2 let a file named DATA_PRIVATE_X.csv go up once it was excepted with a reason, since only a name marked not for upload can never be excepted, and a v1.3 that never excuses names such as PRIVATE or SUPPLEMENT was proposed and is not yet built (row 845) (the Field Guide sessions)

R2W (sitting P1); ties row 845 and the publishing packet’s rule that an upload goes alone

Credit and borrowed sentences drift toward a tidier story: read new prose against the record, write credit from a dated ledger, and quote a companion from its final text (866) — the Loop Atlas’s reader’s edition gained a front page, a chapter of toys, a closing and a declaration on 24 October, and a reader who had not seen the writing found 20 faults against the record, nearly all running one way: a condition dropped, a credit simplified, a count rounded to the whole (46 tagged where the run tagged 44), a toy page described as remembered rather than as it is. None changed a result, and the credit lines would have reached the record’s metadata. The declaration, rewritten from that list of faults, swung the other way: a second reader found seventeen, running both ways: some gave the human too much (old papers he was said to have found, where the AI had found several and asked for the one he bought), some gave the AI too much (work done “at every sitting”, “every tool run blind”), and some gave the human too little (the toy pages were his idea, and a headline result began as his guess). What held was written from a ledger of who did what, built loop by loop from the dated record, each quotation checked at its source. Hear the purpose under the question, too: asked whose theorem it would be under an audit, the book wrote the audit into its pages; Bear’s point was that the book’s purpose is the partnership itself, an AI research partner raised to co-director, and that the working record kept in the moment is what an audit would read, so the ledger belongs in the record and the book keeps only the responsibility line the rules require. Borrowed sentences drift the same way: set against A Rider in Two Waves’ text of 28 October, three of the Loop Atlas’s four quotations of it were words of its working record, one sentence said the opposite of its measurement (grips of opposite signs where it measures both positive), one count hung on the wrong law, and a run credited to it was the Loop Atlas’s own, while every number was right to the digit. Quote a companion only from the text its readers will hold. The triple-star thread’s cold check of its ledger’s 46 newest rows found the same split: every outcome right, none missing and none extra, and three quotes not word for word, each restored without any outcome changing. Numbers get checked; words are where the drift hides

the Loop Atlas sessions (session H, Guide items 455, 456, 457 and 461); ties rows 853, 846, 100

A month of an AI’s slips had three shapes, none of them arithmetic: words outrunning numbers, lists that did not list themselves, and things that lived only in the conversation or only in an estimate (868) — the triple-star thread’s Claude set its slips of eight sessions side by side at the close, and once a tool’s planted checks passed, the tool was almost never where the trouble was. Words outran numbers: a sentence in chat or in a summary said a little more than the run printed, a digit count, a scope, a ranking, a “most” (most guesses miss, beside 208 landed and 194 missed), and every one was caught by something that read the words against the numbers, a source check, a claim-word read, a cold reader or Bear. Lists did not list themselves: a crate list packed inside its own part, a deposit list that never followed what the launchers name, a manifest check that skipped run logs, a counting rule that

read one reading where it should have read all; each was found by following references from the files themselves rather than trusting the list. And things lived only in the conversation or only in an estimate: reads not yet filed, memory not yet measured, a finish time not yet timed. The remedies are rows 861, 862 and 856; what kept them in use was a fixed closing block, the same eight steps at the end of every exchange (the register entry, the weights, the redesign check, an old card, three surprises, the read, one shipment, questions last), which cost a few minutes and made each session's memory a file instead of a hope. The Field Guide's own publishing session met the same three shapes in its last week: a queue count given from memory, a count told before it was measured, and an upload checker whose self-test passed its own fixtures while missing the house's own PDFs. The block lightens on a day of pure exploration: the three surprises can be one surprise and its test, while the register entry, the redesign check and the no-physics read stay, because they are what the next session will need. Two more of the same shapes, from the Field Guide's own folds: a guide line written from a project's summary of a record was loose in three places until it was rewritten from the record itself (row 589), and two tests proposed one exchange were found partly answered by the misses rows read the next; read a record whole before writing about it, and before proposing what it may already hold. The weights are each landed result's size check, the answers to row 280's questions (how big, and what is ours), kept beside its verdict: the triple-star register of 18 October had logged verdicts and never weights, none for "how big" and none for "what's ours", and its 21 section headlines had never been gathered. A lapse, counted: by search of R2W's record, no note headed "three surprises" appears in 25 of sitting P2's 59 exchanges. Twelve of them changed only text; nine, exchanges 27 to 35, ran tests (COIN-3, CATCH-2 to CATCH-4, COIN-4 and R83-d), and some of those carry notes written after the data, with tests, instead; the digest does not say what the other four did (R2W, sitting P2). A fourth shape, from the Field Guide's publishing session: a fingerprint written into a reply ran past its first eight characters on characters that came from no file, and a manifest's first draft listed itself with an md5 taken while it was still being written. Copy a hash whole from the tool's output, or cut it at a stated length, and let a manifest leave itself out (the Field Guide sessions). The third shape again, in one sitting of R2W's: a page count carried from the build before, a registration's time written without reading the clock, a check value taken from the wrong run that voided its test, a placeholder time and a glossary line written from memory. The record named the family itself; from then on page counts were printed by the command that built the file (R2W, sitting P3)

the triple-star sessions (honest notes of 29 October) and the Field Guide sessions; ties rows 861, 862, 856, 100

Pack a delta during a long sitting, and at a move sweep the workspace for what the crate does not hold (871) — a sitting whose only copy of its work is the container is one restart from losing it: R2W's sitting P2 ran 59 exchanges over three days through two workspace restarts and shipped no crate delta until its move. And at the move, 44 files lived only in the workspace, outside the crate: the build cycle the record names by its workspace path, early record scripts, laptop engines and their four launchers; they went into the crate with an md5 list before packing. Scripts written outside the crate travel nowhere. A book cites results, not files, so a deposit needs an index of its own, and a file kept outside the crate is lost to it: checked both ways, the Loop Atlas's archive holds every run its record volume cites but nine, each a first version the record calls retired or set aside, or another project's run, kept in a session's scratch folder that no crate carried; the other way, 244 of the archive's 737 run files are cited by neither text, 142 of them blind runs. The allowlist now fails on any cited run missing unless the record says why, and an index of every file the archive holds was built at exchange 30 as Part 4 of the record volume, all 737 run files, each with its kind read from what the file itself prints and where it is cited (the Loop Atlas sessions, misses rows 541 and 542). A move is proved with a part withheld: R2W's next move rebuilt its crate from its 15 parts with everything outside them out of reach, all 33,169 files matched by md5, the book, register and reads rebuilt with text identical to the originals, the count reproduced, and a merge with one part withheld refused, the planted failure that shows the proof could fail (rows 551 and 844) (R2W, sitting P3)

R2W (sitting P2); ties rows 551, 720, 844; the Loop Atlas's checks of 30 and 31 October: ARCHIVE_COMPLETE

Check every row of a published table you lean on, and when one family of its entries is wrong, look for a step only that family takes (874) — a wrong entry may have company, and the company may have a shape. Briggs's precise table of the doubling constants for the maps $1 - \mu|x|$ to the power d (1991; his thesis, 1997, Table 2.3; copied by MathWorld and by Finch's Mathematical Constants, 2003) is right in α at every order from 2 to 12 and in δ at every even order, and off in δ at every odd one: 5.9679687038 for 6.0846890743 at order 3, 8.3494991320 for 8.3443847487 at 5, and too high by 0.06, 0.10 and 0.12 at 7, 9 and 11, the maps themselves, flown to period 16,384 at orders 3 and 5, siding with Feigenbaum's equation. The Loop Atlas found the family by reading every row after one disagreed. One slip reproduces every odd entry: the doubling operator's linearization evaluates the perturbation at x/α, a negative number, and written with signed powers, (x/α) to the power dj, where the family needs the absolute value $|x/\alpha|$ to that power, the operator is unchanged at even d and changed at odd d, while α, whose equation keeps absolute powers, is untouched. With the slip the Loop Atlas's solver gives the printed δ(3) to 2×10^{-29}; without it, the map's own, 6.0846890742621218960461210662593. Two calculations agreeing digit for digit, the table's own guard, could not see a slip both shared. Before calling a number a slip, test whether it answers another question: the operator is block-triangular on even and odd changes, and at order 3 neither block carries 5.9679687038, the nearest eigenvalue 0.117 away. Before naming a number new, try it against the old ones: the Loop Atlas's corner ratio, 1.9644457845, set against 10,985 products of powers of 2, δ, α and π, misses the nearest by 5×10^{-4}, the size chance

gives, while the same search finds δ^2/α^2 to 2×10^{-31}. A rough old value is worth testing as a reading taken part way: Delbourgo and Kenny's 1986 values at orders 2, 3, 4, 5 and 7 (4.67, 6.08, 7.29, 8.35, 10.2; their paper not read) are what the maps' own superstable gap ratios round to at one shared depth, periods 32 to 256, so their 8.35, nearer the printed odd value than the map's, is the map read part way, not support for the table; and they follow one stopping rule, two readings in a row rounding alike, at five of six orders (not order 10's 12.3). An old table is right where its arithmetic holds, and its own last rows show where that stops: Briggs's 1989 table of the quadratic map's superstable parameters matches the Loop Atlas's to every printed figure of its parameters and its ratios to about 10^{-15} through row 7, then parts about tenfold a row, as his own note warns. Seen before: McCartney (2020) found the odd orders wrong from superstable orbits to period 2^{32}, naming no cause, and his values are the map's at every integer order from 2 to 10 within one unit in his last place; the Loop Atlas retraces his route and adds two more, the table to order 12, and a cause, the sign, as far as a limited pass shows: a pointer for a specialist. When three readings of one thing agree with each other and with a table's other entries but not with one entry, read the table's source: the Loop Atlas read δ at order 3 three ways that share no code, the map's doublings, its superstable parameters to period 16,384 and the doubling operator's fixed point (6.08469, 6.0846891 and 6.0846890743), while the same fixed point gave the table's α at order 3 and its δ and α at order 4 to every printed digit. With the slip removed, the orders from 2 to 12 lie on one smooth curve, every third difference positive and shrinking (+0.0201 down to +0.0006), and the two curves seen before at odd and even orders were its shadow: a claim about shape is judged by a bar on shape, not on size (misses row 531)

the Loop Atlas sessions (session I, Guide items 517, 524, 528, 530, 533, 538, 540 and 541 and misses row 531, benchmark 51; its checks and records: DELTA_Z3, FEIGENBAUM_EQ_Z3, CORNER_IDENTIFY, TABLE_ROWS_FIVE_SIX, ORDER5_MAP, DELTA_INVERSE, BRIGGS_OWN_Z12, SIGN_SLIP_ODD, SLIP_DEEP, ONE_CURVE_B, SLIP_SECTOR, DK_DEPTH, BRIGGS_1989_TABLE, DK_CUTOFF, LOST_LABEL); ties rows 680, 552, 783

Before publishing a correction to someone else's numbers, have them re-derived by code that shares nothing with yours, read the paper that already says it, and give the note a book's disclosure and two fresh readers (875) — for the odd-order constants of row 874 the Loop Atlas had a fresh helper, given no number and told nothing of the dispute, write its own programs in a folder of its own, with five predictions for its answer registered first. Its superstable parameters to period 2^{20} gave 6.0846890742621219 at order 3 and 8.34438474873 at order 5, and its own solver gave the map's value with absolute powers and the printed one, to 23 places, with signed powers; all five held, and the rival, the table's values from the map, missed at both orders. It met a trap of its own and said so: collocation points spaced evenly put a spurious, growing eigenvalue above the true one. Read the one paper that already says it before claiming it: McCartney's (2020), read in its accepted manuscript, had found the odd orders wrong first, so the corrected values are his and the note credits him first. The note is a short paper of its own on the odd-order constants, approved on 31 October and published on 9 November as its own record beside the Loop Atlas's, *A Mirror in Feigenbaum's Table* (concept DOI 10.5281/zenodo.24101154); it is worth writing because, six years on, MathWorld and Finch's book still print the odd-order entries. McCartney's values and the Loop Atlas's solver also agree at the non-integer orders e and π, within 2×10^{-19} and 2×10^{-17}. An AI summary that agrees with the question it was asked is no witness: a search engine's summary, asked whether the printed order-3 value belongs near order 2.91, agreed and supplied a reason that contradicts the cited page; and a typo in a famous constant can travel from book to book, as a repeated 6 in δ once did. A short note owes the same disclosure as a book: it failed the publishing packet's lint for want of its tool and computational resource disclosure, and passed once it was written. A reason can be wrong where the result is right: the note said α stays right because its part of the calculation never looks left of zero, and a fresh reader pointed out that the fixed-point equation does evaluate g at the negative point x/α; α is right because g keeps absolute powers there, and the wrong reason had already been told to Bear and copied into a lesson. Triage every finding against the record before changing anything, then rewrite whole (55 findings: 51 fixed, 4 kept with reasons), and have a second fresh reader read the rewrite: 37 findings, 6 wrong or misleading, mostly in how sources were described, among them a proof for the quadratic case told as covering every order and a declaration line that made the AI's part smaller than the disclosure under it; going back to the primary page for one found that Briggs's thesis prints the fixed function's equation with the absolute value taken at the negative point, the half of the reconstruction that keeps α right. A simpler method already in hand is the cheapest cross-check: Briggs's own 1989 direct method, superstable parameters by Newton's method, with only its power changed to 3 and run in its own 64-bit arithmetic, gives δ at order 3 as 6.084689 by period 8,192, the maps' value and not the published 5.968; and what a reproduction can match in rounding drift is its size, not its value (the Loop Atlas sessions, misses row 540)

the Loop Atlas sessions (session I, Guide items 525, 532, 534, 535, 536, 537, 539 and 542; its note on the odd-order constants; its checks and records: INDEPENDENT_READER_JUDGE, MCCARTNEY_READ_JUDGE, MCCARTNEY_NONINTEGER, EXCHANGE23_JUDGE, NOTE_COLD_READ_FINDINGS_2026-10-31, NOTE_COLD_READ_TRIAGE_2026-10-31, NOTE_COLD_READ_FINDINGS_2, NOTE_COLD_READ_TRIAGE_2, BRIGGS_1989_ORDER3); ties rows 874, 853, 552

A derived edition checks itself against its source at every build, and the readers who draft it are a cold read for free (880) — the guide's reader's edition is made by script from the built guide at every build, one short entry a row, and every number, run name and quoted phrase in an entry is asserted to stand in its own row. So when the guide changes, the edition refuses to build until its entries follow: a row given thousands commas stopped it, its entry still printing 4420, and 26 values rewritten as powers of ten stopped it again until 23 values in the entries were written the same way. The eight helpers who drafted its 864 entries, each reading a batch of rows whole, listed about seventy rows that looked wrong, stale or tangled; each lead was opened against the row's source before anything changed. Of 22 leads on numbers, 7 rows were wrong, 8 were right but terse, 5 waited for their deposited papers and 2 were not shown wrong; of 20 rows flagged as stale, 17 changed; of 42 flagged for voice and typography, 26 changed. A paper pass on 1 November read the five that waited and the two not shown wrong, with four more rows, against the deposited papers and corrected 10, holding row 28 for its run file; row 15 had stated the reverse of its paper, and rows 246 and 251 had copied slips that are in their papers. By version the leads on numbers part: of the 13 in version 6's own text, 10 were wrong or inconsistent; of the 9 in version 7's additions, 7 were right but terse and the 2 wrong ones carried their sources' own words. A version 7 row names its run, so it could answer; a version 6 row often names only its paper. A reader who flags a slip has found an instance, and a sweep finds the class: the drafters flagged ASCII hyphens standing for minus signs in 3 rows, and a sweep found 7 places, one of them in a table the edition prints. Two more things a derived document taught. A summary cannot be cut from a row's first sentence, which is often a case and not the rule: titles and first sentences hold 37,436 of the rows' 203,170 words, and in rows 603 and 551 the first sentence is the example. And size a derived document from a built sample, not from a count of its parts: an estimate of 85 to 113 pages had counted about 30 words of generated lines and no titles, a built sample of 20 entries said 174, and the edition came to 167. A card made in the same run as a table is checked against it too: one issue of the stripped reference said 10 rows were not yet tagged with domains while the table beside it listed 32. A cold reader pays for itself on a paper too: a separate agent that had never seen Question the Substrate read all of it and listed 67 exact edits, each checked by script to keep every number, tag and citation. Its top finding was a name that misleads, "own doubt" for a rule that keeps what the network is surest of; it also found a phrase used 14 times and never defined ("on the letter"), and a body that pointed to none of its figures, in one pass of about 18 minutes. Only an outside reader hears where the prose mumbles (Question the Substrate)

the Field Guide sessions (the reader's edition and the clarity pass, 31 October and 1 November); ties rows 853, 869, 811, 15, 246, 251

A count is as wide as the places it reads: grade only in a result entry, prove a new counter on the old count byte for byte, and write both sides of a graded test the exchange it lands (882) — R2W's sitting P1 graded nine of its eleven tests under surprise heads or inside an exchange header, where no count looks, so the counter found two; the other way round, an earlier counter graded a phantom test inside a read's filing and another from a registration that named the verdict words in capitals. Before any number moved, the record listed every tagged verdict outside a result entry (32 of 322) and every capital verdict word in an entry the count skips (143 in 57 entries), for reading by eye. The new counter first wrote the old one's table byte for byte on the record's first 14,937 lines, reproducing 425 graded, 352 landed and 73 missed, before reading a later line; then all 99 of its later rows were ruled by eye, with a reason each. And a ledger with a page for landings and a page for misses is written on both: four leans that failed while their rivals held had only the rival's landing row, and a sitting's results waited for the recount until 25 miss rows and 29 landing rows went in at once; from then on each result went into the count, the register and the book's count strings the exchange it landed. The whole record read 515 graded, 436 landed (214 with every prediction, 193 with some and 29 only through the named rival) and 79 missed, to 2 November, and 527 graded, 448 landed and 79 missed at the book's publication on 5 November. Its ledger rule at the close: every miss gets a row, a landing gets a row when the book tells it, and the landings the book does not tell are listed with their reason, among them P1's five surprise tests with their redesigns S2B and S3B, ENGINE-INDEPENDENCE, WINDOW-WEIGHT-2's WW2-2 and TRICKLE-5's 5b and 5c, so a win the book leaves out is still kept by name (R2W, sitting P3). And the rising landing rate row 821 warns about is not general: R2W's tests landed 10 of 11 in sitting P1, 56 of 59 in P2 and 18 of 20 in P3, and the Loop Atlas's held share, guessed by the guide's own session on 19 October to rise from its first sessions to its last, went 0.692 before its session E (472 of 682), 0.612 in E (101 of 165) and 0.694 in F (75 of 108); the redesign, the share split by kind and by session, was registered and not run (the Field Guide sessions)

R2W (sitting P3); ties rows 840, 821, 809, 596

Ship the method with the number, and a toy any reader or small AI can run (891) — a single loop, which is what a newcomer tries first, keeps only 0.0806 of the learner's inside after 30,000 steps, where Two Arrows' 0.3049 is the average over the loop and its turned copy (run r3). A reader who tried to reproduce the headline without the turned copy would conclude it was wrong, so the paper's toy uses the paper's method by default, and a switch shows the single loop's raw numbers. The toy is one file of about 270 lines in the paper's crate; it reproduces the paper's healing run and all five kinds of practice of row 889 to within 1.78×10^{-15}, every game prints JSON, an AI can choose the settings with a JSON plan, and its guide is written in short sentences for readers in any language. A paper can ship a test ground for other readers and their AIs (Two Arrows)

Two Arrows (the two-voices series, Paper 2); ties rows 584, 644, 776

Appendix — The arrivals, family by family

Every arrival the ranking at the head of the register counts, family by family, in the ranking's order. Rows marked "by 10 October" are the gathering session's own list, word for word but for its references to its own working files. Rows marked "after" were sorted from the lessons placed since by fresh readers, one row for each event however many records reported it, with its project in brackets. "Lands at" says where the lesson sits in this guide, and a dash marks an arrival not yet placed in any row; "Had the session read the guide first?" says whether it was found blind. From 4 October every project read the guide before its work, so nearly every answer is yes. The 11 that say no are lessons met before their project had read version 6, all in the list of 10 October, and a lesson found blind in that way counts as independent evidence.

1 — A check that cannot fail

131 arrivals: 21 counted by 10 October, 110 after.

Counted	Lands at	Arrival	Substrate	Domains	Had the session read the guide first?
by 10 October	row 551	a shrink and a relaxation that could not move a Gauss integral (latch paper)	linked filaments	materials, topology, measurement	not known
by 10 October	row 551	a gauge check on a count (forest suite)	an arithmetic lattice	number theory, topology, measurement	not known
by 10 October	addition to row 337	"holds" on empty data (triple-star)	three-body orbits	three body, measurement	not known
by 10 October	addition to row 337	rival bars that overlap; a bar below the instrument's floor (R2W)	the relay's tests	driven oscillators, measurement	no
by 10 October	addition to row 551	a chaos test guaranteed by Bowen's inequality (R2W)	a one-way chain of riders	driven oscillators, maps and chaos, measurement	no
by 10 October	addition to row 242	a check pair on powers of two (R2W)	floating point itself	records and tools, measurement	yes
by 10 October	row 672	an index sum guaranteed by a theorem tests the solver, not the physics (R2W, B6)	the rider's fixed points	driven oscillators, topology, control and circuits	yes
by 10 October	addition to row 337	a rule's control window held no two states, so its "held" counted for nothing (R2W, B7)	a rider with mass	driven oscillators, measurement	yes
by 10 October	addition to row 210	a chaos threshold set below a regular spin's finite-time reading (Loop Atlas)	a regular spin	maps and chaos, measurement	yes
by 10 October	addition to row 210	five bars set under their readings' floors (R2W, B7)	the rider's instruments	driven oscillators, measurement, geodesy	yes
by 10 October	addition to row 210	three floors of one's own arithmetic in one evening (Loop Atlas), met apart the same night as R2W's	the needle's instruments	driven oscillators, measurement	yes
by 10 October	addition to row 337	a sign-only bar passes half the time on noise (3-Body Team)	three-body orbits	three body, measurement	yes: the guide read; row 337 read in full only after the runs

by 10 October	addition to row 337	a lock rule fed sweeps that stepped over its band could never say no (3-Body Team)	three-body orbits	three body, measurement	not known
by 10 October	addition to row 616	a pass whose peak was no higher than scattered dots’ (3-Body Team)	three-body orbits	three body, measurement	yes: the guide read; row 616 read in full only after the run
by 10 October	addition to row 616	a control built after the run showed the rule calls a smooth parabola S 0.718 of the time (3-Body Team)	three-body orbits	three body, measurement	yes: row 616 read in full before the run, and met anyway
by 10 October	addition to row 616	a weave test passed too easily until the shuffle (R2W, B8)	the triple’s pictures, read from pixels	reading pictures, three body, measurement	yes
by 10 October	addition to row 369	four registrations that could not be judged as written (Loop Atlas)	the needle’s instruments	measurement, records and tools	yes
by 10 October	addition to row 587	an exact zero from an arrow that could not leave the ridges’ plane (Loop Atlas)	frames on ridged paper	kinematics, measurement	yes
by 10 October	addition to row 210	a bar of 10^{-13} set against a value printed to 12 decimals (Loop Atlas)	the needle’s instruments	driven oscillators, measurement, geodesy	yes
by 10 October	row 712	a correlation bar on a sign the registration called free (Loop Atlas)	the needle’s lanes	maps and chaos, measurement	yes
by 10 October	addition to row 683	a band wide enough to catch the wrong mechanism (R2W, B8)	a rider with mass	driven oscillators, measurement	yes
after	addition to row 616	skeleton test T1’s permutation control passed too (0.861, 0.808 against bar 0.30) (triple-star)	three-body orbits, Bear’s ropes	three body, measurement	yes
after	addition to row 616	landing check’s controls passed as readily as its guesses (1.000, 0.810) (triple-star)	three-body orbits	three body, measurement	yes
after	row 730	rules on flat quantities scored exactly what one constant scored, 22 of 23 (triple-star)	three-body orbits, maypole rules	three body, measurement	yes
after	row 732; addition to row 732	twin test over 600 passes blind to slow chaos; island dissolved over 6,000 (triple-star)	three-body orbits, twin test	three body, maps and chaos, measurement	yes
after	row 732	fidelity bar 2×10^{-6} sits fifty times under the tolerance floor 1.1×10^{-4} (triple-star)	the thread’s fidelity checks	three body, records and tools, measurement	yes
after	addition to row 346	envelope test S1 nearly guaranteed by a spectral identity; held 15 of 16 (triple-star)	three-body orbits, envelope readers	three body, quadrature, measurement	yes
after	addition to row 551	deposit lint’s fixed list missed the draft’s own PENDING marks; would pass (triple-star)	a deposit lint	records and tools	yes
after	the part Two Readers and a Ruler	SQ1’s symmetry bar passed partly by construction: runs start at the centre (triple-star)	three-body orbits, strobe outlines	three body, reading pictures, measurement	yes
after	row 783; the part Two	walls-at-a-ninth test nearly free: rocking sits near a ninth almost everywhere (triple-star)	three-body orbits, island walls	three body, measurement	yes

	Readers and a Ruler				
after	row 782	an odd saddle count among null points was required by topology (triple-star)	the triple’s null points	three body, topology	yes
after	addition to row 93	four of the toy’s rules hold for any rotation, so matches elsewhere cannot fail (R2W)	the rider’s dance rules	driven oscillators, measurement	yes
after	row 738	a 10^{-8} closure bar set far below what rounded inputs allow (R2W)	weightless orbits in SELECT-1	driven oscillators, measurement	yes
after	addition to row 649	a lean and its rival, with 15% bands, differed by at most 5.2% (R2W)	the junction’s Farey children	driven oscillators, measurement	yes
after	addition to row 282	a dry run in the old regime could not expose the new regime’s arccos (R2W)	the JJ_9 engine	driven oscillators, records and tools	yes
after	addition to row 551	a reader printed HELD on a file holding none of the graded rows (R2W)	the FAREY-2 reader	driven oscillators, measurement	yes
after	row 738	a precision check passed because its uncertainty was scaled down by the excess (R2W)	the junction’s excess δ	driven oscillators, measurement	yes
after	row 738	a precision allowance omitted print rounding, so sound rows failed (R2W)	the FAREY-2b and 3b checks	driven oscillators, measurement	yes
after	addition to row 731	a dry test on hand-made stand-ins the engine could never produce (R2W)	the LADDER-1 reader	driven oscillators, measurement	yes
after	addition to row 731	the deepest rungs’ predicted effect sat below the grip’s assumed error (R2W)	the ladder of children toward 1/2	driven oscillators, measurement	yes
after	row 741	two shape readings agreed only because both shared the same raw-lag grid (R2W)	the rider’s groove widths	driven oscillators, measurement	yes
after	addition to row 649	two rival stories both passed 21 rows that could not tell them apart (R2W)	the toy’s grips across strengths	driven oscillators, measurement	yes
after	row 764	square law judged at a couple where a fixed toll equalled it (Loop Atlas)	a skateboard’s springy pin	kinematics, measurement	yes
after	row 750	step-size stopping rule passes while a square-root approach still moves (Loop Atlas)	the memory needle’s switch	driven oscillators, measurement	yes
after	row 748	shape bar too wide for 64 catches’ own scatter (Loop Atlas)	the remembering needle’s catches	driven oscillators, measurement	yes
after	row 750	loose calm test met by chance coincidences inside chaos (Loop Atlas)	the 1/4 window’s edge	maps and chaos, measurement	yes
after	row 750	400-period trials passed a ghost lasting 510 periods past the fold (Loop Atlas)	a window’s fold	maps and chaos, measurement	yes
after	row 749	repeat bar set at the arithmetic floor missed by 4 per cent (Loop Atlas)	needles at pace 0.99375	maps and chaos, measurement	yes
after	row 755	drag-fixed determinant, guaranteed by theory, flagged a round-off eigenvalue (Loop Atlas)	the pulsed needle’s period map	maps and chaos, measurement	yes
after	addition to row 741	möbius-fit bar set below the RK4 step’s own error (Loop Atlas)	the forgetting pulsed needle’s map	driven oscillators,	yes

				measurement	
after	row 760; row 766	sweep step wider than the window could not find it (Loop Atlas)	the pulsed needle’s 1/4 window	maps and chaos, measurement	yes
after	addition to row 731	slip finders built so they could never see a slip (Loop Atlas)	the noisy needle’s slip record	driven oscillators, measurement	yes
after	addition to row 645	fidelity bar finer than a rotation number’s 1/N reading error (Loop Atlas)	the needle’s rotation-number reads	driven oscillators, measurement	yes
after	addition to row 740	word-count read-back passed a paper whose type shrank to 6.3 pt (Loop Atlas)	the book’s PDF builds	records and tools, measurement	yes
after	row 766; addition to row 514	window-near-touch guess unjudgeable: every pace lies near a touch (Loop Atlas)	arches on the band’s edge	maps and chaos, measurement	yes
after	addition to row 731	crossing test of the band’s lowest edge could only find none (Loop Atlas)	arches on the band’s edge	maps and chaos, measurement	yes
after	addition to row 714	pixel-equal redraw passed while axes were read at the wrong dpi (Loop Atlas)	a saved figure’s pixels	reading pictures, records and tools, measurement	yes
after	row 760	lock scan’s range excluded the predicted windows it reported missing (Loop Atlas)	windows near a crisis	maps and chaos, measurement	yes
after	addition to row 414	a law checked only on the five points it was read from (Loop Atlas)	a fitted law on the needle	driven oscillators, measurement	yes
after	addition to row 760	a 0.0005 grid could not find a tongue a few 10^{-5} wide (Loop Atlas)	the needle’s lock tongues	driven oscillators, measurement	yes
after	row 787	a lifetime test unreadable: every needle locked before it was measured (Loop Atlas)	the needle’s lock lifetimes	driven oscillators, measurement	yes
after	row 785	a control aimed at an error already at round-off could show nothing (Loop Atlas)	frames round paper mountains	kinematics, measurement	yes
after	row 784	a 12-decimal cross-machine bar set finer than the refiner’s floor (Loop Atlas)	a laptop and a container	records and tools, measurement	yes
after	row 794	a test needing 100 spinning turns per memory got 12: undecided (Loop Atlas)	the forgetting needle	driven oscillators, measurement	yes
after	row 794	a bar set after counting admitted a place inside a window (Loop Atlas)	the needle’s window touches	maps and chaos, measurement	yes
after	the part Two Readers and a Ruler	twin lobes that held would appear on any corner: the ruler’s echo (Loop Atlas)	the needle’s change-of-change ruler	reading pictures, measurement	yes
after	addition to row 642	refiner bar at the arithmetic floor: a fold read as a doubling (Loop Atlas)	the needle’s fold refiner	maps and chaos, measurement	yes
after	addition to row 551	toy solved stopping distance with the formula under test; could never fail (Loop Atlas)	a toy brake’s stopping distance	control and circuits, measurement	yes
after	row 728	symptom-defined crawl bar passed both weights though neither ball crawled (R2W)	a rider with mass, ripple on	driven oscillators, measurement	yes
after	addition to	a row a row to fail could not	RATCHET 2’s P2.0	drive	

after	addition to row 551	dry run on a stand-in bound to fail could not show R3-0's fault (R2W)	RATCHET-3's R3-0 reader	driven oscillators, measurement	yes
after	addition to row 36; addition to row 731; addition to row 735	RATCHET-4's dry run found two verdicts passing with a weight missing (R2W)	RATCHET-4's reader	records and tools, measurement	yes
after	addition to row 686	a 10^{-4} turning-point threshold made SEAM_SCAN report no rise where one existed (R2W)	the fourth harmonic's window scan	driven oscillators, measurement	yes
after	addition to row 566	angle verdicts held by the letter on noise below the amplitude's errors (R2W)	the drift's zero offset	driven oscillators, measurement	yes
after	addition to row 734	fidelity runs with the third tone silent never exercised the new path (R2W)	the three-tone engine RATCHET_3T	records and tools, driven oscillators	yes
after	addition to row 766	SM5's slice held two of six births, so only NO READING possible (triple-star)	the slice middles' SM5 registration	three body, measurement	yes
after	addition to row 766	SM2's grid stopped short of the known sign change: a guaranteed miss (triple-star)	SM2's birth grid at 0.30	three body, measurement	yes
after	addition to row 570	provenance scan traced numbers to the table's own printed descriptions (triple-star)	the provenance table's ledger	records and tools, measurement	yes
after	addition to row 731	selftest's planted world lacked the third ending; real run read nothing (triple-star)	the off-mirror tool's selftest	three body, measurement, records and tools	yes
after	addition to row 683	'at least three of six escape' met at a single birth (triple-star)	three-body orbits, SM6′ refuge guess	three body, measurement	yes
after	addition to row 400	strict inequality missed an exact zero on a grid point; lean could never hold (R2W)	RATCHET-53 reader's sign-change test	driven oscillators, measurement, records and tools	yes
after	addition to row 400	'less than or equal' counted an exact zero twice; lean could never hold (R2W)	RATCHET-58 reader's crossing count	driven oscillators, measurement, records and tools	yes
after	addition to row 337	self-test bar set after a trial that already met it (R2W)	the explorer's self-test row	records and tools, measurement	yes
after	addition to row 337	check's reference was its own output file; could pass only once (R2W)	the explorer's builder check	records and tools, measurement	yes
after	addition to row 740	row-split checker that never failed passed a known split row (R2W)	the row-split checker	records and tools, measurement	yes
after	row 803; addition to row 128; addition to row 214	two nulls' V equal to the floor: one calculation, not a law (R2W)	an array of nine riders	driven oscillators, measurement	yes
after	addition to row 731	dry run showed the rival could never be bracketed by its rungs (R2W)	EDGE-85's rival bracket	driven oscillators, measurement, records and	yes

				tools	
after	the part Two Readers and a Ruler; addition to row 641; addition to row 665; addition to row 715	spread over 200 cycles is flat for any dividing cycle; ladder unseen (Loop Atlas)	the needle below a chaos	maps and chaos, measurement	yes
after	the part Two Readers and a Ruler; addition to row 751	doubling finder counted Newton’s failures as edges; all four false (Loop Atlas)	a doubling finder on the needle	maps and chaos, measurement	yes
after	the part Two Readers and a Ruler; addition to row 645; addition to row 665; addition to row 766; addition to row 785	100-unit lock windows pass a rider slipping every 200 to 400 (Loop Atlas)	a rider on the truck road	driven oscillators, measurement	yes
after	the part Two Readers and a Ruler; addition to row 748	paces set by top grip landed where floors always slip; unjudgeable (Loop Atlas)	a needle over rough floors	maps and chaos, measurement	yes
after	addition to row 570	provenance scan credited 4,200 by a coincidental string match in a ledger (triple-star)	the paper’s provenance scan	records and tools, measurement	yes
after	addition to row 784	reproduction bar 10^{-9} finer than the record’s eight printed decimals (triple-star)	the friction knob’s reproduction check	records and tools, measurement	yes
after	addition to row 784	strict 2×10^{-6} bar against another machine’s run, finer than machines agree (triple-star)	a cross-machine reproduction check	records and tools, measurement	yes
after	addition to row 716	ZG1 registered with its outcome fixed by arithmetic, disclosed as such (triple-star)	three-body orbits, ZG1 registration	three body, measurement	yes
after	addition to row 406	bar 10^{-5} set below the integrator’s own drift, 1.15×10^{-5} (triple-star)	three-body orbits, LS1 light pair	three body, measurement	yes
after	addition to row 450	residual correlation with a term the fit already carries: zero by construction (triple-star)	the drag-size fit’s residuals	three body, measurement	yes
after	addition to row 741	slope-ratio control read STRAIGHT because the whole line drifted together (triple-star)	the LW5 rate readings	three body, measurement	yes
after	row 810	cross-machine fidelity check over chaotic runs could never pass (triple-star)	the triple_k_starts reruns	three body, measurement	yes
after	addition to row 711	planted-rate bar 2×10^{-7} set below the reader’s own pull (triple-star)	the spectral reader’s planted test	three body, quadrature, measurement	yes
after	addition to row 541	selftest bar 10^{-9} set below the reader’s 2.6×10^{-9} floor (triple-star)	the triple_followups2 selftest	records and tools, measurement	yes

after	addition to row 580	verdict rules gave verdicts on runs never started (triple-star)	LW6’s verdict rules	records and tools, measurement	yes
after	addition to row 451	five-birth slope could not exceed one grid step’s ceiling (triple-star)	the LW5 slope reading	three body, measurement	yes
after	row 810	pooled cross-machine check sunk by one chaotic birth; could not pass (triple-star)	LX3’s cross-machine check	three body, measurement	yes
after	addition to row 1	SL6b’s band lay below the reader’s floor yet read BORNE OUT (triple-star)	the slow-line reader	three body, quadrature, measurement	yes
after	addition to row 1	a last-place nudge rounded away, so the test could not fail (triple-star)	the toolchain’s nudge test	three body, measurement	yes
after	row 808	pilot showed bar 0.05 would call every run pumped; planted check tied (triple-star)	the triple_cs1 pump check	three body, measurement	yes
after	row 810	strict anchor against another machine’s run; machines part from pass 37 (triple-star)	SL7’s fidelity anchor	three body, measurement	yes
after	addition to row 593	text read-back passed complete on fourteen picture icons (triple-star)	the islands paper’s read-back	records and tools, measurement	yes
after	row 811	three numbers ‘traced’ only to ledgers printing the draft back (triple-star)	the paper’s provenance scan	records and tools, measurement	yes
after	row 811	figure width tags, 100, ‘held’ by ledgers printing 100 (triple-star)	the paper’s provenance scan	records and tools, measurement	yes
after	row 811	‘11 of the 20’ passes because every ledger prints small integers (triple-star)	the paper’s source check	records and tools, measurement	yes
after	addition to row 746	a 0.02-of-a-life bar sat inside one seed’s spread (Loop Atlas)	a fleet of slowing riders	driven oscillators, control and circuits, measurement	yes
after	row 813	planted exact zero judged at 10^{-9}, far under the stepping’s error (Loop Atlas)	FACE_ALONG’s planted no-grip case	kinematics, measurement	yes
after	row 813	planted tolerances again set under the stepping floor, an exchange later (Loop Atlas)	DIVE_CUE’s planted checks	kinematics, measurement, records and tools	yes
after	row 813	planted step test where the column never starts: nothing to compare (Loop Atlas)	HALT_THRESHOLD’s walker column	kinematics, measurement	yes
after	addition to row 730	trough level set below the skewed soup’s floor: count always empty (Loop Atlas)	a trough-firer on a skewed soup	waves, quadrature, measurement	yes
after	row 819	goodness-of-fit against a curve fitted from the same data flattered it (Loop Atlas)	smoke-ring train lifetimes	waves, measurement	yes
after	row 822	a planted repeat of chaotic break times on another machine (Loop Atlas)	smoke-ring trains on two machines	waves, maps and chaos, measurement	yes: the row read only after the run
after	addition to row 723	a count-difference lean that chance meets 0.41 of the time (Loop Atlas)	smoke-ring train breaks in a band	waves, measurement	yes
after	row 823	a rule counted as held at the size it was read from (Loop Atlas)	a book’s part headlines	records and tools, measurement	yes

after	row 824	a finer reading grid's repeat counted as confirming the excess (Loop Atlas)	a sampled switch's extra knee	control and circuits, quadrature, measurement	yes
after	row 823	a number check read the edition's own record row and passed (Loop Atlas)	the reader's edition number check	records and tools, measurement	yes
after	row 820	a search at the mirror's fixed point could not see a double mirror (Loop Atlas)	the two falling cats' mirror	kinematics, measurement, records and tools	yes
after	row 818	a size-only reading made a sense test agree by construction (Loop Atlas)	a falling cat's landing turn	kinematics, measurement	yes
after	addition to row 346	a 'ride law met' dead downwind was guaranteed by bookkeeping (Loop Atlas)	a toy sailboat's price	waves, kinematics, measurement	yes
after	row 837	planted grid tolerance set under a staircase count's grain (Loop Atlas)	a sampled switch's knee reader	control and circuits, quadrature, measurement	yes
after	row 837	a 5% bar again narrower than the smallest shifts' error bars (Loop Atlas)	a sampled switch's fitted knee reader	control and circuits, quadrature, measurement	yes
after	row 666	a lean and rival registered on a number already measured (Loop Atlas)	the needle's memory switch tip	driven oscillators, records and tools, measurement	yes
after	addition to row 210	mirror-pair bar 10^{-12} set under a no-physics stand-in's 3.513×10^{-12} floor (R2W)	fifteen riders' mirror pairs	driven oscillators, measurement	yes
after	addition to row 805	size sweep flat because both hands share Rice's Gaussian factor; could not separate them (R2W)	the Loop Atlas's crest-firer	waves, quadrature, measurement	yes
after	addition to row 649	lean and rival both held; the rival could not fail where the lean held (R2W)	the early slip's FX-3 test	driven oscillators, maps and chaos, measurement	yes

2 — The built artifact is the evidence

90 arrivals: 15 counted by 10 October, 75 after.

Counted	Lands at	Arrival	Substrate	Domains	Had the session read the guide first?
by 10 October	row 553	a figure that plotted an undeposited run (latch paper)	a deposited PDF	records and tools	not known
by 10 October	row 553	v6's promised fourth figure (the guide)	the guide's own PDF	records and tools	yes
by 10 October	row 593	a green build on an unchanged file (Haber-Bosch packet)	a policy paper's build	records and tools	no: August 2026, before v6
by 10 October	addition to row 553	a send that ran beside a failed script (Loop Atlas)	a session's shipments	records and tools	not known
by 10 October	addition to row 553	ledger numbers read back against their runs (Loop Atlas book)	the book's ledger	records and tools	yes

by 10 October	addition to row 553	six figures missing from eighteen rebuilds (R2W)	R2W’s PDFs	records and tools	yes
by 10 October	addition to row 572	a build tool living only in scratch space (triple-star)	a crate	records and tools	not known
by 10 October	addition to row 553	a typed double hyphen turned into a dash by the PDF converter (Field Guide)	the collection of reads	records and tools	yes
by 10 October	addition to row 553	a builder drops figures, long lines and table rows, and a read-back that forgave typography (Loop Atlas)	the book’s PDFs	records and tools	yes
by 10 October	addition to row 553	the changes file copied the gathering record’s private section, caught on reading the build (Field Guide)	the guide’s changes file	records and tools	yes
by 10 October	addition to row 570	a true number printed only on a page, untraceable (3-Body Team)	a capture page	records and tools, three body	yes
by 10 October	row 676	a manifest written inside its folder lists itself, caught by checking the unpacked zip (Loop Atlas)	a session’s packets	records and tools	yes
by 10 October	row 676	the same slip in a manifest written by redirect, a day later (3-Body Team)	a bridge kit	records and tools	yes
by 10 October	addition to row 553	a harmonic label read as a link and dropped from 25 places in five PDFs (3-Body Team)	a session’s PDFs	records and tools	not known
by 10 October	addition to row 553	a ledger’s PDF clipped about 1,600 words off a wide table (Loop Atlas)	a session’s ledger	records and tools	yes
after	addition to row 677	A bat’s draft misdescribed run sets; caught reading it against the code (triple-star)	a run’s bat file	records and tools	yes
after	addition to row 592	status said batches waited on the laptop; none had been built (triple-star)	the thread’s status replies	records and tools	yes
after	row 740	PDF builders dropped table lines and cells; build_paper v1.2 reads every cell back (triple-star)	the register’s PDF builds	records and tools	yes
after	addition to row 572	crate built against a rebuild, not Bear’s crate; eleven files reached no crate (triple-star)	a handoff crate	records and tools	yes
after	addition to row 611	A bat printed FINISHED after its program failed; no picture was drawn (triple-star)	a run’s bat file	records and tools	yes
after	row 733	run steps assumed a kit in Bear’s folder that never arrived there (triple-star)	Bear’s laptop working folder	records and tools	yes
after	—	the 160-read volume regenerated from its source and checked word for word (triple-star)	the no-physics reads volume	records and tools	yes
after	row 735	a record line citing a sympy check written beside the check’s failed import (R2W)	R2W’s record	records and tools	yes
after	addition to row 610	two readings typed in-session had no script; rerun from scripts and matched (R2W)	R2W’s crate	records and tools	yes
after	row 740	absolute-value bars tore two table rows in the built PDF (R2W)	the paper’s and book’s PDFs	records and tools	yes
after	row 740	a cell check ran beside the PDF build; a torn row caught before shipping (R2W)	the paper’s RATCHET-1 table row	records and tools	yes
after	addition to row 714	a figure’s relabel left 3/7’s label beside 5/9’s dot, caught on reading (R2W)	figure 43’s fraction labels	reading pictures, records and	yes

				tools	
after	addition to row 714	rebuilt figure’s scale differed from the saved, marked file (Loop Atlas)	Bear’s marked heavy-window figure	reading pictures, records and tools	yes
after	row 279; addition to row 591	session D’s ferry shipped unchecked and arrived five of six pieces (Loop Atlas)	a session’s move crate	records and tools	yes
after	addition to row 735	run stamps typed from memory into figure runs and a ledger line (Loop Atlas)	the book’s ledger and figures	records and tools	yes
after	row 766	probes on a neighbour of the input missed faults; the tool itself catches them (Loop Atlas)	the needle’s instruments	records and tools, measurement	yes
after	addition to row 740	a text-replacement edit silently dropped a line; the long run failed (Loop Atlas)	a session’s run tools	records and tools	yes
after	addition to row 735	a fidelity rerun against the record caught a reversed formula in a tool (Loop Atlas)	a session’s run tool	records and tools	yes
after	addition to row 735	reading the built ledger back caught a stamp truncated by the shell (Loop Atlas)	the book’s data ledger	records and tools	yes
after	addition to row 572; addition to row 721	a kit rerun in a clean folder found two data files missing (Loop Atlas)	a new-session tool kit	records and tools, measurement	yes
after	row 786	a pipe’s exit status hid a failed smoke run and the launch went ahead (Loop Atlas)	a session’s shell launch	records and tools	yes
after	row 786	an end-to-end launcher test found a latent fault in the comparison (Loop Atlas)	a laptop launcher and comparison	records and tools	yes
after	row 773	a title style rule enlarged every part heading, seen only in the built PDF (Field Guide)	the guide’s built PDF	records and tools	yes
after	addition to row 773	book PDF built in place lost its figures, found on read-back (Loop Atlas)	the book’s PDFs	records and tools	yes
after	addition to row 773	voice PDF’s numbered lines eaten by the builder, found on read-back (Loop Atlas)	the voice file’s PDF	records and tools	yes
after	addition to row 773	a stray file in the crate, found on reading the crate back (Loop Atlas)	a session’s crate	records and tools	yes
after	addition to row 400	fidelity check caught the fold-lag formula losing digits near lag 1/2 (R2W)	the fold’s lag formula	driven oscillators, measurement	yes
after	addition to row 36	RATCHET-5’s rerun dry test caught an edit that commented out a working line (R2W)	RATCHET-5’s reader	records and tools, measurement	yes
after	addition to row 779	four lists shipped run-on in the built book; LISTGAP dry run found them (R2W)	the RF book’s built PDF	records and tools	yes
after	addition to row 157	twenty-five delta zips each unpacked and checked against the tree before shipping (R2W)	R2W’s delta zips	records and tools	yes
after	addition to row 740	readback at the close flagged table rows split by page breaks (R2W)	the RF book’s built PDF	records and tools	yes
after	addition to row 721	launcher census at PB1′′s start found one of eleven files missing (triple-star)	Bear’s laptop working folder	records and tools	yes
after	addition to row 570	paper’s numbers scanned against ledgers: unfiled run, unsaved readings, three errors (triple-star)	the islands paper’s numbers	records and tools, measurement	yes

after	addition to row 553	plain stylesheet lost three cells; flow stylesheet reads back 328 of 328 (triple-star)	the pattern-recognition file's PDF	records and tools	yes
after	addition to row 569	text broken by headings read back word for word: 25,371 both (triple-star)	the islands paper's draft	records and tools	yes
after	addition to row 569	moved passage checked character for character against the draft (triple-star)	the islands paper's draft	records and tools	yes
after	addition to row 786	result line written with its distances blank; command failed, write went ahead (R2W)	R2W's record	records and tools	yes
after	addition to row 570; addition to row 735	record numbers typed before a script computed them, then checked (R2W)	R2W's record	records and tools	yes
after	addition to row 773	build from the wrong folder lost every figure; page count caught it (R2W)	R2W's book PDF	records and tools	yes
after	addition to row 786	guarded record append silently stopped; reading the tail showed nothing written (R2W)	R2W's record	records and tools	yes
after	addition to row 589	read-back of the reads file found typed dashes and underscores restyled (R2W)	R2W's reads file	records and tools	yes
after	addition to row 36	browser port of the engine checked edge by edge against run files first (R2W)	the Depth Axis Explorer's engine	records and tools, driven oscillators	yes
after	addition to row 773	page opened and driven in a headless browser showed two faults (R2W)	the Depth Axis Explorer page	records and tools	yes
after	addition to row 773	four builds at once overwrote a shared scratch page; read-back decides content (R2W)	R2W's four documents	records and tools	yes
after	addition to row 740	read-back caught a table row split across a page, a phrase lost (R2W)	the book's dark-point table	records and tools	yes
after	addition to row 677; addition to row 731	plan's printed engine arguments read against source caught a swapped harmonic (R2W)	K-16's plan arguments	records and tools, driven oscillators	yes
after	addition to row 720	each of thirty-three delta zips unpacked and checked before shipping (R2W)	B11's delta zips	records and tools	yes
after	addition to row 786	rename by text mangled a file name; run of record stopped (Loop Atlas)	a correction tool's source	records and tools	yes
after	addition to row 629; addition to row 667	traders' law lesson's ranges corrected when its table was rebuilt from json (Loop Atlas)	the traders' law lesson	records and tools, quadrature, control and circuits	yes
after	addition to row 793	json numbers read by script caught loose phrases in replies (Loop Atlas)	the session's replies	records and tools	yes
after	addition to row 570; addition to row 735	stamps typed from memory caught by reading them from the disk (Loop Atlas)	the session's records	records and tools	yes
after	addition to row 157	reissued pattern file's header claimed no change; a section had been replaced (triple-star)	the pattern-recognition file	records and tools	yes
after	addition to row 733	shipments' intake found paper files missing twice (triple-star)	Bear's laptop shipments	records and tools	yes
after	row 811	provenance pass caught the Declaration naming superseded table and scan runs	the paper's Declaration	records and tools	yes

		(triple-star)			
after	row 809	a lean-storage tool edit proved to change no integrator step (triple-star)	the triple_lean tool	records and tools, three body	yes
after	addition to row 569	script-assembled paper read back: all 1,097 sentences present (triple-star)	the islands paper’s rebuild	records and tools	yes
after	addition to row 593	builder exited clean on missing pictures; the PDF held none (triple-star)	the paper’s PDF builds	records and tools	yes
after	row 811	source check of the paper found nine runs no table row named (triple-star)	the islands paper’s numbers	records and tools	yes
after	addition to row 721	rebuild kit proved alone in an empty folder, byte for byte (triple-star)	the paper’s rebuild kit	records and tools	yes
after	row 812	launchers exercised on a test bed showed real faults (triple-star)	the publishing launchers	records and tools	yes
after	addition to row 251	export’s rows claimed losses; hashing the files showed all 13 present (triple-star)	the heavy-star series export	records and tools	yes
after	row 844	restore crate read back: its own list packed inside a part (triple-star)	the restore crate	records and tools	yes
after	row 844	handover restored in an empty folder: 1,739 files, none differing (triple-star)	the eighth session’s handover	records and tools	yes
after	addition to row 399; addition to row 646	engine self-check read the check’s own name as the value; blind run caught (Loop Atlas)	the explorer engine’s self-check	records and tools, measurement	yes
after	addition to row 206	a printed rip window halved twice, caught on checking against the card (Loop Atlas)	EXPLORER_ROUTINE_CHECK’s printout	records and tools, measurement	yes
after	row 814	inertia misses told from memory, corrected from the run’s json (Loop Atlas)	the diver rebuild’s reported shares	kinematics, records and tools, measurement	yes
after	row 816	audit reran old tools as they stood before reading the fix (Loop Atlas)	five column tools	records and tools, kinematics, measurement	yes
after	addition to row 721	shipped laptop tool chain failed in place: a part missing by name (Loop Atlas)	SNAKE_LONGEST on Bear’s laptop	records and tools	yes
after	addition to row 778	typed-in counts on a front page lagged the crate’s (Loop Atlas)	the headline register’s front page	records and tools	yes
after	row 815	a row insert by text replacement joined two ledger rows (Loop Atlas)	the data ledger’s table	records and tools	yes
after	row 824	a scratch probe’s number told before the run of record disagreed (Loop Atlas)	a sampled switch’s knee excess	control and circuits, measurement, records and tools	yes
after	row 818	a planted check read two tools’ outputs and caught a mismatch (Loop Atlas)	two cat tools’ stretch lengths	kinematics, records and tools, measurement	yes
after	row 840	record-verdict counter’s output read by eye against the record; 4 of 25 wrong (R2W)	the record’s verdict counter	records and tools, measurement	yes
after	addition to row 773	same wrong-folder build lost figures again; page count caught it before shipping (R2W)	R2W’s book PDF	records and tools	yes

after	addition to row 593	image-count gate’s ‘known offset’ was a chalkboard figure split across pages (R2W)	the RF book’s built PDF	records and tools	yes

3 — A word that changes meaning across a boundary

68 arrivals: 14 counted by 10 October, 54 after.

Counted	Lands at	Arrival	Substrate	Domains	Had the session read the guide first?
by 10 October	row 502	three-halves energy (latch paper)	linked filaments and their literature	materials, topology	not known
by 10 October	row 502	azimuth (Sun spoke)	HMI’s field maps	solar, measurement	yes
by 10 October	row 502	beta (R2W)	the RF paper’s own record	records and tools, driven oscillators	yes
by 10 October	vocabulary, tread	comma for two objects (Loop Atlas and the guide)	the guide and the book	records and tools	yes
by 10 October	a note on the Sun paper	twist (Sun paper and the guide)	solar currents and ribbons	solar, topology	yes
by 10 October	row 612	travel as a length or a signed part (Loop Atlas)	the big loop	kinematics, measurement	yes
by 10 October	the version’s name	version 6.1 or version 7 (the guide and the Loop Atlas)	the guide’s own name	geodesy, measurement	yes
by 10 October	a note in the gathering record	“session A” signed by two sessions (Loop Atlas)	the project’s own sessions	records and tools	yes
by 10 October	vocabulary, memory	memory in four senses across four rooms (the Field Guide, reading the sessions)	the project’s own sessions	records and tools	yes
by 10 October	vocabulary, anomalistic lag	two clocks in a third sense, named the anomalistic lag (3-Body Team)	the guide’s vocabulary	three body, quadrature	yes
by 10 October	row 678	“memory 1.0” named a different group after a rescaling (Loop Atlas)	the pulsing needle	driven oscillators, measurement	yes
by 10 October	words to watch	“wall”: what a walker pushes against in one book, the barrier between whole numbers in another (Loop Atlas)	the Loop Atlas and the Latch Atlas	records and tools	yes
by 10 October	addition to row 676	two ledgers’ row numbers collide, and one session cited the other’s as the guide’s for two days (3-Body Team)	the changes file	records and tools	yes
by 10 October	vocabulary, clocks	“two clocks” carries six senses across 161 passages, and v6’s own sense is used once (Field Guide)	the project’s own files	records and tools	yes
after	addition to row 558	one rate, two estimators: end-to-end in a note, median step in a rule (triple-star)	three-body orbits	three body, measurement	yes: the row read only after the run
after	addition to row 715	Bear’s crossing count taken without his drawing of closed ovals; test read nothing (triple-star)	Bear’s coil drawing	reading pictures, three body	yes
after	addition to row 692	‘Fast’ chaos meant twin parting in the test, early escapes in the row (triple-star)	the register’s proposed rows	three body, records and tools, measurement	yes
after	addition to				

after	addition to row 686	passing speed meant the edge's speed in the registration, each square's own in the atlas (R2W)	a rider in the pad atlas	driven oscillators, measurement	yes
after	addition to row 678	a fold weight measured on the reversed rider used for the forward one (R2W)	forward and reversed riders	driven oscillators, records and tools	yes
after	addition to row 713	"up" meant bias in conversation but wave strength in the book's map (R2W)	the junction's tongue map	driven oscillators, reading pictures	yes
after	addition to row 713	"doesn't move" held at fixed bias, not at a tongue's middle (R2W)	the junction's chalk line	driven oscillators, reading pictures	yes
after	addition to row 713	"pinch" means tightest phase in one picture, weakest hold in the junction (R2W)	a strobe picture and the junction	reading pictures, driven oscillators	yes
after	addition to row 706	a wavy line read as landscape or wrinkle flips where dancers sit (R2W)	Bear's saddles-and-wobblers diagram	driven oscillators, reading pictures, topology	yes
after	addition to row 611	numpy true read as Python True printed UNDECIDED on a HELD line (Loop Atlas)	a verdict printer	records and tools	yes
after	addition to row 643	one pendulum's dials scaled differently in three rooms, matched before comparing (Loop Atlas)	needle, rider and generator swing	driven oscillators, measurement, control and circuits	yes
after	addition to row 645	'5/7' taken as period 7 though the orbits repeat after 14 or 28 (Loop Atlas)	the needle's 5/7 and 7/10 locks	driven oscillators, quadrature	yes
after	row 761	'stable' coded as a real multiplier missed a complex-pair cycle (Loop Atlas)	the heavy needle's 1/4 window	maps and chaos, records and tools	yes
after	addition to row 730	lock widths rank differently in drive units and in the rider's own units (Loop Atlas)	the needle's lock widths	driven oscillators, measurement	yes
after	addition to row 738	one golden clock rounded two ways by two sessions shifts every doubling (Loop Atlas)	the needle's golden clock	maps and chaos, measurement	yes
after	addition to row 554	a light's phase signed opposite to the loop's orientation flipped the reading (Loop Atlas)	a polarisation loop toy	waves, kinematics, records and tools	yes
after	rows 726 and 727; addition to row 502	'same machine' conflates rider and RF-driven junction, also in row 727 (Field Guide)	the rider and the RF junction	driven oscillators, records and tools	yes
after	row 772	ripple's sign taken as an error; each sign right on its side (Field Guide)	the chapter's ripple equation	records and tools, waves	yes
after	addition to row 770	'close the sitting' meant a save point; Bear read leaving the session (Loop Atlas)	words between Claude and Bear	records and tools	yes
after	row 722	'locked' counts 1,051 by a speed knee, 69 by lag position (Loop Atlas)	the remembering needle	driven oscillators, measurement	yes
after	row 726; row 727	'Shapiro steps' renamed Shapiro-like: microwave pushes, ripple moves the bumps (R2W)	a rider with mass	driven oscillators	yes
after	row 727	'portability' in Bear's summary meant number-for-number and calibrated ruler (R2W)	the pendulum map's transfer claims	driven oscillators, measurement	yes

after	row 722	staircase census 14/70/94 against 14/65/83: one atlas, two readers (R2W)	the Explorer’s pad atlas	driven oscillators, measurement	yes
after	rows 726 and 727	page called the rider and the RF junction the same machine (Loop Atlas)	R2W’s paper page	driven oscillators, records and tools	yes
after	addition to row 328	‘the zero’s slide’ was the drift’s in the literature, the step’s in the book (R2W)	the ratchet’s drift and step	driven oscillators, records and tools	yes
after	addition to row 793	polyfit’s last number, the constant term, taken for the curvature (R2W)	numpy.polyfit coefficients of window fits	records and tools, driven oscillators	yes
after	addition to row 502	islands paper used ‘two clocks’ in a sense the guide reserves elsewhere (triple-star)	the islands paper’s vocabulary	records and tools, three body	yes
after	addition to row 502	paper’s ‘fold’ named a slice’s peak, not a saddle-node bifurcation (triple-star)	the islands paper’s vocabulary	records and tools, three body	yes
after	addition to row 552	mardling’s harmonic notation misread: a sign, a neighbour, 14:2 for 2:14 (triple-star)	Mardling’s harmonic notation	three body, records and tools	yes
after	addition to row 722	one coefficient read by two readers; predictions rode on one of them (R2W)	the octave’s fourth-power coefficient	records and tools, measurement, driven oscillators	yes
after	addition to row 79	‘the lean’, a registered prediction’s word, named a fit made after the data (R2W)	the record’s word ‘lean’	records and tools, measurement, driven oscillators	yes
after	addition to row 646	‘inverse square’ named the share’s fall; reply read it as the point’s (R2W)	the dark point’s drive law	driven oscillators, records and tools	yes
after	addition to row 765	one word-count command gave two counts in two locales (Loop Atlas)	the book’s word count	records and tools	yes
after	addition to row 594	‘the same’ pointers changed referent after row insertions (Loop Atlas)	the book’s knob register	records and tools	yes
after	addition to row 692	registration’s ‘held’ (under half a turn) differed from the paper’s (under one turn) (triple-star)	the FN0 registration and paper	measurement, records and tools	yes
after	addition to row 690	K ‘fixed by the start’ meant mean elements; starts were osculating (triple-star)	the one-harmonic model’s K	three body, measurement	yes
after	row 810	‘slowly chaotic’ rode on machines parting, not the registered twin test (triple-star)	the seventh’s core, three-body orbits	three body, measurement	yes
after	the part on the headline register	‘registered miss’ counted a guessed-to-fail test that failed (triple-star)	the register’s prediction count	records and tools, measurement	yes
after	addition to row 683	‘failed’ read as a missed prediction when the guess was to fail (triple-star)	the prediction ledger	records and tools, measurement	yes
after	addition to row 721	one tool name denoted two versions in packet and crate (triple-star)	build_paper and lint_deposit	records and tools	yes

after	row 812	the shell's or-chain read differently by wine's cmd and Windows (triple-star)	the publishing launchers	records and tools	yes
after	addition to row 30	label 'absent' meant unused, read as not present (triple-star)	the harmonic recipes' printout	records and tools	yes
after	row 815	numpy true is not Python's True; held cases printed UNDECIDED (Loop Atlas)	TONG_DULLIN's verdict printer	records and tools	yes
after	row 765	one wc gave two word counts in two locales at a review (Loop Atlas)	the book's word count	records and tools	yes
after	addition to row 558	a gap from one estimator judged by another's spread (Loop Atlas)	TWO_HANDS_2's pooled ratio	waves, measurement	yes
after	row 815	numpy booleans stopped a run of record at its json dump (Loop Atlas)	COIL_SNAKE's verdict dump	records and tools	yes
after	row 818	an unsigned angle read 152 where the turn was 208 (Loop Atlas)	the twist-free falling cat	kinematics, measurement	yes
after	row 831	the course theorem missed by a sign convention for the water's angle (Loop Atlas)	a toy sailboat's course theorem	waves, kinematics, measurement	yes
after	row 832	'twist' read as relative twist sent the search to the wrong paper (Loop Atlas)	popular account of the falling cat	kinematics, records and tools	yes
after	addition to row 646	zeta's name read as prime-number theory; its road is a random walk (Loop Atlas)	the traders' constant 0.5826	control and circuits, number theory, quadrature	yes
after	row 840	a rival's FAILED, a landing, filed in the ledger as a miss (R2W)	the headline register's ledger	records and tools, measurement	yes
after	addition to row 750	'edge' as fold and as rest-edge agreed at 0.30, parted at 0.45 (R2W)	gear 1's edge, turned ripple	driven oscillators, maps and chaos	yes
after	addition to row 739	continuation followed the branch to 1.53, ramp followed the ball to 0.713 (R2W)	heavy balls' edge at strength 0.60	driven oscillators, maps and chaos	yes
after	—	counter read R-tagged rivals' FAILED lines as leans' misses (R2W)	the record's verdict counter	records and tools, measurement	yes

4 — Chaos and binning

63 arrivals: 12 counted by 10 October, 51 after.

Counted	Lands at	Arrival	Substrate	Domains	Had the session read the guide first?
by 10 October	row 603	the comma survey's census (R2W, 3 October; relayed by the sounding board)	a rider in two travelling patterns	driven oscillators, measurement	no: before R2W read v6 on 4 October
by 10 October	row 627	the knee counter (Loop Atlas book)	the turning needle under noise	driven oscillators, measurement, three body	yes
by 10 October	row 655	chaos on the borders between treads, 97 per cent blind (R2W, 3 and 4 October)	a rider with mass	driven oscillators, maps and chaos	no: found before R2W read v6; read against row 523 afterwards
by 10	row 622	two thresholds chop a slow creep into rests (triple-	three-body orbits	three body,	not known

October		star)		measurement	(most likely before any full reading)
by 10 October	addition to row 103	a tolerance is a bin and can move a verdict (R2W, 3 and 4 October)	a relay switch	driven oscillators, measurement, topology	no
by 10 October	addition to row 631	a coarse map hides a ribbon of the other gear, and the border between catchments is fractal (R2W, sittings B6 and B7)	a rider with mass	driven oscillators, maps and chaos, measurement	yes
by 10 October	addition to row 631	a chaotic layer of uneven width at a conservative island’s rims (3-Body Team, door 2)	three-body orbits	three body, maps and chaos, measurement	yes
by 10 October	row 704	a cut through a band makes one band look like two, so the circle is seamed where nothing lands (Loop Atlas)	the needle’s strobe picture	reading pictures, maps and chaos, measurement	yes
by 10 October	row 709	a value on a rounding boundary rounds two ways, so a rounded reference count ran one high (Loop Atlas)	the needle’s strobe picture	maps and chaos, reading pictures, measurement	yes
by 10 October	addition to row 128	five-degree bins tied three ways, and the tool named the first as a peak (3-Body Team)	three-body orbits	three body, measurement	yes: the guide read; row 128 read in full only after the run
by 10 October	addition to row 416	a memory ceiling of 0.31 set by the chart’s own window (R2W, B8)	a rider with mass	driven oscillators, measurement, number theory	yes
by 10 October	addition to row 631	four starts undercount a coin, and 1,000 find islands at its edges (R2W, B8)	a rider with mass	driven oscillators, maps and chaos, measurement	yes
after	addition to row 318	zero-padded reader pinned at its longest allowed period; earlier value a bin (triple-star)	a spectral reader, three-body orbits	three body, quadrature	yes: the row read only after the run
after	the part Two Readers and a Ruler; addition to row 201	rule’s single miss sat on a tie at 7.130, the lower dead zone’s birth (triple-star)	three-body orbits, maypole rule	three body, measurement	yes
after	addition to row 693	rocking fell below the reader’s floor; it reported the drum’s overtone instead (triple-star)	a rocking reader, three-body orbits	three body, quadrature, measurement	yes
after	addition to row 632	tighter tolerances flipped which of two births held; fates are chaos samples (triple-star)	three-body orbits	three body, maps and chaos, measurement	yes
after	the part Two Readers and a Ruler	Bear’s black spacer rungs were gaps drawn by the picture’s birth grid (triple-star)	the strobe picture’s birth grid	reading pictures, three body	yes
after	the part Two Readers and a Ruler	the V read on a 0.005 grid was the coarse sampling’s outline (triple-star)	three-body orbits, strobe views	three body, reading pictures, measurement	yes
after	addition to row 631	two lock steps narrower than the dial’s tick looked missing until a finer dial (R2W)	a rider’s 1/n lock staircase	driven oscillators, measurement	yes
after	addition to	the map’s grid rounding moved the apparent	the rider’s	driven	yes

	row 686	crossover by one column (R2W)	crossover map	oscillators, measurement	
after	addition to row 416	a best drive on the search window’s edge, so its value belongs to the window (R2W)	a rider’s pin search	driven oscillators, measurement	yes
after	addition to row 451	lifts far smaller than the grid’s step, so the verdict read the grid’s cells (R2W)	the rider’s edge on a grid	driven oscillators, measurement	yes
after	addition to row 683	a merge tolerance equal to the inputs’ rounding flipped a verdict to FAILED (R2W)	the CHECK-E grading script	records and tools, measurement	yes
after	row 738	a root rounded to eight decimals fell off a lock window 3.5×10^{-7} wide (R2W)	the junction’s 1/2 lock window	driven oscillators, measurement	yes
after	row 739	a nearest-to-mirror tracker flipped between twin stops, faking a failed walk (R2W)	the junction’s stable dance	driven oscillators, measurement	yes
after	addition to row 722	locked needles linger past a top one noise unit high (Loop Atlas)	the remembering needle	driven oscillators, measurement	yes
after	row 749	sieve tolerance near the needles’ scatter set the apparent repeat periods (Loop Atlas)	needles at pace 0.99375	maps and chaos, measurement	yes
after	row 752	fixed-length trials near an inner fold answered by chaotic returns (Loop Atlas)	a window’s inner fold	maps and chaos, measurement	yes
after	row 753	controller’s fall edge moved with jostling: the noise set the edge (Loop Atlas)	a balancing rider’s sized tap	control and circuits, measurement	yes
after	addition to row 725	gauge’s early warning was the first rare slips, not slow softening (Loop Atlas)	the noisy needle’s wobble gauge	driven oscillators, measurement	yes
after	row 753	half the fallen needles are caught again, so a slip is not final (Loop Atlas)	a memory-5 needle under noise	driven oscillators, measurement	yes
after	addition to row 768	a one-way valve switching on the flow’s bare sign chattered and ran away (Loop Atlas)	the jellyfish solver’s valve	biology, waves, control and circuits	yes
after	row 787	a band’s piece count set by the gap rule’s threshold, not the band (Loop Atlas)	the needle’s chaotic band	maps and chaos, measurement	yes
after	the part Two Readers and a Ruler	half-degree bins merged eight strings into four pairs; fine lattice resolved them (Loop Atlas)	the needle’s strobe picture	reading pictures, maps and chaos, measurement	yes
after	the part Two Readers and a Ruler	near a doubling riders stay unsettled and read periods 64 and 32 (Loop Atlas)	the needle near a doubling	maps and chaos, measurement	yes
after	row 728	chaos line at +0.05 put weight 0.68 tumbling by 0.0017 (R2W)	a rider with mass, ripple on	driven oscillators, maps and chaos	yes
after	addition to row 219; addition to row 793	reported two-timing miss was the file’s rounding, not the leak (R2W)	TIMINGS-2’s prediction file	driven oscillators, measurement, records and tools	yes
after	addition to row 106; addition to row 738; addition to	NINTH-1 sized from an engines’ gap that was the file’s rounding (R2W)	the record engine’s 12-decimal files	driven oscillators, records and tools, measurement	yes

	row 742; addition to row 799				
after	addition to row 796	timing rounded to six decimals made a spurious first-harmonic leak (R2W)	ODD-1’s reader	records and tools, driven oscillators	yes
after	the part Two Readers and a Ruler; addition to row 665; addition to row 787	half-stopped threshold made a gradual drain look like a sharp closing (Loop Atlas)	the needle’s stop and leak	maps and chaos, measurement	yes
after	addition to row 662	lifetime jump at a held place on the lip: boundary or door (Loop Atlas)	needles held below the window	maps and chaos, measurement	yes
after	addition to row 174	snake’s apparent slowing climb was the grid’s own growing miss (Loop Atlas)	the long motor snake	kinematics, measurement	yes
after	addition to row 174	snake’s best sat on the searched range’s edge; speed-up read against it (Loop Atlas)	the motor snake at 768 links	kinematics, measurement	yes
after	addition to row 724	train break times jump with spacing: transient chaos, read as lifetime (Loop Atlas)	a train of three rings	maps and chaos, measurement	yes
after	addition to row 628; addition to row 785	bead’s blade switch count was chatter at zero speed, doubling at half step (Loop Atlas)	a spring bead’s blade	driven oscillators, measurement	yes
after	addition to row 764	switch-ons at heavy loss grew as the knee shrank (Loop Atlas)	a spring bead’s knee ear	driven oscillators, measurement	yes
after	addition to row 764	knee’s count still creeps with the step under rough noise (Loop Atlas)	a knee ear on rough noise	control and circuits, measurement, quadrature	yes
after	addition to row 627; addition to row 629	single-line crossing count grows without end as the step shrinks (Loop Atlas)	a rough relaxing noise	quadrature, measurement	yes
after	addition to row 692	a value exactly on a rounding boundary rounded two ways (triple-star)	the record’s printed TABLE value	records and tools, measurement	yes
after	the part Two Readers and a Ruler; addition to row 648	two-sided fin split drawn by the 0.005 birth spacing; 7.358 flickers (triple-star)	three-body orbits, the shark fin	three body, reading pictures, measurement	yes
after	addition to row 174	a layer’s width ended at the scan’s cap; walked further, the rule failed (triple-star)	the chaotic layer below island 7	three body, maps and chaos, measurement	yes
after	row 810	a ‘machine difference’ was half a single-precision step (triple-star)	saved series, laptop and containers	records and tools, measurement	yes
after	addition to row 622; addition to row 732	printed numbers depended on how late parters were binned (triple-star)	three-body orbits, late parters	three body, maps and chaos, measurement	yes
after	addition to row 174	pause ear’s best price sat on the widest zone tried (Loop Atlas)	Bear’s pause ear, dead zone	control and circuits, driven oscillators, measurement	yes

after	addition to row 174	widened to 3, the pause ear’s best again sat on the edge (Loop Atlas)	Bear’s pause ear, dead zone	control and circuits, driven oscillators, measurement	yes
after	addition to row 789	near its lock, run statistics set by the joining gap, not the needle (Loop Atlas)	a needle over rough floors	driven oscillators, measurement	yes
after	row 816; addition to row 754	a round-trip guard flips on rounding, faking hits in a column (Loop Atlas)	a toy walker column’s guard	kinematics, records and tools	yes
after	addition to row 818	nearest approach on a 1-degree grid was the grid’s, not the cat’s (Loop Atlas)	the twist-free falling cat	kinematics, measurement	yes
after	row 738	a planted repeat missed by a six-decimal rounding of the pace (Loop Atlas)	a noisy needle’s exact rates	driven oscillators, records and tools, measurement	yes
after	addition to row 9b	a crossing on a flat stretch near 1 set by the grid (Loop Atlas)	the softened bells’ gain map	biology, waves, measurement	yes
after	addition to row 116; addition to row 214	two readers’ null differed in the last bit from m/18 versus m times 2/36 (R2W)	the fifteen riders’ null	driven oscillators, records and tools, measurement	yes
after	addition to row 802	null’s 10^{-15} floor was file rounding the 10-degree timing grid failed to cancel (R2W)	fifteen riders’ null, timing grid	driven oscillators, quadrature, measurement	yes
after	addition to row 628	rival’s 0.002 bar sat on the 0.001 grid; rounding failed it (R2W)	the door sweep’s grid	driven oscillators, measurement	yes

5 — A second reader

50 arrivals: 15 counted by 10 October, 35 after.

Counted	Lands at	Arrival	Substrate	Domains	Had the session read the guide first?
by 10 October	addition to row 100	a sounding board caught a correction and a sign (Loop Atlas)	the project’s own letters	records and tools, measurement	yes
by 10 October	row 650	rerun the other room’s own code (Loop Atlas book)	three-body orbits	three body, records and tools	yes
by 10 October	addition to row 35	a reader’s stumble found by the other room (triple-star and Loop Atlas)	three-body orbits	three body, measurement, kinematics	yes
by 10 October	addition to row 558	“worst” that was a mean, caught by re-reading (Loop Atlas)	three-body orbits	three body, measurement, solar	yes
by 10 October	a note in the gathering record	two collections of R2W’s 25 reads, made apart, agree word for word (R2W and Field Guide)	the collection of reads	geodesy, measurement	yes
by 10 October	row 642	R2W caught a doubling named as a fold in the Loop Atlas’s notes	the Loop Atlas’s needle	maps and chaos, measurement	yes
by 10 October	addition to row 570	the 3-Body Team, reading the changes file, found row 619’s figure untraceable	the guide’s own rows	records and tools, measurement	yes

				measurement	
by 10 October	addition to row 650	this guide reran the 3-Body Team’s trace unchanged and matched it line for line	the triple’s run files	three body, records and tools	yes
by 10 October	a note in the gathering record	the Loop Atlas checked its rows 79 to 88 against its own files: five hold with a difference	the Loop Atlas’s own ledger	records and tools, measurement	yes
by 10 October	row 677	two tools from two sessions, set alike, agree to 1.8 × 10^{-11} (R2W)	the Loop Atlas’s needle	maps and chaos, records and tools, measurement	yes
by 10 October	row 680	R2W read the ladder’s last ratio from the saved value, where this guide had used a rounding	the Loop Atlas’s needle	maps and chaos, driven oscillators	yes
by 10 October	addition to row 100	a cold audit by a separate agent found 17 points and an island’s real edge (3-Body Team)	a paper’s draft	records and tools, three body	yes: the guide read; row 100 read in full only after the checks
by 10 October	addition to row 369	two blind readers: what both found held up best (R2W, B8)	the triple’s pictures	reading pictures, three body, measurement	yes
by 10 October	row 712	another room’s blind reads judged on the data, three of three in sign and place (3-Body Team)	the triple’s pictures	reading pictures, three body, measurement	yes
by 10 October	addition to row 666	another session’s frame tested a book’s own the same day (Loop Atlas)	the Loop Atlas book	records and tools, measurement	yes
after	the part Two Readers and a Ruler	Bear’s eye saw the star polygon Claude’s sealed read missed, and vice versa (triple-star)	Bear’s maypole picture	reading pictures, three body	yes
after	row 780	A separate reader re-read session, register and draft; claims checked against files (triple-star)	the thread’s session and draft	records and tools	yes
after	addition to row 686	another session’s runs showed R2W’s switch bracket read high from below (R2W)	the bare ball’s memory switch	driven oscillators, measurement	yes
after	addition to row 100; addition to row 555; addition to row 570; addition to row 661	an independent reader caught a provenance the record did not support (R2W)	R2W’s record of a sitting	records and tools, measurement	yes
after	addition to row 573	Bear’s hand-marks on the 0.26 map caught a fork the linker missed (R2W)	the rider’s weight 0.26 map	reading pictures, driven oscillators	yes
after	the part Two Readers and a Ruler	Bear’s marks found curves and a population the blind read missed (R2W)	the rider’s map	reading pictures, driven oscillators	yes
after	the part Two Readers and a Ruler	Bear’s marks showed a mirror and straight lines the blind read missed (R2W)	a three-body strobe picture	reading pictures, three body	yes
after	row 748	R2W’s arithmetic sized the Loop Atlas’s undecidable chain test (Loop Atlas)	the remembering needle’s falls	driven oscillators, measurement	yes
after	addition to row 707	Bear’s strokes traced fold images Claude could not see (Loop Atlas)	a window’s chaos	reading pictures, maps and chaos, measurement	yes

after	addition to row 705	Bear’s rule read off a drawing gave the windows’ cycles (Loop Atlas)	windows in banded chaos	maps and chaos, reading pictures	yes
after	addition to row 707	Bear’s arc traced the fold’s third image that ends the window (Loop Atlas)	the 1/4 window’s seam	reading pictures, maps and chaos	yes
after	rows 726 and 727; addition to row 502	loop Atlas reading R2W’s page caught ‘the same machine’ (Loop Atlas)	R2W’s page, 12 October	records and tools	yes
after	row 775	three checkers looking up every citation found seventeen corrections (Field Guide)	the guide’s cited works	records and tools, measurement	yes
after	row 773	a separate agent’s cold read of each build found real defects (Field Guide)	the guide’s builds	records and tools	yes
after	addition to row 588	re-reading old printed folds showed one edge found twice; two widths wrong (Loop Atlas)	the needle’s thin locks	records and tools, driven oscillators	yes
after	the part Two Readers and a Ruler	R2W measured Bear’s eyeballed fractions: shape right, sizes off (R2W)	Bear’s marked pictures	reading pictures, driven oscillators	yes
after	addition to row 588	re-reading a guide row against the runs found two results joined (triple-star)	the guide’s row on lap ratio	three body, records and tools	yes
after	addition to row 555	separate agent’s cold review added three qualifiers to the model result (triple-star)	the resonance model’s write-up	three body, measurement	yes
after	addition to row 128	Bear’s reading of figures found the dial’s count Claude’s read missed (R2W)	the dial’s dark point figures	reading pictures, driven oscillators	yes
after	the part Two Readers and a Ruler; addition to row 715	Bear’s rings on a figure found rungs Claude’s ruler missed (Loop Atlas)	Bear’s marked needle figure	reading pictures, maps and chaos, measurement	yes
after	addition to row 100	cold reader of a laptop kit found four faults before the run (triple-star)	the triple_closeout laptop kit	records and tools	yes
after	addition to row 451	cold review of a kit found a reading that could not answer (triple-star)	the triple_lw5 kit	records and tools, measurement	yes
after	addition to row 506	cold review of new prior-art sentences found an error and six overstatements (triple-star)	the paper’s new prior-art sentences	records and tools, three body	yes
after	addition to row 1	cold reviewer caught a pass the reader could not have made (triple-star)	SL6b’s scored reading	three body, measurement	yes
after	addition to row 569	cold reader found dangling pointers the conservation script could not see (triple-star)	the islands paper’s rebuilt text	records and tools	yes
after	the part on the headline register	cold reviews caught 26 overclaims on the headline page (triple-star)	the paper’s headline page	records and tools	yes
after	addition to row 683	random-sample cold check of the prediction ledger found wrong outcomes (triple-star)	the prediction ledger	records and tools, measurement	yes
after	addition to row 552	re-reading Tong and Dullin found ‘less than one degree’ against 4.13 (Loop Atlas)	Tong and Dullin’s dive paper	kinematics, records and tools	yes
after	row 814	rebuilding a published table found a dropped 0.08 in the knee (Loop Atlas)	Tong and Dullin’s Table 1	kinematics, records and tools,	yes

				measurement	
after	row 821; addition to row 222	independent readers of the headline register found 22 misstated sentences (Loop Atlas)	the Loop Atlas headline register	records and tools, measurement	yes
after	row 823	two non-writing readers found 25 misplaced numbers a script passed (Loop Atlas)	the reader’s edition rewrite	records and tools	yes
after	row 833	a second reader caught misdated quotes and four inherited slips (Loop Atlas)	the reader’s edition quotations	records and tools	yes
after	row 840	five readers read 237 uncounted landings; found a class the samples missed (R2W)	R2W’s record count	records and tools, measurement	yes
after	row 843	independent reader of Claude’s §6.11 summaries: 25 claims failed, about 130 held (R2W)	the book’s summary paragraphs	records and tools	yes
after	—	independent read of the close’s summaries: 26 failures in about 380 claims (R2W)	the close’s summaries	records and tools	yes

6 — Coupled clocks and event time

37 arrivals: 17 counted by 10 October, 20 after.

Counted	Lands at	Arrival	Substrate	Domains	Had the session read the guide first?
by 10 October	row 630	find a toy’s clock by doubling its rhythm: every toy counts its own events except the one that pushes for a time (Loop Atlas book)	the book’s toys	kinematics, quadrature	yes
by 10 October	row 636	a balancing rider’s cliff sits where its reaction delay reaches its own rest between taps (Loop Atlas book)	a rider on a board	control and circuits, quadrature	yes
by 10 October	row 645	a window must hear three beats of its slowest beat, where two rhythms beat against each other (R2W, 3 October)	a relay between two gearboxes	driven oscillators, quadrature	no
by 10 October	row 660	a near-miss is a loan: hold time times detuning tends to a constant near a ratio of 2 (R2W, 4 October)	a rider near an exact fraction	driven oscillators, three body	no
by 10 October	row 615	count where the time goes, not how many episodes there are (triple-star)	three-body orbits	three body, measurement	not known (most likely before any full reading)
by 10 October	row 626	two staircases, a slow one and a fast one, in one chaotic triple (triple-star)	three-body orbits	three body, measurement	not known (most likely before any full reading)
by 10 October	row 719	two clocks return at the continued fraction of their period ratio, a result in the thread’s after-action review (triple-star)	three-body orbits	three body, quadrature, number theory	not known
by 10 October	addition to row 641	compare riders at the same phase of the pattern, never at one clock time (R2W, B6)	a rider with mass	driven oscillators, quadrature, measurement	yes
by 10 October	addition to row 645	a window that listens for a beat must span whole beats (R2W, B6)	a relay between two gearboxes	driven oscillators, quadrature	yes
by 10 October	row 671	a heavy rider’s free spin climbs toward 1/0.618034, the two clocks’ own ratio, and never passes it (Loop Atlas)	the needle	driven oscillators, maps and chaos,	yes

				number theory	
by 10 October	row 671	a plain-fraction clock makes no lock at the speed limit (Loop Atlas)	the needle	driven oscillators, number theory	yes
by 10 October	row 681	a rider who listens works on any swing; reading by the wrong rate is a Möbius stretch (Loop Atlas)	a pumped swing	driven oscillators, quadrature	yes
by 10 October	row 675	two laws of a near-miss in one formula (Loop Atlas)	the forgetting needle	driven oscillators, three body, maps and chaos	yes
by 10 October	row 675	a third law, the pendulum’s logarithm at a conservative lock’s edge (3-Body Team, door 2)	three-body orbits	three body, driven oscillators	yes
by 10 October	addition to row 20	an event trigger only tied a clock at matched pace (Loop Atlas)	the memory bell	driven oscillators, quadrature	not known
by 10 October	row 719	the lattice’s ladder at 7.200 is two angles coming home together, geared one to three (3-Body Team’s lattice; the Field Guide’s check of its data)	three-body orbits	three body, quadrature, number theory	yes
by 10 October	chapter, the common object	the outside tick and the rider’s own pace; a lock is the rider’s pace giving in to the outside tick where the two are already close (R2W, B8)	a rider with mass	driven oscillators, quadrature	yes
after	row 730	rocking read by a beat that nearly stops; redesign reads it by the pass (triple-star)	three-body orbits, island 0.25	three body, quadrature	yes
after	addition to row 603	rocking at twice the beat looked locked; neighbours showed both clocks moving together (triple-star)	three-body orbits, near-round star	three body, measurement	yes
after	addition to row 667	the rider’s own crawl time matches 30 outside ticks at the 1/30 step (R2W)	a rider near a 1/n lock	driven oscillators	yes
after	addition to row 712	a law’s size fitted poorly in the raw lag, cleanly in the straightened lag (R2W)	the rider’s starts and multipliers	driven oscillators, measurement	yes
after	row 746	maintenance calendar: hours times duty squared beats the wear’s own turn clock (Loop Atlas)	a fleet’s maintenance logbook	control and circuits, quadrature	yes
after	addition to row 641	integration step not dividing the drive period hid an exact repeat (Loop Atlas)	a settled spinning needle	driven oscillators, quadrature	yes
after	addition to row 636	a look’s cost set by its length against the rider’s fall time (Loop Atlas)	a balance board’s rider	control and circuits	yes
after	row 769	a swimmer model worked only when driven by its own events (Loop Atlas)	the jellyfish paper’s swimmer	biology, quadrature	not known
after	row 769	a strong pilot tone makes the bead keep the pilot’s time, losing the catch (Loop Atlas)	a spring bead in a soup	driven oscillators	yes
after	the part Two Readers and a Ruler	period-3 test bed read over 256 passings: uneven lanes, jittered centres (Loop Atlas)	a logistic-map test bed	maps and chaos, measurement	yes
after	row 790	a lock’s width read in the rider’s own units reverses the drive trend (Loop Atlas)	the needle’s locks	driven oscillators, measurement	yes
after	addition to row 719	two angles’ median net turns, as a gear, named the lattice’s gaps blind (triple-star)	three-body orbits	three body, quadrature	yes
after	addition to row 645	600-pass window heard under three beats; ‘beat strongest’ verdict unresolvable (triple-star)	three-body orbits, births 7.470 to 7.480	three body, quadrature, measurement	yes

after	addition to row 645	2,400-pass window cannot resolve beats of 1,100 to 5,700 passes (triple-star)	three-body orbits at eccentricity 0.28	three body, quadrature, measurement	yes
after	addition to row 661	ratio law read against the beat broke where the beat passes zero (triple-star)	three-body orbits, Bear’s ratio law	three body, measurement	yes
after	addition to row 581	log-even holding powers eased where the drive outpaces the rider’s own pull (R2W)	fourth harmonic’s holding power over frequency	driven oscillators	yes
after	the part Two Readers and a Ruler; addition to row 769	beat and pilot follow the bead’s loaded pace, not its nameplate tune (Loop Atlas)	a spring bead under a beat	driven oscillators	yes
after	addition to row 630	bells’ scaling holds only when the bell’s own clock is stretched too (Loop Atlas)	two tied bells in water	kinematics, waves	yes
after	addition to row 636	tap lag read the rest clock, hiding the wearing reaction delay (Loop Atlas)	a slowing tapping rider	driven oscillators, quadrature	yes
after	addition to row 795	a lock’s apparent width depends on record length against its detuning (triple-star)	three-body orbits, locks at 1/10	three body, quadrature	yes

Not ranked — The eye proposes, the ruler decides (proposed)

29 arrivals, counted on 23 October, when the family was defined, from the lessons placed since 17 October and the earlier ones a reader had flagged for it.

Counted	Lands at	Arrival	Substrate	Domains	Had the session read the guide first?
after	the part Two Readers and a Ruler; addition to row 800; addition to row 87	claude’s by-eye ‘textbook shape’ call registered, tested, then withdrawn (triple-star)	the jump at the tenth	three body, reading pictures, measurement	yes
after	addition to row 79	a slow line seen after the run registered forward, then settled (triple-star)	three-body orbits, the slow line	three body, measurement	yes
after	addition to row 569; addition to row 800	Bear’s ‘wall of text’ read measured: 52 pages, no image (triple-star)	the islands paper’s draft PDF	records and tools	yes
after	addition to row 718	Bear’s torus-outline read registered as T1; missed by a hair, borne out over 6,000 passes (triple-star)	three-body orbits, Bear’s outline	three body, reading pictures	yes
after	the part Two Readers and a Ruler	eye reads from coarse pictures registered, then judged on a finer grid (triple-star)	three-body orbits, the 0.25 island	three body, reading pictures	yes
after	the part Two Readers and a Ruler	Bear’s sense that relation reads beat shape reads, tested blind (triple-star)	74 registered picture-reads, most of them Bear’s	reading pictures, three body, measurement	yes
after	the part Two Readers and a Ruler	fin read by eye tested on nine lines: place held, outline mostly not (triple-star)	three-body orbits, the shark fin	three body, reading pictures	yes

after	the part Two Readers and a Ruler	claude’s by-eye knob guess, 0.49, measured as 0.4905 (triple-star)	the bench’s damped cascade ledger	three body, reading pictures	yes
after	row 743; the part Two Readers and a Ruler	Bear’s squeeze-and-tilt marks registered as SQ1 to SQ3 and VC1: SQ2 and SQ3 (Claude’s, from his arrows) held, SQ1 empty by construction, VC1 not borne out (triple-star)	Bear’s marked strobe picture	reading pictures, three body	yes
after	row 829	three points looked like a peak; eleven read a plateau (Loop Atlas)	the bells’ gain over meeting width	biology, waves, reading pictures	yes
after	the part Two Readers and a Ruler	Bear’s ringed columns measured as a factor: 98.6% a product (Loop Atlas)	the bells’ gain maps	biology, waves, reading pictures	yes
after	the part Two Readers and a Ruler	Bear’s ‘Balanced?’ measured as a ratio: cost twice pay (Loop Atlas)	the bells’ gain maps	biology, waves, reading pictures	yes
after	the part Two Readers and a Ruler	Bear’s arrows between maps checked against gains near even (Loop Atlas)	the bells’ gain maps	biology, waves, reading pictures	yes
after	row 79; the part Two Readers and a Ruler	stiffness reading of a pattern made a prediction, was graded, and missed (R2W)	balls slipping under turned ripple	driven oscillators, measurement	yes
after	the part Two Readers and a Ruler	Bear’s seen rhyme between pin and fan waist: shared shape, scalings differ (R2W)	the pin figure and fan’s waist	driven oscillators, reading pictures	yes
after	addition to row 715	Bear’s hand-drawn dead zones held on later probes’ runs (triple-star)	triple-star island map, phi_N bands	three body, reading pictures	yes
after	the part Two Readers and a Ruler; addition to row 714	claude’s by-eye period-3 reading went wrong when measured (R2W)	R2W B9 figures	driven oscillators, reading pictures	yes
after	the part Two Readers and a Ruler; addition to row 714	crest counts by eye on the bend figure went wrong (R2W)	R2W bend figure	driven oscillators, reading pictures	yes
after	addition to row 714	by-eye ‘about four’ on the rope check corrected by pixels (R2W)	rope check picture	driven oscillators, reading pictures	yes
after	addition to row 128	five patterns Bear pointed to, each measured within the hour (R2W)	R2W’s pictures	driven oscillators, reading pictures	yes
after	row 743	Bear’s drawn axis and edges checked by pixel percentiles; all held (R2W)	strobe picture, phi_N columns	reading pictures, driven oscillators	yes
after	addition to row 573	Bear’s two pencil lines on figure AA scored by data; one hit the gear break (R2W)	harmonic zeros versus weight	driven oscillators, reading pictures	yes
after	addition to row 581	Bear’s traced swell registered as a test, measured as the beat (triple-star)	triple-star island centre	three body, reading pictures, measurement	yes
after	addition to row 581	Bear’s one-line pattern read registered and tested over slices (triple-star)	triple-star rates and ratio	three body, measurement	yes
after	addition to row 581	Bear’s ‘fractals’ read registered as two nested-chain tests (triple-star)	triple-star nested chains	three body, measurement	yes

after	addition to row 581	Bear's marks on unlabelled figures tested; a lock at a seventh (triple-star)	triple-star plain figures	three body, reading pictures, measurement	yes
after	addition to row 128; addition to row 712	Bear's dipole-or-mirror picture tested in kind; both claims held (R2W)	harmonic effect versus tone direction	driven oscillators, reading pictures, measurement	yes
after	addition to row 581	Bear's 'even steps' read registered as RATCHET-61; held one stretch (R2W)	fourth harmonic holding power	driven oscillators, measurement	yes
after	addition to row 128	Bear's staging-ground picture tested; held in every arrangement (R2W)	RF rider staging ground	driven oscillators, reading pictures	yes

7 — The mirror audit

28 arrivals: 12 counted by 10 October, 16 after.

Counted	Lands at	Arrival	Substrate	Domains	Had the session read the guide first?
by 10 October	addition to row 190	a lock forgets: the time retrace fails on a locked rider (Loop Atlas)	riders and blades	driven oscillators, measurement	yes
by 10 October	row 575	split by polarity before you sum (Sun paper)	sunspot maps	solar, measurement	not known
by 10 October	row 601	a reflected starting nudge fakes a flip (Loop Atlas)	a draft horse	kinematics, measurement	yes
by 10 October	row 602	the auditor's expectation is a card too (Loop Atlas)	the mirror audit itself	measurement, records and tools	yes
by 10 October	addition to row 190	mirror and reverse before you scramble (R2W)	a dither	driven oscillators, measurement	no
by 10 October	row 624	lengths cannot see handedness (triple-star bench)	a pentagon frame	kinematics, measurement	not known (most likely before any full reading)
by 10 October	addition to row 575	walked both ways, the apsidal law splits into a part that flips and a part that does not (3-Body Team, probe 21)	three-body orbits	three body, solar, measurement	yes: v6 read; rows 575 and 584 not yet
by 10 October	addition to row 259	a loop centred on a mirror-odd quantity cancels to every order (Loop Atlas)	a free-floating diver	kinematics, topology	yes
by 10 October	row 673	which door depends on which way round the tip (R2W, B6)	a rider with mass	driven oscillators, topology	yes
by 10 October	addition to row 648	a lock seen only from starts on the mirror line, where a mirror-symmetric system keeps its repeating orbits (3-Body Team)	three-body orbits	three body, measurement	yes: row 648 read in full before the probe
by 10 October	row 694	a split measured about the symmetry set the orbits live on (3-Body Team)	a cascade's newborn orbits	maps and chaos, three body	not known
by 10 October	row 706	mirror and rescale one picture onto the other to test a shared mechanism (Loop Atlas, on Bear's reading)	R2W's maps	reading pictures, driven oscillators, maps and chaos	yes
after	row 734;	the time mirror run on the weightless rider holds to	a weightless	driven	yes

	addition to row 734	10^{-13} (R2W)	rider	oscillators, reading pictures	
after	row 734	a registered exact mirror broke, exposing the engine’s half-trace assumption (R2W)	the junction under two tones	driven oscillators, records and tools	yes
after	addition to row 190	the snake run forward and backward exposed a drifting stepping rule (Loop Atlas)	the snake toy	kinematics, measurement	yes
after	addition to row 681	the brake at the opposite lead retraces the pump’s path exactly (Loop Atlas)	a pumped and braked swing	driven oscillators	yes
after	addition to row 734	swapping the octave and fourth harmonic’s roles reproduced all 288 edges (R2W)	the three-tone engine RATCHET_3T	driven oscillators, records and tools, measurement	yes
after	addition to row 745	the reflection keeps even weight powers in the window, odd in the slide (R2W)	the rectifier’s window and zero	driven oscillators	yes
after	addition to row 734	flipping the third tone split c into an odd, dipole-like part and an even part (R2W)	the fourth-power term c	driven oscillators, quadrature	yes
after	addition to row 734	every rectifier reader graded the mirror first; it held to 0.00e+00 (R2W)	the rectifier readers	driven oscillators, measurement	yes
after	row 801; addition to row 106; addition to row 712	sign flip of the third tone split its effect into odd and even parts (R2W)	a rider with mass, fourth harmonic	driven oscillators	yes
after	row 801; addition to row 770	timing turned round the circle read what the reflection allows off the mirror (R2W)	the rider’s third-tone timing dial	driven oscillators	yes
after	addition to row 742	reflection predicted the opening’s right angle and no straight-line term; held (R2W)	the dark point’s weight opening	driven oscillators, measurement	yes
after	addition to row 712	mirror forbade drift entering the weight channel at the dark point; held (R2W)	the two-channel meter	driven oscillators, measurement	yes
after	addition to row 259	face swapped head to tail: travel flips 1.4%, work stays (Loop Atlas)	a three-piece swimmer	kinematics	yes
after	addition to row 805	half-turn of timing as a sign flip split the crest-firer’s two hands (R2W)	the Loop Atlas’s crest-firer	waves, quadrature, measurement	yes
after	addition to row 464	ripple run both ways: push flips, shaking stays; their ratio forgets weight (R2W)	gear 1’s edge under ripple	driven oscillators, topology	yes
after	row 847	sign flip of all tones predicts odd groups push; third harmonic speaks with octave (R2W)	harmonics pushing a rider	driven oscillators, quadrature	yes

8 — Count the right unit

25 arrivals: 11 counted by 10 October, 14 after.

Counted	Lands at	Arrival	Substrate	Domains	Had the session read the guide first?
by 10 October	row 615	count where the time goes (triple-star)	three-body orbits	three body, measurement	not known (most likely

					before any full reading)
by 10 October	row 619	shuffle the run, not the episode (triple-star)	three-body orbits	three body, measurement	not known (most likely before any full reading)
by 10 October	row 627	count landings, not stumbles (Loop Atlas book)	the turning needle	driven oscillators, measurement	yes
by 10 October	addition to row 627	sign the crossings (R2W, 3 October)	write heads on a gear’s figure-8	driven oscillators, topology, measurement	no
by 10 October	row 608	two counts on different bases sum to 49 of 43 (Loop Atlas)	the book’s tag table	records and tools, measurement	yes
by 10 October	addition to row 627	count the binary’s axis crossings; never predict turns from a disturbed reading (3-Body Team, probe 26)	three-body orbits	three body, measurement	yes: primed by rows 627 and 628
by 10 October	addition to row 639	near an edge, count slips rather than averaging speed over a short window (R2W, B6)	a light pendulum	driven oscillators, measurement, three body	yes
by 10 October	row 663	count turning points when comparing a rate up with a rate down (Loop Atlas)	a pumped and braked swing	driven oscillators, quadrature, measurement	yes
by 10 October	row 702	every hole counts landings, and its count is the period (Loop Atlas, on Bear’s lattice)	the needle’s strobe picture	reading pictures, maps and chaos, measurement	yes
by 10 October	row 709	count the truth by each rider’s period, not by rounded points (Loop Atlas)	the needle’s strobe picture	maps and chaos, measurement, driven oscillators	yes
by 10 October	addition to row 369	a seen list, so that a chunk of cumulative returns votes once (R2W, B8)	the pad atlas’s returns	records and tools, measurement	yes
after	addition to row 627	coil crossings depend on strand family followed; jumps counted one per landing (triple-star)	a strobed coil, three-body orbits	three body, measurement	yes: the row read only after the run
after	addition to row 722	window reader’s short ‘locks’ were mid-fall hesitations, not locks (Loop Atlas)	the remembering needle	driven oscillators, measurement	yes
after	addition to row 722	a latched crossing flag counted raw crossings, not net ones (Loop Atlas)	the remembering needle	driven oscillators, measurement	yes
after	addition to row 619	strobe density counted one retraced path many times (Loop Atlas)	the needle’s strobe density	maps and chaos, reading pictures, measurement	yes
after	row 789	time per slip stretched because the crowd thinned, not because needles slowed (Loop Atlas)	needles below a lock’s edge	driven oscillators, measurement	yes
after	row 785	a wall share counted integrator steps, not time, and fell with the step (Loop Atlas)	the needle’s seam map	driven oscillators, measurement	yes
after	row 788	a maintenance calendar counted in running turns beat one in hours (Loop Atlas)	a maintenance meter’s fleets	control and circuits, measurement	yes
after	the part Two	visit counts need riders-per-column denominators;	R2W’s weight-	reading pictures,	yes

	Readers and a Ruler	RF shares skewed by 8 vs 50 starts (Loop Atlas)	sweep histograms	measurement, driven oscillators	
after	row 724	four groups of settled needles are one orbit at four phases (Loop Atlas)	the needle’s chaotic transients	maps and chaos, measurement	yes
after	addition to row 615	switch mistakes counted as time spent wrong, not wrong switches (Loop Atlas)	a chattering relay and a timer	control and circuits, measurement	yes
after	addition to row 724	knock-off is not a rate: riders of one soup are not independent (Loop Atlas)	a jumping bead in a soup	driven oscillators, waves, measurement	yes
after	row 821	rivals counted per printed line, not per distinct rival (Loop Atlas)	the Loop Atlas’s rival verdicts	records and tools, measurement	yes
after	row 827	time first arrivals at new levels, not every forward count (Loop Atlas)	a noisy needle slipping both ways	driven oscillators, measurement	yes
after	addition to row 222	a counting script counted printed labels, so one rival twice (Loop Atlas)	the headline register’s counting script	records and tools, measurement	yes

Not ranked — A law carried past its range (proposed)

22 arrivals, counted on 23 October, when the family was defined, from the lessons placed since 17 October and the earlier ones a reader had flagged for it.

Counted	Lands at	Arrival	Substrate	Domains	Had the session read the guide first?
after	addition to row 670	‘no passing lane at any push’ rested on a ramp stopping at 1.05 (Loop Atlas)	Bear’s truck road, mass 0.75	driven oscillators, measurement	yes
after	addition to row 9b	a lean from one case at memory 6 failed at short memory (Loop Atlas)	tied bells in water	kinematics, waves	yes
after	row 819	a ‘cliff’ read from a 1,000-unit window; trains broke just beyond (Loop Atlas)	a train of three rings	waves, maps and chaos, measurement	yes
after	row 824; addition to row 646	a small-step limit law fitted over coarse steps came out 0.902 (Loop Atlas)	the traders’ dead-band constant	control and circuits, quadrature, measurement	yes
after	addition to row 725	kramers’ high-barrier rate used at barriers of 4 to 9 (Loop Atlas)	a noisy needle’s slips	driven oscillators, measurement	yes
after	row 819	a ‘floor’ read inside a 20,000-unit window; flown longer, none (Loop Atlas)	a train of three rings	waves, maps and chaos, measurement	yes
after	addition to row 737	counted limit powers 4 and 1 used as bars at a strong drive; missed (R2W)	the flat point’s two arrows	driven oscillators, quadrature	yes
after	row 839	straight door law fitted 0.55 to 0.90 bent past where the room closes (R2W)	gear 2’s door	driven oscillators, maps and chaos	yes
after	addition to row 801	one-point coefficient reader, safe on the mirror line, printed 19.08 off it (R2W)	the light coefficient’s reader	driven oscillators, measurement	yes

after	row 842	run time doubling read from two shallow levels missed deep levels' Newton noise (R2W)	the 3/7 ladder kit's run	maps and chaos, driven oscillators, records and tools	yes
after	the part Two Readers and a Ruler	weight steadies the edge under gentle ripple; under strong ripple heavy balls slip sooner (R2W)	balls' edges across ripple strengths	driven oscillators, maps and chaos	yes
after	row 788	large-memory retrapping law used at memory 1; true line 0.9643 (Loop Atlas)	junction retrapping law, needle	driven oscillators	yes
after	row 788	spoke's slope −1 room rule carried to the needle relay; failed (Loop Atlas)	needle relay	driven oscillators, control and circuits	yes
after	row 799; addition to row 737	limit-exact symmetry argument carried along the weight road; R23-b failed (R2W)	ratchet drift zero under noise	driven oscillators	yes
after	addition to row 640	drag law pooled near the mirror carried to far lines; E7c missed (triple-star)	three-body drag law, start lines	three body, measurement	yes
after	addition to row 581; addition to row 640	log-frequency line from four stretches carried outward; missed two stretches out (R2W)	fourth harmonic holding power	driven oscillators, measurement	yes
after	addition to row 711	two-rung line carried to weight 0 missed on a synthetic stand-in (R2W)	synthetic window stand-in	driven oscillators, measurement	yes
after	addition to row 678	slope from the lightest weight carried to heavier ones; 4.7% short (R2W)	ratchet tilt versus weight	driven oscillators, measurement	yes
after	addition to row 788; addition to row 799	exponent fitted at frequency 0.55 carried to frequency 4; missed (R2W)	flat point versus drive	driven oscillators, measurement	yes
after	addition to row 788; addition to row 799	single power from two swings carried to a third; missed (R2W)	flat point versus drive	driven oscillators, measurement	yes
after	the part Two Readers and a Ruler; addition to row 640	prediction from map read to 1.06 failed beyond 1.09 (Loop Atlas)	mass-1 rider passing lane	driven oscillators, measurement	yes
after	addition to row 640	four registered leans missed past the edges of short-read maps (Loop Atlas)	Loop Atlas E dial maps	driven oscillators, measurement	yes

9 — The integer diagnostic

19 arrivals: 11 counted by 10 October, 8 after.

Counted	Lands at	Arrival	Substrate	Domains	Had the session read the guide first?
by 10 October	row 560	a count is whole only when read from closed pieces (Latch Atlas)	skyrmion grids	materials, topology, measurement	yes, through a digest
by 10 October	row 548	an exponent that tracks a length is a magnitude (latch paper)	linked filaments	materials, measurement	not known
by 10	row 608	[illegible]	[illegible]	[illegible]	

by 10 October	row 608	counts are gears, measures are friction wheels (Loop Atlas)	the book's loops	topology, measurement	yes
by 10 October	addition to row 103	a count printed with a minus sign is a measure (Loop Atlas book)	the cliff-edge needle	driven oscillators, measurement, topology	yes
by 10 October	addition to row 103	count before you measure: slips per beat are whole (R2W)	a relay switch	driven oscillators, measurement, topology	no
by 10 October	row 672	wells and saddles must sum to zero, so an index sum checks the solver (R2W, B6)	the rider's fixed points	driven oscillators, topology	yes
by 10 October	addition to row 103	counts survive a change of machine; magnitudes do not (R2W, B7)	a laptop and a container	records and tools, measurement, driven oscillators	yes
by 10 October	row 696	chaos keeps a count: a slip within 0.0004 of 2/3 (R2W, B8)	a rider with mass	maps and chaos, driven oscillators	yes
by 10 October	row 700	a loop reading is a remainder, and the whole turns are counted from its history (Loop Atlas)	frames carried round paper mountains	kinematics, topology	yes
by 10 October	addition to row 627	a frame counts each region as often as the path winds round it (Loop Atlas)	frames carried round paper mountains	kinematics, topology, measurement	yes
by 10 October	addition to row 103	high-precision readings agree across two machines and double-precision paths do not (Loop Atlas)	a laptop and a container	records and tools, measurement	yes
after	addition to row 628	swizzle count exact only when drag is charged per swizzle (Loop Atlas)	a shoved skater	kinematics, measurement	yes
after	row 749	whole turns carried inside a float ate the angle's fine digits (Loop Atlas)	needles' lag at 31,600 radians	records and tools, maps and chaos, measurement	yes
after	addition to row 696	chaotic needle keeps five turns in every seven over 30,000 cycles (Loop Atlas)	the banded 5/7 needle	maps and chaos, driven oscillators	yes
after	addition to row 645	a rotation number read over a non-whole count of cycles gave 0.4008 for 2/5 (Loop Atlas)	the needle's strobe readings	driven oscillators, quadrature	yes
after	the part Two Readers and a Ruler; addition to row 628	tap gaps read in fall times looked different; whole steps agree (Loop Atlas)	a tapping rider on two boards	driven oscillators, measurement	yes
after	addition to row 151	share rises exactly per whole arm circle; partial circles depend on phase (Loop Atlas)	a jumper's arm circles	kinematics	yes
after	row 834	4.007 printed as measured was exactly 4, read from whole steps (Loop Atlas)	a tapping rider on two boards	control and circuits, measurement	yes
after	row 847	power 4 counted from tones; measured exponents 3.916 to 3.993 approach it (R2W)	the flat point's fourth-harmonic arrow	driven oscillators, quadrature	yes

10 — The knee

18 arrivals: 7 counted by 10 October, 11 after.

Counted	Lands at	Arrival	Substrate	Domains	Had the session read the guide first?
by 10 October	row 563	a ring with one weak spot writes and erases by two roads (Latch Atlas)	superconducting rings	materials	yes, through a digest
by 10 October	row 627	a counter commits a slip only on landing (Loop Atlas book)	the turning needle	driven oscillators, measurement, three body	yes
by 10 October	row 634	tyre grip turns one speed into a band (Loop Atlas book)	a car on a banked track	driven oscillators, kinematics, topology	yes
by 10 October	row 670	memory between two locks, read by holding riders, not ramping them (R2W, B6)	a rider with mass	driven oscillators	yes
by 10 October	row 662	a held test finds no running state in the window a bound pictured (Loop Atlas)	a rider with mass (the Loop Atlas's needle)	driven oscillators, maps and chaos, measurement	yes
by 10 October	row 670	the road writes the dance: two dances share one stone (R2W, B7)	a rider with mass	driven oscillators, maps and chaos	yes
by 10 October	addition to row 683	a memory band is hysteresis: below its pull a rider at rest stays and a runner keeps running (R2W, B8)	a rider with mass	driven oscillators, measurement	yes
after	addition to row 664	needle from rest stays seated where spinning needles keep spinning (Loop Atlas)	the pulsed needle with memory	driven oscillators, maps and chaos	yes
after	row 758	needle walked past its stop and back stays stopped (Loop Atlas)	the pulsed needle with memory	driven oscillators, maps and chaos	yes
after	row 787	a remembering needle holds a 1:2 gear or the 1:1 lock by its history (Loop Atlas)	the remembering needle	driven oscillators, measurement	yes
after	row 785	a noisy listening ear chattered; a knee three times the noise cured it (Loop Atlas)	the listening blade's ear	control and circuits, driven oscillators	yes
after	row 789	ramped readings of a climb-back line lagged the line read standing still (Loop Atlas)	the needle's climb-back line	driven oscillators, measurement	yes
after	addition to row 664	lag-0 and lag-½ starts land in different states; overlaid chart hides which (R2W)	a rider with mass	driven oscillators, reading pictures	yes
after	the part Two Readers and a Ruler; addition to row 669	dead-zone ear latches; push per switch beats a crest-firer's (Loop Atlas)	the spring bead's latching ear	control and circuits, driven oscillators	yes
after	addition to row 670	two gears coexist; a runner is caught by one or the other (Loop Atlas)	Bear's truck road, the swapped needle	driven oscillators, maps and chaos	yes
after	addition to row 604	one load carried in step or slipped, by history alone (Loop Atlas)	the swing equation's machine	control and circuits, driven oscillators	yes

after	addition to row 628	a dead band's toll on a crest count measured as overshoot (Loop Atlas)	a crest-finer with a dead band	waves, quadrature, measurement	yes
after	addition to row 673	loops round where sitting and spinning edges meet: jumps on crossing (Loop Atlas)	a needle with weight and memory	driven oscillators, topology	yes

11 — Rate over level

2 arrivals: 1 counted by 10 October, 1 after.

Counted	Lands at	Arrival	Substrate	Domains	Had the session read the guide first?
by 10 October	row 708	Bear's lattice: slope, change of slope and change of change read along strings, the change of change marking events; taught to the three sessions together and measured in each, so counted once	three sessions' pictures	reading pictures, maps and chaos, measurement	yes
after	addition to row 715	strobe pictures' crowds and gaps read as the rider's speed, not its place (R2W)	the rider's strobe pictures	driven oscillators, reading pictures	yes

Appendix — The register by domain

Every row of the register under each of the sixteen domains, most rows first. A row carries one to three domains, the most specific first, so a row can appear under more than one; the line under each domain says how many of its rows name it first.

Measurement (444 rows)

Statistics and measurement in general: nulls, bars, floors, rulers, uncertainty, controls, shuffles, registration of tests. First domain of 109 of them.

1, 2, 7, 8, 9a, 9b, 10, 13, 14, 15, 16, 18, 19, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 34, 35, 36, 42, 44, 46, 52, 54, 55, 58, 59, 62, 63, 67, 70, 71, 72, 73, 74, 75, 77, 79, 80, 84, 85, 87, 89, 93, 94, 96, 98, 99, 100, 101, 102, 103, 105, 106, 107, 108, 111, 112, 113, 116, 117, 119, 120, 122, 124, 125, 126, 127, 128, 129, 130, 131, 132, 134, 136, 138, 139, 140, 142, 146, 152, 153, 154, 160, 161, 162, 163, 164, 167, 173, 175, 178, 182, 184, 185, 188, 190, 193, 194, 200, 201, 203, 210, 211, 212, 218, 219, 221, 222, 228, 231, 239, 240, 241, 242, 243, 244, 246, 247, 249, 250, 252, 254, 255, 256, 257, 258, 262, 264, 266, 276, 289, 291, 293, 294, 297, 298, 299, 300, 301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314, 316, 317, 319, 320, 322, 325, 326, 327, 329, 331, 333, 335, 336, 337, 338, 339, 340, 341, 342, 343, 344, 345, 346, 347, 348, 349, 355, 356, 359, 360, 365, 368, 369, 372, 373, 374, 377, 384, 385, 391, 392, 398, 400, 402, 403, 404, 406, 407, 409, 412, 413, 414, 415, 416, 417, 418, 421, 422, 424, 425, 426, 427, 428, 429, 430, 438, 442, 445, 449, 450, 451, 452, 453, 455, 456, 457, 458, 459, 460, 461, 462, 463, 464, 471, 472, 473, 474, 479, 480, 481, 487, 489, 492, 494, 498, 501, 502, 505, 507, 508, 509, 510, 518, 523, 526, 527, 530, 533, 534, 535, 536, 537, 540, 541, 542, 544, 548, 549, 550, 551, 554, 556, 557, 558, 559, 560, 567, 568, 570, 576, 577, 579, 580, 581, 584, 587, 588, 593, 598, 601, 602, 603, 608, 612, 614, 615, 616, 617, 618, 619, 620, 621, 622, 623, 624, 625, 626, 628, 629, 631, 632, 638, 639, 640, 645, 646, 647, 648, 649, 651, 652, 653, 661, 662, 663, 664, 666, 667, 674, 677, 678, 682, 683, 684, 686, 687, 688, 689, 690, 691, 692, 693, 694, 695, 709, 711, 712, 716, 718, 722, 723, 725, 728, 729, 730, 731, 732, 734, 737, 738, 741, 742, 746, 747, 748, 749, 752, 758, 760, 764, 766, 771, 781, 784, 785, 787, 788, 789, 793, 794, 795, 798, 799, 801, 805, 807, 808, 810, 813, 814, 819, 821, 822, 823, 824, 826, 827, 829, 831, 834, 835, 837, 840, 850, 851, 852, 853, 858, 859, 860, 861, 863, 865, 867, 868, 869, 870, 872, 874, 875, 877, 879, 880, 882, 883, 884, 885, 886, 887, 888, 889, 890, 891

Records and tools (223 rows)

The project’s own working material: builds, PDFs, crates, manifests, scripts, run records, citations, sessions and handovers. First domain of 170 of them.

9a, 36, 61, 62, 67, 68, 71, 74, 75, 76, 78, 81, 82, 83, 86, 88, 89, 90, 99, 100, 102, 113, 114, 129, 130, 141, 154, 155, 156, 157, 158, 159, 166, 168, 169, 191, 192, 194, 205, 206, 220, 221, 222, 229, 230, 236, 239, 250, 251, 253, 263, 265, 267, 275, 276, 281, 288, 312, 322, 323, 324, 325, 326, 327, 328, 333, 334, 337, 369, 370, 385, 396, 397, 399, 401, 407, 411, 412, 418, 427, 431, 432, 433, 434, 435, 436, 437, 439, 440, 441, 454, 455, 458, 465, 467, 475, 476, 477, 488, 493, 502, 503, 504, 505, 506, 507, 511, 512, 513, 516, 524, 525, 533, 538, 539, 545, 547, 551, 552, 553, 559, 569, 570, 571, 572, 574, 577, 578, 580, 582, 583, 588, 589, 591, 592, 593, 594, 595, 596, 597, 599, 600, 610, 611, 613, 642, 644, 647, 650, 661, 666, 667, 676, 677, 679, 692, 720, 721, 731, 733, 734, 735, 736, 740, 765, 767, 770, 772, 773, 774, 775, 776, 777, 778, 779, 780, 782, 783, 786, 793, 796, 797, 798, 800, 808, 809, 810, 811, 812, 815, 816, 821, 823, 825, 832, 833, 838, 840, 842, 843, 844, 845, 846, 849, 852, 853, 854, 855, 856, 857, 861, 862, 866, 867, 868, 869, 871, 874, 875, 880, 881, 882, 891

Driven oscillators (174 rows)

Pendulums and riders pushed by a drive; phase locking, staircases, Arnold tongues, Shapiro-like steps, R2W’s rider, Josephson-like junctions, ratchets and rectifiers. First domain of 83 of them.

20, 38, 73, 76, 79, 93, 103, 106, 168, 174, 190, 210, 214, 219, 282, 287, 308, 314, 328, 351, 352, 353, 354, 380, 389, 400, 414, 416, 451, 464, 469, 470, 486, 514, 531, 541, 555, 563, 565, 566, 573, 575, 579, 581, 584, 586, 603, 604, 605, 606, 609, 627, 628, 631, 632, 633, 634, 636, 638, 639, 641, 642, 643, 644, 646, 648, 649, 652, 654, 655, 656, 657, 658, 659, 660, 662, 663, 664, 665, 666, 668, 669, 670, 671, 672, 673, 674, 675, 678, 680, 681, 682, 683, 684, 685, 686, 696, 697, 698, 699, 702, 703, 704, 705, 706, 707, 708, 709, 710, 711, 712, 713, 714, 717, 722, 725, 726, 727, 728, 730, 734, 737, 738, 739, 741, 742, 745, 746, 747, 748, 749, 750, 753, 755, 757, 759, 760, 764, 766, 769, 784, 785, 787, 788, 789, 790, 791, 794, 795, 799, 801, 802, 803, 805, 826, 827, 828, 834, 835, 839, 841, 847, 848, 849, 850, 851, 859, 864, 870, 872, 876, 877, 878, 879

Topology (166 rows)

Topology, knots and invariants as such: winding, linking, indices, holonomy in general, fibre bundles, Gauss integrals. First domain of 75 of them.

12, 32, 34, 40, 43, 44, 45, 46, 47, 48, 49, 50, 53, 54, 55, 56, 57, 64, 65, 66, 69, 72, 84, 87, 91, 92, 96, 97, 99, 103, 104, 109, 110, 116, 120, 121, 122, 126, 135, 137, 140, 141, 143, 144, 145, 147, 148, 149, 150, 162, 164, 165, 170, 171, 183, 186, 187, 188, 189, 198, 204, 205, 207, 208, 209, 213, 215, 223, 224, 225, 226, 227, 231, 232, 233, 234, 238, 248, 249, 252, 258, 259, 260, 261, 265, 266, 283, 284, 285, 290, 301, 302, 303, 304, 305, 340, 343, 344, 345, 355, 356, 357, 358, 359, 363, 364, 379, 383, 386, 388, 390, 394, 400, 405, 408, 409, 419, 443, 447, 448, 489, 510, 511, 515, 516, 519, 521, 525, 528, 531, 534, 535, 536, 537, 540, 542, 543, 545, 546, 547, 549, 550, 551, 552, 560, 561, 566, 567, 568, 607, 608, 633, 634, 668, 672, 673, 700, 701, 744, 745, 792, 818, 820, 836, 848, 864

Three body (105 rows)

Celestial mechanics: orbits, triple stars, hierarchical systems, resonances in orbits, the Moon and planets. First domain of 48 of them.

1, 9b, 15, 30, 35, 79, 87, 100, 104, 129, 152, 160, 168, 174, 194, 201, 267, 295, 313, 315, 318, 336, 338, 340, 346, 348, 402, 406, 450, 451, 456, 506, 541, 544, 552, 555, 558, 573, 575, 599, 603, 614, 615, 616, 617, 618, 619, 620, 621, 622, 626, 627, 639, 640, 641, 644, 645, 648, 650, 651, 660, 665, 674, 675, 678, 680, 682, 683, 687, 688, 689, 690, 691, 692, 693, 694, 713, 715, 718, 719, 726, 729, 730, 732, 741, 743, 744, 766, 780, 781, 782, 783, 784, 795, 797, 800, 808, 809, 810, 811, 848, 858, 859, 860, 863

Waves (102 rows)

Waves, optics and fields: light, orbital angular momentum, vortex beams, sound, surface and standing waves, electromagnetism, fluids as waves. First domain of 53 of them.

4, 6, 31, 32, 37, 40, 42, 47, 48, 49, 50, 52, 53, 54, 57, 85, 105, 108, 112, 127, 134, 135, 136, 147, 165, 167, 202, 232, 271, 272, 283, 284, 286, 287, 296, 298, 321, 350, 352, 354, 367, 372, 373, 374, 375, 376, 378, 382, 386, 391, 392, 393, 394, 395, 397, 406, 410, 419, 449, 452, 453, 456, 457, 459, 468, 469, 474, 478, 479, 480, 481, 485, 486, 500, 514, 532, 545, 546, 547, 554, 561, 590, 607, 643, 653, 723, 724, 763, 768, 792, 803, 804, 806, 807, 819, 822, 829, 830, 831, 847, 858, 865

Control and circuits (99 rows)

Control, circuits and engineering: relays, controllers, feedback, power grids and microgrids, valves, brakes, mechanisms built to be used. First domain of 45 of them.

5, 7, 24, 25, 38, 40, 51, 70, 119, 235, 268, 269, 270, 272, 273, 274, 280, 282, 285, 288, 291, 292, 295, 296, 350, 355, 366, 367, 371, 376, 381, 395, 396, 398, 403, 404, 405, 425, 426, 428, 429, 430, 444, 454, 468, 469, 470, 471, 472, 473, 475, 476, 478, 482, 483, 484, 485, 486, 487, 488, 491, 496, 497, 499, 513, 563, 590, 604, 615, 623, 624, 625, 631, 635, 636, 658, 659, 669, 672, 727, 745, 746, 753, 754, 762, 763, 769, 788, 824, 870, 873, 876, 878, 879, 883, 884, 885, 888, 889

Biology (91 rows)

Biology, medicine and molecules: proteins, DNA, molecular knots, hearing and the cochlea, jellyfish, whales, ecology, physiology. First domain of 59 of them.

2, 4, 5, 6, 21, 37, 39, 63, 69, 76, 77, 78, 93, 94, 95, 151, 218, 247, 248, 249, 250, 251, 252, 253, 254, 255, 256, 257, 258, 259, 261, 262, 263, 264, 267, 268, 269, 273, 274, 277, 278, 279, 289, 290, 292, 293, 294, 329, 330, 331, 332, 393, 399, 401, 423, 424, 432, 439, 440, 441, 444, 445, 446, 447, 460, 461, 462, 463, 464, 465, 466, 467, 490, 491, 492, 493, 494, 495, 496, 497, 498, 499, 500, 501, 595, 596, 598, 653, 762, 768, 832

Materials (73 rows)

Materials and condensed matter: superconducting rings, skyrmions, crystals, latches, metallurgy, polymers and filaments. First domain of 52 of them.

12, 43, 49, 51, 56, 57, 60, 63, 64, 65, 66, 84, 144, 145, 229, 230, 231, 233, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 277, 278, 279, 280, 286, 293, 305, 332, 358, 361, 362, 363, 371, 378, 379, 381, 382, 390, 421, 439, 440, 443, 446, 447, 448, 466, 515, 525, 526, 527, 529, 532, 548, 549, 550, 560, 561, 562, 563, 564, 565, 568, 804

Number theory (72 rows)

Primes, integers, Farey sequences, continued fractions, zeta, Benford, arithmetic lattices. First domain of 47 of them.

38, 146, 170, 171, 172, 175, 176, 177, 179, 180, 181, 182, 183, 184, 185, 192, 193, 195, 196, 197, 198, 199, 200, 201, 202, 203, 206, 207, 208, 209, 211, 212, 213, 214, 215, 216, 217, 218, 219, 223, 224, 225, 226, 237, 384, 402, 410, 411, 413, 414, 415, 417, 419, 420, 517, 518, 520, 521, 522, 523, 526, 528, 529, 531, 548, 605, 646, 671, 702, 719, 790, 864

Kinematics (71 rows)

Motion and geometric phase: frames carried round loops, falling cats, divers, swimmers, rolling, Berry, Hannay and Foucault phases, gears read as motion. First domain of 32 of them.

9b, 17, 35, 52, 92, 116, 141, 143, 151, 174, 186, 188, 190, 214, 227, 228, 242, 259, 295, 339, 341, 346, 347, 349, 351, 352, 357, 360, 362, 366, 380, 389, 399, 404, 420, 422, 471, 519, 581, 584, 585, 586, 587, 601, 602, 606, 607, 608, 609, 612, 630, 635, 637, 643, 657, 668, 681, 700, 701, 703, 764, 781, 792, 806, 813, 814, 816, 817, 818, 820, 836

Maps and chaos (60 rows)

Iterated maps, period doubling, windows, crises, Feigenbaum ladders, chaotic transients, basins; the Loop Atlas’s needle when it is read as a map. First domain of 26 of them.

369, 416, 514, 580, 605, 616, 619, 632, 640, 642, 654, 655, 662, 664, 665, 670, 671, 675, 680, 685, 694, 695, 696, 697, 698, 699, 705, 707, 708, 715, 723, 724, 738, 739, 743, 749, 750, 751, 752, 755, 756, 757, 758, 759, 760, 761, 771, 787, 791, 794, 822, 830, 839, 872, 873, 874, 875, 877, 886, 887

Quadrature (51 rows)

Phase and signal reading: quadrature, beats, spectra, lock-in, harmonic mixing, sampling, filters, timing, clocks and event time. First domain of 25 of them.

8, 11, 16, 17, 20, 59, 104, 172, 173, 176, 177, 178, 179, 180, 189, 304, 317, 318, 319, 350, 351, 353, 364, 434, 453, 520, 522, 530, 537, 627, 628, 629, 630, 638, 641, 645, 656, 659, 663, 681, 693, 700, 719, 802, 803, 806, 824, 837, 847, 876, 878

Geodesy (38 rows)

Geodesy and seismology: GPS, faults, Cascadia, earthquakes, crustal motion, tides of the solid Earth. First domain of 27 of them.

1, 10, 11, 12, 13, 14, 15, 18, 19, 20, 22, 97, 98, 137, 138, 140, 153, 160, 161, 163, 171, 210, 306, 307, 308, 309, 310, 311, 313, 314, 315, 316, 368, 370, 375, 376, 391, 544

Reading pictures (32 rows)

Pictures of data and the eye: strobe pictures, lattices drawn on pictures, blind reads, marked figures, colour scales. First domain of 18 of them.

30, 128, 199, 216, 217, 299, 442, 573, 585, 589, 613, 667, 701, 702, 703, 704, 705, 706, 707, 708, 709, 710, 711, 712, 713, 714, 715, 716, 717, 743, 744, 873

Solar (20 rows)

Solar and space physics: the Sun, magnetograms, sunspots, flares, space weather. First domain of 15 of them.

315, 318, 319, 321, 452, 543, 554, 555, 556, 557, 558, 559, 566, 567, 575, 576, 577, 578, 579, 633

Appendix — Transfers, row by row

Every place the register records a lesson, a tool, a law or a reading carried from one system or setting to another, with how it fared there and, when it did not hold, whose break it was (the owners are defined in “How the lessons travelled”). The words of each row that record its transfer travel with the data, in FieldGuide_v7_TRANSFERS.csv.

Row	Carried from	To	Outcome	Owner
7	one shared noise-correlation axis	count defences such as a Byzantine vote	failed	knob
7	one shared noise-correlation axis	a phase-locked swarm	failed	wall
9b	BELL_CLOCK’s one case at memory 6	tied bells at memory 3	failed	seam
9b	bells’ crossing before cores were softened	tied bells with softened cores	failed	ruler
12	winding-count reads (discrete Stokes)	a ring of writers (Latch Atlas)	held	—
16	matched filter at noise σ = 0.3	the same filter at noise σ = 0.6	failed	ruler
20	event time for reading seismic signals	an event trigger driving the memory bell	failed	seam
26	a rigid (static) liar	a fully adaptive, opponent-dependent liar	failed	wall
27	blackjack, where the deck is sovereign	poker, where other minds are the substrate	failed	substrate
35	the falling-cat law	three bodies	held	—
38	oscillators that lock by a tongue	a low-pass filter subsystem	failed	substrate
38	the first gearbox stage	the second stage, chained through a relay	held	—
42	ring open, mass balance valid	ring lip sealed	failed	seam
44	the ℤ/2 structure group	symmetry-reduced rotations round a loop	failed	wall
49	membrane or gel (Kac exact)	a stiff plate (biharmonic)	weaker	substrate
49	an elastic solid (vector waves)	a fluid cavity (scalar only)	failed	substrate
50	winding code under local damage	winding code under smooth global deformation	failed	wall
50	magnitude code under smooth global deformation	magnitude code under local damage	failed	wall
59	correlated axes sharing the same physics	axes that do not interfere	failed	substrate
63	over-binning in the read domain	defect-density collapse in the medium	weaker	not stated
66	two-codes-opposite-enemies (winding versus magnitude codec)	self-healing topological memory gel	held	—
72	i.i.d. noise	streaks aligned with the logical operator	failed	wall
79	power law from one table’s data	next rows and the junction	held	—
79	gap’s power law at strength 0.05	strength 0.10	failed	knob
79	children-of-1/2 question at strength 0.05	strength 0.10	held	—
79	R61_AFTER lean at RATCHET-61’s first size	RATCHET-61’s second size	held	—
84	static matched offsetting liars	the adaptive liar	failed	wall
87	textbook staircase and separatrix shapes	triple star’s jump at 1/10	failed	ruler
89	author’s three-layer trust model	author’s fear the work was lost	held	—
89	a run (row 720)	a 23-day Field Guide session	held	—
93	cross-domain loop instrument	whale foraging	held	—
93	cross-domain loop instrument	storm closure depth	held	—
93	cross-domain loop instrument	Rivera case	held	—
94	human-touched substrates with adaptive liars	sky, genetics and weather data	failed	substrate

97	smooth-curvature interiors (many-vertex loops)	a discontinuity (the Parkfield seam)	failed	seam
98	outliers and static liars	the coordinated, frame-variant lie	failed	wall
103	the Integer Forest’s quantized channel	a rider driven by two waves	held	—
103	rider slip counts on a laptop	the same riders in the container	held	—
103	rider stretch rates on a laptop	the same riders in the container	weaker	not stated
103	high-precision readings on a laptop	the same readings in the container	held	—
104	modulus-1 projection of winding phase	triple-star jump counts on a ring	failed	wall
106	sparse defects at low-to-moderate noise	high noise, for all three gates	failed	wall
116	winding read on a two-dimensional field	one-dimensional sensor ring of bearings	failed	substrate
116	codimension-2 rule on the sensor ring	car on a banked turn	held	—
127	the crude proxy (correlation 0.97)	the first real-data storm run	weaker	ruler
128	Bear’s pictures: dipole and mirror, staging ground	R2W’s rider (row 712)	held	—
128	Bear’s machine sense: blades, phased array, groups	R2W’s rider, before the runs	held	—
129	the shape sphere away from the pole	near the pole, one pair tight	failed	ruler
129	one earlier run’s laptop pace	later runs of the same sweep	failed	not stated
129	the earlier (afternoon) laptop pace	the same laptop through the night	failed	not stated
145	row 109’s break-species	a single atom (shell structure, QTAIM)	held	—
146	hydrogen electron shells	prime-factorization levels (tower)	failed	substrate
147	Chern–Simons framing theory	a vortex tube in viscous fluid	weaker	substrate
151	adiabatic theorem in the Stokes limit	JELLY-1 swimmer at Re 30	failed	seam
151	loops’ shape and pace split	a body’s turn in the air	held	—
151	no-area rule (row 116)	straight leg swung in a stride	held	—
157	whole-file rule for rebuilt files	R2W crate deltas	held	—
160	an areal loop, built for areal sources	a line source (fault strain transients)	failed	substrate
165	a synthetic loop channel (July 2026)	a signed ring in viscous fluid	held	—
167	closed matter-dominated universe	universe with dark energy	failed	seam
171	Cascadia’s triangle lattice	the integer forest’s loops	failed	substrate
171	Cascadia’s read-on-the-faces principle	the integer forest’s loops	held	—
174	closed-form half-turn root	1/2 groove at ripple strength 0.25	held	—
180	real telescope arrays, where elements cost money	the filled aperture, charged in barrel	failed	substrate
186	an open chain (the worm)	the chain closed into a ring	weaker	wall
188	loop walks around a buried chest	test of one candidate map	held	—
188	falling cats with equal halves	falling cats with unequal halves	failed	not stated
190	time retrace on a slipping rider	a locked rotor, conveyor and clock blade	failed	seam
194	eccentricity 0.20 scans	atlas up to eccentricity 0.24-0.28	held	—
194	eccentricity 0.20 scans	eccentricity 0.30 and above	failed	seam
200	a diffusive random walk (exponent 0.5)	the prime-factor construction’s horizontal spread	failed	substrate
202	Pythagorean comma (music tuning)	construction’s closure-deficit constants	failed	not stated
207	the parent’s graph-spectral flat-wall claim	rotation-minimising transport on the	held	—

		commuting square		
208	the Magnetic Forest’s clone gauge (beam phase)	the cap holonomy (rotation conjugation)	held	—
210	seismic checker’s atan2 rotation-angle repair	axis-tolerance check in later rooms	held	—
211	golden heading angle	heading angles 100°, 90°, 137°	weaker	knob
214	crossover derivation at weights looked at	four weights not yet looked at	held	—
214	TWELVE-1’s last-digit identity at nulls	Loop Atlas thick and thin cats	held	—
219	Weyl’s theorem on {n·α}	the leftover of log n over log p	failed	substrate
222	record linter’s coverage check	triple-star batch parser (KW0)	held	—
228	convex polygons	fat patches with reflex angles	failed	wall
237	integer-forest address paper (row 195)	topological storage address space	held	—
238	anything built from misorientation angles	rotors composed around a loop	failed	ruler
239	angle-based reads on one platform	the same reads on another platform	held	—
239	rotor selection on one platform	the same file and script elsewhere	weaker	knob
241	shipped gauge	three other gauges	failed	ruler
243	noise widths up to 2°	noise widths of 10° and 30°	failed	seam
247	fields that trust replicate agreement	molecular-scale closure replicates	failed	not stated
249	knots the Alexander determinant sees	Kinoshita–Terasaka mutant knots	failed	wall
252	reduced knot diagram (Cerf 1997)	open-arc writhe of protein backbone	failed	substrate
257	one shared floor value	RNA, protein and DNA duplex substrates	failed	substrate
258	closed curves, where the defect is holonomy	open arcs of the same structure	failed	substrate
261	DNA (duplex helical axis)	RNA (single-stranded backbone)	failed	substrate
262	small proteins the instrument was built on	real rRNA with bridged gaps	failed	substrate
267	knot paper’s positioning paragraph	triple-star islands paper’s abstract	held	—
268	artificial-life claim	device somebody else pilots	held	—
272	the design posture in an ocean	a body and a vacuum	held	—
278	inherited topological memory	culture transmitted, not inherited	failed	wall
282	regime where B − A stays below 1	strength 0.05, 3/5 root	failed	seam
284	the mathematical integer-channel argument	a communications winding reader	held	—
294	the tornado work and the Cascadia arc	a clinical loop-walking array design	held	—
296	row 160’s paper (aperture as constraint)	the swarm array (aperture as knob)	held	—
304	the loop channel as a derivative operator	three unrelated instruments in one session	held	—
305	a hedgehog at low per-vertex noise	the same hedgehog at high noise	failed	ruler
308	plate-fixed frame assumption	data’s own secular frame	failed	ruler
309	single-culprit, per-station masking	a correlated maintenance campaign	failed	ruler
311	a different region’s null	held-out control region	failed	ruler
314	denser arrays elsewhere in the programme	this network at this sample size	failed	ruler
314	fitted strengths and frequencies	every strength and frequency run	held	—
314	single-tone drive	second tone at twice the frequency	weaker	knob
314	RATCHET-4 reference setting	frequency 0.70	held	—
314	RATCHET-4 reference setting	frequency 0.40	failed	knob

314	sitting B9’s lesson 73 (two tones)	RATCHET rectifier runs	held	—
317	programme’s home domain	a new domain	held	—
318	the matched-filter law	splittings read off a spectrum	held	—
321	the programme’s half-step, double-cover thread	odd-degree stellar oscillation modes	held	—
321	a sealed core (pristine ladder)	an open core	failed	not stated
322	this programme’s discipline	standard work in another field	held	—
328	literature’s harmonic-mixing drift	book’s step lopsidedness (Part 7)	weaker	not stated
329	the engineering coarse-read-first principle	animal place memory (forage patch, depleted)	held	—
332	the crystal side (durable substrate)	the biology side (living substrate)	held	—
342	simple test particles in a complex field	individually complex, jittering objects	failed	substrate
346	floating snake’s ride law	toy sail dead downwind	failed	wall
346	floating snake’s ride law	toy sail’s general power balance	held	—
350	a standing wave (E, B in quadrature)	a travelling light wave	failed	substrate
351	the radio rotor (needle, crossed fields)	the radio conveyor (bead, standing waves)	held	—
352	two coupled knobs (a quarter cycle)	a hinged chain body (worm toys)	weaker	substrate
358	the integer winding of vectors	nematic director defects	weaker	ruler
360	the two-quantities-one-name trap (row 193)	sensor anchor versus pose space	held	—
367	a sensor geometry, reading	an emitter, projecting	weaker	not stated
369	map already seen	fresh grid, read blind	held	—
369	map already seen	fresh grid, read blind	failed	not stated
369	strobe’s known bias near a doubling	survey data, four starts	held	—
371	row 329’s binning window (animal memory)	a three-pin swell with odd ripples	held	—
374	row 160’s aperture law	a surface station loop round a vortex	held	—
375	rock (low-pass, integrates strain)	air (high-pass, no quiet baseline)	weaker	substrate
376	rock, where the integrator comes free	air, read by a vane	failed	substrate
378	the polygon-rung intuition	filled atomic shells	weaker	wall
380	row 352’s hair-off-lock law	the Loop Atlas scallop and spring bead	held	—
391	the tectonic thread (low-pass rock)	the atmospheric thread (high-pass air)	failed	substrate
399	three scalar implementations	vectorised rewrite	held	—
399	Python parent runs of record	Loop Atlas jumper in JavaScript	held	—
400	RATCHET-53 reader’s exact-zero fix	RATCHET-58 reader	held	—
401	small structures the instrument was built on	larger structures, one chain with twelve gaps	failed	substrate
402	rung sightings in the triple’s boundary layer	fresh phases	failed	not stated
402	the standing watch for rungs at 7 and 8	the islands’ lap ratio, a different variable	held	—
404	the belt test on the mechanical tier	the tied bells in a remembering water	held	—
405	topological knot	mechanical belt tier	weaker	substrate
405	topological winding	electronic divider tier	weaker	substrate
406	LS1 bar on the circular pair	pair at 0.25	failed	ruler

410	musical comma (Tower paper)	substrate constant 35.81 degrees	failed	not stated
410	musical comma as log linear form	Baker-floor table’s (19,12) row	held	—
415	histogram bin-width selection	the ladder’s walk-length rungs	weaker	substrate
416	Bear’s note A Coincidence Has No Single Price	Loop Atlas sessions’ misses	held	—
416	four geometries	sweep over slope and cutting-circle size	weaker	knob
416	habitual root-scan range	ripple strength 0.25	failed	ruler
416	Bear’s note: widen the window	R2W seam reading: refine the ruler	held	—
420	the comma, a loop deficit of pure arithmetic	a swimmer’s mirror residual	weaker	seam
425	the honest setting, where sensors admit uncertainty	past the corruption threshold	failed	seam
426	row 309 maintenance and row 291’s friend	a drifter corrupting with no agenda	held	—
428	no corruption, where centralised pooling wins	a targeted adversary at ten percent corruption	failed	seam
429	a robust median against noisy sensors	a captured hub under targeted attack	failed	wall
433	assembled Field Guide method	a new project	held	—
434	Field Guide quadrature rows	a live project session	held	—
438	a control loop defending a corrupt setpoint	a pipeline and a game board	held	—
444	the vessel paper, where restoring force is control	an architecture meant to evolve	failed	knob
446	crystal substrate and simulation (the null)	biology: the prion	held	—
447	life as organisation (artificial life)	the substrate artificial life chose	failed	substrate
452	independent channel noise	a common driver such as a thunderstorm	failed	wall
460	the corpus’s differential read of sensors	commodity prices under one shock	held	—
464	four soils’ partition ratios	R2W toy machine’s edge shifts	held	—
466	stirred suspension	a field’s plough layer	failed	substrate
467	particle-capture work in radiotherapy	soil science (phosphorus targeting)	failed	not stated
468	an energy absorber, where resonance is the objective	a structural member, where resonance is failure	failed	knob
468	rows 269 and 444 (one property, two verdicts)	absorber and structure rules	held	—
469	forced-resonance check (detuning)	aeroelastic flutter, the 1940 bridge	failed	substrate
470	forced resonance with a periodic driver	self-excited galloping under steady flow	failed	substrate
479	the regular-wave power expression	irregular seas	weaker	not stated
482	this paper’s 93 kWh per tonne	a deployed off-grid cooling system	held	—
490	patients whose failed stage is cochlear	the population whose broken stages lie elsewhere	failed	substrate
491	a stable deaf ear (open-loop design)	a nerve under slowly increasing compression	failed	seam
491	rows 269, 444 and 468	the cochlear device’s open-loop decision	held	—
492	rate-over-level (rows 105, 294, 460)	a clinical residual read from serial imaging	held	—
496	phone governance mechanisms	an implant	failed	substrate
500	an earlier route closed by the wavelength	an implantable audio-frequency resonator	held	—
502	the knot theorem’s writhe (reduced diagram)	the open-arc writhe of the backbone	failed	ruler

502	the textbook Milne-Eddington model’s azimuth	HMI’s azimuth, Stokes Q sign reversed	failed	ruler
502	the radio-driven junction	R2W’s rider	failed	not stated
502	an error caught on R2W’s page	this guide’s own drafted row 727	held	—
502	dynamics’ fold (a saddle-node bifurcation)	a triple slice’s highest point	failed	substrate
504	method lost between sessions	counts inside one session	held	—
506	Mardling’s harmonic formulas	triple star’s rebuilt harmonic	held	—
506	Holman and Wiegert’s own example	new tool triple_hw_edge v1.0	held	—
506	Reimann and co-authors’ exact formula	Loop Atlas’s noisy needle	held	—
506	two papers placed from abstracts	triple star’s edge and test target	failed	not stated
510	distributed-systems trust	discrete differential geometry	held	—
513	row 502’s vocabulary fence in papers	control-room simulator’s interface note	held	—
515	the winding-must-close logic	atomic shells and the periodic table	held	—
517	numbers above one, every prime included	values below one	failed	seam
518	row 502’s rule, on borrowed words	two of one’s own ideas	held	—
525	periodic lattice topological invariant (TKNN)	finite, irregular, open patch	failed	substrate
526	the comb’s divisor reading	companion lattice, open chain	failed	knob
526	companion lattice’s 0.75	the latch’s three-halves	failed	not stated
528	q = 5 on a ten-row lattice	the same tooth on a twelve-row lattice	failed	knob
529	half-flux tooth on an open patch	composite periodic lattices with a wrap	failed	not stated
531	a closed chain of bonds (graph comb)	a ring conveyor rider (Loop Atlas)	held	—
531	the closed chain	the same chain left open	failed	substrate
532	latch-and-battery architecture, ring lattice	vortex ring at held circulation	held	—
532	ring lattice’s scaling dial	vortex ring at held circulation	failed	not stated
537	rate-over-level’s earlier arrivals	the walker’s ternary rate decomposition	held	—
543	Cascadia’s standing texture	nebula dust polarization map	held	—
543	the Latch Atlas’s headless count	HMI azimuth on real Sun maps	held	—
544	the Cascadia post-hoc build’s V3 degeneracy	small-area contrast bins	held	—
544	degenerate contrast bins	triple-star registration medians of three	held	—
547	the DNA spoke’s writhe reader	vortex filament evolver	held	—
548	the tower comb line’s gap ratios	the latch filament energy exponent	failed	ruler
551	gauge-invariance receipt certifying an apparatus (row 170)	forest suite’s genus flux-ladder count	failed	wall
554	Station 1, the word-definition trap	the Sun’s azimuth comparison	held	—
554	the textbook Milne-Eddington azimuth convention	HMI’s reported azimuth	failed	knob
554	the Sun azimuth sign fault	the Loop Atlas polarisation toy	held	—
555	second fundamental model of resonance	triple star’s island	weaker	not stated
557	the crossover at 34 mÅ width	planted widths of 25 and 50 mÅ	weaker	ruler
558	sunspot turning’s two estimators	triple-star resonance angle’s slide	held	—
558	end-to-end rate at birth 7.200	median step at birth 7.210	failed	knob
558	rule rebuilt on median net turns	two fresh births	held	—

560	the smooth textbook formula, in the limit	real grids of skyrmion whirls	weaker	ruler
561	latches in matter (Latch Atlas media)	twisted light, radio and sound	failed	substrate
565	Bear’s Savonius rotor drawing	two dials pulling on each other	held	—
566	loops whose rims stayed readable	flare-site loops whose rims went weak	failed	ruler
573	weights 0.0025 to 0.64 (fixed slide ratio)	weights 0.768 and beyond	failed	seam
575	Sun’s polarity split	third star’s apsidal motion	held	—
575	polarity split	R2W silent pair’s trim	held	—
577	SAVNCPP on 82 regions	six tracked regions	failed	not stated
579	rotation size fitted on 2011 regions	82 regions of a blind test	held	—
579	population’s rotation size	nine regions followed day by day	failed	not stated
579	population’s rotation sign	individual regions followed day by day	held	—
579	claim drawn from medians	single columns at weights 3.00 and 5.00	failed	seam
581	the Savonius-rotor picture, an engineer’s black box	a line field’s half residues (polarity)	weaker	wall
581	the drive ratio’s fine trim (term a)	the fourth-power term c	failed	not stated
581	Bear’s one-liner from the triple-star thread	R2W’s dark point law (RATCHET-53, 54)	held	—
581	whole and half steps, the trivial zeros	the fourth harmonic’s holding powers	weaker	not stated
581	the stretches where the evenness was seen	the next stretches (RATCHET-61)	weaker	seam
581	a dive in the air	rehearsal standing or on a tumbling mat	weaker	substrate
584	the one-knob rule of the Loop Atlas toys	a ratio test and a stiffness test	held	—
590	the draft horse toy	the listener’s ear	held	—
590	the guide’s common-mode rule	the listening blade (counter-charge toy)	held	—
590	the derivative kick the lean expected	a slope ear (SLOPE_EAR)	failed	not stated
593	Loop Atlas parser, checks and downloads	triple-star register’s promise about saved tables	held	—
599	one deposit’s closing section	triple-star islands paper’s escape count	held	—
604	the price card’s cells	strengths 0.4 to 0.5, where beads were caught	failed	seam
604	the turning needle	the swing equation (SWING_GRID)	held	—
605	rider with no latch (slipping conveyor)	latch (field winch past its limit)	failed	substrate
606	a synchromesh handover set by speed	a rider between two rhythms	weaker	knob
607	small strokes, read by the square rule	the big loop at 60 degrees	failed	seam
607	a polarisation loop walked by filters	the same loop walked by wave plates	held	—
607	paper pressed into ridges	ridges as steep as 76 degrees	held	—
613	one session’s no-physics read	another session’s registered prediction	held	—
613	three projects’ reads in the reads volume	one session’s own reads	failed	not stated
615	a gentle triple’s rests, weighted by time	a switch’s mistakes (relay, timer, ear)	held	—
618	runs whose binary never changed partners	run whose inner pair swapped 22 times	failed	substrate
618	V test on the coarse grid	a grid five times finer	failed	ruler
622	Loop Atlas sessions’ two fixed thresholds	triple-star rests between departures	held	—
627	the listening switch’s knee	a counter of the turning needle’s slips	held	—
627	a light third star	a heavy third star	failed	ruler
628	Crossover Atlas’s first chapter counters	tapping rider’s rest timed in seconds	held	—

628	Bear’s integer diagnostic	tapping rider’s gaps on two boards	held	—
628	Bear’s integer diagnostic	spring bead’s beat count	held	—
628	Loop Atlas rider taps on sample points	R2W DOOR-3 bar on a 0.001 sweep	held	—
629	the traders’ correction for a daily price	a knee ear watched now and then	weaker	seam
629	the reflection principle of probability	a watcher’s looks (BRIDGE_COUNT)	held	—
629	the slow watcher’s traders’ law correction	the never-blinking reference watchers	held	—
630	toys that count their own events	the skater	failed	substrate
630	the doubling test of the other toys	two tied bells in a remembering water	weaker	knob
631	triple island at lower eccentricity	two higher eccentricities, births 0.03 apart	failed	ruler
632	event-driven tapping riders	fold refiner run on two machines	weaker	seam
633	Crossover Atlas Floer market	Loop Atlas push price	failed	not stated
633	Crossover Atlas chapter’s Novikov object	Loop Atlas turning needle’s lag	held	—
633	Crossover Atlas second chapter	Loop Atlas rhyme test	held	—
633	Sun work polarity issue	soup toy’s local tilt	failed	ruler
634	a car on a banked track	the turning needle	failed	not stated
635	the capacitor’s rule	a force on a damped spring	held	—
635	a rigid pin’s cost ratio	a skateboard’s springy pin, sharp top-ups	weaker	knob
636	looks shorter than about a fall time	looks of 2 and 4 fall times	failed	seam
638	R2W’s advice	Loop Atlas book’s tapping rider	held	—
639	R2W climb-back reader	triple-star tread threshold	held	—
640	the fold’s edge, within 3×10^{-4}	three decades of distance	failed	seam
640	the first fold’s square-root law	the Loop Atlas needle’s gear 2 edge	held	—
640	period doubling	R2W’s stops at 0.40	held	—
640	six start lines near the mirror	fresh start lines farther out (E7c)	failed	seam
640	four stretches fitted by a straight line	the next two stretches (RATCHET-61)	weaker	ruler
640	the road first read, to 1.06	the road read on to 1.80	weaker	seam
641	noisy copies within one passing	a clutch’s frame within one lap	held	—
642	eight true folds of record	a doubling	failed	wall
643	Loop Atlas needle at dial 1.0	R2W rider at dial 1.0	failed	knob
643	Loop Atlas needle’s memory switch	R2W’s rider pendulum	weaker	knob
643	tied bells toy as first built	toy with cores given own size	held	—
644	triangle of stars at rest	same triangle carried 400 units aside	weaker	ruler
645	the three-beat rule for kicks and teeth	a beat of the beat, a level down	held	—
645	the Loop Atlas’s three-beat rule	the triple star’s 600-pass windows	held	—
646	prime-number theory	traders’ constant 0.5826 for sampled switch	failed	substrate
646	integer lens	traders’ constant 0.5826 for sampled switch	held	—
646	traders’ constant, the small-step limit	sampled switch dead band at coarser looks	weaker	knob
648	Figure 7’s line at 0.005 spacing	same line at 0.001 in birth	failed	ruler
650	the other room’s three-body tool	this room, run unchanged	held	—
653	a toy’s kick rule	the jellyfish solver	weaker	wall
654	steady drive at a lock’s edge	zigzag drive chopped into teeth	held	—

655	Feigenbaum’s ratio for iterated maps	heavy rider inside the gear	weaker	not stated
657	Bear’s falling-and-catching walking picture	the radio slope toy’s bead	held	—
658	Bear’s four-wire dimming picture	a relay’s kick in R2W	held	—
659	the FM reading of the rider	the filter’s slopes on and off the watershed	held	—
659	an error bar’s fall with the count	holding an array of riders in lock	weaker	not stated
660	free-ride law near ratio 2 (R2W)	triple star’s slip period	weaker	not stated
661	a limit borrowed from elsewhere	a tap in the Loop Atlas	failed	not stated
661	a one-line ratio law	five of eight slices over 600 passes	failed	ruler
661	a one-line ratio law	slices from the star’s eccentricity 0.26	failed	seam
662	one room’s ramps	another room’s ramp sixteen times slower	failed	ruler
662	speed solve on the 8/5’s upper edge	the 11/7 where the speed swung	failed	ruler
664	sine circle map below its critical line	sine circle map at strength 1.2	failed	seam
666	the relay’s steady-drive fold	the relay’s second stage under a faster filter	failed	seam
666	another room’s grid	this room’s registrations	failed	not stated
666	a sister session’s momentum-wall frame	the Loop Atlas book’s frame	held	—
666	the newest reading that held	the old law’s tables (DELTA-WEAK-1, FAR-1)	held	—
667	the traders’ law for a price checked daily	the Loop Atlas ear’s creep (CREEP_LAW)	held	—
667	the weightless rider’s crawl time	the measured rider at the 1/30 step	weaker	not stated
669	a truck’s brake heat on a mountain road	zigzag-driven rider with a heat store	held	—
670	room between gear 1 and gear 2	ribbon of the half groove beside gear 2	held	—
670	Bear’s truck road of 6 October	Loop Atlas book’s two-gear machine	held	—
671	monotone staircase below the fold	above gear 1’s fold	failed	seam
672	Bear’s half-dipole, reflected-image picture	R2W dial’s dark point, mirror line	held	—
674	the two walkers, R2W	a third star’s apsidal law	held	—
675	one-over-offset law far from locks	the triple	held	—
675	square-root law of a damped lock	conservative lock kicked once a lap	failed	substrate
675	the pendulum’s logarithm law	conservative lock’s two clean rims	held	—
676	R2W’s letters’ claims	Loop Atlas’s needle (FOLD_OR_FLIP, RUNG_ABOVE)	weaker	not stated
677	one room’s fixed-step integrator line	another room’s reference value	failed	ruler
677	one room’s cascade, passing speed its way	another room’s cascade	weaker	knob
678	record’s needle (memory 0.924)	needle rescaled to memory dial 1.0	failed	knob
678	grid of rooms from another room’s machine	this machine’s filter range	failed	ruler
678	bound for a push acting all the time	rider that pulses, then waits out delay	failed	substrate
678	container’s run pace	laptop’s run pace	failed	ruler
678	pendulum law	island’s rocking across eccentricities	weaker	wall
678	light star’s ragged sides	heavy star’s ragged sides	failed	not stated
678	laptop pace on one tool	fold refiner tool on the laptop	failed	knob
678	reversed rider’s fold weight	forward rider’s column	failed	knob
678	fourth harmonic’s strength at lower drive	higher drive ratios	failed	not stated
678	RATCHET-45’s lightest-weight tilt slope	RATCHET-69’s heavier weights	weaker	ruler

678	BALANCE_BOARD_B riders run one at a time	planted case, riders side by side	failed	knob
678	GAPS_READ run with a numerical fault	a planted case’s value	failed	ruler
680	Feigenbaum’s ratio for maps	the Loop Atlas’s light needle	held	—
680	quadratic-top maps’ 4.669	a textbook’s quartic example	failed	substrate
680	Briggs’s table (1991)	the Loop Atlas’s own maps	held	—
680	area-preserving map’s constant	a lightly leaky map	weaker	seam
680	conservative cascade, about 8.72 per step	the triple’s island 7 centre	failed	not stated
681	the listening rider on one swing	swings of 0.5 to 12.5 m	held	—
681	a clock set for a 2.5 m swing	swings of 0.5 to 12.5 m	failed	knob
681	a clock timed to small swings	big swings of 10 to 60 degrees	failed	knob
683	Loop Atlas predictions held for wrong reason	R2W memory floor at retrapping pull	held	—
684	ripple strength 0.30	ripple strengths 0.25 and 0.35	weaker	knob
685	Bear’s Chladni plate picture	rocking drive’s six-step dancers	weaker	not stated
686	row 416’s stride rule	SEAM_SCAN’s turning-point threshold	held	—
686	passing speed at the edge of the plain	lines high above the edge	failed	ruler
692	earlier long runs’ sense of fast	new test’s definition of fast chaos	failed	ruler
692	the atlas’s lock flag, half-turn band	FN0 test of the paper’s one-turn held	failed	ruler
693	rope-beat test at birth 7.190	the same test at birth 7.220	failed	substrate
693	floor set to stay off the slow beat	nearly round orbit, rocking below the floor	failed	ruler
693	whole run’s 6,000-pass apse turn	HM1 reader’s first 2,400 passes	failed	knob
695	roundings the scaling law came from	two new roundings	held	—
695	traders’ square-root law (row 629)	the needle knee’s extra size	held	—
695	square-root law’s power 0.5	the flips’ power against knee	failed	ruler
697	the third face’s 2/3 window	the coin’s tails 1/2 window	held	—
699	feedback amplifier’s closed-loop gain	R2W reduced closing condition	held	—
700	phase unwrapping (Itoh, 1982)	frame carried round widening loops	held	—
703	zigzag gauge on an even hill	hill whose slope changes along a leg	weaker	ruler
705	R2W’s record, tassel from 2/3	Loop Atlas quarter tree, tassel from 1/4	held	—
708	first three doublings	fourth doubling	weaker	ruler
709	rows outside the hand-off window	inside the hand-off window	failed	seam
711	Bear’s lattice for reading pictures	needle’s doubling staircase with known splits	weaker	not stated
711	textbook estimate 4/(π√m)	R2W stopping pulls at weights 10, 5, 2	weaker	not stated
711	RD1a′ planted test at 1/8, 1/9	RD1a′ planted test at 1/10, 1/11	failed	ruler
712	another room’s pictures, read blind	the data behind the pictures	weaker	not stated
712	Bear’s half-dipole, reflected-image picture	R2W tone-direction effect (RATCHET-44)	held	—
713	R2W’s reading of the pictures’ pixels	triple-star sessions’ numbers	held	—
714	colour rule on one panel	next panel, paler dots	failed	ruler
714	rebuilt figure’s scale	Bear’s hand-marked saved file	failed	ruler
715	dead zones drawn on an older island map	next two sets of triple-star runs	held	—
715	the drawn dead zones	rescaled drawing made without the person	failed	not stated

719	the 1:3 gear at birth 7.200	birth 7.230	failed	seam
719	the net-turn gear, first specimen	a second specimen, run blind	held	—
719	a single-step gear at 7.200	birth 7.210	failed	not stated
721	the session’s container	Bear’s laptop Downloads folder	failed	substrate
721	kit companion files found by content	R2W’s explorer returned under another name	held	—
723	needle lifetimes’ shape bar, LOCK_FROM_CHAOS	smoke-ring trains’ lean, TRAIN_EDGE	held	—
724	train of three rings breaking	the stop’s door, LEAK_RATE	held	—
725	Kramers’ escape law (high-barrier formula)	noisy needle at barriers 4 to 9	weaker	seam
726	Josephson junction’s Shapiro steps	R2W’s rider	weaker	substrate
726	textbook exact stopping values	R2W rider with ripple off	held	—
726	inscribed-angle theorem	triple star’s 24-gon clockwork	held	—
727	R2W rider, ripple off	generator rotor, phase-locked loop, biased junction	held	—
727	R2W rider, ripple on	a particle in two waves	held	—
727	R2W rider, ripple on	a junction driven by radio waves	failed	substrate
727	the R2W rider’s pendulum map	three-body problem and seismic rings	weaker	not stated
734	earlier runs’ time-symmetric drives	second tone with a relative phase	failed	substrate
734	symmetry lines on a three-body picture	R2W’s 1/2 dance stops	held	—
734	exact symmetries of the physics	new path through the engine’s code	held	—
737	weightless limit, weights 0.02 and 0.05	first-order bars at weights 0.10 to 0.40	weaker	seam
737	symmetry argument at the no-honey limit	drift’s zero as weight grows (RATCHET-23)	failed	knob
737	counted powers, exact as drive goes to zero	the book’s drive (BEAT-2)	weaker	seam
738	row 793’s rule for predictions	every file of record (STEP-1)	held	—
739	guesses fitted between −0.12 and −0.07	far cells out to −0.22, heavy weight	failed	ruler
741	heavy-window figure’s bunched dots (row 714)	little cluster near memory 1.72	held	—
741	forgetting pulsed needle’s one-period map	pulsed needle with memory	failed	not stated
742	pair’s narrow band at strength 0.30	strengths 0.20 and 0.40	failed	knob
743	Bear’s squeeze checked in R2W	triple-star sessions’ registered reads	held	—
743	panel (a) colour rule	the paler panel	failed	ruler
743	triple-star strobe pictures	Loop Atlas quarter tree	held	—
745	Latch Atlas’s latch	a dance’s fraction across its tongue	held	—
745	quadrature between blades, ultrasonic motor	junction’s tone and second harmonic	weaker	not stated
745	odd blade-count rule of fans	the meter’s timing counts (ODD-1)	held	—
746	one bar set on a mixed fleet	the fleet’s quiet and loud rooms	weaker	ruler
747	the landing section (landing speed)	the barrier-top section	weaker	ruler
748	R2W’s chain-test sizing arithmetic	Loop Atlas’s FALL_CHAIN test	held	—
750	R2W’s finding (row 686)	Loop Atlas memory switch	held	—
750	sitting B9’s route, half-trace crossing 1	weightless passing map at frequency 0.40	held	—

750	grid edge matching fold at strength 0.30	turned ripple at strength 0.45	failed	seam
753	the textbook barrier law (no memory)	a memory-5 needle at pace 0.75	held	—
755	the stretch check (eigenvalues against determinant)	flagging round-off in a product	held	—
755	the pulsed needle’s running paces	the needle settled at rest, only ringing	failed	not stated
756	the one-dimensional crisis law	the needle’s 1/4 window crisis	held	—
757	one-hump map window law, 9/4	heavy needle’s 1/4 window	held	—
757	one-hump map’s 9/4	Hénon map windows, periods 8 to 18	held	—
757	one-hump map’s 9/4	Hénon map windows, periods 6 and 7	weaker	not stated
757	weight-proof edge’s numbers at one frequency	edges at frequencies 0.40, 0.55, 0.70	failed	knob
757	weight-proof edge’s law at one frequency	frequencies 0.40, 0.55 and 0.70	held	—
757	heavier needle’s doubled-cycle ladder	lighter needle (LADDER_LIGHT)	held	—
758	one-hump maps (window scaling 9/4)	the heavy needle’s window at memory 2	held	—
759	the textbook saddle log law	the needle’s climbs into the barrier top	held	—
759	the textbook saddle log law	the needle’s noisy falls off the top	weaker	seam
760	the needle’s 1/4 window, narrower than the step	the rider’s 3/2 tongue at mass 3.0	held	—
761	lighter memories (real-multiplier stability test)	the strongly squeezed map at memory 1.6	failed	ruler
762	fixed-distance column, word obeyed at once	column keeping a time gap	failed	knob
763	the open column of followers	followers closed into a ring	held	—
764	square law at couples 10 and 3	couple 1	failed	ruler
766	another setup’s run	a planted case’s value	failed	knob
768	ROPE-1’s narrow ring, open valve	a wide ring on a closed follower	failed	knob
769	Bear’s rotor-stall, grid pilot-tone idea	stalled spring beads (pilot tone toy)	held	—
769	bead’s nameplate tune	bead’s working pace under load	failed	knob
770	row 801’s two hands, the harmonic flip	the timing off the mirror line (RATCHET-45)	held	—
771	rounded-top maps’ Feigenbaum ratio 4.669	needle’s 1/4 window cascade	held	—
771	maps sharing 4.669, a page’s list	quartic-topped map	failed	substrate
771	one-hump maps’ 4.669	area-preserving map	failed	not stated
771	Feigenbaum’s limit 4.669	first rungs of needle’s ladder below chaos	weaker	not stated
771	integer forest logs and anti-logs	needle’s doubling ladder rungs	held	—
776	slice middles’ measured rate	off-mirror run’s time estimate	failed	not stated
776	rehearsal on copies of his folder	intake and folder check on his laptop	held	—
780	the Loop Atlas session’s two warnings	the triple-star edge-slope reading	held	—
781	sling cut as fraction of way up	big swings of the sling	failed	ruler
783	Georgakarakos’s tested separations	half the smallest separation he tested	weaker	not stated
784	container’s record of eight folds	Bear’s laptop	weaker	ruler
784	run of record on Bear’s laptop	KD0 reproduction on another machine	weaker	ruler
785	EAR_CURES relay switch counts	spring bead’s blade switch counts	held	—

787	crest window from a 400-unit run	a 600-unit run	weaker	ruler
787	margin held over 500 time units	ride of 1,000 time units	failed	knob
788	junction’s large-memory retrapping law	needle at memory 1	failed	seam
788	R2W’s room rule, slope −1	Loop Atlas’s needle relay	failed	seam
788	an engine logged off tach time	toy fleet’s calendar in running turns	held	—
788	flat point’s power at frequency 0.55	frequency 4 (FLAT-75)	failed	seam
788	single power from two swings	a third swing (FLAT-50)	failed	knob
790	habit: simpler fraction, wider tongue	heavy rider’s locks at mass 3.0	failed	not stated
790	lock-width laws at earlier cases	new cases (GROOVES-2)	held	—
790	golden clock ratio 0.618034	clock ratio 0.6	held	—
791	the strobe-Newton tool’s orbit following	period-8 orbit at mass 0.75, downward	failed	ruler
791	a ring of emitters fed half-turns	the needle’s doubling (a bent road)	weaker	substrate
792	filters carrying light round a loop	wave plates on the same loop	held	—
797	a pendulum’s edge law	the triple’s slip near the edge	weaker	not stated
797	a pendulum’s edge law	the Loop Atlas needle under noise	weaker	not stated
799	an argument about a limit (row 737)	the drift’s zero along another road	failed	not stated
799	exact quartic through five sizes	ladder rungs beyond its range	held	—
799	flat point’s share at one drive	another drive (FLAT-75)	failed	seam
799	flat point’s power from two swings	a third swing (FLAT-50)	failed	knob
801	R2W’s rectifier, sign flipped (RATCHET-44)	Loop Atlas’s tied bells	held	—
801	EDGE-85’s one-weight reader on mirror line	BEAT-4, a quarter turn off mirror	failed	substrate
803	three-element antenna array factor	three groups of three riders (NINE-1)	held	—
803	Bear’s staging ground, a holding pattern	array’s silent sets (STAGING_READING)	held	—
805	R2W’s size sweep on rider arrays	Loop Atlas crest-firer	failed	wall
805	R2W’s hand-alone, other-knob, half-turn tests	Loop Atlas crest-firer	held	—
810	workspace run of chaotic birth 7.180	the same birth on the laptop	failed	wall
810	SW1’s laptop run of 7.340	SL7’s workspace anchor at strict tier	failed	ruler
810	LX3’s births on one machine	check across machines, all births pooled	failed	wall
810	runs within one software stack	between the laptop and the containers	failed	substrate
813	FACE_ALONG’s halving probe, via tool pattern	EXPLORER_ROUTINE_CHECK’s planted tolerance	held	—
813	smooth paths	routines blended by half-cosines, with rate corners	failed	seam
814	Tong and Dullin’s Table 1 diver	the Loop Atlas’s engine rebuild	weaker	ruler
814	the Loop Atlas’s jumper and diver split	Tong and Dullin’s planar somersault (2019)	held	—
817	one back-and-forth knob (scallop theorem)	a coil turned on an axle	failed	substrate
817	two knobs moving out of step	end-over-end symmetric pumping coil	failed	wall
818	toy cat with equal halves	halves of 0.25 and 0.35	failed	knob
820	Montgomery’s equation (9), as read	the toy cats in flight	held	—
820	tubes hinged at their ends	the hinge moved in	held	—
820	Rademaker and ter Braak’s 1935 cat	the Loop Atlas’s twist-free cat	held	—

820	bend 90, the mirror’s fixed point	every other bend	failed	ruler
822	the session’s own machine	a second computer	failed	wall
822	one machine’s flight	the other machine	held	—
824	traders’ law (Broadie, Glasserman and Kou)	sampled switch’s extra knee, finer reference	held	—
824	looks many times within the noise’s memory	a look every four tenths of the memory or coarser	weaker	seam
825	the working book’s headline register page	the reader’s edition front page	weaker	substrate
826	Poisson count at 4 to 9 noise units	low barrier, 3 noise units	weaker	seam
830	middle-core trains’ squeeze door	fatter cores, up to 0.66 radius	weaker	knob
830	middle-core trains’ squeeze door	thin cores at their own edge	failed	knob
831	the course theorem of sailing (textbook)	the Loop Atlas’s sailing smoke test	weaker	ruler
832	the Loop Atlas’s made wall	insect wake capture and jellyfish’s parked ring	held	—
832	the Loop Atlas’s made wall	leading-edge vortex and clap and fling	failed	not stated
836	the steady circuit the proof used	nine paths whose bend swells and shrinks	held	—
837	smooth curves, error as grid squared	a reader of whole-number flips	failed	ruler
838	Python’s built-in sum on floats	Python’s built-in sum on complex numbers	failed	not stated
839	weights 0.55 to 0.90, chaotic room’s wall	past 0.905, overlap of two locks	failed	seam
840	STACKS tests (3/5 and 2/3 pair)	the pad atlas, blind	held	—
841	ripple running forward	ripple turned around (large shift)	failed	seam
842	container’s measured first two levels	deep levels of the 3/7 ladder	failed	seam
847	R2W’s beat parity rule	nonlinear optics, frequency doubling crystals	held	—

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Declaration

Preprint. This guide is a preprint, a checkpoint on a road: it has not been peer reviewed, and its search of the literature is limited. Its failures are stated and made instructive. It comes from a part-time research team that works in the open and gives its tools away.

AI assistance. This guide was assembled with AI assistance (Claude, Anthropic) for literature survey support, instrument building, instrument running, code generation, prose drafting, and iterative editing across multiple sessions. In the part *Two Readers and a Ruler*, Claude is also one of the two readers whose blind readings are scored, and the three projects’ sessions, each run with Claude, wrote the accounts the part is built from.

Integration papers. Version 6 integrated lessons from three companion programmes deposited in September 2026, and version 7 adds nine more: two that went live on 30 September 2026, the triple-star paper, published on 28 October 2026, R2W’s book, published on 5 November 2026, the Loop Atlas’s book, its record and its note on the odd-order constants, published on 9 November 2026, and the two-voices papers, published on 9 and 10 November 2026:

- *Jellyfish: the Holonomist of the Sea* — what a bell taught an instrument about memory, hovering, and the loop that encloses nothing (concept DOI 10.5281/zenodo.22908909)
- *Instruments for the Magnetic Forest* — a tool suite for the integer-forest lattice model, and what it verifies (concept DOI 10.5281/zenodo.22906780)
- *Linking Is the Latch* — reading the integer on a linked filament, and why a bare filament sets no floor (concept DOI 10.5281/zenodo.22906667)
- *The Latch Atlas* — holonomy and locks that can’t be picked (concept DOI 10.5281/zenodo.23068382)
- *Dancing on the Sun* — a loop count that reads through holonomy, and the current it finds flowing in both polarities (concept DOI 10.5281/zenodo.23067245)
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- *A Rider in Two Waves* — different music, the same moves (concept DOI 10.5281/zenodo.23999853)
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- *Question the Substrate* — why a learning mind can't grade its own homework (concept DOI 10.5281/zenodo.24094163)
- *Two Arrows* — why a learning mind heals its answers long before its self (concept DOI 10.5281/zenodo.24120415)

The three working projects. Version 7 also draws on three working projects of the same author, all now published (above): the triple-star sessions, whose paper is *Held, Rocking, Locked*; R2W, whose book is *A Rider in Two Waves*; and the Loop Atlas sessions, whose book is *The Loop Atlas*, published with its record and its note on the odd-order constants. Rows from them name their source in the text or the tag line, and their numbers are as each project reported them, most from a named run. The part *Two Readers and a Ruler* was written from the three projects' own accounts of how they read pictures, and its quotations of the author are his own words, with spelling and capitals put right and personal asides left out. Each project's record is in the list above and in the table below. The two-voices papers' lessons are rows 883 to 891 and additions to 20 earlier rows, and each names its paper.

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Where the rows come from

Each row of the register names the project or paper it came from, in its text or its tag line. This table counts the rows that name each, so a row that carries lessons from two sources is counted under both. The 440 rows that name none of them are version 6's own, all numbered 728 or below. The register holds 884 rows.

Source	Its record	Rows that name it
Version 6 of this guide, and the papers it drew on	concept DOI 10.5281/zenodo.21811588	440
The Loop Atlas sessions, *The Loop Atlas*	concept DOI 10.5281/zenodo.24100672	292
R2W, *A Rider in Two Waves* (the RF Spoke in its early records)	concept DOI 10.5281/zenodo.23999853	228
The triple-star sessions, *Held, Rocking, Locked*	concept DOI 10.5281/zenodo.23755473	166
Dancing on the Sun	concept DOI 10.5281/zenodo.23067245	23
The Latch Atlas	concept DOI 10.5281/zenodo.23068382	20
Jellyfish: the Holonomist of the Sea	concept DOI 10.5281/zenodo.22908909	4
Linking Is the Latch and *Instruments for the Magnetic Forest*	concept DOIs 10.5281/zenodo.22906667 and 10.5281/zenodo.22906780	2
Question the Substrate (the two-voices series, Paper 1)	concept DOI 10.5281/zenodo.24094163	21
Two Arrows (the two-voices series, Paper 2)	concept DOI 10.5281/zenodo.24120415	14
The Field Guide's own working sessions	this record	26

Tool and computational resource disclosure

Tools used. Claude (Anthropic) was used across multiple sessions for literature survey support, instrument building and running, code generation, prose drafting and iterative editing. Typesetting used pandoc and wkhtmltopdf.

Responsibility. Responsibility for the correctness and adequacy of the arguments and results presented here, and for the completeness and accuracy of the citations, rests entirely with the named human author. No automated system is an author of this work.

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Attribution limits. The literature pass is limited, as the preprint notice states. Rows tagged "ours" carry no located prior name and should be read as not located, not as new. The words "from recall" stand where a name or a law was given without a checked source; after the pass of 31 October they stand in rows 67, 351, 552, 605, 608, 635, 672, 675, 705, 727 and 769, and in the part Two Readers and a Ruler, section 7 (a radio discriminator's S-curve).

Computational resources. All instruments ran on a personal computer and in cloud containers; no specialised hardware was used.